

axiomTM



The 30 Year Horizon

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Volume BugList: Axiom Bugs

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New Foreword

On October 1, 2001 Axiom was withdrawn from the market and ended life as a commercial product. On September 3, 2002 Axiom was released under the Modified BSD license, including this document. On August 27, 2003 Axiom was released as free and open source software available for download from the Free Software Foundation's website, Savannah.

Work on Axiom has had the generous support of the Center for Algorithms and Interactive Scientific Computation (CAISS) at City College of New York. Special thanks go to Dr. Gilbert Baumslag for his support of the long term goal.

The online version of this documentation is roughly 1000 pages. In order to make printed versions we've broken it up into three volumes. The first volume is tutorial in nature. The second volume is for programmers. The third volume is reference material. We've also added a fourth volume for developers. All of these changes represent an experiment in print-on-demand delivery of documentation. Time will tell whether the experiment succeeded.

Axiom has been in existence for over thirty years. It is estimated to contain about three hundred man-years of research and has, as of September 3, 2003, 143 people listed in the credits. All of these people have contributed directly or indirectly to making Axiom available. Axiom is being passed to the next generation. I'm looking forward to future milestones.

With that in mind I've introduced the theme of the "30 year horizon". We must invent the tools that support the Computational Mathematician working 30 years from now. How will research be done when every bit of mathematical knowledge is online and instantly available? What happens when we scale Axiom by a factor of 100, giving us 1.1 million domains? How can we integrate theory with code? How will we integrate theorems and proofs of the mathematics with space-time complexity proofs and running code? What visualization tools are needed? How do we support the conceptual structures and semantics of mathematics in effective ways? How do we support results from the sciences? How do we teach the next generation to be effective Computational Mathematicians?

The "30 year horizon" is much nearer than it appears.

Tim Daly
CAISS, City College of New York
November 10, 2003 ((iHy))

Chapter 1

Introduction

1.1 The Numbering Scheme

bug 7324:
todo 342:
wish 1012:
meh 5:
errors 10016:
warnings 20574:
typos 40363:
dup 50006:
nonextend 60077:

Chapter 2

book0 Jenks

todo 341: merge endpapers with text

The books/endpaper.pamphlet should be added to the Jenks book

Chapter 3

book5 Interpreter

bug 7322:)co dh)constructor DHMATRIX)functions identity

```
)co dh )constructor DHMATRIX )functions identity
```

```
Compiling AXIOM source code from dh.spad
DHMATRIX abbreviates domain DenavitHartenbergMatrix
processing macro definition minrow ==> One
processing macro definition mincolumn ==> One
processing macro definition nx ==> ::(x(One,One),R)
processing macro definition ny ==> ::(x(2,One),R)
processing macro definition nz ==> ::(x(3,One),R)
processing macro definition ox ==> ::(x(One,2),R)
processing macro definition oy ==> ::(x(2,2),R)
processing macro definition oz ==> ::(x(3,2),R)
processing macro definition ax ==> ::(x(One,3),R)
processing macro definition ay ==> ::(x(2,3),R)
processing macro definition az ==> ::(x(3,3),R)
processing macro definition px ==> ::(x(One,4),R)
processing macro definition py ==> ::(x(2,4),R)
processing macro definition pz ==> ::(x(3,4),R)
processing macro definition row ==> Vector R
processing macro definition col ==> Vector R
processing macro definition radians ==> /(pi,180)
processing macro definition SIG ==> -- the constructor category
processing macro definition CODE ==> -- the constructor capsule
```

```
-----
initializing nrlib DHMATRIX for DenavitHartenbergMatrix
compiling into nrlib DHMATRIX
***** Domain: R already in scope
compiling exported identity : () -> $
```

Error: ERROR "Caught fatal error [memory may be damaged]"

Fast links are on: do (si::use-fast-links nil) for debugging
 Signalled by |COMPILERDOITWITHSCREENEDLISPLIB|.
 ERROR "Caught fatal error [memory may be damaged]"

Broken at APPLY. Type :H for Help.
 1 Return to top level.
 BOOT>>

bug 7321: missing spaces in)show output

```
)show INTABL
InnerTable(Key: SetCategory,Entry: SetCategory,addDom)where
  addDom: TableAggregate(Key,Entry)with
    finiteAggregate is a domain constructor
Abbreviation for InnerTable is INTABL
This constructor is not exposed in this frame.
Issue )edit bookvol10.3.pamphlet to see algebra source code for INTABL
```

todo 334: eliminate bcString2HyString

```
eliminate bcString2HyString
```

todo 333: eliminate bcFindString, replace with position

```
eliminate bcFindString, replace with position
```

todo 332: rename linkGen to bcGen in all source files

```
rename linkGen to bcGen in all source files. linkGen is obsolete.
```

bug 7250: print(typeOf(1)::OutputForm) Value Stack Overflow

```
(1) -> typeOf(1)
```

```
(1) PositiveInteger
```

Type: Domain


```

(2) -> typeOf(1)::OutputForm

      (2)  PositiveInteger()
                                                    Type: OutputForm
(3) -> print(typeOf(1)::OutputForm)

>> System error:
Value stack overflow.

```

bug 7262: bad expression parse

reported by Ralf Hemmecke)

$\sqrt{2}$

bug 7265: interpreter does early retract to Taylor series

```

Z ==> Integer
Q ==> Fraction Z
z:Symbol := 'z
L ==> UnivariateLaurentSeries(Q,z,0)
q:L := z::Variable(z)::L
t1:L := (1/((1-q)*(1-q^2)))
t2 := t1-1
recip t2
t3 := recip t2
t4:Union(L,"failed") := recip t2 -- bug in interpreter
t5:Union(L,"failed") := (recip t2)$L
t6:Union(L,"failed") := t3

```

bug 7269: hash does not work for Record

```

(1) -> T:=Record(f1:Integer)

      (1)  Record(f1: Integer)
                                                    Type: Domain
(2) -> hash(t)

      (2)  116
                                                    Type: SingleInteger
(3) -> t:=[0]@T

```

```

(3) [f1= 0]
                                                    Type: Record(f1: Integer)
(4) -> hash(t)
  There are 8 exposed and 1 unexposed library operations named hash
    having 1 argument(s) but none was determined to be applicable.
    Use HyperDoc Browse, or issue
                                )display op hash
    to learn more about the available operations. Perhaps
    package-calling the operation or using coercions on the arguments
    will allow you to apply the operation.

  Cannot find a definition or applicable library operation named hash
    with argument type(s)
                                Record(f1: Integer)

  Perhaps you should use "@" to indicate the required return type,
  or "$" to specify which version of the function you need.

```

bug 7268: Typechecker gets confused by flow control

Spad typechecker is supposed to track type tests, and allow operation which are safe due to dynamic tests. However, tis works for 'if' but fails for loops, as in:

```

a:R
b:R
c:=a exquo b
while c case R repeat
  a:=c
  c:=a exquo b

```

Spad compiler report type error on assignment to a.

But both of these work:

```

a:R
b:R
c:=a exquo b
if c case R then
  while c case R repeat
    a:=c
    c:=a exquo b

```

and

```

a:R

```

```

b:R
c:=a exquo b
while c case R repeat
  if c case R then
    a:=c
    c:=a exquo b

```

bug 7272:)savesystem does not build a restorable session

Raymond Rogers <raymond.rogers72@gmail.com> reports that

```

a:=4
)savesystem foo

./foo
a <=== should be 4 but is a Variable

```

todo 336: getdatabase should handle special forms

This code should not exist. Fix getdatabase to return the right answer.

```

((BOOT-EQUAL name '|Union|)'(|Union| (|:| |a| A) (|:| |b| B)))
((BOOT-EQUAL name '|UntaggedUnion|)'(|Union| A B))
((BOOT-EQUAL name '|Record|)'(|Record| (|:| |a| A) (|:| |b| B)))
((BOOT-EQUAL name '|Mapping|)'(|Mapping| T S))
((BOOT-EQUAL name '|Enumeration|)'(|Enumeration| |a| |b|))
(t (GETDATABASE name 'CONSTRUCTORFORM)))

```

bug 7285: tracing of Kernel failed

```

)trace Kernel )math

Parameterized constructors traced:
  KERNEL
(22) -> sin(1)
1<enter Kernel.kernel,30 :
  arg1= pi
  arg2= []
  arg3= 1
1<enter Kernel.setPosition,26 :
  arg1= %pi

```

```

arg2= 1024
1>exit Kernel.setPosition,26
>> System error:
1024 is not of type CHARACTER.

```

bug 7282: Cannot convert from type Matrix(Integer)...

```

K := Fraction Integer
g := matrix [[1,0,0,0], [0,-1,0,0], [0,0,-1,0], [0,0,0,-1]]
D := CliffordAlgebra(4, K, g)
D::InputForm
unparse %
parse(%)@InputForm
1::D

```

bug 7281:)show TYPE gives duplicate output

```

)show TYPE
Type is a category constructor
Abbreviation for Type is TYPE
This constructor is exposed in this frame.
Issue )edit bookvol10.2.pamphlet to see algebra source code for TYPE

```

```

----- Operations -----
Type is a category constructor.
Abbreviation for Type is TYPE
This constructor is exposed in this frame.
Issue )edit bookvol10.2.pamphlet to see algebra source code for TYPE
----- Operations -----

```

bug 7280: Enumeration fails

```

Insn == Enumeration(New, Move, Line, Close)
Insn

```

bug 7279: subscripting "1"::Symbol fails

```
subscript("s"::Symbol,["x"::Symbol::OutputForm])
```

```
(1)  s
      x
```

Type: Symbol

```
(2) -> subscript("1"::Symbol,["x"::Symbol::OutputForm])
```

```
>> System error:
The index, 4, is too large.
```

(reported by Ralf Hemmecke)

bug 7284: outputFixed seems wrong

```
outputFixed(0)
```

Type: Void

```
(7) -> 0.0
```

```
(7)  0.0
```

Type: Float

bug 7283: printing of typeOf failed

```
print(typeOf(1)::OutputForm)
```

```
>> System error:
Caught fatal error [memory may be damaged]
```

bug 7293:)trace EXPR)math

```
)trace EXPR )math
```

```
Parameterized constructors traced:
EXPR
```

```
(1) -> exp(x)
```

```
1<enter Expression.coerce,360 :
```

```
arg1= x
```

```
...
```

```
arg1= x
```

```
>> System error:  
Bind stack overflow.
```

bug 7292:)with API fails

```
)with API

>> System error:
The function |with| is undefined.

(1) -> )library API
)library cannot find the file API.
```

bug 7291: cos(2/3@Float) parses wrong

```
--S 8 of 51
cos(2/3@Float)
```

```

Daly Bug
  An expression involving @ Float actually evaluated to one of type
    PositiveInteger . Perhaps you should use :: Float .
--R
--R
--R (8)  0.7858872607 7694800072
--R
--R
--R                                         Type: Float
--E 8

```

```
--S 9 of 51
cos((2/3)@Float)

(8) 0.7858872607 7694800072

Type: Float

--R
--R
--R (9) 0.7858872607 7694800072
--R
Type: Float
--E 9
```

bug 7290:)d op coerce fails

```
)d op coerce
```

There are 195 exposed functions called coerce :

```
[1] List(D2) -> D from D if D2 has FIELD and D has AFSPCAT(D2)
[2] D -> List(D2) from D if D has AFSPCAT(D2) and D2 has FIELD
[3] D1 -> D from D if D has ALGEBRA(D1) and D1 has COMRING
```

```
>> System error:
D2 is not of type SEQUENCE.
```

bug 7303: Duplicate signature in)show ALIST

```
parts : % -> List(Record(key: Key,entry: Entry))
occurs twice in )show ALIST
```

todo 338: Convert to Float fails

```
(1) -> eval(integrate(x*exp(-(x-u)^2/2)/(sqrt(2*%pi)),x=a..%plusInfinity),[a=1,u=0.5])

(1) - 0.25 erf(0.3535533905 932737622) + 0.6020653267 6429947778
                                     Type: Expression(Float)
(2) -> eval(integrate(x*exp(-(x-u)^2/2)/(sqrt(2*%pi)),x=a..%plusInfinity),[a=1,u=0.5]):Float

Cannot convert from type Expression(Float) to Float for value
- 0.25 erf(0.3535533905 932737622) + 0.6020653267 6429947778
```

bug 7302: rich12b.input 457 failed

```
The storage for STRING is exhausted.
Currently, 710 pages are allocated.
Use ALLOCATE to expand the space.
```


Chapter 4

book7 Hyperdoc

bug 7320: bookvol7.1 ugInOutFortranPage fails

Attempting to format page crashes hyperdoc

Reference

- > Axiom Book
- > Input Files and Output Styles
- > Fortran Format

\end{chunk}

%%%

\subsection{todo 331: in bookvol7.1 this is commented out}

\begin{verbatim}

```
todo 331: in bookvol7.1 this is commented out but appears to be implemented
      be sure to update bookvol11, search for "Do a summation"
      be sure to generate v5bcproduct.eps and uncomment the minipage
      in bookvol15
```

%\menulispdownlink{Compute a product}{(|bcProduct|)}\space{}

bug 7277: Hyperdoc fails to reconnect after failure

Hyperdoc crash

)hd

Hyperdoc -> Referecne -> Language -> block

page: Not Connected to Axiom

Hyperdoc isn't connected to Axiom, therefore cannot execute the button you pressed.

binding UNIX server socket: Address already in use
(HyperDoc) Warning: Not connected to AXIOM Server!

bug 7308: Hyperdoc page Menuexlap fails to format

bookvol5: bcLinearSolveMatrixInhomo

unknown itemType

(inputStrings ({\em Coefficient 1:}\space{2} 30 0 c1 P) ({\em Coefficient 2:}\space{2} 30 0 c2

Hyperdoc ->

Solve ->

A System of Linear Equations ->

In Matrix Form ->

Continue ->

Not Zero ->

FAIL

Chapter 5

book8 Graphics

bug 7319: ignoring return values

```
edin.c: In function do_reading:
edin.c:632:11: warning: ignoring return value of write, declared with attribute warn_unused_result [-Wunused-result]
    write(contNum, &in_buff[num_proc], num_read - num_proc);
    ^

edin.c:634:13: warning: ignoring return value of write, declared with attribute warn_unused_result [-Wunused-result]
    write(contNum, "\n", 1);
    ^

edin.c:641:8: warning: ignoring return value of write, declared with attribute warn_unused_result [-Wunused-result]
    write(contNum, "\n", 1);
    ^

edin.c:659:10: warning: ignoring return value of write, declared with attribute warn_unused_result [-Wunused-result]
    write(contNum, "\n", 1);
    ^

edin.c: In function send_line_to_child:
edin.c:718:8: warning: ignoring return value of write, declared with attribute warn_unused_result [-Wunused-result]
    write(contNum, converted_buffer, converted_num);
    ^

edin.c: In function send_function_to_child:
edin.c:1126:8: warning: ignoring return value of write, declared with attribute warn_unused_result [-Wunused-result]
    write(contNum, buff, buff_pntr);
    ^

edin.c: In function send_buff_to_child:
edin.c:1145:10: warning: ignoring return value of write, declared with attribute warn_unused_result [-Wunused-result]
    write(chann, buff, buff_pntr);
    ^

fnct-key.c: In function get_str:
fnct-key.c:406:9: warning: ignoring return value of read, declared with attribute warn_unused_result [-Wunused-result]
    read(fd, &c, 1);
    ^
```

```

fnct-key.c:408:13: warning: ignoring return value of read, declared with attribute warn_unused_result
    read(fd, &c, 1);
    ^
fnct-key.c: In function handle_function_key:
fnct-key.c:497:18: warning: ignoring return value of write, declared with attribute warn_unused_result
    write(fd, buff, buff_ptr);
    ^
fnct-key.c:504:22: warning: ignoring return value of write, declared with attribute warn_unused_result
    write(fd, buff, buff_ptr);
    ^
sockio-c.c: In function remote_stdio:
sockio-c.c:837:12: warning: ignoring return value of fgets, declared with attribute warn_unused_result
    fgets(buf,1024,stdin);
    ^
ar: creating libspad.a
edible.c: In function main:
edible.c:550:7: warning: missing sentinel in function call [-Wformat=]
    execlp( program,program, 0);
    ^
edible.c:626:16: warning: ignoring return value of write, declared with attribute warn_unused_result
    write(1,out_buff, num_read);
    ^
edible.c:629:19: warning: ignoring return value of write, declared with attribute warn_unused_result
    else write(1,out_buff, num_read);
    ^
edible.c:642:16: warning: ignoring return value of write, declared with attribute warn_unused_result
    write(contNum, in_buff, num_read);
    ^

```

bug 7245: graphs differ from CRC book

v81crp50-2.6.5.2 and v81crp50-2.6.5.3 graphs differ from CRC
 (see ps/v81crp50-2.6.5.1-3.eps)

bug 7242: calling title on an empty viewport shows no title

```

)clear all
f(c,x,n) == c/x^n
lineColorDefault(green())
viewport1:=draw(f(0.01,x,1),x=-2..2,adaptive==true,unit==[1.0,1.0])
graph2121:=getGraph(viewport1,1)
lineColorDefault(blue())
viewport2:=draw(f(0.01,x,3),x=-2..2,adaptive==true,unit==[1.0,1.0])
graph2122:=getGraph(viewport2,1)
lineColorDefault(red())

```

```

viewport3:=draw(f(0.01,x,5),x=-2..2,adaptive==true,unit==[1.0,1.0])
graph2123:=getGraph(viewport3,1)
viewall:=viewport2D()$TwoDimensionalViewport
title(viewall,"p27-2.1.2.1-3")
putGraph(viewall,graph2122,1)
putGraph(viewall,graph2122,2)
putGraph(viewall,graph2123,3)
units(viewall,1,"on")
points(viewall,1,"off")
points(viewall,2,"off")
points(viewall,3,"off")
makeViewport2D(viewall,title=="test")
close(viewport1)
close(viewport2)
close(viewport3)

```

bug 7244: graphs differ from CRC book

v81crp50-2.5.5.2 and v81crp50-2.5.5.3 graphs differ from CRC
(see ps/v81crp50-2.5.5.1-3.eps)

bug 7243: graphs differ from CRC book

v81crp27-2.1.2.1-3.eps differs from CRC (see bookvol8.1)

bug 7266: integration failure for $1/(\sin(x)^4+1)$

```

f:=1/(sin(x)^4+1)
a:=integrate(f,x)
b:=differentiate(a,x,1)
draw(b,x=0..4)
draw(f,x=0..4)

```

reported by Yanyang Xiao <xyy82148@sjtu.edu.cn> 30 Nov 2014

bug 7296: connect from VIEW2D is not graph specific

The connect function in VIEW2D globally turns lines on or off.

This doesn't allow for overlaying lines on a set of points.

```

)clear all
FIN:=FiniteField(67,1) -- declare a field p**n, p=67, n=1
f(x:INT,y:INT):FIN == (y^2-x^3-7)::FIN -- compute a finite point
m():LIST(POINT(DFLOAT)) ==
  lli:LIST(LIST(INT)) := []
  for x in 0..66 repeat
    for y in 0..66 repeat
      a:FIN:=f(x,y)
      if a = 0 then lli:=concat(lli,[[x,y]])
  ld:=lli::LIST(LIST(DFLOAT))
  lpd:=[point(p) for p in ld]
m()
line(x1:INT,y1:INT,x2:INT,y2:INT):EXPR(INT) ==
  -- y-y1 = m(x-x1) where m = (y2-y1)/x2-x1
  (y2-y1)/(x2-x1)*(x-x1) + y1
line1:=draw(line(2,22,6,25),x=0..67)
line1graph:=getGraph(line1,1)
line2:=draw(line(30,26,47,39),x=0..67)
line2graph:=getGraph(line2,1)
ldo:=title("BitCoin Modular Points")
vp:=draw(m(),[ldo])$DRAWPT
putGraph(vp,line1graph,2)
putGraph(vp,line2graph,3)
axes(vp,1,"on")$VIEW2D -- turn on the axes
units(vp,1,"on")$VIEW2D -- turn on the units
connect(vp,2,"on")$VIEW2D -- turn on the lines
connect(vp,3,"on")$VIEW2D -- turn on the lines
connect(vp,1,"off")$VIEW2D -- turn off the lines
makeViewport2D(vp)

```

Chapter 6

book9 Compiler

Chapter 7

book10.2 Algebra Categories

7.1 U

bug 7318: ULSCAT undefined variables

```
finalizing nrlib ULSCAT
; (DEFUN |UnivariateLaurentSeriesCategory| ...) is being compiled.
;; The variable |UnivariateLaurentSeriesCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |UnivariateLaurentSeriesCategory;| ...) is being compiled.
;; The variable |UnivariateLaurentSeriesCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

bug 7316: UPOLYC undefined variable

```
finalizing nrlib UPOLYC
; (DEFUN |UnivariatePolynomialCategory| ...) is being compiled.
;; The variable |UnivariatePolynomialCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |UnivariatePolynomialCategory;| ...) is being compiled.
;; The variable |UnivariatePolynomialCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

bug 7317: UPOLYC- not known that (Ring) is of mode...

```
finalizing nrlib UPOLYC-
Warnings:
  [1] solveLinearPolynomialEquation: not known that (Ring) is of mode...
  [2] factorPolynomial: not known that (Ring) is of mode...
  [3] factor: not known that (IntegralDomain) is of mode...
  [4] elt: not known that (IntegralDomain) is of mode...
```

bug 7314: URAGG undefined variables

```
finalizing nrlib URAGG
; (DEFUN |UnaryRecursiveAggregate| ...) is being compiled.
;; The variable |UnaryRecursiveAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |UnaryRecursiveAggregate;| ...) is being compiled.
;; The variable |UnaryRecursiveAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

Chapter 8

book10.3 Algebra Domains

8.1 E

bug 7311: EQ undefined variable

```
>compiling EQ.spad to EQ.nrllib  
  
; (DEFUN |Equation;| ...) is being compiled.  
;; The variable IDENTITY is undefined.
```

8.2 F

bug 7312: FDIV undefined variable

```
>compiling FDIV.spad to FDIV.nrllib  
  
-----constructor-----  
; (DEFUN |FiniteDivisor;| ...) is being compiled.  
;; The variable IDENTITY is undefined.  
;; The compiler will assume this variable is a global.
```

```
radix(10^100,32)::TEX
(1)
["$$",
"4#\I9#\L#\K#\I#\P9#\G#\R#\S#\T#\C5#\I#\F164#\P#\05#\V72#\M#\E827226#\J#\S#
\L#\A#\P462585#\Q7#\H00000000000000000000",
,
"$$"]
Type: TexFormat
```

bug 7287: TexFormat is incorrect

```
)set output fraction horizontal
```

```
(1) -> (1/2)::TEX
```

```
(1) ["$$","SLASH ","\left(","{1, \: 2} ","\right)","$$"]
```

```
Type: TexFormat
```

8.6 U**bug 7286: UPXS arithmetic fails**

```
a := series(z, z = 0)
```

```
(1) z
```

```
Type: UnivariatePuisseuxSeries(Expression(Integer),z,0)
```

```
(2) -> b := series(1, x = 0)
```

```
(2) 1
```

```
Type: UnivariatePuisseuxSeries(Expression(Integer),x,0)
```

```
(3) -> a + b
```

```
>> Error detected within library code:
division by zero
```


Chapter 9

book10.4 Algebra Packages

9.1 A

bug 7313: AXSERV undefined variables

```
>compiling AXSERV.spad to AXSERV.nrlib
```

```
; (DEFUN |AXSERV;getCommand| ...) is being compiled.
;; The variable |tmpmathml| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |tmpalgebra| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |savemathml| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$texOutputStream| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |savealgebra| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |resultmathml| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |resultalgebra| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |AXSERV;getShow| ...) is being compiled.
;; The variable SAVESTREAM is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |AXSERV;lastType| ...) is being compiled.
;; The variable |first| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |stepSav| is undefined.
;; The compiler will assume this variable is a global.
```

9.2 N

bug 7309: NAGE04 undefined variable

```
>compiling NAGE04.spad to NAGE04.nrlib
```

```
; (DEFUN |NAGE04;e04dgm;I2DfIDfB2Df4IMIUR;1| ...) is being compiled.
;; The variable OBJFUN is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NAGE04;e04fdf;4IMIUR;2| ...) is being compiled.
;; The variable LSFUN1 is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NAGE04;e04gcf;4IMIUR;3| ...) is being compiled.
;; The variable LSFUN2 is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NAGE04;e04jaf;4I3MIUR;4| ...) is being compiled.
;; The variable FUNCT1 is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NAGE04;e04naf;8IDf6M3B2IMMIUR;6| ...) is being compiled.
;; The variable QPHESS is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NAGE04;e04ucf;6I3M2IBDfI2DfB4DfB5I3Df5IM4MIUUR;7| ...) is being compiled.
;; The variable CONFUN is undefined.
;; The compiler will assume this variable is a global.
```

bug 7310: NAGD02 undefined variables

```
>compiling NAGD02.spad to NAGD02.nrlib
```

```
; (DEFUN |NAGD02;d02bbf;Df3IDfMDfIUUR;1| ...) is being compiled.
;; The variable FCN is undefined.
;; The compiler will assume this variable is a global.
;; The variable OUTPUT is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NAGD02;d02bhf;Df2I2DfMDfIUUR;2| ...) is being compiled.
;; The variable G is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NAGD02;d02ejf;Df2ISIDfMDfIUUUUR;4| ...) is being compiled.
;; The variable PEDERV is undefined.
;; The compiler will assume this variable is a global.
```



```

; (DEFUN |NAGD02;d02gbf;2DfIDf3I4M2IUUR;6| ...) is being compiled.
;; The variable FCNF is undefined.
;; The compiler will assume this variable is a global.
;; The variable FCNG is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NAGD02;d02kef;M2IDf2I2DfM2IUUR;7| ...) is being compiled.
;; The variable COEFFN is undefined.
;; The compiler will assume this variable is a global.
;; The variable BDYVAL is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NAGD02;d02raf;4IDf6I2MDfIUUR;9| ...) is being compiled.
;; The variable JACOBf is undefined.
;; The compiler will assume this variable is a global.
;; The variable JACEPS is undefined.
;; The compiler will assume this variable is a global.
;; The variable JACOBG is undefined.
;; The compiler will assume this variable is a global.
;; The variable JACGEP is undefined.
;; The compiler will assume this variable is a global.

```

bug 7273: SOLVERAD wester algebra radicalSolve bug

```
--S 26 of 63
```

```
radicalSolve(x**3 + x**2 - 7 = 0, x)
```

```

(26)
[
  x =
    +-----+2
    | +----+   +-+
    |9\|1295  + 187\|3
    |-----
    3|      +-+
    \|      54\|3
  +
    +-----+
    | +----+   +-+
    |9\|1295  + 187\|3
    |----- - 2
    3|      +-+
    \|      54\|3
  /
    +-----+
    | +----+   +-+
    |9\|1295  + 187\|3

```

```

(9\|- 3 + 9) |-----
      3|      +-+
      \|      54\|3
,
x =
      +-----+2
      | +-----+ +-+
      |9\|1295 + 187\|3
      +-----+
      3|      +-+
      \|      54\|3
+
      +-----+
      | +-----+ +-+
      |9\|1295 + 187\|3
      +-----+ + 2
      3|      +-+
      \|      54\|3
/
      +-----+
      | +-----+ +-+
      |9\|1295 + 187\|3
      +-----+
      3|      +-+
      \|      54\|3
,
      +-----+2      +-----+
      | +-----+ +-+      | +-----+ +-+
      |9\|1295 + 187\|3      |9\|1295 + 187\|3
      +-----+      +-----+ + 1
      3|      +-+      3|      +-+
      \|      54\|3      \|      54\|3
x= -----]
      +-----+
      | +-----+ +-+
      |9\|1295 + 187\|3
      +-----+
      3|      +-+
      \|      54\|3
Type: List(Equation(Expression(Integer)))

--R
--R
--R (26)
--R
--R      +-----+2      +-----+
--R      | +-----+ +-+      | +-----+
--R      | |1295
--R      |9 |---- + 187      |9 |---- + 187
--R      +-----+      +-----+
--R      | \| 3      | \| 3
--R      (- \| - 3 + 1) 3|----- + (- \| - 3 - 1) 3|----- - 2

```

```

--R      \ |      2      \ |      2
--R      [x= -----,
--R                                     +-----+
--R                                     | +----+
--R                                     | |1295
--R                                     |9 |---- + 187
--R                                     +---+ | \ | 3
--R                                     (3\|- 3 + 3) 3|-----
--R                                     \ |      2
--R
--R      +-----+2      +-----+
--R      | +----+      | +----+
--R      | |1295      | |1295
--R      |9 |---- + 187      |9 |---- + 187
--R      | \ | 3      | \ | 3
--R      +---+      +---+
--R      (- \|- 3 - 1) 3|----- + (- \|- 3 + 1) 3|----- + 2
--R      \ |      2      \ |      2
--R
--R      x= -----,
--R                                     +-----+
--R                                     | +----+
--R                                     | |1295
--R                                     |9 |---- + 187
--R                                     +---+ | \ | 3
--R                                     (3\|- 3 - 3) 3|-----
--R                                     \ |      2
--R
--R      +-----+2      +-----+
--R      | +----+      | +----+
--R      | |1295      | |1295
--R      |9 |---- + 187      |9 |---- + 187
--R      | \ | 3      | \ | 3
--R      3|----- - 3|----- + 1
--R      \ |      2      \ |      2
--R
--R      x= -----]
--R
--R      +-----+
--R      | +----+
--R      | |1295
--R      |9 |---- + 187
--R      | \ | 3
--R      3 3|-----
--R      \ |      2
--R
--R
--R      Type: List Equation Expression Integer
--E 26

```

```
--S 27 of 63
reduce(*, map(e +> lhs(e) - rhs(e), %))
```

$$(27) \quad \frac{(9x^3 + 9x^2 - 63)\sqrt{3}\sqrt{1295} + 561x^3 + 561x^2 - 3927}{\sqrt{3}\sqrt{1295} + 561x^2}$$

$$9\sqrt{3}\sqrt{1295} + 187\sqrt{3}$$

Type: Expression(Integer)

--R

--R

$$(27) \quad x^3 + x^2 - 7$$

--R

Type: Expression Integer

--E 27

Chapter 10

book10.5 Algebra Numerics

Chapter 11

Build System

bug 7278: make complains on TESTSET=notests

reported by "leonbaum" <notifications@github.com>

Chapter 12

CATS

bug 7246: kamke tests missing

```
kamke1 seems to be missing ode97
kamke2 missing 104, 105
kamke6 missing 332
kamke7 missing 357, 359, 360, 362 and many others
kamke7 repeats 776 with different equations
```


Chapter 13

Testing

bug 7240: in richtrig800-899 there is an occasional failure

```
--S 349 of 526
a0864:= integrate(t0864,x)
--R
--R
--R >> Error detected within library code:
--R (1 . failed) cannot be coerced to mode (SparseUnivariatePolynomial (Integer))
--R
--R Continuing to read the file...
--R
--E 349
```

bug 7238: en.regress failures

```
int/input/en.regress

MISMATCH
expected:"      ,"
got:"      4503599627370496 4503599627370496 576460752303423488 576460752303423488"
MISMATCH
expected:"      1152921504606846976"
got:"      [-----,-----,-----,- -----],"
```

bug 7241: missing algebra

in `li.input.pamphlet`, chunks 2, 6, 34, 38, 42 could be integrated and were not.

Chapter 14

Documentation

todo 330: fix the inheritance graph for these

```
todo 330: fix the inheritance graph for these
```

```
"AHYP" [color="#4488FF",href="bookvol10.2.pdf#nameddest=AHYP"]
"AHYP" -> "Category"
```

```
"ATTREG" [color="#4488FF",href="bookvol10.2.pdf#nameddest=ATTREG"]
"ATTREG" -> "Category"
```

```
/* nobody seems to go to bastype by itself */
/* we combine these two to minimize edges in the graph */
/* note that koerce is duplicated */
"BASTYPE/KOERCE" [color="blue",href="bookvol10.2.pdf#nameddest=BASTYPE"]
"BASTYPE/KOERCE" -> "Category"
```

```
"KOERCE" [color="#4488FF",href="bookvol10.2.pdf#nameddest=KOERCE"]
"KOERCE" -> "Category"
```

```
"BASTYPE-" [color="#88FF44",href="bookvol10.3.pdf#nameddest=BASTYPE"]
"BASTYPE-" -> "Domain"
```

```
"CFCAT" [color="#4488FF",href="bookvol10.2.pdf#nameddest=CFCAT"]
"CFCAT" -> "Category"
```

```
"ELTAB" [color="#4488FF",href="bookvol10.2.pdf#nameddest=ELTAB"]
"ELTAB" -> "Category"
```

```
"ESCONT1" [color="#FF4488",href="bookvol10.4.pdf#nameddest=ESCONT1"]
"ESCONT1" -> "Package"
/*"ESCONT1" -> "DFLOAT"*/
```

```

/*"ESCONT1" -> "BOOLEAN"*/

"GRDEF" [color="#FF4488",href="bookvol10.4.pdf#nameddest=GRDEF"]
"GRDEF" -> "Package"
/*"GRDEF" -> "BOOLEAN"*/

"INTBIT" [color="#FF4488",href="bookvol10.4.pdf#nameddest=INTBIT"]
"INTBIT" -> "Package"
/*"INTBIT" -> "INT"*/

"KONVERT" [color="#4488FF",href="bookvol10.2.pdf#nameddest=KONVERT"]
"KONVERT" -> "Category"

"MAGCDOC" [color="#4488FF",href="bookvol10.2.pdf#nameddest=MAGCDOC"]
"MAGCDOC" -> "Category"

"MSYSCMD" [color="#FF4488",href="bookvol10.4.pdf#nameddest=MSYSCMD"]
"MSYSCMD" -> "Package"

"ODEIFTBL" [color="#88FF44",href="bookvol10.3.pdf#nameddest=ODEIFTBL"]
"ODEIFTBL" -> "Domain"

"OM" [color="#4488FF",href="bookvol10.2.pdf#nameddest=OM"]
"OM" -> "Category"

"OMCONN" [color="#88FF44",href="bookvol10.3.pdf#nameddest=OMCONN"]
"OMCONN" -> "Domain"

"OMDEV" [color="#88FF44",href="bookvol10.3.pdf#nameddest=OMDEV"]
"OMDEV" -> "Domain"

"PRIMCAT" [color="#4488FF",href="bookvol10.2.pdf#nameddest=PRIMCAT"]
"PRIMCAT" -> "Category"

"PRINT" [color="#FF4488",href="bookvol10.4.pdf#nameddest=PRINT"]
"PRINT" -> "Package"

"PTRANFN" [color="#4488FF",href="bookvol10.2.pdf#nameddest=PTRANFN"]
"PTRANFN" -> "Category"

"RFDIST" [color="#FF4488",href="bookvol10.4.pdf#nameddest=RFDIST"]
"RFDIST" -> "Package"
/*"RFDIST" -> {"INT"; "PI"; "NNI"; "BOOLEAN"; "SINT"}*/

"RIDIST" [color="#FF4488",href="bookvol10.4.pdf#nameddest=RIDIST"]
"RIDIST" -> "Package"
/*"RIDIST" -> {"SINT"; "NNI"; "INT"}*/

"SPFCAT" [color="#4488FF",href="bookvol10.2.pdf#nameddest=SPFCAT"]
"SPFCAT" -> "Category"

```

```
"TYPE" [color="#4488FF",href="bookvol10.2.pdf#nameddest=TYPE"]  
"TYPE" -> "Category"
```


Chapter 15

Unclassified

warning 20573: rec has no value

```
finalizing nrlib CWMMT
Warnings:
[1] test6:  rec has no value
[2] test7:  rec has no value
[3] test8:  rec has no value
[4] test9:  rec has no value
[5] test10: rec has no value
```

bug 7249: radicalSolve($z^7=1$, z)

Obviously, all the roots of the equation $z^7 = 1$ can be expressed in radicals, and Mathematica can easily produce the explicit expressions in terms of radicals.

```
Solve[z^7 == 1, z]
```

```
{{z -> 1}, {z -> -(-1)^(1/7)}, {z -> (-1)^(2/7)}, {z -> -(-1)^(3/7)},
{{z -> {z -> (-1)^(4/7)}, {z -> -(-1)^(5/7)}, {z -> (-1)^(6/7)}}
```

To save the space, below the only example is given.

```
FunctionExpand[ComplexExpand[-(-1)^(1/7)]]
```

```
(1/2)*((1/3)*((1/2)*(-1 + I*Sqrt[7]) + ((-1 + I*Sqrt[3]))*((1/2)*(-1 +
I*Sqrt[7]) + (1/2)*(-1 - I*Sqrt[7]))*((1/2)*(-1 + I*Sqrt[3]) +
```

$$\begin{aligned}
& (1/4)*(-1 + \text{I*Sqrt}[3])^2))/2*(6 + (3/4)*(-1 + \text{I*Sqrt}[3])*(-1 + \\
& \text{I*Sqrt}[7]) + (1/2)*(-1 - \text{I*Sqrt}[7])*(1 + (3/4)*(-1 + \\
& \text{I*Sqrt}[3])^2))^{\wedge}(1/3)) + (1/4)*(-1 + \text{I*Sqrt}[3])^2*(6 + (3/4)*(-1 + \\
& \text{I*Sqrt}[3])*(-1 + \text{I*Sqrt}[7]) + (1/2)*(-1 - \text{I*Sqrt}[7])*(1 + (3/4)*(-1 + \\
& \text{I*Sqrt}[3])^2))^{\wedge}(1/3)) + (1/3)*((1/2)*(1 + \text{I*Sqrt}[7]) - ((-1 + \\
& \text{I*Sqrt}[3])^2*((1/2)*(-1 - \text{I*Sqrt}[7]) + (1/2)*(-1 + \\
& \text{I*Sqrt}[7])*((1/2)*(-1 + \text{I*Sqrt}[3]) + (1/4)*(-1 + \text{I*Sqrt}[3])^2)))/4*(6 \\
& + (3/4)*(-1 + \text{I*Sqrt}[3])*(-1 - \text{I*Sqrt}[7]) + (1/2)*(-1 + \text{I*Sqrt}[7])*(1 \\
& + (3/4)*(-1 + \text{I*Sqrt}[3])^2))^{\wedge}(1/3)) - (1/2)*(-1 + \text{I*Sqrt}[3])*(6 + \\
& (3/4)*(-1 + \text{I*Sqrt}[3])*(-1 - \text{I*Sqrt}[7]) + (1/2)*(-1 + \text{I*Sqrt}[7])*(1 + \\
& (3/4)*(-1 + \text{I*Sqrt}[3])^2))^{\wedge}(1/3))) + (1/2)*((1/3)*((1/2)*(-1 + \\
& \text{I*Sqrt}[7]) + ((-1 + \text{I*Sqrt}[3])*((1/2)*(-1 + \text{I*Sqrt}[7]) + (1/2)*(-1 - \\
& \text{I*Sqrt}[7])*((1/2)*(-1 + \text{I*Sqrt}[3]) + (1/4)*(-1 + \text{I*Sqrt}[3])^2)))/2*(6 \\
& + (3/4)*(-1 + \text{I*Sqrt}[3])*(-1 + \text{I*Sqrt}[7]) + (1/2)*(-1 - \text{I*Sqrt}[7])*(1 \\
& + (3/4)*(-1 + \text{I*Sqrt}[3])^2))^{\wedge}(1/3)) + (1/4)*(-1 + \text{I*Sqrt}[3])^2*(6 + \\
& (3/4)*(-1 + \text{I*Sqrt}[3])*(-1 + \text{I*Sqrt}[7]) + (1/2)*(-1 - \text{I*Sqrt}[7])*(1 + \\
& (3/4)*(-1 + \text{I*Sqrt}[3])^2))^{\wedge}(1/3)) + (1/3)*((1/2)*(-1 - \text{I*Sqrt}[7]) \\
& + ((-1 + \text{I*Sqrt}[3])^2*((1/2)*(-1 - \text{I*Sqrt}[7]) + (1/2)*(-1 + \\
& \text{I*Sqrt}[7])*((1/2)*(-1 + \text{I*Sqrt}[3]) + (1/4)*(-1 + \text{I*Sqrt}[3])^2)))/4*(6 \\
& + (3/4)*(-1 + \text{I*Sqrt}[3])*(-1 - \text{I*Sqrt}[7]) + (1/2)*(-1 + \text{I*Sqrt}[7])*(1 \\
& + (3/4)*(-1 + \text{I*Sqrt}[3])^2))^{\wedge}(1/3)) + (1/2)*(-1 + \text{I*Sqrt}[3])*(6 + \\
& (3/4)*(-1 + \text{I*Sqrt}[3])*(-1 - \text{I*Sqrt}[7]) + (1/2)*(-1 + \text{I*Sqrt}[7])*(1 + \\
& (3/4)*(-1 + \text{I*Sqrt}[3])^2))^{\wedge}(1/3)))
\end{aligned}$$

According to the AXIOM Book

AXIOM Book> Use radicalSolve if you want your solutions expressed in
 AXIOM Book> terms of radicals.

However, already for $z^7 = 1$ this is not so,

```
-> radicalSolve(z^7=1, z)
```

```
[z= 1]
```

and the problem exists for 11, 13, 14, 15, 17, 19 etc

```
-> for i in 1..20 repeat print([i,#radicalSolve(z^i=1,z)])
```

```

[1,1]
[2,2]
[3,3]
[4,4]
[5,5]
[6,6]
[7,1]  <-- not good
[8,8]
[9,9]
[10,10]
```

```

[11,1] <-- not good
[12,12]
[13,1] <-- not good
[14,2] <-- not good
[15,7] <-- not good
[16,16]
[17,1] <-- not good
[18,18]
[19,1] <-- not good
[20,20]

```

Subject: Re: [Q] radicalSolve fails to find all roots ?

These are NOT bugs! But the following may be! Consider the equation $z^n=1$ for $n = 7$:

```
(1) -> radicalSolve(z^7=2)
```

```
(1)
```

```

      7+++      +---+7+++      2%pi      7+++      2%pi
[z= \|- 1 \|2 sin(-----) + \|2 cos(-----),
                        7                        7

```

```

      +---+7+++      4%pi      7+++      4%pi
z= \|- 1 \|2 sin(-----) + \|2 cos(-----),
                        7                        7

```

```

      +---+7+++      6%pi      7+++      6%pi
z= \|- 1 \|2 sin(-----) + \|2 cos(-----),
                        7                        7

```

```

      +---+7+++      8%pi      7+++      8%pi
z= \|- 1 \|2 sin(-----) + \|2 cos(-----),
                        7                        7

```

```

      +---+7+++      10%pi      7+++      10%pi
z= \|- 1 \|2 sin(-----) + \|2 cos(-----),
                        7                        7

```

```

      +---+7+++      12%pi      7+++      12%pi
z= \|- 1 \|2 sin(-----) + \|2 cos(-----)]
                        7                        7

```

Type: List Equation Expression Integer

----- comments

Of course, these are correct solutions by Euler's Formula. A bit surprising that radicalSolve invokes these for $z^7=2$ and not for $z^7=1$; when n is 7, these trigonometric values are not embeddable in a tower of "solvable" extensions. That is, these are not solutions expressible in terms of radicals (of *real* numbers) and arithmetic alone. Put another way, the regular 7-gon is not constructible by compass and ruler alone. From:

<http://mathworld.wolfram.com/ConstructiblePolygon.html>

A necessary and sufficient condition that a regular n -gon be constructible is that $\phi(n)$ be a power of 2, where $\phi(n)$ is the totient function (Krzek 2001, p. 34).

n =	7	11	13	14	15	17	19
-----	---	----	----	----	----	----	----

So if you compare the constructible regular n -gons, you can see why Axiom's results are reasonable: radicalSolve only finds solutions that are expressible in terms of radicals and arithmetic operations. It did not find those for $n = 15$ and 17 probably (I am guessing) because at the time of implementation, these constructions were not known (at least to the programmer). On the other hand, for $n = 9, 18$, the solutions are expressible in radicals only if radicals of **complex** numbers are allowed and Axiom found those (perhaps it shouldn't?). The expansion for $(-1)^{1/7}$ that Vladimir gave involves radicals of complex numbers, as theory predicts.

When Axiom cannot find solutions, it is (presumably) a PROOF that the other solutions are NOT solvable by radicals (using *real* numbers), or at least, there is no known proof that it is solvable at the time of implementation=2E (That is why I am surprised at the above result for $z^7=2$).

In other words, rather than viewing the answer for $z^7=1$ as a bug, we should view the answers for $z^7=2$, $z^7=3$ (and may be even $z^9=1$, $z^{18}=1$) as bugs!

Still, the package should be upgraded.

```

(1) -> radicalSolve(z^9=1,z)

(1)

```

```

      +---+ +---+      | - \|- 3 - 1      +-----+
      (- \|- 1 \|3 - 1) 3|-----      | +---+
      \|      2      \| - 3 - 1
z= -----, z= 3|-----,
      2      \|      2
      +-----+      +-----+
      | +---+      | +---+
      +---+ +---+      \| - 3 - 1      +---+ +---+      \| - 3 - 1
      (\|- 1 \|3 - 1) 3|-----      (- \|- 1 \|3 - 1) 3|-----
      \|      2      \|      2
z= -----, z= -----
---,
      2      2
      +---+      +---+
      - \|- 3 - 1      \|- 3 - 1
z= -----, z= -----, z= 1]
      2      2
Type: List Equation Expression Integer
=

```

(2) -> radicalSolve(z^7=3)

```

(2)
      7+-+      +---+7+-+      2%pi      7+-+      2%pi
[z= \|3 , z= \|- 1 \|3 sin(-----) + \|3 cos(-----),
      7      7
      +---+7+-+      4%pi      7+-+      4%pi
z= \|- 1 \|3 sin(-----) + \|3 cos(-----),
      7      7
      +---+7+-+      6%pi      7+-+      6%pi
z= \|- 1 \|3 sin(-----) + \|3 cos(-----),
      7      7
      +---+7+-+      8%pi      7+-+      8%pi
z= \|- 1 \|3 sin(-----) + \|3 cos(-----),
      7      7
      +---+7+-+      10%pi      7+-+      10%pi
z= \|- 1 \|3 sin(-----) + \|3 cos(-----),
      7      7
      +---+7+-+      12%pi      7+-+      12%pi
z= \|- 1 \|3 sin(-----) + \|3 cos(-----)]
      7      7
Type: List Equation Expression Integer
=

```

(3) -> radicalSolve(z^7=1.)

```

      7
WARNING (genufact): No known algorithm to factor ? - 1.0
, trying square-free.

```

```

(3)
                                     +-----+
[z= 1.0, z= 0.7818314824 6802980871\|- 1.0  + 0.6234898018 5873353=
053,
                                     +-----+
z= 0.9749279121 8182360702\|- 1.0  - 0.2225209339 5631440428,
                                     +-----+
z= 0.4338837391 1755812048\|- 1.0  - 0.9009688679 0241912624,
                                     +-----+
z= - 0.4338837391 1755812048\|- 1.0  - 0.9009688679 0241912625,
                                     +-----+
z= - 0.9749279121 8182360702\|- 1.0  - 0.2225209339 563144043,
                                     +-----+
z= - 0.7818314824 6802980872\|- 1.0  + 0.6234898018 5873353052]
                                     Type: List Equation Expression Float
                                     =

```

```

(4) -> radicalSolve(z^6+z^5+z^4+z^3+z^2+z+1=0)

```

```

(4) []
                                     Type: List Equation Expression Integer

```

bug 7248: differentiation bug in D(1,z)

```

(94) -> D(sin(z)^2+cos(z)^2, z)

```

```

(94)  0

```

```

(98) -> D(simplify(sin(z)^2+cos(z)^2), z)

```

```

(98)  0

```

BUT

```

(95) -> D(1, z)

```

```

There are 5 exposed and 0 unexposed library operations named D
having 2 argument(s) but none was determined to be applicable.
Use HyperDoc Browse, or issue
                                     )display op D
to learn more about the available operations. Perhaps
package-calling the operation or using coercions on the arguments
will allow you to apply the operation.

```

```

Cannot find a definition or applicable library operation named D

```

```

with argument type(s)
                                PositiveInteger
                                Variable z

```

Perhaps you should use "@" to indicate the required return type,
or "\$" to specify which version of the function you need.

bug 7142: $\exp^{\log}(x)$ Cannot take first of an empty list

```

WS> ElementaryFunctionCategory(): Category == with
WS>   log : $ -> $      ++ log(x) returns the natural logarithm of x.
WS>   exp : $ -> $      ++ exp(x) returns %e to the power x.
WS>   "**": ($, $) -> $ ++ x**y returns x to the power y.
WS> add
WS>   if $ has Monoid then
WS>     x ** y == exp(y * log x)

```

The error apparently comes from EXPR where the implementation is
`x**y == exp(y * log(x))`

so $\exp^{\log}(x)$ is computed as

```
exp(log(x) * log(exp))
```

which should be ok, but then the system handles `log exp` and tries to
simplify it to the identity map, and not finding the argument.

```
(3) -> exp^log(x)
```

```
>> System error:
Cannot take first of an empty list

```

```
(1) -> integrate(%i^log(z), z)
```

```

      log(%i)log(z)
z %e
(1)  -----
      log(%i) + 1

```

Type: Union(Expression(Complex(Integer)),...)

```
(2) -> integrate(f^log(z), z)
```

```

      log(f)log(z)
z %e
(2)  -----
      log(f) + 1

```

Type: Union(Expression(Integer),...)

```
(3) -> integrate(sin^log(z), z)
```

```

      log(sin)log(z)
      z %e
(3)  -----
      log(sin) + 1
Type: Union(Expression(Integer),...)

(4) -> integrate(tan^log(z), z)

      log(tan)log(z)
      z %e
(4)  -----
      log(tan) + 1
Type: Union(Expression(Integer),...)

(5) -> integrate(sqrt^log(z), z)

      log(sqrt)log(z)
      z %e
(5)  -----
      log(sqrt) + 1
Type: Union(Expression(Integer),...)

(6) -> integrate(atan^log(z), z)

      log(atan)log(z)
      z %e
(6)  -----
      log(atan) + 1
Type: Union(Expression(Integer),...)

(7) -> integrate(asinh^log(z), z)

      log(asinh)log(z)
      z %e
(7)  -----
      log(asinh) + 1
Type: Union(Expression(Integer),...)

(8) -> integrate(exp^log(z), z)

>> System error:
Cannot take first of an empty list

(8) -> integrate(log^log(z), z)

>> System error:
Cannot take first of an empty list

```

bug 7259: in taylor expansion

```
x:=taylor 'x
```



```
y:=taylor 'y
sinh(x)*cosh(y)
```

wish 1011: $\text{sum}(1/(k+a), k=1..n)$ by Gosper's method

```
(1) -> sum(1/(k+a), k=1..n)
```

```
      n
  --+  1
(1)  > ----
  --+  k + a
    k= 1
```

```
Type: Union(Expression(Integer),...)
```

but evaluating by Gosper's method should give

```
digamm(n+a+1)-digamma(a+1)
```

bug 7258: $\text{acosh}(0.0)$ invalid argument to acosh

```
(1) -> acosh(0.0)
```

```
>> Error detected within library code:
invalid argument to acosh
```

```
(1) -> acosh(0)::Complex Float
```

```
(1)  0.6776263578 0344027125 E -20 + 1.5707963267 948966192 %i
                                           Type: Complex(Float)
```

bug 7257: $\text{normalize}(\text{subst}(\text{asin}(z), z=-1))$ division by zero

```
(1) -> normalize(subst(asin(z), z=-1))
```

```
>> Error detected within library code:
catdef: division by zero
```

```
(1) -> asin(-1)
```

```
      %pi
(1)  - ---
      2
```

```

(2) -> subst(asin(z),z=-1)
                                         Type: Expression(Integer)

(2)  asin(- 1)
                                         Type: Expression(Integer)
(3) -> normalize %

>> Error detected within library code:
catdef: division by zero

(3) -> normalize %(1)

      %pi
(3)  - ---
      2
                                         Type: Expression(Integer)
(4) -> normalize(asin(- 1))

      %pi
(4)  - ---
      2
                                         Type: Expression(Integer)

```

bug 7256: `acot(-1)` values differ

```

(1) -> normalize(acot(-1))

      3%pi
(1)  ----
      4
                                         Type: Expression(Integer)
(2) -> normalize(subst(acot(z),z=-1))

      %pi
(2)  - ---
      4
                                         Type: Expression(Integer)

```

bug 7254: `f==n+-i sum(sum(1/i,i=1..j),j=1..n)` complains

```

f==n+->sum(sum(1/i,i=1..j),j=1..n)
                                         Type: Void
(2) -> f(1)
There are 6 exposed and 2 unexposed library operations named sum
having 2 argument(s) but none was determined to be applicable.

```

Use HyperDoc Browse, or issue
)display op sum
 to learn more about the available operations. Perhaps
 package-calling the operation or using coercions on the arguments
 will allow you to apply the operation.
 Cannot find a definition or applicable library operation named sum
 with argument type(s)
 Union(Fraction(Polynomial(Integer)),Expression(Integer))
 SegmentBinding(PositiveInteger)

Perhaps you should use "@" to indicate the required return type,
 or "\$" to specify which version of the function you need.
 AXIOM will attempt to step through and interpret the code.

(2) 1
 Type: Expression(Integer)
 (3) ->

bug 7253: There are no library operations named 'when'

(1) -> ch : Integer -> UnivariatePolynomial(x,FRAC(INT))
 Type: Void
 (2) -> ch(0) == 1
 Type: Void
 (3) -> ch(1) == x
 Type: Void
 (4) -> ch(n) == 2 * x * ch(n-1) - ch(n-2) when n > 1
 Type: Void
 (5) -> ch(2)
 There are no library operations named when
 Use HyperDoc Browse or issue
)what op when
 to learn if there is any operation containing " when " in its
 name.
 Cannot find a definition or applicable library operation named when
 with argument type(s)
 Integer

 Perhaps you should use "@" to indicate the required return type,
 or "\$" to specify which version of the function you need.
 AXIOM will attempt to step through and interpret the code.
 Compiling function ch with type Integer -> UnivariatePolynomial(x,
 Fraction(Integer))
 There are no library operations named when
 Use HyperDoc Browse or issue
)what op when
 to learn if there is any operation containing " when " in its

name.

Cannot find a definition or applicable library operation named when
with argument type(s)

Integer

Perhaps you should use "@" to indicate the required return type,
or "\$" to specify which version of the function you need.

bug 7252: argument(-%i)

```
> AXIOM ->      complexForm(log(%i) - log(-%i))
>
>      0
```

could you please file this as a bug on issuetracker? The problem is in the
operation "argument":

```
(8) -> argument(%i)
```

```
      %pi
(8)  ---
      2
```

Type: Expression Integer

```
(9) -> argument(-%i)
```

```
      %pi
(9)  ---
      2
```

Type: Expression Integer

the latter should be $-\pi/2$, of course. The problem is in gaussian.spad,
COMPCAT:

```
if R has TranscendentalFunctionCategory then
  half := recip(2::R)::R

if R has RealNumberSystem then
  atan2loc(y: R, x: R): R ==
    pi1 := pi()$R
    pi2 := pi1 * half
    x = 0 => if y >= 0 then pi2 else -pi2

-- Atan in (-pi/2,pi/2]
theta := atan(y * recip(x)::R)
while theta <= -pi2 repeat theta := theta + pi1
while theta > pi2 repeat theta := theta - pi1
```

```

x >= 0 => theta      -- I or IV

if y >= 0 then
  theta + pi1        -- II
else
  theta - pi1        -- III

argument x == atan2loc(imag x, real x)

else
  -- Not ordered so dictate two quadrants
  argument x ==
    zero? real x => pi()$R * half
    atan(imag(x) * recip(real x)::R)

```

so the problem occurs if R does not have "RealNumberSystem":

```
(15) -> argument(-%i*1.0)
```

```
(15) - 1.5707963267 948966192
```

since Float does have RNS, but INT does not. I'm not quite sure what the condition should be. OrderedSet?

bug 7251: integrate(1/(1+z⁴), z=0..1) :: Complex Float

```
integrate(1/(1+z^4), z=0..1) :: Complex Float
```

```
(1) - 0.2437477471 9968052418
```

Type: Complex(Float)

```
(2) -> integrate(1/(1+z^4), z=0..1)
```

```
(2)
```

$$\frac{\sqrt{2} \log(\sqrt{2} + 2) - \sqrt{2} \log(-\sqrt{2} + 2) - 2\sqrt{2} \operatorname{atan}\left(\frac{1}{\sqrt{2} - 1}\right)}{8}$$

$$+ \frac{-2\sqrt{2} \operatorname{atan}\left(\frac{1}{\sqrt{2} + 1}\right)}{8}$$

/

8

Type: Union(f1: OrderedCompletion(Expression(Integer)),...)

should be

(4) -> integrate(1/(1+z^4),z=0..1)

```
(4)
      +-+
      1
- 4\|2 atan(-----)
      +-----+
      +-+ | +-+      +-+
      \|2 \|- \|2  + 2  + \|2  - 1
+
      +-+
      1
- 4\|2 atan(-----) + \|2 log(\|2  + 2)
      +-----+
      +-+ | +-+      +-+
      \|2 \|\|2  + 2  + \|2  + 1
+
      +-+      +-+      +-+      1      +-+      1
      - \|2 log(- \|2  + 2) + 4\|2 atan(-----) + 4\|2 atan(-----)
                                  +-+      +-+
                                  \|2  - 1      \|2  + 1
/
8
Type: Union(f1: OrderedCompletion(Expression(Integer)),...)
```

(5) -> integrate(1/(1+z^4),z=0..1)::Complex Float

(5) 0.8669729873_3991103758

Type: Complex(Float)

bug 7261: wrong integral

reported by Thomas Baruchel <baruchel@gmx.com>

```
t1:=(16*x^14-125*x^10+150*x^6+375*x^2)/(256*x^16+480*x^12+1025*x^8+750*x^4+625)
integrate(t1,x)
```

bug 7267: src/input/liu is not always **

This makes it clear that the exponential operation semantics is different for the use of “\verb|^|” and “\verb|**|” in some cases.

Does exponential operation “\verb|^|” and “\verb|**|” of a differential operator, say L, means repeating multiplications of L in Axiom ? If

so, it seems the following code produced an unexpected result:

bug 7270: integral confused by branches

```
(1) -> integrate(x*(asin(sin(x)))^3,x)

      5
      x
(1)  --
      5
                                     Type: Union(Expression(Integer),...)
(2) -> (%pi/2)*integrate((asin(sin(x)))^3,x=0..%pi)

      5
      %pi
(2)  ----
      8
                                     Type: Expression(Integer)
(3) -> integrate((asin(sin(x)))^3,x=0..%pi)

      4
      %pi
(3)  ----
      4
                                     Type: Union(f1: OrderedCompletion(Expression(Integer)),...)
(4) -> integrate((asin(sin(x)))^3,x=0..%pi/2)

      4
      %pi
(4)  ----
      64
                                     Type: Union(f1: OrderedCompletion(Expression(Integer)),...)
(5) -> integrate(x^3,x=0..%pi/2)

      4
      %pi
(5)  ----
      64
                                     Type: Union(f1: OrderedCompletion(Expression(Integer)),...)
(6) -> numeric %

      (6)  1.5220170474 062880818
                                                     Type: Float
(7) -> integrate(x^3,x)

      1  4
(7)  - x
```

```

      4
                                         Type: Polynomial(Fraction(Integer))
(8) -> numeric(%pi^4/4)

      (8)  24.3522727585 00609309
                                         Type: Float

```

bug 7275: typos in axbook

(reported by Doug Stewart <doug.dastew@gmail.com>)

section-0.3.xhtml

```

"but integer division isn't quite so obvious, For example, if one types:
23
Type: Fraction Integer
a fractional result is obtained. The function used to display fractions
attempts to produce the most readable answer. In the example:

```

probably should read 2/3

Also

```

"4.6
Type: Float
Although Axiom can convert this back to a fraction it might not be
the same fraction you started with as due to rounding errors. For example,
the following conversion appears to be without error but others might not:

```

```

235
Type: Fraction Integer

```

bug 7276: The axbook does not display correctly in Chrome

(reported by Doug Stewart <doug.dastew@gmail.com>)

bug 7306: bad counter in)regress function

There are 27 stanzas in src/input/intalgorithm.input.pamphlet
 All 27 stanzas pass regression tests but...

regression result passed 25 of 25 stanzas file intalgorithm

bug 7305: series should simplify

-> series(sin(x),x=0)

$$(1) \quad x - \frac{1}{6}x^3 + \frac{1}{120}x^5 - \frac{1}{5040}x^7 + \frac{1}{362880}x^9 - \frac{1}{39916800}x^{11} + 0(x^{12})$$

(2) -> %/x

$$(2) \quad -\frac{1}{x} + \frac{1}{6x}x^3 - \frac{1}{120x}x^5 + \frac{1}{5040x}x^7 - \frac{1}{362880x}x^9 + \frac{1}{39916800x}x^{11} + 0(x^{12})$$

(3) -> %-1

$$(3) \quad -1 + \frac{1}{x} - \frac{1}{6x}x^3 + \frac{1}{120x}x^5 - \frac{1}{5040x}x^7 + \frac{1}{362880x}x^9 + 0(x^{11})$$

todo 340: exponential-linear

t1:=(1.6)^x + 122.35*x - 5054.4

Type: Expression(Float)

solve(t1,x) fails

see: Corl93.pdf On the Lambert W Function

see: Kalm01.pdf A Generalized Logarithm for Exponential-Linear Equations

todo 339: missing side conditions

integrate((x-b)^(-1),x)

(1) log(x - b)

Type: Union(Expression(Integer),...)

should show the side-condition $x > b$ or should be $\log(\text{abs}(x-b))$

todo 337: erf does not evaluate

```

(2) -> - 0.25* erf(0.3535533905932737622) + 0.60206532676429947778
      (2)  - 0.25 erf(0.3535533905 932737622) + 0.6020653267 6429947778
                                           Type: Expression(Float)
(3) -> erf(0.3535533905932737622)
      (3)  erf(0.3535533905 932737622)

```

bug 7301: rich12c.input 310 failed

Error detected within library code: index out of range

Regression computation shows prior correct output

bug 7300: outputDomainConstructor failure

```

outputDomainConstructor(Integer)$Lisp

>> System error:
Caught fatal error [memory may be damaged]

```

bug 7295: guessRat : (1 . arbitrary) cannot be coerced

```

t1:=guessRat(q)([1,q,q^2], [])
--R
--R
--R  >> Error detected within library code:
--R  (1 . arbitrary) cannot be coerced to mode (NonNegativeInteger)
--R
--R  Continuing to read the file...
--R

```

bug 7294: GuessOption: displayAsGF not set

```

--S 7 of 102

```

```

t1:=[0,1,0,1,1,0,0,0,0,0,0,0,0,0]
--R
--R
--R (1) [0,1,0,1,1,0,0,0,0,0,0,0,0,0]
--R                                         Type: List(NonNegativeInteger)
--E 7

--S 8 of 102
t2:=first guessADE first(t1,10)
--R
--R
--R >> Error detected within library code:
--R   GuessOption: displayAsGF not set
--R
--R   Continuing to read the file...
--R
--E 8

```

warnings 20572: IDPAM

```

finalizing nrlib IDPAM
Warnings:
  [1] +: res has no value
  [2] +: endcell has no value

```

warnings 20571: INBFF

```

>compiling INBFF.spad to INBFF.nrlib

Warnings:
  [1] dAndcExp: erg has no value

```

warnings 20570: DFLOAT

```

>compiling DFLOAT.spad to DFLOAT.nrlib

Warnings:
  [1] mantissa: MANTISSA has no value
  [2] exponent: EXPONENT has no value
  [3] hash: signature of lhs not unique: (Integer)$ chosen

```

warnings 20569: LIST

```
>compiling LIST.spad to LIST.nrlib

Warnings:
  [1] setIntersection:  u has no value
  [2] setDifference:   lu has no value
```

warnings 20568: MFLOAT

```
>compiling MFLOAT.spad to MFLOAT.nrlib

Warnings:
  [1] normalise:  man has no value
  [2] changeBase: newMan has no value
  [3] changeBase: newExp has no value
  [4] changeBase:  f has no value
```

nonextend 60076: AFFSP

```
>compiling AFFSP.spad to AFFSP.nrlib

-----non extending category-----
.. AffineSpace(#1,#2) of cat
(|AffineSpaceCategory| |#2|)  has no
(|ListAggregate| |#2|)      finalizing nrlib AFFSP
```

nonextend 60075: BSTREE

```
>compiling BSTREE.spad to BSTREE.nrlib
-----non extending category-----
.. BinarySearchTree(#1) of cat
(|Join| (|BinaryTreeCategory| |#1|) (CATEGORY |domain| (ATTRIBUTE |shallowlyMutable|) (ATTRIBUTE
```

nonextend 60074: BTOURN

```
>compiling BTOURN.spad to BTOURN.nrlib
```

```
-----non extending category-----
```

```
.. BinaryTournament(#1) of cat
```

```
(|Join| (|BinaryTreeCategory| |#1|) (CATEGORY |domain| (ATTRIBUTE |shallowlyMutable|) (SIGNATURE |binaryTournament|)))
```

nonextend 60073: BBTREE

```
>compiling BBTREE.spad to BBTREE.nrlib
```

```
-----non extending category-----
```

```
.. BalancedBinaryTree(#1) of cat
```

```
(|Join| (|BinaryTreeCategory| |#1|) (CATEGORY |domain| (ATTRIBUTE |finiteAggregate|) (ATTRIBUTE |shallowlyMutable|)))
```

nonextend 60072: D01GBFA

```
>compiling D01GBFA.spad to D01GBFA.nrlib
```

```
-----non extending category-----
```

```
.. d01gbfAnnaType of cat
```

```
(|NumericalIntegrationCategory|) has no
```

```
(|TableAggregate| (|Symbol|) (|Any|)) finalizing nrlib D01GBFA
```

nonextend 60071: D02EJFA

```
>compiling D02EJFA.spad to D02EJFA.nrlib
```

```
-----non extending category-----
```

```
.. d02ejfAnnaType of cat
```

```
(|OrdinaryDifferentialEquationsSolverCategory|) has no
```

```
(|TableAggregate| (|Symbol|) (|Any|)) finalizing nrlib D02EJFA
```

nonextend 60070: D03FAFA

```
>compiling D03FAFA.spad to D03FAFA.nrlib
```

```
-----non extending category-----
```

```

.. d03fafAnnaType of cat
(|PartialDifferentialEquationsSolverCategory|) has no
(|TableAggregate| (|Symbol|) (|Any|)) finalizing nrllib D03FAFA

```

nonextend 60069: D01FCFA

```
>compiling D01FCFA.spad to D01FCFA.nrllib
```

```

-----non extending category-----
.. d01fcfAnnaType of cat
(|NumericalIntegrationCategory|) has no
(|TableAggregate| (|Symbol|) (|Any|)) finalizing nrllib D01FCFA

```

nonextend 60068: LPOLY

```
>compiling LPOLY.spad to LPOLY.nrllib
```

```

-----non extending category-----
.. LiePolynomial(#1,#2) of cat
(|Join| (|FreeLieAlgebra| |#1| |#2|) (|FreeModuleCat| |#2| (|LyndonWord| |#1|)) (CATEGORY |domain|

```

nonextend 60067: DBASE

```
>compiling DBASE.spad to DBASE.nrllib
```

```

-----non extending category-----
.. Database(#1) of cat
(|Join| (|SetCategory|) (CATEGORY |domain| (SIGNATURE |elt| ($ $ (|QueryEquation|))) (SIGNATURE
(|ListAggregate| |#1|) finalizing nrllib DBASE

```

nonextend 60066: DHMATRIX

```
>compiling DHMATRIX.spad to DHMATRIX.nrllib
```

```

-----non extending category-----
.. DenavitHartenbergMatrix(#1) of cat
(|Join| (|MatrixCategory| |#1| (|Vector| |#1|) (|Vector| |#1|)) (CATEGORY |domain| (SIGNATURE *

```

nonextend 60065: D02BBFA

```
>compiling D02BBFA.spad to D02BBFA.nrlib
-----non extending category-----
.. d02bbfAnnaType of cat
(|OrdinaryDifferentialEquationsSolverCategory|) has no
(|TableAggregate| (|Symbol|) (|Any|)) finalizing nrlib D02BBFA
```

nonextend 60064: D02BHFA

```
>compiling D02BHFA.spad to D02BHFA.nrlib
-----non extending category-----
.. d02bhfAnnaType of cat
(|OrdinaryDifferentialEquationsSolverCategory|) has no
(|TableAggregate| (|Symbol|) (|Any|)) finalizing nrlib D02BHFA
```

nonextend 60063: D02CJFA

```
>compiling D02CJFA.spad to D02CJFA.nrlib
-----non extending category-----
.. d02cjfAnnaType of cat
(|OrdinaryDifferentialEquationsSolverCategory|) has no
(|TableAggregate| (|Symbol|) (|Any|)) finalizing nrlib D02CJFA
```

nonextend 60062: FNLA

```
>compiling FNLA.spad to FNLA.nrlib
-----non extending category-----
.. FreeNilpotentLie(#1,#2,#3) of cat
(|Join| (|NonAssociativeAlgebra| |#3|) (CATEGORY |domain| (SIGNATURE |dimension| ((|NonNegativeInteger|))))
(|IndexedDirectProductCategory| |#3| (|OrdSetInts|)) finalizing nrlib FNLA
Processing FreeNilpotentLie for Browser database:
```

nonextend 60061: POINT

```
>compiling POINT.spad to POINT.nrlib
-----non extending category-----
.. Point(#1) of cat
(|PointCategory| |#1|)    has no  vector : List(#1) -> %
```

nonextend 60060: STRING

```
>compiling STRING.spad to STRING.nrlib
-----non extending category-----
.. String of cat
(|StringCategory|)    has no  hash : % -> Integer
```

nonextend 60059: BINARY

```
>compiling BINARY.spad to BINARY.nrlib
-----non extending category-----
.. BinaryExpansion of cat
(|Join| (|QuotientFieldCategory| (|Integer|)) (CATEGORY |domain| (SIGNATURE |coerce| ((|Fraction
```

nonextend 60058: DECIMAL

```
>compiling DECIMAL.spad to DECIMAL.nrlib
-----non extending category-----
.. DecimalExpansion of cat
(|Join| (|QuotientFieldCategory| (|Integer|)) (CATEGORY |domain| (SIGNATURE |coerce| ((|Fraction
    finalizing nrlib DECIMAL
```

nonextend 60057: E04DGFA

```
>compiling E04DGFA.spad to E04DGFA.nrlib
-----non extending category-----
.. e04dgfAnnaType of cat
(|NumericalOptimizationCategory|)    has no
(|TableAggregate| (|Symbol|) (|Any|))    finalizing nrlib E04DGFA
```


Processing e04dgfAnnaType for Browser database:

nonextend 60056: E04FDFA

```
>compiling E04FDFA.spad to E04FDFA.nrlib
-----non extending category-----
.. e04fdfAnnaType of cat
(|NumericalOptimizationCategory|)  has no
(|TableAggregate| (|Symbol|) (|Any|))  finalizing nrlib E04FDFA
```

nonextend 60055: E04GCFA

```
>compiling E04GCFA.spad to E04GCFA.nrlib
-----non extending category-----
.. e04gcfAnnaType of cat
(|NumericalOptimizationCategory|)  has no
(|TableAggregate| (|Symbol|) (|Any|))  finalizing nrlib E04GCFA
```

nonextend 60054: E04JAFA

```
>compiling E04JAFA.spad to E04JAFA.nrlib
-----non extending category-----
.. e04jafAnnaType of cat
(|NumericalOptimizationCategory|)  has no
(|TableAggregate| (|Symbol|) (|Any|))  finalizing nrlib E04JAFA
    Processing e04jafAnnaType for Browser database:
```

nonextend 60053: E04UCFA

```
>compiling E04UCFA.spad to E04UCFA.nrlib
-----non extending category-----
.. e04ucfAnnaType of cat
(|NumericalOptimizationCategory|)  has no
(|TableAggregate| (|Symbol|) (|Any|))  finalizing nrlib E04UCFA
```

nonextend 60052: FRAC

```
>compiling FRAC.spad to FRAC.nrlib
```

```
-----non extending category-----
```

```
.. Fraction(#1) of cat
```

```
(|Join| (|QuotientFieldCategory| |#1|) (CATEGORY |package| (IF (|has| |#1| (|IntegerNumberSystem|
(IF (|has| |#1| (|OrderedRing|)) (ATTRIBUTE (|OrderedRing|)) |noBranch|) finalizing nrlib FRA
```

nonextend 60051: FTEM

```
>compiling FTEM.spad to FTEM.nrlib
```

```
-----non extending category-----
```

```
.. FortranTemplate of cat
```

```
(|Join| (|FileCategory| (|FileName|) (|String|)) (CATEGORY |package| (SIGNATURE |processTemplate
```

nonextend 60050: GMODPOL

```
>compiling GMODPOL.spad to GMODPOL.nrlib
```

```
(|Module| |#6|) extends
```

```
(|BiModule| |#6| |#6|) but not
```

```
(|BiModule| |#2| |#2|) -----non extending category-----
```

```
.. GeneralModulePolynomial(#1,#2,#3,#4,#5,#6) of cat
```

```
(|Join| (|Module| |#6|) (|Module| |#2|) (CATEGORY |domain| (SIGNATURE |leadingCoefficient| (|#2|
```

```
(|IndexedDirectProductCategory| |#2| (|ModuleMonomial| |#3| |#4| |#5|)) finalizing nrlib GMOD
```

nonextend 60049: HEXADEC

```
>compiling HEXADEC.spad to HEXADEC.nrlib
```

```
-----non extending category-----
```

```
.. HexadecimalExpansion of cat
```

```
(|Join| (|QuotientFieldCategory| (|Integer|)) (CATEGORY |domain| (SIGNATURE |coerce| ((|Fraction
```

nonextend 60048: MINT

```
>compiling MINT.spad to MINT.nrlib
```

```
(|RealConstant|)      extends
(|ConvertibleTo| (|DoubleFloat|))      but not
(|ConvertibleTo| (|String|)) -----non extending category-----
.. MachineInteger of cat
(|Join| (|FortranMachineTypeCategory|) (|IntegerNumberSystem|) (CATEGORY |domain| (SIGNATURE |maxint| ((|Pos
(|ConvertibleTo| (|String|))      finalizing nrlib MINT
```

nonextend 60047: NSUP

```
>compiling NSUP.spad to NSUP.nrlib
```

```
-----non extending category-----
.. NewSparseUnivariatePolynomial(#1) of cat
(|Join| (|UnivariatePolynomialCategory| |#1|) (|CoercibleTo| (|SparseUnivariatePolynomial| |#1|)) (|Retract
```

nonextend 60046: SMTS

```
>compiling SMTS.spad to SMTS.nrlib
```

```
-----non extending category-----
.. SparseMultivariateTaylorSeries(#1,#2,#3) of cat
(|Join| (|MultivariateTaylorSeriesCategory| |#1| |#2|) (CATEGORY |domain| (SIGNATURE |coefficient| (|#3| $
```

nonextend 60045: SUP

```
>compiling SUP.spad to SUP.nrlib
```

```
-----non extending category-----
.. SparseUnivariatePolynomial(#1) of cat
(|Join| (|UnivariatePolynomialCategory| |#1|) (CATEGORY |domain| (SIGNATURE |outputForm| ((|OutputForm|) $
(IF (|has| |#1| (|IntegralDomain|)) (IF (|has| (|NonNegativeInteger|) (|CancellationAbelianMonoid|)) (SIGNA
```

nonextend 60044: SUEXPR

```
>compiling SUEXPR.spad to SUEXPR.nrllib
-----non extending category-----
.. SparseUnivariatePolynomialExpressions(#1) of cat
(|Join| (|UnivariatePolynomialCategory| |#1|) (CATEGORY |package| (IF (|has| |#1| (|Transcendent
```

nonextend 60043: UP

```
>compiling UP.spad to UP.nrllib
-----non extending category-----
.. UnivariatePolynomial(#1,#2) of cat
(|Join| (|UnivariatePolynomialCategory| |#2|) (CATEGORY |domain| (SIGNATURE |coerce| ($ (|Variab
```

nonextend 60042: EXPUPXS

```
>compiling EXPUPXS.spad to EXPUPXS.nrllib
-----non extending category-----
.. ExponentialOfUnivariatePuisseuxSeries(#1,#2,#3) of cat
(|Join| (|UnivariatePuisseuxSeriesCategory| |#1|) (|OrderedAbelianMonoid|) (CATEGORY |domain| (SI
(|UnivariatePuisseuxSeriesConstructorCategory| |#1| (|UnivariateLaurentSeries| |#1| |#2| |#3|))
```

nonextend 60041: BSD

```
>compiling BSD.spad to BSD.nrllib
-----non extending category-----
.. BasicStochasticDifferential of cat
(|Join| (|OrderedSet|) (CATEGORY |domain| (ATTRIBUTE (|ConvertibleTo| (|Symbol|))) (SIGNATURE |c
(|ConvertibleTo| (|InputForm|)) finalizing nrllib BSD
```

nonextend 60040: ANTISYM

```
>compiling ANTISYM.spad to ANTISYM.nrllib
-----non extending category-----
```

```
.. AntiSymm(#1,#2) of cat
(|Join| (|LeftAlgebra| |#1|) (|RetractableTo| |#1|) (CATEGORY |domain| (SIGNATURE |leadingCoefficient| (|#1|
(|BiModule| |#1| |#1|)      finalizing nrlib ANTISYM
```

nonextend 60039: FEXPR

```
>compiling FEXPR.spad to FEXPR.nrlib
```

```
-----non extending category-----
.. FortranExpression(#1,#2,#3) of cat
(|Join| (|ExpressionSpace|) (|Algebra| |#3|) (|RetractableTo| |#3|) (|PartialDifferentialRing| (|Symbol|))
(|FunctionSpace| |#3|)      finalizing nrlib FEXPR
```

nonextend 60038: GDMP

```
>compiling GDMP.spad to GDMP.nrlib
```

```
-----non extending category-----
.. GeneralDistributedMultivariatePolynomial(#1,#2,#3) of cat
(|Join| (|PolynomialCategory| |#2| |#3| (|OrderedVariableList| |#1|)) (CATEGORY |domain| (SIGNATURE |reorder
(IF (|has| |#2| (|IntegralDomain|)) (IF (|has| |#3| (|CancellationAbelianMonoid|)) (SIGNATURE |fmecg| ($ $
```

nonextend 60037: HACKPI

```
>compiling HACKPI.spad to HACKPI.nrlib
```

```
-----non extending category-----
.. Pi of cat
(|Join| (|Field|) (|CharacteristicZero|) (|RetractableTo| (|Integer|)) (|RetractableTo| (|Fraction| (|Integer|
(|QuotientFieldCategory| (|SparseUnivariatePolynomial| (|Integer|)))      finalizing nrlib HACKPI
```

nonextend 60036: LIB

```
>compiling LIB.spad to LIB.nrlib
```

```
-----non extending category-----
.. Library of cat
```

```
(|Join| (|TableAggregate| (|String|) (|Any|)) (CATEGORY |domain| (SIGNATURE |library| ($ (|FileM
(|FileCategory| (|FileName|) (|Record| (|:| |key| (|String|)) (|:| |entry| (|Any|))))    finaliz
```

nonextend 60035: LODO

```
>compiling LODO.spad to LODO.nrlib
```

```
-----non extending category-----
.. LinearOrdinaryDifferentialOperator(#1,#2) of cat
(|LinearOrdinaryDifferentialOperatorCategory| |#1|)    has no  outputForm : (% ,OutputForm) -> Ou
```

nonextend 60034: M3D

```
>compiling M3D.spad to M3D.nrlib
```

```
-----non extending category-----
.. ThreeDimensionalMatrix(#1) of cat
(|Join| (|HomogeneousAggregate| |#1|) (CATEGORY |domain| (IF (|has| |#1| (|Ring|)) (PROGN (SIGNA
(|OneDimensionalArrayAggregate| (|PrimitiveArray| (|PrimitiveArray| |#1|))))    finalizing nrlib
```

nonextend 60033: NSDPS

```
>compiling NSDPS.spad to NSDPS.nrlib
```

```
-----non extending category-----
.. NeitherSparseOrDensePowerSeries(#1) of cat
(|Join| (|LocalPowerSeriesCategory| |#1|) (|LazyStreamAggregate| (|Record| (|:| |k| (|Integer|))
(ATTRIBUTE |shallowlyMutable|)    finalizing nrlib NSDPS
```

nonextend 60032: RESULT

```
>compiling RESULT.spad to RESULT.nrlib
```

```
-----non extending category-----
.. Result of cat
(|Join| (|TableAggregate| (|Symbol|) (|Any|)) (CATEGORY |package| (SIGNATURE |showScalarValues|
(ATTRIBUTE |finiteAggregate|)    finalizing nrlib RESULT
```

nonextend 60031: RMATRIX

>compiling RMATRIX.spad to RMATRIX.nrlib

```
-----non extending category-----
.. RectangularMatrix(#1,#2,#3) of cat
(|Join| (|RectangularMatrixCategory| |#1| |#2| |#3| (|DirectProduct| |#2| |#3|) (|DirectProduct| |#1| |#3|))
(|MatrixCategory| |#3| (|Vector| |#3|) (|Vector| |#3|))    finalizing nrlib RMATRIX
```

nonextend 60030: ROMAN

>compiling ROMAN.spad to ROMAN.nrlib

```
(|RealConstant|)    extends
(|ConvertibleTo| (|DoubleFloat|))    but not
(|ConvertibleTo| (|String|)) -----non extending category-----
.. RomanNumeral of cat
(|Join| (|IntegerNumberSystem|) (CATEGORY |domain| (ATTRIBUTE |canonical|) (ATTRIBUTE |canonicalsClosed|) (
(|ConvertibleTo| (|String|))    finalizing nrlib ROMAN
```

nonextend 60029: ROUTINE

>compiling ROUTINE.spad to ROUTINE.nrlib

```
-----non extending category-----
.. RoutinesTable of cat
(|Join| (|TableAggregate| (|Symbol|) (|Any|)) (CATEGORY |domain| (SIGNATURE |concat| ($ $ $)) (SIGNATURE |r
```

nonextend 60028: SQMATRIX

>compiling SQMATRIX.spad to SQMATRIX.nrlib

```
-----non extending category-----
.. SquareMatrix(#1,#2) of cat
(|Join| (|SquareMatrixCategory| |#1| |#2| (|DirectProduct| |#1| |#2|) (|DirectProduct| |#1| |#2|)) (|Coerci
(|MatrixCategory| |#2| (|Vector| |#2|) (|Vector| |#2|))    finalizing nrlib SQMATRIX
```

nonextend 60027: ALGSC

```
>compiling ALGSC.spad to ALGSC.nrlib
```

```
-----non extending category-----
.. AlgebraGivenByStructuralConstants(#1,#2,#3,#4) of cat
(|Join| (|FramedNonAssociativeAlgebra| |#1|) (|LeftModule| (|SquareMatrix| |#2| |#1|)) (CATEGORY
(|DirectProductCategory| |#2| |#1|)) finalizing nrlib ALGSC
Processing AlgebraGivenByStructuralConstants for Browser database:
```

nonextend 60026: AN

```
>compiling AN.spad to AN.nrlib
```

```
(|RealConstant|) extends
(|ConvertibleTo| (|DoubleFloat|)) but not
(|ConvertibleTo| (|Complex| (|Float|))) -----non extending category-----
.. AlgebraicNumber of cat
(|Join| (|ExpressionSpace|) (|AlgebraicallyClosedField|) (|RetractableTo| (|Integer|)) (|Retract
```

nonextend 60025: D03EEFA

```
>compiling D03EEFA.spad to D03EEFA.nrlib
```

```
-----non extending category-----
.. d03eefAnnaType of cat
(|PartialDifferentialEquationsSolverCategory|) has no
(|TableAggregate| (|Symbol|) (|Any|)) finalizing nrlib D03EEFA
Processing d03eefAnnaType for Browser database:
-----constructor-----
```

nonextend 60024: D01AGNT

```
>compiling D01AGNT.spad to D01AGNT.nrlib
```

```
-----non extending category-----
.. d01AgentsPackage of cat
(CATEGORY |package| (SIGNATURE |rangeIsFinite| ((|Union| (|:| |finite| "The range is finite") (|
```


nonextend 60023: DERHAM

```
>compiling DERHAM.spad to DERHAM.nrlib
```

```
-----non extending category-----
```

```
.. DeRhamComplex(#1,#2) of cat
```

```
(|Join| (|LeftAlgebra| (|Expression| |#1|)) (|RetractableTo| (|Expression| |#1|)) (CATEGORY |domain| (SIGNA
```

nonextend 60022: D01TRNS

```
>compiling D01TRNS.spad to D01TRNS.nrlib
```

```
-----non extending category-----
```

```
.. d01TransformFunctionType of cat
```

```
(|NumericalIntegrationCategory|) has no
```

```
(|TableAggregate| (|Symbol|) (|Any|)) finalizing nrlib D01TRNS
```

nonextend 60021: E04NAFA

```
>compiling E04NAFA.spad to E04NAFA.nrlib
```

```
-----non extending category-----
```

```
.. e04nafAnnaType of cat
```

```
(|NumericalOptimizationCategory|) has no
```

```
(|TableAggregate| (|Symbol|) (|Any|)) finalizing nrlib E04NAFA
```

```
Processing e04nafAnnaType for Browser database:
```

nonextend 60020: EXPEXPAN

```
>compiling EXPEXPAN.spad to EXPEXPAN.nrlib
```

```
-----non extending category-----
```

```
.. ExponentialExpansion(#1,#2,#3,#4) of cat
```

```
(|Join| (|QuotientFieldCategory| (|UnivariatePuisseuxSeriesWithExponentialSingularity| |#1| |#2| |#3| |#4|))
```

```
(IF (|has| (|UnivariatePuisseuxSeriesWithExponentialSingularity| |#1| |#2| |#3| |#4|) (|IntegerNumberSystem|
```

nonextend 60019: GCNAALG

```
>compiling GCNAALG.spad to GCNAALG.nrlib
```

```
(|Module| (|Fraction| (|Polynomial| |#1|))) extends
(|LeftModule| (|Fraction| (|Polynomial| |#1|))) but not
(|LeftModule| (|SquareMatrix| |#2| (|Fraction| (|Polynomial| |#1|)))) finalizing nrlib GCNAALG
```

nonextend 60018: IAN

```
>compiling IAN.spad to IAN.nrlib
```

```
-----non extending category-----
.. InnerAlgebraicNumber of cat
(|Join| (|ExpressionSpace|) (|AlgebraicallyClosedField|) (|RetractableTo| (|Integer|)) (|Retract
(|FunctionSpace| (|Integer|)) finalizing nrlib IAN
```

nonextend 60017: LSQM

```
>compiling LSQM.spad to LSQM.nrlib
```

```
-----non extending category-----
.. LieSquareMatrix(#1,#2) of cat
(|Join| (|SquareMatrixCategory| |#1| |#2| (|DirectProduct| |#1| |#2|) (|DirectProduct| |#1| |#2|)
(|CoercibleTo| (|SquareMatrix| |#1| |#2|)) finalizing nrlib LSQM
```

nonextend 60016: MCMPLEX

```
>compiling MCMPLEX.spad to MCMPLEX.nrlib
```

```
-----non extending category-----
.. MachineComplex of cat
(|Join| (|FortranMachineTypeCategory|) (|ComplexCategory| (|MachineFloat|)) (CATEGORY |domain| (
```

nonextend 60015: MYUP

```
>compiling MYUP.spad to MYUP.nrlib
```

```
-----non extending category-----
.. MyUnivariatePolynomial(#1,#2) of cat
```

```
(|Join| (|UnivariatePolynomialCategory| |#2|) (CATEGORY |domain| (ATTRIBUTE (|RetractableTo| (|Symbol|)))) (
```

nonextend 60014: MYEXPR

```
>compiling MYEXPR.spad to MYEXPR.nrllib
```

```
-----non extending category-----
.. MyExpression(#1,#2) of cat
(|Join| (|FunctionSpace| |#2|) (|IntegralDomain|) (|RetractableTo| (|MyUnivariatePolynomial| |#1| |#2|)) (|
(IF (|has| |#2| (|IntegralDomain|)) (PROGN (ATTRIBUTE (|AlgebraicallyClosedFunctionSpace| |#2|)) (ATTRIBUTE
```

nonextend 60013: MYEXPR

```
>compiling MYEXPR.spad to MYEXPR.nrllib
```

```
-----non extending category-----
.. MyUnivariatePolynomial(#1,#2) of cat
(|Join| (|UnivariatePolynomialCategory| |#2|) (CATEGORY |domain| (ATTRIBUTE (|RetractableTo| (|Symbol|)))) (
-----non extending category-----
.. MyExpression(#1,#2) of cat
(|Join| (|FunctionSpace| |#2|) (|IntegralDomain|) (|RetractableTo| (|MyUnivariatePolynomial| |#1| |#2|)) (|
(IF (|has| |#2| (|IntegralDomain|)) (PROGN (ATTRIBUTE (|AlgebraicallyClosedFunctionSpace| |#2|)) (ATTRIBUTE
```

nonextend 60012: SD

```
>compiling SD.spad to SD.nrllib
```

```
-----non extending category-----
.. StochasticDifferential(#1) of cat
(|Join| (|Rng|) (|Module| (|Expression| |#1|)) (CATEGORY |domain| (ATTRIBUTE (|RetractableTo| (|BasicStochas
(|PolynomialCategory| (|Expression| |#1|) (|IndexedExponents| (|BasicStochasticDifferential|)) (|BasicStoch
```

nonextend 60011: SUTS

```
>compiling SUTS.spad to SUTS.nrllib
```

```
(|UnivariateTaylorSeriesCategory| |#1|) extends
```

```
(|UnivariatePowerSeriesCategory| |#1| (|NonNegativeInteger|)) but not
(|UnivariatePowerSeriesCategory| |#1| (|Integer|)) -----non extending category-----
.. SparseUnivariateTaylorSeries(#1,#2,#3) of cat
(|Join| (|UnivariateTaylorSeriesCategory| |#1|) (CATEGORY |domain| (SIGNATURE |coerce| ($ (|Univ
(|UnivariatePowerSeriesCategory| |#1| (|Integer|)) finalizing nrlib SUTS
```

nonextend 60010: UPXSSING

```
>compiling UPXSSING.spad to UPXSSING.nrlib
```

```
-----non extending category-----
.. UnivariatePuisseuxSeriesWithExponentialSingularity(#1,#2,#3,#4) of cat
(|Join| (|FiniteAbelianMonoidRing| (|UnivariatePuisseuxSeries| |#2| |#3| |#4|) (|ExponentialOfUni
(IF (|has| (|UnivariatePuisseuxSeries| |#2| |#3| |#4|) (|IntegralDomain|)) (IF (|has| (|Exponenti
```

nonextend 60009: ESCONT

```
>compiling ESCONT.spad to ESCONT.nrlib
```

```
-----non extending category-----
.. ExpertSystemContinuityPackage of cat
(CATEGORY |package| (SIGNATURE |getlo| ((|DoubleFloat|) (|Segment| (|OrderedCompletion| (|Double
```

nonextend 60008: GSERIES

```
>compiling GSERIES.spad to GSERIES.nrlib
```

```
-----non extending category-----
.. GeneralUnivariatePowerSeries(#1,#2,#3) of cat
(|Join| (|UnivariatePuisseuxSeriesCategory| |#1|) (CATEGORY |domain| (SIGNATURE |coerce| ($ (|Var
(|UnivariatePuisseuxSeriesConstructorCategory| |#1| (|UnivariateLaurentSeries| |#1| |#2| |#3|))
```

nonextend 60007: RGCHAIN

```
>compiling RGCHAIN.spad to RGCHAIN.nrlib
```

```
-----non extending category-----
.. RegularChain(#1,#2) of cat
```

```
(|Join| (|RegularTriangularSetCategory| |#1| (|IndexedExponents| (|OrderedVariableList| |#2|)) (|OrderedVar
```

nonextend 60006: SULS

```
>compiling SULS.spad to SULS.nrlib
```

```
-----non extending category-----
.. SparseUnivariateLaurentSeries(#1,#2,#3) of cat
(|Join| (|UnivariateLaurentSeriesConstructorCategory| |#1| (|SparseUnivariateTaylorSeries| |#1| |#2| |#3|))
```

nonextend 60005: UTS

```
>compiling UTS.spad to UTS.nrlib
```

```
-----non extending category-----
.. UnivariateTaylorSeries(#1,#2,#3) of cat
(|Join| (|UnivariateTaylorSeriesCategory| |#1|) (CATEGORY |domain| (SIGNATURE |coerce| ($ (|UnivariatePolyn
```

nonextend 60004: UTSZ

```
>compiling UTSZ.spad to UTSZ.nrlib
```

```
-----non extending category-----
.. UnivariateTaylorSeriesCZero(#1,#2) of cat
(|Join| (|UnivariateTaylorSeriesCategory| |#1|) (CATEGORY |domain| (SIGNATURE |coerce| ($ (|UnivariatePolyn
```

nonextend 60003: INFCLSPT

```
>compiling INFCLSPT.spad to INFCLSPT.nrlib
```

```
-----non extending category-----
.. InfinitelyClosePoint(#1,#2,#3,#4,#5,#6,#7,#8,#9) of cat
(|Join| (|InfinitelyClosePointCategory| |#1| |#2| |#3| |#4| |#5| |#6| |#7| |#8| |#9|) (CATEGORY |domain| (SI
(|RecordCategory| (|:| |point| |#5|) (|:| |curve| (|DistributedMultivariatePolynomial| (|construct| X Y) |#
```

nonextend 60002: GUESSUP

```
>compiling GUESSUP.spad to GUESSUP.nrlib
```

```
-----non extending category-----
.. GuessUnivariatePolynomial(#1) of cat
(CATEGORY |package| (SIGNATURE |guess| ((|List| (|Record| (|:| |function| (|MyExpression| |#1| (
(IF (|has| (|Fraction| (|MyUnivariatePolynomial| |#1| (|Integer|)))) (|RetractableTo| (|Symbol|))
```

nonextend 60001: NNI

```
>compiling NNI.spad to NNI.nrlib
```

```
-----non extending category-----
.. NonNegativeInteger of cat
(|Join| (|OrderedAbelianMonoidSup|) (|Monoid|) (CATEGORY |domain| (SIGNATURE |quo| ($ $ $)) (SIG
(|IntegerNumberSystem|)      finalizing nrlib NNI
```

nonextend 60000: PI

```
>compiling PI.spad to PI.nrlib
```

```
-----non extending category-----
.. PositiveInteger of cat
(|Join| (|AbelianSemiGroup|) (|OrderedSet|) (|Monoid|) (CATEGORY |domain| (SIGNATURE |gcd| ($ $
(|OrderedAbelianMonoidSup|)      finalizing nrlib PI
```

dup 50004: duplication definition

```
Warning: RPOLCAT-;exactQuo has a duplicate definition in this file
Warning: RPOLCAT-;ZToR has a duplicate definition in this file
Warning: RPOLCAT-;PZToPR has a duplicate definition in this file
```

dup 50003: duplication definition

```
Warning: PF0;cmult has a duplicate definition in this file
```



```
--->bookvol10.3.pamphlet-->None(constructor): Mismatch: left pren matches right brace
"\spadtype{None} implements a type with no objects. It is mainly used in technical situations w
```

typos 40348: NUMINT

```
>compiling NUMINT.spad to NUMINT.nrlib

----- (measure ((Record (: measure (Float)) (: explanations (String)) (: extra (Result))) (Rou
----- (measure ((Record (: measure (Float)) (: explanations (String)) (: extra (Result))) (Rou
```

typos 40343: SETCATD

```
>compiling SETCATD.spad to SETCATD.nrlib

--->bookvol10.2.pamphlet-->SetCategoryWithDegree((degree ((PositiveInteger) %)): Not documented
```

typos 40337: AFSPCAT

```
>compiling AFSPCAT.spad to AFSPCAT.nrlib

--->bookvol10.2.pamphlet-->AffineSpaceCategory((origin (%)): Not documented!!!!

--->bookvol10.2.pamphlet-->AffineSpaceCategory((definingField (K %)): Not documented!!!!
--->bookvol10.2.pamphlet-->AffineSpaceCategory((coerce ((List K) %)): Not documented!!!!
```

typos 40328: PlacesCategory

```
--->bookvol10.2.pamphlet-->PlacesCategory((+ ((Divisor %) % %)): Not documented!!!!
--->bookvol10.2.pamphlet-->PlacesCategory((+ ((Divisor %) (Divisor %) %)): Not documented!!!!
--->bookvol10.2.pamphlet-->PlacesCategory((+ ((Divisor %) % (Divisor %))): Not documented!!!!
--->bookvol10.2.pamphlet-->PlacesCategory((- ((Divisor %) % %)): Not documented!!!!
--->bookvol10.2.pamphlet-->PlacesCategory((- ((Divisor %) (Divisor %) %)): Not documented!!!!
--->bookvol10.2.pamphlet-->PlacesCategory((- ((Divisor %) % (Divisor %))): Not documented!!!!
```



```

--->bookvol10.2.pamphlet-->PlacesCategory((- ((Divisor %) %)): Not documented!!!!
--->bookvol10.2.pamphlet-->PlacesCategory(((* ((Divisor %) (Integer) %)): Not documented!!!!
--->bookvol10.2.pamphlet-->PlacesCategory((reduce ((Divisor %) (List %)): Not documented!!!!
--->bookvol10.2.pamphlet-->PlacesCategory((create (% (List K)): Not documented!!!!
--->bookvol10.2.pamphlet-->PlacesCategory((create (% (Symbol)): Not documented!!!!

```

typos 40327: PRSPCAT

```
>compiling PRSPCAT.spad to PRSPCAT.nrllib
```

```
--->bookvol10.2.pamphlet-->ProjectiveSpaceCategory((lastNonNul (INT %)): Not documented!!!!
```

```
--->bookvol10.2.pamphlet-->ProjectiveSpaceCategory((definingField (K %)): Not documented!!!!
```

typos 40324: IFAMON

```
>compiling IFAMON.spad to IFAMON.nrllib
```

```

.. InnerFreeAbelianMonoid(#1,#2,#3) of cat
(|FreeAbelianMonoidCategory| |#1| |#2|)   has no   outputForm : (%,(OutputForm,OutputForm) -> OutputForm),

```

typos 40321a: BSTREE

```
>compiling BSTREE.spad to BSTREE.nrllib
```

```
-----non extending category-----
```

```
.. BinarySearchTree(#1) of cat
```

```
(|Join| (|BinaryTreeCategory| |#1|) (CATEGORY |domain| (ATTRIBUTE |shallowlyMutable|) (ATTRIBUTE |finiteAgg|
```

typos 40317: FACTFUNC

```
>compiling FACTFUNC.spad to FACTFUNC.nrllib
```

```

----- (nthRoot ((Record (: exponent N) (: coef M) (: radicand (List M))) (Factored M) N))-----
----- (log ((List (Record (: coef N) (: logand M))) (Factored M)))-----

```

typos 40301: DIRRING

```
>compiling DIRRING.spad to DIRRING.nrlib

--->bookvol10.3.pamphlet-->DirichletRing((coerce (% FUN))): Not documented!!!!
--->bookvol10.3.pamphlet-->DirichletRing((coerce (FUN %))): Not documented!!!!
--->bookvol10.3.pamphlet-->DirichletRing((coerce (% (Stream Coef)))): Not documented!!!!
--->bookvol10.3.pamphlet-->DirichletRing((coerce ((Stream Coef) %))): Not documented!!!!
```

typos 40300: DIV

```
>compiling DIV.spad to DIV.nrlib
--->bookvol10.3.pamphlet-->Divisor((head (PT %))): Not documented!!!!
--->bookvol10.3.pamphlet-->Divisor((reductum (% %))): Not documented!!!!
```

typos 40299: D01GBFA

```
>compiling D01GBFA.spad to D01GBFA.nrlib

-----non extending category-----
.. d01gbfAnnaType of cat
(|NumericalIntegrationCategory|) has no
(|TableAggregate| (|Symbol|) (|Any|)) finalizing nrlib D01GBFA
Processing d01gbfAnnaType for Browser database:
```

typos 40294: LIECAT

```
>compiling LIECAT.spad to LIECAT.nrlib

--->-->LieAlgebra&(): Spurious comments: \indented{1}{\axiom{JacobiIdentity} means that} \axiom{
-----constructor-----
--->bookvol10.2.pamphlet-->LieAlgebra(): Spurious comments: \indented{1}{\axiom{JacobiIdentity}
```

typos 40288: PACPERC

```
>compiling PACPERC.spad to PACPERC.nrllib
```

```
--->bookvol10.2.pamphlet-->PseudoAlgebraicClosureOfPerfectFieldCategory((definingPolynomial ((SUP %))): Not documented!!!!
--->bookvol10.2.pamphlet-->PseudoAlgebraicClosureOfPerfectFieldCategory((definingPolynomial ((SUP %) %))): Not documented!!!!
--->bookvol10.2.pamphlet-->PseudoAlgebraicClosureOfPerfectFieldCategory((lift ((SUP %) %))): Not documented!!!!
--->bookvol10.2.pamphlet-->PseudoAlgebraicClosureOfPerfectFieldCategory((lift ((SUP %) % %))): Not documented!!!!
--->bookvol10.2.pamphlet-->PseudoAlgebraicClosureOfPerfectFieldCategory((reduce (%) (SUP %))): Not documented!!!!
-----distinguishedRootsOf ((List %) (SparseUnivariatePolynomial %) %))-----
--->bookvol10.2.pamphlet-->PseudoAlgebraicClosureOfPerfectFieldCategory((ground? ((Boolean) %))): Not documented!!!!
--->bookvol10.2.pamphlet-->PseudoAlgebraicClosureOfPerfectFieldCategory((vectorise ((Vector %) % %))): Not documented!!!!
--->bookvol10.2.pamphlet-->PseudoAlgebraicClosureOfPerfectFieldCategory((conjugate (%) %))): Not documented!!!!
--->bookvol10.2.pamphlet-->PseudoAlgebraicClosureOfPerfectFieldCategory((newElement (%) (SUP %) % (Symbol))) : Not documented!!!!
--->bookvol10.2.pamphlet-->PseudoAlgebraicClosureOfPerfectFieldCategory((newElement (%) (SUP %) (Symbol))) : Not documented!!!!
--->bookvol10.2.pamphlet-->PseudoAlgebraicClosureOfPerfectFieldCategory((setTower! ((Void) %))): Not documented!!!!
--->bookvol10.2.pamphlet-->PseudoAlgebraicClosureOfPerfectFieldCategory((fullOutput ((OutputForm) %))): Not documented!!!!
```

typos 40265a: LOCPOWC

```
>compiling LOCPOWC.spad to LOCPOWC.nrllib
```

```
--->bookvol10.2.pamphlet-->LocalPowerSeriesCategory((findCoef (K % (Integer)))): Not documented!!!!
--->bookvol10.2.pamphlet-->LocalPowerSeriesCategory((coerce (%) SER)): Not documented!!!!
--->bookvol10.2.pamphlet-->LocalPowerSeriesCategory((coerce (SER %))): Not documented!!!!
--->bookvol10.2.pamphlet-->LocalPowerSeriesCategory((orderIfNegative ((Union (Integer) failed) %))): Not documented!!!!
--->bookvol10.2.pamphlet-->LocalPowerSeriesCategory((removeFirstZeroes (%) %)): Not documented!!!!
--->bookvol10.2.pamphlet-->LocalPowerSeriesCategory((sbt (%) % %)): Not documented!!!!
--->bookvol10.2.pamphlet-->LocalPowerSeriesCategory(constructor): Not documented!!!!
--->bookvol10.2.pamphlet-->LocalPowerSeriesCategory(): Missing Description
```

typos 40248a: LOP

```
>compiling LOP.spad to LOP.nrllib
```

```
--->bookvol10.4.pamphlet-->LinesOpPack((rowEchWoZeroLinesWOVectorise ((Matrix K) (Matrix K)))): Not documented!!!!
--->bookvol10.4.pamphlet-->LinesOpPack((rowEchWoZeroLines ((Matrix K) (Matrix K)))): Not documented!!!!
```

typos 40246: PERMAN

```
>compiling PERMAN.spad to PERMAN.nrllib
```

```
--->bookvol10.4.pamphlet--->Permanent((commutative (attribute *))) : Not documented!!!!
```

typos 40242: ASP24

```
>compiling ASP24.spad to ASP24.nrllib
```

```
--->bookvol10.3.pamphlet--->Asp24((coerce ($ FEXPR))): (coerce ($ FEXPR)) has varying indentation
```

typos 40241: AXSERV

```
>compiling AXSERV.spad to AXSERV.nrllib
```

```
--->bookvol10.4.pamphlet--->AxiomServer((axServer ((Void) (Integer) (Mapping (Void) (SEExpression))
```

```
--->bookvol10.4.pamphlet--->AxiomServer((multiServ ((Void) (SEExpression)))): Not documented!!!!
```

```
--->bookvol10.4.pamphlet--->AxiomServer((getDatabase ((String) (String) (String)))): Not document
```

typos 40224: FFSQFR

```
>compiling FFSQFR.spad to FFSQFR.nrllib
```

```
--->bookvol10.4.pamphlet--->FiniteFieldSquareFreeDecomposition((Musser ((Factored PolK) PolK))):
```

```
--->bookvol10.4.pamphlet--->FiniteFieldSquareFreeDecomposition((Yun ((Factored PolK) PolK))): Not
```

typos 40212a: GOSPER

```
>compiling GOSPER.spad to GOSPER.nrllib
```

```
--->bookvol10.4.pamphlet--->GosperSummationMethod((coerce (% P))): Not documented!!!!
```

```
--->bookvol10.4.pamphlet--->GosperSummationMethod((numer (P %))): Not documented!!!!
```

```
--->bookvol10.4.pamphlet--->GosperSummationMethod((denom (P %))): Not documented!!!!
```

typos 40205: INTERGB**typos 40202: LISYSER****typos 40193: PARAMP**

typos 40192: PFORP

```
>compiling PFORP.spad to PFORP.nrlib

-->bookvol10.4.pamphlet-->PackageForPoly((mapExponents (PolyRing (Mapping E E) PolyRing))): Not documented
-->bookvol10.4.pamphlet-->PackageForPoly((degree (NNI PolyRing (Integer)))): Not documented!!!!
-->bookvol10.4.pamphlet-->PackageForPoly((univariate ((SparseUnivariatePolynomial R) PolyRing))): Not documented
-->bookvol10.4.pamphlet-->PackageForPoly((totalDegree (NNI PolyRing))): Not documented!!!!
```

```

--->bookvol10.4.pamphlet-->PackageForPoly((subs1stVar (PolyRing PolyRing PolyRing))): Not docume
--->bookvol10.4.pamphlet-->PackageForPoly((subs2ndVar (PolyRing PolyRing PolyRing))): Not docume
--->bookvol10.4.pamphlet-->PackageForPoly((subsInVar (PolyRing PolyRing PolyRing (Integer)))): N

--->bookvol10.4.pamphlet-->PackageForPoly((listVariable ((List PolyRing)))): Not documented!!!!
--->bookvol10.4.pamphlet-->PackageForPoly((monomials ((List PolyRing) PolyRing))): Not documente

```

typos 40189: POLYCATQ

```

>compiling POLYCATQ.spad to POLYCATQ.nrllib

--->bookvol10.4.pamphlet-->PolynomialCategoryQuotientFunctions((coerce (% P))): Not documented!!!
--->bookvol10.4.pamphlet-->PolynomialCategoryQuotientFunctions((numer (P %))): Not documented!!!
--->bookvol10.4.pamphlet-->PolynomialCategoryQuotientFunctions((denom (P %))): Not documented!!!

```

typos 40188: POLYLIFT

```

>compiling POLYLIFT.spad to POLYLIFT.nrllib

--->bookvol10.4.pamphlet-->PolynomialCategoryLifting((+ (% % %))): Not documented!!!!
--->bookvol10.4.pamphlet-->PolynomialCategoryLifting((* (% % %))): Not documented!!!!
--->bookvol10.4.pamphlet-->PolynomialCategoryLifting((* (% % (NonNegativeInteger)))): Not docum

```

typos 40184: RINTERP

```

>compiling RINTERP.spad to RINTERP.nrllib

--->bookvol10.4.pamphlet-->RationalInterpolation((interpolate ((Fraction (Polynomial F)) (List F

```

typos 40180: SOLVEFOR

```

>compiling SOLVEFOR.spad to SOLVEFOR.nrllib

--->bookvol10.4.pamphlet-->PolynomialSolveByFormulas((* (% % (Fraction (Integer))))): Not docum

```

typos 40166: FACTRN

```
>compiling FACTRN.spad to FACTRN.nrlib
```

```
--->bookvol10.4.pamphlet-->FactorisationOverPseudoAlgebraicClosureOfRationalNumber((factor ((Factored UP) U)
--->bookvol10.4.pamphlet-->FactorisationOverPseudoAlgebraicClosureOfRationalNumber((factorSqFree ((Factored
```

typos 40165: FFFACTSE

```
>compiling FFFACTSE.spad to FFFACTSE.nrlib
```

```
--->bookvol10.4.pamphlet-->FiniteFieldFactorizationWithSizeParseBySideEffect((factorSquareFree ((List PolK)
--->bookvol10.4.pamphlet-->FiniteFieldFactorizationWithSizeParseBySideEffect((factorCantorZassenhaus ((List
--->bookvol10.4.pamphlet-->FiniteFieldFactorizationWithSizeParseBySideEffect((factor ((Factored PolK) PolK)
--->bookvol10.4.pamphlet-->FiniteFieldFactorizationWithSizeParseBySideEffect((factorUsingYun ((Factored PolK)
--->bookvol10.4.pamphlet-->FiniteFieldFactorizationWithSizeParseBySideEffect((factorUsingMusser ((Factored PolK)
--->bookvol10.4.pamphlet-->FiniteFieldFactorizationWithSizeParseBySideEffect((irreducible? ((Boolean) PolK))
```

typos 40163: INTFRSP

```
>compiling INTFRSP.spad to INTFRSP.nrlib
```

```
--->bookvol10.4.pamphlet-->InterpolateFormsPackage((basisOfInterpolateForms ((List (Vector K)) DIVISOR (List
--->bookvol10.4.pamphlet-->InterpolateFormsPackage((basisOfInterpolateFormsForFact ((List (Vector K)) DIVISOR
--->bookvol10.4.pamphlet-->InterpolateFormsPackage((interpolateFormsForFact ((List PolyRing) DIVISOR (List PolyRing))
```

typos 40161: NPOLYGON

```
>compiling NPOLYGON.spad to NPOLYGON.nrlib
```

```
--->bookvol10.4.pamphlet-->NewtonPolygon((newtonPolygon ((List (List PolyRing)) PolyRing (Integer) (Integer)
--->bookvol10.4.pamphlet-->NewtonPolygon((multiplicity ((NonNegativeInteger) (List (List PolyRing))))): Not
--->bookvol10.4.pamphlet-->NewtonPolygon((negAndPosEdge ((List (List PolyRing)) PolyRing (List (List PolyRing)
--->bookvol10.4.pamphlet-->NewtonPolygon((slope (recSlope PolyRing PolyRing))): Not documented!!!!
--->bookvol10.4.pamphlet-->NewtonPolygon((slope (recSlope (List PolyRing))))): Not documented!!!!
```

typos 40158: AFALGGRO

```
>compiling AFALGGRO.spad to AFALGGRO.nrllib
```

```
--->bookvol10.4.pamphlet-->AffineAlgebraicSetComputeWithGroebnerBasis((affineAlgSet ((Union (List
--->bookvol10.4.pamphlet-->AffineAlgebraicSetComputeWithGroebnerBasis((affineSingularPoints ((Un
```

typos 40157: AFALGRES

```
>compiling AFALGRES.spad to AFALGRES.nrllib
```

```
--->bookvol10.4.pamphlet-->AffineAlgebraicSetComputeWithResultant((affineSingularPoints ((Union
--->bookvol10.4.pamphlet-->AffineAlgebraicSetComputeWithResultant((affineSingularPoints ((Union
--->bookvol10.4.pamphlet-->AffineAlgebraicSetComputeWithResultant((affineAlgSetLocal ((Union (Li
--->bookvol10.4.pamphlet-->AffineAlgebraicSetComputeWithResultant((affineAlgSet ((Union (List Pr
--->bookvol10.4.pamphlet-->AffineAlgebraicSetComputeWithResultant((polyRing2UPUP (UPUP PolyRing)
--->bookvol10.4.pamphlet-->AffineAlgebraicSetComputeWithResultant((allPairsAmong ((List (List UP
--->bookvol10.4.pamphlet-->AffineAlgebraicSetComputeWithResultant((affineRationalPoints ((Union
```

typos 40134b: BlowUpPackage

```
--->bookvol10.4.pamphlet-->BlowUpPackage((newtonPolySlope ((List (List NNI)) BUpRing))) : Not do
--->bookvol10.4.pamphlet-->BlowUpPackage((polyRingToBUpRing (BUpRing PolyRing BLMET))) : Not do
--->bookvol10.4.pamphlet-->BlowUpPackage((biringToPolyRing (PolyRing BUpRing BLMET))) : Not docu
```

typos 40126b: Float

```
--->bookvol10.3.pamphlet-->Float(constructor): Not documented!!!!
--->bookvol10.3.pamphlet-->Float(): Missing Description
```

typos 40120a: INFCLCT

```
>compiling INFCLCT.spad to INFCLCT.nrllib
```

```
--->bookvol10.2.pamphlet-->InfinitelyClosePointCategory((create (% ProjPt PolyRing))) : Not docume
--->bookvol10.2.pamphlet-->InfinitelyClosePointCategory((setpoint! (ProjPt % ProjPt))) : Not docum
```



```

--->bookvol10.2.pamphlet-->InfinitlyClosePointCategory((setcurve! (BUpRing % BUpRing))): Not documented!!!
--->bookvol10.2.pamphlet-->InfinitlyClosePointCategory((setlocalPoint! (AFP % AFP))): Not documented!!!!
--->bookvol10.2.pamphlet-->InfinitlyClosePointCategory((setsubmult! ((NonNegativeInteger) % (NonNegativeInteger))): Not documented!!!!
--->bookvol10.2.pamphlet-->InfinitlyClosePointCategory((setmult! ((NonNegativeInteger) % (NonNegativeInteger))): Not documented!!!!
--->bookvol10.2.pamphlet-->InfinitlyClosePointCategory((setchart! (BLMET % BLMET))): Not documented!!!!
--->bookvol10.2.pamphlet-->InfinitlyClosePointCategory((setexcpDiv! (DIVISOR % DIVISOR))): Not documented!!!!
--->bookvol10.2.pamphlet-->InfinitlyClosePointCategory((setlocalParam! ((List PCS) % (List PCS))): Not documented!!!!
--->bookvol10.2.pamphlet-->InfinitlyClosePointCategory((setsymbName! ((Symbol) % (Symbol))): Not documented!!!!
--->bookvol10.2.pamphlet-->InfinitlyClosePointCategory((subMultV ((NonNegativeInteger) %))): Not documented!!!!
--->bookvol10.2.pamphlet-->InfinitlyClosePointCategory((localParamV ((List PCS) %))): Not documented!!!!
--->bookvol10.2.pamphlet-->InfinitlyClosePointCategory((symbNameV ((Symbol) %))): Not documented!!!!

--->bookvol10.2.pamphlet-->InfinitlyClosePointCategory((actualExtensionV (K %))): Not documented!!!!

```

typos 40115: NAGD02

```
>compiling NAGD02.spad to NAGD02.nrllib
```

```

--->bookvol10.4.pamphlet-->NagOrdinaryDifferentialEquationsPackage((d02kef ((Result) (Matrix (DoubleFloat)) (DoubleFloat))
"d02kef(xpoint,m,k,tol,maxfun,match,elam,delam, \\\indented{7}{hmax,maxit,ifail,coeffn,bdyval})} finds a spec

--->bookvol10.4.pamphlet-->NagOrdinaryDifferentialEquationsPackage((d02kef ((Result) (Matrix (DoubleFloat)) (DoubleFloat))
"d02kef(xpoint,m,k,tol,maxfun,match,elam,delam, \\\indented{7}{hmax,maxit,ifail,coeffn,bdyval,monit,report})}

--->bookvol10.4.pamphlet-->NagOrdinaryDifferentialEquationsPackage((d02raf ((Result) (Integer) (Integer) (Integer) (Integer))
"d02raf(n,mnp,numbeg,nummix,tol,init,iy,ijac,lwork, \\\indented{7}{liwork,np,x,y,deleps,ifail,fcn,g})} solves

```

typos 40114: NAGD03

```
>compiling NAGD03.spad to NAGD03.nrllib
```

```

--->bookvol10.4.pamphlet-->NagPartialDifferentialEquationsPackage((d03faf ((Result) (DoubleFloat) (DoubleFloat) (DoubleFloat) (DoubleFloat))
"d03faf(xs,xf,l,lbdend,bdxs,bdxf,ys,yf,m,mbdend,bdys,bdyf,zs, \\\indented{7}{zf,n,nbdend,bdys,bdzf,lambdas,ldf})}

```

typos 40112: NAGE04

```
>compiling NAGE04.spad to NAGE04.nrllib
```

```

--->bookvol10.4.pamphlet-->NagOptimisationPackage((e04mbf ((Result) (Integer) (Integer) (Integer) (Integer) (Integer) (Integer))
"e04mbf(itmax,msglvl,n,nclin,nctotl,nrowa,a,bl,bu, \\\indented{7}{cvec,linobj,liwork,lwork,x,ifail})} is an e

```

```

--->bookvol10.4.pamphlet-->NagOptimisationPackage((e04naf ((Result) (Integer) (Integer) (Integer)
"e04naf(itmax,msglvl,n,nclin,nctotl,nrowa,nrowh,ncolh,bigbnd,a,bl, bu,cvec,featol,hess,cold,lpp,

--->bookvol10.4.pamphlet-->NagOptimisationPackage((e04ucf ((Result) (Integer) (Integer) (Integer)
"e04ucf(n,nclin,ncnln,nrowa,nrowj,nrowr,a,bl,bu,liwork,lwork,sta, \\indented{7}{cra,der,fea,fun,

```

typos 70111: NSDPS

```
>compiling NSDPS.spad to NSDPS.nrlib
```

```

--->bookvol10.3.pamphlet-->NeitherSparseOrDensePowerSeries((findTerm (TERM % (Integer)))): Not d
-----constructor-----

```

typos 40107: PACOFF

```
>compiling PACOFF.spad to PACOFF.nrlib
```

```

--->bookvol10.3.pamphlet-->PseudoAlgebraicClosureOfFiniteField((fullOutput ((OutputForm) %))): N

```

typos 40106: PACRAT

```
>compiling PACRAT.spad to PACRAT.nrlib
```

```

--->bookvol10.3.pamphlet-->PseudoAlgebraicClosureOfRationalNumber((fullOutput ((OutputForm) %)))
--->bookvol10.3.pamphlet-->PseudoAlgebraicClosureOfRationalNumber((newElement (% (SUP %) (SUP %)

```

typos 40096: SAOS

```
>compiling SAOS.spad to SAOS.nrlib
```

```

--->bookvol10.3.pamphlet-->SingletonAsOrderedSet((create (%))): Not documented!!!!
--->bookvol10.3.pamphlet-->SingletonAsOrderedSet((convert ((Symbol) %))): Not documented!!!!

```

typos 40082a: DTP

```
>compiling DTP.spad to DTP.nrlib
```

```
--->bookvol10.4.pamphlet-->DesingTreePackage((blowUp ((List InfClsPoint) InfClsPoint))): Not documented!!!!
```

```
--->bookvol10.4.pamphlet-->DesingTreePackage((blowUpWithExcpDiv ((Void) DesTree))): Not documented!!!!
```

```
--->bookvol10.4.pamphlet-->DesingTreePackage((inBetweenExcpDiv (DIVISOR DesTree))): Not documented!!!!
```

nonextend 40081: D01TRNS

```
>compiling D01TRNS.spad to D01TRNS.nrlib
```

```
--->bookvol10.3.pamphlet-->d01TransformFunctionType(): Missing Description
```

typos 40078: EXPRODE

```
>compiling EXPRODE.spad to EXPRODE.nrlib
```

```
--->bookvol10.4.pamphlet-->ExpressionSpaceODESolver((seriesSolve ((Any) (List F) (List OP) EQ (List EQ)))):
"seriesSolve([eq1,...,eqn], [y1,...,yn], \spad{x} = \spad{a,[y1] a = b1,..., \spad{yn} a = bn}) is equiv
```

typos 40069: MYUP

```
>compiling MYUP.spad to MYUP.nrlib
```

```
-----fme cg (% % (NonNegativeInteger) R %))-----
```

```
--->bookvol10.3.pamphlet-->MyUnivariatePolynomial((coerce (% R))): Not documented!!!!
```

```
--->bookvol10.3.pamphlet-->MyUnivariatePolynomial((coerce (% (Polynomial R))): Not documented!!!!
```

typos 40068: MYEXPR

```
>compiling MYEXPR.spad to MYEXPR.nrlib
```

```
--->bookvol10.3.pamphlet-->MyExpression((* (% % %))): Not documented!!!!
```

typos 40067: MYEXPR**typos 40066: NAGF02**

typos 40065: NAGF04

```
>compiling NAGF04.spad to NAGF04.nrllib
```

```
--->bookvol10.4.pamphlet-->NagLinearEquationSolvingPackage((f04maf ((Result) (Integer) (Integer) (Matrix (D
"f04maf(n,nz,avals,licn,irn,lirn,icn,wkeep,ikeep, \\indented{7}{inform,b,acc,noits,ifail)} \\spad{e} a spar
----- (f04mbf ((Result) (Integer) (Matrix (DoubleFloat)) (Boolean) (DoubleFloat) (Integer) (Integer) (Int

--->bookvol10.4.pamphlet-->NagLinearEquationSolvingPackage((f04qaf ((Result) (Integer) (Integer) (DoubleFloa
"f04qaf(m,n,damp,atol,btol,conlim,itnlim,msglvl, \\indented{7}{lrwork,liwork,b,ifail,aproduct}) solves sparse
```

typos 40064: OMEXPR

```
>compiling OMEXPR.spad to OMEXPR.nrllib
```

```
--->bookvol10.4.pamphlet-->ExpressionToOpenMath((OMwrite ((String) (Expression R)))): Not documented!!!!
--->bookvol10.4.pamphlet-->ExpressionToOpenMath((OMwrite ((String) (Expression R) (Boolean)))): Not document
--->bookvol10.4.pamphlet-->ExpressionToOpenMath((OMwrite ((Void) (OpenMathDevice) (Expression R)))): Not do
--->bookvol10.4.pamphlet-->ExpressionToOpenMath((OMwrite ((Void) (OpenMathDevice) (Expression R) (Boolean)))):
```

typos 40058: TRMANIP

```
>compiling TRMANIP.spad to TRMANIP.nrllib
```

```
--->bookvol10.4.pamphlet-->TranscendentalManipulations((expand (F F))): \item appears outside a \begin-\end
--->bookvol10.4.pamphlet-->TranscendentalManipulations((expand (F F))): \item appears outside a \begin-\end
--->bookvol10.4.pamphlet-->TranscendentalManipulations((expand (F F))): \item appears outside a \begin-\end
"\spad{expand(f)} performs the following expansions on f:\begin{items} \item 1. logs of products are exp

--->bookvol10.4.pamphlet-->TranscendentalManipulations((simplify (F F))): \item appears outside a \begin-\end
--->bookvol10.4.pamphlet-->TranscendentalManipulations((simplify (F F))): \item appears outside a \begin-\end
--->bookvol10.4.pamphlet-->TranscendentalManipulations((simplify (F F))): \item appears outside a \begin-\end
--->bookvol10.4.pamphlet-->TranscendentalManipulations((simplify (F F))): \item appears outside a \begin-\end
"\spad{simplify(f)} performs the following simplifications on f:\begin{items} \item 1. rewrites trigs and
```

typos 40057: UPXSSING

```
>compiling UPXSSING.spad to UPXSSING.nrllib
```

```
-----non extending category-----
.. UnivariatePuisseuxSeriesWithExponentialSingularity(#1,#2,#3,#4) of cat
(|Join| (|FiniteAbelianMonoidRing| (|UnivariatePuisseuxSeries| |#2| |#3| |#4|) (|ExponentialOfUnivariatePuis
(IF (|has| (|UnivariatePuisseuxSeries| |#2| |#3| |#4|) (|IntegralDomain|)) (IF (|has| (|ExponentialOfUnivari
```

typos 40056: UTSSOL

```
>compiling UTSSOL.spad to UTSSOL.nrllib
```

```
--->bookvol10.4.pamphlet-->TaylorSolve((seriesSolve (UTSF (Mapping UTSSUPF UTSSUPF) (List F)))):
```

typos 40050: EXPRSOL

```
>compiling EXPRSOL.spad to EXPRSOL.nrllib
```

```
--->bookvol10.4.pamphlet-->ExpressionSolve((seriesSolve (UTSF F OP SY (List F)))): Not documented
```

```
--->bookvol10.4.pamphlet-->ExpressionSolve((replaceDiffs (F F OP (Symbol)))): Not documented!!!!
```

typos 40049: FACTEXT

```
>compiling FACTEXT.spad to FACTEXT.nrllib
```

```
--->bookvol10.4.pamphlet-->FactorisationOverPseudoAlgebraicClosureOfAlgExtOfRationalNumber((fact
```

```
--->bookvol10.4.pamphlet-->FactorisationOverPseudoAlgebraicClosureOfAlgExtOfRationalNumber((fact
```

typos 40047: GSERIES

```
>compiling GSERIES.spad to GSERIES.nrllib
```

```
-----non extending category-----
```

```
.. GeneralUnivariatePowerSeries(#1,#2,#3) of cat
```

```
(|Join| (|UnivariatePuisseuxSeriesCategory| |#1|) (CATEGORY |domain| (SIGNATURE |coerce| ($ (|Var
```

```
(|UnivariatePuisseuxSeriesConstructorCategory| |#1| (|UnivariateLaurentSeries| |#1| |#2| |#3|))
```

typos 40046: REGSET

```
>compiling REGSET.spad to REGSET.nrllib
```

Warning: REGSET;decompose has a duplicate definition in this file

typos 40044: RSDCMPK

>compiling RSDCMPK.spad to RSDCMPK.nrlib

```

--->bookvol10.4.pamphlet-->RegularSetDecompositionPackage((KrullNumber (N LP Split))): Not documented!!!!
--->bookvol10.4.pamphlet-->RegularSetDecompositionPackage((numberOfVariables (N LP Split))): Not documented
--->bookvol10.4.pamphlet-->RegularSetDecompositionPackage((algebraicDecompose ((Record (: done Split) (: tod
--->bookvol10.4.pamphlet-->RegularSetDecompositionPackage((transcendentalDecompose ((Record (: done Split)
--->bookvol10.4.pamphlet-->RegularSetDecompositionPackage((transcendentalDecompose ((Record (: done Split)
--->bookvol10.4.pamphlet-->RegularSetDecompositionPackage((internalDecompose ((Record (: done Split) (: tod
--->bookvol10.4.pamphlet-->RegularSetDecompositionPackage((internalDecompose ((Record (: done Split) (: tod
--->bookvol10.4.pamphlet-->RegularSetDecompositionPackage((internalDecompose ((Record (: done Split) (: tod
--->bookvol10.4.pamphlet-->RegularSetDecompositionPackage((decompose (Split LP Split B B))): Not documented
--->bookvol10.4.pamphlet-->RegularSetDecompositionPackage((decompose (Split LP Split B B B B B))): Not docu
--->bookvol10.4.pamphlet-->RegularSetDecompositionPackage((updateBranches ((List LpWT) LP Split (List LpWT)
--->bookvol10.4.pamphlet-->RegularSetDecompositionPackage((convert ((String) (Record (: val (List P)) (: to
--->bookvol10.4.pamphlet-->RegularSetDecompositionPackage((printInfo ((Void) (List (Record (: val (List P))

```

typos 40040: UFPS1

>compiling UFPS1.spad to UFPS1.nrlib

```

--->bookvol10.4.pamphlet-->UnivariateFormalPowerSeriesFunctions((hadamard (UFPS UFPS UFPS))): Not documented

```

typos 40037a: GPAFF

>compiling GPAFF.spad to GPAFF.nrlib

```

--->bookvol10.4.pamphlet-->GeneralPackageForAlgebraicFunctionField((reset ((Void)))): Not documented!!!!
--->bookvol10.4.pamphlet-->GeneralPackageForAlgebraicFunctionField((setCurve (PolyRing PolyRing))): Not doc
--->bookvol10.4.pamphlet-->GeneralPackageForAlgebraicFunctionField((homogenize (PolyRing PolyRing (Integer))
--->bookvol10.4.pamphlet-->GeneralPackageForAlgebraicFunctionField((genusNeg (INT))): Not documented!!!!
--->bookvol10.4.pamphlet-->GeneralPackageForAlgebraicFunctionField((findOrderOfDivisor ((Record (: ord (Int
--->bookvol10.4.pamphlet-->GeneralPackageForAlgebraicFunctionField((interpolateFormsForFact ((List PolyRing)

```

```

--->bookvol10.4.pamphlet--->GeneralPackageForAlgebraicFunctionField((placesAbove ((List Plc) ProjPt
--->bookvol10.4.pamphlet--->GeneralPackageForAlgebraicFunctionField((rationalPoints ((List ProjPt

```

typos 40036: PACEXT

```
>compiling PACEXT.spad to PACEXT.nrllib
```

```

--->bookvol10.3.pamphlet--->PseudoAlgebraicClosureOfAlgExtOfRationalNumber((fullOutput ((OutputFo
--->bookvol10.3.pamphlet--->PseudoAlgebraicClosureOfAlgExtOfRationalNumber((retractToGrn (K %)))

```

typos 40035: RECOP

```
>compiling RECOP.spad to RECOP.nrllib
```

```

--->bookvol10.4.pamphlet--->RecurrenceOperator((numberOfValuesNeeded ((Integer) (Integer) (BasicO
--->bookvol10.4.pamphlet--->RecurrenceOperator((getShiftRec ((Union (Integer) failed) (BasicOpera
--->bookvol10.4.pamphlet--->RecurrenceOperator((shiftInfoRec ((Record (: max (Union (Integer) fai

```

typos 40034: SFRGCD

```
>compiling SFRGCD.spad to SFRGCD.nrllib
```

```

--->bookvol10.4.pamphlet--->SquareFreeRegularTriangularSetGcdPackage((startTableGcd! ((Void) S S
--->bookvol10.4.pamphlet--->SquareFreeRegularTriangularSetGcdPackage((stopTableGcd! ((Void))) : M
--->bookvol10.4.pamphlet--->SquareFreeRegularTriangularSetGcdPackage((startTableInvSet! ((Void) S
--->bookvol10.4.pamphlet--->SquareFreeRegularTriangularSetGcdPackage((stopTableInvSet! ((Void)))
--->bookvol10.4.pamphlet--->SquareFreeRegularTriangularSetGcdPackage((stosePrepareSubResAlgo ((Li
--->bookvol10.4.pamphlet--->SquareFreeRegularTriangularSetGcdPackage((stoseInternalLastSubResulta
--->bookvol10.4.pamphlet--->SquareFreeRegularTriangularSetGcdPackage((stoseInternalLastSubResulta
--->bookvol10.4.pamphlet--->SquareFreeRegularTriangularSetGcdPackage((stoseIntegralLastSubResulta
--->bookvol10.4.pamphlet--->SquareFreeRegularTriangularSetGcdPackage((stoseLastSubResultant ((Li
--->bookvol10.4.pamphlet--->SquareFreeRegularTriangularSetGcdPackage((stoseInvertible? (B P TS)))
--->bookvol10.4.pamphlet--->SquareFreeRegularTriangularSetGcdPackage((stoseInvertible?sqfreg ((Li
--->bookvol10.4.pamphlet--->SquareFreeRegularTriangularSetGcdPackage((stoseInvertibleSetsqfreg (S
--->bookvol10.4.pamphlet--->SquareFreeRegularTriangularSetGcdPackage((stoseInvertible?reg ((List
--->bookvol10.4.pamphlet--->SquareFreeRegularTriangularSetGcdPackage((stoseInvertibleSetreg (Spli
--->bookvol10.4.pamphlet--->SquareFreeRegularTriangularSetGcdPackage((stoseInvertible? ((List BWT
--->bookvol10.4.pamphlet--->SquareFreeRegularTriangularSetGcdPackage((stoseInvertibleSet (Split P
--->bookvol10.4.pamphlet--->SquareFreeRegularTriangularSetGcdPackage((stoseSquareFreePart ((List

```


typos 40033: SRDCMPK

>compiling SRDCMPK.spad to SRDCMPK.nrlib

```

--->bookvol10.4.pamphlet-->SquareFreeRegularSetDecompositionPackage((KrullNumber (N LP Split))): Not documented
--->bookvol10.4.pamphlet-->SquareFreeRegularSetDecompositionPackage((numberOfVariables (N LP Split))): Not documented
--->bookvol10.4.pamphlet-->SquareFreeRegularSetDecompositionPackage((algebraicDecompose ((Record (: done Split))))): Not documented
--->bookvol10.4.pamphlet-->SquareFreeRegularSetDecompositionPackage((transcendentalDecompose ((Record (: done Split))))): Not documented
--->bookvol10.4.pamphlet-->SquareFreeRegularSetDecompositionPackage((transcendentalDecompose ((Record (: done Split))))): Not documented
--->bookvol10.4.pamphlet-->SquareFreeRegularSetDecompositionPackage((internalDecompose ((Record (: done Split))))): Not documented
--->bookvol10.4.pamphlet-->SquareFreeRegularSetDecompositionPackage((internalDecompose ((Record (: done Split))))): Not documented
--->bookvol10.4.pamphlet-->SquareFreeRegularSetDecompositionPackage((internalDecompose ((Record (: done Split))))): Not documented
--->bookvol10.4.pamphlet-->SquareFreeRegularSetDecompositionPackage((decompose (Split LP Split B B))): Not documented
--->bookvol10.4.pamphlet-->SquareFreeRegularSetDecompositionPackage((decompose (Split LP Split B B B B))): Not documented
--->bookvol10.4.pamphlet-->SquareFreeRegularSetDecompositionPackage((updateBranches ((List LpWT) LP Split (List LpWT)))): Not documented
--->bookvol10.4.pamphlet-->SquareFreeRegularSetDecompositionPackage((convert ((String) (Record (: val (List LpWT))))): Not documented
--->bookvol10.4.pamphlet-->SquareFreeRegularSetDecompositionPackage((printInfo ((Void) (List (Record (: val (List LpWT))))): Not documented

```

typos 40031: GUESS

>compiling GUESS.spad to GUESS.nrlib

```

--->bookvol10.4.pamphlet-->Guess((* (% % %))): Not documented!!!!
--->bookvol10.4.pamphlet-->Guess(/ (% % %))): Not documented!!!!
--->bookvol10.4.pamphlet-->Guess(** (% % %))): Not documented!!!!
--->bookvol10.4.pamphlet-->Guess((numerator (% %))): Not documented!!!!
--->bookvol10.4.pamphlet-->Guess((denominator (% %))): Not documented!!!!
--->bookvol10.4.pamphlet-->Guess((ground? ((Boolean) %))): Not documented!!!!

```

typos 40029a: PAFF

>compiling PAFF.spad to PAFF.nrlib

```

--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionField((homogenize (PolyRing PolyRing (Integer)))): Not documented
--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionField((interpolateFormsForFact ((List PolyRing) DIVISOR DIVISOR))): Not documented
--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionField((fullDesTree ((Void)))): Not documented!!!!
--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionField((fullInfClsPt ((Void)))): Not documented!!!!
--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionField((setCurve (PolyRing PolyRing))): Not documented
--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionField((projectivePoint (ProjPt (List K)))): Not documented
----- (pointDominateBy (ProjPt Plc)) -----
--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionField((placesAbove ((List Plc) ProjPt))): Not documented
--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionField((setSingularPoints ((List ProjPt) (List ProjPt)))): Not documented
--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionField((goppaCode ((Matrix K) DIVISOR DIVISOR))): Not documented

```

```

--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionField((goppaCode ((Matrix K) DIVISOR (List
----- (rationalPlaces ((List Plc)))-----
----- (theCurve (PolyRing))-----
----- (genus (NNI))-----
--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionField((genusNeg ((Integer)))): Not documente

--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionField((rationalPoints ((List ProjPt)))): N

--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionField((lBasis ((List FRACPOLY) DIVISOR NNI

--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionField((findOrderOfDivisor ((Record (: ord

```

typos 40028a: PAFFF

>compiling PAFFFF.spad to PAFFFF.nrllib

```

--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionFieldOverFiniteField((homogenize (PolyRing
--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionFieldOverFiniteField((fullDesTree ((Void)))
--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionFieldOverFiniteField((fullInfClsPt ((Void)))
--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionFieldOverFiniteField((setCurve (PolyRing P
--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionFieldOverFiniteField((translateToOrigin (P
--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionFieldOverFiniteField((goppaCode ((Matrix K
--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionFieldOverFiniteField((goppaCode ((Matrix K

--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionFieldOverFiniteField((placesAbove ((List P
--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionFieldOverFiniteField((projectivePoint (Pro
--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionFieldOverFiniteField((setSingularPoints ((

--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionFieldOverFiniteField((genusNeg ((Integer)))

--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionFieldOverFiniteField((rationalPoints ((List

--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionFieldOverFiniteField((lBasis ((List FracPo
--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionField((homogenize (PolyRing PolyRing (Inte
--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionField((interpolateFormsForFact ((List Poly
--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionField((fullDesTree ((Void)))): Not documente
--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionField((fullInfClsPt ((Void)))): Not docume
--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionField((setCurve (PolyRing PolyRing)))): Not
--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionField((projectivePoint (ProjPt (List K))))

--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionField((placesAbove ((List Plc) ProjPt)))):
--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionField((setSingularPoints ((List ProjPt) (L
--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionField((goppaCode ((Matrix K) DIVISOR DIVIS
--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionField((goppaCode ((Matrix K) DIVISOR (List

--->bookvol10.4.pamphlet--->PackageForAlgebraicFunctionField((genusNeg ((Integer)))): Not documente

```

```

--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionField((rationalPoints ((List ProjPt)))): Not documented
--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionField((lBasis ((List FRACPOLY) DIVISOR NNI))): Not documented
--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionField((findOrderOfDivisor ((Record (: ord (Integer)) DIVISOR NNI))): Not documented
--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionFieldOverFiniteField((interpolateFormsForFact ((List LPolynomial) DIVISOR NNI))): Not documented
--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionFieldOverFiniteField((homogenize (PolyRing PolyRing (List LPolynomial) DIVISOR NNI))): Not documented
--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionFieldOverFiniteField((fullDesTree ((Void)))): Not documented
--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionFieldOverFiniteField((fullInfClsPt ((Void)))): Not documented
--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionFieldOverFiniteField((setCurve (PolyRing PolyRing (List LPolynomial) DIVISOR NNI))): Not documented
--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionFieldOverFiniteField((translateToOrigin (PolyRing2 PolyRing (List LPolynomial) DIVISOR NNI))): Not documented
--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionFieldOverFiniteField((goppaCode ((Matrix K) DIVISOR NNI))): Not documented
--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionFieldOverFiniteField((goppaCode ((Matrix K) DIVISOR NNI))): Not documented

--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionFieldOverFiniteField((placesAbove ((List Plc) ProjPt))): Not documented
--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionFieldOverFiniteField((projectivePoint (ProjPt (List Divisor))): Not documented
--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionFieldOverFiniteField((setSingularPoints ((List ProjPt))): Not documented

--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionFieldOverFiniteField((genusNeg ((Integer)))): Not documented

--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionFieldOverFiniteField((rationalPoints ((List ProjPt)))): Not documented

PAFFFF abbreviates package
PackageForAlgebraicFunctionFieldOverFiniteField

--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionFieldOverFiniteField((lBasis ((List FracPoly) DIVISOR NNI))): Not documented
--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionFieldOverFiniteField((findOrderOfDivisor ((Record (: ord (Integer)) DIVISOR NNI))): Not documented
--->bookvol10.4.pamphlet-->PackageForAlgebraicFunctionFieldOverFiniteField((interpolateFormsForFact ((List LPolynomial) DIVISOR NNI))): Not documented

```

warnings 20566: MAPPKG1

```
>compiling MAPPKG1.spad to MAPPKG1.nrllib
```

```
Warnings:
```

```

[1] Use: import (MappingPackageInternalHacks1 A)
[2] fixedPoint: :A -- should replace by pretend
[3] **: a1 has no value
[4] recur: n1 has no value
[5] recur: a2 has no value

```

warnings 20565: MAPPKG2

```
>compiling MAPPKG2.spad to MAPPKG2.nrllib
```

Warnings:

- [1] Use: import (MappingPackageInternalHacks2 A C)
- [2] const: a1 has no value
- [3] constant: a1 has no value
- [4] diag: a1 has no value

warnings 20564: MAPPKG3

>compiling MAPPKG3.spad to MAPPKG3.nrlib

Warnings:

- [1] Use: import (MappingPackageInternalHacks3 A B C)
- [2] curryRight: a has no value
- [3] curryLeft: b has no value
- [4] constantRight: a has no value
- [5] constantLeft: b has no value
- [6] twist: a has no value
- [7] twist: b has no value
- [8] *: a has no value

warnings 20563: MKBCFUNC

>compiling MKBCFUNC.spad to MKBCFUNC.nrlib

Warnings:

- [1] binaryFunction: d1 has no value
- [2] binaryFunction: d2 has no value

warnings 20562: MKUCFUNC

>compiling MKUCFUNC.spad to MKUCFUNC.nrlib

Warnings:

- [1] unaryFunction: d1 has no value

warnings 20561: NIPROB

>compiling NIPROB.spad to NIPROB.nrlib

```

Warnings:
[1] coerce:  nia has no value
[2] coerce:  mdnia has no value
[3] retract:  nia has no value
[4] retract:  mdnia has no value

```

warnings 20560: OPTPROB

```
>compiling OPTPROB.spad to OPTPROB.nrlib
```

```

Warnings:
[1] coerce:  noa has no value
[2] coerce:  lsa has no value
[3] retract:  noa has no value
[4] retract:  lsa has no value

```

warnings 20559: PATTERN1

```
>compiling PATTERN1.spad to PATTERN1.nrlib
```

```

Warnings:
[1] predicate:  d1 has no value
[2] satisfy?:  var has no value
[3] satisfy?:  pred has no value

```

warnings 20558: ASP29

```
>compiling ASP29.spad to ASP29.nrlib
```

```

Warnings:
[1] ISTATE has no value
[2] NEXTIT has no value
[3] NEVALS has no value
[4] NVECS has no value
[5] K has no value
[6] F has no value
[7] D has no value

```

warnings 20557: BEZIER

```
>compiling BEZIER.spad to BEZIER.nrllib
```

```
Warnings:
```

```
[1] linearBezier:  t has no value  
[2] quadraticBezier:  t has no value  
[3] cubicBezier:  t has no value
```

warnings 20556: FCOMP

```
>compiling FCOMP.spad to FCOMP.nrllib
```

```
Warnings:
```

```
[1] elt:  y has no value  
[2] eval:  y has no value
```

warnings 20555: IDPAG

```
>compiling IDPAG.spad to IDPAG.nrllib
```

```
Warnings:
```

```
[1] -:  res has no value  
[2] -:  endcell has no value
```

warnings 20554: IDPAM

```
>compiling IDPAM.spad to IDPAM.nrllib
```

```
Warnings:
```

```
[1] +:  res has no value  
[2] +:  endcell has no value
```

warnings 20553: INCRMAPS

```
>compiling INCRMAPS.spad to INCRMAPS.nrllib
```

```
Warnings:
[1] increment:  x has no value
[2] incrementBy:  x has no value
```

warnings 20552: MONADWU

```
>compiling MONADWU.spad to MONADWU.nrlib
```

```
Warnings:
[1] rightPower: signature of lhs not unique: SS(NonNegativeInteger) chosen
[2] leftPower: signature of lhs not unique: SS(NonNegativeInteger) chosen
```

warnings 20551: PATMATCH

```
>compiling PATMATCH.spad to PATMATCH.nrlib
```

```
Warnings:
[1] ist: not known that (SetCategory) is of mode (CATEGORY package (SIGNATURE is? ((Boolean) Subject I
```

warnings 20550: AUTOMOR

```
>compiling AUTOMOR.spad to AUTOMOR.nrlib
```

```
Warnings:
[1] inv:  r1 has no value
[2] inv:  i2 has no value
[3] **: signature of lhs not unique: $$ (Integer) chosen
[4] **:  r1 has no value
[5] **:  i2 has no value
[6] morphism:  i2 has no value
[7] morphism:  r1 has no value
[8] *:  u1 has no value
[9] *:  v1 has no value
[10] *:  i2 has no value
[11] *:  r1 has no value
```

warnings 20549: CHARPOL

```
>compiling CHARPOL.spad to CHARPOL.nrlib
```

```
Warnings:
```

```
[1] characteristicPolynomial: :(PositiveInteger) -- should replace by pretend
```

warnings 20548: ELAGG

```
>compiling ELAGG.spad to ELAGG.nrlib
```

```
Warnings:
```

```
[1] remove!: y has no value
```

warnings 20547: ES2

```
>compiling ES2.spad to ES2.nrlib
```

```
Warnings:
```

```
[1] map: IN has no value
```

```
[2] map: x has no value
```

warnings 20546: MKCHSET

```
>compiling MKCHSET.spad to MKCHSET.nrlib
```

```
Warnings:
```

```
[1] coerce: x has no value
```

```
[2] <: x1 has no value
```

```
[3] =: x1 has no value
```

warnings 20545: IFAMON

```
>compiling IFAMON.spad to IFAMON.nrlib
```

```
Warnings:
```

```
[1] *: x has no value
```

```
[2] coerce: y has no value
```

```
[3] coerce: x has no value
```


warnings 20544: CARD

```
>compiling CARD.spad to CARD.nrlib
```

```
Warnings:
```

- ```
[1] **: signature of lhs not unique: $$$ chosen
[2] **: :(NonNegativeInteger) -- should replace by pretend
```

**warnings 20543: FACTFUNC**

```
>compiling FACTFUNC.spad to FACTFUNC.nrlib
```

```
Warnings:
```

- ```
[1] nthRoot: radi has no value
```

warnings 20541: AMR

```
>compiling AMR.spad to AMR.nrlib
```

```
Warnings:
```

- ```
[1] *: x1 has no value
```

**warnings 20540: DEGRED**

```
>compiling DEGRED.spad to DEGRED.nrlib
```

```
Warnings:
```

- ```
[1] reduce: :(PositiveInteger) -- should replace by pretend
```

warnings 20539: DLP

```
>compiling DLP.spad to DLP.nrlib
```

```
Warnings:
```

- ```
[1] shanksDiscLogAlgorithm: found has no value
[2] shanksDiscLogAlgorithm: disclog has no value
[3] shanksDiscLogAlgorithm: pretend(Integer) -- should replace by @
```

**warnings 20538: FAGROUP**

```
>compiling FAGROUP.spad to FAGROUP.nrlib
```

```
Warnings:
 [1] <: exp has no value
 [2] <: gen has no value
```

**warnings 20537: FIELD**

```
>compiling FIELD.spad to FIELD.nrlib
```

```
Warnings:
 [1] squareFree: not known that (IntegralDomain) is of mode (CATEGORY domain (SIGNATURE / (
```

**warnings 20536: FLAGG**

```
>compiling FLAGG.spad to FLAGG.nrlib
```

```
Warnings:
 [1] sorted?: x has no value
 [2] sorted?: y has no value
```

**warnings 20535: FLINEXP**

```
>compiling FLINEXP.spad to FLINEXP.nrlib
```

```
Warnings:
 [1] reducedSystem: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE reducedSys
```

**warnings 20534: MAPPKG4**

```
>compiling MAPPKG4.spad to MAPPKG4.nrlib
```

```
Warnings:
[1] +: c has no value
[2] -: c has no value
[3] *: c has no value
[4] /: c has no value
```

## warnings 20533: OMLO

```
>compiling OMLO.spad to OMLO.nrlib
```

```
Warnings:
[1] op: :$ -- should replace by @
[2] po: :P -- should replace by pretend
[3] *: :P -- should replace by pretend
[4] coerce: :P -- should replace by pretend
```

## warnings 20532: PRODUCT

```
>compiling PRODUCT.spad to PRODUCT.nrlib
```

```
Warnings:
[1] count: n has no value
```

## warnings 20531: ARR2CAT

```
>compiling ARR2CAT.spad to ARR2CAT.nrlib
```

```
Warnings:
[1] count: num has no value
[2] count: x has no value
```

## warnings 20530: ASP34

```
>compiling ASP34.spad to ASP34.nrlib
```

```
Warnings:
[1] IFLAG has no value
```

```

[2] N has no value
[3] X has no value
[4] Y has no value
[5] LRWORK has no value
[6] LIWORK has no value
[7] RWORK has no value
[8] IWORK has no value
[9] coerce: I has no value
[10] coerce: J has no value
[11] coerce: W1 has no value
[12] coerce: W2 has no value
[13] coerce: MS has no value

```

### warnings 20529: BFUNCT

```
>compiling BFUNCT.spad to BFUNCT.nrlib
```

```

Warnings:
[1] bf: x1 has no value

```

### warnings 20528: BTREE

```
>compiling BTREE.spad to BTREE.nrlib
```

```

Warnings:
[1] empty: pretend$ -- should replace by @
[2] node: :Rep -- should replace by pretend
[3] setvalue!: :Rep -- should replace by pretend
[4] setleft!: :Rep -- should replace by pretend
[5] setright!: :(List (Tree S)) -- should replace by pretend

```

### warnings 20527: CRAPACK

```
>compiling CRAPACK.spad to CRAPACK.nrlib
```

```

Warnings:
[1] chineseRemainder: a has no value
[2] chineseRemainder: d has no value
[3] chineseRemainder: b has no value
[4] chineseRemainder: c has no value

```

## warnings 20526: DIRRING

```
>compiling DIRRING.spad to DIRRING.nrllib
```

Warnings:

```
[1] per: pretend$ -- should replace by @
[2] rep: pretendRep -- should replace by @
[3] coerce: n has no value
[4] One: n has no value
[5] +: n has no value
[6] -: n has no value
[7] *: n has no value
[8] *: a has no value
[9] *: b has no value
[10] recip: n has no value
[11] multiplicative?: not known that (IntegralDomain) is of mode (CATEGORY domain (IF (has Coef (Comm
[12] multiplicative?: r has no value
[13] multiplicative?: s has no value
[14] additive?: r has no value
[15] additive?: s has no value
```

## warnings 20525: DIV

```
>compiling DIV.spad to DIV.nrllib
```

Warnings:

```
[1] reductum: pretendRep -- should replace by @
[2] head: pretendRep -- should replace by @
[3] coefficient: r has no value
[4] collect: inList? has no value
[5] collect: t has no value
[6] +: pretendRep -- should replace by @
[7] *: pretendRep -- should replace by @
```

## warnings 20524: D01GBFA

```
>compiling D01GBFA.spad to D01GBFA.nrllib
```

Warnings:

```
[1] numericalIntegration: x has no value
```

**warnings 20523: FAMR**

```
>compiling FAMR.spad to FAMR.nrllib
```

```
Warnings:
```

- [1] pomopo!: x1 has no value
- [2] /: x1 has no value
- [3] exquo: signature of lhs not unique: (Union S failed)SR chosen

**warnings 20522: FM1**

```
>compiling FM1.spad to FM1.nrllib
```

```
Warnings:
```

- [1] listOfTerms: :(List (Record (: k S) (: c R))) -- should replace by pretend

**warnings 20521: FMONOID**

```
>compiling FMONOID.spad to FMONOID.nrllib
```

```
Warnings:
```

- [1] overlap: l has no value
- [2] divide: l has no value
- [3] hcllf: h has no value
- [4] lquo: gen has no value
- [5] lquo: exp has no value

**warnings 20520: IPADIC**

```
>compiling IPADIC.spad to IPADIC.nrllib
```

```
Warnings:
```

- [1] padic: pretend\$ -- should replace by @
- [2] =: st has no value
- [3] intToPAAdic: digit has no value
- [4] intToPAAdic: carry has no value
- [5] intPlusPAAdic: digit has no value
- [6] intPlusPAAdic: carry has no value
- [7] intMinusPAAdic: digit has no value
- [8] intMinusPAAdic: carry has no value

```

[9] plusAux: digit has no value
[10] plusAux: carry has no value
[11] minusAux: digit has no value
[12] minusAux: carry has no value
[13] intMult: digit has no value
[14] intMult: carry has no value
[15] timesAux: digit has no value
[16] timesAux: carry has no value
[17] coerce: pretend(Integer) -- should replace by @
[18] coerce: 1 has no value

```

### warnings 20519: LIST2MAP

```
>compiling LIST2MAP.spad to LIST2MAP.nrlib
```

```

Warnings:
 [1] match: z1 has no value

```

### warnings 20518: LMOPS

```
>compiling LMOPS.spad to LMOPS.nrlib
```

```

Warnings:
 [1] mapExpon: ans has no value
 [2] outputForm: The conditional modes (List (OutputForm)) and $ conflict

```

### warnings 20517: MESH

```
>compiling MESH.spad to MESH.nrlib
```

```

Warnings:
 [1] ptFunc: z1 has no value
 [2] ptFunc: z2 has no value
 [3] meshFun2Var: z1 has no value
 [4] meshFun2Var: x2 has no value

```

### warnings 20516: MOEBIUS

```
>compiling MOEBIUS.spad to MOEBIUS.nrlib
```

```
Warnings:
```

- [1] recip: signature of lhs not unique: \$\$ chosen
- [2] proportional?: f1 has no value
- [3] proportional?: g1 has no value

## warnings 20515: MRING

```
>compiling MRING.spad to MRING.nrlib
```

```
Warnings:
```

- [1] index: ans has no value
- [2] retractIfCan: coef has no value
- [3] retractIfCan: monom has no value
- [4] recip: monom has no value
- [5] recip: coef has no value
- [6] coerce: coef has no value
- [7] coerce: monom has no value
- [8] =: coef has no value
- [9] =: monom has no value
- [10] sortAndAdd: m has no value
- [11] sortAndAdd: res has no value

## warnings 20514: ODR

```
>compiling ODR.spad to ODR.nrlib
```

```
Warnings:
```

- [1] not known that (IntegralDomain) is of mode (CATEGORY domain (IF (has R (Field)) (ATTRI

## warnings 20513: ORDCOMP

```
>compiling ORDCOMP.spad to ORDCOMP.nrlib
```

```
Warnings:
```

- [1] retract: fin has no value
- [2] retractIfCan: fin has no value
- [3] coerce: fin has no value
- [4] coerce: inf has no value
- [5] whatInfinity: inf has no value



```

[6] =: inf has no value
[7] =: fin has no value
[8] *: inf has no value
[9] *: fin has no value
[10] -: inf has no value
[11] -: fin has no value
[12] +: inf has no value
[13] +: fin has no value
[14] recip: fin has no value
[15] <: inf has no value
[16] <: fin has no value

```

## warnings 20512: PARTPERM

>compiling PARTPERM.spad to PARTPERM.nrlib

Warnings:

```

[1] partitions: l1 has no value
[2] shuffle: l1 has no value
[3] shuffle: l2 has no value
[4] shufflein: l1 has no value

```

## warnings 20511: PENDTREE

>compiling PENDTREE.spad to PENDTREE.nrlib

Warnings:

```

[1] ptree: pretend$ -- should replace by @
[2] ptree: :Rep -- should replace by pretend
[3] ptree: :$ -- should replace by @
[4] =: :Rep -- should replace by pretend

```

## warnings 20510: PLOTTOOL

>compiling PLOTTOOL.spad to PLOTTOOL.nrlib

Warnings:

```

[1] xRange: u1 has no value
[2] xRange: v1 has no value
[3] yRange: u1 has no value
[4] yRange: v1 has no value

```

**warnings 20509: PFR**

```
>compiling PFR.spad to PFR.nrllib
```

```
Warnings:
[1] LessThan: :(Boolean) -- should replace by pretend
[2] multiplyFracTerms: :Rep -- should replace by pretend
[3] multiplyFracTerms: c has no value
[4] compactFraction: :(NonNegativeInteger) -- should replace by pretend
[5] compactFraction: s has no value
[6] compactFraction: bf has no value
```

**warnings 20508: PMDOWN**

```
>compiling PMDOWN.spad to PMDOWN.nrllib
```

```
Warnings:
[1] fixPredicate: a1 has no value
[2] patternMatch: a1 has no value
[3] patternMatch: not known that (SetCategory) is of mode (CATEGORY package (SIGNATURE fix
[4] fixPredicates: l1 has no value
```

**warnings 20507: PMINS**

```
>compiling PMINS.spad to PMINS.nrllib
```

```
Warnings:
[1] patternMatchInner: i1 has no value
[2] patternMatchInner: i2 has no value
```

**warnings 20506: PMLSAGG**

```
>compiling PMLSAGG.spad to PMLSAGG.nrllib
```

```
Warnings:
[1] match: not known that (SetCategory) is of mode (CATEGORY package (SIGNATURE patternMat
```

## warnings 20505: PMTOOLS

```
>compiling PMTOOLS.spad to PMTOOLS.nrlib
```

```
Warnings:
```

```
[1] patternMatchTimes: l1 has no value
[2] patternMatch: p1 has no value
[3] patternMatch: z2 has no value
[4] patternMatch: z1 has no value
[5] patternMatch: p2 has no value
```

## warnings 20504: PRTITION

```
>compiling PRTITION.spad to PRTITION.nrlib
```

```
Warnings:
```

```
[1] coerce: signature of lhs not unique: (List (Integer))$ chosen
[2] partition: i2 has no value
[3] partition: i1 has no value
[4] conjugate: pretendRep -- should replace by @
[5] coerce: pretendRep -- should replace by @
[6] pdct: pretendRep -- should replace by @
```

## warnings 20503: PSCAT

```
>compiling PSCAT.spad to PSCAT.nrlib
```

```
Warnings:
```

```
[1] *: r1 has no value
[2] -: r1 has no value
[3] /: r1 has no value
```

## warnings 20502: SCACHE

```
>compiling SCACHE.spad to SCACHE.nrlib
```

```
Warnings:
```

```
[1] enterInCache: pos has no value
```

**warnings 20500: STTAYLOR**

```
>compiling STTAYLOR.spad to STTAYLOR.nrlib
```

```
Warnings:
```

```
[1] +: z has no value
[2] -: z has no value
[3] *: z has no value
[4] exquo: z has no value
[5] /: z has no value
[6] nnintegers: y has no value
[7] integers: y has no value
[8] oddintegers: y has no value
[9] int: y has no value
[10] eval: y has no value
[11] eval: z has no value
[12] lagrange: y has no value
[13] generallambert: x has no value
[14] generallambert: y has no value
[15] comps: y has no value
[16] nlde: y has no value
[17] smult: y has no value
[18] powern: order has no value
[19] powern: y has no value
[20] power: y has no value
```

**warnings 20499: TABLBUMP**

```
>compiling TABLBUMP.spad to TABLBUMP.nrlib
```

```
Warnings:
```

```
[1] bumptab1: s1 has no value
[2] bumptab1: s2 has no value
[3] lex: s1 has no value
[4] lex: s2 has no value
[5] slex: s1 has no value
[6] slex: s2 has no value
[7] inverse: s1 has no value
[8] inverse: s2 has no value
[9] untab: s2 has no value
[10] untab: s1 has no value
```

**warnings 20498: UDPO**

```
>compiling UDPO.spad to UDPO.nrlib
```

```
Warnings:
```

```
[1] more?: y has no value
[2] more?: z has no value
[3] largest: y has no value
[4] largest: z has no value
```

**warnings 20497: UNISEG**

```
>compiling UNISEG.spad to UNISEG.nrlib
```

```
Warnings:
```

```
[1] expand: x has no value
```

**warnings 20496: XPR**

```
>compiling XPR.spad to XPR.nrlib
```

```
Warnings:
```

```
[1] quasiRegular?: k has no value
[2] *: k has no value
[3] *: c has no value
```

**warnings 20495: AFFSP**

```
>compiling AFFSP.spad to AFFSP.nrlib
```

```
Warnings:
```

```
[1] removeConjugate: conjrem has no value
```

**warnings 20494: COMBINAT**

```
>compiling COMBINAT.spad to COMBINAT.nrlib
```

```
Warnings:
[1] binomial: s has no value
[2] binomial: b has no value
```

### warnings 20493: D01FCFA

```
>compiling D01FCFA.spad to D01FCFA.nrlib
```

```
Warnings:
[1] numericalIntegration: x has no value
```

### warnings 20492: IFARRAY

```
>compiling IFARRAY.spad to IFARRAY.nrlib
```

```
Warnings:
[1] remove!: k has no value
[2] select!: k has no value
```

### warnings 20491: INTHEORY

```
>compiling INTHEORY.spad to INTHEORY.nrlib
```

```
Warnings:
[1] fibonacci: f2 has no value
[2] jacobi: j has no value
[3] eulerPhi: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE bern
[4] divisors: i1 has no value
[5] divisors: i2 has no value
```

### warnings 20490: LWORD

```
>compiling LWORD.spad to LWORD.nrlib
```

```
Warnings:
[1] factor1: d has no value
[2] LyndonWordsList1: :(Integer) -- should replace by pretend
```

```
[3] LyndonWordsList1: lbase1 has no value
```

### warnings 20489: MATCAT

```
>compiling MATCAT.spad to MATCAT.nrlib
```

```
Warnings:
```

```
[1] matrix: rows has no value
[2] -: r1 has no value
[3] *: r1 has no value
[4] columnSpace: basis has no value
[5] PfChar: g has no value
[6] /: r1 has no value
```

### warnings 20488: PLOT3D

```
>compiling PLOT3D.spad to PLOT3D.nrlib
```

```
Warnings:
```

```
[1] select: fp has no value
[2] rangeRefine: c has no value
[3] rangeRefine: q has no value
[4] rangeRefine: NUMFUNVALS has no value
[5] refine: curves has no value
[6] plot: curves has no value
[7] plot: z has no value
```

### warnings 20487: POLYVEC

```
>compiling POLYVEC.spad to POLYVEC.nrlib
```

```
Warnings:
```

```
[1] extendedgcd: swapped has no value
```

### warnings 20486: PR

```
>compiling PR.spad to PR.nrlib
```

Warnings:

```
[1] times!: endcell has no value
[2] times!: res has no value
[3] times: x1 has no value
[4] addm!: newcell has no value
[5] addm!: res has no value
[6] addm!: endcell has no value
[7] **: pretend(NonNegativeInteger) -- should replace by @
[8] ^: pretend(NonNegativeInteger) -- should replace by @
[9] unitNormal: lcf has no value
[10] unitCanonical: lcf has no value
[11] fmecg: rout has no value
```

## warnings 20485: PROJSP

>compiling PROJSP.spad to PROJSP.nrlib

Warnings:

```
[1] definingField: pretendRep -- should replace by @
[2] coerce: pretendRep -- should replace by @
[3] list: pretendRep -- should replace by @
[4] pointValue: pretendRep -- should replace by @
[5] removeConjugate: conjrem has no value
[6] lastNonNull: pretendRep -- should replace by @
[7] lastNonNull: pretend(Integer) -- should replace by @
```

## warnings 20484: STREAM

>compiling STREAM.spad to STREAM.nrlib

Warnings:

```
[1] pretendS -- should replace by @
[2] showAllElements: not known that (LazyStreamAggregate S) is of mode (CATEGORY domain (S
[3] remove: signature of lhs not unique: $(Mapping (Boolean) S)$ chosen
[4] lazyEval: :(Mapping $) -- should replace by pretend
[5] expand!: d has no value
```

## warnings 20483: VECTCAT

>compiling VECTCAT.spad to VECTCAT.nrlib



```
Warnings:
[1] -: x has no value
[2] *: x has no value
```

## warnings 20482: XRPOLY

```
>compiling XRPOLY.spad to XRPOLY.nrlib
```

```
Warnings:
[1] unexpand: z has no value
[2] unexpand: y has no value
```

## warnings 20481: IMATRIX

```
>compiling IMATRIX.spad to IMATRIX.nrlib
```

```
Warnings:
[1] determinant: not known that (CommutativeRing) is of mode (CATEGORY R (ATTRIBUTE (commutative *)))
```

## warnings 20479: ASSOCEQ

```
>compiling ASSOCEQ.spad to ASSOCEQ.nrlib
```

```
Warnings:
[1] uncouplingMatrices: z1 has no value
```

## warnings 20478: CARTEN

```
>compiling CARTEN.spad to CARTEN.nrlib
```

```
Warnings:
[1] index2int: n has no value
[2] permsign!: nTrans has no value
[3] elt: n has no value
```

**warnings 20477: CLIF**

```
>compiling CLIF.spad to CLIF.nrllib
```

```
Warnings:
```

```
[1] addMonomProd: c has no value
```

**warnings 20476: CLIP**

```
>compiling CLIP.spad to CLIP.nrllib
```

```
Warnings:
```

```
[1] iClipParametric: x has no value
[2] iClipParametric: y has no value
[3] iClipParametric: z has no value
[4] discardAndSplit: ans has no value
[5] clip: x has no value
[6] clip: y has no value
[7] clip: yMax has no value
[8] clip: yMin has no value
[9] clipWithRanges: x has no value
```

**warnings 20475: COORDSYS**

```
>compiling COORDSYS.spad to COORDSYS.nrllib
```

```
Warnings:
```

```
[1] elliptic: x has no value
[2] prolateSpheroidal: x has no value
[3] oblateSpheroidal: x has no value
[4] bipolar: x has no value
[5] toroidal: x has no value
[6] conical: x has no value
```

**warnings 20474: DHMATRIX**

```
>compiling DHMATRIX.spad to DHMATRIX.nrllib
```

```
Warnings:
```

```
[1] *: signature of lhs not unique: (Point R)$(Point R) chosen
```

**warnings 20473: DIOSP**

```
>compiling DIOSP.spad to DIOSP.nrllib
```

```
Warnings:
 [1] dioSolve: c has no value
```

**warnings 20472: DIRPCAT**

```
>compiling DIRPCAT.spad to DIRPCAT.nrllib
```

```
Warnings:
 [1] reducedSystem: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE coerce (S (Fraction (
```

**warnings 20471: D02CJFA**

```
>compiling D02CJFA.spad to D02CJFA.nrllib
```

```
Warnings:
 [1] ODESolve: i has no value
```

**warnings 20470: FAXF**

```
>compiling FAXF.spad to FAXF.nrllib
```

```
Warnings:
 [1] extensionDegree: signature of lhs not unique: (PositiveInteger) chosen
 [2] minimalPolynomial: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE charthRoot ((Union
 [3] degree: signature of lhs not unique: (PositiveInteger)S chosen
```

**warnings 20469: FFPOLY2**

```
>compiling FFPOLY2.spad to FFPOLY2.nrllib
```

```

Warnings:
[1] rootOfIrreduciblePoly: basispointer has no value
[2] rootOfIrreduciblePoly: beta has no value

```

### warnings 20468: FNLA

```
>compiling FNLA.spad to FNLA.nrllib
```

```

Warnings:
[1] have: hi has no value

```

### warnings 20466: HB

```
>compiling HB.spad to HB.nrllib
```

```

Warnings:
[1] lfunc: sum has no value
[2] generate: newNumComms has no value

```

### warnings 20464: IRSN

```
>compiling IRSN.spad to IRSN.nrllib
```

```

Warnings:
[1] aIdInverse: :(NonNegativeInteger) -- should replace by @
[2] signum: s has no value
[3] signum: z has no value
[4] testPermutation: n has no value
[5] testPermutation: ok has no value

```

### warnings 20463: LOP

```
>compiling LOP.spad to LOP.nrllib
```

```

Warnings:
[1] rowEchWoZeroLines: pretend(NonNegativeInteger) -- should replace by @
[2] rowEchWoZeroLinesWOVectorise: pretend(NonNegativeInteger) -- should replace by @

```

## warnings 20462: MHROWRED

```
>compiling MHROWRED.spad to MHROWRED.nrlib
```

```
Warnings:
[1] nonzero?: s has no value
[2] non0: allZero has no value
[3] non0: ans has no value
[4] mkMat: STEP has no value
[5] mkMat: i has no value
[6] diagSubMatrix: IN has no value
[7] diagSubMatrix: r has no value
[8] diagSubMatrix: STEP has no value
[9] diagSubMatrix: z has no value
[10] vconc: s has no value
[11] rowEchelonLocal: pivord has no value
```

## warnings 20461: NUMQUAD

```
>compiling NUMQUAD.spad to NUMQUAD.nrlib
```

```
Warnings:
[1] romberg: change has no value
[2] simpson: change has no value
[3] trapezoidal: change has no value
[4] rombergo: change has no value
[5] simpsono: change has no value
[6] trapezoidal0: change has no value
```

## warnings 20460: ODESYS

```
>compiling ODESYS.spad to ODESYS.nrlib
```

```
Warnings:
[1] solve: f1 has no value
[2] triangulate: f1 has no value
[3] backsolve: part has no value
[4] solveInField: l1 has no value
[5] solveInField: f2 has no value
```

**warnings 20459: ODETOOLS**

```
>compiling ODETOOLS.spad to ODETOOLS.nrlib
```

```
Warnings:
```

```
[1] wronskianMatrix: f1 has no value
```

**warnings 20458: PERMAN**

```
>compiling PERMAN.spad to PERMAN.nrlib
```

```
Warnings:
```

```
[1] permanent3: a has no value
[2] permanent: :(Integer) -- should replace by pretend
[3] permanent: :(PositiveInteger) -- should replace by pretend
[4] permanent: a has no value
[5] permanent2: :(Integer) -- should replace by pretend
[6] permanent2: :(PositiveInteger) -- should replace by pretend
[7] permanent2: :(NonNegativeInteger) -- should replace by pretend
[8] permanent2: a has no value
```

**warnings 20457: PFECAT**

```
>compiling PFECAT.spad to PFECAT.nrlib
```

```
Warnings:
```

```
[1] gcdPolynomial: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE charthRoot
[2] charthRoot: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE charthRoot ((
[3] solveLinearPolynomialEquation: not known that (Ring) is of mode (CATEGORY domain (SIGN
```

**warnings 20456: PSEUDLIN**

```
>compiling PSEUDLIN.spad to PSEUDLIN.nrlib
```

```
Warnings:
```

```
[1] changeBase: k1 has no value
[2] normalForm0: B has no value
[3] normalForm0: Binv has no value
```

**warnings 20455: REP2**

```
>compiling REP2.spad to REP2.nrlib
```

```
Warnings:
```

- [1] cyclicSubmodule: basis has no value
- [2] standardBasisOfCyclicSubmodule: standardBasis has no value
- [3] splitInternal: submoduleRepresentation has no value
- [4] splitInternal: factormoduleRepresentation has no value
- [5] areEquivalent?: x0 has no value
- [6] areEquivalent?: x1 has no value
- [7] areEquivalent?: foundResult has no value
- [8] areEquivalent?: result has no value
- [9] areEquivalent?: transitionM has no value
- [10] isAbsolutelyIrreducible?: result has no value
- [11] meatAxe: :(PositiveInteger) -- should replace by pretend
- [12] meatAxe: x has no value
- [13] meatAxe: foundResult has no value
- [14] meatAxe: result has no value
- [15] meatAxe: signature of lhs not unique: (List (List (Matrix R)))(List (Matrix R))(Boolean) chosen

**warnings 20454: SETMN**

```
>compiling SETMN.spad to SETMN.nrlib
```

```
Warnings:
```

- [1] elements: 1 has no value
- [2] replaceKthElement: found has no value
- [3] replaceKthElement: i has no value

**warnings 20453: STRING**

```
>compiling STRING.spad to STRING.nrlib
```

```
Warnings:
```

- [1] OMwrite: pretend(String) -- should replace by @

**warnings 20452: ASP1**

```
>compiling ASP1.spad to ASP1.nrlib
```

```
Warnings:
[1] X has no value
```

### warnings 20451: ASP10

```
>compiling ASP10.spad to ASP10.nrlib
```

```
Warnings:
[1] P has no value
[2] Q has no value
[3] DQDL has no value
[4] X has no value
[5] ELAM has no value
[6] JINT has no value
```

### warnings 20450: ASP24

```
>compiling ASP24.spad to ASP24.nrlib
```

```
Warnings:
[1] N has no value
[2] XC has no value
[3] FC has no value
[4] coerce: FC has no value
```

### warnings 20449: ASP4

```
>compiling ASP4.spad to ASP4.nrlib
```

```
Warnings:
[1] NDIM has no value
[2] X has no value
```

### warnings 20448: ASP50

```
>compiling ASP50.spad to ASP50.nrlib
```



```

Warnings:
[1] M has no value
[2] N has no value
[3] XC has no value
[4] FVECC has no value
[5] I has no value
[6] coerce: FVECC has no value

```

### warnings 20447: ASP6

```
>compiling ASP6.spad to ASP6.nrlib
```

```

Warnings:
[1] N has no value
[2] X has no value
[3] FVEC has no value
[4] IFLAG has no value
[5] coerce: FVEC has no value

```

### warnings 20446: ASP73

```
>compiling ASP73.spad to ASP73.nrlib
```

```

Warnings:
[1] X has no value
[2] Y has no value
[3] ALPHA has no value
[4] BETA has no value
[5] GAMMA has no value
[6] DELTA has no value
[7] EPSOLN has no value
[8] PHI has no value
[9] PSI has no value

```

### warnings 20445: AXSERV

```
>compiling AXSERV.spad to AXSERV.nrlib
```

```

Warnings:
[1] getFile: contentType has no value

```

```
[2] getContentType: contentType has no value
```

### warnings 20444: BALFACT

```
>compiling BALFACT.spad to BALFACT.nrllib
```

```
Warnings:
```

```
[1] balancedFactorisation: z1 has no value
[2] balancedFactorisation: not known that (IntegralDomain) is of mode (CATEGORY package (S
```

### warnings 20443: BOUNDZRO

```
>compiling BOUNDZRO.spad to BOUNDZRO.nrllib
```

```
Warnings:
```

```
[1] bringDown: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE integer
[2] integerBound: z1 has no value
[3] qbound: bound has no value
```

### warnings 20442: CDEN

```
>compiling CDEN.spad to CDEN.nrllib
```

```
Warnings:
```

```
[1] clearDenominator: x has no value
[2] splitDenominator: x has no value
[3] commonDenominator: x has no value
```

### warnings 20441: CHVAR

```
>compiling CHVAR.spad to CHVAR.nrllib
```

```
Warnings:
```

```
[1] eval: s has no value
[2] eval: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE mkIntegr
[3] chvar: s has no value
[4] rootPoly: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE mkIn
```

## warnings 20440: CONTFRAC

```
>compiling CONTFRAC.spad to CONTFRAC.nrllib
```

Warnings:

```
[1] =: whole has no value
[2] =: den has no value
[3] partialNumerators: x1 has no value
[4] partialDenominators: x1 has no value
[5] partialQuotients: whole has no value
[6] extend: fract has no value
[7] complete: fract has no value
[8] iGenApproximants: num has no value
[9] iGenApproximants: den has no value
[10] iGenConvergents: num has no value
[11] iGenConvergents: den has no value
[12] +: x has no value
[13] +: y has no value
[14] -: x has no value
[15] -: y has no value
[16] *: x has no value
[17] *: y has no value
[18] /: x has no value
[19] /: y has no value
[20] recip: x has no value
[21] coerce: 1 has no value
```

## warnings 20439: DDFACT

```
>compiling DDFACT.spad to DDFACT.nrllib
```

Warnings:

```
[1] notSqFr: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE factor ((Factored
[2] exptMod: :FP -- should replace by pretend
[3] ddffact1: ddfact has no value
[4] separateFactors: n1 has no value
[5] separateFactors: ris has no value
[6] separateFactors: newaux has no value
[7] distdfact: factlist has no value
```

## warnings 20438: DIRPROD

```
>compiling DIRPROD.spad to DIRPROD.nrllib
```

```

Warnings:
[1] same?: x has no value
[2] Zero: pretend$ -- should replace by @
[3] One: pretend$ -- should replace by @
[4] *: x has no value
[5] *: STEP has no value
[6] *: i has no value
[7] *: pretend$ -- should replace by @
[8] subtractIfCan: pretend$ -- should replace by @
[9] recip: pretend$ -- should replace by @
[10] unitVector: pretend$ -- should replace by @

```

### warnings 20437: DISPLAY

```
>compiling DISPLAY.spad to DISPLAY.nrlib
```

```

Warnings:
[1] sayLength: sum has no value

```

### warnings 20436: DPOLCAT

```
>compiling DPOLCAT.spad to DPOLCAT.nrlib
```

```

Warnings:
[1] makeVariable: n has no value
[2] degree: d has no value
[3] weights: ws has no value
[4] initial: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE coerce (A A)) (S
[5] eval: IN has no value
[6] eval: s has no value
[7] eval: e has no value

```

### warnings 20435: DSTREE

```
>compiling DSTREE.spad to DSTREE.nrlib
```

```

Warnings:
[1] tree: :(Record (: value S) (: args (List $))) -- should replace by @
[2] tree: :$ -- should replace by pretend
[3] value: :(Record (: value S) (: args (List $))) -- should replace by pretend

```

```

[4] children: :(Record (: value S) (: args (List $))) -- should replace by pretend
[5] children: :(List $) -- should replace by @
[6] setchildren!: :(Record (: value S) (: args (List $))) -- should replace by pretend
[7] setchildren!: pretend$ -- should replace by @
[8] setvalue!: :(Record (: value S) (: args (List $))) -- should replace by pretend

```

### warnings 20434: D01AMFA

```
>compiling D01AMFA.spad to D01AMFA.nrlib
```

```

Warnings:
[1] numericalIntegration: x has no value

```

### warnings 20433: D01APFA

```
>compiling D01APFA.spad to D01APFA.nrlib
```

```

Warnings:
[1] numericalIntegration: x has no value

```

### warnings 20432: D01AQFA

```
>compiling D01AQFA.spad to D01AQFA.nrlib
```

```

Warnings:
[1] numericalIntegration: x has no value

```

### warnings 20431: EQ

```
>compiling EQ.spad to EQ.nrlib
```

```

Warnings:
[1] factorAndSplit: not known that (IntegralDomain) is of mode (CATEGORY domain (SIGNATURE factorAndSplit))
[2] factorAndSplit: not known that (IntegralDomain) is of mode (CATEGORY S (SIGNATURE factor ((Factor

```

**warnings 20430: MDDFACT**

```
>compiling MDDFACT.spad to MDDFACT.nrllib
```

```
Semantic Errors:
```

- [1] ddfactor: degree is BOTH a variable and a literal
- [2] ddfact: degree is BOTH a variable and a literal
- [3] ddFact: degree is BOTH a variable and a literal
- [4] sepfact: degree is BOTH a variable and a literal
- [5] separateFactors: degree is BOTH a variable and a literal
- [6] sepFact1: degree is BOTH a variable and a literal

**warnings 20429: MMAP**

```
>compiling MMAP.spad to MMAP.nrllib
```

```
Warnings:
```

- [1] map: x has no value

**warnings 20428: MODMON**

```
>compiling MODMON.spad to MODMON.nrllib
```

```
Warnings:
```

- [1] :(NonNegativeInteger) -- should replace by pretend
- [2] lift: pretendRep -- should replace by @
- [3] not known that (SetCategory) is of mode (CATEGORY domain (SIGNATURE frobenius (\$ \$)))
- [4] not known that (Ring) is of mode (CATEGORY domain (SIGNATURE frobenius (\$ \$)))
- [5] not known that (IntegralDomain) is of mode (CATEGORY domain (SIGNATURE frobenius (\$ \$)))

**warnings 20427: MONOTOOL**

```
>compiling MONOTOOL.spad to MONOTOOL.nrllib
```

```
Warnings:
```

- [1] splitSquarefree: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE frobenius (\$ \$)))
- [2] splitSquarefree: s has no value

**warnings 20426: MPCPF**

```
>compiling MPCPF.spad to MPCPF.nrlib
```

```
Warnings:
```

```
[1] factor: not known that (SIGNATURE variable ((Union $ failed) (Symbol))) is of mode (CATEGORY domain)
```

**warnings 20425: MPC2**

```
>compiling MPC2.spad to MPC2.nrlib
```

```
Warnings:
```

```
[1] map: not known that (Ring) is of mode (CATEGORY package (SIGNATURE map (PS (Mapping S R) PR)) (SIGNATURE))
```

**warnings 20424: MPC3**

```
>compiling MPC3.spad to MPC3.nrlib
```

```
Warnings:
```

```
[1] map: not known that (Ring) is of mode (CATEGORY package (SIGNATURE map (PR2 (Mapping Vars2 Vars1) (SIGNATURE)))
```

**warnings 20423: MPRFF**

```
>compiling MPRFF.spad to MPRFF.nrlib
```

```
Warnings:
```

```
[1] factor: not known that (PolynomialCategory (Fraction (Integer)) (IndexedExponents (Symbol)) (Symbol))
```

```
[2] factor: pretend(Factored (Polynomial R)) -- should replace by @
```

```
[3] factor: flist has no value
```

```
[4] pushdown: not known that (Ring) is of mode (CATEGORY package (SIGNATURE totalfract ((Record (: sup
```

```
[5] pushup: y has no value
```

**warnings 20422: MULTSQFR**

```
>compiling MULTSQFR.spad to MULTSQFR.nrlib
```

```
Warnings:
```

```

[1] nsqfree: lcf has no value
[2] nsqfree: z2 has no value
[3] nsqfree: z1 has no value
[4] nsqfree: leadpol has no value
[5] nsqfree: sqlc has no value
[6] nsqfree: :(NonNegativeInteger) -- should replace by pretend
[7] nsqfree: sqlead has no value
[8] nsqfree: unitsq has no value
[9] nsqfree: sqdec has no value
[10] squareFree: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE s
[11] squareFree: fctr has no value
[12] squareFree: xpnt has no value
[13] squareFree: p has no value
[14] squareFree: result1 has no value
[15] squareFree: lvar has no value
[16] squareFree: not known that (Ring) is of mode (CATEGORY package (SIGNATURE squareFree
[17] intChoose: :R -- should replace by @
[18] intChoose: d1 has no value
[19] coefChoose: lcoef has no value
[20] lift: leadlist has no value
[21] lift: p0 has no value
[22] lift: p1 has no value

```

## warnings 20421: NPCOEF

>compiling NPCOEF.spad to NPCOEF.nrllib

### Warnings:

```

[1] npcoef: :(NonNegativeInteger) -- should replace by pretend
[2] npcoef: ndet has no value
[3] npcoef: detufact has no value
[4] npcoef: n2 has no value
[5] npcoef: n1 has no value
[6] npcoef: ltodel has no value
[7] npcoef: detcoef has no value
[8] check: :(NonNegativeInteger) -- should replace by pretend
[9] check: doit has no value
[10] check: cfu has no value
[11] check: poselt has no value
[12] check: pp has no value
[13] buildtable: table has no value
[14] modify: :(NonNegativeInteger) -- should replace by pretend
[15] modify: lterase has no value
[16] modify: n2 has no value
[17] modify: n1 has no value

```



**warning 20420: NSUP**

```
>compiling NSUP.spad to NSUP.nrlib
```

```
Warnings:
```

```
[1] subResultantGcd: not known that (UnivariatePolynomialCategory R) is of mode (CATEGORY domain (SIG
```

**warnings 20419: ODEPRIM**

```
>compiling ODEPRIM.spad to ODEPRIM.nrlib
```

```
Warnings:
```

```
[1] UP2UP2: f1 has no value
[2] indicialEquations: eq has no value
[3] NPMulambda: lamb has no value
[4] NPMulambda: lf has no value
```

**warnings 20418: ODEPRRIC**

```
>compiling ODEPRRIC.spad to ODEPRRIC.nrlib
```

```
Warnings:
```

```
[1] lambda: z has no value
[2] infLambda: z has no value
[3] infmax: z has no value
[4] dmax: z has no value
[5] padicsol: z has no value
[6] leadingDenomRicDE: ind has no value
[7] leadingDenomRicDE: z2 has no value
[8] leadingDenomRicDE: z1 has no value
[9] constantCoefficientOperator: ans has no value
[10] innermax: ans has no value
[11] leadingCoefficientRicDE: ind has no value
[12] leadingCoefficientRicDE: z2 has no value
[13] leadingCoefficientRicDE: z1 has no value
[14] innerlb: lb has no value
[15] polysol: z has no value
```

**warnings 20417: OMPKG**

```
>compiling OMPKG.spad to OMPKG.nrllib
```

```
Warnings:
```

- [1] OMreadStr: pretend(String) -- should replace by @
- [2] OMlistCDs: pretend(List (String)) -- should replace by @
- [3] OMlistSymbols: pretend(List (String)) -- should replace by @

## warnings 20416: PADICRC

```
>compiling PADICRC.spad to PADICRC.nrllib
```

```
Warnings:
```

- [1] coerce: 1 has no value

## warnings 20415: PFBR

```
>compiling PFBR.spad to PFBR.nrllib
```

```
Warnings:
```

- [1] hensel: z1 has no value
- [2] hensel: not known that (Ring) is of mode (CATEGORY package (SIGNATURE solveLinearPolyn
- [3] hensel: foundFactors has no value
- [4] factorSFBRLcUnitInner: z1 has no value
- [5] factorSFBRLcUnitInner: not known that (Ring) is of mode (CATEGORY package (SIGNATURE s
- [6] factorSFBRLcUnitInner: fctr has no value
- [7] bivariateSLPEBR: z1 has no value
- [8] chooseFSQViableSubstitutions: z1 has no value
- [9] chooseFSQViableSubstitutions: ppR has no value
- [10] chooseSLPEViableSubstitutions: z1 has no value
- [11] chooseSLPEViableSubstitutions: lpolysR has no value
- [12] raise: z1 has no value
- [13] raise: :R -- should replace by pretend
- [14] raise: :R -- should replace by @
- [15] lower: z1 has no value
- [16] SLPEBR: z1 has no value
- [17] factorByRecursion: not known that (IntegralDomain) is of mode (CATEGORY package (SIGN
- [18] factorByRecursion: z1 has no value
- [19] factorByRecursion: :S -- should replace by @
- [20] factorSquareFreeByRecursion: lcppPow has no value

## warnings 20414: PFBRU

```
>compiling PFBRU.spad to PFBRU.nrlib
```

```
Warnings:
```

```
[1] solveLinearPolynomialEquationByRecursion: z1 has no value
[2] solveLinearPolynomialEquationByRecursion: not known that (Ring) is of mode (CATEGORY package (SIGNATURE solveLinearPolynomialEquationByRecursion))
[3] solveLinearPolynomialEquationByRecursion: answer has no value
[4] hensel: z1 has no value
[5] hensel: foundFactors has no value
[6] chooseFSQViableSubstitutions: z1 has no value
[7] chooseFSQViableSubstitutions: ppR has no value
[8] raise: z1 has no value
[9] raise: :R -- should replace by pretend
[10] raise: :R -- should replace by @
[11] lower: z1 has no value
[12] factorSFBRLcUnitInner: z1 has no value
[13] factorSFBRLcUnitInner: not known that (Ring) is of mode (CATEGORY package (SIGNATURE solveLinearPolynomialEquationByRecursion))
[14] factorSFBRLcUnitInner: fctr has no value
[15] factorByRecursion: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE solveLinearPolynomialEquationByRecursion))
[16] factorByRecursion: z1 has no value
[17] factorByRecursion: :S -- should replace by pretend
[18] factorByRecursion: :S -- should replace by @
[19] factorSquareFreeByRecursion: lcppPow has no value
```

## warnings 20413: PFORP

```
>compiling PFORP.spad to PFORP.nrlib
```

```
Warnings:
```

```
[1] translateLocal: The conditional modes (List R) and (Mapping E (Integer)) conflict
[2] minimalForm: lminForm has no value
[3] listAllMonoExp: pretend(Integer) -- should replace by @
```

## warnings 20412: PFRPAC

```
>compiling PFRPAC.spad to PFRPAC.nrlib
```

```
Warnings:
```

```
[1] makeSup: not known that (Ring) is of mode (CATEGORY package (SIGNATURE partialFraction ((Any) (Fraction)))
[2] makeSup: z1 has no value
```

**warnings 20411: PGCD**

```
>compiling PGCD.spad to PGCD.nrllib
```

```
Warnings:
```

```
[1] gcdPrimitive: b has no value
[2] gcdPrimitive: a has no value
[3] localgcd: s has no value
[4] gcdTermList: not known that (Ring) is of mode (CATEGORY package (SIGNATURE gcd (P P P)
[5] lift: p0 has no value
[6] gcd: z1 has no value
[7] gcd: z2 has no value
```

**warnings 20410: PLEQN**

```
>compiling PLEQN.spad to PLEQN.nrllib
```

```
Warnings:
```

```
[1] regime: wcd has no value
[2] regime: yzero has no value
[3] bsolve: rksoln has no value
[4] bsolve: lrec3 has no value
[5] ParCondList: covered has no value
[6] ParCondList: zro has no value
[7] ParCondList: npc has no value
[8] ParCondList: done has no value
[9] ParCondList: rcl has no value
[10] pr2dmp: pretendGR -- should replace by @
[11] dmp2rfi: v1 has no value
[12] dmp2rfi: r1 has no value
[13] sqfree: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE psolv
[14] ParCond: found has no value
[15] ParCond: DetEqn has no value
[16] ParCond: z1 has no value
[17] ParCond: z2 has no value
```

**warnings 20409: PMPLCAT**

```
>compiling PMPLCAT.spad to PMPLCAT.nrllib
```

```
Warnings:
```

```
[1] patternMatch: l1 has no value
[2] patternMatch: p1 has no value
```

```
[3] patternMatch: p2 has no value
[4] patternMatch: p3 has no value
```

## warnings 20408: PNTHEORY

```
>compiling PNTHEORY.spad to PNTHEORY.nrlib
```

```
Warnings:
[1] cyclotomic: s has no value
[2] cyclotomic: c has no value
```

## warnings 20407: POLUTIL

```
>compiling POLUTIL.spad to POLUTIL.nrlib
```

```
Warnings:
[1] sylvesterSequence: res has no value
[2] sturmVariationsOf: ll has no value
```

## warnings 20406: POLYCATQ

```
>compiling POLYCATQ.spad to POLYCATQ.nrlib
```

```
Warnings:
[1] isPower: var has no value
[2] P2UP: z has no value
[3] P2UP: not known that (Ring) is of mode (CATEGORY package (SIGNATURE variables ((List V) F)) (SIGNATURE
```

## warnings 20405: POLYLIFT

```
>compiling POLYLIFT.spad to POLYLIFT.nrlib
```

```
Warnings:
[1] map: not known that (Ring) is of mode (CATEGORY package (SIGNATURE map (S (Mapping S Vars) (Mapping
```

**warnings 20404: POLYROOT**

```
>compiling POLYROOT.spad to POLYROOT.nrllib
```

```
Warnings:
```

```
[1] zroot: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE rroot (
```

**warnings 20403: POLY2**

```
>compiling POLY2.spad to POLY2.nrllib
```

```
Warnings:
```

```
[1] map: x1 has no value
```

```
[2] map: x2 has no value
```

**warnings 20402: POLY2UP**

```
>compiling POLY2UP.spad to POLY2UP.nrllib
```

```
Warnings:
```

```
[1] univariate: not known that (Ring) is of mode (CATEGORY package (SIGNATURE univariate (
```

```
[2] univariate: x1 has no value
```

**warnings 20401: PRS**

```
>compiling PRS.spad to PRS.nrllib
```

```
Warnings:
```

```
[1] exquo: p1 has no value
```

```
[2] resultantnaif: a has no value
```

```
[3] resultantEuclideanif: a has no value
```

```
[4] semiResultantEuclideanif: a has no value
```

```
[5] chainSubResultants: L has no value
```

```
[6] schema: L has no value
```

## warnings 20400: PSQFR

```
>compiling PSQFR.spad to PSQFR.nrllib
```

```
Warnings:
```

```
[1] pPolRoot: not known that (Ring) is of mode (CATEGORY package (SIGNATURE squareFree ((Factored P) I
[2] pthPower: isSq has no value
[3] pthPower: fctr has no value
[4] pthPower: xpnt has no value
[5] finSqFr: :(NonNegativeInteger) -- should replace by pretend
[6] finSqFr: cont1 has no value
```

## warnings 20399: PUSHVAR

```
>compiling PUSHVAR.spad to PUSHVAR.nrllib
```

```
Warnings:
```

```
[1] map: x has no value
[2] map: not known that (Ring) is of mode (CATEGORY package (SIGNATURE pushdown (PPR PPR OV)) (SIGNAT
[3] pushup: y has no value
```

## warnings 20398: QALGSET

```
>compiling QALGSET.spad to QALGSET.nrllib
```

```
Warnings:
```

```
[1] npoly: not known that (OrderedAbelianMonoid) is of mode (CATEGORY domain (IF (has (NonNegativeInt
[2] idealSimplify: not known that (OrderedAbelianMonoid) is of mode (CATEGORY domain (IF (has (NonNeg
[3] idealSimplify: not known that (OrderedAbelianMonoidSup) is of mode (CATEGORY domain (IF (has (Nonl
[4] idealSimplify: not known that (PolynomialCategory R (Product (NonNegativeInteger) Expon) Var) is c
[5] idealSimplify: gb has no value
```

## warnings 20397: RADIX

```
>compiling RADIX.spad to RADIX.nrllib
```

```
Warnings:
```

```
[1] fractionPart: signature of lhs not unique: (Fraction (Integer))$ chosen
[2] coerce: le has no value
[3] radixFrac: c has no value
```

```
[4] unknown Functor code (error Radix base must be at least 2)
```

### warnings 20396: RCFIELD

```
>compiling RCFIELD.spad to RCFIELD.nrlib
```

```
Warnings:
```

```
[1] rootOf: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE sqrt (S (Integer)
[2] allRootsOf: z has no value
[3] allRootsOf: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE sqrt (S (Integer)
```

### warnings 20395: RDETR

```
>compiling RDETR.spad to RDETR.nrlib
```

```
Warnings:
```

```
[1] polyRDE: eq has no value
[2] SPDEncancel1: q has no value
[3] SPDEncancel2: q has no value
```

### warnings 20394: RDETRS

```
>compiling RDETRS.spad to RDETRS.nrlib
```

```
Warnings:
```

```
[1] baseRDEsys: z1 has no value
[2] baseRDEsys: z2 has no value
[3] baseRDEsys: z3 has no value
[4] baseRDEsys: z4 has no value
[5] baseRDEsys: z5 has no value
[6] baseRDEsys: z6 has no value
[7] DSPDEsys: z1 has no value
[8] DSPDEsys: z2 has no value
[9] DSPDEsys: z3 has no value
[10] DSPDEsys: z4 has no value
[11] DSPDEsys: z5 has no value
```



**warnings 20393: REAL0**

```
>compiling REAL0.spad to REAL0.nrlib
```

```
Warnings:
```

```
[1] makeSqfr: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE realZeros ((List
[2] Zero1: G has no value
[3] rootBound: i has no value
[4] transMultInv: :(NonNegativeInteger) -- should replace by pretend
[5] var: i has no value
```

**warnings 20392: REALSOLV**

```
>compiling REALSOLV.spad to REALSOLV.nrlib
```

```
Warnings:
```

```
[1] prn2rfi: x has no value
```

**warnings 20391: RESRING**

```
>compiling RESRING.spad to RESRING.nrlib
```

```
Warnings:
```

```
[1] unknown Functor code (error the residue ring is the zero ring)
```

**warnings 20390: RETSOL**

```
>compiling RETSOL.spad to RETSOL.nrlib
```

```
Warnings:
```

```
[1] PQ2P: q1 has no value
[2] PQIfCan: not known that (Ring) is of mode (CATEGORY package (SIGNATURE solveRetract ((List (List
```

**warnings 20389: RF**

```
>compiling RF.spad to RF.nrlib
```

```
Warnings:
[1] peval: z1 has no value
[2] peval: z2 has no value
```

## warnings 20388: RFFACTOR

```
>compiling RFFACTOR.spad to RFFACTOR.nrlib
```

```
Warnings:
[1] factorFraction: not known that (PolynomialCategory (Fraction (Integer)) (IndexedExponentiation
```

## warnings 20387: SHP

```
>compiling SHP.spad to SHP.nrlib
```

```
Warnings:
[1] subresultantSequenceNext: :(List (UnivariatePolynomial x R)) -- should replace by @
[2] subresultantSequenceInner: :(UnivariatePolynomial x R) -- should replace by @
[3] subresultantSequenceInner: :(List (UnivariatePolynomial x R)) -- should replace by @
[4] subresultantSequence: :(List (UnivariatePolynomial x R)) -- should replace by @
[5] polsth1: :(List (UnivariatePolynomial x R)) -- should replace by @
[6] polsth1: Listf has no value
[7] polsth2: :(List (UnivariatePolynomial x R)) -- should replace by @
[8] polsth3: :(List (UnivariatePolynomial x R)) -- should replace by @
[9] wfunctaux: :(List R) -- should replace by @
[10] wfunctaux: :(NonNegativeInteger) -- should replace by @
[11] wfunct: :(List R) -- should replace by @
[12] SturmHabicht: :(UnivariatePolynomial x R) -- should replace by @
[13] SturmHabichtMultiple: :(UnivariatePolynomial x R) -- should replace by @
[14] SturmHabichtMultiple: not known that (IntegralDomain) is of mode (CATEGORY domain (SI
```

## warnings 20386: SIGNRF

```
>compiling SIGNRF.spad to SIGNRF.nrlib
```

```
Warnings:
[1] psign: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE sign ((
```

**warnings 20385: SMITH**

```
>compiling SMITH.spad to SMITH.nrlib
```

```
Warnings:
```

- [1] elRow2: x has no value
- [2] elColumn2: x has no value
- [3] ijDivide: x has no value
- [4] findEqMat: x has no value

**warnings 20384: SMP**

```
>compiling SMP.spad to SMP.nrlib
```

```
Warnings:
```

- [1] moreThanOneVariable?: x1 has no value
- [2] ^: pretend(NonNegativeInteger) -- should replace by @
- [3] \*\*: pretend(NonNegativeInteger) -- should replace by @
- [4] exquo: The conditional modes (SparseUnivariatePolynomial \$) and D conflict
- [5] map: x1 has no value
- [6] evalSortedVarlist: x1 has no value
- [7] eval: x2 has no value
- [8] eval: x1 has no value
- [9] eval: IN has no value
- [10] eval: val has no value
- [11] differentiate: x1 has no value

**warnings 20382: SOLVEFOR**

```
>compiling SOLVEFOR.spad to SOLVEFOR.nrlib
```

```
Warnings:
```

- [1] mapSolve: x has no value
- [2] intsolve: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE solve ((List F)

**warnings 20381: SPLTREE**

```
>compiling SPLTREE.spad to SPLTREE.nrlib
```

```
Warnings:
```

```
[1] splitNodeOf!: 1a has no value
```

### warnings 20380: STINPROD

```
>compiling STINPROD.spad to STINPROD.nrlib
```

```
Warnings:
```

```
[1] applyOverQF: z1 has no value
[2] infiniteProduct: z1 has no value
[3] evenInfiniteProduct: z1 has no value
[4] oddInfiniteProduct: z1 has no value
[5] generalInfiniteProduct: z1 has no value
```

### warnings 20379: STTF

```
>compiling STTF.spad to STTF.nrlib
```

```
Warnings:
```

```
[1] exp: y has no value
[2] sincos: y has no value
[3] tan: y has no value
[4] cot: y has no value
[5] sinhcos: y has no value
[6] tanh: y has no value
```

### warnings 20378: SUBRESP

```
>compiling SUBRESP.spad to SUBRESP.nrlib
```

```
Warnings:
```

```
[1] primitivePart: x1 has no value
```

### warnings 20377: SUMRF

```
>compiling SUMRF.spad to SUMRF.nrlib
```

```
Warnings:
```

```
[1] sum: z has no value
```

## warnings 20376: SUP

```
>compiling SUP.spad to SUP.nrlib
```

```
Warnings:
[1] **: pretend(NonNegativeInteger) -- should replace by @
[2] ^: pretend(NonNegativeInteger) -- should replace by @
[3] **: y has no value
[4] pomopo!: rout has no value
[5] exquo: signature of lhs not unique: (Union $ failed)$$ chosen
[6] fmecg: rout has no value
[7] pseudoRemainder: :(Integer) -- should replace by pretend
[8] pseudoRemainder: :(NonNegativeInteger) -- should replace by pretend
[9] elt: :(NonNegativeInteger) -- should replace by pretend
```

## warnings 20375: SUPFRACF

```
>compiling SUPFRACF.spad to SUPFRACF.nrlib
```

```
Warnings:
[1] squareFree: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE factor ((Fact
```

## warnings 20374: TEX

```
>compiling TEX.spad to TEX.nrlib
```

```
Warnings:
[1] display: pretend(Integer) -- should replace by @
[2] coerce: pretend(Integer) -- should replace by @
[3] stringify: pretend(String) -- should replace by @
[4] splitLong1: l has no value
[5] splitLong1: ls has no value
[6] splitLong1: s has no value
[7] formatSpecial: form has no value
[8] formatSpecial: prescript has no value
[9] formatPlex: s has no value
[10] formatTex: pretend(Boolean) -- should replace by @
```

**warnings 20373: TEXTFILE**

```
>compiling TEXTFILE.spad to TEXTFILE.nrlib
```

```
Warnings:
```

```
[1] endOfFile?: pretend(Boolean) -- should replace by @
```

**warnings 20372: TREE**

```
>compiling TREE.spad to TREE.nrlib
```

```
Warnings:
```

```
[1] children: node has no value
[2] setchildren!: node has no value
[3] setchildren!: pretend$ -- should replace by @
[4] setvalue!: node has no value
[5] count: signature of lhs not unique: (NonNegativeInteger)$ chosen
[6] value: node has no value
[7] node?: t has no value
[8] multipleOverbar: The conditional modes (String) and S conflict
```

**warnings 20371: TWOFAC**

```
>compiling TWOFAC.spad to TWOFAC.nrlib
```

```
Warnings:
```

```
[1] pthRoot: x has no value
[2] generalSqFr: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE g
[3] generalSqFr: flg has no value
[4] generalSqFr: xpnt has no value
[5] generalSqFr: :(NonNegativeInteger) -- should replace by pretend
[6] generalSqFr: fctr has no value
[7] generalTwoFactor: flg has no value
[8] generalTwoFactor: fctr has no value
[9] generalTwoFactor: xpnt has no value
[10] generalTwoFactor: ll has no value
[11] generalTwoFactor: x has no value
[12] generalTwoFactor: unitPart has no value
[13] twoFactor: x has no value
[14] twoFactor: n has no value
[15] twoFactor: look has no value
[16] twoFactor: umv has no value
[17] twoFactor: umex has no value
```

**warnings 20370: UNIFACT**

```
>compiling UNIFACT.spad to UNIFACT.nrllib
```

```
Warnings:
```

```
[1] eisenstein: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE factor ((Fact
[2] negShiftp: x has no value
[3] bound: :(NonNegativeInteger) -- should replace by pretend
[4] bound: cbound has no value
[5] numFactors: pretend(Integer) -- should replace by @
[6] numFactors: ans has no value
[7] choose: qSave has no value
[8] choose: ddSave has no value
[9] quadratic: beta has no value
[10] quadratic: alpha has no value
[11] subMinusX: :ZP -- should replace by pretend
[12] henselFact: c has no value
[13] henselFact: factorlist has no value
[14] henselFact: :(List ZP) -- should replace by pretend
```

**warnings 20369: UPCDEN**

```
>compiling UPCDEN.spad to UPCDEN.nrllib
```

```
Warnings:
```

```
[1] clearDenominator: x has no value
[2] splitDenominator: x has no value
```

**warnings 20368: UPDECOMP**

```
>compiling UPDECOMP.spad to UPDECOMP.nrllib
```

```
Warnings:
```

```
[1] leftFactorIfCan: g has no value
```

**warnings 20367: UPDIVP**

```
>compiling UPDIVP.spad to UPDIVP.nrlib
```

```
Warnings:
```

```
[1] divideIfCan: x has no value
```

## warnings 20366: UPSQFREE

```
>compiling UPSQFREE.spad to UPSQFREE.nrlib
```

```
Warnings:
```

```
[1] BumInSepFFE: :(NonNegativeInteger) -- should replace by @
```

```
[2] squareFree: lffe has no value
```

```
[3] squareFree: dunit has no value
```

## warnings 20365: VIEWDEF

```
>compiling VIEWDEF.spad to VIEWDEF.nrlib
```

```
Warnings:
```

```
[1] viewWriteDefault: thingsToWrite has no value
```

## warnings 20364: WEIER

```
>compiling WEIER.spad to WEIER.nrlib
```

```
Warnings:
```

```
[1] transback: :(Stream (Stream (Polynomial R))) -- should replace by pretend
```

```
[2] transback: :(TaylorSeries R) -- should replace by pretend
```

```
[3] transback: :(Stream (TaylorSeries R)) -- should replace by pretend
```

```
[4] clikeUniv: p has no value
```

```
[5] clikeUniv: not known that (Ring) is of mode (CATEGORY package (SIGNATURE crest ((Mappi
```

```
[6] streamlikeUniv: :(NonNegativeInteger) -- should replace by pretend
```

```
[7] sts2stst: x has no value
```

```
[8] sts2stst: y has no value
```

```
[9] sts2stst: :(Stream (NonNegativeInteger)) -- should replace by pretend
```

```
[10] weier: :(TaylorSeries R) -- should replace by pretend
```

```
[11] cfirst: s has no value
```

```
[12] crest: s has no value
```

```
[13] qq: s has no value
```

```
[14] wei: :(Stream (Polynomial R)) -- should replace by pretend
```



**warnings 20363: WP**

```
>compiling WP.spad to WP.nrlib
```

```
Warnings:
```

- ```
[1] innercoerce:  ans has no value
[2] unknown Functor code (error incompatible length lists in WeightedPolynomial)
```

warnings 20362: DIAGG

```
>compiling DIAGG.spad to DIAGG.nrlib
```

```
Warnings:
```

- ```
[1] select!: x has no value
```

**warnings 20361: DSMP**

```
>compiling DSMP.spad to DSMP.nrlib
```

```
Warnings:
```

- ```
[1] retractIfCan:  x has no value
[2] retractIfCan:  y has no value
[3] coerce:  x has no value
[4] coerce:  y has no value
```

warnings 20360: FACTRN

```
>compiling FACTRN.spad to FACTRN.nrlib
```

```
Warnings:
```

- ```
[1] factor: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE factor ((Factored
[2] norm: not known that (Ring) is of mode (CATEGORY package (SIGNATURE factor ((Factored (SparseUniv
```

**warnings 20359: FFFACTSE**

```
>compiling FFFACTSE.spad to FFFACTSE.nrlib
```

Warnings:

- [1] exp: Q has no value
- [2] internalFactorCZ: result has no value
- [3] internalFactorCZ: listOfFactors has no value
- [4] internalFactorSquareFree: S has no value

## warnings 20358: FRAMALG

>compiling FRAMALG.spad to FRAMALG.nrlib

Warnings:

- [1] coordinates: signature of lhs not unique: (Matrix R)(Vector S) chosen
- [2] characteristicPolynomial: y has no value

## warnings 20357: INTFRSP

>compiling INTFRSP.spad to INTFRSP.nrlib

Warnings:

- [1] sbSpcOfCurve: pretend(Integer) -- should replace by @

## warnings 20356: LPARSPT

>compiling LPARSPT.spad to LPARSPT.nrlib

Warnings:

- [1] valuationAndMore: pretend(Integer) -- should replace by @
- [2] localize: pretend(Integer) -- should replace by @

## warnings 20355: NPOLYGON

>compiling NPOLYGON.spad to NPOLYGON.nrlib

Warnings:

- [1] slope: pretend(Integer) -- should replace by @
- [2] slope: remainder has no value

**warnings 20354: PLOT**

```
>compiling PLOT.spad to PLOT.nrlib
```

```
Warnings:
```

- [1] listBranches: newl has no value
- [2] listBranches: outList has no value
- [3] rangeRefine: c has no value
- [4] rangeRefine: q has no value
- [5] rangeRefine: NUMFUNEVALS has no value
- [6] refine: curves has no value
- [7] plot: curves has no value
- [8] plot: u1 has no value
- [9] plot: z1 has no value
- [10] plot: t has no value
- [11] plotPolar: u1 has no value
- [12] plotPolar: v1 has no value
- [13] coerce: l has no value

**warnings 20353: RFP**

```
>compiling RFP.spad to RFP.nrlib
```

```
Warnings:
```

- [1] distinguishedRootsOf: not known that (Ring) is of mode (CATEGORY package (SIGNATURE distinguishedRootsOf))
- [2] distinguishedRootsOf: pretend(List (Integer)) -- should replace by @
- [3] distinguishedRootsOf: not known that (UnivariatePolynomialCategory (Fraction (Integer))) is of mode (CATEGORY package (SIGNATURE distinguishedRootsOf))

**warnings 20352: ROIRC**

```
>compiling ROIRC.spad to ROIRC.nrlib
```

```
Warnings:
```

- [1] invert: z has no value

**warnings 20351: SMATCAT**

```
>compiling SMATCAT.spad to SMATCAT.nrlib
```

Warnings:

```
[1] equation2R: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE ** (S S (Inte
```

### warnings 20350: AFALGGRO

```
>compiling AFALGGRO.spad to AFALGGRO.nrllib
```

Warnings:

```
[1] affineAlgSetLocal: listPtsId1 has no value
```

### warnings 20349: AFALGRES

```
>compiling AFALGRES.spad to AFALGRES.nrllib
```

Warnings:

```
[1] affineAlgSetLocal: listPtsId1 has no value
```

### warnings 20348: FFINTBAS

```
>compiling FFINTBAS.spad to FFINTBAS.nrllib
```

Warnings:

```
[1] squaredFactors: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATUR
```

### warnings 20347: FRIDEAL

```
>compiling FRIDEAL.spad to FRIDEAL.nrllib
```

Warnings:

```
[1] upmat: s has no value
[2] ret?: s has no value
[3] inv: s has no value
```

### warnings 20346: FRIDEAL2

```
>compiling FRIDEAL2.spad to FRIDEAL2.nrlib
```

```
Warnings:
 [1] map: s has no value
```

## warnings 20345: FRMOD

```
>compiling FRMOD.spad to FRMOD.nrlib
```

```
Warnings:
 [1] vectProd: pretend(Vector A) -- should replace by @
```

## warnings 20344: INTFACT

```
>compiling INTFACT.spad to INTFACT.nrlib
```

```
Warnings:
 [1] PollardSmallFactor: G has no value
```

## warnings 20343: MONOGEN

```
>compiling MONOGEN.spad to MONOGEN.nrlib
```

```
Warnings:
 [1] derivationCoordinates: x has no value
```

## warnings 20342: NFINTBAS

```
>compiling NFINTBAS.spad to NFINTBAS.nrlib
```

```
Warnings:
 [1] wildPrimes: ans has no value
 [2] tameProduct: ans has no value
 [3] integralBasis: runningRbden has no value
 [4] iWildLocalIntegralBasis: rbden has no value
```

**warnings 20341: CCLASS**

```
>compiling CCLASS.spad to CCLASS.nrlib
```

```
Warnings:
```

```
[1] remove!: signature of lhs not unique: $(Character)$ chosen
```

**warnings 20340: CPIMA**

```
>compiling CPIMA.spad to CPIMA.nrlib
```

```
Warnings:
```

```
[1] XtoY: x has no value
```

**warnings 20339: GALFACT**

```
>compiling GALFACT.spad to GALFACT.nrlib
```

```
Warnings:
```

```
[1] eisensteinIrreducible?: not known that (IntegralDomain) is of mode (CATEGORY package (
[2] modularFactor: s has no value
[3] completeFactor: g0 has no value
[4] completeFactor: degg has no value
[5] completeFactor: g has no value
[6] completeFactor: x has no value
[7] completeFactor: ltrue has no value
[8] completeFactor: level has no value
[9] divideSet: l has no value
[10] btwFactor: reverse? has no value
[11] btwFactor: negativelc? has no value
[12] btwFactor: x has no value
[13] btwFactor: lf has no value
[14] henselFact: c has no value
[15] henselFact: factorlist has no value
[16] henselFact: z has no value
[17] btwFact: x has no value
[18] btwFact: c has no value
[19] btwFact: factorlist has no value
[20] btwFact: z has no value
```

## warnings 20338: IALGFACT

```
>compiling IALGFACT.spad to IALGFACT.nrllib
```

```
Warnings:
```

```
[1] factor: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE factor ((Factored
[2] x has no value
```

## warnings 20337: IBACHIN

```
>compiling IBACHIN.spad to IBACHIN.nrllib
```

```
Warnings:
```

```
[1] applyFrobToMatrix: k1 has no value
[2] listConjugateBases: k1 has no value
[3] approximateExtendedEuclidean: r1 has no value
[4] approximateExtendedEuclidean: z1 has no value
```

## warnings 20336: MMLFORM

```
>compiling MMLFORM.spad to MMLFORM.nrllib
```

```
Warnings:
```

```
[1] exprex: s has no value
[2] formatSub: j has no value
[3] formatSub: s has no value
[4] formatSub1: s has no value
[5] formatPlex: checkarg has no value
[6] formatPlex: s has no value
[7] formatNaryNoGroup: checkargs has no value
```

## warnings 20335: NORMMA

```
>compiling NORMMA.spad to NORMMA.nrllib
```

```
Warnings:
```

```
[1] PolR2SUP: x has no value
```

**warnings 20334: ODERED**

```
>compiling ODERED.spad to ODERED.nrlib
```

```
Warnings:
```

- [1] matF2L: f1 has no value
- [2] reduceLODE: f1 has no value

**warnings 20333: PERM**

```
>compiling PERM.spad to PERM.nrlib
```

```
Warnings:
```

- [1] rotateCycle: minpos has no value
- [2] coerceToCycle: nextCycle has no value
- [3] coercePreimagesImages: preImage has no value
- [4] coercePreimagesImages: image has no value

**warnings 20332: PERMGRP**

```
>compiling PERMGRP.spad to PERMGRP.nrlib
```

```
Warnings:
```

- [1] shortenWord: newlw has no value
- [2] orbitInternal: pos has no value
- [3] orbitInternal: orbitList has no value
- [4] bsgs1: j has no value
- [5] bsgs1: out has no value
- [6] bsgs1: outword has no value
- [7] bsgs1: baseOfGroup has no value
- [8] reduceGenerators: wordlist has no value
- [9] bsgs: gpbases has no value
- [10] bsgs: baseOfGroup has no value
- [11] bsgs: wordProblem has no value
- [12] bsgs: wordlist has no value
- [13] bsgs: gporb has no value
- [14] bsgs: noresult has no value
- [15] bsgs: newBasePoint has no value
- [16] bsgs: basePoint has no value
- [17] memberInternal: supp has no value
- [18] memberInternal: sgs has no value
- [19] memberInternal: baseOfGroup has no value
- [20] memberInternal: gporb has no value



```
[21] coerce: :(Symbol) -- should replace by pretend
```

## warnings 20331: PRIMES

```
>compiling PRIMES.spad to PRIMES.nrlib
```

```
Warnings:
```

```
[1] rabinProvesComposite: rootsMinus1 has no value
```

## warnings 20330: PRJALGPK

```
>compiling PRJALGPK.spad to PRJALGPK.nrlib
```

```
Warnings:
```

```
[1] singularPoints: listPtsIdl2 has no value
```

```
[2] algebraicSet: listPtsIdl2 has no value
```

```
[3] rationalPoints: listPtsIdl has no value
```

## warnings 20329: PWFFINTB

```
>compiling PWFFINTB.spad to PWFFINTB.nrlib
```

```
Warnings:
```

```
[1] reducedDiscriminant: r1 has no value
```

```
[2] compLocalBasisOverExt: k1 has no value
```

```
[3] listSquaredFactors: ans has no value
```

## warnings 20328: SAE

```
>compiling SAE.spad to SAE.nrlib
```

```
Warnings:
```

```
[1] lift: pretendRep -- should replace by @
```

```
[2] coordinates: y has no value
```

```
[3] index: ans has no value
```

```
[4] unknown Functor code (error Modulus cannot be made monic)
```

**warnings 20327: SGCF**

```
>compiling SGCF.spad to SGCF.nrlib
```

```
Warnings:
```

```
[1] unrankImproperPartitions0: 1 has no value
[2] unrankImproperPartitions1: nonZeros has no value
[3] subSet: 1 has no value
[4] nextLatticePermutation: ready has no value
[5] listYoungTableaus: lattice has no value
```

**warnings 20326: SPACE3**

```
>compiling SPACE3.spad to SPACE3.nrlib
```

```
Warnings:
```

```
[1] coerce: llliPt has no value
```

**warnings 200325: TBAGG**

```
>compiling TBAGG.spad to TBAGG.nrlib
```

```
Warnings:
```

```
[1] map!: signature of lhs not unique: S(Mapping Entry Entry)S chosen
```

**warnings 20324: VIEW3D**

```
>compiling VIEW3D.spad to VIEW3D.nrlib
```

```
Warnings:
```

```
[1] viewpoint: Theta has no value
```

**warnings 20323: WFFINTBS**

```
>compiling WFFINTBS.spad to WFFINTBS.nrlib
```

```
Warnings:
```

```
[1] listSquaredFactors: ans has no value
[2] iLocalIntegralBasis: rbdn has no value
[3] integralBasis: runningRbdn has no value
```

## warnings 20322: ALIST

```
>compiling ALIST.spad to ALIST.nrlib
```

```
Warnings:
[1] latex: s has no value
[2] remove!: key has no value
```

## warnings 20321: HASHTBL

```
>compiling HASHTBL.spad to HASHTBL.nrlib
```

```
Warnings:
[1] remove!: pretendEntry -- should replace by @
[2] search: pretendEntry -- should replace by @
```

## warnings 20320: INTPACK

```
>compiling INTPACK.spad to INTPACK.nrlib
```

```
Warnings:
[1] integrateConstantList: x has no value
[2] integrateConstantList: y has no value
[3] preAnalysis: nia has no value
[4] measureSpecific: nia has no value
[5] measureSpecific: mdnia has no value
[6] integrateSpecific: nia has no value
[7] integrateSpecific: mdnia has no value
[8] recoverAfterFail: nia has no value
[9] integrateArgs: mdnia has no value
[10] integrateArgs: nia has no value
```

## warnings 20319: IPF

```
>compiling IPF.spad to IPF.nrllib
```

```
Warnings:
```

```
[1] tableForDiscreteLogarithm: pretend(Table (PositiveInteger) (NonNegativeInteger)) -- sh
[2] initializeLog: n has no value
[3] coordinates: signature of lhs not unique: (Vector $)$ chosen
[4] charthRoot: signature of lhs not unique: $$ chosen
```

## warnings 20318: ACF

```
>compiling ACF.spad to ACF.nrllib
```

```
Warnings:
```

```
[1] zeroOf: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE zerosOf ((List S)
```

## warnings 20317: ACPLLOT

```
>compiling ACPLLOT.spad to ACPLLOT.nrllib
```

```
Warnings:
```

```
[1] listPtsOnHorizBdry: pointList has no value
[2] listPtsOnVertBdry: pointList has no value
[3] listPtsInRect: pointList has no value
[4] newX: x has no value
[5] newY: x has no value
[6] makeOneVarSketch: sketchRoots has no value
[7] makeOneVarSketch: lf has no value
[8] makeOneVarSketch: rt has no value
[9] makeOneVarSketch: bt has no value
[10] makeOneVarSketch: tp has no value
[11] makeLineSketch: branch has no value
[12] makeLineSketch: lf has no value
[13] makeLineSketch: rt has no value
[14] makeLineSketch: bt has no value
[15] makeLineSketch: tp has no value
[16] makeRatFcnSketch: s has no value
[17] makeRatFcnSketch: outList has no value
[18] makeRatFcnSketch: lf has no value
[19] makeRatFcnSketch: rt has no value
[20] makeRatFcnSketch: bt has no value
[21] makeRatFcnSketch: tp has no value
[22] makeRatFcnSketch: topList has no value
[23] makeRatFcnSketch: botList has no value
[24] makeGeneralSketch: s has no value
```

```
[25] listPtsOnLoop: pointList has no value
[26] computeNextPt: xPointList has no value
[27] computeNextPt: yPointList has no value
```

## warnings 20316: ANTISYM

```
>compiling ANTISYM.spad to ANTISYM.nrllib
```

Warnings:

```
[1] retractable?: k has no value
[2] retractIfCan: k has no value
[3] *: k has no value
[4] *: c has no value
[5] *: z has no value
[6] coerce: c has no value
[7] coerce: k has no value
```

## warnings 20315: ASP12

```
>compiling ASP12.spad to ASP12.nrllib
```

Warnings:

```
[1] MAXIT has no value
[2] IFLAG has no value
[3] ELAM has no value
[4] FINFO has no value
```

## warnings 20314: ASP27

```
>compiling ASP27.spad to ASP27.nrllib
```

Warnings:

```
[1] IFLAG has no value
[2] N has no value
[3] LRWORK has no value
[4] LIWORK has no value
[5] Z has no value
[6] W has no value
[7] RWORK has no value
[8] IWORK has no value
```

**warnings 20313: ASP28**

```
>compiling ASP28.spad to ASP28.nrlib
```

```
Warnings:
```

- [1] IFLAG has no value
- [2] N has no value
- [3] LRWORK has no value
- [4] LIWORK has no value
- [5] Z has no value
- [6] W has no value
- [7] RWORK has no value
- [8] IWORK has no value

**warnings 20312: ASP30**

```
>compiling ASP30.spad to ASP30.nrlib
```

```
Warnings:
```

- [1] MODE has no value
- [2] M has no value
- [3] N has no value
- [4] LRWORK has no value
- [5] LIWORK has no value
- [6] X has no value
- [7] Y has no value
- [8] RWORK has no value
- [9] IWORK has no value
- [10] IFAIL has no value
- [11] coerce: A has no value

**warnings 20311: ASP33**

```
>compiling ASP33.spad to ASP33.nrlib
```

```
Warnings:
```

- [1] JINT has no value
- [2] X has no value
- [3] V has no value

**warnings 20310: ASP49**

```
>compiling ASP49.spad to ASP49.nrlib
```

```
Warnings:
```

- [1] MODE has no value
- [2] N has no value
- [3] X has no value
- [4] OBJF has no value
- [5] OBJGRD has no value
- [6] NSTATE has no value
- [7] IUSER has no value
- [8] USER has no value

**warnings 20309: ASP55**

```
>compiling ASP55.spad to ASP55.nrlib
```

```
Warnings:
```

- [1] MODE has no value
- [2] NCNLN has no value
- [3] N has no value
- [4] NROWJ has no value
- [5] NEEDC has no value
- [6] X has no value
- [7] C has no value
- [8] CJAC has no value
- [9] NSTATE has no value
- [10] IUSER has no value
- [11] USER has no value
- [12] makeCond: NEEDC has no value

**warnings 20308: ASP7**

```
>compiling ASP7.spad to ASP7.nrlib
```

```
Warnings:
```

- [1] X has no value
- [2] Y has no value
- [3] F has no value
- [4] coerce: F has no value

**warnings 20307: ASP78**

```
>compiling ASP78.spad to ASP78.nrlib
```

```
Warnings:
[1] X has no value
[2] G has no value
[3] coerce: G has no value
```

**warnings 20306: ASP8**

```
>compiling ASP8.spad to ASP8.nrlib
```

```
Warnings:
[1] COUNT has no value
[2] XSOL has no value
[3] N has no value
[4] Y has no value
[5] FORWRD has no value
[6] RESULT has no value
[7] M has no value
[8] coerce: POINTS has no value
[9] coerce: X02ALF has no value
[10] coerce: RESULT has no value
[11] coerce: COUNT has no value
[12] coerce: XSOL has no value
```

**warnings 20305: ASP9**

```
>compiling ASP9.spad to ASP9.nrlib
```

```
Warnings:
[1] X has no value
[2] Y has no value
[3] coerce: CHDUM1 has no value
[4] coerce: CD02EJ has no value
[5] coerce: AD02CJ has no value
[6] coerce: GOPT1 has no value
[7] coerce: GOPT2 has no value
```



## warnings 20304: BLUPPACK

```
>compiling BLUPPACK.spad to BLUPPACK.nrlib
```

```
Warnings:
```

- [1] stepBlowUp: not known that (FiniteAbelianMonoidRing K (DirectProduct (call LENGTH (LIST (QUOTE X)
- [2] stepBlowUp: listRec has no value
- [3] blowExp: pretend(Integer) -- should replace by @

## warnings 20303: BOP

```
>compiling BOP.spad to BOP.nrlib
```

```
Warnings:
```

- [1] display: x1 has no value

## warnings 20302: BOP1

```
>compiling BOP1.spad to BOP1.nrlib
```

```
Warnings:
```

- [1] evaluate: ll has no value
- [2] derivative: ll has no value
- [3] constOp: ll has no value
- [4] constantOperator: ll has no value

## warnings 20301: COMPCAT

```
>compiling COMPCAT.spad to COMPCAT.nrlib
```

```
Warnings:
```

- [1] solveLinearPolynomialEquation: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE differentiate (S S))
- [2] normPolynomial: z has no value
- [3] factorPolynomial: not known that (IntegralDomain) is of mode (CATEGORY domain (SIGNATURE differentiate (S S))
- [4] factorPolynomial: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE differentiate (S S))
- [5] factorSquareFreePolynomial: z has no value
- [6] reducedSystem: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE differentiate (S S))
- [7] patternMatch: not known that (SetCategory) is of mode (CATEGORY domain (SIGNATURE differentiate (S S))

**warnings 20300: DRAWCFUN**

```
>compiling DRAWCFUN.spad to DRAWCFUN.nrlib
```

```
Warnings:
```

```
[1] drawToScaleRanges: x has no value
[2] drawPlot: brans has no value
[3] drawPlot: xValues has no value
[4] drawPlot: yValues has no value
[5] drawPlot: x has no value
[6] draw: x has no value
[7] draw: fcn has no value
[8] makeObject: x has no value
[9] makeObject: fcn has no value
[10] makeObject: pointsColored? has no value
[11] recolor: f1 has no value
[12] recolor: f2 has no value
[13] makeObject: y has no value
[14] makeObject: col2 has no value
[15] makeObject: col3 has no value
```

**warnings 20299: D01ANFA**

```
>compiling D01ANFA.spad to D01ANFA.nrlib
```

```
Warnings:
```

```
[1] numericalIntegration: x has no value
```

**warnings 20298: D01ASFA**

```
>compiling D01ASFA.spad to D01ASFA.nrlib
```

```
Warnings:
```

```
[1] numericalIntegration: x has no value
```

**warnings 20297: EP**

```
>compiling EP.spad to EP.nrlib
```

```
Warnings:
```

```

[1] not known that (SIGNATURE variable ((Union $ failed) (Symbol))) is of mode (CATEGORY domain (SIGN
[2] charpol: :(PositiveInteger) -- should replace by pretend
[3] eigenvalues: lrat has no value
[4] eigenvalues: lsym has no value
[5] intAlgEig: not known that (Ring) is of mode (CATEGORY package (SIGNATURE characteristicPolynomial
[6] eigenvectors: ratSol has no value
[7] eigenvectors: algSol has no value

```

### warnings 20296: E04AGNT

```
>compiling E04AGNT.spad to E04AGNT.nrlib
```

Warnings:

```

[1] optAttributes: noa has no value
[2] optAttributes: lsa has no value
[3] optAttributes: s has no value

```

### warnings 20295: FEXPR

```
>compiling FEXPR.spad to FEXPR.nrlib
```

Warnings:

```

[1] checkSymbols: IN has no value
[2] checkForNagOperators: X01AAF has no value
[3] pi: X01AAF has no value

```

### warnings 20294: FFCAT

```
>compiling FFCAT.spad to FFCAT.nrlib
```

Warnings:

```

[1] rationalPoint?: not known that (IntegralDomain) is of mode (CATEGORY domain (SIGNATURE differenti
[2] UP2P: s has no value
[3] UPUP2P: s has no value
[4] nonSingularModel: s has no value
[5] primitivePart: STEP has no value
[6] primitivePart: i has no value
[7] integralAtInfinity?: s has no value
[8] repOrder: nostart has no value
[9] kmin: nostart has no value

```

```
[10] kmin: k has no value
[11] integral?: den has no value
[12] differentiate: x has no value
```

### warnings 20293: FFCGP

```
>compiling FFCGP.spad to FFCGP.nrllib
```

```
Warnings:
[1] pretend(NonNegativeInteger) -- should replace by @
[2] +: pretend(SingleInteger) -- should replace by @
[3] +: pretendRep -- should replace by @
[4] retractIfCan: pretendRep -- should replace by @
[5] inGroundField?: pretendRep -- should replace by @
[6] -: pretendRep -- should replace by @
[7] extensionDegree: signature of lhs not unique: (PositiveInteger) chosen
[8] unknown Functor code (error field too large for this representation)
```

### warnings 20292: FFNBP

```
>compiling FFNBP.spad to FFNBP.nrllib
```

```
Warnings:
[1] degree: signature of lhs not unique: (PositiveInteger)$ chosen
[2] coerce: 1 has no value
[3] initializeLog: n has no value
[4] extensionDegree: signature of lhs not unique: (PositiveInteger) chosen
```

### warnings 20291: FFP

```
>compiling FFP.spad to FFP.nrllib
```

```
Warnings:
[1] degree: signature of lhs not unique: (PositiveInteger)$ chosen
[2] tableForDiscreteLogarithm: pretend(Table (PositiveInteger) (NonNegativeInteger)) -- sh
[3] initializeLog: n has no value
[4] extensionDegree: signature of lhs not unique: (PositiveInteger) chosen
```

**warnings 20290: FLOAT**

```
>compiling FLOAT.spad to FLOAT.nrlib
```

```
Warnings:
```

- [1] OMwrite: pretend(String) -- should replace by @
- [2] cos: s has no value
- [3] exp: e1 has no value
- [4] exp1: E has no value
- [5] normalize: e has no value
- [6] power: y has no value
- [7] power10: y has no value

**warnings 20289: FPARFRAC**

```
>compiling FPARFRAC.spad to FPARFRAC.nrlib
```

```
Warnings:
```

- [1] UP2SUP: z1 has no value
- [2] fullPartialFraction: not known that (IntegralDomain) is of mode (CATEGORY domain (SIGNATURE + (\$ I
- [3] fullParFrac: ans has no value

**warnings 20288: FR**

```
>compiling FR.spad to FR.nrlib
```

```
Warnings:
```

- [1] zero?: fctr has no value
- [2] eval: IN has no value
- [3] eval: e has no value
- [4] convert: fctr has no value
- [5] convert: xpnt has no value
- [6] exquo: associate has no value
- [7] exquo: goodQuotient has no value

**warnings 20287: FRNAALG**

```
>compiling FRNAALG.spad to FRNAALG.nrlib
```

Warnings:

- [1] leftRankPolynomial: xx has no value
- [2] rightRankPolynomial: xx has no value
- [3] coordinates: signature of lhs not unique: (Matrix R)(Vector S) chosen

## warnings 20286: FS

>compiling FS.spad to FS.nrlib

Warnings:

- [1] coerce: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE \* (S (Positi
- [2] subs: IN has no value
- [3] subs: x has no value
- [4] variables: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE \* (S (Po
- [5] eval: y has no value
- [6] eval: IN has no value
- [7] eval: f has no value
- [8] smprep: IN has no value
- [9] smprep: a has no value
- [10] smprep: y has no value
- [11] smprep: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE \* (S (PositiveIn
- [12] diffdiff0: ans has no value
- [13] diffEval: k has no value
- [14] opderiv: x has no value
- [15] opderiv: y has no value
- [16] smpderiv: s has no value
- [17] coerce: s has no value
- [18] coerce: r has no value
- [19] smp20: x has no value
- [20] smpsubst: x has no value
- [21] smpsubst: s has no value
- [22] smpsubst: z has no value
- [23] smpsubst: y has no value
- [24] smpeval: x has no value
- [25] smpeval: s has no value
- [26] smpeval: z has no value
- [27] smpeval: y has no value
- [28] smpunq: x has no value
- [29] smpunq: y has no value
- [30] smpret: x has no value
- [31] smpret: y has no value
- [32] univariate: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE \* (S (Positi
- [33] convert: not known that (IntegralDomain) is of mode (CATEGORY domain (SIGNATURE \* (S
- [34] coerce: x has no value
- [35] coerce: y has no value

**warnings 20285: FST**

```
>compiling FST.spad to FST.nrlib
```

```
Warnings:
```

- [1] coerce: REAL has no value
- [2] coerce: INTEGER has no value
- [3] coerce: COMPLEX has no value
- [4] coerce: CHARACTER has no value
- [5] coerce: LOGICAL has no value
- [6] coerce: real has no value
- [7] coerce: integer has no value
- [8] coerce: complex has no value
- [9] coerce: character has no value
- [10] coerce: logical has no value
- [11] coerce: upperDoubleCOMplexSymbol has no value

**warnings 20284: GDMP**

```
>compiling GDMP.spad to GDMP.nrlib
```

```
Warnings:
```

- [1] zero?: :Rep -- should replace by pretend
- [2] listCoef: :Rep -- should replace by pretend
- [3] numberOfMonomials: :Rep -- should replace by pretend
- [4] monomial?: :Rep -- should replace by pretend
- [5] maxNorm: m has no value
- [6] variables: :(PositiveInteger) -- should replace by pretend
- [7] reorder: z2 has no value
- [8] reorder: z1 has no value
- [9] univariate: deg has no value
- [10] evalSortedVarlist: x has no value
- [11] eval: y has no value
- [12] eval: x has no value
- [13] coerce: l has no value

**warnings 20283: HACKPI**

```
>compiling HACKPI.spad to HACKPI.nrlib
```

```
Warnings:
```

- [1] pi: pretend\$ -- should replace by @
- [2] p2sf: x has no value

```
[3] p2f: x has no value
```

### warnings 20282: IDEAL

```
>compiling IDEAL.spad to IDEAL.nrllib
```

```
Warnings:
```

```
[1] not known that (OrderedAbelianMonoidSup) is of mode (CATEGORY domain (IF (has (NonNegat
[2] leadterm: not known that (Ring) is of mode (CATEGORY package (SIGNATURE * ($ $ $)) (SI
[3] contractGrob: not known that (PolynomialCategory F newExpon VarSet) is of mode (CATEGO
[4] monomDim: b has no value
[5] monomDim: a has no value
[6] monomDim: monvar has no value
[7] **: :(NonNegativeInteger) -- should replace by pretend
[8] relationsIdeal: not known that (OrderedAbelianMonoid) is of mode (CATEGORY domain (IF
[9] relationsIdeal: not known that (OrderedAbelianMonoidSup) is of mode (CATEGORY domain (
[10] relationsIdeal: not known that (PolynomialCategory F nExponent VarSet1) is of mode (C
[11] relationsIdeal: :(Vector (NonNegativeInteger)) -- should replace by pretend
[12] relationsIdeal: The conditional modes (Polynomial F) and nPoly conflict
```

### warnings 20281: INFORM

```
>compiling INFORM.spad to INFORM.nrllib
```

```
Warnings:
```

```
[1] conv: pretend$ -- should replace by @
```

### warnings 20280: IR

```
>compiling IR.spad to IR.nrllib
```

```
Warnings:
```

```
[1] nesimp: IN has no value
[2] nesimp: ne has no value
[3] *: x1 has no value
[4] *: x2 has no value
[5] differentiate: x1 has no value
```



## warnings 20279: ISUPS

```
>compiling ISUPS.spad to ISUPS.nrllib
```

Warnings:

```
[1] map: y has no value
[2] map: z has no value
[3] differentiate: n has no value
[4] differentiate: y has no value
[5] differentiate: z has no value
[6] multiplyCoefficients: n has no value
[7] multiplyCoefficients: y has no value
[8] multiplyCoefficients: z has no value
[9] multiplyExponents: y has no value
[10] multiplyExponents: z has no value
[11] iPlus1: x1 has no value
[12] iPlus1: z has no value
[13] iPlus2: z has no value
[14] +: xi has no value
[15] +: yi has no value
[16] -: xi has no value
[17] -: yi has no value
[18] -: x has no value
[19] -: z has no value
[20] *: z has no value
[21] productByTerm: y has no value
[22] productByTerm: z has no value
[23] iTimes: a has no value
[24] iTimes: b has no value
[25] iDivide: a has no value
[26] iDivide: b has no value
[27] iExquo: y1 has no value
[28] iExquo: z has no value
[29] taylorQuoByVar: y has no value
[30] taylorQuoByVar: z has no value
[31] compose0: s has no value
[32] compose0: m has no value
[33] compose0: a has no value
[34] compose0: b has no value
[35] integrate: n has no value
[36] integrate: y has no value
[37] integrate: z has no value
[38] cPower: s has no value
[39] iExp: s has no value
[40] iSincos: s has no value
[41] tan0: a has no value
[42] tan0: b has no value
[43] iTan: s has no value
[44] cSin: %sin has no value
```

```

[45] cCos: %cos has no value
[46] sinhCosh: %sin has no value
[47] sinhCosh: %cos has no value
[48] cSinh: %sinh has no value
[49] cCosh: %cosh has no value

```

### warnings 20278: LMDICT

```
>compiling LMDICT.spad to LMDICT.nrlib
```

```

Warnings:
[1] substitute: z1 has no value
[2] select!: z has no value

```

### warnings 20277: LODOOPS

```
>compiling LODOOPS.spad to LODOOPS.nrlib
```

```

Warnings:
[1] nonTrivial?: x1 has no value
[2] killer: not known that (Ring) is of mode (CATEGORY package (SIGNATURE symmetricProduct

```

### warnings 20276: MATRIX

```
>compiling MATRIX.spad to MATRIX.nrlib
```

```

Warnings:
[1] positivePower: pretend(Matrix R) -- should replace by @
[2] determinant: not known that (CommutativeRing) is of mode (CATEGORY R (ATTRIBUTE (commu

```

### warnings 20275: MKFLCFN

```
>compiling MKFLCFN.spad to MKFLCFN.nrlib
```

```

Warnings:
[1] mkLisp: pretend(InputForm) -- should replace by @

```

**warnings 20274: MSET**

```
>compiling MSET.spad to MSET.nrlib
```

```
Warnings:
```

- [1] select!: s1 has no value
- [2] intersect: n has no value
- [3] difference: n has no value

**warnings 20273: M3D**

```
>compiling M3D.spad to M3D.nrlib
```

```
Warnings:
```

- [1] matrixConcat3D: retVal has no value

**warnings 20272: NAGC02**

```
>compiling NAGC02.spad to NAGC02.nrlib
```

```
Warnings:
```

- [1] c02aff: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [2] c02agf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @

**warnings 20271: NAGC05**

```
>compiling NAGC05.spad to NAGC05.nrlib
```

```
Warnings:
```

- [1] c05adf: fn has no value
- [2] c05adf: fp has no value
- [3] c05adf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [4] c05nbf: fn has no value
- [5] c05nbf: fp has no value
- [6] c05nbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [7] c05pbf: fn has no value
- [8] c05pbf: fp has no value
- [9] c05pbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @

**warnings 20270: NAGC06**

```
>compiling NAGC06.spad to NAGC06.nrlib
```

```
Warnings:
```

```
[1] c06eaf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[2] c06ebf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[3] c06ecf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[4] c06ekf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[5] c06fpf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[6] c06fqf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[7] c06frf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[8] c06fuf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[9] c06gbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[10] c06gcf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[11] c06gqf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[12] c06gsf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
```

**warnings 20269: NAGD01**

```
>compiling NAGD01.spad to NAGD01.nrlib
```

```
Warnings:
```

```
[1] d01ajf: fn has no value
[2] d01ajf: fp has no value
[3] d01ajf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[4] d01akf: fn has no value
[5] d01akf: fp has no value
[6] d01akf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[7] d01alf: fn has no value
[8] d01alf: fp has no value
[9] d01alf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[10] d01amf: fn has no value
[11] d01amf: fp has no value
[12] d01amf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[13] d01anf: fn has no value
[14] d01anf: fp has no value
[15] d01anf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[16] d01apf: fn has no value
[17] d01apf: fp has no value
[18] d01apf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[19] d01aqf: fn has no value
[20] d01aqf: fp has no value
[21] d01aqf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[22] d01asf: fn has no value
[23] d01asf: fp has no value
```

```

[24] d01asf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[25] d01bbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[26] d01fcf: fn has no value
[27] d01fcf: fp has no value
[28] d01fcf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[29] d01gaf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[30] d01gbf: fn has no value
[31] d01gbf: fp has no value
[32] d01gbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @

```

## warnings 20268: NAGD02

>compiling NAGD02.spad to NAGD02.nrlib

Warnings:

```

[1] d02bbf: fn has no value
[2] d02bbf: fp has no value
[3] d02bbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[4] d02bhf: fn has no value
[5] d02bhf: fp has no value
[6] d02bhf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[7] d02cjf: fn has no value
[8] d02cjf: fp has no value
[9] d02cjf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[10] d02ejf: fn has no value
[11] d02ejf: fp has no value
[12] d02ejf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[13] d02gaf: fn has no value
[14] d02gaf: fp has no value
[15] d02gaf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[16] d02gbf: fn has no value
[17] d02gbf: fp has no value
[18] d02gbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[19] d02kef: fn has no value
[20] d02kef: fp has no value
[21] d02kef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[22] d02raf: fn has no value
[23] d02raf: fp has no value
[24] d02raf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @

```

## warnings 20267: NAGD03

>compiling NAGD03.spad to NAGD03.nrlib

Warnings:

```
[1] d03edf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[2] d03eef: fn has no value
[3] d03eef: fp has no value
[4] d03eef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[5] d03faf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
```

## warnings 20266: NAGE01

>compiling NAGE01.spad to NAGE01.nrlib

Warnings:

```
[1] e01baf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[2] e01bef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[3] e01bff: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[4] e01bgf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[5] e01bhf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[6] e01daf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[7] e01saf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[8] e01sbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[9] e01sef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[10] e01sff: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
```

## warnings 20265: NAGE02

>compiling NAGE02.spad to NAGE02.nrlib

Warnings:

```
[1] e02adf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[2] e02aef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[3] e02agf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[4] e02ahf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[5] e02ajf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[6] e02akf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[7] e02baf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[8] e02bbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[9] e02bcf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[10] e02bdf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[11] e02bef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[12] e02daf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[13] e02dcf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[14] e02ddf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[15] e02def: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
```

```
[16] e02dff: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[17] e02gaf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[18] e02zaf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
```

## warnings 20264: NAGE04

>compiling NAGE04.spad to NAGE04.nrlib

Warnings:

```
[1] e04dgm: fn has no value
[2] e04dgm: fp has no value
[3] e04dgm: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[4] e04fdf: fn has no value
[5] e04fdf: fp has no value
[6] e04fdf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[7] e04gcm: fn has no value
[8] e04gcm: fp has no value
[9] e04gcm: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[10] e04jaf: fn has no value
[11] e04jaf: fp has no value
[12] e04jaf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[13] e04mbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[14] e04naf: fn has no value
[15] e04naf: fp has no value
[16] e04naf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[17] e04ucf: fn has no value
[18] e04ucf: fp has no value
[19] e04ucf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[20] e04ycf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
```

## warnings 20263: NAGF07

>compiling NAGF07.spad to NAGF07.nrlib

Warnings:

```
[1] f07adf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[2] f07aef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[3] f07fdf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[4] f07fef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
```

**warnings 20262: NAGS**

```
>compiling NAGS.spad to NAGS.nrllib
```

```
Warnings:
```

```
[1] s01eaf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[2] s13aaf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[3] s13acf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[4] s13adf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[5] s14aaf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[6] s14abf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[7] s14baf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[8] s15adf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[9] s15aef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[10] s17acf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[11] s17adf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[12] s17aef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[13] s17aff: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[14] s17agf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[15] s17ahf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[16] s17ajf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[17] s17akf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[18] s17dcf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[19] s17def: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[20] s17dgf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[21] s17dhf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[22] s17dlf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[23] s18acf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[24] s18adf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[25] s18aef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[26] s18aff: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[27] s18dcf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[28] s18def: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[29] s19aaf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[30] s19abf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[31] s19acf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[32] s19adf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[33] s20acf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[34] s20adf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[35] s21baf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[36] s21bbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[37] s21bcf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[38] s21bdf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
```

**warnings 20261: NAGSP**



```
>compiling NAGSP.spad to NAGSP.nrlib
```

```
Warnings:
```

```
[1] getUniqueId: uniqueId has no value
```

## warnings 20260: NSDPS

```
>compiling NSDPS.spad to NSDPS.nrlib
```

```
Warnings:
```

```
[1] removeZeroes: c has no value
[2] iDiv: c has no value
[3] iDiv: k has no value
[4] sbt: k has no value
[5] sbt: c has no value
[6] shift: k has no value
[7] shift: c has no value
[8] +: k has no value
[9] +: c has no value
[10] multC: k has no value
[11] multC: c has no value
[12] *: k has no value
[13] *: c has no value
```

## warnings 20259: NUMFMT

```
>compiling NUMFMT.spad to NUMFMT.nrlib
```

```
Warnings:
```

```
[1] check: c1 has no value
[2] check: c2 has no value
[3] ScanRoman: nprens has no value
```

## warnings 20258: ODERAT

```
>compiling ODERAT.spad to ODERAT.nrlib
```

```
Warnings:
```

```
[1] ratDsolve1: not known that (Ring) is of mode (CATEGORY package (SIGNATURE ratDsolve ((Record (: p
[2] ratDsolve0: b has no value
[3] candidates: l1 has no value
```

```
[4] infMuLambda: lamb has no value
[5] infMuLambda: lf has no value
```

## warnings 20257: OMERRK

```
>compiling OMERRK.spad to OMERRK.nrllib
```

Warnings:

```
[1] OMParseError?: parseError has no value
[2] OMUnknownCD?: unknownCD has no value
[3] OMUnknownSymbol?: unknownSymbol has no value
[4] OMReadError?: readError has no value
[5] coerce: OMParseError has no value
[6] coerce: OMUnknownCD has no value
[7] coerce: OMUnknownSymbol has no value
[8] coerce: OMReadError has no value
```

## warnings 20256: OPTPACK

```
>compiling OPTPACK.spad to OPTPACK.nrllib
```

Warnings:

```
[1] constant: noa has no value
[2] measureSpecific: noa has no value
[3] measureSpecific: lsa has no value
[4] optimizeSpecific: noa has no value
[5] optimizeSpecific: lsa has no value
[6] goodnessOfFit: lsa has no value
[7] goodnessOfFit: method has no value
[8] goodnessOfFit: nameOfRoutine has no value
[9] goodnessOfFit: w has no value
[10] goodnessOfFit: objf has no value
```

## warnings 20255: OSI

```
>compiling OSI.spad to OSI.nrllib
```

Warnings:

```
[1] value: :Rep -- should replace by pretend
```

## warnings 20254: PACOFF

>compiling PACOFF.spad to PACOFF.nrlib

Warnings:

```
[1] repPolynomial: pretend(Record (: recEl (SparseUnivariatePolynomial $)) (: recTower (SparseUnivariatePolynomial $)))
[2] vectorise: pretend(Record (: recEl (SparseUnivariatePolynomial $)) (: recTower (SparseUnivariatePolynomial $)))
[3] localRandom: recTower has no value
[4] +: recEl has no value
[5] *: recEl has no value
[6] *: recTower has no value
[7] -: pretend(Record (: recEl (SparseUnivariatePolynomial $)) (: recTower (SparseUnivariatePolynomial $)))
[8] coerce: pretend(Record (: recEl (SparseUnivariatePolynomial $)) (: recTower (SparseUnivariatePolynomial $)))
[9] fullOutput: pretend(Record (: recEl (SparseUnivariatePolynomial $)) (: recTower (SparseUnivariatePolynomial $)))
[10] definingPolynomial: pretend(Record (: recEl (SparseUnivariatePolynomial $)) (: recTower (SparseUnivariatePolynomial $)))
[11] extDegree: pretend(Record (: recEl (SparseUnivariatePolynomial $)) (: recTower (SparseUnivariatePolynomial $)))
[12] previousTower: pretend(Record (: recEl (SparseUnivariatePolynomial $)) (: recTower (SparseUnivariatePolynomial $)))
[13] name: pretend(Record (: recEl (SparseUnivariatePolynomial $)) (: recTower (SparseUnivariatePolynomial $)))
[14] down: pretend(Record (: recEl (SparseUnivariatePolynomial $)) (: recTower (SparseUnivariatePolynomial $)))
```

## warnings 20253: PACRAT

>compiling PACRAT.spad to PACRAT.nrlib

Warnings:

```
[1] down: pretend(Record (: recEl (SparseUnivariatePolynomial $)) (: recTower (SparseUnivariatePolynomial $)))
[2] lift: pretend(Record (: recEl (SparseUnivariatePolynomial $)) (: recTower (SparseUnivariatePolynomial $)))
[3] reduce: pretend(Record (: recEl (SparseUnivariatePolynomial $)) (: recTower (SparseUnivariatePolynomial $)))
[4] vectorise: pretend(Record (: recEl (SparseUnivariatePolynomial $)) (: recTower (SparseUnivariatePolynomial $)))
[5] +: recEl has no value
[6] *: recEl has no value
[7] *: recTower has no value
[8] inv: pretend(Record (: recEl (SparseUnivariatePolynomial $)) (: recTower (SparseUnivariatePolynomial $)))
[9] -: pretend(Record (: recEl (SparseUnivariatePolynomial $)) (: recTower (SparseUnivariatePolynomial $)))
[10] coerce: pretend(Record (: recEl (SparseUnivariatePolynomial $)) (: recTower (SparseUnivariatePolynomial $)))
[11] fullOutput: pretend(Record (: recEl (SparseUnivariatePolynomial $)) (: recTower (SparseUnivariatePolynomial $)))
[12] definingPolynomial: pretend(Record (: recEl (SparseUnivariatePolynomial $)) (: recTower (SparseUnivariatePolynomial $)))
[13] extDegree: pretend(Record (: recEl (SparseUnivariatePolynomial $)) (: recTower (SparseUnivariatePolynomial $)))
[14] previousTower: pretend(Record (: recEl (SparseUnivariatePolynomial $)) (: recTower (SparseUnivariatePolynomial $)))
[15] name: pretend(Record (: recEl (SparseUnivariatePolynomial $)) (: recTower (SparseUnivariatePolynomial $)))
```

## warnings 20252: PATTERN

```
>compiling PATTERN.spad to PATTERN.nrlib
```

```
Warnings:
```

```
[1] quoted?: sym has no value
[2] generic?: sym has no value
[3] multiple?: sym has no value
[4] optional?: sym has no value
[5] =: pretend(Boolean) -- should replace by @
[6] isOp: ker has no value
[7] variables: exp has no value
[8] variables: qot has no value
[9] variables: ker has no value
[10] PAT20: ret has no value
[11] PAT20: sym has no value
[12] PAT20: exp has no value
[13] PAT20: qot has no value
[14] PAT20: ker has no value
[15] patcopy: ret has no value
[16] patcopy: sym has no value
[17] patcopy: ker has no value
[18] patcopy: qot has no value
[19] patcopy: exp has no value
[20] pateq?: ret has no value
[21] pateq?: qot has no value
[22] pateq?: sym has no value
[23] pateq?: ker has no value
[24] pateq?: exp has no value
```

## warnings 20251: PLCS

```
>compiling PLCS.spad to PLCS.nrlib
```

```
Warnings:
```

```
[1] ActualSetOfPlacesName has no value
[2] =: The conditional modes (String) and $ conflict
[3] setFoundPlacesToEmpty: ActualSetOfPlacesName has no value
[4] create: SIMPLE has no value
```

## warnings 20250: PMKERNEL

```
>compiling PMKERNEL.spad to PMKERNEL.nrlib
```

```
Warnings:
```

```
[1] patternMatchArg: not known that (SetCategory) is of mode (CATEGORY package (SIGNATURE
```

```
[2] patternMatchInner: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE patternMat
```

## warnings 20249: PRIMELT

```
>compiling PRIMELT.spad to PRIMELT.nrlib
```

```
Warnings:
```

```
[1] multi: f1 has no value
[2] incl?: s1 has no value
[3] triangularLinearIfCan: not known that (Ring) is of mode (CATEGORY package (SIGNATURE primitiveEle
[4] triangularLinearIfCan: z1 has no value
[5] triangularLinearIfCan: f1 has no value
```

## warnings 20248: QALGSET2

```
>compiling QALGSET2.spad to QALGSET2.nrlib
```

```
Warnings:
```

```
[1] npoly: z1 has no value
[2] npoly: z2 has no value
[3] npoly: not known that (PolynomialCategory (Fraction (Integer)) (DirectProduct nv (NonNegativeInte
[4] oldpoly: z1 has no value
[5] oldpoly: z2 has no value
[6] radicalSimplify: not known that (PolynomialCategory (Fraction (Integer)) (DirectProduct nv (NonNeg
[7] radicalSimplify: gb has no value
```

## warnings 20247: RECLOS

```
>compiling RECLOS.spad to RECLOS.nrlib
```

```
Warnings:
```

```
[1] *: z has no value
```

## warnings 20246: REP1

```
>compiling REP1.spad to REP1.nrlib
```

```
Warnings:
```

```
[1] antisymmetricTensors: not known that (Ring) is of mode (CATEGORY R (ATTRIBUTE (commuta
```

### warnings 20245: RFFACT

```
>compiling RFFACT.spad to RFFACT.nrlib
```

```
Warnings:
```

```
[1] likuniv: y has no value
[2] likuniv: not known that (Ring) is of mode (CATEGORY package (SIGNATURE factor ((Factor
[3] factor: x has no value
```

### warnings 20244: RPOLCAT

```
>compiling RPOLCAT.spad to RPOLCAT.nrlib
```

```
Warnings:
```

```
[1] mainCoefficients: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE coerce
[2] convert: stpol has no value
```

### warnings 20243: SWITCH

```
>compiling SWITCH.spad to SWITCH.nrlib
```

```
Warnings:
```

```
[1] NULL has no value
```

### warnings 20242: SYMTAB

```
>compiling SYMTAB.spad to SYMTAB.nrlib
```

```
Warnings:
```

```
[1] typeList: fst has no value
[2] typeList2: fst has no value
[3] newTypeLists: fst has no value
[4] oForm2: S has no value
[5] oForm2: P has no value
[6] oForm: bounds has no value
```

## warnings 20241: SYSSOLP

```
>compiling SYSSOLP.spad to SYSSOLP.nrllib
```

```
Warnings:
```

- [1] makeEq: not known that (Ring) is of mode (CATEGORY package (SIGNATURE solve ((List (List (Equation
- [2] triangularSystems: not known that (SIGNATURE variable ((Union \$ failed) (Symbol))) is of mode (CA
- [3] triangularSystems: not known that (PolynomialCategory (Polynomial R) DP OV) is of mode (CATEGORY d
- [4] triangularSystems: not known that (PolynomialCategory (Polynomial R) (DirectProduct (# lv) (NonNeg
- [5] triangularSystems: not known that (PolynomialCategory (Polynomial R) (DirectProduct (call LENGTH

## warnings 20240: UTSCAT

```
>compiling UTSCAT.spad to UTSCAT.nrllib
```

```
Warnings:
```

- [1] coerce: 1 has no value

## warnings 20239: ACFS

```
>compiling ACFS.spad to ACFS.nrllib
```

```
Warnings:
```

- [1] zeroOf: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE zerosOf ((List S) S (Symbol))
- [2] zeroOf: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE zerosOf ((List S) S (S
- [3] not known that (Ring) is of mode (CATEGORY domain (SIGNATURE zerosOf ((List S) S (Symbol))) (SIGN

## warnings 20238: AF

```
>compiling AF.spad to AF.nrllib
```

```
Warnings:
```

- [1] rootOf: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE rootOf (F (SparseUnivari
- [2] rootOf: not known that (Ring) is of mode (CATEGORY package (SIGNATURE rootOf (F (SparseUnivariatel
- [3] dvalg: s has no value
- [4] inrootof: not known that (Ring) is of mode (CATEGORY package (SIGNATURE rootOf (F (SparseUnivariat
- [5] hackroot: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE iroot (F R (Integer))
- [6] inroot0: rn has no value
- [7] inroot0: coef has no value

```

[8] inroot0: radicand has no value
[9] inroot0: exponent has no value
[10] inroot0: rd has no value

```

## warnings 20237: ALGFACT

```
>compiling ALGFACT.spad to ALGFACT.nrllib
```

Warnings:

```

[1] allk: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE factor ((Fact
[2] liftpoly: x has no value
[3] downpoly: x has no value
[4] extend: not known that (Ring) is of mode (CATEGORY package (SIGNATURE factor ((Factore
[5] fact: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE factor ((Fac
[6] fact: x has no value
[7] fact: not known that (SIGNATURE coerce ($ (SparseMultivariatePolynomial (Integer) (Ker
[8] fact: not known that (SIGNATURE numer ((SparseMultivariatePolynomial (Integer) (Kernel
[9] fact: not known that (SIGNATURE denom ((SparseMultivariatePolynomial (Integer) (Kernel
[10] fact: y has no value

```

## warnings 20236: ALGFF

```
>compiling ALGFF.spad to ALGFF.nrllib
```

Warnings:

```
[1] getInfBasis: s has no value
```

## warnings 20235: ALGMANIP

```
>compiling ALGMANIP.spad to ALGMANIP.nrllib
```

Warnings:

```

[1] ratDenom: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE rootSplit
[2] algekernels: z1 has no value
[3] rootkernels: z1 has no value
[4] ratPoly: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE rootSplit
[5] ratPoly: not known that (SIGNATURE coerce ($ (SparseMultivariatePolynomial R (Kernel F
[6] ratPoly: not known that (SIGNATURE numer ((SparseMultivariatePolynomial R (Kernel F))
[7] ratPoly: not known that (SIGNATURE denom ((SparseMultivariatePolynomial R (Kernel F))
[8] innerRF: z2 has no value
[9] innerRF: z1 has no value

```



```

[10] innerRF: z3 has no value
[11] innerRF: z4 has no value
[12] innerRF: not known that (Ring) is of mode (CATEGORY package (SIGNATURE rootSplit (F F)) (SIGNATURE
[13] innerRF: z5 has no value
[14] innerRF: z6 has no value

```

## warnings 20234: ALGMFACT

```
>compiling ALGMFACT.spad to ALGMFACT.nrlib
```

```

Warnings:
 [1] factor: z1 has no value

```

## warnings 20233: ALGSC

```
>compiling ALGSC.spad to ALGSC.nrlib
```

```

Warnings:
 [1] elt: :Rep -- should replace by pretend
 [2] coerce: le has no value

```

## warnings 20232: APPRULE

```
>compiling APPRULE.spad to APPRULE.nrlib
```

```

Warnings:
 [1] splitRules: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE applyRules (F (List
 [2] rewrite: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE applyRules (F (List
 [3] isitwithpred: not known that (SetCategory) is of mode (CATEGORY package (SIGNATURE applyRules (F

```

## warnings 20231: ASP19

```
>compiling ASP19.spad to ASP19.nrlib
```

```

Warnings:
 [1] M has no value
 [2] N has no value

```

```

[3] LJC has no value
[4] XC has no value
[5] FVECC has no value
[6] FJACC has no value
[7] localAssign1: not known that (FiniteLinearAggregate (FortranExpression (construct) (co
[8] coerce: FJACC has no value

```

### warnings 20230: ASP20

```
>compiling ASP20.spad to ASP20.nrllib
```

Warnings:

```

[1] N has no value
[2] NROWH has no value
[3] NCOLH has no value
[4] JTHCOL has no value
[5] HESS has no value
[6] X has no value
[7] HX has no value

```

### warnings 20229: ASP31

```
>compiling ASP31.spad to ASP31.nrllib
```

Warnings:

```

[1] X has no value
[2] Y has no value
[3] localAssign: not known that (FiniteLinearAggregate (FortranExpression (construct (QUOT
[4] coerce: PW has no value

```

### warnings 20228: ASP35

```
>compiling ASP35.spad to ASP35.nrllib
```

Warnings:

```

[1] N has no value
[2] X has no value
[3] FVEC has no value
[4] LDFJAC has no value
[5] FJAC has no value
[6] IFLAG has no value

```

```
[7] localAssign1: not known that (FiniteLinearAggregate (FortranExpression (construct) (construct (QU
```

### warnings 20227: ASP41

```
>compiling ASP41.spad to ASP41.nrllib
```

```
Warnings:
```

```
[1] N has no value
[2] X has no value
[3] EPS has no value
[4] Y has no value
[5] F has no value
[6] localAssign1: not known that (FiniteLinearAggregate (FortranExpression (construct (QUOTE X) (QUOT
[7] makeCodeThree: EPS has no value
```

### warnings 20226: ASP42

```
>compiling ASP42.spad to ASP42.nrllib
```

```
Warnings:
```

```
[1] EPS has no value
[2] N has no value
[3] YA has no value
[4] YB has no value
[5] BC has no value
[6] AJ has no value
[7] BJ has no value
[8] BCEP has no value
[9] localAssign1: not known that (FiniteLinearAggregate (FortranExpression (construct (QUOTE EPS)) (c
[10] makeCodeThree: EPS has no value
```

### warnings 20225: ASP74

```
>compiling ASP74.spad to ASP74.nrllib
```

```
Warnings:
```

```
[1] X has no value
[2] Y has no value
[3] A has no value
[4] B has no value
[5] C has no value
```

```
[6] IBND has no value
```

### warnings 20224: ASP77

```
>compiling ASP77.spad to ASP77.nrllib
```

```
Warnings:
```

```
[1] X has no value
[2] localAssign: not known that (FiniteLinearAggregate (FortranExpression (construct (QUOTE
[3] coerce: F has no value
```

### warnings 20223: ASP80

```
>compiling ASP80.spad to ASP80.nrllib
```

```
Warnings:
```

```
[1] XL has no value
[2] XR has no value
[3] ELAM has no value
[4] YL has no value
[5] YR has no value
```

### warnings 20222: CINTSLPE

```
>compiling CINTSLPE.spad to CINTSLPE.nrllib
```

```
Warnings:
```

```
[1] solveLinearPolynomialEquation: slpePrime has no value
[2] solveLinearPolynomialEquation: oldtable has no value
```

### warnings 20221: COMBF

```
>compiling COMBF.spad to COMBF.nrllib
```

```
Warnings:
```

```
[1] **: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE belong? ((Bool
[2] facts: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE belong? ((B
```

```

[3] summand: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE belong? ((Boolean) (I
[4] smpfact: x has no value
[5] smpfact: y has no value
[6] ipow: n has no value

```

## warnings 20220: COMPFACT

```
>compiling COMPFACT.spad to COMPFACT.nrlib
```

```

Warnings:
[1] conv: pris has no value
[2] conv: dris has no value
[3] backConv: fctr has no value
[4] backConv: xpnt has no value
[5] backConv: flg has no value

```

## warnings 20219: COMPLEX

```
>compiling COMPLEX.spad to COMPLEX.nrlib
```

```

Warnings:
[1] OMwrite: pretend(String) -- should replace by @

```

## warnings 20218: CPMATCH

```
>compiling CPMATCH.spad to CPMATCH.nrlib
```

```

Warnings:
[1] patternMatch: not known that (SetCategory) is of mode (CATEGORY $ (SIGNATURE patternMatch ((Patter

```

## warnings 20217: CRFP

```
>compiling CRFP.spad to CRFP.nrlib
```

```

Warnings:
[1] pleskenSplit: split has no value
[2] startPolynomial: centerIsRoot has no value
[3] startPolynomial: fp has no value

```

```
[4] factor: result has no value
```

### warnings 20216: CTRIGMNP

```
>compiling CTRIGMNP.spad to CTRIGMNP.nrlib
```

```
Warnings:
```

```
[1] complexForm: not known that (RadicalCategory) is of mode (CATEGORY domain (IF (has R (
[2] complexForm: not known that (TranscendentalFunctionCategory) is of mode (CATEGORY doma
[3] real?: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE complexNorm
[4] complexElementary: x has no value
[5] complexElementary: y has no value
[6] complexElementary: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE
[7] complexNormalize: y has no value
```

### warnings 20215: D01WGTS

```
>compiling D01WGTS.spad to D01WGTS.nrlib
```

```
Warnings:
```

```
[1] inRest?: x has no value
[2] findCommonFactor: x has no value
[3] exprIsLogarithmicWeight: not known that (OrderedSet) is of mode (CATEGORY package (SIG
[4] functionIsQuotient: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATUR
```

### warnings 20214: D02AGNT

```
>compiling D02AGNT.spad to D02AGNT.nrlib
```

```
Warnings:
```

```
[1] stiffnessAndStabilityFactor: b has no value
[2] stiffnessAndStabilityFactor: e has no value
[3] stiffnessAndStabilityOfODE: Y has no value
```

### warnings 20213: DBLRESP

```
>compiling DBLRESP.spad to DBLRESP.nrlib
```

Warnings:

- [1] UP22: x has no value
- [2] UP23: x has no value

## warnings 20212: D01AGNT

>compiling D01AGNT.spad to D01AGNT.nrlib

Warnings:

- [1] continuousAtPoint?: not known that (AlgebraicallyClosedField) is of mode (CATEGORY domain (IF (has
- [2] continuousAtPoint?: not known that (TranscendentalFunctionCategory) is of mode (CATEGORY domain (
- [3] functionIsOscillatory: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE rangeI
- [4] singularitiesOf: str has no value

## warnings 20211: DFSFUN

>compiling DFSFUN.spad to DFSFUN.nrlib

Warnings:

- [1] En: The conditional modes (DoubleFloat) and (Complex (DoubleFloat)) conflict
- [2] fresnelC: The conditional modes (Integer) and (DoubleFloat) conflict
- [3] fresnelS: The conditional modes (Integer) and (DoubleFloat) conflict

## warnings 20210: DRAWCURV

>compiling DRAWCURV.spad to DRAWCURV.nrlib

Warnings:

- [1] draw: s has no value
- [2] draw: xRangeFloat has no value
- [3] draw: yRangeFloat has no value
- [4] draw: floatRanges has no value

## warnings 20209: DTP

>compiling DTP.spad to DTP.nrlib

Warnings:

```

[1] blowUp: I has no value
[2] adjunctionDivisorForHamburgeNoether: pretend(Integer) -- should replace by @
[3] adjunctionDivisorForQuadTrans: pretend(Integer) -- should replace by @

```

## warnings 20208: D01TRNS

```
>compiling D01TRNS.spad to D01TRNS.nrlib
```

Warnings:

```

[1] transformFunction: not known that (TranscendentalFunctionCategory) is of mode (CATEGORY package (SIGNATURE exp (F F))) (F F)
[2] transformFunction: x has no value

```

## warnings 20207: EF

```
>compiling EF.spad to EF.nrlib
```

Warnings:

```

[1] pi: Pie has no value
[2] iisqrt1: isqrt1 has no value
[3] iisqrt2: isqrt2 has no value
[4] iisqrt3: isqrt3 has no value
[5] kernel: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE exp (F F))) (F F)
[6] ilog: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE exp (F F))) (F F)
[7] x has no value

```

## warnings 20206: EFSTRUC

```
>compiling EFSTRUC.spad to EFSTRUC.nrlib
```

Warnings:

```

[1] realElem: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE normalize (F F))) (F F)
[2] rtNormalize: func has no value
[3] toR: s has no value
[4] tanQ: not known that (Ring) is of mode (CATEGORY package (SIGNATURE normalize (F F))) (F F)
[5] rootNormalize0: s has no value
[6] rootNormalize0: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE normalize (F F))) (F F)
[7] ktoY: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE normalize (F F))) (F F)
[8] smpElem: x has no value
[9] smpElem: y has no value
[10] k2Elem: ez has no value
[11] k2Elem: iez has no value

```



```

[12] k2Elem: tz2 has no value
[13] tannosimp: x has no value
[14] tannosimp: den has no value
[15] tannosimp: num has no value
[16] expnosimp: x has no value
[17] expnosimp: den has no value
[18] expnosimp: num has no value
[19] rischNormalize: vec has no value
[20] rootKernelNormalize: vec has no value
[21] validExponential: IN has no value
[22] validExponential: g has no value

```

## warnings 20205: ELFUTS

```
>compiling ELFUTS.spad to ELFUTS.nrllib
```

```

Warnings:
 [1] sncndn: x has no value

```

## warnings 20204: ESTOOLS

```
>compiling ESTOOLS.spad to ESTOOLS.nrllib
```

```

Warnings:
 [1] isQuotient: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE f2df ((DoubleFloat
 [2] numberOfOperations1: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE f2df ((D

```

## warnings 20203: EXPRODE

```
>compiling EXPRODE.spad to EXPRODE.nrllib
```

```

Warnings:
 [1] diffRhs: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE seriesSolve ((Any) (E
 [2] k2exquo: IN has no value
 [3] k2exquo: f has no value
 [4] smp2exquo: x has no value
 [5] div2exquo: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE seriesSolve ((Any)
 [6] diffRhsK: not known that (Ring) is of mode (CATEGORY package (SIGNATURE seriesSolve ((Any) (Equat
 [7] findEq: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE seriesSolve ((Any) (E

```

```
[8] seriesSolve: IN has no value
[9] seriesSolve: deq has no value
[10] seriesSolve: y has no value
```

## warnings 20202: EXPRTUBE

```
>compiling EXPRTUBE.spad to EXPRTUBE.nrlib
```

```
Warnings:
 [1] constantToUnaryFunction: s has no value
```

## warnings 20201: EXPR2

```
>compiling EXPR2.spad to EXPR2.nrlib
```

```
Warnings:
 [1] map: x1 has no value
 [2] map: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE map ((Express
```

## warnings 20200: FC

```
>compiling FC.spad to FC.nrlib
```

```
Warnings:
 [1] getElseIf: conditionalBranch has no value
 [2] getCode: assignmentBranch has no value
 [3] getCode: arrayAssignmentBranch has no value
 [4] getCode: conditionalBranch has no value
 [5] getCode: returnBranch has no value
 [6] getCode: blockBranch has no value
 [7] getCode: commentBranch has no value
 [8] getCode: callBranch has no value
 [9] getCode: forBranch has no value
 [10] getCode: labelBranch has no value
 [11] getCode: loopBranch has no value
 [12] getCode: printBranch has no value
 [13] getCode: commonBranch has no value
```

**warnings 20199: FDIV**

```
>compiling FDIV2.spad to FDIV2.nrlib
```

```
Warnings:
```

```
[1] map: s has no value
```

**warnings 20198: FSPRMELT**

```
>compiling FSPRMELT.spad to FSPRMELT.nrlib
```

```
Warnings:
```

```
[1] F2P: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE primitiveElement ((Record
[2] F2P: k1 has no value
[3] F2P: r1 has no value
[4] K2P: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE primitiveElement ((Record
[5] primitiveElement: not known that (Ring) is of mode (CATEGORY package (SIGNATURE primitiveElement
[6] primitiveElement: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE primitiveEl
[7] F2UP: not known that (Ring) is of mode (CATEGORY $ (SIGNATURE primitiveElement ((Record (: primel
[8] primitiveElement: z1 has no value
```

**warnings 20197: FSRED**

```
>compiling FSRED.spad to FSRED.nrlib
```

```
Warnings:
```

```
[1] bringDown: not known that (Ring) is of mode (CATEGORY package (SIGNATURE bringDown ((Fraction (Int
[2] bringDown: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE bringDown ((Fracti
```

**warnings 20196: FSUPFACT**

```
>compiling FSUPFACT.spad to FSUPFACT.nrlib
```

```
Warnings:
```

```
[1] UPAN2F: x has no value
[2] UPQ2AN: x has no value
[3] anfactor: overq has no value
[4] anfactor: overan has no value
[5] UPQ2F: x has no value
[6] ffactor0: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE ffactor ((Factored U
```

```

[7] ffactor0: x has no value
[8] ffactor0: not known that (Ring) is of mode (CATEGORY package (SIGNATURE ffactor ((Fact
[9] UPQ2UP: x has no value
[10] PQ2F: x has no value
[11] PQ2F: y has no value
[12] qfactor: overq has no value
[13] P2QifCan: x has no value
[14] P2QifCan: y has no value

```

### warnings 20195: FSPECF

```
>compiling FSPECF.spad to FSPECF.nrlib
```

Warnings:

```

[1] iiabs: not known that (Ring) is of mode (CATEGORY R (SIGNATURE abs (R R)))
[2] not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE belong? ((Boolean)
[3] x has no value

```

### warnings 20194: FS2

```
>compiling FS2.spad to FS2.nrlib
```

Warnings:

```

[1] smpmap: z has no value
[2] smpmap: x has no value
[3] smpmap: y has no value
[4] map: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE map (B (Mappi

```

### warnings 20193: FS2UPS

```
>compiling FS2UPS.spad to FS2UPS.nrlib
```

Warnings:

```

[1] exprToUPS: not known that (Ring) is of mode (CATEGORY package (SIGNATURE exprToUPS ((U
[2] exprToUPS: y1 has no value
[3] exprToUPS: y2 has no value
[4] listToUPS: %series has no value
[5] powerToUPS: %series has no value
[6] kernelToUPS: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE exprT
[7] nthRootToUPS: %series has no value
[8] logToUPS: %series has no value

```

```

[9] atancotToUPS: %series has no value
[10] applyIfCan: %series has no value
[11] powToUPS: %series has no value
[12] newElem: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE exprToUPS ((Union (
[13] smpElem: x1 has no value
[14] iExprToGenUPS: y1 has no value
[15] iExprToGenUPS: y2 has no value
[16] opsInvolvingX: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE exprToUPS ((U
[17] powerToGenUPS: %series has no value
[18] nthRootToGenUPS: %series has no value
[19] logToGenUPS: %series has no value
[20] expToGenUPS: %series has no value
[21] expGenUPS: %series has no value
[22] atancotToGenUPS: %problem has no value
[23] atancotToGenUPS: %series has no value
[24] genUPSApplyIfCan: %series has no value
[25] applyBddIfCan: %problem has no value
[26] applyBddIfCan: %series has no value
[27] powToGenUPS: %series has no value

```

## warnings 20192: GAUSSFAC

```
>compiling GAUSSFAC.spad to GAUSSFAC.nrlib
```

Warnings:

```

[1] intfactor: unity has no value
[2] factor: :(NonNegativeInteger) -- should replace by pretend
[3] factor: result has no value
[4] factor: unity has no value

```

## warnings 20191: GCNAALG

```
>compiling GCNAALG.spad to GCNAALG.nrlib
```

Warnings:

```

[1] leftRankPolynomial: signature of lhs not unique: (SparseUnivariatePolynomial (Fraction (Polynomial
[2] rightRankPolynomial: signature of lhs not unique: (SparseUnivariatePolynomial (Fraction (Polynomial
[3] genericLeftMinimalPolynomial: x has no value
[4] genericRightMinimalPolynomial: x has no value
[5] genericLeftNorm: rf has no value
[6] genericRightNorm: rf has no value

```

**warnings 20190: GENUFACT**

```
>compiling GENUFACT.spad to GENUFACT.nrllib
```

```
Warnings:
```

```
[1] factor: not known that (UnivariatePolynomialCategory (Integer)) is of mode (CATEGORY d
[2] factor: not known that (UnivariatePolynomialCategory (Fraction (Integer))) is of mode
[3] factor: not known that (UnivariatePolynomialCategory (Complex (Integer))) is of mode (
[4] factor: not known that (UnivariatePolynomialCategory (Complex (Fraction (Integer)))) i
[5] factor: not known that (UnivariatePolynomialCategory (AlgebraicNumber)) is of mode (CA
[6] factor: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE factor
```

**warnings 20189: GENUPS**

```
>compiling GENUPS.spad to GENUPS.nrllib
```

```
Warnings:
```

```
[1] taylor: i has no value
[2] laurent: i has no value
[3] puiseux: i has no value
[4] puiseux: r has no value
[5] series: r has no value
[6] series: x has no value
[7] series: i has no value
```

**warnings 20188: IAN**

```
>compiling IAN.spad to IAN.nrllib
```

```
Warnings:
```

```
[1] makeUnivariate: x has no value
[2] norm: x has no value
[3] convert: y has no value
```

**warnings 20187: INEP**

```
>compiling INEP.spad to INEP.nrllib
```

```
Warnings:
```

```
[1] charpol: :(PositiveInteger) -- should replace by pretend
```

```
[2] unknown Functor code (error unsupported matrix type)
```

## warnings 20186: INFSP

```
>compiling INFSP.spad to INFSP.nrlib
```

Warnings:

```
[1] evaluate: not known that (Ring) is of mode (CATEGORY package (SIGNATURE innerSolve1 ((List F) (Sp
[2] innerSolve1: not known that (UnivariatePolynomialCategory (Complex (Integer))) is of mode (CATEGOR
[3] innerSolve1: pretend(List F) -- should replace by @
[4] innerSolve: pretenddmp -- should replace by @
[5] innerSolve: lq has no value
[6] innerSolve: not known that (PolynomialCategory K (DirectProduct (# lv) (NonNegativeInteger)) OV) :
[7] innerSolve: not known that (PolynomialCategory K (DirectProduct (call LENGTH lv) (NonNegativeInte
[8] innerSolve: pretend(List dmp) -- should replace by @
[9] innerSolve: listGen has no value
[10] innerSolve: result has no value
```

## warnings 20185: INPRODPF

```
>compiling INPRODPF.spad to INPRODPF.nrlib
```

Warnings:

```
[1] applyOverZ: z1 has no value
[2] generalInfiniteProduct: z1 has no value
```

## warnings 20184: INTAF

```
>compiling INTAF.spad to INTAF.nrlib
```

Warnings:

```
[1] rootintegrate: not known that (Ring) is of mode (CATEGORY package (SIGNATURE algint ((Integration
[2] rootintegrate: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE algint ((Integ
[3] rootintegrate: x1 has no value
[4] algintegrate: x1 has no value
[5] UP2UPUP: x1 has no value
[6] UP2UPUP: not known that (Ring) is of mode (CATEGORY package (SIGNATURE algint ((IntegrationResult
[7] UPUP2F: x1 has no value
```

**warnings 20183: INTALG**

```
>compiling INTALG.spad to INTALG.nrlib
```

```
Warnings:
```

```
[1] algintexp: x1 has no value
[2] UPQ2F: x has no value
[3] UP2SUP: x has no value
[4] SUP2UP: x has no value
[5] R2UP: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE algintegrate
[6] R2UP: f1 has no value
[7] R2UP: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE alginteg
[8] R2UP: x1 has no value
[9] R2UP: not known that (Ring) is of mode (CATEGORY package (SIGNATURE algintegrate ((Int
[10] univ: not known that (Ring) is of mode (CATEGORY package (SIGNATURE algintegrate ((In
[11] rlift: x1 has no value
[12] palgintegrate: x1 has no value
[13] alglogint: x1 has no value
```

**warnings 20182: INTEF**

```
>compiling INTEF.spad to INTEF.nrlib
```

```
Warnings:
```

```
[1] tanint: not known that (Ring) is of mode (CATEGORY package (SIGNATURE lfextendedint ((
[2] tanint: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE lfextended
[3] tanint: x1 has no value
[4] tanint: x2 has no value
[5] tanint: x3 has no value
[6] tanint: x4 has no value
[7] tanint: x5 has no value
[8] tanint: x6 has no value
[9] tanint: x7 has no value
[10] tanint: x8 has no value
[11] tanint: x9 has no value
[12] unknownint: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE lfext
[13] droponex: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE lfexten
[14] alglfint: not known that (Ring) is of mode (CATEGORY package (SIGNATURE lfextendedint
[15] alglfint: x1 has no value
[16] alglflimint: x1 has no value
[17] lfintegrate0: x1 has no value
[18] tryChangeVar: x1 has no value
[19] algexpint: x1 has no value
[20] algexpint: x2 has no value
[21] algprimint: x1 has no value
[22] algprimint: x2 has no value
```



```

[23] lfextendedint: x1 has no value
[24] lflimitedint: x1 has no value
[25] primextint: x1 has no value
[26] primextint: x2 has no value
[27] primextint: x3 has no value
[28] expextint: x1 has no value
[29] expextint: x2 has no value
[30] expextint: x3 has no value
[31] expextint: x4 has no value
[32] expextint: x5 has no value
[33] expextint: x6 has no value
[34] expextint: x7 has no value
[35] expextint: x8 has no value
[36] primint: x1 has no value
[37] primint: x2 has no value
[38] primint: x3 has no value
[39] expint: x1 has no value
[40] expint: x2 has no value
[41] expint: x3 has no value
[42] expint: x4 has no value
[43] expint: x5 has no value
[44] expint: x6 has no value
[45] expint: x7 has no value
[46] expint: x8 has no value
[47] primlimint: x1 has no value
[48] primlimint: x2 has no value
[49] primlimint: x3 has no value
[50] explimint: x1 has no value
[51] explimint: x2 has no value
[52] explimint: x3 has no value
[53] explimint: x4 has no value
[54] explimint: x5 has no value
[55] explimint: x6 has no value
[56] explimint: x7 has no value
[57] explimint: x8 has no value

```

## warnings 20181: INTG0

```
>compiling INTG0.spad to INTG0.nrlib
```

Warnings:

```

[1] kerdiff: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE palgint0 ((Integrati
[2] univ: not known that (Ring) is of mode (CATEGORY package (SIGNATURE palgint0 ((IntegrationResult I
[3] univ: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE palgint0 ((IntegrationR
[4] lift: x1 has no value
[5] palgint0: not known that (Ring) is of mode (CATEGORY package (SIGNATURE palgint0 ((IntegrationRes
[6] palgint0: f1 has no value

```

```

[7] palgint0: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE palg
[8] palgint0: x1 has no value
[9] palgextint0: x1 has no value
[10] palglimint0: x1 has no value
[11] multivariate: x1 has no value
[12] palgextint0: f1 has no value
[13] palglimint0: f1 has no value
[14] palgRDE0: var has no value
[15] palgRDE0: coeff has no value

```

### warnings 20180: INTHERAL

```
>compiling INTHERAL.spad to INTHERAL.nrllib
```

Warnings:

```

[1] HermiteIntegrate: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNAT
[2] localsolve: x has no value
[3] localsolve: coef1 has no value

```

### warnings 20179: INTPAF

```
>compiling INTPAF.spad to INTPAF.nrllib
```

Warnings:

```

[1] UPUP2F0: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE palgint (
[2] chv: not known that (Ring) is of mode (CATEGORY package (SIGNATURE palgint ((Integrati
[3] RF2UPUP: z1 has no value
[4] RF2UPUP: x1 has no value
[5] linearInXIfCan: not known that (Ring) is of mode (CATEGORY package (SIGNATURE palgint
[6] linearInXIfCan: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATUR
[7] prootintegrate: f1 has no value
[8] prootintegrate1: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE p
[9] prootintegrate1: x1 has no value
[10] candidates: l has no value
[11] prootlimint: x1 has no value
[12] prootextint: x1 has no value
[13] palgintegrate: x1 has no value

```

### warnings 20178: INTPM

```
>compiling INTPM.spad to INTPM.nrlib
```

```
Warnings:
```

- [1] x1 has no value
- [2] splitConstant: more than 1 modemap for: (One) with dc=F ==>(((F F) ((has R (SemiGroup)) (CONST F
- [3] matcherfei: not known that (SetCategory) is of mode (CATEGORY domain (SIGNATURE pmlntegrate ((Union
- [4] matchdilog: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE pmlntegrate ((Union
- [5] matchdilog: x1 has no value
- [6] matchdilog0: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE pmlntegrate ((Union (Re
- [7] matchdilog0: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE pmlntegrate ((Union
- [8] matchli: x1 has no value
- [9] matchli0: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE pmlntegrate ((Union (Record
- [10] matchsici: ci0? has no value
- [11] pmlntegrate: const has no value
- [12] pmlntegrate: nconst has no value
- [13] pmComplexintegrate: const has no value
- [14] pmComplexintegrate: nconst has no value
- [15] formula1: not known that (OrderedSet) is of mode (CATEGORY \$ (SIGNATURE pmlntegrate ((Union F fa
- [16] pmlntegrate: x1 has no value

## warnings 20177: INTTOOLS

```
>compiling INTTOOLS.spad to INTTOOLS.nrlib
```

```
Warnings:
```

- [1] varselect: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE varselect ((List (I
- [2] vark: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE varselect ((List (Kerne
- [3] removeConstantTerm: not known that (OrderedSet) is of mode (CATEGORY \$ (SIGNATURE removeConstantT
- [4] removeConstantTerm: ans has no value
- [5] cont: not known that (Ring) is of mode (CATEGORY \$ (SIGNATURE mkPrim (F F (Symbol))))
- [6] cont: unit has no value
- [7] linearLog?: not known that (Ring) is of mode (CATEGORY \$ (SIGNATURE mkPrim (F F (Symbol))))
- [8] intPatternMatch: nl has no value

## warnings 20176: ITRIGMNP

```
>compiling ITRIGMNP.spad to ITRIGMNP.nrlib
```

```
Warnings:
```

- [1] FG2F: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE GF2FG (FG (Complex F)))
- [2] F2FG: x has no value
- [3] ker2explogs: IN has no value

```

[4] ker2explogs: v has no value
[5] ker2explogs: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE GF2FG
[6] ker2explogs: a has no value
[7] ker2trigs: IN has no value
[8] smp2trigs: x has no value
[9] explogs2trigs: not known that (Ring) is of mode (CATEGORY package (SIGNATURE GF2FG (FG
[10] smp2explogs: x has no value
[11] smp2explogs: y has no value

```

### warnings 20175: KOVACIC

```
>compiling KOVACIC.spad to KOVACIC.nrllib
```

Warnings:

```
[1] kovacic: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE kovacic
```

### warnings 20174: LF

```
>compiling LF.spad to LF.nrllib
```

Warnings:

```

[1] integrand: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE belong?
[2] eqint: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE belong? ((B
[3] z1 has no value

```

### warnings 20173: LODOF

```
>compiling LODOF.spad to LODOF.nrllib
```

Warnings:

```

[1] factor: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE factor
[2] rfactor: z has no value
[3] zro1: z1 has no value
[4] zro1: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE factor1 ((List (Lin
[5] factor: p has no value
[6] factor1: p has no value

```

**warnings 20172: LSQM**

```
>compiling LSQM.spad to LSQM.nrllib
```

```
Warnings:
```

```
[1] not known that (NonAssociativeAlgebra R) is of mode (CATEGORY domain (SIGNATURE transpose ($ $)))
```

**warnings 20171: MULTFACT**

```
>compiling MULTFACT.spad to MULTFACT.nrllib
```

```
Warnings:
```

```
[1] factor: not known that (OrderedSet) is of mode (CATEGORY OV (SIGNATURE convert ((Symbol) $)))
```

**warnings 20170: MYUP**

```
>compiling MYUP.spad to MYUP.nrllib
```

```
Warnings:
```

```
[1] coerce: z1 has no value
```

**warnings 20169: MYEXPR**

```
>compiling MYEXPR.spad to MYEXPR.nrllib
```

```
Warnings:
```

```
[1] iunivariate: z1 has no value
```

**warnings 20168: MYEXPR**

```
>compiling MYEXPR.spad to MYEXPR.nrllib
```

```
Warnings:
```

```
[1] coerce: z1 has no value
```

```
Warnings:
```

```
[1] iunivariate: z1 has no value
```

**warnings 20167: NAGF01**

```
>compiling NAGF01.spad to NAGF01.nrllib
```

```
Warnings:
```

```
[1] f01brf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[2] f01bsf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[3] f01maf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[4] f01mcf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[5] f01qcf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[6] f01qdf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[7] f01qef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[8] f01rcf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[9] f01rdf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[10] f01ref: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
```

**warnings 20166: NAGF02**

```
>compiling NAGF02.spad to NAGF02.nrllib
```

```
Warnings:
```

```
[1] f02aaf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[2] f02abf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[3] f02adf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[4] f02aef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[5] f02aff: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[6] f02agf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[7] f02ajf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[8] f02akf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[9] f02awf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[10] f02axf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[11] f02bbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[12] f02bjf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[13] f02fjf: fn has no value
[14] f02fjf: fp has no value
[15] f02fjf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[16] f02wef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[17] f02xef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
```

## warnings 20165: NAGF04

>compiling NAGF04.spad to NAGF04.nrllib

Warnings:

```
[1] f04adf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[2] f04arf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[3] f04asf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[4] f04atf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[5] f04axf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[6] f04faf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[7] f04jgf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[8] f04maf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[9] f04mbf: fn has no value
[10] f04mbf: fp has no value
[11] f04mbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[12] f04mcf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[13] f04qaf: fn has no value
[14] f04qaf: fp has no value
[15] f04qaf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
```

## warnings 20164: NLINSOL

>compiling NLINSOL.spad to NLINSOL.nrllib

Warnings:

```
[1] allRoots: not known that (Ring) is of mode (CATEGORY package (SIGNATURE solveInField ((List (List
```

## warnings 20163: NSMP

>compiling NSMP.spad to NSMP.nrllib

Warnings:

```
[1] localMonicModulo: a1 has no value
[2] monicModulo: mM has no value
[3] exactQuotient: a1 has no value
[4] exactQuotient!: a1 has no value
[5] primitivePart!: a1 has no value
```

**warnings 20162: ODERTRIC**

```
>compiling ODERTRIC.spad to ODERTRIC.nrlib
```

```
Warnings:
```

- [1] UP2SUP: z has no value
- [2] ricDsolve: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE ric
- [3] mapeval: z has no value
- [4] ratsln: ls has no value
- [5] ratsln: lv has no value
- [6] ratsol: ans has no value
- [7] polyRicDE: deg has no value
- [8] polyRicDE: ans has no value
- [9] newtonSolution: m has no value
- [10] nopoly: z has no value
- [11] ricDsolve: z has no value
- [12] zro1: z has no value
- [13] zro1: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE ricDsolve ((List (

**warnings 20161: OMEXPR**

```
>compiling OMEXPR.spad to OMEXPR.nrlib
```

```
Warnings:
```

- [1] outputOMFunction: %defint has no value
- [2] outputOMFunction: %defsum has no value
- [3] outputOMFunction: %defprod has no value
- [4] outputOMFunction: %power has no value
- [5] outputOMExpr: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE OMwr
- [6] OMwrite: pretend(String) -- should replace by @

**warnings 20160: PADE**

```
>compiling PADE.spad to PADE.nrlib
```

```
Warnings:
```

- [1] padeInner: plist has no value
- [2] padeInner: alist has no value



## warnings 20159: PAN2EXPR

```
>compiling PAN2EXPR.spad to PAN2EXPR.nrllib
```

```
Warnings:
```

- [1] coerce: x has no value
- [2] coerce: y has no value

## warnings 20158: PFO

```
>compiling PFO.spad to PFO.nrllib
```

```
Warnings:
```

- [1] klist: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE order ((Union (NonNegat
- [2] UPQ2F: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE order ((Union (NonNegat
- [3] UP22UP: p1 has no value
- [4] UP32UPUP: p1 has no value
- [5] commonDen: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE order ((Union (Non
- [6] alglis: k1 has no value
- [7] krmod: z1 has no value
- [8] raise2: z1 has no value
- [9] alcurve: z1 has no value
- [10] alcurve: q1 has no value
- [11] rat: q1 has no value
- [12] selectIntegers: not known that (Ring) is of mode (CATEGORY package (SIGNATURE order ((Union (Non
- [13] toQ1: p1 has no value
- [14] toQ2: p1 has no value
- [15] kpmmod: p1 has no value
- [16] goodRed: z1 has no value

## warnings 20157: PFOQ

```
>compiling PFOQ.spad to PFOQ.nrllib
```

```
Warnings:
```

- [1] rat: z1 has no value

## warnings 20156: PICOERCE

```
>compiling PICOERCE.spad to PICOERCE.nrllib
```

```
Warnings:
 [1] p2e: x1 has no value
```

## warnings 20155: PMASSFS

```
>compiling PMASSFS.spad to PMASSFS.nrlib
```

```
Warnings:
 [1] ass: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE assert (F F (
```

## warnings 20154: PMFS

```
>compiling PMFS.spad to PMFS.nrlib
```

```
Warnings:
 [1] patternMatch: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE patt
 [2] patternMatch: l1 has no value
 [3] patternMatch: var has no value
```

## warnings 20153: PMPREDFS

```
>compiling PMPREDFS.spad to PMPREDFS.nrlib
```

```
Warnings:
 [1] suchThat: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE suchThat
```

## warnings 20152: PSETPK

```
>compiling PSETPK.spad to PSETPK.nrlib
```

```
Warnings:
 [1] selectPolynomials: gps has no value
 [2] selectPolynomials: bps has no value
 [3] selectOrPolynomials: gps has no value
 [4] selectOrPolynomials: bps has no value
 [5] selectAndPolynomials: gps has no value
 [6] selectAndPolynomials: bps has no value
```

```

[7] certainlySubVariety?: polnum has no value
[8] autoRemainder: newlp has no value
[9] rewriteSetByReducingWithParticularGenerators: rs has no value
[10] squareFreeFactors: not known that (IntegralDomain) is of mode (CATEGORY domain (SIGNATURE remove
[11] univariatePolynomialsGcds: p has no value
[12] univariatePolynomialsGcds: pInV has no value
[13] univariatePolynomialsGcds: stop has no value
[14] univariatePolynomialsGcds: lg has no value
[15] removeRoughlyRedundantFactorsInContents: newcp has no value
[16] removeRedundantFactorsInContents: newcp has no value
[17] removeRedundantFactorsInPols: newcp has no value
[18] removeRedundantFactorsInPols: newp has no value
[19] unprotectedRemoveRedundantFactors: d has no value
[20] removeRedundantFactors: toSee has no value
[21] removeRedundantFactors: toSave has no value

```

## warnings 20151: RADFF

```
>compiling RADFF.spad to RADFF.nrlib
```

Warnings:

```

[1] fullVector: s has no value
[2] charPintbas: s has no value

```

## warnings 20150: RDEEF

```
>compiling RDEEF.spad to RDEEF.nrlib
```

Warnings:

```

[1] rischDEalg: z1 has no value
[2] rischDEalg: z2 has no value
[3] rischDEalg: z3 has no value
[4] rischDEalg: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE rischDE ((Record
[5] rischDEalg: not known that (Ring) is of mode (CATEGORY package (SIGNATURE rischDE ((Record (: ans
[6] normalise0: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE rischDE ((Record
[7] normalise0: not known that (Ring) is of mode (CATEGORY package (SIGNATURE rischDE ((Record (: ans
[8] normalise: z1 has no value
[9] normalise: z2 has no value
[10] polyDElog: z1 has no value
[11] polyDElog: limitedlogs has no value
[12] polyDElog: more than 1 modemap for: (Zero) with dc=F ==>(((F F) ((has R (AbelianSemiGroup)) (CO
[13] polyDElog: eq has no value
[14] gpolDEexp: z1 has no value

```

```

[15] gpolDEexp: z2 has no value
[16] polyDEexp: z1 has no value
[17] polyDEexp: z2 has no value
[18] polyDEexp: eq has no value
[19] logdegrad: z1 has no value
[20] expdegrad: z1 has no value
[21] expdegrad: z2 has no value

```

### warnings 20149: RDEEFS

```
>compiling RDEEFS.spad to RDEEFS.nrllib
```

Warnings:

```

[1] basecase: not known that (Ring) is of mode (CATEGORY package (SIGNATURE rischDEsys ((U
[2] basecase: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE rischDEs
[3] rischDEsys: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE rischD

```

### warnings 20148: RSETCAT

```
>compiling RSETCAT.spad to RSETCAT.nrllib
```

Warnings:

```
[1] intersect: toSave has no value
```

### warnings 20147: RSETGCD

```
>compiling RSETGCD.spad to RSETGCD.nrllib
```

Warnings:

```

[1] toseInvertible?: x has no value
[2] toseInvertible?: y has no value
[3] toseInvertibleSet: toSave has no value
[4] prepareSubResAlgo: toSave has no value
[5] internalLastSubResultant: toReturn has no value

```

### warnings 20146: RULE

```
>compiling RULE.spad to RULE.nrllib
```

Warnings:

- [1] F2Symbol: z has no value
- [2] F2Symbol: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE rule (\$ F F)) (SIGNA
- [3] retractIfCan: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE rule (\$ F F)) (S

## warnings 20145: SIGNEF

>compiling SIGNEF.spad to SIGNEF.nrllib

Warnings:

- [1] sign: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE sign ((Union (Integer) :
- [2] sign: z has no value
- [3] smpsign: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE sign ((Union (In

## warnings 20144: SIMPAN

>compiling SIMPAN.spad to SIMPAN.nrllib

Warnings:

- [1] simplify: not known that (TranscendentalFunctionCategory) is of mode (CATEGORY domain (IF (has (I

## warnings 20143: SOLVESER

>compiling SOLVESER.spad to SOLVESER.nrllib

Warnings:

- [1] unvectorise: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE decomposeFunc ((I
- [2] unvectorise: not known that (Ring) is of mode (CATEGORY package (SIGNATURE decomposeFunc ((Fracti

## warnings 20142: SOLVETRA

>compiling SOLVETRA.spad to SOLVETRA.nrllib

Warnings:

- [1] solveInner: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE solve ((List (Equa

```

[2] solveInner: not known that (Ring) is of mode (CATEGORY package (SIGNATURE solve ((List
[3] tryToTrans: not known that (TranscendentalFunctionCategory) is of mode (CATEGORY domain
[4] tryToTrans: not known that (AlgebraicallyClosedField) is of mode (CATEGORY domain (IF
[5] subsTan: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE solve
[6] buildnexpr: anscoeff has no value
[7] buildnexpr: ansmant has no value
[8] combineLog: ans has no value
[9] funcinv: not known that (OrderedSet) is of mode (CATEGORY R (ATTRIBUTE complex))
[10] smp2Poly: x has no value
[11] smp2Poly: y has no value

```

### warnings 20141: SUMFS

```
>compiling SUMFS.spad to SUMFS.nrllib
```

Warnings:

```

[1] sum: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE sum (F F (Sym
[2] notRF?: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE sum (F F (

```

### warnings 20140: TOOLSIGN

```
>compiling TOOLSIGN.spad to TOOLSIGN.nrllib
```

Warnings:

```

[1] nonQsign: pretend(AlgebraicNumber) -- should replace by @
[2] nonQsign: not known that (AlgebraicallyClosedField) is of mode (CATEGORY domain (IF (h
[3] nonQsign: not known that (TranscendentalFunctionCategory) is of mode (CATEGORY domain

```

### warnings 20139: TRIGMNIP

```
>compiling TRIGMNIP.spad to TRIGMNIP.nrllib
```

Warnings:

```

[1] K2KG: not known that (RadicalCategory) is of mode (CATEGORY domain (IF (has (Complex R
[2] K2KG: not known that (TranscendentalFunctionCategory) is of mode (CATEGORY domain (IF
[3] K2KG: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE complexNorma
[4] real?: ker has no value
[5] complexKernels: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE co
[6] complexKernels: lk has no value
[7] complexKernels: lv has no value
[8] locexplogs: x has no value

```

```

[9] complexNormalize: y has no value
[10] complexNormalize: not known that (AlgebraicallyClosedField) is of mode (CATEGORY domain (IF (has
[11] complexNormalize: not known that (TranscendentalFunctionCategory) is of mode (CATEGORY domain (I
[12] complexNormalize: x has no value
[13] complexElementary: y has no value
[14] complexElementary: x has no value

```

## warnings 20138: TRMANIP

>compiling TRMANIP.spad to TRMANIP.nrlib

### Warnings:

```

[1] logArgs: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE expand (F F)) (SIGNA
[2] logArgs: sum has no value
[3] logArgs: arg has no value
[4] simplifyLog1: exprs has no value
[5] simplifyLog1: terms has no value
[6] simplifyLog1: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE expand (F F)) (S
[7] expandpow: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE expand (F F)) (SIG
[8] termexp: z has no value
[9] termexp: exponent has no value
[10] expandPower: z has no value
[11] powersimp: z has no value
[12] cos2sec: z1 has no value
[13] sin2csc: z1 has no value
[14] tan2cot: z1 has no value
[15] cot2tan: z1 has no value
[16] cosh2sech: z1 has no value
[17] sinh2csch: z1 has no value
[18] tanh2coth: z1 has no value
[19] coth2tanh: z1 has no value
[20] removeCosSq: z1 has no value
[21] removeCoshSq: z1 has no value
[22] smpexpand: r1 has no value
[23] smplog: r1 has no value
[24] smp2htrigs: k1 has no value
[25] smp2htrigs: r1 has no value
[26] htrigs: not known that (Ring) is of mode (CATEGORY package (SIGNATURE expand (F F)) (SIGNATURE s
[27] exlog: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE expand (F F)) (SIG
[28] logexpand: IN has no value
[29] logexpand: x has no value
[30] kerexpand: IN has no value
[31] kerexpand: x has no value

```

**warnings 20137: UPXSSING**

```
>compiling UPXSSING.spad to UPXSSING.nrlib
```

```
Warnings:
```

```
[1] retractIfCan: signature of lhs not unique: (Union (UnivariatePuisseuxSeries FE var cen)
[2] sortAndDiscardTerms: zeroTerms has no value
[3] sortAndDiscardTerms: infiniteTerms has no value
[4] sortAndDiscardTerms: failedTerms has no value
[5] dominantTermOnList: x has no value
[6] dominantTermOnList: y has no value
[7] dominantTermOnList: %list has no value
[8] dominantTermOnList: %order has no value
```

**warnings 20136: UTSODE**

```
>compiling UTSODE.spad to UTSODE.nrlib
```

```
Warnings:
```

```
[1] stFunc1: s has no value
[2] stFunc2: s1 has no value
[3] stFunc2: s2 has no value
[4] stFuncN: ls has no value
[5] divloop: s has no value
[6] i0de1: s has no value
[7] i0de2: s has no value
[8] i0de: ls has no value
[9] iMpsode: ls has no value
```

**warnings 20135: UTSODETL**

```
>compiling UTSODETL.spad to UTSODETL.nrlib
```

```
Warnings:
```

```
[1] LOD02FUN: l1 has no value
```

**warnings 20134: UTSSOL**

```
>compiling UTSSOL.spad to UTSSOL.nrlib
```



Warnings:

- [1] seriesSolve: x has no value
- [2] seriesSolve: res has no value

## warnings 20133: WUTSET

>compiling WUTSET.spad to WUTSET.nrlib

Warnings:

- [1] medialSetWithTrace: rs has no value
- [2] medialSetWithTrace: contradiction has no value
- [3] medialSetWithTrace: bs has no value
- [4] characteristicSetUsingTrace: contradiction has no value
- [5] characteristicSetUsingTrace: rs has no value
- [6] characteristicSetUsingTrace: ms has no value
- [7] zeroSetSplit: newlts has no value

## warnings 20132: DEFINTEF

>compiling DEFINTEF.spad to DEFINTEF.nrlib

Warnings:

- [1] checkForPole: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE integrate ((Union
- [2] polyIfCan: not known that (Ring) is of mode (CATEGORY package (SIGNATURE integrate ((Union (: f1
- [3] polyIfCan: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE integrate ((Union

## warnings 20131: DEFINTRF

>compiling DEFINTRF.spad to DEFINTRF.nrlib

Warnings:

- [1] nopole: not known that (TranscendentalFunctionCategory) is of mode (CATEGORY domain (IF (has R (I
- [2] nopole: not known that (AlgebraicallyClosedFunctionSpace R) is of mode (CATEGORY domain (IF (has I
- [3] integrate: x has no value

## warnings 20130: DFINTTLS

>compiling DFINTTLS.spad to DFINTTLS.nrlib

Warnings:

- [1] findLimit: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE ignore? (
- [2] mkLogPos: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE ignore? (
- [3] checkForZero: s has no value
- [4] checkForZero: not known that (Ring) is of mode (CATEGORY package (SIGNATURE ignore? (
- [5] findRealZero: fin has no value
- [6] findRealZero: s has no value
- [7] findRealZero: halfinf has no value
- [8] var: i has no value

## warnings 20129: EFULS

>compiling EFULS.spad to EFULS.nrlib

Warnings:

- [1] tanIfCan: not known that (Algebra (Fraction (Integer))) is of mode (CATEGORY Coef (SIG

## warnings 20128: ESCONT

>compiling ESCONT.spad to ESCONT.nrlib

Warnings:

- [1] zerosOf: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE getlo ((
- [2] zerosOf: t1 has no value
- [3] zerosOf: t2 has no value
- [4] singularitiesOf: t3 has no value

## warnings 20127: EXPR

>compiling EXPR.spad to EXPR.nrlib

Warnings:

- [1] not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE simplifyPower (\$ \$
- [2] simplifyPower: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE simp
- [3] \*\*: pretend(Integer) -- should replace by @
- [4] \*\*: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE simplifyPower (
- [5] <: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE simplifyPower (\$
- [6] numer: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE simplifyPowe
- [7] alskernels: x has no value
- [8] topnat: not known that (Field) is of mode (CATEGORY domain (SIGNATURE simplifyPower (\$

```

[9] toprat: not known that (ExpressionSpace) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[10] toprat: not known that (SIGNATURE number ((SparseMultivariatePolynomial R (Kernel $)) $)) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[11] toprat: not known that (SIGNATURE denom ((SparseMultivariatePolynomial R (Kernel $)) $)) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[12] toprat: not known that (SIGNATURE coerce ($ (SparseMultivariatePolynomial R (Kernel $)))) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[13] reducedSystem: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[14] commonk0: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[15] rootOf: not known that (FunctionSpace R) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[16] rootOf: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[17] pi: not known that (FunctionSpace R) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[18] pi: not known that (RadicalCategory) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[19] abs: not known that (FunctionSpace R) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[20] **: not known that (FunctionSpace R) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[21] erf: not known that (FunctionSpace R) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[22] erf: not known that (RadicalCategory) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[23] erf: not known that (TranscendentalFunctionCategory) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[24] operator: not known that (FunctionSpace R) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[25] operator: not known that (ExpressionSpace) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[26] evl0: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[27] gcdPolynomial: not known that (GcdDomain) is of mode (CATEGORY domain (SIGNATURE squareFreePolynomial ($ $ (Integer)))
[28] factorPolynomial: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE squareFreePolynomial ($ $ (Integer)))
[29] factorPolynomial: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE squareFreePolynomial ($ $ (Integer)))
[30] coerce: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[31] retract: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[32] retractIfCan: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[33] k2expr: x has no value
[34] k2expr: not known that (ExpressionSpace) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[35] smp2expr: x has no value
[36] smp2expr: not known that (SetCategory) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[37] smp2expr: not known that (SIGNATURE + ($ $ $)) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[38] smp2expr: not known that (SIGNATURE * ($ $ $)) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[39] smp2expr: not known that (SIGNATURE ** ($ $ (NonNegativeInteger))) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[40] smp2an: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[41] convert: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[42] eval: not known that (ConvertibleTo (InputForm)) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[43] patternMatch: not known that (FunctionSpace R) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[44] patternMatch: not known that (ConvertibleTo (Pattern (Integer))) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[45] patternMatch: not known that (PatternMatchable (Integer)) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[46] patternMatch: not known that (RetractableTo (Kernel $)) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[47] patternMatch: not known that (SetCategory) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[48] patternMatch: not known that (ConvertibleTo (Pattern (Float))) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[49] patternMatch: not known that (PatternMatchable (Float)) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer)))
[50] kereval: x1 has no value
[51] kereval: x2 has no value
[52] subeval: x has no value
[53] isPlus: gen has no value
[54] eval: x2 has no value
[55] eval: x1 has no value
[56] subst: x1 has no value
[57] not known that (Ring) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer))))
[58] not known that (IntegralDomain) is of mode (CATEGORY domain (SIGNATURE simplifyPower ($ $ (Integer))))

```

**warnings 20126: EXPRSOL**

```
>compiling EXPRSOL.spad to EXPRSOL.nrllib
```

```
Warnings:
```

- [1] replaceDiffs: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE seri
- [2] replaceDiffs: %diff has no value
- [3] seriesSolve: x has no value

**warnings 20125: EXPR2UPS**

```
>compiling EXPR2UPS.spad to EXPR2UPS.nrllib
```

```
Warnings:
```

- [1] iTaylor: %problem has no value
- [2] iTaylor: %series has no value
- [3] taylor: more than 1 modemap for: (Zero) with dc=FE ==>(((FE FE) ((has R (AbelianSemiG
- [4] iLaurent: %problem has no value
- [5] iLaurent: %series has no value
- [6] laurent: more than 1 modemap for: (Zero) with dc=FE ==>(((FE FE) ((has R (AbelianSemi
- [7] iPuisseux: %problem has no value
- [8] iPuisseux: %series has no value
- [9] puisseux: more than 1 modemap for: (Zero) with dc=FE ==>(((FE FE) ((has R (AbelianSemi
- [10] iSeries: %problem has no value
- [11] iSeries: %series has no value
- [12] series: more than 1 modemap for: (Zero) with dc=FE ==>(((FE FE) ((has R (AbelianSemi

**warnings 20124: FACTEXT**

```
>compiling FACTEXT.spad to FACTEXT.nrllib
```

```
Warnings:
```

- [1] factor: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE factor
- [2] norm: not known that (Ring) is of mode (CATEGORY package (SIGNATURE factor ((Factored

**warnings 20123: FDIV**

```
>compiling FDIV.spad to FDIV.nrlib
```

```
Warnings:
```

- [1] \*: signature of lhs not unique: \$(Integer)\$ chosen
- [2] reduce: s has no value

## warnings 20122: FSCINT

```
>compiling FSCINT.spad to FSCINT.nrlib
```

```
Warnings:
```

- [1] K2KG: not known that (RadicalCategory) is of mode (CATEGORY domain (IF (has (Complex R) (Integral
- [2] K2KG: not known that (TranscendentalFunctionCategory) is of mode (CATEGORY domain (IF (has (Compl
- [3] K2KG: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE internalIntegrate ((Int
- [4] internalIntegrate: x1 has no value
- [5] internalIntegrate: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE internalInt
- [6] internalIntegrate: not known that (AlgebraicallyClosedField) is of mode (CATEGORY domain (IF (has
- [7] internalIntegrate: not known that (TranscendentalFunctionCategory) is of mode (CATEGORY domain (I

## warnings 20121: FSINT

```
>compiling FSINT.spad to FSINT.nrlib
```

```
Warnings:
```

- [1] K2KG: not known that (RadicalCategory) is of mode (CATEGORY domain (IF (has (Complex R) (Integral
- [2] K2KG: not known that (TranscendentalFunctionCategory) is of mode (CATEGORY domain (IF (has (Compl
- [3] K2KG: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE integrate ((Union F (Lis
- [4] trans?: x1 has no value
- [5] postSubst: not known that (Ring) is of mode (CATEGORY package (SIGNATURE integrate ((Union F (Lis
- [6] postSubst: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE integrate ((Union I
- [7] integrate: x1 has no value
- [8] integrate: not known that (AlgebraicallyClosedField) is of mode (CATEGORY domain (IF (has (Comple
- [9] integrate: not known that (TranscendentalFunctionCategory) is of mode (CATEGORY domain (IF (has (

## warnings 20120: FS2EXPXP

```
>compiling FS2EXPXP.spad to FS2EXPXP.nrlib
```

```
Warnings:
```

- [1] newElem: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE exprToXXP ((Union (:
- [2] smpElem: x1 has no value

```

[3] k2Elem: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE exprToXXP
[4] iExprToXXP: %series has no value
[5] iExprToXXP: y1 has no value
[6] iExprToXXP: y2 has no value
[7] listToXXP: %expansion has no value
[8] powerToXXP: %expansion has no value
[9] nthRootXXPIfCan: %problem has no value
[10] nthRootXXPIfCan: %series has no value
[11] nthRootToXXP: %problem has no value
[12] nthRootToXXP: %expansion has no value
[13] genPowerToXXP: %expansion has no value
[14] genExp: %series has no value
[15] exponential: %problem has no value
[16] exponential: %series has no value
[17] expToXXP: %expansion has no value
[18] logToXXP: %expansion has no value
[19] logToXXP: %problem has no value
[20] logToXXP: %series has no value
[21] applyIfCan: %expansion has no value
[22] applyBddIfCan: %problem has no value
[23] applyBddIfCan: %expansion has no value
[24] opsInvolvingX: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE ex
[25] atancotToXXP: %problem has no value
[26] atancotToXXP: %series has no value

```

## warnings 20119: GSERIES

```
>compiling GSERIES.spad to GSERIES.nrllib
```

Warnings:

```

[1] differentiate: x has no value
[2] integrate: x has no value

```

## warnings 20118: HELLFDIV

```
>compiling HELLFDIV.spad to HELLFDIV.nrllib
```

Warnings:

```
[1] unknown Functor code (error HyperellipticFiniteDivisor: curve must be hyperelliptic)
```

**warnings 20117: INTDIVP**

```
>compiling INTDIVP.spad to INTDIVP.nrlib
```

```
Warnings:
```

```
[1] intersectionDivisor: pretend(Integer) -- should replace by @
```

**warnings 20116: INVLAPLA**

```
>compiling INVLAPLA.spad to INVLAPLA.nrlib
```

```
Warnings:
```

```
[1] ilt: not known that (Ring) is of mode (CATEGORY package (SIGNATURE inverseLaplace ((Union F failed
[2] ilt: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE inverseLaplace ((Union F
[3] iltsqfr: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE inverseLaplace (
[4] iltirred: not known that (Ring) is of mode (CATEGORY package (SIGNATURE inverseLaplace ((Union F
```

**warnings 20115: IR2F**

```
>compiling IR2F.spad to IR2F.nrlib
```

```
Warnings:
```

```
[1] evenRoots: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE split ((Integration
[2] ilog: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE split ((IntegrationResu
[3] ilog: not known that (Ring) is of mode (CATEGORY package (SIGNATURE split ((IntegrationResult F)
[4] lg2func: not known that (Ring) is of mode (CATEGORY package (SIGNATURE split ((IntegrationResult
```

**warnings 20114: IRRF2F**

```
>compiling IRRF2F.spad to IRRF2F.nrlib
```

```
Warnings:
```

```
[1] toEF: z1 has no value
[2] expand: not known that (AlgebraicallyClosedFunctionSpace R) is of mode (CATEGORY domain (IF (has R
[3] expand: not known that (TranscendentalFunctionCategory) is of mode (CATEGORY domain (IF (has R (I
[4] integrate: not known that (AlgebraicallyClosedField) is of mode (CATEGORY domain (IF (has R (Integ
[5] integrate: not known that (TranscendentalFunctionCategory) is of mode (CATEGORY domain (IF (has R
```

**warnings 20113: LAPLACE**

```
>compiling LAPLACE.spad to LAPLACE.nrllib
```

```
Warnings:
```

- [1] algebraic?: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE laplace (F F (
- [2] isLinear: not known that (Ring) is of mode (CATEGORY package (SIGNATURE laplace (F F (
- [3] isLinear: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE laplace (F F (
- [4] atn: d has no value
- [5] mkPlus: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE laplace (F F (
- [6] locallaplace: const has no value
- [7] locallaplace: nconst has no value

**warnings 20112: LIMITPS**

```
>compiling LIMITPS.spad to LIMITPS.nrllib
```

```
Warnings:
```

- [1] firstNonLogPtr: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE li
- [2] complLimit: %series has no value
- [3] realLimit: %problem has no value
- [4] realLimit: %series has no value
- [5] realLimit: func has no value
- [6] realLimit: prob has no value
- [7] xxpLimit: %expansion has no value
- [8] limitPlus: %problem has no value
- [9] limitPlus: %series has no value
- [10] limitPlus: func has no value
- [11] limitPlus: prob has no value

**warnings 20111: LODEEF**

```
>compiling LODEEF.spad to LODEEF.nrllib
```

```
Warnings:
```

- [1] algSolve: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE solve ((Union (
- [2] algSolve: not known that (Ring) is of mode (CATEGORY package (SIGNATURE solve ((Union (
- [3] algSolve: f1 has no value
- [4] doVarParams: f1 has no value
- [5] xpart: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE solve ((Union (
- [6] kovode: z1 has no value
- [7] ulodo: not known that (Ring) is of mode (CATEGORY package (SIGNATURE solve ((Union (Re
- [8] solve: f1 has no value



## warnings 20110: NODE1

```
>compiling NODE1.spad to NODE1.nrllib
```

```
Warnings:
```

- [1] solve: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE solve ((Union F failed
- [2] checkBernoulli: not known that (Ring) is of mode (CATEGORY package (SIGNATURE solve ((Union F fai

## warnings 20109: ODECONST

```
>compiling ODECONST.spad to ODECONST.nrllib
```

```
Warnings:
```

- [1] constDsolve: f1 has no value
- [2] basisSqfr: not known that (Ring) is of mode (CATEGORY package (SIGNATURE constDsolve ((Record (: p

## warnings 20108: ODEEF

```
>compiling ODEEF.spad to ODEEF.nrllib
```

```
Warnings:
```

- [1] solve: z1 has no value
- [2] solve: z2 has no value
- [3] solve: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE solve ((Union (Record
- [4] parseODE: n has no value
- [5] parseODE: c has no value
- [6] parseODE: k has no value
- [7] getcoeff: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE solve ((Union (Reco
- [8] getcoeff: not known that (Ring) is of mode (CATEGORY package (SIGNATURE solve ((Union (Record (: p

## warnings 20107: ODEINT

```
>compiling ODEINT.spad to ODEINT.nrllib
```

```
Warnings:
```

- [1] diff: f1 has no value
- [2] mkprod: z1 has no value

```

[3] expint: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE int (F F (
[4] expint: lrec has no value
[5] expint: exponent has no value
[6] isQlog: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE int (F F (

```

## warnings 20106: QCMPACK

>compiling QCMPACK.spad to QCMPACK.nrlib

Warnings:

```

[1] removeSuperfluousCases: x has no value
[2] removeSuperfluousCases: y has no value
[3] removeSuperfluousCases: maxcases has no value
[4] removeSuperfluousCases: lpwt1 has no value
[5] removeSuperfluousCases: headmaxcases has no value
[6] removeSuperfluousCases: toSave has no value
[7] removeSuperfluousQuasiComponents: maxlts has no value
[8] removeSuperfluousQuasiComponents: headmaxlts has no value
[9] removeSuperfluousQuasiComponents: toSave has no value
[10] branchIfCan: polnum has no value
[11] prepareDecompose: branches has no value
[12] prepareDecompose: x has no value
[13] prepareDecompose: y has no value

```

## warnings 20105: REGSET

>compiling REGSET.spad to REGSET.nrlib

Warnings:

```

[1] construct: ts has no value
[2] preprocess: lp1 has no value
[3] preprocess: lp2 has no value
[4] preprocess: lts has no value

```

## warnings 20104: REP

>compiling REP.spad to REP.nrlib

Warnings:

```

[1] evalvect: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE radicalE
[2] gramschmidt: :(PositiveInteger) -- should replace by pretend

```

```
[3] gramschmidt: :RMR -- should replace by pretend
[4] gramschmidt: :(Matrix (Expression (Integer))) -- should replace by pretend
```

### warnings 20103: RSDCMPK

```
>compiling RSDCMPK.spad to RSDCMPK.nrlib
```

```
Warnings:
[1] algebraicDecompose: llpwt has no value
[2] internalDecompose: llpwt has no value
[3] internalDecompose: lts has no value
[4] decompose: toSave has no value
[5] upDateBranches: branches1 has no value
[6] upDateBranches: branches2 has no value
```

### warnings 20102: SOLVERAD

```
>compiling SOLVERAD.spad to SOLVERAD.nrlib
```

```
Warnings:
[1] quadratic: not known that (SIGNATURE ** ($ $ (Fraction (Integer)))) is of mode (CATEGORY domain (
[2] findGenZeros: not known that (Ring) is of mode (CATEGORY package (SIGNATURE radicalSolve ((List (
[3] findGenZeros: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE radicalSolve ((
[4] findZeros: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE radicalSolve ((List
[5] radicalRoots: result has no value
[6] solveInner: solutions has no value
```

### warnings 20101: SULS

```
>compiling SULS.spad to SULS.nrlib
```

```
Warnings:
[1] exp: not known that (UnivariateLaurentSeriesConstructorCategory Coef (SparseUnivariateTaylorSeries
```

### warnings 20100: UFPS1

```
>compiling UFPS1.spad to UFPS1.nrlib
```

```

Warnings:
[1] hadamard: z1 has no value
[2] hadamard: z2 has no value

```

### warnings 20099: ULSCONS

```
>compiling ULSCONS.spad to ULSCONS.nrllib
```

```

Warnings:
[1] multiplyCoefficients: z1 has no value
[2] differentiate: z1 has no value
[3] integrate: z has no value
[4] integrate: z1 has no value
[5] termsToOutputForm: 1 has no value

```

### warnings 20098: UPXS

```
>compiling UPXS.spad to UPXS.nrllib
```

```

Warnings:
[1] termsToOutputForm: 1 has no value

```

### warnings 20097: UPXSCONS

```
>compiling UPXSCONS.spad to UPXSCONS.nrllib
```

```

Warnings:
[1] terms: t1 has no value
[2] series: t1 has no value
[3] +: z1 has no value
[4] +: z2 has no value
[5] -: z1 has no value
[6] -: z2 has no value
[7] *: z1 has no value
[8] *: z2 has no value
[9] /: z1 has no value
[10] /: z2 has no value
[11] differentiate: z1 has no value
[12] integrate: z1 has no value

```

## warnings 20096: UTS

```
>compiling UTS.spad to UTS.nrllib
```

```
Warnings:
```

- [1] differentiate: y has no value
- [2] integrate: y has no value

## warnings 20095: GPAFF

```
>compiling GPAFF.spad to GPAFF.nrllib
```

```
Warnings:
```

- [1] lBasis: pretend(Integer) -- should replace by @
- [2] genus: theGenus has no value
- [3] genusNeg: theGenus has no value
- [4] findOrderOfDivisor: pretend(Integer) -- should replace by @
- [5] desingTreeWoFullParam: theTree has no value
- [6] adjunctionDivisor: theAdjDiv has no value
- [7] singularPoints: lesPtsSing has no value
- [8] rationalPoints: lesRatPts has no value
- [9] ZetaFunction: zf has no value
- [10] ZetaFunction: pretend(UnivariateTaylorSeriesCZero (Integer) t) -- should replace by @
- [11] numberOfPlacesOfDegreeUsingZeta: pretend(UnivariateTaylorSeriesCZero (Integer) t) -- should repl
- [12] numberOfPlacesOfDegreeUsingZeta: serdel has no value
- [13] numberOfPlacesOfDegree: calculatedNP has no value
- [14] numberPlacesDegExtDeg: res has no value

## warnings 20094: IRURPK

```
>compiling IRURPK.spad to IRURPK.nrllib
```

```
Warnings:
```

- [1] makeLinearAndMonic: toSave has no value
- [2] rur: toSave has no value

## warnings 20093: LEXTRIPK

```
>compiling LEXTRIPK.spad to LEXTRIPK.nrllib
```

Warnings:

- [1] trueVariables: truels has no value
- [2] lexTriangular: polnum has no value
- [3] lexTriangular: toSave has no value
- [4] squareFreeLexTriangular: polnum has no value
- [5] squareFreeLexTriangular: toSave has no value

## warnings 20092: NORMPK

>compiling NORMPK.spad to NORMPK.nrlib

Warnings:

- [1] normalizedAssociate: mp has no value
- [2] normalizedAssociate: tp has no value

## warnings 20091: PACEXT

>compiling PACEXT.spad to PACEXT.nrlib

Warnings:

- [1] retractToGrn: pretend(Record (: recEl (SparseUnivariatePolynomial \$)) (: recTower (SparseUnivariatePolynomial \$)))
- [2] down: pretend(Record (: recEl (SparseUnivariatePolynomial \$)) (: recTower (SparseUnivariatePolynomial \$)))
- [3] lift: pretend(Record (: recEl (SparseUnivariatePolynomial \$)) (: recTower (SparseUnivariatePolynomial \$)))
- [4] reduce: pretend(Record (: recEl (SparseUnivariatePolynomial \$)) (: recTower (SparseUnivariatePolynomial \$)))
- [5] vectorise: pretend(Record (: recEl (SparseUnivariatePolynomial \$)) (: recTower (SparseUnivariatePolynomial \$)))
- [6] +: recEl has no value
- [7] \*: recEl has no value
- [8] \*: recTower has no value
- [9] inv: pretend(Record (: recEl (SparseUnivariatePolynomial \$)) (: recTower (SparseUnivariatePolynomial \$)))
- [10] -: pretend(Record (: recEl (SparseUnivariatePolynomial \$)) (: recTower (SparseUnivariatePolynomial \$)))
- [11] coerce: pretend(Record (: recEl (SparseUnivariatePolynomial \$)) (: recTower (SparseUnivariatePolynomial \$)))
- [12] fullOutput: pretend(Record (: recEl (SparseUnivariatePolynomial \$)) (: recTower (SparseUnivariatePolynomial \$)))
- [13] definingPolynomial: pretend(Record (: recEl (SparseUnivariatePolynomial \$)) (: recTower (SparseUnivariatePolynomial \$)))
- [14] extDegree: pretend(Record (: recEl (SparseUnivariatePolynomial \$)) (: recTower (SparseUnivariatePolynomial \$)))
- [15] previousTower: pretend(Record (: recEl (SparseUnivariatePolynomial \$)) (: recTower (SparseUnivariatePolynomial \$)))
- [16] name: pretend(Record (: recEl (SparseUnivariatePolynomial \$)) (: recTower (SparseUnivariatePolynomial \$)))

## warnings 20090: RECOP

>compiling RECOP.spad to RECOP.nrlib

Warnings:

```
[1] operatorName: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE evalRec (F (Bas
[2] getShiftRec: z has no value
[3] getShiftRec: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE evalRec (F (Bas
[4] getShiftRec: not known that (Ring) is of mode (CATEGORY package (SIGNATURE evalRec (F (BasicOperat
[5] shiftInfoRec: maxShift has no value
[6] shiftInfoRec: minShift has no value
[7] shiftInfoRec: nextKernel has no value
[8] evalRec: ord has no value
[9] evalRec: ker has no value
[10] evalRec: z has no value
[11] getOrder: %diff has no value
```

## warnings 20089: RURPK

>compiling RURPK.spad to RURPK.nrlib

Warnings:

```
[1] rur: lq has no value
[2] rur: q has no value
[3] rur: toSave has no value
[4] rur: toReturn has no value
```

## warnings 20088: SFRGCD

>compiling SFRGCD.spad to SFRGCD.nrlib

Warnings:

```
[1] stosePrepareSubResAlgo: toSave has no value
[2] stoseInternalLastSubResultant: toReturn has no value
[3] stoseInvertible?sqfreg: x has no value
[4] stoseInvertible?sqfreg: y has no value
[5] stoseInvertibleSetsqfreg: toSave has no value
[6] stoseInvertible?reg: x has no value
[7] stoseInvertible?reg: y has no value
[8] stoseInvertibleSetreg: toSave has no value
```

## warnings 20087: SFQCMPPK

>compiling SFQCMPPK.spad to SFQCMPPK.nrlib

Warnings:

- [1] removeSuperfluousCases: x has no value
- [2] removeSuperfluousCases: y has no value
- [3] removeSuperfluousCases: maxcases has no value
- [4] removeSuperfluousCases: lpwt1 has no value
- [5] removeSuperfluousCases: headmaxcases has no value
- [6] removeSuperfluousCases: toSave has no value
- [7] removeSuperfluousQuasiComponents: maxlts has no value
- [8] removeSuperfluousQuasiComponents: headmaxlts has no value
- [9] removeSuperfluousQuasiComponents: toSave has no value
- [10] branchIfCan: polnum has no value
- [11] prepareDecompose: branches has no value
- [12] prepareDecompose: x has no value
- [13] prepareDecompose: y has no value

## warnings 20086: SRDCMPK

>compiling SRDCMPK.spad to SRDCMPK.nrlib

Warnings:

- [1] algebraicDecompose: lts has no value
- [2] internalDecompose: llpwt has no value
- [3] internalDecompose: lts has no value
- [4] decompose: toSave has no value
- [5] upDateBranches: branches1 has no value
- [6] upDateBranches: branches2 has no value

## warnings 20085: SREGSET

>compiling SREGSET.spad to SREGSET.nrlib

Warnings:

- [1] construct: ts has no value
- [2] internalAugment: lts has no value
- [3] preprocess: lp1 has no value
- [4] preprocess: lp2 has no value
- [5] preprocess: lts has no value

## warnings 20084: ZDSOLVE



```
>compiling ZDSOLVE.spad to ZDSOLVE.nrlib
```

```
Warnings:
```

- [1] squareFree: toSave has no value
- [2] realSolve: not known that (Ring) is of mode (CATEGORY package (SIGNATURE triangSolve ((List (Regu
- [3] realSolve: toSave has no value
- [4] positiveSolve: toSave has no value
- [5] univariateSolve: lq2 has no value

## warnings 20083: GUESS

```
>compiling GUESS.spad to GUESS.nrlib
```

```
Warnings:
```

- [1] guessExpRatAux: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE guess ((L
- [2] guessExpRatAux: not known that (SIGNATURE variable ((Union \$ failed) (Symbol))) is of mode (CATEG
- [3] guessExpRatAux: not known that (PolynomialCategory F (IndexedExponents (OrderedVariableList (const
- [4] guessExpRatAux: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE guess ((List
- [5] guessExpRatAux0: zeros has no value
- [6] guessExpRatAux0: newlist has no value
- [7] guessExpRatAux0: xValues has no value
- [8] guessBinRatAux0: zeros has no value
- [9] guessBinRatAux0: newlist has no value
- [10] guessBinRatAux0: xValues has no value
- [11] diffHP: signature of lhs not unique: (Record (: guessStream (Mapping (Stream (UnivariateFormalPo
- [12] shiftHP: signature of lhs not unique: (Record (: guessStream (Mapping (Stream (UnivariateFormalPo
- [13] guessInterpolate: not known that (FiniteAbelianMonoidRing (Fraction S) (NonNegativeInteger)) is o
- [14] testInterpolant: nonZeroCoefficient has no value
- [15] guessHPaux: guessDegree has no value
- [16] guessHPaux: maxEta has no value
- [17] guessHPaux: reslist has no value
- [18] guessHPaux: eta has no value
- [19] guess: res has no value

## warnings 20082: INFCLST

```
>compiling INFCLSPT.spad to INFCLSPT.nrlib
```

```
Warnings:
```

- [1] create: P has no value
- [2] create: chartA has no value
- [3] subMultV: :Rep -- should replace by pretend
- [4] setsubmult!: :Rep -- should replace by pretend

```

[5] pointV: :Rep -- should replace by pretend
[6] symbNameV: :Rep -- should replace by pretend
[7] curveV: :Rep -- should replace by pretend
[8] localPointV: :Rep -- should replace by pretend
[9] multV: :Rep -- should replace by pretend
[10] chartV: :Rep -- should replace by pretend
[11] excpDivV: :Rep -- should replace by pretend
[12] localParamV: :Rep -- should replace by pretend
[13] actualExtensionV: :Rep -- should replace by pretend
[14] setpoint!: :Rep -- should replace by pretend
[15] setcurve!: :Rep -- should replace by pretend
[16] setlocalPoint!: :Rep -- should replace by pretend
[17] setmult!: :Rep -- should replace by pretend
[18] setchart!: :Rep -- should replace by pretend
[19] setlocalParam!: :Rep -- should replace by pretend
[20] setexcpDiv!: :Rep -- should replace by pretend
[21] setsymbName!: :Rep -- should replace by pretend

```

## warnings 20081: GUESSAN

```
>compiling GUESSAN.spad to GUESSAN.nrlib
```

Warnings:

```

[1] not known that (RetractableTo (AlgebraicNumber)) is of mode (CATEGORY domain (IF (has
[2] not known that (CombinatorialOpsCategory) is of mode (CATEGORY domain (IF (has (Intege
[3] not known that (SIGNATURE ** ($ $ $)) is of mode (CATEGORY domain (IF (has (Integer) (

```

## warnings 20080: GUESSINT

```
>compiling GUESSINT.spad to GUESSINT.nrlib
```

Warnings:

```

[1] not known that (CombinatorialOpsCategory) is of mode (CATEGORY domain (IF (has (Intege
[2] not known that (SIGNATURE ** ($ $ $)) is of mode (CATEGORY domain (IF (has (Integer) (

```

## warnings 20079: GUESSP

```
>compiling GUESSP.spad to GUESSP.nrlib
```

Warnings:

```

[1] not known that (CombinatorialOpsCategory) is of mode (CATEGORY domain (IF (has (Intege

```

```
[2] not known that (SIGNATURE ** ($ $ $)) is of mode (CATEGORY domain (IF (has (Integer) (IntegralDomain
```

## warnings 20078: GUESSUP

```
>compiling GUESSUP.spad to GUESSUP.nrlib
```

```
Warnings:
```

```
[1] not known that (RetractableTo (Fraction (MyUnivariatePolynomial q (Integer)))) is of mode (CATEGORY domain (IF (has (Integer) (IntegralDomain
```

## warnings 20077: GUESSEF

```
>compiling GUESSEF.spad to GUESSEF.nrlib
```

```
Warnings:
```

```
[1] not known that (CombinatorialOpsCategory) is of mode (CATEGORY domain (IF (has (Integer) (IntegralDomain
```

```
[2] not known that (SIGNATURE ** ($ $ $)) is of mode (CATEGORY domain (IF (has (Integer) (IntegralDomain
```

## warnings 20076: PAFF

```
>compiling PAFF.spad to PAFF.nrlib
```

```
Warnings:
```

```
[1] homogenize: not known that (FiniteAbelianMonoidRing K (DirectProduct (call LENGTH symb) (NonNegativeIntegerRing
```

## warnings 20075: PAFFFF

```
>compiling PAFFFF.spad to PAFFFF.nrlib
```

```
Warnings:
```

```
[1] homogenize: not known that (FiniteAbelianMonoidRing K (DirectProduct (call LENGTH symb) (NonNegativeIntegerRing
```

```
(1) -> copying PAFFFF.nrlib to PAFFFF.o
```

```
c11 making /research/test/mnt/ubuntu/algebra/PAFF.o from /research/test/int/algebra/cliique2.spad
```

```
c11 making /research/test/mnt/ubuntu/algebra/PAFFFF.o from /research/test/int/algebra/cliique2.spad
```

```
Warnings:
```

```
[1] homogenize: not known that (FiniteAbelianMonoidRing K (DirectProduct (call LENGTH symb) (NonNegativeIntegerRing
```

**warnings 20074: CLAGG**

```
>compiling CLAGG.spad to CLAGG.nrlib
```

```
Warnings:
 [1] remove: y has no value
```

**warnings 20072: ES**

```
>compiling ES.spad to ES.nrlib
```

```
Warnings:
 [1] tower: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE odd? ((Boole
 [2] freeOf?: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE odd? ((Boole
 [3] eval: y has no value
 [4] eval: IN has no value
 [5] eval: f has no value
 [6] eval: s has no value
 [7] map: IN has no value
 [8] map: x has no value
```

**warnings 20071: EUCDOM**

```
>compiling EUCDOM.spad to EUCDOM.nrlib
```

```
Warnings:
 [1] principalIdeal: coef1 has no value
 [2] principalIdeal: coef2 has no value
```

**warnings 20070: FFIELDC**

```
>compiling FFIELDC.spad to FFIELDC.nrlib
```

```
Warnings:
 [1] conditionP: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE order ((Posit
 [2] order: signature of lhs not unique: (PositiveInteger)S chosen
 [3] order: ord has no value
 [4] discreteLog: disc1 has no value
 [5] discreteLog: disclog has no value
 [6] discreteLog: not known that (IntegralDomain) is of mode (CATEGORY domain (SIGNATURE or
```

```
[7] gcdPolynomial: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE order ((PositiveInteg
```

## warnings 20069: GCDDOM

```
>compiling GCDDOM.spad to GCDDOM.nrlib
```

```
Warnings:
```

```
[1] gcdPolynomial: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE gcdPolynomial ((Spars
```

## warnings 20068: HOAGG

```
>compiling HOAGG.spad to HOAGG.nrlib
```

```
Warnings:
```

```
[1] eval: x has no value
```

```
[2] count: y has no value
```

```
[3] member?: x has no value
```

## warnings 20067: ILIST

```
>compiling ILIST.spad to ILIST.nrlib
```

```
Warnings:
```

```
[1] latex: s has no value
```

## warnings 20066: INS

```
>compiling INS.spad to INS.nrlib
```

```
Warnings:
```

```
[1] factor: not known that (IntegralDomain) is of mode (CATEGORY domain (SIGNATURE invmod (S S S)) (S
```

```
[2] patternMatch: not known that (SetCategory) is of mode (CATEGORY domain (SIGNATURE invmod (S S S))
```

```
[3] powmod: y has no value
```

**warnings 20065: INT**

```
>compiling INT.spad to INT.nrlib
```

```
Warnings:
```

- [1] OMwrite: pretend(String) -- should replace by @
- [2] hash: signature of lhs not unique: \$\$ chosen
- [3] factorPolynomial: not known that (UnivariatePolynomialCategory (Integer)) is of mode (CAT
- [4] factorPolynomial: x1 has no value
- [5] gcdPolynomial: not known that (UnivariatePolynomialCategory (Integer)) is of mode (CAT

**warnings 20064: ISTRING**

```
>compiling ISTRING.spad to ISTRING.nrlib
```

```
Warnings:
```

- [1] split: j has no value

**warnings 20062: LSAGG**

```
>compiling LSAGG.spad to LSAGG.nrlib
```

```
Warnings:
```

- [1] removeDuplicates!: x has no value

**warnings 20061: POLYCAT**

```
>compiling POLYCAT.spad to POLYCAT.nrlib
```

```
Warnings:
```

- [1] eval: IN has no value
- [2] coefficient: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE gcdPolynomia
- [3] totalDegree: w has no value
- [4] reducedSystem: IN has no value
- [5] reducedSystem: r has no value
- [6] reducedSystem: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE gcdPolynom
- [7] solveLinearPolynomialEquation: not known that (Ring) is of mode (CATEGORY domain (SIGN
- [8] factorPolynomial: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE gcdPoly
- [9] factor: not known that (IntegralDomain) is of mode (CATEGORY domain (SIGNATURE gcdPoly

```
[10] conditionP: :(Integer) -- should replace by pretend
[11] patternMatch: not known that (SetCategory) is of mode (CATEGORY domain (SIGNATURE gcdPolynomial
```

## warnings 20060: PSETCAT

```
>compiling PSETCAT.spad to PSETCAT.nrlib
```

Warnings:

```
[1] variables1: z2 has no value
[2] variables1: z1 has no value
[3] variables2: z2 has no value
[4] variables2: z1 has no value
[5] collectUnder: lq has no value
[6] collectUpper: lq has no value
[7] collect: lq has no value
[8] sort: us has no value
[9] sort: vs has no value
[10] sort: ws has no value
[11] localTriangular?: z2 has no value
[12] localTriangular?: z1 has no value
[13] localTriangular?: q has no value
[14] headRemainder: r has no value
[15] rewriteIdealWithHeadRemainder: rs has no value
[16] rewriteIdealWithRemainder: rs has no value
```

## warnings 20059: QFCAT

```
>compiling QFCAT.spad to QFCAT.nrlib
```

Warnings:

```
[1] reducedSystem: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE < ((Boolean) A A)) (S
[2] patternMatch: not known that (SetCategory) is of mode (CATEGORY domain (SIGNATURE < ((Boolean) A A
```

## warnings 20058: RNS

```
>compiling RNS.spad to RNS.nrlib
```

Warnings:

```
[1] patternMatch: not known that (SetCategory) is of mode (CATEGORY domain (SIGNATURE round (S S)) (S
```

**warnings 20057: SINT**

```
>compiling SINT.spad to SINT.nrllib
```

```
Warnings:
```

```
[1] hash: signature of lhs not unique: $$ chosen
```

**warnings 20056: SYMBOL**

```
>compiling SYMBOL.spad to SYMBOL.nrllib
```

```
Warnings:
```

```
[1] latex: s has no value
```

```
[2] latex: sc has no value
```

**warnings 20055: TSETCAT**

```
>compiling TSETCAT.spad to TSETCAT.nrllib
```

```
Warnings:
```

```
[1] basicSet: p has no value
```

```
[2] basicSet: gps has no value
```

```
[3] basicSet: bps has no value
```

```
[4] initials: lip has no value
```

```
[5] initiallyReduced?: red has no value
```

```
[6] reduce: signature of lhs not unique: PPS(Mapping P P P)(Mapping (Boolean) P P) chosen
```

```
[7] rewriteSetWithReduction: rs has no value
```

```
[8] select: signature of lhs not unique: (Union P failed)SV chosen
```

```
[9] collectQuasiMonic: newlp has no value
```

**warnings 20054: UFD**

```
>compiling UFD.spad to UFD.nrllib
```

```
Warnings:
```

```
[1] squareFreePart: not known that (IntegralDomain) is of mode (CATEGORY domain (SIGNATURE
```



Warnings:

```
[1] monomial: x1 has no value
[2] solveLinearPolynomialEquation: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE coerce (S S))) (SIGNATURE coerce (S S)) (SIGNATURE coerce (S S))
[3] factorPolynomial: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE coerce (S S))) (SIGNATURE coerce (S S)) (SIGNATURE coerce (S S))
[4] factor: not known that (IntegralDomain) is of mode (CATEGORY domain (SIGNATURE coerce (S S))) (SIGNATURE coerce (S S)) (SIGNATURE coerce (S S))
[5] elt: not known that (IntegralDomain) is of mode (CATEGORY domain (SIGNATURE coerce (S S))) (SIGNATURE coerce (S S)) (SIGNATURE coerce (S S))
```

Warnings:

```
[1] generalCoefficient: res has no value
[2] generalInterpolation: x has no value
[3] generalInterpolation: y has no value
[4] nextVector!: x has no value
[5] interpolate: s has no value
[6] interpolate: u has no value
[7] fffg: M has no value
```

Warnings:

```
[1] generalInterpolation: x has no value
[2] generalInterpolation: y has no value
```

Warnings:

```
[1] compare: equal has no value
[2] initialize: mat has no value
[3] unknown Functor code (error FFHOM: one extension degree must divide the other one)
```

**warnings 20049: FFPOLY**

```
>compiling FFPOLY.spad to FFPOLY.nrlib
```

```
Warnings:
```

```
[1] leastAffineMultiple: coeffVector has no value
[2] primitive?: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE pr
[3] nextSubset: restOfs has no value
[4] nextSubset: noGap has no value
[5] nextSubset: i has no value
[6] nextIrreduciblePoly: fcopy has no value
[7] nextIrreduciblePoly: s has no value
[8] nextPrimitivePoly: fcopy has no value
[9] nextPrimitivePoly: term has no value
[10] nextPrimitivePoly: noGenerator has no value
[11] nextPrimitivePoly: c has no value
[12] nextPrimitivePoly: weight has no value
[13] nextPrimitivePoly: s has no value
[14] nextPrimitivePoly: l has no value
[15] nextNormalPoly: fcopy has no value
[16] nextNormalPoly: l has no value
[17] nextNormalPoly: a has no value
[18] nextNormalPoly: s has no value
[19] nextNormalPrimitivePoly: fcopy has no value
[20] nextNormalPrimitivePoly: term has no value
[21] nextNormalPrimitivePoly: noGenerator has no value
[22] nextNormalPrimitivePoly: c has no value
[23] nextNormalPrimitivePoly: la has no value
[24] nextNormalPrimitivePoly: a has no value
[25] nextNormalPrimitivePoly: middlelookuplist has no value
[26] nextNormalPrimitivePoly: middlepol has no value
[27] nextNormalPrimitivePoly: weight has no value
[28] nextNormalPrimitivePoly: s has no value
[29] nextNormalPrimitivePoly: lc has no value
[30] random: polRepr has no value
```

**warnings 20048: FFSQFR**

```
>compiling FFSQFR.spad to FFSQFR.nrlib
```

```
Warnings:
```

```
[1] rawMusser: A has no value
[2] rawMusser: decomposition has no value
```

```
[3] oneYunStep: B has no value
```

### warnings 20047: FFSLPE

```
>compiling FFSLPE.spad to FFSLPE.nrlib
```

```
Warnings:
```

```
[1] solveLinearPolynomialEquation: slpePrime has no value
[2] solveLinearPolynomialEquation: oldtable has no value
```

### warnings 20046: FGLMICPK

```
>compiling FGLMICPK.spad to FGLMICPK.nrlib
```

```
Warnings:
```

```
[1] zeroDim?: not known that (Ring) is of mode (CATEGORY package (SIGNATURE zeroDimensional? ((Boolean
[2] zeroDim?: lv has no value
```

### warnings 20045: FFF

```
>compiling FFF.spad to FFF.nrlib
```

```
Warnings:
```

```
[1] createLowComplexityTable: k has no value
[2] createLowComplexityTable: t1 has no value
[3] createLowComplexityTable: a has no value
[4] createLowComplexityTable: pretend(NonNegativeInteger) -- should replace by @
[5] createLowComplexityTable: l has no value
[6] createMultiplicationTable: l has no value
```

### warnings 20044: FORMULA

```
>compiling FORMULA.spad to FORMULA.nrlib
```

```
Warnings:
```

```
[1] display: pretend(Integer) -- should replace by @
[2] coerce: pretend(Integer) -- should replace by @
[3] stringify: pretend(String) -- should replace by @
```

```

[4] splitLong1: l has no value
[5] splitLong1: ls has no value
[6] splitLong1: s has no value
[7] formatSpecial: form has no value
[8] formatPlex: s has no value
[9] formatFormula: pretend(Boolean) -- should replace by @
[10] formatFormula: :(Integer) -- should replace by pretend

```

### warnings 20043: FORT

```
>compiling FORT.spad to FORT.nrllib
```

```

Warnings:
[1] linkToFortran: fst has no value

```

### warnings 20042: FRAC

```
>compiling FRAC.spad to FRAC.nrllib
```

```

Warnings:
[1] OMwrite: pretend(String) -- should replace by @
[2] gcdPolynomial: x has no value
[3] factorPolynomial: x has no value
[4] factorPolynomial: den1 has no value
[5] factorSquareFreePolynomial: x has no value
[6] factorSquareFreePolynomial: den1 has no value

```

### warnings 20041: FTEM

```
>compiling FTEM.spad to FTEM.nrllib
```

```

Warnings:
[1] processTemplate: active has no value

```

### warnings 20040: GENEEZ

```
>compiling GENEEZ.spad to GENEEZ.nrllib
```

```
Warnings:
[1] reduction: x has no value
[2] liftSol: x has no value
[3] solveid: x has no value
```

## warnings 20039: GENMFACT

```
>compiling GENMFACT.spad to GENMFACT.nrlib
```

```
Warnings:
[1] factor: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE factor ((Factored
```

## warnings 20038: GENPGCD

```
>compiling GENPGCD.spad to GENPGCD.nrlib
```

```
Warnings:
[1] gcdPolynomial: canonical has no value
[2] gcdSameVariables: x has no value
[3] gcdSameVariables: not known that (Ring) is of mode (CATEGORY package (SIGNATURE gcdPolynomial ((S
[4] gcdSameVariables: y has no value
[5] gcdSameVariables: z has no value
[6] gcdSameVariables: x1 has no value
[7] gcdSameVariables: y1 has no value
[8] gcdSameVariables: up1 has no value
[9] gcdSameVariables: up2 has no value
[10] lift: x has no value
[11] lift: y has no value
[12] lift: x1 has no value
[13] lift: x2 has no value
[14] lift: not known that (Ring) is of mode (CATEGORY package (SIGNATURE gcdPolynomial ((SparseUnivar
[15] recursivelyGCDCoefficients: x has no value
[16] flatten: x has no value
```

## warnings 20037: GALFACTU

```
>compiling GALFACTU.spad to GALFACTU.nrlib
```

```
Warnings:
[1] singleFactorBound: not known that (Ring) is of mode (CATEGORY F (SIGNATURE Gamma (F F)))
```

**warnings 20036: GB**

```
>compiling GB.spad to GB.nrllib
```

```
Warnings:
```

```
[1] groebner: y has no value
```

**warnings 20035: GBEUCLID**

```
>compiling GBEUCLID.spad to GBEUCLID.nrllib
```

```
Warnings:
```

```
[1] strongGbasis: z2 has no value
```

```
[2] strongGbasis: z1 has no value
```

**warnings 20034: GBF**

```
>compiling GBF.spad to GBF.nrllib
```

```
Warnings:
```

```
[1] createGroebnerBases: nP has no value
```

```
[2] createGroebnerBases: allReducedFactors has no value
```

```
[3] createGroebnerBases: y has no value
```

```
[4] createGroebnerBases: x has no value
```

```
[5] createAllFactors: x has no value
```

```
[6] createAllFactors: y has no value
```

```
[7] factorGroebnerBasis: fctr has no value
```

```
[8] groebnerFactorize: signature of lhs not unique: (List (List Dpol))(List Dpol)(List Dpol)
```

```
[9] groebnerFactorize: x has no value
```

```
[10] groebnerFactorize: y has no value
```

**warnings 20033: GBINTERN**

```
>compiling GBINTERN.spad to GBINTERN.nrllib
```

```
Warnings:
```

```
[1] gbasis: z2 has no value
```

```
[2] gbasis: z1 has no value
```

```
[3] redPo: m has no value
```

## warnings 20032: GHENSEL

```
>compiling GHENSEL.spad to GHENSEL.nrllib
```

```
Warnings:
```

- [1] reduction: x has no value
- [2] mQuo: x has no value
- [3] genFact: auxfl has no value
- [4] HenselLift: constp has no value
- [5] HenselLift: fln has no value
- [6] completeHensel: factlist has no value
- [7] completeHensel: finallist has no value
- [8] completeHensel: aux has no value
- [9] completeHensel: auxfl has no value

## warnings 20031: GOSPER

```
>compiling GOSPER.spad to GOSPER.nrllib
```

```
Warnings:
```

- [1] PQ2R: x has no value
- [2] PQ2R: y has no value
- [3] UP2QIfCan: x has no value
- [4] UP2QIfCan: y has no value
- [5] GosperF: not known that (Ring) is of mode (CATEGORY package (SIGNATURE GopersMethod ((Union Q fa

## warnings 20030: GRIMAGE

```
>compiling GRIMAGE.spad to GRIMAGE.nrllib
```

```
Warnings:
```

- [1] makeGraphImage: signature of lhs not unique: \$(List (List (Point (DoubleFloat)))) chosen

## warnings 20029: GROEBSOL

```
>compiling GROEBSOL.spad to GROEBSOL.nrllib
```

```
Warnings:
```

- [1] testPower: not known that (Ring) is of mode (CATEGORY package (SIGNATURE groebSolve ((List (List
- [2] testGenPos: newlpol has no value

```
[3] groebSolve: result has no value
```

### warnings 20028: HEUGCD

```
>compiling HEUGCD.spad to HEUGCD.nrlib
```

```
Warnings:
```

```
[1] lintgcd: z1 has no value
[2] lintgcd: z2 has no value
[3] localgcd: flag has no value
[4] localgcd: result has no value
[5] internal: Cgcd has no value
[6] internal: contgcd has no value
[7] internal: ans has no value
```

### warnings 20027: HTMLFORM

```
>compiling HTMLFORM.spad to HTMLFORM.nrlib
```

```
Warnings:
```

```
[1] exprex: s has no value
[2] outputTree: allString has no value
[3] formatNaryNoGroup: tags has no value
```

### warnings 20026: ICARD

```
>compiling ICARD.spad to ICARD.nrlib
```

```
Warnings:
```

```
[1] elt: pretend(String) -- should replace by @
```

### warnings 20025: ICDEN

```
>compiling ICDEN.spad to ICDEN.nrlib
```

```
Warnings:
```

```
[1] clearDenominator: x has no value
[2] splitDenominator: x has no value
```



## warnings 20024: IDECOMP

```
>compiling IDECOMP.spad to IDECOMP.nrllib
```

Warnings:

- [1] internalForm: not known that (PolynomialCategory (Fraction (Integer)) (DirectProduct nv (NonNegat
- [2] internalForm: not known that (PolynomialCategory (Fraction (Polynomial (Integer))) (DirectProduct
- [3] externalForm: not known that (PolynomialCategory (Fraction (Polynomial (Integer))) (DirectProduct
- [4] externalForm: not known that (PolynomialCategory (Fraction (Integer)) (DirectProduct nv (NonNegat
- [5] rearrange: z2 has no value
- [6] rearrange: z1 has no value
- [7] zeroRadComp: not known that (PolynomialCategory (Fraction (Polynomial (Integer))) (DirectProduct n
- [8] zeroRadComp: not known that (Ring) is of mode (CATEGORY package (SIGNATURE zeroDimPrime? ((Boolea
- [9] findvar: lmonicvar has no value
- [10] zeroPrimDecomp: not known that (PolynomialCategory (Fraction (Polynomial (Integer))) (DirectProd
- [11] testPower: :(Integer) -- should replace by pretend
- [12] primaryDecomp: not known that (PolynomialCategory (Fraction (Integer)) (DirectProduct nv (NonNeg
- [13] contract: IN has no value
- [14] contract: vv has no value
- [15] contract: not known that (PolynomialCategory (Fraction (Integer)) (DirectProduct nv (NonNegative

## warnings 20023: IIARRAY2

```
>compiling IIARRAY2.spad to IIARRAY2.nrllib
```

Warnings:

- [1] latex: s has no value

## warnings 20022: IMATLIN

```
>compiling IMATLIN.spad to IMATLIN.nrllib
```

Warnings:

- [1] nullSpace: basis has no value
- [2] determinant: ans has no value
- [3] generalizedInverse: not known that (MatrixCategory FSUP VFSUP VFSUP) is of mode (CATEGORY domain
- [4] generalizedInverse: r1 has no value
- [5] generalizedInverse: z1 has no value

**warnings 20021: IMATQF**

```
>compiling IMATQF.spad to IMATQF.nrllib
```

```
Warnings:
```

- ```
[1] qfMat:  r1 has no value
[2] nullSpace: not known that (FiniteLinearAggregate QF) is of mode (CATEGORY Col2 (ATTRIB
```

warnings 20020: INMODGCD

```
>compiling INMODGCD.spad to INMODGCD.nrllib
```

```
Warnings:
```

- ```
[1] reduction: r1 has no value
```

**warnings 20019: INNMFAC**

```
>compiling INNMFAC.spad to INNMFAC.nrllib
```

```
Warnings:
```

- ```
[1] supFactor:  factorlist has no value
[2] supFactor:  irr has no value
[3] supFactor:  pow has no value
[4] varChoose: not known that (Ring) is of mode (CATEGORY package (SIGNATURE factor ((Fact
[5] intChoose:  unifact has no value
[6] intChoose:  int has no value
[7] simplify:  pol1 has no value
[8] simplify:  factorlist has no value
[9] intfact:  unifact has no value
[10] intfact:  lpol has no value
[11] intfact:  factfin has no value
[12] monicMfpol:  z1 has no value
[13] mFactor: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE fact
[14] mFactor:  flead has no value
[15] mFactor:  irr has no value
[16] mFactor:  pow has no value
[17] mFactor:  x has no value
[18] mFactor:  ffactor has no value
[19] mFactor:  lvar has no value
[20] mFactor:  factorlist has no value
```

warnings 20018: INTERGB

```
>compiling INTERGB.spad to INTERGB.nrlib
```

```
Warnings:
```

- [1] coerceRtoDwithPF: not known that (PolynomialCategory (PrimeField q) E OV) is of mode (CATEGORY domain (SIGNATURE reord
- [2] coerceDtoRwithPF: not known that (PolynomialCategory (PrimeField q) E OV) is of mode (CATEGORY domain (SIGNATURE reord
- [3] coerceRtoD: not known that (PolynomialCategory K E OV) is of mode (CATEGORY domain (SIGNATURE reord
- [4] coerceDtoR: not known that (PolynomialCategory K E OV) is of mode (CATEGORY domain (SIGNATURE reord
- [5] groebner: not known that (PolynomialCategory (PrimeField q) E OV) is of mode (CATEGORY domain (SIGNATURE reord
- [6] groebner: not known that (PolynomialCategory K E OV) is of mode (CATEGORY domain (SIGNATURE reord

warnings 20017: INTRF

```
>compiling INTRF.spad to INTRF.nrlib
```

```
Warnings:
```

- [1] infieldIntegrate: x1 has no value
- [2] internalIntegrate: x1 has no value
- [3] extendedIntegrate: x1 has no value
- [4] limitedIntegrate: x1 has no value

warnings 20016: INTSLPE

```
>compiling INTSLPE.spad to INTSLPE.nrlib
```

```
Warnings:
```

- [1] solveLinearPolynomialEquation: slpePrime has no value
- [2] solveLinearPolynomialEquation: oldtable has no value

warnings 20015: INTTR

```
>compiling INTTR.spad to INTTR.nrlib
```

```
Warnings:
```

- [1] UP2UP2: x has no value
- [2] UP2UPR: x has no value
- [3] UP22UPR: x has no value
- [4] monomialIntegrate: x1 has no value
- [5] expintegratepoly: coef0 has no value

```

[6] primplimintfrac: x1 has no value
[7] explimintfrac: x1 has no value
[8] explimintfrac: logand has no value
[9] explimintfrac: coeff has no value
[10] tanintegrate: x has no value

```

warnings 20014: ISUMP

```
>compiling ISUMP.spad to ISUMP.nrlib
```

Warnings:

```

[1] pmul: x has no value
[2] sum: not known that (Ring) is of mode (CATEGORY package (SIGNATURE sum ((Record (: num

```

warnings 20013: LAUPOL

```
>compiling LAUPOL.spad to LAUPOL.nrlib
```

Warnings:

```

[1] not known that (Ring) is of mode (CATEGORY domain (SIGNATURE separate ((Record (: poly

```

warnings 20012: LEADCDET

```
>compiling LEADCDET.spad to LEADCDET.nrlib
```

Warnings:

```

[1] distFact: c has no value

```

warnings 20011: LGROBP

```
>compiling LGROBP.spad to LGROBP.nrlib
```

Warnings:

```

[1] tototex: result has no value
[2] minPol: :(NonNegativeInteger) -- should replace by pretend
[3] intcompBasis: part has no value
[4] linGenPos: :(PositiveInteger) -- should replace by pretend
[5] linGenPos: result has no value

```

```
[6] groebgen: :(NonNegativeInteger) -- should replace by pretend
[7] groebgen: not known that (Ring) is of mode (CATEGORY package (SIGNATURE linGenPos ((Record (: gbl
```

warnings 20010: LIMITRF

```
>compiling LIMITRF.spad to LIMITRF.nrlib
```

```
Warnings:
[1] limit: y has no value
[2] complexLimit: y has no value
```

warnings 20009: LINDEP

```
>compiling LINDEP.spad to LINDEP.nrlib
```

```
Warnings:
[1] linearlyDependent?: not known that (Ring) is of mode (CATEGORY package (SIGNATURE linearlyDepender
[2] solveLinear: z has no value
[3] solveLinear: z1 has no value
```

warnings 20008: LISYSER

```
>compiling LISYSER.spad to LISYSER.nrlib
```

```
Warnings:
[1] finiteSeries2LinSys: pretend(NonNegativeInteger) -- should replace by @
[2] finiteSeries2LinSysW0Vectorise: pretend(NonNegativeInteger) -- should replace by @
```

warnings 20007: LPEFRAC

```
>compiling LPEFRAC.spad to LPEFRAC.nrlib
```

```
Warnings:
[1] solveLinearPolynomialEquationByFractions: x has no value
[2] solveLinearPolynomialEquationByFractions: not known that (Ring) is of mode (CATEGORY package (SIG
```

warnings 20006: LSPP

```
>compiling LSPP.spad to LSPP.nrllib
```

```
Warnings:
```

```
[1] poly2vect: not known that (Ring) is of mode (CATEGORY package (SIGNATURE linSolve ((Re
```

warnings 20005: MATLIN

```
>compiling MATLIN.spad to MATLIN.nrllib
```

```
Warnings:
```

```
[1] minorDet:  ans has no value
[2] elRow2!:   r1 has no value
[3] elColumn2!: r1 has no value
[4] nullSpace: r2 has no value
[5] nullSpace: r1 has no value
[6] rowEchelon: xnj has no value
```

warnings 20004: MCDEN

```
>compiling MCDEN.spad to MCDEN.nrllib
```

```
Warnings:
```

```
[1] clearDenominator: x has no value
[2] splitDenominator: x has no value
```

warnings 20003: MDDFACT

```
>compiling MDDFACT.spad to MDDFACT.nrllib
```

```
Warnings:
```

```
[1] reduction:  i1 has no value
[2] ddfact:     ans has no value
[3] ddfact:     :(Integer) -- should replace by @
[4] sepFact1:   ans has no value
[5] sepFact1:   stack has no value
[6] probSplit:  :(NonNegativeInteger) -- should replace by pretend
```

warnings 20002: MFINFACT

```
>compiling MFINFACT.spad to MFINFACT.nrlib
```

```
Warnings:
```

```
[1] pretendOV -- should replace by @
[2] supFactor: p1 has no value
[3] supFactor: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE factor ((Factor
[4] supFactor: p has no value
[5] factor: p1 has no value
[6] mFactor: irr has no value
[7] mFactor: pow has no value
[8] mFactor: not known that (Ring) is of mode (CATEGORY package (SIGNATURE factor ((Factored PG) PG))
[9] mFactor: ffactor has no value
[10] mFactor: factorlist has no value
[11] pushdcoef: f1 has no value
[12] intfact: unifact has no value
[13] intfact: lpol has no value
[14] intfact: factfin has no value
[15] intChoose: newm has no value
[16] intChoose: unifact has no value
[17] intChoose: int has no value
[18] simplify: pol1 has no value
[19] simplify: factorlist has no value
```

warnings 20000: MLIFT

```
>compiling MLIFT.spad to MLIFT.nrlib
```

```
Warnings:
```

```
[1] Use: import (GenExEuclid R (SparseUnivariatePolynomial R))
[2] Use: import (NPCoef (SparseUnivariatePolynomial R) E OV R P)
[3] Use: import (IntegerCombinatoricFunctions (Integer))
[4] corrPoly: p1 has no value
[5] corrPoly: p has no value
[6] lifting1: p1 has no value
[7] lifting: nplist has no value
[8] normalDerivM: not known that (Ring) is of mode (CATEGORY package (SIGNATURE corrPoly ((Union (List
```

errors 10000: FT

```
>compiling FT.spad to FT.nrlib
```

Semantic Errors:

[1] void is not a known type

errors 10001: ASP34

>compiling ASP34.spad to ASP34.nrlib

Semantic Errors:

[1] coerce: code is BOTH a variable and a literal

errors 10002: MOEBIUS

>compiling MOEBIUS.spad to MOEBIUS.nrlib

Semantic Errors:

[1] a is BOTH a variable and a literal
[2] b is BOTH a variable and a literal
[3] c is BOTH a variable and a literal
[4] d is BOTH a variable and a literal
[5] eval: a is BOTH a variable and a literal
[6] eval: b is BOTH a variable and a literal
[7] eval: c is BOTH a variable and a literal
[8] eval: d is BOTH a variable and a literal
[9] proportional?: a is BOTH a variable and a literal
[10] proportional?: b is BOTH a variable and a literal
[11] proportional?: c is BOTH a variable and a literal
[12] proportional?: d is BOTH a variable and a literal

errors 10003: POLYVEC

>compiling POLYVEC.spad to POLYVEC.nrlib

Semantic Errors:

[1] vectorcombination: i has two modes:

errors 10004: EQ

>compiling EQ.spad to EQ.nrlib

Semantic Errors:

[1] factorAndSplit: rcf has two modes:

errors 10005: MULTSQFR

>compiling MULTSQFR.spad to MULTSQFR.nrlib

Semantic Errors:

[1] squareFree: ff has two modes:

errors 10006: ASP30

>compiling ASP30.spad to ASP30.nrlib

Semantic Errors:

[1] coerce: code is BOTH a variable and a literal

errors 10007: ASP8

>compiling ASP8.spad to ASP8.nrlib

Semantic Errors:

[1] coerce: code is BOTH a variable and a literal

errors 10008: ASP9

>compiling ASP9.spad to ASP9.nrlib

Semantic Errors:

[1] coerce: code is BOTH a variable and a literal

errors 10009: ISUPS

>compiling ISUPS.spad to ISUPS.nrlib

Semantic Errors:

[1] compose0: k is BOTH a variable and a literal

errors 10010: SYMS

>compiling SYMS.spad to SYMS.nrllib

Semantic Errors:

[1] Domain is not a known type

[2] void is not a known type

errors 10011: ASP31

>compiling ASP31.spad to ASP31.nrllib

Semantic Errors:

[1] coerce: code is BOTH a variable and a literal

errors 10012: ASP77

>compiling ASP77.spad to ASP77.nrllib

Semantic Errors:

[1] coerce: code is BOTH a variable and a literal

errors 10013: FC

>compiling FC.spad to FC.nrllib

Semantic Errors:

[1] common: name is BOTH a variable and a literal

[2] common: contents is BOTH a variable and a literal

[3] forLoop: body is BOTH a variable and a literal

errors 10014: ODERTRIC

```
>compiling ODERTRIC.spad to ODERTRIC.nrllib
```

```
Semantic Errors:
```

```
[1] RatODETools is not a known type
```

errors 10015: NORMPK

```
>compiling NORMPK.spad to NORMPK.nrllib
```

```
Semantic Errors:
```

```
[1] recip: hesrg has two modes:
```

bug 7233: fill! operation from U8Vector does not show up

```
)d op fill!
```

```
There are 2 exposed functions called fill! :
```

```
[1] (D,D1) -> D from D
      if D has ARR2CAT(D1,D2,D3) and D1 has TYPE and D2 has FLAGG
      (D1) and D3 has FLAGG(D1)
[2] (D,D1) -> D from D
      if D has shallowlyMutable and D has IXAGG(D2,D1) and D2 has
      SETCAT and D1 has TYPE
```

```
Examples of fill! from TwoDimensionalArrayCategory
```

```
arr : ARRAY2 INT := new(5,4,0)
fill!(arr,10)
```

```
Examples of fill! from IndexedAggregate
```

bug 7232: Cannot convert from type Symbol to PI for value #3

```
)clear all
foo(E:PI,totient:INT,N:INT):LIST(INT) ==
```

```

modN:=IntegerMod(N)
modtotient:=IntegerMod(totient)
D:=[x for x in 1..totient | (x*E)::modtotient = 1].1
secret:=(42^E)::modN
clear:=(secret^D)::modN
[secret,D,E,N]

P:=11
Q:=17
N:=P*Q
totient:=(P-1)*(Q-1)
t1:=[x for x in 1..totient | gcd(totient,x) = 1]
[foo(t1.y,totient,N) for y in 1..#t1]
-- fails with
-- Cannot convert from type Symbol to PositiveInteger for value #3

```

bug 7231: D2 is not of type SEQUENCE

```

see dop.input.pamphlet S 69
)d op coerce

```

bug 7230: richhyper1000-1098 S 272 failed

```

Bisect to see when this started

```

bug 7229:)show FortranCode has NIL for the containing file

```

similarly
FortranProgram, FortranType, ROIRC

)show InfClsPt simply fails
)show MyExpression simply fails
)show MyUnivariatePolynomial simply fails

)show PackageForAlgebraicFunctionField fails with:
  listSort: second argument must be a list

)show BSD lists the source file as Makefile.pamphlet

```

bug 7228: graphics crash running space3

The src/input/space3.input causes an early exit

bug 7227: Unexpected end of #input stream

see exampleagcode.output, test 3. Now gives the result

Unexpected end of

```
#<input stream "/research/test/mnt/ubuntu/algebra/interp.daase">
```

This is caused by the removal of PAFFFF from the current system.

bug 7226:)d op coerce regression caused by waldek format update.

correct in git 95402C

see src/input/dop test 69

appears to be in ALGSC

```
[4] Vector D2 -> AlgebraGivenByStructuralConstants(D2,D3,D4,D5)
    from AlgebraGivenByStructuralConstants(D2,D3,D4,D5)
```

bug 7225: Lasagna Larga Doppia Riccia does not plot 0.0

See bookvol8.1, section Lasagna Larga Doppia Riccia

given:

```
X(i,j) == _
  if ((8 <= i) and (i <= 42)) _
    then 5.0/6.0+(5.0*i-40.0)/34.0 _
  else if (i <= 8) _
    then 5.0*i/48.0 _
    else 5.0/6.0*(7+(i-42.0)/8)
Y(i,j) == j/15.0
Z(i,j) == _
  if ((8 <= i) and (i <= 42)) _
    then 0.0 _
  else if (i <= 8) _
    then (8.0-i)/32.0*cos((j+3)*%pi/6) _
```

```

        else 0.25*(i-42)/8.0*cos((j+9)*%pi/6)
v3d:=draw(surface(X(i,j),Y(i,j),Z(i,j)),i=0..50,j=0..150,_,
            style=="smooth",title=="Lasagna Larga Doppia Riccia")
colorDef(v3d,yellow(),yellow())
axes(v3d,"off")
zoom(v3d,3.0,3.0,3.0)

```

The middle part of the image should generate 0.0 for the Z value but the drawn image does not show a flat plane. Calling Z(30.0,30.0) shows that the values of 0.0 is generated.

bug 7224: Axiom2D PS save function clips right side of image

```

draw(sin(4*t/7),t=0..14*%pi,coordinates==elliptic(1$DFLOAT))

menu -> ps
use gimp to look at ps

```

bug 7223: Bind stack overflow

```

-- (defun |*2;dot;5;frame1394| is being compiled
-- The variable |*2;dot;5;frame1394;MV is undefined
-- The compiler will assume this variable is a global
-- Internal error
-- The function coerce with signature OrderedCompletion Integer ->
--   Expression OrderedCompletion Integer is missing from domain
--   Expression(OrderedCompletion(Integer))
--
-- Internal Error
-- The function coerce with signature hashcode is missing from domain
--   Expression(OrderedCompletion (Integer))
dot(f,g) == integrate(f*g*x^2,x=-1..1)
proj(f,g) == dot(f,g)*f/dot(f,f)
p0:=1
p1:=x-proj(p0,x)
p2:=x^2-proj(p0,x^2)-proj(p1,x^2)
-- this give bind stack overflow
p(0) == 1
p(n | n > 0) == x^n-sum(proj(p(k),x^n),k=0..n-1)
p(1)

(7) -> )clear all
(1) -> dot(f,g) == integrate(f*g*x^2,x=-1..1)

```

Type: Void


```
(7) -> p(1)
      Cannot compile map: proj
      We will attempt to interpret the code.
      Cannot compile map: p
      We will attempt to interpret the code.

      >> System error:
      Bind stack overflow.

(7) ->
```

bug 7222: errors while compiling bookvol7

```
make[3]: Entering directory '/research/test/int/hyper'
1 making /research/test/mnt/ubuntu/lib/spadbuf from /research/test/books/bookvol7.pamphlet
spadbuf.c: In function interpI0:
spadbuf.c:89: warning: ignoring return value of write, declared with attribute warn_unused_result
2 making /research/test/mnt/ubuntu/lib/ex2ht from /research/test/books/bookvol7.pamphlet
3 making /research/test/mnt/ubuntu/bin/htadd from /research/test/books/bookvol7.pamphlet
htadd.c: In function buildHtFilename:
htadd.c:1314: warning: ignoring return value of getcwd, declared with attribute warn_unused_result
4 making /research/test/mnt/ubuntu/bin/hthits from /research/test/books/bookvol7.pamphlet
hthits.c: In function handleFile:
hthits.c:65: warning: ignoring return value of fgets, declared with attribute warn_unused_result
hthits.c: In function handlePage:
hthits.c:166: warning: ignoring return value of fread, declared with attribute warn_unused_result
5 making /research/test/mnt/ubuntu/bin/htsearch from /research/test/books/bookvol7.pamphlet
6 making /research/test/mnt/ubuntu/lib/presea from /research/test/books/bookvol7.pamphlet
7 /research/test/mnt/ubuntu/bin/hypertex from /research/test/books/bookvol7.pamphlet
hypertex.c: In function buildHtFilename:
hypertex.c:2029: warning: ignoring return value of getcwd, declared with attribute warn_unused_result
hypertex.c: In function handleKey:
hypertex.c:8748: warning: ignoring return value of system, declared with attribute warn_unused_result
hypertex.c: In function issueUnixcommand:
hypertex.c:13808: warning: ignoring return value of system, declared with attribute warn_unused_result
10 making /research/test/mnt/ubuntu/doc/axbook/xhtml from /research/test/books/axbook.tgz
8 making /research/test/mnt/ubuntu/doc/bigbayou.png from /research/test/books/bigbayou.png
9 making /research/test/mnt/ubuntu/doc/doctitle.png from /research/test/books/d
```

bug 7221: The variable IDENTITY is undefined.

```
; (DEFUN |Pattern;| ...) is being compiled.
```



```
;; The variable IDENTITY is undefined.
;; The compiler will assume this variable is a global.
```

```
-----
Pattern is now explicitly exposed in frame initial
Pattern will be automatically loaded when needed from
/research/test/int/algebra/PATTERN.nrlib/code
```

bug 7220: Manicotti only draws 1/2 of image

See bookvol8.1.pamphlet vs Pasta by Design book

Not clear if the problem is the equations or the graphics

bug 7219: Funghini draws only lines

See bookvol8.1.pamphlet vs Pasta by Design book

Not clear if the problem is the equations or the graphics

bug 7218: makeViewport3D closed unexpectedly

given:

```
A(i) == sin(i*pi/2000)^0.5
X1(i,j) == _
  if (i <= 2000) _
    then 2.1*cos((2*A(i)+1)*pi)+0.65*cos((2*A(i)+0.5)*pi)+_
      2.5*sin((A(i)+1.83)*pi)^500 _
    else -2.1
Y1(i,j) == _
  if (i <= 2000) _
    then 2.5*sin((2*A(i)+1)*pi)+0.1*sin(A(i)*2*pi)+_
      3*sin((A(i)+1.83)*pi)^500 _
    else 0.0
Z(i,j) == j/4.0
vsp:=createThreeSpace()
makeObject(surface(X1(i,j),Y1(i,j),Z(i,j)),i=0..3000,j=0..4,space==vsp)
vp:=makeViewport3D(vsp,style=="smooth",title=="Racchette")
axes(vp,"off")
```

```
>> Error detected within library code:
This viewport has already been closed
```

See section{Racchette} in bookvol8.1.pamphlet

Note that if the whole example is run the viewport is still open.
Only when the subset involves just (X1,Y1,Z) does it fail.

Note the same error occurs in Rotelle.

```
X1(i,j) == _
  if (i <= 666) _
    then 2*cos(3*i*%pi/1000)+0.03*cos(93*i*%pi/1000) _
    else 2.03
Y1(i,j) == _
  if (i <= 666) _
    then 2.05*sin(3*i*%pi/1000)+0.03*sin(93*i*%pi/1000) _
    else 0.0
Z(i,j) == j/5.0
vsp:=createThreeSpace()
makeObject(surface(X1(i,j),Y1(i,j),Z(i,j)),i=0..2000,j=0..5,space==vsp)
vp:=makeViewport3D(vsp,style=="smooth",title=="Rotelle")
axes(vp,"off")
```

```
>> Error detected within library code:
This viewport has already been closed
```

```
draw(surface(X1(i,j),Y1(i,j),Z(i,j)),i=0..2000,j=0..5)
```

```
succeeds with a wildly strange float
"float(149787561878521559122,-66,2)/float(1,0,2)
```

so I suspect the equations are wrong in some way. --tpd

bug 7216: Sorry - cannot handle that integrand yet

```
t0207:= -a^3*(a+b*x)*Ei(log(c*(a+b*x)^n)/n)/((c*(a+b*x)^n)^(1/n))/b^4/n+3*a^2*(a+b*x)^2*Ei(2*log
```

(32)

$$\frac{(b^4 x^4 + 4 a b^3 x^3 + 6 a^2 b^2 x^2 + 4 a^3 b x + a^4)(c(b x + a))^{\frac{1}{n}}(c(b x + a))^{\frac{2}{n}}}{3}$$

$$\begin{aligned}
& \frac{-n^n 4 \log(c(bx+a))}{(c(bx+a))^n \operatorname{Ei}\left(\frac{-n}{n}\right)} \\
& + \frac{(-3a^3bx^3 - 9a^2b^2x^2 - 9a^3bx - 3a^4)(c(bx+a))^{\frac{1}{n}}(c(bx+a))^{\frac{2}{n}}}{*} \\
& \frac{-n^n 3 \log(c(bx+a))}{(c(bx+a))^n \operatorname{Ei}\left(\frac{-n}{n}\right)} \\
& + \frac{(3a^2bx^2 + 6a^3bx + 3a^4)(c(bx+a))^{\frac{1}{n}}(c(bx+a))^{\frac{3}{n}}(c(bx+a))^{\frac{4}{n}}}{*} \\
& \frac{n^n 2 \log(c(bx+a))}{\operatorname{Ei}\left(\frac{-n}{n}\right)} \\
& + \frac{(-a^3bx^3 - a^4)(c(bx+a))^{\frac{2}{n}}(c(bx+a))^{\frac{3}{n}}(c(bx+a))^{\frac{4}{n}}}{*} \\
& \frac{n^n \log(c(bx+a))}{\operatorname{Ei}\left(\frac{-n}{n}\right)} \\
& / \\
& \frac{b^n (c(bx+a))^{\frac{1}{n}}(c(bx+a))^{\frac{2}{n}}(c(bx+a))^{\frac{3}{n}}(c(bx+a))^{\frac{4}{n}}}{\text{Type: Expression Integer}}
\end{aligned}$$

a0207:= integrate(t0207,x)

>> Error detected within library code:
 Sorry - cannot handle that integrand yet

Continuing to read the file...

bug 7215: integration bug?

t0317:= (-1+tanh(x)^2)^(3/2)

$$(85) \quad (\tanh(x)^2 - 1) \sqrt{\tanh(x)^2 - 1}$$

Type: Expression Integer

r0317:= -1/2*cosh(x)*(-sech(x)^2)^(1/2)*(atan(sinh(x))+sech(x)*tanh(x))

$$(86) \quad \frac{(-\cosh(x)\operatorname{atan}(\sinh(x)) - \cosh(x)\operatorname{sech}(x)\tanh(x))\sqrt{-\operatorname{sech}(x)^2}}{2}$$

Type: Expression Integer

a0317:= integrate(t0317,x)

$$(87) \quad 0$$

Type: Union(Expression Integer,...)

bug 7214: integration bug?

(1) -> t0062:= x*(sin(x)^2)^(1/2)

$$(1) \quad x \sqrt{\sin(x)^2}$$

(2) -> r0062:= -csc(x)*(x*cos(x)-sin(x))*(sin(x)^2)^(1/2)

$$(2) \quad (\csc(x)\sin(x) - x \cos(x)\csc(x))\sqrt{\sin(x)^2}$$

(3) -> a0062:= integrate(t0062,x)

$$(3) \quad \sin(x) - x \cos(x)$$

Axiom's answer seems simpler

(4) -> t1:=D(a0062,x)

$$(4) \quad x \sin(x)$$

But it is not equivalent to the integrand

$$(5) \rightarrow t2:=t0062-t1$$

$$(5) \quad x \sqrt{\sin(x)^2} - x \sin(x)$$

And it does not differ by a constant

$$(6) \rightarrow t3:=D(t2,x)$$

$$(6) \quad \frac{(-\sin(x) - x \cos(x)) \sqrt{\sin(x)^2} + \sin(x)^2 + x \cos(x) \sin(x)}{\sqrt{\sin(x)^2}}$$

Of course, Rich's answer does not differentiate to the closed form

$$(7) \rightarrow t4:=D(r0062,x)$$

$$(7) \quad \frac{(-\cot(x) + x) \csc(x) \sin(x)^3 + (x \cos(x) \cot(x) + \cos(x)) \csc(x) \sin(x)^2 - x \cos(x) \csc(x) \sin(x)}{\sqrt{\sin(x)^2}}$$

And it does not differ by a constant

$$(8) \rightarrow t5:=t4-t1$$

$$(8) \quad \frac{-x \sin(x) \sqrt{\sin(x)^2} + (-\cot(x) + x) \csc(x) \sin(x)^3 + (x \cos(x) \cot(x) + \cos(x)) \csc(x) \sin(x)^2 - x \cos(x) \csc(x) \sin(x)}{\sqrt{\sin(x)^2}}$$


```

--R      +-----+ +-+      +-----+ +-+
--R      - log(- 2\|x - 1 \|x - 2x + 1) + 2\|x - 1 \|x
--R  (23) -----
--R                        2
--R                                     Type: Union(Expression Integer,...)

```

another example:

```

tt:=sqrt(x)/sqrt(x+1)
integrate(tt,x)

```

```

      +-+ +-----+      +-+ +-----+
--R      - log(- 2\|x \|x + 1 - 2x - 1) + 2\|x \|x + 1
--R  (22) -----
--R                        2
--R                                     Type: Union(Expression Integer,...)

```

but sometimes we get:

```

--R  (18)
--R      +-+ +-----+      +-----+      +-+      +-+ +-----+
--R      (- 2\|x \|x + 1 - 2x - 1)log(\|x + 1 + \|x ) + (2x + 1)\|x \|x + 1
--R      +
--R      2
--R      2x + 2x
--R      /
--R      +-+ +-----+
--R      2\|x \|x + 1 + 2x + 1
--R
--R                                     Type: Union(Expression Integer,...)

```

todo 329: axiom reference info update

get all <http://axiom-portal.newsynthesis.org/refs/articles>

bug 7210: but in partial fraction

Axiom gets a different answer than Fracas.

Is the following close to what you have in mind? (two problems: you need to know the extension in advance, and I don't see a way to factor over extensions of degree higher than one right now. Possibly Waldek

knows.)

(1) -> SAEs5 := SAE(FRAC INT,UP(s5,FRAC INT),s5^2-5)

(1)
SimpleAlgebraicExtension(Fraction(Integer),UnivariatePolynomial(s5,Fraction(Integer)),s5^2+-5)

Type: Type

(2) -> p:UP(x,SAEs5) :=(x^5-1)*(x^2-1)*(x-1)

(2)
$$x^8 - x^7 - x^6 + x^5 - x^3 + x^2 + x - 1$$

Type:

UnivariatePolynomial(x,SimpleAlgebraicExtension(Fraction(Integer),UnivariatePolynomial(s5,Fraction(Integer)),s5^2+-5))

(3) -> factor p

(3)
$$(x^3 - 1)(x + 1)(x^2 + (-\frac{1}{2}s5 + -)x + 1)(x^2 + (-\frac{1}{2}s5 + -)x + 1)$$

Type:

Factored(UnivariatePolynomial(x,SimpleAlgebraicExtension(Fraction(Integer),UnivariatePolynomial(s5,Fraction(Integer)),s5^2+-5)))

(4) -> partialFraction(1/p, x)

(4)
$$\frac{\frac{13}{40}x^2 - \frac{9}{10}x + \frac{27}{40}}{(x-1)^3} - \frac{\frac{1}{8}}{x+1} + \frac{\frac{1}{50}(-\frac{1}{10}s5 - \frac{1}{10})x + \frac{1}{50}s5 - \frac{1}{10}}{x^2 + (-\frac{1}{2}s5 + -)x + 1}$$

+
$$\frac{\frac{1}{50}(-\frac{1}{10}s5 - \frac{1}{10})x - \frac{1}{50}s5 - \frac{1}{10}}{x^2 + (-\frac{1}{2}s5 + -)x + 1}$$

Type:

PartialFraction(UnivariatePolynomial(x,Fraction(Polynomial(SimpleAlgebraicExtension(Fraction(Integer),UnivariatePolynomial(s5,Fraction(Integer)),s5^2+-5))),s5^2+-5))

Apart from that: Grbner bases are in Axiom (FriCAS, OpenAxiom), and a rather good implementation, too.

bug 7209: Factoring in LODOF is incorrect


```

rfactor(op, r, zeros, ezfactor, adj?) ==
--   degree r > 1 or not one? leadingCoefficient r =>
     degree r > 1 or not ((leadingCoefficient r) = 1) =>
       recurfactor(op, r, zeros, ezfactor, adj?)
op1 := opeval(op, dd - coefficient(r, 0)::L)
map_!((z:L):L-->opeval(z,r), recurfactor(op1, dd, zeros, ezfactor, adj?))

```

Waldek claims this should read:

```

rfactor(op, r, zeros, ezfactor, adj?) ==
  recurfactor(op, r, zeros, ezfactor, adj?)

```

Martin comments:

I reworked lodof.input into the unittest framework, including additionally a simplified version of the bug and a new product of two irreducible degree 2 factors, which fails, of course. Unfortunately, your patched rfactor seems to go into an infinite loop with

```

-- reducible order 3 operator (1-1-1)
testFactor [(t^9 + t^3)*d + (-sqrt 91+7)*t^8+(-sqrt 91 + 1)*t^2, _
             d + ((sqrt 91 + 6)*t^6 + sqrt 91)/(t^7+t), _
             d + (5*t^6 - 1)/(t^7 + t)]

```

Waldek comments:

The real problem seems to be that

```

op2 := (t^9+t^3)*d^2 + (13*t^8+t^2)*d - 55*t^7 - 91*t
factor(op2)

```

is extremely slow. I am looking at this -- one problem is that we spend a lot of time computing gcd-s of polynomials with algebraic coefficients

bug 7207: $\text{integrate}(1/(2-3*x)^{1/2}/(2+3*x)^{1/2}, x)$

should be $1/3*\text{asin}(3/2*x)$
but causes an infinite loop in integration

bug 7206: from <http://bugs.debian.org/349877>

X-Debbugs-CC: Jiri Palecek <jpalecek@web.de>
Package: axiom

Version: 20050901-4
Severity: normal

When I enter in axiom

```
> p:(NonNegativeInteger)->Polynomial AlgebraicNumber
> p i==if i=0 then 1 else x^i-reduce(+,[integrate(x^i*p k,x=0..1)*p k for
> k in 0..i-1])
```

and then

```
> p 1
```

I get this error:

```
Internal Error
Interpreter code generation failed for expression
  (IF (= |#1| 0) 1
    (- (^ |x| |#1|)
      (|reduce| +
        (|construct|
          (COLLECT (STEP |k| 0 1 (- |#1| 1))
            (* (|integrate|
              (* (^ |x| |#1|) (|p| |k|))
              (= |x| (SEGMENT 0 1)))
              (|p| |k|)))))))
```

-- System Information:

```
Debian Release: testing/unstable
APT prefers testing
APT policy: (500, 'testing'), (500, 'stable')
Architecture: i386 (i686)
Shell: /bin/sh linked to /bin/dash
Kernel: Linux 2.6.15
Locale: LANG=C, LC_CTYPE=C (charmap=ISO-8859-2) (ignored: LC_ALL set to
cs_CZ)
```

Versions of packages axiom depends on:

```
ii axiom-databases      20050901-4 A general purpose computer
algebra
ii libc6                2.3.5-8    GNU C Library: Shared
libraries an
ii libgmp3c2            4.1.4-10   Multiprecision arithmetic
library
ii libncurses5          5.5-1      Shared libraries for terminal
hand
ii libreadline5         5.1-5      GNU readline and history
libraries
```

Versions of packages axiom recommends:

```
ii axiom-doc                20050901-4 A general purpose computer
algebra
ii axiom-graphics           20050901-4 A general purpose computer
algebra
ii axiom-hypertext           20050901-4 A general purpose computer
algebra
pn axiom-source              <none>      (no description available)
```

-- no debconf information

--

Using Opera's revolutionary e-mail client: <http://www.opera.com/mail/>

bug 7205: interval is not recognized:

)d op interval

There are 3 exposed functions called interval :

- [1] Fraction Integer -> D from D
 - if D has INTCAT D2 and D2 has Join(FloatingPointSystem, TranscendentalFunctionCategory)
- [2] D1 -> D from D
 - if D has INTCAT D1 and D1 has Join(FloatingPointSystem, TranscendentalFunctionCategory)
- [3] (D1,D1) -> D from D
 - if D has INTCAT D1 and D1 has Join(FloatingPointSystem, TranscendentalFunctionCategory)

Examples of interval from IntervalCategory

(31) -> fi:=1/4::FRAC(INT)

```
      1
(31)  -
      4
```

Type: Fraction Integer

(32) -> interval(fi)\$INTERVAL(FRAC(INT))

There are 2 exposed and 0 unexposed library operations named interval having 1 argument(s) but none was determined to be applicable. Use HyperDoc Browse, or issue

)display op interval

to learn more about the available operations. Perhaps package-calling the operation or using coercions on the arguments will allow you to apply the operation.

```

Cannot find a definition or applicable library operation named
interval with argument type(s)
      Fraction Integer

```

Perhaps you should use "@" to indicate the required return type,
or "\$" to specify which version of the function you need.

bug 7204: bug in this rule form?

```

-- bug in this rule form?  int(x+ :y,z) == int(x,z)+int(y,z)
-- and also this one?    int(x+y,z) == int(x,z)+int(y,z)
intRules:=rule
  int(x+y,z) == int(x,z)+int(y,z)
  int(k*x | freeOf?(k,z),z) == k*int(x,z)
  int(y | integer? y, z) == y*z
  int(x^(?p | D(p,x)=0),x) == x^(p+1)/(p+1)
intRules int(a^2*b+a^b+3*a-5,a)
intRules int(a^(a+1),a)

```

bug 7203: in GRAS the line

```

vType:SINT := xor(op2type,uType) -- remaining ^factors

```

fails to compile. It expects op2type to be boolean even though
op2type and uType are SINT and the target vType is SINT. The line

```

XOR(x:SINT,y:SINT):SINT == LOGXOR(x,y)$Lisp

```

was temporarily added to work around the bug.

bug 7201: sin(x) drawing error

drawing tubeplots does not always work as expected

```

s:=create3Space()$(ThreeSpace DFLOAT)
draw(curve(x,sin(x),0),x=0..4,style="shade",tubeRadius=0.3,tubePoints=20,
      space==s)
draw(curve(x,sin(x),1),x=-4..4,style="shade",tubeRadius=0.3,tubePoints=20,
      space==s)

```

bug 7200: Improper syntax after browse

```

in hyperdoc click on browse
enter draw
click operations
select descriptions
enter any command in main window: e.g. 1+1
and you get

line 1: 1+1
Error A: Improper syntax
Error A: Improper syntax
Error A: syntax error at top level
...

```

bug 7199: real/imag giving wrong answers

```

a:=(-1*%i)^(1/2)
b:=(%i)^(3/2)
real(a)
imag(a)
real(b)
imag(b)

```

bug 7198: missing coerce function in OrderedCompletion Integer

```

-- (defun |*2;dot;5;frame1394| is being compiled
-- The variable |*2;dot;5;frame1394;MV is undefined
-- The compiler will assume this variable is a global
-- Internal error
-- The function coerce with signature OrderedCompletion Integer ->
--   Expression OrderedCompletion Integer is missing from domain
--   Expression(OrderedCompletion(Integer))
--
-- Internal Error
-- The function coerce with signature hashCode is missing from domain
--   Expression(OrderedCompletion(Integer))
dot(f,g) == integrate(f*g*x^2,x=-1..1)
proj(f,g) == dot(f,g)*f/dot(f,f)
p0:=1

```

```

p1:=x-proj(p0,x)
p2:=x^2-proj(p0,x^2)-proj(p1,x^2)
-- this give bind stack overflow
p(0) == 1
p(n | n > 0) == x^n-sum(proj(p(k),x^n),k=0..n-1)
p(1)

```

todo 326: Guess.tex needs to be used to document the guess package

```

todo 326: Guess.tex needs to be used to document the guess package

```

todo 326: in LODOOPS the fricas version

```

todo 326: in LODOOPS the fricas version of the killer function differs
          and the nonTrivial function is missing

```

todo 325: permute list of anything

```

n:=[1,2,3,4]
permutations n
p:=[x.[i] for i in 1..4]
permutations p

```

bug 7197: hyperdoc/graphics failure

```

Basic Commands -> Draw -> exit

```

bug 7196: factor is partial

```

n:= 1303 * 16927 * 157543
factor(n)

```

occasionally only partially factors, returning

1303 * 2666730361

todo 324: integrate: implementation incomplete (constant residues)

```
integrate(
  (x^2+2*x+1*(3*x+1)*sqrt(x+log(x)))/
  (x*sqrt(x+log(x))*(x+sqrt(x+log(x)))),x)
```

the antiderivative is

```
2*(sqrt(x+log(x))+log(x+sqrt(x+log(x))))
```

or, a shorter example is:

```
integrate(sqrt(x+log(x)),x)
```

bug 7195: draw bug

```
i tried to draw a surface, typing 'z' instead of 'y':
  draw(surface(x,x*x,y),x=0..5,z=0..3)
axiom responds with
Loading .... opensus/algebra/PALETTE.o for domain Palette
Unrecoverable error: Segmentation violation.

)set function compile off
  draw(surface(x,x*x,y),x=0..5,z=0..3)
generates
>> System error:
The variable |y| is unbound.
```

bug 7194: .fn files are not used during first compile phase

should be loaded from previous build cache into compile image?

bug 7193: src/algebra/*.pamphlet files are not copied.

bug 7193: src/algebra/*.pamphlet files are not copied.

7190: integrate(sqrt(1+sec(x)),x) returns 0

7190: integrate(sqrt(1+sec(x)),x) returns 0

MMA returns a huge expression

Bug poster claims the Maple and cac1101.com return closed forms

bug 7189: browsing 'eigenMatrix' works only with 'Operations'

bug 7189: browsing 'eigenMatrix' works only with 'Operations'

Michael Becker <Michael.Becker@coconet.de>

bug 7188: hyperdoc browsing with wildcards does not work

bug 7188: hyperdoc browsing with wildcards does not work

Michael Becker <Michael.Becker@coconet.de>

bug 7187: hyperdoc cat* bug

Michael Becker <Michael.Becker@coconet.de>

Entering 'cat*' in the input area of 'Glossary' gives:

'no glossary items match d*'

bug 7186: ACPLLOT missing functions

```
sketch:=makeSketch(x+y,x,y,-1/2..1/2,-1/2..1/2)$ACPLLOT
```

```
xRange(sketch)
```

```
Internal Error
```

```
The function xRange with signature hashcode is missing from domain
```

```
PlaneAlgebraicCurvePlot
```

```
yRange(sketch)
```

```
Internal Error
```


The function `yRange` with signature `hashCode` is missing from domain
`PlaneAlgebraicCurvePlot`

bug 7181: `SQMATRIX(2,INT)` has coerce `Integer - > %`

bug 7181: `SQMATRIX(2,INT)` has coerce `Integer -> %`

is not a valid type

even if:

`)expose SQMATRIX`

bug 7169: Matrix printing/parsing bug (notice the - in E21)

`E21:=matrix([[1,0,0],[-2,1,-],[0,0,1]])`

```
(6)  + 1  0  0+
      |      |
      |- 2  1  -|
      |      |
      + 0  0  1+
```

Type: Matrix Polynomial Integer

(7) `-> E32:=matrix([[1,0,0],[0,1,0],[0,-5,1]])`

```
(7)  +1  0  0+
      |      |
      |0  1  0|
      |      |
      +0 - 5  1+
```

Type: Matrix Integer

(8) `-> E32*E21`

```
(8)  + 1      0      0      +
      |      |      |
      |- 2      1      -      |
      |      |      |
      +10     - 5     - 5- + 1+
```

Type: Matrix Polynomial Integer

(9) `->`

todo 322: use charts.jpg to guide axiom graphing

```
todo 322: use charts.jpg to guide axiom graphing
```

todo 321: try to use raphael to scale graphs

```
todo 321: try to use raphael to scale graphs
```

todo 320: pick up FLINT package

```
todo 320: pick up FLINT package
```

bug 7169: algebra bug

```
)co RESLATC
```

```
RESLATC;coerce;ES;2 is replaced by errorBug:
  Should not be able to obtain value of type Exit
```

bug 7167: KeyedAccessFile fails because

- (a) readable? in defstream is checking the directory name, not the file
- (b) BINARY_OPEN_INPUT does not exist

bug 7166: library doesn't seem to work

```
stuff := library "/tmp/Neat.stuff"
stuff.int      := 32**2
stuff."poly"   := x**2 + 1
stuff.str      := "Hello"
keys stuff
stuff.poly
stuff("poly")
```

```
)system rm -rf /tmp/Neat.stuff
```

bug 7165: this rule does not match

```
--S
sininv:=rule(sin(%i*x) == %i*sinh(x))
--R
--R (1) sin(%i x) == %i sinh(x)
--R Type: RewriteRule(Integer,Complex Integer,Expression Complex Integer)
--E

--S
t1:=sin(x%i) - %i*sinh(x)
--R
--R (2) - %i sinh(x) + sin(%i x)
--R Type: Expression Complex Integer
--E

--S
t2:=sininv t1
--R
--R (3) - %i sinh(x) + sin(%i x)
--R Type: Expression Complex Integer
--E

(1) -> cosinv:=rule(cos(x%i) == cosh(x))
(2) -> t1:=cos(x%i) - cosh(x)
(3) -> t2:=cosinv t1

(1) -> taninv:=rule(tan(x%i) == %i*tanh(x))
(2) -> t1:=tan(x%i) - %i*tanh(x)
(3) -> t2:=taninv t1

(1) -> cscinv:=rule(csc(x%i) == -%i*csch(x))
(2) -> t1:=csc(x%i) - -%i*csch(x)
(3) -> t2:=cscinv t1

(1) -> secinv:=rule(sec(x%i) == sech(x))
(2) -> t1:=sec(x%i) - sech(x)
(3) -> t2:=secinv t1

(1) -> cotinv:=rule(cot(x%i) == -%i*coth(x))
(2) -> t1:=cot(x%i) - -%i*coth(x)
(3) -> t2:=cotinv t1

(1) -> sinhinv:=rule(sinh(x%i) == %i*sin(x))
(2) -> t1:=sinh(x%i) - %i*sin(x)
```

```

(3) -> t2:=sinhinv t1

(1) -> coshinv:=rule(cosh(x%i) == cos(x))
(2) -> t1:=cosh(x%i) - cos(x)
(3) -> t2:=coshinv t1

(1) -> tanhinv:=rule(tanh(x%i) == %i*tan(x))
(2) -> t1:=tanh(x%i) - %i*tan(x)
(3) -> t2:=tanhinv t1

(1) -> cschinv:=rule(x%i == -%i*csc(x))
(2) -> t1:=x%i - -%i*csc(x)
(3) -> t2:=cschinv t1

(1) -> sechinv:=rule(sech(x*i) == sec(x))
(2) -> t1:=sech(x*i) - sec(x)
(3) -> t2:=sechinv t1

(1) -> cothinv:=rule(coth(x%i) == -%i*cot(x))
(2) -> t1:=coth(x%i) - -%i*cot(x)
(3) -> t2:=cothinv t1

(1) -> coshperiod:=rule(cosh(x+2*k*pi%i) == cosh(x))
(2) -> t1:=cosh(x+2*k*pi%i) - cosh(x)
(3) -> t2:=coshperiod t1

(1) -> tanhperiod:=rule(tanh(x+k*pi%i) == tanh(x))
(2) -> t1:=tanh(x+k*pi%i) - tanh(x)
(3) -> t2:=tanhperiod t1

(1) -> sechperiod:=rule(sech(x+2*k*pi%i) == sech(x))
(2) -> t1:=sech(x+2*k*pi%i) - sech(x)
(3) -> t2:=sechperiod t1

(1) -> sinsinh:=rule(sin(%i*x)^(-1) == %i*sinh(x)^(-1))
(2) -> t1:=sin(%i*x)^(-1) - %i*sinh(x)^(-1)
(3) -> t2:=sinsinh t1

(1) -> sinhsin:=rule(sinh(%i*x)^(-1) == %i*sin(x)^(-1))
(2) -> t1:=sinh(%i*x)^(-1) - %i*sin(x)^(-1)
(3) -> t2:=sinhsin t1

(1) -> coscosh:=rule(cos(x)^(-1) == %i*cosh(x)^(-1))
(2) -> t1:=cos(x)^(-1) - %i*cosh(x)^(-1)
(3) -> t2:=coscosh t1

(1) -> coscosh2:=rule(cos(x)^(-1) == -%i*cosh(x)^(-1))
(2) -> t1:=cos(x)^(-1) - -%i*cosh(x)^(-1)
(3) -> t2:=coscosh2 t1

```

```

(1) -> coshcos:=rule(cosh(x)^(-1) == %i*cos(x)^(-1))
(2) -> t1:=cosh(x)^(-1) - %i*cos(x)^(-1)
(3) -> t2:=coshcos t1

(1) -> coshcos2:=rule(cosh(x)^(-1) == -%i*cos(x)^(-1))
(2) -> t1:=cosh(x)^(-1) - -%i*cos(x)^(-1)
(3) -> t2:=coshcos2 t1

(1) -> tantanh:=rule(tanh(%i*x)^(-1) == %i*tanh(x)^(-1))
(2) -> t1:=tanh(%i*x)^(-1) - %i*tanh(x)^(-1)
(3) -> t2:=tantanh t1

(1) -> tanhtan:=rule(tanh(%i*x)^(-1) == %i*tan(x)^(-1))
(2) -> t1:=tanh(%i*x)^(-1) - %i*tan(x)^(-1)
(3) -> t2:=tanhtan t1

```

bug 7164: this rule does not match

```

--S
cschinv:=rule(csch(x)^(-1) == sinh(1/x)^(-1))
--R
--R
--R      1      1
--R  (1)  ----- == -----
--R      csch(x)      1
--R                  sinh(-)
--R                  x
--R                                     Type: RewriteRule(Integer,Integer,Expression Integer)
--E

--S
t1:=csch(x)^(-1) - sinh(1/x)^(-1)
--R
--R
--R      1
--R      sinh(-) - csch(x)
--R      x
--R  (2)  -----
--R      1
--R      csch(x)sinh(-)
--R      x
--R
--R                                     Type: Expression Integer
--E

--S
t2:=cschinv t1
--R
--R
--R      1
--R      sinh(-) - csch(x)

```

```

--R          x
--R  (3)  -----
--R          1
--R      csch(x)sinh(-)
--R          x
--R
--R                                          Type: Expression Integer
--E

```

bug 7163: tanhneg rule fails

The first 5 succeed, but the 6th (tanhneg t3) loops forever

```

(1) -> tanhadd:=rule(tanh(x+y) == (tanh(x)+tanh(y))/(1+tanh(x)*tanh(y)))
(2) -> t1:=tanh(x+y) - (tanh(x)+tanh(y))/(1+tanh(x)*tanh(y))
(3) -> t2:=tanhadd t1
(4) -> tanhneg:=rule(tanh(x-y) == (tanh(x)-tanh(y))/(1-tanh(x)*tanh(y)))
(5) -> t3:=tanh(x-y) - (tanh(x)-tanh(y))/(1-tanh(x)*tanh(y))
(6) -> t4:=tanhneg t3

(1) -> cothadd:=rule(coth(x+y) == (coth(x)*coth(y)+1)/(coth(y)+coth(x)))
(2) -> t1:=coth(x+y) - (coth(x)*coth(y)+1)/(coth(y)+coth(x))
(3) -> t2:=cothadd t1
(4) -> cothneg:=rule(coth(x-y) == (coth(x)*coth(y)-1)/(coth(y)-coth(x)))
(5) -> t3:=coth(x-y) - (coth(x)*coth(y)-1)/(coth(y)-coth(x))
(6) -> t4:=cothneg t3

```

bug 7162: (p,q):UP(x,INT) fails only in fixed.input.pamphlet

```

p:UnivariatePolynomial(x,Integer); q:UnivariatePolynomial(x,Integer)
                                          Type: Void

```

Daly Bug

Category, domain or package constructor : is not available.

The expression

```
(p,q):UP(x,INT)
```

generates the error

Daly Bug

Category, domain or package constructor : is not available.

works fine at the top of the fixed.input file but not further down.
 Which means that some property is not being cleared properly.
 The symbol ‘p’ does get properly declared despite the error message.

bug 7160: numeric(erf(0.1))

the erf function is defined as:

```
erf(x) == 1/sqrt(2*pi)*integrate(exp(-1/2*t^2),t=0..x)
```

ref: CRC Standard Mathematical Tables and Formulae p547

bug 7159: rootsOf(2*z⁴+z³-z-2) is wrong

```
(1) [%z0,%z1,%z1,- %z1]
                                         Type: List AlgebraicNumber

(2) -> (z^2-1)*(2*z^2+z+2)

      4      3
(2)  2z  + z  - z - 2
                                         Type: Polynomial Integer

(3) -> rootsOf(z^2-1)

(3)  [%z3,- %z3]
                                         Type: List AlgebraicNumber

(4) -> rootsOf(2*z^2+z+2)

      - 2%z5 - 1
(4)  [%z5,-----]
              2
                                         Type: List AlgebraicNumber

so no two roots are equal

(5) -> f(z)==(z^2-1)*(2*z^2+z+2)
                                         Type: Void

(6) -> m1:=%(3).1

(6)  %z3
                                         Type: AlgebraicNumber

(7) -> m2:=%(3).2

(7)  - %z3
```

```

(8) -> m3:=%%(4).1
                                     Type: AlgebraicNumber
(8)  %z5
                                     Type: AlgebraicNumber
(9) -> m4:=%%(4).2
      - 2%z5 - 1
(9)  -----
      2
                                     Type: AlgebraicNumber
(10) -> f(m1)
      Compiling function f with type AlgebraicNumber -> AlgebraicNumber
(10)  0
                                     Type: AlgebraicNumber
(11) -> f(m2)
(11)  0
                                     Type: AlgebraicNumber
(12) -> f(m3)
(12)  0
                                     Type: AlgebraicNumber
(13) -> f(m4)
(13)  0
                                     Type: AlgebraicNumber
(14) -> n1:=%(1).1
(14)  %z0
                                     Type: AlgebraicNumber
(15) -> n2:=%(1).2
(15)  %z1
                                     Type: AlgebraicNumber
(16) -> n3:=%(1).3
(16)  %z1
                                     Type: AlgebraicNumber
(17) -> n4:=$$(1).4
      Category, domain or package constructor 1 is not available.
(17) -> n4:=%(1).4
(17)  - %z1
                                     Type: AlgebraicNumber
(18) -> f(n1)
(18)  0

```



```

Type: AlgebraicNumber
(19) -> f(n2)

(19) 0
Type: AlgebraicNumber
(20) -> f(n3)

(20) 0
Type: AlgebraicNumber
(21) -> f(n4)

(21) (2%z0 + 1)%z12 + (2%z02 + %z0 + 2)%z1 + 2%z03 + %z02 - 1
Type: AlgebraicNumber

```

so %(1).4 is not a root?

```

rootsOf(2*z^4 + z^3 - z - 2) wrong
rootsOf(2*z^4 + z^3 - z - 2, z) correct
rootsOf(z^4 + z^3 + z - 1) wrong
rootsOf(z^4 + z^3 + z - 1, z) correct

```

Sat 22 May 2004 01:48:36 PM GMT, comment #1:

Maybe I should have added why rootsOf(2*z^4 + z^3 - z - 2) is wrong:
 $2z^4 + z^3 - z - 2 = (z^2 - 1)(2z^2 + z + 2)$,
 thus no two roots are equal, contrary to axioms result.

Martin
 Martin Rubey <kratt6>

Wed 10 Mar 2004 09:06:43 AM GMT, original submission:

a fresh Axiom says

```

(1) -> rootsOf(2*z^4 + z^3 - z - 2, z)

- 2%z2 - 2%z1 - 2%z0 - 1
(1) [%z0,%z1,%z2,-----]
2
Type: List Expression Integer

(2) -> rootsOf(2*z^4 + z^3 - z - 2)

(2) [%z4,%z5,%z5,- %z5]
Type: List AlgebraicNumber

```

which is wrong.

Same thing for zerosOf, same thing for z^4+z^3+z-1

bug 7158: patch 3127 is wrong fix

currently Axiom gives:

```
product(summation(i*j,i=a..b),j=c..d)
```

```

      d      b
  +-+ +-+  +-+
(1)  | | >   i j
      | |  +-+
      j= c  i= a

```

Type: Expression Integer

but NAG version gives

```

      b      d - c + 1
  +-+      2
  >      i
  +-+
  i=a

```

bug 7157: 0::CARDÔ::CARD should be 1?

```
(1) -> 0::CARD
```

```
(1) 0
```

Type: CardinalNumber

```
(2) -> 0::CARD^0::CARD
```

```
>> Error detected within library code:
0**0 not defined for cardinal numbers.
```

```
(2) -> complex(0,0)^complex(0,0.0)
```

```
>> Error detected within library code:
log 0 generated
```

```
(2) -> complex(0,0)^complex(2,2.0)
```

```
>> Error detected within library code:
```

log 0 generated

(2) ->

bug 7156: nonsense result

(1) -> matrix([[0,0],[0,0]])^0.1

(1) 0.0

Type: Float

(2) -> matrix([[0,0],[0,0]])^x

(2) 0

Type: Expression Integer

(3) -> matrix([[0,0],[0,1]])^0.1

There are 4 exposed and 0 unexposed library operations named ^
having 2 argument(s) but none was determined to be applicable.
Use HyperDoc Browse, or issue

)display op ^

to learn more about the available operations. Perhaps
package-calling the operation or using coercions on the arguments
will allow you to apply the operation.

Cannot find a definition or applicable library operation named ^
with argument type(s)

Matrix Integer
Float

Perhaps you should use "@" to indicate the required return type,
or "\$" to specify which version of the function you need.

(3) -> matrix([[0,0],[0,1]])^x

There are 4 exposed and 0 unexposed library operations named ^
having 2 argument(s) but none was determined to be applicable.
Use HyperDoc Browse, or issue

)display op ^

to learn more about the available operations. Perhaps
package-calling the operation or using coercions on the arguments
will allow you to apply the operation.

Cannot find a definition or applicable library operation named ^
with argument type(s)

Matrix Integer
Variable x

Perhaps you should use "@" to indicate the required return type,

or "\$" to specify which version of the function you need.

bug 7155: fatal error

```
(1) -> a:=series sin(x)
      Loading /research2/test/mnt/fedora5/algebra/EXPR2UPS.o for package
      ExpressionToUnivariatePowerSeries

      1 3    1 5    1 7    1 9    1 11    12
(1)  x - - x + --- x - ---- x + ----- x - ----- x + 0(x )
      6    120   5040  362880  39916800
                                Type: UnivariatePuisseuxSeries(Expression Integer,x,0)

(2) -> a*1.0
      Loading /research2/test/mnt/fedora5/algebra/EXPR2.o for package
      ExpressionFunctions2

>> System error:
      Caught fatal error [memory may be damaged]
```

bug 7154: strange handling of args outside $-\pi/2 \leq x \leq \pi/2$

```
(1) -> atan(tan(3))

(1)  3
                                           Type: Expression Integer

(2) -> atan(tan(3.0))

(2)  - 0.1415926535 8979323846
                                           Type: Float

(3) -> atan(tan(1))

(3)  1
                                           Type: Expression Integer

(4) -> atan(tan(1.0))

(4)  1.0
                                           Type: Float
```

bug 7153: division by zero due to definition problem

```
asin(complex(1.0,0.0))
```

```
>> Error detected within library code:
      catdef: division by zero
```

```
trigcat.spad.pamphlet =>
asin x == atan(x/sqrt(1-x**2))
```

```
asin(1.0) => %pi/2.0
but not for Complex(Float)
```

bug 7152: integration bug

```
integrate((1/x)::UTS(FRAC POLY INT, x, 0),x)
integrate((1/y)::UTS(FRAC POLY INT, x, 0),z)
```

The first bug is an indication that $1/x$ should not be of type `UTS(x,FRAC POLY INT)`, `UP(x, FRAC POLY INT)` and so on. The second is an indication that QFCAT should have a function variables: % -> List Symbol.

Sat 02 Oct 2004 12:09:22 PM GMT, comment #2:

I disagree with the previous comment; x should be captured and not allowed in the `FRAC POLY INT`. (I agree that the second example in the original report is not a bug.)

But surely we can all agree that

```
integrate((1/x)::ULS(FRAC POLY INT,x, 0),x)
is a bug? The problem is the same: the 1/x is being treated
in the coefficient domain. Note that
integrate(1/(x::ULS(FRAC POLYINT, x, 0)),x)
produces a correct result.
```

Dylan (dpt@math.harvard.edu)

Sat 02 Oct 2004 08:33:01 AM GMT, comment #1:

Neither of these is a bug. In the first one, Axiom coerced $1/x$ into `FRAC POLY INT` correctly: the only `/` operation available in `UTS` is one induced from the coefficient domain, which requires the denominator to be in the coefficient domain, and the division is done termwise to the coefficients of the series. So $1/x$ ends up in `FRAC POLY INT`. Note that to obtain a Taylor series at $x = 0$ is mathematically wrong, since $1/x$ is not defined at $x=0$. Also the way to obtain a Taylor series is `taylor(func, x=a)`. If you do `integrate(taylor(1/x,x=1),x)`, that would cause no problems. Note that the domain of this command is `UTS(EXPR`

INT, x,1), so such towers are valid and necessary in Axiom. Note also there are only two exported[coerce] in UTS and they do NOT apply to 1/x. The x in UTS is like the x in UP and is different from the x in FRAC POLY INT. The representation is Stream Coef (no variable specified because it is univariate).

For the same reason, in the second command, 1/y is correctly coerced into FRAC POLY INT. However, in UTS(x,), the only integrations allowed are with respect to x. If you want to do integration in FRAC POLY INT, then you should do so without coercing 1/y into UTS.

So your examples do not illustrate the problem about mixed up variables. In fact, it supports use of towers like UTS(EXPR INT, x,a).

William (wyscc@cunyvm.cuny.edu)

Sat 11 Sep 2004 12:34:23 PM GMT, original submission:

```
integrate((1/x)::UTS(FRAC POLY INT, x, 0),x)
```

```
1
```

```
(11) - x
```

```
x
```

```
Type: UnivariateTaylorSeries(Fraction Polynomial Integer,x,0)
```

```
integrate((1/y)::UTS(FRAC POLY INT, x, 0),z)
```

```
Cannot find a definition or applicable library operation named
integrate with argument type(s)
```

```
UnivariateTaylorSeries(Fraction Polynomial Integer,x,0)
```

```
Variable z
```

bug 7151: bind stack overflow

```
(1) -> ?::Symbol
```

```
(1) ?
```

Type: Symbol

```
(2) -> ?::Symbol::SAOS
```

```
>> System error:
Bind stack overflow.
```

```
(2) -> x::Symbol::SAOS
```

```
(2) "?"
```

Type: SingletonAsOrderedSet

bug 7150: exquo cannot detect zero equivalence?

```
gcd((A-2^a)::UP(A,EXPR INT),(A^2-2^(2*a))::UP(A,EXPR INT))
```

Cause:

In EXPR INT, 2^a and $2^{(2*a)}$ are treated as two variables without relations in EXPR INT. Therefore exquo in `gcdPrimitive(p1:SUPP,p2:SUPP)$PGCD` fails.

(+) Discussion

Thu 30 Sep 2004 09:31:59 AM GMT, comment #3:

>> Sorry, this is not correct. It is in FIELD (for EXPR INT)

Excuse me, I was too quick again. Here is the (hopefully correct) analysis:

```
exquo(simplify((A-2^a)*(A+2^a))::UP(A,EXPR INT),(A-2^a)::UP(A,EXPR INT))
```

calls `exquo$SUP(EXPR INT)`. This implements exact division of polynomials p_1 by p_2 as usual. After each subtraction - done via `fmcg$SUP` - the result is again stored in p_1 . `exquo` terminates when p_1 is the empty list - note that SUPs are stored as lists of pairs (degree, coefficient) - or the degree of p_2 is larger than p_1 . In the latter case, `exquo` fails.

Thus, in our case, at one point p_1 is $4^a - 2^{(2*a)}$, which is zero mathematically, but axiom does not know it. In particular, p_1 is not the empty list, but rather a constant polynomial...

It would be interesting to see how MuPAD or Aldor handle this.
Martin Rubey <kratt6>

Wed 29 Sep 2004 04:20:16 PM GMT, comment #2:

>> The instance of `exquo` involved is the one in SMP.

Sorry, this is not correct. It is in FIELD (for EXPR INT)
Martin Rubey <kratt6>

Wed 29 Sep 2004 04:02:03 PM GMT, comment #1:

I should have added:

```
(1) -> exquo(normalize(simplify(((A-2^a)*(A+2^a)))::EXPR INT),normalize((A-2^a)::EXPR INT))
```

```

a log(4) 2
  %e  - A
(1) -----
a log(2)
  %e  - A
Type: Union(Expression Integer,...)

(2) -> exquo(simplify((A-2^a)*(A+2^a))::UP(A,EXPR INT),(A-2^a)::UP(A,EXPR INT))

(2) "failed"
Type: Union("failed",...)

```

I'm afraid that this cannot be fixed easily, since there is no general mechanism to determine whether an expression is zero or not, which is needed in exquo. The instance of exquo involved is the one in SMP.
 Martin Rubey <kratt6>

Wed 29 Sep 2004 03:36:55 PM GMT, original submission:

```

gcd((x-2^a)::UP(x, EXPR INT), simplify((x-2^a)*(x+2^a))::UP(x, EXPR INT))

gives 1, while the correct answer should be x-2^a, as given by

gcd((x-2^a)::UP(x, EXPR INT),((x-2^a)*(x+2^a))::UP(x, EXPR INT))

A workaround is presented on
http://page.axiom-developer.org/zope/mathaction/EXPR\_GCD

```

bug 7149: bad choice of signature by interpreter

```

y x == (1+x)^x/x
                                                    Type: Void
interpolate(x,[1,2,3],[1/z(i) for i in 1..3])$PINTERP(x,FRAC POLY INT)
  Compiling function y with type PositiveInteger -> Polynomial Integer

  Conversion failed in the compiled user function y .
  Loading /research2/test/mnt/fedora5/algebra/UNISEG.o for domain
    UniversalSegment

  Cannot convert from type Float to Polynomial Integer for value
    4.5

y(2)
  Conversion failed in the compiled user function y .

```


Cannot convert from type Float to Polynomial Integer for value
4.5

HOWEVER:

$z(x) == (1+x)^x/x$

Type: Void

$z(2)$

Compiling function z with type PositiveInteger -> Fraction Integer

$$(3) \quad -\frac{9}{2}$$

Type: Fraction Integer

$\text{interpolate}(x, [1, 2, 3], [1/z(i) \text{ for } i \text{ in } 1..3])\$PINTERP(x, \text{FRAC POLY INT})$

$$(4) \quad \frac{59}{1152}x^2 - \frac{497}{1152}x + \frac{169}{192}$$

Type: UnivariatePolynomial(x, Fraction Polynomial Integer)

)clear all

$y(x:\text{PositiveInteger}):\text{Fraction}(\text{Integer}) == (1+x)^x/x$

Function declaration y : PositiveInteger -> Fraction Integer has
been added to workspace.

Type: Void

$\text{interpolate}(x, [1, 2, 3], [1/y(i) \text{ for } i \text{ in } 1..3])\$PINTERP(x, \text{FRAC POLY INT})$

Compiling function y with type PositiveInteger -> Fraction Integer

$$(2) \quad \frac{59}{1152}x^2 - \frac{497}{1152}x + \frac{169}{192}$$

Type: UnivariatePolynomial(x, Fraction Polynomial Integer)

bug 7147: $\text{integrate}(1/(1+x^4), x=\%minusInfinity..\%plusInfinity)$

bug 7147: $\text{integrate}(1/(1+x^4), x=\%minusInfinity..\%plusInfinity)$

NOTE: fixed in Fricas prior to fork

returns 0

should return $\pi \sqrt{2}/2$

```

integrate(1/(4+x^4),x=%minusInfinity..%plusInfinity)
returns %pi/4
should return %pi/4

integrate(1/(1+x^6),x=%minusInfinity..%plusInfinity)
returns 0
should return 2*%pi/3

integrate(1/(1+x^2),x=%minusInfinity..%plusInfinity)
returns %pi
should return %pi

```

bug 7145: fxiedPointExquo bug

```

see:
http://groups.google.com/group/fricas-commit/browse\_thread/thread/1befef363e8c02be?hl=en#

exprode.spad.pamphlet
  opex := operator("exquo"::Symbol)$OP
  opex := operator("fixedPointExquo"::Symbol)$OP

a:=operator 'a
sol:=seriesSolve((2*x-2)*D(a(x),x)+4*a(x),a,x=0,[1])

[coefficient(sol,i) for i in 0..7] = [1,2,3,4,5,6,7,8]?

```

todo 319: construct coerce graph

```

todo 319: construct coerce graph

```

bug 7144: bug in eval

```

http://groups.google.com/group/fricas-devel/browse\_thread/thread/964c156e5925db15

y:=operator 'y
(1) y
deq:=differentiate(y x, x) + y x
,
(2) y (x) + y(x)
res:=first(solve(deq, y, x).basis)

```

```

      - x
(3) %e
eval(deq, y, x+>res)
      - x
(4) %e

instead:

y:=operator 'y
(1) y
deq:=differentiate(y x, x) + y x
,
(2) y (x) + y(x)
res:=first(solve(deq, y, x).basis)
      - x
(3) %e
function(res,f::Symbol,variables(res))
(4) f
f(1)
      1
(5) --
      %e
eval(deq,y,f)
(6) 0

```

bug 7143: asq cannot find AHYP

bug 7143: asq cannot find AHYP

todo 318: do a plucker/grassman example

todo 318: do a plucker/grassman example

Hi,

The Grassmannian $G(d,n)$ is the set of all d -dimensional subspaces of k^n . Such a subspace can be represented as a $d \times n$ matrix whose d rows are a basis for the subspace. The $d \times d$ minors of this matrix are the Plucker coordinates for the subspace, which gives an embedding of the Grassmannian into projective space.

(For example, a 2-dimensional subspace of k^4 can be represented as a 2×4 matrix, and the minors are the determinants of the 2×2 matrices we get by taking pairs of columns 12, 13, 14, 23, 24, 34. The values of

these minors can be used for form a point $[p_{12}:p_{13}:p_{14}:p_{23}:p_{24}:p_{34}]$ in projective 5-space, where p_{ij} is the ij th Plucker coordinate.)

It is claimed that the Grassmannian is a variety - in other words, the zero set of a set of polynomials. How do I go about listing these polynomials (ie generators for the ideal), in terms of the Plucker coordinates? (In other words, what relations must hold between the minors of a matrix, given that the rows are linearly independent?)

(I'm aware that there's a way to think about this in terms of the exterior algebra, but I'm hoping that there's a more straightforward way to see it.)

Thanks, David

Daniel Lichtblau

Newsgroups: sci.math.symbolic

From: Daniel Lichtblau <d...@wolfram.com>

Date: Wed, 10 Sep 2008 15:05:51 -0700 (PDT)

Local: Wed, Sep 10 2008 6:05 pm

Subject: Re: Defining relations for Grassmannian in Plucker coordinates?

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On Sep 10, 3:56 pm, DavidA <polyom...@f2s.com> wrote:

- Hide quoted text -

- Show quoted text -

> Hi,

> The Grassmannian $G(d,n)$ is the set of all d -dimensional subspaces of k^n . Such a subspace can be represented as a $d \times n$ matrix whose d rows are a basis for the subspace. The $d \times d$ minors of this matrix are the Plucker coordinates for the subspace, which gives an embedding of the Grassmannian into projective space.

> (For example, a 2-dimensional subspace of k^4 can be represented as a 2×4 matrix, and the minors are the determinants of the 2×2 matrices we get by taking pairs of columns 12, 13, 14, 23, 24, 34. The values of these minors can be used for form a point $[p_{12}:p_{13}:p_{14}:p_{23}:p_{24}:p_{34}]$ in projective 5-space, where p_{ij} is the ij th Plucker coordinate.)

> It is claimed that the Grassmannian is a variety - in other words, the zero set of a set of polynomials. How do I go about listing these polynomials (ie generators for the ideal), in terms of the Plucker coordinates? (In other words, what relations must hold between the minors of a matrix, given that the rows are linearly independent?)

> (I'm aware that there's a way to think about this in terms of the exterior algebra, but I'm hoping that there's a more straightforward way to see it.)

> Thanks, David

I don't know if this helps, but such a generating set can be computed explicitly for fixed (d,n) . One uses an elimination order on polynomials of the form $\{y_1-p_1, y_2-p_2, \dots\}$ and the result comprises polynomial relations between the y_j .

Here is code for this in Mathematica. It is simple enough that it should not be too hard to translate to other languages if so desired (I mean by you; I do not intend to do that).

```
relationsGrassman[dims_List] := Module[
  {x, xmat, y, yvars, pcoords},
  xmat = Array[x, dims];
  pcoords = First[Minors[xmat, dims[[1]]]];
  yvars = Array[y, Length[pcoords]];
  GroebnerBasis[yvars-pcoords, yvars,
    Flatten[xmat], MonomialOrder->EliminationOrder]
]

In[5]:= relationsGrassman[{2,4}]
Out[5]= {y$19[3] y$19[4] - y$19[2] y$19[5] + y$19[1] y$19[6]}

In[6]:= relationsGrassman[{2,5}]
Out[6]= {y$20[7] y$20[8] - y$20[6] y$20[9] + y$20[5] y$20[10],
  y$20[4] y$20[8] - y$20[3] y$20[9] + y$20[2] y$20[10],
  y$20[4] y$20[6] - y$20[3] y$20[7] + y$20[1] y$20[10],
  y$20[4] y$20[5] - y$20[2] y$20[7] + y$20[1] y$20[9],
  y$20[3] y$20[5] - y$20[2] y$20[6] + y$20[1] y$20[8]}
```

I note that use of Minors probably changes ordering of the p_{jk} minors from what you have in mind, so the order of the y variables would need to be altered accordingly.

Daniel Lichtblau
Wolfram Research

David L. Johnson

More options Sep 10, 10:37 pm
Newsgroups: sci.math.symbolic
From: "David L. Johnson" <david.john...@lehigh.edu>
Date: Wed, 10 Sep 2008 22:37:18 -0400
Local: Wed, Sep 10 2008 10:37 pm
Subject: Re: Defining relations for Grassmannian in Plucker coordinates?
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- Hide quoted text -

- Show quoted text -

DavidA wrote:

> Hi,

> The Grassmannian $G(d,n)$ is the set of all d -dimensional subspaces of
 > k^n . Such a subspace can be represented as a $d \times n$ matrix whose d rows
 > are a basis for the subspace. The $d \times d$ minors of this matrix are the
 > Plucker coordinates for the subspace, which gives an embedding of the
 > the Grassmannian into projective space.

> (For example, a 2-dimensional subspace of k^4 can be represented as a
 > 2×4 matrix, and the minors are the determinants of the 2×2 matrices we
 > get by taking pairs of columns 12, 13, 14, 23, 24, 34. The values of
 > these minors can be used for form a point $[p_{12}:p_{13}:p_{14}:p_{23}:p_{24}:p_{34}]$ in
 > projective 5-space, where p_{ij} is the ij th Plucker coordinate.)

> It is claimed that the Grassmannian is a variety - in other words, the
 > zero set of a set of polynomials. How do I go about listing these
 > polynomials (ie generators for the ideal), in terms of the Plucker
 > coordinates? (In other words, what relations must hold between the
 > minors of a matrix, given that the rows are linearly independent?)

> (I'm aware that there's a way to think about this in terms of the
 > exterior algebra, but I'm hoping that there's a more straightforward
 > way to see it.)

Well, that is what I was going to mention. It is also the source of the
 Pluecker coordinates. A k -plane in R^n is a unit, decomposable k -vector
 in $\Lambda_k(R^n)$. For $G(2,n)$, the relationship is simpler than for
 larger subspace dimensions, because a bivector $\xi \in \Lambda_2(R^n)$
 is decomposable if and only if $\xi \wedge \xi = 0$. You can then write
 that out explicitly as equations in the coordinates in $\Lambda_2(R^n)$,
 and they will be quadratics. For $k > 2$, certainly $\xi \wedge \xi = 0$, but
 that is not sufficient for ξ to be decomposable. Instead, the kernel
 of the map $R^n \rightarrow \Lambda_{k+1}(R^n)$ given by $v \mapsto \xi \wedge v$ has
 to be k -dimensional, but that still is polynomial in the coordinates.

From that description, though, it is not clear that $G(2,n)$ is smooth,
 and of the right dimension, since it is not a complete intersection, but
 of course it is.

--

David L. Johnson

When you are up to your ass in alligators, it's hard to remember that
 your initial objective was to drain the swamp.

-- LBJ

DavidA

More options Sep 11, 8:55 am
 Newsgroups: sci.math.symbolic
 From: DavidA <polyom...@f2s.com>
 Date: Thu, 11 Sep 2008 05:55:51 -0700 (PDT)
 Local: Thurs, Sep 11 2008 8:55 am
 Subject: Re: Defining relations for Grassmannian in Plucker coordinates?
 Reply | Reply to author | Forward | Print | Individual message | Show original | Report this message | Find
 Ah yes, thanks, that should have been obvious.

However, that still leaves a couple of questions:

- Clearly the Grassmannian is contained in the zero-set of those relations, but how do we know that they coincide? That is, how do we know that there aren't other zeroes of those relations, which are not the image of a $d \times n$ matrix under the Pluecker embedding.
- Is there any way we could have written down the relations a priori, without Groebner basis computations (and without exterior algebra)?

David L. Johnson

More options Sep 11, 10:45 am
 Newsgroups: sci.math.symbolic
 From: "David L. Johnson" <david.john...@lehigh.edu>
 Date: Thu, 11 Sep 2008 10:45:21 -0400
 Local: Thurs, Sep 11 2008 10:45 am
 Subject: Re: Defining relations for Grassmannian in Plucker coordinates?
 Reply | Reply to author | Forward | Print | Individual message | Show original | Report this message | Find

DavidA wrote:

> Ah yes, thanks, that should have been obvious.

> However, that still leaves a couple of questions:

- > - Clearly the Grassmannian is contained in the zero-set of those relations, but how do we know that they coincide? That is, how do we know that there aren't other zeroes of those relations, which are not the image of a $d \times n$ matrix under the Pluecker embedding.

Because of the nature of the Pluecker embedding. These coordinates are only the exterior algebra with respect to an explicit basis. In particular, as Lichtblau wrote, in the case of 2-planes, the condition $\xi \wedge \xi = 0$ is exactly equivalent to ξ being decomposable, $\xi = v \wedge w$. In higher dimensions the conditions are more complicated, but nevertheless completely describe the Grassmannian.

- > - Is there any way we could have written down the relations a priori,

> without Groebner basis computations (and without exterior algebra)?

I can't imagine it, but I also can't imagine why you would want it. Exterior algebra is the natural setting for this question. For the price of making sense of the mechanics, it clarifies exactly these relationships and makes clear what the relations are.

--

David L. Johnson

Some people used to claim that, if enough monkeys sat in front of enough typewriters and typed long enough, eventually one of them would reproduce the collected works of Shakespeare. The internet has proven this not to be the case.

DavidA

More options Sep 11, 4:14 pm

Newsgroups: sci.math.symbolic

From: DavidA <polyom...@f2s.com>

Date: Thu, 11 Sep 2008 13:14:36 -0700 (PDT)

Local: Thurs, Sep 11 2008 4:14 pm

Subject: Re: Defining relations for Grassmannian in Plucker coordinates?

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Perhaps the problem is that I don't understand the connection between the exterior algebra description of the Grassmannian and the Plucker coordinates (matrix minors) description. Please can someone spell it out.

Maarten Bergvelt

More options Sep 11, 4:35 pm

Newsgroups: sci.math.symbolic

From: Maarten Bergvelt <be...@math.uiuc.edu>

Date: Thu, 11 Sep 2008 20:35:47 +0000 (UTC)

Local: Thurs, Sep 11 2008 4:35 pm

Subject: Re: Defining relations for Grassmannian in Plucker coordinates?

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On 2008-09-11, DavidA <polyom...@f2s.com> wrote:

> Perhaps the problem is that I don't understand the connection between
> the exterior algebra description of the Grassmannian and the Plucker
> coordinates (matrix minors) description. Please can someone spell it
> out.

The matrix minors are the coefficients of the exterior products of the basis elements.

So let v, w be vectors in C^4 , and consider the wedge product v^w . It can be expanded in terms of the standard basis $e_i^e_j$, for $i < j$ of the second exterior power of C^4 , where e_i are the standard basis vectors of C^4 . Just write it out and you will see the 2x2 minors appear. In some detail:
 $v = v_1e_1 + v_2e_2 + v_3e_3 + v_4e_4$, $w = w_1e_1 + w_2e_2 + w_3e_3 + w_4e_4$,
 so
 $v^w = (v_1w_2 - v_2w_1) e_1^e_2 + \dots$

The pattern still holds for k vectors in C^n .

Hope that helps.

--

Maarten Bergvelt

todo 317: update INFORM notes

```
PRETTYPRINT((matrix [[1,2],[3,4]]):INFORM)$Lisp
```

todo 316: CircularList domain

make a new CircularList domain that handles length (#) properly.
 Move the destructive operations out of list and into here.

bug 7140: this returns the wrong answers

```
p:=(x:EXPR(INT)):EXPR(INT)+->3*x
q:=(x:EXPR(INT)):EXPR(INT)+->2*x+3
(p+q)(4) => 23      (correct)
(p+q)(x) => 5x + 3  (correct)
(p-q)(4) => 1       (correct)
(p-q)(x) => x-3     (correct)
(p*q)(4) => 33      (wrong)
(p*q)(x) => 6x+9    (wrong)
```

```
\subsection{package FUNMAPS FunctionMaps}
<<package FUNMAPS FunctionMaps>>=
)abbrev package FUNMAPS FunctionMaps
++ Description: Functional composition
++ Given functions f and g, returns the function composition (f op g)
```

```

FunctionMaps(A:SetCategory, B:IntegralDomain): cat == def where
EXP ==> Expression(Integer)
cat == with
  "+": (A->B, A->B) -> (A->B)
      ++ \spad(+) does functional addition
  "-": (A->B, A->B) -> (A->B)
      ++ \spad(-) does functional addition
  "*": (A->B, A->B) -> (A->B)
      ++ \spad(*) does functional addition
def == add
  funcAdd: (A->B, A->B, A) -> B
  funcSub: (A->B, A->B, A) -> B
  funcMul: (A->B, A->B, A) -> B

  funcAdd(g,h,x) == ((g x) + (h x))$B
  funcSub(g,h,x) == ((g x) - (h x))$B
  funcMul(g,h,x) == ((g x) * (h x))$B

  (a:(A -> B))+(b:(A -> B)) == funcAdd(a,b,#1)
  (a:(A -> B))-(b:(A -> B)) == funcSub(a,b,#1)
  (a:(A -> B))*(b:(A -> B)) == funcMul(a,b,#1)

```

bug 7139: this crashes Axiom when compiled

```

\subsection{package FUNMAPS FunctionMaps}
<<package FUNMAPS FunctionMaps>>=
)abbrev package FUNMAPS FunctionMaps
++ Description: Functional composition
++ Given functions f and g, returns the function composition (f op g)
FunctionMaps(A:SetCategory, B:IntegralDomain): cat == def where
EXP ==> Expression(Integer)
cat == with
  "+": (A->B, A->B) -> (A->B)
      ++ \spad(+) does functional addition
      ++
      ++E f:=(x:INT):INT +-> 3*x
      ++E g:=(x:INT):INT +-> 2*x+3
      ++E (f+g)(4)
  "-": (A->B, A->B) -> (A->B)
      ++ \spad(+) does functional addition
      ++
      ++E f:=(x:INT):INT +-> 3*x
      ++E g:=(x:INT):INT +-> 2*x+3
      ++E (f-g)(4)
  "*": (A->B, A->B) -> (A->B)
      ++ \spad(+) does functional addition
      ++

```

```

++E f:=(x:INT):INT +-> 3*x
++E g:=(x:INT):INT +-> 2*x+3
++E (f*g)(4)
"/": (A->EXP, A->EXP) -> (A->EXP)
++ \spad(+) does functional addition
++
++E p:=(x:EXPR(INT)):EXPR(INT)+->3*x
++E q:=(x:EXPR(INT)):EXPR(INT)+->2*x+3
++E (p/q)(4)
++E (p/q)(x)
def == add
fab ==> (A -> B)
faei ==> (A -> Expression(Integer))
EXP ==> Expression(Integer)

funcAdd: (A->B, A->B, A) -> B
funcSub: (A->B, A->B, A) -> B
funcMul: (A->B, A->B, A) -> B
funcDiv: (A -> EXP, A -> EXP, A) -> EXP
funcAdd(g,h,x) == ((g x) + (h x))$B
funcSub(g,h,x) == ((g x) - (h x))$B
funcMul(g,h,x) == ((g x) * (h x))$B
funcDiv(g,h,x) == ((g x) / (h x))$EXP

(a:fab)+(b:fab) == funcAdd(a,b,#1)
(a:fab)-(b:fab) == funcSub(a,b,#1)
(a:fab)*(b:fab) == funcMul(a,b,#1)
(a:faei)/(b:faei) == funcDiv(a,b,#1)

```

bug 7138: unravel in CartesianTensor does not work

```

n:SquareMatrix(2,Integer):=matrix [[2,3],[0,1]]
tn:CartesianTensor(1,2,Integer):=n
unravel ravel tn

```

bug 7137: {}\$(List INT)

The function SEQ is not implemented in List Integer .

meh 4: as of aug 2008, changecount/name/percent in changelog

tpd	4485	Tim Daly	82.55
gdr	452	Gabriel Dos Reis	8.31
wxh	216	Waldek Hebisch	3.97
mxm	46	Mark Murray	0.84
wsp	28	Bill Page	0.51
mxr	24	Martin Rubey	0.44
acr	21	Arthur C. Ralfs	0.38
gxv	19	Gregory Vanuxem	0.34
sxw	19	Stephen Wilson	0.34
rhx	13	Ralf Hemmecke	0.23
pab	11	Peter Broadbery	0.20
cxc	8	Christophe Conil	0.01
cxm	8	Camm Maguire	0.01
dxm	8	David Mentre	0.01
mxt	6	Mike Thomas	0.01
fxl	5	Frederic Lehouby	0.00
rxr	5	Renaud Rioboo	
cys	4	Chris Yapp	
jap	3	Jose Alfredo Portes	
jxw	3	Juergen Weiss	
mmm	3	Marc Moreno Maza	
wys	3	William Sit	
axh	2	Antoine Hersen	
bmt	2	Barry Trager	
bsm	2	Bob McElrath	
bxt	2	Baldir Thomas	
hoz	2	Humberto Ortiz-Zuazaga	
exs	2	Eugene Surowitz	
dxh	2	Dan Hatton	
mvz	2	Matijs van Zuijlen	
rxh	2	Richard Harke	
tst	2	Themos Tsikas	
axr	1	Anatoly Raportirenko	
dpt	1	Dylan Thurston	
fxm	1	Francois Maltey	
kxo	1	Kostas Oikonomou	

bug 7136: uncommenting `--output("")` changes behavior

bug 7136: uncommenting `--output("")` changes behavior (greg vanuxem)

Further investigation shows that it is a bug in PARSE-Conditional

This appears to be a problem only when you have a nested one-armed if-statement (gaby)

```
)abb package FOO Foo
Foo(R,VCR): Exports == Implementation where
  R : FloatingPointSystem
  VCR : VectorCategory R
  Exports == with
    bar: (Boolean,VCR) -> Integer
    ++ bar(b,v)
    bad: (Boolean,VCR) -> Integer
    ++ bad(b,v)
  Implementation == add
    import OutputPackage
    bar(b,v)==
      if b then
        --output("")
        for i in minIndex v..maxIndex v repeat
          if v.i <= 0 then return -1
      else
        for i in minIndex.v..maxIndex.v repeat
          v.i:=1
    output(v::OutputForm)
    0

    bad(b,v)==
      if b then
        output("")
        for i in minIndex v..maxIndex v repeat
          if v.i <= 0 then return -1
      else
        for i in minIndex.v..maxIndex.v repeat
          v.i:=1
    output(v::OutputForm)
    0
```

```
v:=new(5,0)$Vector(SF)
bar(false,v)
[0.,0.,0.,0.,0.]
```

```
(7) 0
```

Type: NonNegativeInteger

```
v:=new(5,0)$Vector(SF)
bad(false,v)
[1.,1.,1.,1.,1.]
```

```
(3) 0
```

meh 3: from Bill Page

I just wanted to mention something here that I just learned about how the Axiom interpreter handles option arguments. When discovering how the 'draw' operations in Axiom processes optional arguments, e.g.

```
draw(sin(x), x=1..10, title=="sin function")
```

I realized the interpreter does something rather clever but probably quite unexpected. The problem is that (unlike Aldor) SPAD does not provide built-in support for calling functions with optional arguments. Instead what it does is collect all the arguments containing == into a List *and* then it applies the name to the left of == as a function to the value on the right. So much to my surprise:

```
(1) -> (x+>x)(sin==x,cos==y)
```

```
(1) [sin(x),cos(y)]
```

Type: List Expression Integer

is equivalent to:

```
(2) -> (x+>x)([sin(x),cos(y)])
```

```
(2) [sin(x),cos(y)]
```

Type: List Expression Integer

The list of optional arguments is always passed as the last argument to the function. This is completely general so one can write for example:

```
(3) -> ((x,y)+>[x,y])(sin==x,n,cos==y)
```

```
(3) [n,[sin(x),cos(y)]]
```

Type: List Any

What do you think? Do you like that, or is this something that should be implemented in a deeper way?

todo 316: EGADS Evolutionary Gaussian Automatic Decomposition

look at implementing such an algorithm for axiom

todo 315: missing input files

/research/research/Axiom.NAG/development/int/doc/htex/input

LAZM3PK.input
bios.input
copyright.input
cover.input
foreword.input
intro.input
redo.input
sum.input
tecintro.input
ug00.input
ug01.input
ug02.input
ug03.input
ug04.input
ug05.input
ug06.input
ug07.input
ug08.input
ug10.input
ug11.input
ug12.input
ug13.input
ug14.input
ug15.input
ug16.input
ug17.input
ug18.input
ug19.input
ug20.input
ug21.input

/research/research/Axiom.NAG/development/int/input

.input.record
ANNA-ES.input
CPHelp.input
HTXAdvPage1.input
HTXAdvPage2.input
HTXAdvPage3.input
HTXAdvPage4.input
HTXAdvPage5.input

HTXAdvPage6.input
HTXAdvTopPage.input
HTXFormatPage1.input
HTXFormatPage2.input
HTXFormatPage3.input
HTXFormatPage4.input
HTXFormatPage5.input
HTXFormatPage6.input
HTXFormatPage7.input
HTXFormatPage8.input
HTXFormatTopPage.input
HTXIntroPage1.input
HTXIntroPage2.input
HTXIntroPage3.input
HTXIntroTopPage.input
HTXLinkPage1.input
HTXLinkPage2.input
HTXLinkPage3.input
HTXLinkPage4.input
HTXLinkPage5.input
HTXLinkPage6.input
HTXLinkTopPage.input
HTXTopPage.input
HTXTryPage.input
HTXplay.input
LAZM3PK.input
Link.input
algebra.input
aspex.input
basic.input
bmcatt.input
coverex.input
exmatrix.input
expose.input
gloss.input
hilbert.input
hilbert0.input
hyperdoc.input
images6a.input
man0.input
mapping.input
matops.input
nagaux.input
nagc.input
nagd.input
nage.input
nagf.input
nagm.input
nags.input
nagx.input

newuser.input
patch.input
r2.0_bugs.input
r2.0a_bugs.input
r2.1_bugs.input
r2.1_bugs_big.input
record.input
romnum.input
rootpage.input
srchkey.input
topics.input
type.input
ug.input
ug00.input
ug01.input
ug02.input
ug03.input
ug04.input
ug05.input
ug06.input
ug07.input
ug08.input
ug10.input
ug11.input
ug12.input
ug13.input
ug14.input
ug15.input
ug16.input
ug20.input
ug21.input
union.input
util.input
xmpexp.input

/research/research/Axiom.NAG/development/int/newpaste/

ANNA-ES.input
CPHelp.input
HTXAdvPage1.input
HTXAdvPage2.input
HTXAdvPage3.input
HTXAdvPage4.input
HTXAdvPage5.input
HTXAdvPage6.input
HTXAdvTopPage.input
HTXFormatPage1.input
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HTXFormatPage4.input

HTXFormatPage5.input
HTXFormatPage6.input
HTXFormatPage7.input
HTXFormatPage8.input
HTXFormatTopPage.input
HTXIntroPage1.input
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HTXplay.input
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algebra.input
asug.input
asug01.input
asug02.input
asug03.input
asug04.input
asug05.input
asug06.input
asug07.input
asug08.input
asug0A.input
asug0B.input
asug0C.input
asutil.input
basic.input
bmcats.input
coverex.input
exmatrix.input
expose.input
gloss.input
hyperdoc.input
man0.input
mapping.input
nagaux.input
nagc.input
nagd.input
nage.input
nagf.input
nagm.input
nags.input

nagx.input
newuser.input
patch.input
record.input
redo.input
rootpage.input
srchkey.input
topics.input
type.input
ug.input
ug00.input
ug01.input
ug02.input
ug03.input
ug04.input
ug05.input
ug06.input
ug07.input
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ug12.input
ug13.input
ug14.input
ug15.input
ug20.input
union.input
util.input
xmpexp.input

/research/research/Axiom.NAG/development/int/paste/

ANNA-ES.input
CPHelp.input
HTXAdvPage1.input
HTXAdvPage2.input
HTXAdvPage3.input
HTXAdvPage4.input
HTXAdvPage5.input
HTXAdvPage6.input
HTXAdvTopPage.input
HTXFormatPage1.input
HTXFormatPage2.input
HTXFormatPage3.input
HTXFormatPage4.input
HTXFormatPage5.input
HTXFormatPage6.input
HTXFormatPage7.input
HTXFormatPage8.input
HTXFormatTopPage.input

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HTXIntroPage2.input
HTXIntroPage3.input
HTXIntroTopPage.input
HTXLinkPage1.input
HTXLinkPage2.input
HTXLinkPage3.input
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HTXLinkPage5.input
HTXLinkPage6.input
HTXLinkTopPage.input
HTXTopPage.input
HTXTryPage.input
HTXplay.input
LAZM3PK.input
Link.input
algebra.input
aspex.input
basic.input
bmcats.input
mapping.input
redo.input
topics.input
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ug01.input
ug02.input
ug03.input
ug04.input
ug05.input
ug06.input
ug07.input
ug08.input
ug10.input
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ug12.input
ug13.input
ug14.input
ug15.input
ug16.input
ug21.input

/research/research/Axiom.NAG/development/src/algebra_marc

benchs.input

/research/research/Axiom.NAG/development/src/htex/input.awk

input.awk
mktinput.c
tinput.awk
```

tinput.awk.old

/research/research/Axiom.NAG/development/src/input

as-eg7.input
hilbert.input
hilbert0.input
images6a.input
matops.input
romnum.input

/research/research/gold/src/input

images6a.input.pamphlet

/research/research/integration/src/input

1.input
10.input
100.input
101.input
102.input
103.input
104.input
105.input
106.input
107.input
108.input
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183.input
184.input
185.input
186.input
187.input
188.input
189.input
19.input
190.input
191.input
192.input
193.input
194.input
195.input
196.input
197.input
198.input
199.input
2.input
20.input
200.input
201.input
202.input
203.input
204.input
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206.input
207.input
208.input
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210.input
211.input
212.input
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215.input
216.input
217.input
218.input

219.input
22.input
220.input
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222.input
223.input
224.input
225.input
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227.input
228.input
229.input
23.input
230.input
231.input
232.input
24.input
25.input
26.input
27.input
28.input
29.input
3.input
30.input
31.input
32.input
33.input
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38.input
39.input
4.input
40.input
41.input
42.input
43.input
44.input
45.input
46.input
47.input
48.input
49.input
5.input
50.input
51.input
52.input
53.input
54.input

55.input
56.input
57.input
58.input
59.input
6.input
60.input
61.input
62.input
63.input
64.input
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67.input
68.input
69.input
7.input
70.input
71.input
72.input
73.input
74.input
75.input
76.input
77.input
78.input
79.input
8.input
80.input
81.input
82.input
83.input
84.input
85.input
86.input
87.input
88.input
89.input
9.input
90.input
91.input
92.input
93.input
94.input
95.input
96.input
97.input
98.input
99.input
Xinf.input

```
Xinfinite.input
demo.input
f.input
foo.input
foo1.input
redo.input

/research/research/integration/src/provisos/foo.input

/research/research/journal/bailey/cl-f2cl-1.0+cvs.2002.11.11/packages/minpack/lmdif-input.dat

/research/newdatabase/src/input

    images6a.input.pamphlet

/research/newdatabase

    intparfrac.input.pamphlet
    improper.input.pamphlet
    typeconv.input.pamphlet
    doit.input

/research/axiom/src/input

    xpr.input.pamphlet
    images6a.input.pamphlet

/research/may07/wh-sandbox/src/input

    images6a.input.pamphlet

/research/pending

    improper.input.pamphlet
    intparfrac.input.pamphlet
    ioutput.input.pamphlet
    typeconv.input.pamphlet
    doit.input
    bonda.input
    bug361.input
    expint.input
    r.input

/research/pending/ODE/input

    f.input
    keep/f.input
    keep/kamke10.input
    keep/kamke2a.input
    keep/kamke2.input
```

keep/kamke3.input
keep/kamke4.input
keep/kamke5.input
keep/kamke6.input
keep/kamke7.input
keep/kamke8.input
keep/kamke9.input
keep/k.input
keep/kamke3a.input
k.input
ODES.input
ODESzero.input
#ODES.input#
krantz.input
decomp.input
bronson.input

/research/pending/p.input

hyper/input.pamphlet
hyper/parse_input.pamphlet
storevar.input.txt
storevar1.input.txt
des.input
debug.input
strang.input
magma/magmafirst.input
guessrec.input
redblack.input
vb/vb1-20.input
vb/vb1001-1020.input
vb/vb101-120.input
vb/vb1021-1040.input
vb/vb1041-1060.input
vb/vb1061-1080.input
vb/vb1081-1100.input
vb/vb1101-1120.input
vb/vb1121-1140.input
vb/vb1141-1160.input
vb/vb1161-1180.input
vb/vb1181-1200.input
vb/vb1201-1220.input
vb/vb121-140.input
vb/vb1221-1240.input
vb/vb1241-1260.input
vb/vb1261-1280.input
vb/vb1281-1300.input
vb/vb1301-1320.input
vb/vb1321-1340.input
vb/vb1341-1360.input

vb/vb1361-1380.input
vb/vb1381-1400.input
vb/vb1401-1420.input
vb/vb141-160.input
vb/vb1421-1440.input
vb/vb1441-1460.input
vb/vb1461-1480.input
vb/vb1481-1500.input
vb/vb1501-1520.input
vb/vb1521-1540.input
vb/vb1541-1560.input
vb/vb1561-1580.input
vb/vb1581-1600.input
vb/vb1601-1620.input
vb/vb161-180.input
vb/vb1621-1640.input
vb/vb1641-1660.input
vb/vb1661-1680.input
vb/vb1681-1700.input
vb/vb1701-1720.input
vb/vb1721-1740.input
vb/vb1741-1760.input
vb/vb1761-1780.input
vb/vb1781-1800.input
vb/vb1801-1820.input
vb/vb181-200.input
vb/vb1821-1840.input
vb/vb1841-1860.input
vb/vb1861-1880.input
vb/vb1881-1900.input
vb/vb1901-1920.input
vb/vb1921-1940.input
vb/vb1941-1960.input
vb/vb1961-1980.input
vb/vb1981-2000.input
vb/vb2001-2020.input
vb/vb201-220.input
vb/vb2021-2040.input
vb/vb2041-2060.input
vb/vb2061-2080.input
vb/vb2081-2100.input
vb/vb21-40.input
vb/vb2101-2120.input
vb/vb2121-2140.input
vb/vb2141-2160.input
vb/vb2161-2180.input
vb/vb2181-2200.input
vb/vb2201-2220.input
vb/vb221-240.input
vb/vb2221-2240.input

vb/vb2241-2260.input
vb/vb2261-2280.input
vb/vb2281-2300.input
vb/vb2301-2320.input
vb/vb2321-2340.input
vb/vb2341-2360.input
vb/vb2361-2380.input
vb/vb2381-2400.input
vb/vb2401-2420.input
vb/vb241-260.input
vb/vb2421-2440.input
vb/vb2441-2460.input
vb/vb2461-2480.input
vb/vb2481-2500.input
vb/vb2501-2520.input
vb/vb2521-2540.input
vb/vb2541-2560.input
vb/vb2561-2580.input
vb/vb2581-2600.input
vb/vb2601-2620.input
vb/vb261-280.input
vb/vb2621-2640.input
vb/vb2641-2660.input
vb/vb2641-2760.input
vb/vb2661-vb2680.input
vb/vb2661a-2680a.input
vb/vb2681-2700.input
vb/vb2701-2720.input
vb/vb2721-2740.input
vb/vb2761-2780.input
vb/vb2781-2800.input
vb/vb2801-2820.input
vb/vb281-300.input
vb/vb2821-2840.input
vb/vb2841-2860.input
vb/vb2861-2880.input
vb/vb2881-2900.input
vb/vb2901-2920.input
vb/vb2921-2940.input
vb/vb2941-2960.input
vb/vb2961-2980.input
vb/vb2981-3000.input
vb/vb3001-3020.input
vb/vb301-320.input
vb/vb3021-3040.input
vb/vb3041-3060.input
vb/vb3061-3080.input
vb/vb3081-3100.input
vb/vb3101-3120.input
vb/vb3121-3140.input

vb/vb3141-3160.input
vb/vb3161-3180.input
vb/vb3181-3200.input
vb/vb3201-3220.input
vb/vb3221-3240.input
vb/vb3241-3260.input
vb/vb3261-3280.input
vb/vb3281-3300.input
vb/vb3301-3320.input
vb/vb3321-3340.input
vb/vb3341-3360.input
vb/vb3361-3380.input
vb/vb3381-3400.input
vb/vb3401-3420.input
vb/vb3421-3440.input
vb/vb3441-3460.input
vb/vb3461-3480.input
vb/vb3481-3500.input
vb/vb3501-3520.input
vb/vb3521-3540.input
vb/vb3541-3560.input
vb/vb3561-3580.input
vb/vb3581-3600.input
vb/vb3601-3620.input
vb/vb3621-3640.input
vb/vb3641-3660.input
vb/vb3661-3680.input
vb/vb3681-3700.input
vb/vb3701-3720.input
vb/vb3721-3740.input
vb/vb3741-3760.input
vb/vb3761-3780.input
vb/vb3781-3800.input
vb/vb3801-3820.input
vb/vb3821-3840.input
vb/vb3841-3860.input
vb/vb3861-3880.input
vb/vb3881-3900.input
vb/vb3901-3920.input
vb/vb3921-3940.input
vb/vb3941-3960.input
vb/vb3961-3980.input
vb/vb3981-4000.input
vb/vb4001-4020.input
vb/vb4021-4040.input

vb/vb4041-4060.input
vb/vb4061-4080.input
vb/vb4081-4100.input
vb/vb41-60.input
vb/vb4101-4120.input
vb/vb4121-4140.input
vb/vb4141-4160.input
vb/vb4161-4180.input
vb/vb4181-4200.input
vb/vb4201-4220.input
vb/vb421-440.input
vb/vb4221-4440.input
vb/vb4241-4260.input
vb/vb4261-4280.input
vb/vb4281-4300.input
vb/vb4301-4320.input
vb/vb4321-4340.input
vb/vb4341-4360.input
vb/vb4361-4380.input
vb/vb4381-4400.input
vb/vb4401-4420.input
vb/vb441-460.input
vb/vb4422-4440.input
vb/vb4441-4460.input
vb/vb4461-4480.input
vb/vb4481-4500.input
vb/vb4501-4520.input
vb/vb4521-4540.input
vb/vb4541-4560.input
vb/vb4561-4580.input
vb/vb4581-4600.input
vb/vb4601-4620.input
vb/vb461-480.input
vb/vb4621-4640.input
vb/vb4641-4660.input
vb/vb4661-4680.input
vb/vb4681-4700.input
vb/vb4701-4720.input
vb/vb4721-4740.input
vb/vb4741-4760.input
vb/vb4761-4772.input
vb/vb481-500.input
vb/vb501-520.input
vb/vb521-540.input
vb/vb541-560.input
vb/vb561-580.input
vb/vb581-600.input
vb/vb601-620.input
vb/vb61-80.input
vb/vb621-640.input

```
vb/vb641-660.input
vb/vb661-680.input
vb/vb681-700.input
vb/vb701-720.input
vb/vb721-740.input
vb/vb741-760.input
vb/vb761-780.input
vb/vb781-800.input
vb/vb801-820.input
vb/vb81-100.input
vb/vb821-840.input
vb/vb841-860.input
vb/vb861-880.input
vb/vb881-900.input
vb/vb901-920.input
vb/vb921-940.input
vb/vb941-960.input
vb/vb961-980.input
vb/vb980-1000.input
factor/twister.input
groebner.input
testgrob.input
factor.input
gcd.input
twister.input
outofmemory.input
redo.input
kkttest.input
cannotsolve.input
normalizeBug.input
sinaxminusaxBUG.input
cannotsolve.input~
integrationBUG.input
integrateBUG.input
mail.input
mail.input~
mathmlbug.input.pamphlet

/research/int/input

ACPLOT.input
REALSOLV.input
TESTFR.input
VIEW2D.input
images6a.input
```


todo 314: Tiny programs for computing constants

<http://numbers.computation.free.fr/Constants/TinyPrograms/tinycodes.html>

bug 7135: hyperdoc bug

```
topics -> calculus -> integrate -> first integral
(close the integration popup window kills axiom) with:
(1) -> (HyperDoc) Unknown command from SessionServer -1
```

meh 2: Fonts available at www.dafont.com

meh 2: Fonts available at www.dafont.com

bug 7134: lodo.spad LinearOrdinaryDifferentialOperator1

```
% NOTE: the book has a different answer
f := rightLcm(a,b)

% NOTE: the book has a different answer
rightRemainder(f, b)

% NOTE: the book has a different answer
leftRemainder(f, b)
```

bug 7133: files.spad Library

```
)spool Library.output

Starts dribbling to Library.output (2007/8/31, 12:8:10).
)set message test on

)set message auto off

)clear all

All user variables and function definitions have been cleared.
```

```
stuff := library "Neat.stuff"
```

```
Daly Bug
>> Error detected within library code:
File is not readable
"Neat.stuff"
```

bug 7132: numtheor.spad IntegerNumberTheoryFunctions

check the text against the book

bug 7131: – files.spad.pamphlet KeyedAccessFile

```
ey: KeyedAccessFile(Integer) := open("editor.year", "output")
```

```
(1) "editor.year"
                                     Type: KeyedAccessFile Integer
ey."Char" := 1986
```

```
(2) 1986
                                     Type: PositiveInteger
ey."Caviness" := 1985
```

```
(3) 1985
                                     Type: PositiveInteger
ey."Fitch"    := 1984
```

```
(4) 1984
                                     Type: PositiveInteger
ey."Char"
```

```
Daly Bug
>> Error detected within library code:
File is not readable
"editor.year"
```

bug 7130: – table.spad.pamphlet GeneralSparseTable.input

```
)spool GeneralSparseTable.output
)set message test on
)set message auto off
)clear all
patrons: GeneralSparseTable(String, Integer, KeyedAccessFile(Integer), 0) := table() ;
patrons: GeneralSparseTable(String, Integer, KeyedAccessFile(Integer), 0) := table() ;
```

Daly Bug

```
>> Error detected within library code:
File is not readable
"kaf1405.sdata"
```

```
(1) ->
patrons."Smith" := 10500
patrons."Jones" := 22000
patrons."Jones"
patrons."Stingy"
reduce(+, entries patrons)
)system rm -r kaf*.sdata
)spool
)lisp (bye)
```

bug 7129: these take a long time

```
ode97 := x*D(y(x),x) + a*y(x)**2 - y(x) + b*x**2
ode104 := x*D(y(x),x) + a*x*y(x)**2 + 2*y(x) + b*x
ode105 := x*D(y(x),x) + a*x*y(x)**2 + b*y(x) + c*x + d
```

todo 313: put PolynomialCategory, etc into front matter diagram

```
todo 313: put PolynomialCategory, etc into front matter diagram
```

todo 312: maple routines

sharelib.zip
maplenuts.zip

bug 7128: This kills Axiom

```
draw(1/a*log(a*x+b),x=-1..1)

; (DEFUN %A ...) is being compiled.
;; The variable |a| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |b| is undefined.
;; The compiler will assume this variable is a global.
Compiling function %B with type DoubleFloat -> DoubleFloat
```

meh 1: script for exploring domain relationships

```
getDomains(cat: Symbol): Set Symbol ==
  set map(d +-> symbol first destruct first destruct d,
    destruct domainsOf(cat, nil)$Lisp)
```

bug 7127: Schaums 14:472 fails to integrate properly

(1) -> t1:=x*asin(x/a)

$$(1) \quad x \operatorname{asin}\left(\frac{x}{a}\right)$$

Type: Expression Integer

(2) -> t2:=integrate(t1,x)

$$(2) \quad \frac{(-2x^2 + a^2) \operatorname{atan}\left(\frac{2x\sqrt{-x^2 + a^2}}{2x^2 - a^2}\right) + 2x\sqrt{-x^2 + a^2}}{8}$$

Type: Union(Expression Integer,...)

(3) -> t3:=D(t2,x)

$$(3) \quad - \frac{x \operatorname{atan}\left(\frac{2x\sqrt{-x^2+a^2}}{2x^2-a^2}\right)}{2}$$

Type: Expression Integer

(4) -> t4:=(x^2/2-a^2/4)*asin(x/a)+(x*sqrt(a^2-x^2))/4

$$(4) \quad \frac{x\sqrt{-x^2+a^2} + (2x^2-a^2)\operatorname{asin}\left(\frac{x}{a}\right)}{4}$$

Type: Expression Integer

(5) -> t5:=D(t4,x)

(5)

$$\frac{(4ax \operatorname{asin}\left(\frac{x}{a}\right)\sqrt{-x^2+a^2} - 2ax^3 + a^2) \sqrt{-x^2+a^2} + (2x^2-a^2)\sqrt{-x^2+a^2}}{4a\sqrt{-x^2+a^2}}$$

Type: Expression Integer

(6) -> f:=makeFloatFunction(t1,x,a)

Compiling function %BF with type (DoubleFloat,DoubleFloat) ->
DoubleFloat

(6) theMap(MKBCFUNC;binaryFunction;SM;2!0,120)

Type: ((DoubleFloat,DoubleFloat) -> DoubleFloat)

(7) -> axiom:=makeFloatFunction(t3,x,a)

Compiling function %BJ with type (DoubleFloat,DoubleFloat) ->
DoubleFloat

(7) theMap(MKBCFUNC;binaryFunction;SM;2!0,996)

Type: ((DoubleFloat,DoubleFloat) -> DoubleFloat)

(8) -> schaums:=makeFloatFunction(t5,x,a)

Compiling function %BK with type (DoubleFloat,DoubleFloat) ->

```

DoubleFloat

(8)  theMap(MKBCFUNC;binaryFunction;SM;2!0,62)
      Type: ((DoubleFloat,DoubleFloat) -> DoubleFloat)
(9) -> [ [f(i::Float,i::Float+1.0::Float)::Float,axiom(i::Float,i::Float+1.0::Float)::Float,scha
(9)
      [[0.5235987755 9829892668,0.5235987755 9829892668,0.5235987755 9829881566],
      [1.4594553124 539326738,1.4594553124 539326738,1.4594553124 539324518],
      [2.5441862369 444430136,- 2.1682027434 402466604,2.5441862369 444430136],
      [3.7091808720 064496363,- 2.5740044351 731374839,3.7091808720 064500804]]
      Type: List List Float
(10) ->

```

bug 7126: sinaxminusaxBUG.input shows a bug

why does $\sin(a x - ax)$ not become $\sin(0)$ then 0?

bug 7125: normalizeBug.input shows a bug

bug 7125: normalizeBug.input shows a bug

bug 7124: why is the superscript not in smallest terms?

```

--S 100
aa:=integrate(1/(x^2*(x^2-a^2)^(3/2)),x)
--R
--R
--R
--R
--R      1
--R      (1)  - ----
--R              +-----+
--R      3      2      | 2      2      4      2 2
--R      (2x  - a x)\|x  - a  - 2x  + 2a x
--R
--R                                          Type: Union(Expression Integer,...)
--E

--S 101
bb:=-sqrt(x^2-a^2)/(a^4*x)-x/(a^4*sqrt(x^2-a^2))
--R
--R
--R      2      2
--R      - 2x  + a

```

```

--R (2) -----
--R          +-----+
--R          4 | 2 2
--R          a x\|x - a
--R
--R                                          Type: Expression Integer
--E

--S 102      14:228 Schaums and Axiom differ by a constant
cc:=aa-bb
--R
--R          2
--R (3)  - --
--R          4
--R          a
--R
--R                                          Type: Expression Integer
--E

```

todo 311: integerMathLibrary FFLAS_technical_report.ps.gz

todo 311: integerMathLibrary FFLAS_technical_report.ps.gz

todo 310: /research/femlisp (finite element meshes)

todo 310: /research/femlisp (finite element meshes)

bug 7123: cannot simplify forms $(ax+b)(ax+b)^n$ into $(ax+b)^{n+1}$

bug 7123: cannot simplify forms like $(ax+b)(ax+b)^n$ into $(ax+b)^{n+1}$

wish 1010: look at ocalc.tex for big-O notation by Knuth

wish 1010: look at ocalc.tex for big-O notation by Knuth

todo 309: huh?

```
in2360a:=integrate(1/(1-%i*z^2)^(1/2)-csch(z-1), z= -1..1,"noPole")
```

```
(175) [ + infinity, + infinity]
      Type: Union(f2: List OrderedCompletion Expression Complex Integer,...)
```

Note: verified against NAG

todo 309: these won't integrate

```
in13:=integrate(cos(z)^2*sin(z)^n, z)
in160:=integrate((-z^2-z)^sin(z), z)
in163:=integrate(log(z)*abs(exp(z)/z), z)
in166:=integrate((1+z)^z, z)
```

bug 7122: "failed"

```
in106a:=integrate(((%i+z)^%i)^%i, z= 0..1,"noPole")
in1029a:=integrate((%i*z/(%i*z+1))^(1/2), z= 0..%plusInfinity,"noPole")
in1107a:=integrate(-log(z)/(-1+z^(1/2))^3, z= 0..1/2*%i,"noPole")
in1108a:=integrate(log(z)/(-1+z^(1/2))^3, z= 0..1/2*%i,"noPole")
in159a:=integrate(log(z)/z^2, z= -%i..-1,"noPole")
in1859a:=integrate(1/(z^2-1)*(1+1/z^3)^(1/2), z= 0..1,"noPole")
in1867a:=integrate(1/(z^2-1)*(1-(%i*z)^(1/2))^(1/2), z= 0..1,"noPole")
in1869a:=integrate(1/(z^2-1)*(1-1/(%i*z)^(1/2))^(1/2), z= 0..1,"noPole")
in2109a:=integrate((1+%i*z)^(1/2)+cot(z-1), z= 0..1,"noPole")
in2179a:=integrate((z^3-%i*(1-z^4)^(1/2))^4, z= -1..1,"noPole")
in2247a:=integrate(1/(z^2-%i*z+2)^(1/2)+csc(z), z= 0..1,"noPole")
in2340a:=integrate(exp(%i*(%i*z)^(1/2))+cot(z-1), z= -1..1,"noPole")
in2527a:=integrate(1/(1+%i*z^(1/3))+cosh(z), z= %minusInfinity..0,"noPole")
in2567a:=integrate(1/(%i*z)^(1/3)*log(abs(z-1)), z= 0..1/2*%pi,"noPole")
in2597a:=integrate((1+%i/z)^(1/2)-log(1-1/z^2), z= 0..%plusInfinity,"noPole")
in2963a:=integrate(cosh(-1+(1-z)^(1/2))+(z^2+%i*z-3)^(1/2), z= -1..1,"noPole")
```

bug 7121: Imaginary part is nonzero. Cannot retract.

```
in2691a:=integrate(1/(z^2+%i*z-1)^(1/2)+(1-%i*z)^(1/2), z= -1..1,"noPole")
```


bug 7120: Zero divisor

```
in2307a:=integrate(tan(z)-1/(z^2+%i*z-3)^(1/2), z= -1..1,"noPole")
```

bug 7119: Cannot take first of an empty list

```
in1278a:=integrate((1+1/(%i*z)^(1/2))^(1/2), z= 0..%plusInfinity,"noPole")
in1433a:=integrate(-1/((%i*z)^(1/2)-1)*(-2+(%i*z)^(1/2)), z= 0..%plusInfinity,"noPole")
in1435a:=integrate(-1/(1+(%i*z)^(1/2))*(%i*z)^(1/2), z= 0..%plusInfinity,"noPole")
in1436a:=integrate(-((%i*z)^(1/2)-1)/(-2+(%i*z)^(1/2)), z= 0..%plusInfinity,"noPole")
in1463a:=integrate(1-(%i*z)^(1/2)/(1-%i*z), z= 0..%plusInfinity,"noPole")
in1563a:=integrate(log(1-z^3)*(%i*z)^(1/2), z= 0..1,"noPole")
in1638a:=integrate(-z/(z-1)/(1-%i*z)^(1/2), z= 0..1,"noPole")
in1724a:=integrate(-z^2/(z^2-1)*(1-(%i*z)^(1/2))^(1/2), z= 0..1,"noPole")
in183:=integrate(1/z, z= %i..2*i)
in2108a:=integrate((1-%i/(1+%i/z)^2)^(1/2), z= -1..1,"noPole")
in2124a:=integrate(-1/z-1/(%i/(z+%i))^(1/2), z= -1..0,"noPole")
in2815a:=integrate((%i*z)^(1/2)-(1+1/(%i*z)^(1/2))^(1/2), z= -1..1,"noPole")
in2924a:=integrate(1/(%i*z)^(1/2)+(1+1/(%i*z)^(1/2))^(1/2), z= -1..1,"noPole")
in3001a:=integrate(((%i*z)^(1/2)-1)/(-2+(%i*z)^(1/2)), z= 0..%plusInfinity,"noPole")
```

wish 1009: integrate computational geometry algorithms (CGAL)

```
bounding volumes
polyhedral surfaces
boolean operators
triangulations
voronoi diagrams
mesh generation
subdivision
simplification
parametrization
streamlines
ridge detection
neighbor search
kinetic data structures
lower envelope
arrangement
intersection detection
minkowski sum
PCA
polytope distance
```

QP solver

geometric objects in 2d, 3d, nd: point, segment, ray, line, circle,
 sphere, iso-rectangle, predicates (orientation, in_circle),
 constructions (intersection, circumcenter)

delaunay triangles and regular, fully dynamic,

todo 308: combinat

todo 308: combinat

todo 307: elliptic

todo 307: elliptic

figure this stuff out

todo 306: fastmatrix.spad

todo 306: fastmatrix.spad

follow Waldek's lead on this

bug 7118: Factored AlgebraicNumber

sqrt(2^32)

(1) 65536

Type: AlgebraicNumber

(2) -> %^2

(2) 4294967296

Type: AlgebraicNumber

(3) -> factor %

(3) 4294967296

Type: Factored AlgebraicNumber

(4) -> factor 4294967296

32
(4) 2

Type: Factored Integer

(5) ->

todo 305: in bookvol11, in graphviewport.xhtml

make graphics automatically pop up when the image is clicked

todo 304: make graphics be saved as .bmp during build

todo 304: make graphics be saved as .bmp during build

todo 303: remove .Z format

todo 303: remove .Z format

bug 7116: hyperdoc title does not show

bug 7116: topics -> graphics -> 3D graphics -> building object -> cube

makeViewport3D(spaceC,title=="Cube") never appears

wish 1008: ArbitraryConstant domain

$C+3 \Rightarrow C$

or potentially

$C+3+5 \Rightarrow C[3+5]$

bug 7114: "possible missing then" bug

```

insert(h:NODE,key:KEY,value:VALUE):NODE
  if h = null then return node(key,value,red)
  if isRed(h.left) and isRed(h.left.left) then h:=splitFourNode(h)
  if h.key = key
    then h.value:=value
    else if h.key < key then h.left := insert(h.left,key,value)
         else h.right:= insert(h.right,key,value)
  if isRed(h.right) then h:=leanLeft(h)
  h

insert(h:NODE,key:KEY,value:VALUE):NODE
  if h = null then return node(key,value,red)
  if isRed(h.left) and isRed(h.left.left) then h:=splitFourNode(h)
  if h.key = key then h.value:=value
    else if h.key < key then h.left := insert(h.left,key,value)
         else h.right:= insert(h.right,key,value)
  if isRed(h.right) then h:=leanLeft(h)
  h

```

wish 1007: probe-file vs truename

maybe use (defun my-probe-file (path) (ignore-errors (truename path)))

camm writes, the way to test if a directory exists without error...

```

(defun pathname-pop-slash (pn)
  (let ((pn (pathname pn)))
    (if (or (pathname-name pn) (pathname-type pn)) pn
        (merge-pathnames
         (make-pathname :directory (butlast (pathname-directory pn)))
         (merge-pathnames (pathname (car (last (pathname-directory pn)))) pn)))))

(defun directory-exists-p (pm)
  (let ((pn (pathname-pop-slash pm)))
    (and (not (probe-file pn)) (member pn (directory pn) :test 'equal) t)))

```

or, in GCL

```

(defun directory-exists-p (pn)
  (unless (fboundp 'si::stat) (error "no stat"))
  (eq :directory (car (si::stat (namestring pn)))))

```

bug 7112: `t3:=coerce([i for i in 1..10])$Tuple(Integer)`

```
t3:=coerce([i for i in 1..10])$Tuple(Integer)

(1)  (1,2,3,4,5,6,7,8,9,10)
                                     Type: OutputForm

should be Type: Tuple(Integer)
```

bug 7111: NIL is not of type CONS.

```
fun:=(a:IntegralDomain):IntegralDomain +-> a**2

>> System error:
NIL is not of type CONS.
```

todo 302: getdatabase information unit tests

```
todo 302: getdatabase information unit tests
```

bug 7110: ramifiedAtInfinity? not implemented

```
bug 7110: RadicalFunctionField(Integer) ramifiedAtInfinity? not implemented

ramifiedAtInfinity()$R
The function ramifiedAtInfinity is not implemented in
RadicalFunctionField(Integer,UnivariatePolynomial(x,Integer),
UnivariatePolynomial(y,Fraction UnivariatePolynomial(x,Integer)),
x*x,1) .
```

bug 7109: singularAtInfinity? not implemented

```
bug 7109: RadicalFunctionField(Integer) singularAtInfinity? not implemented

singularAtInfinity()$R
Internal Error
The function singularAtInfinity? with signature hashcode is missing
```

```

from domain RadicalFunctionField(Integer)
(UnivariatePolynomial x (Integer))
(UnivariatePolynomial y (Fraction (UnivariatePolynomial x (Integer))))
(((2 . 1)) (0 . 1))1

```

bug 7108: axiom crashes if draw does not include all variables

```

(1) -> draw(5*y^2+(-4*x+4)*y+8*x^2+2*x-1,x=-5..5)
; (DEFUN %A ...) is being compiled.
;; The variable |y| is undefined.
;; The compiler will assume this variable is a global.

```

bug 7107: function xRange with hashcode is missing from ACPLLOT

```

m:=makeSketch(x+y,x,y,-1/2..1/2,-1/2..1/2)$ACPLLOT
xRange m
  Internal Error
    The function xRange with signature hashcode is missing from domain
      PlaneAlgebraicCurvePlot

```

wish 1006: API domain for lisp functions

```
wish 1006: API domain for lisp functions
```

wish 1005: algebra graph

```
wish 1005: algebra graph
```

wish 1004: create)example for all functions in all domains

```

wish 1004: create )example for all functions in all domains

construct set of all domains

```

bug 7105: [print [i,i*exp(-i)*Ei(i/10.0)] for i in 20..70]

```
bug 7105: [print [i,i*exp(-i)*Ei(i/10.0)] for i in 20..70]
Internal Error
The function coerce with signature hashcode is missing from domain
Expression(OnePointCompletion (DoubleFloat))
```

bug 7104: integrate(%e^{a*x}/x²,x)

```
bug 7104: integrate(%e^(a*x)/x^2,x)
```

$$(5) \quad \int \frac{e^{ax}}{x^2} dx$$

Type: Union(Expression Integer,...)

should be

$$-\frac{e^{ax}}{x} + aEi(ax)$$

```
but integrate(%e^(a*x)/x,x)
```

$$(4) \quad Ei(a x)$$

wish 1003:)help limit, integrate, etc

```
wish 1003: )help limit, integrate, etc
```

bug 7103/47: complexForm(log(-%i))

```
bug 7103/47: complexForm(log(-%i))
```

$$(4) \quad -\frac{\pi}{2}i$$

$$2$$

Type: Complex Expression Integer

should be

$$(4) - \frac{\%pi}{2} \%i$$

todo 301: etags

```
etags 'find . -name "*.pamphlet"'
```

bug 7098: numeric(Si(0.01))

```
>> Error detected within library code:
Can only compute the numerical value of a constant, real-valued Expression
```

bug 7097: numeric(Ci(0.01))

```
>> Error detected within library code:
Can only compute the numerical value of a constant, real-valued Expression
```

wish 1000: WHERE form

```
a:= b+c
where
  b = 3x
  c = 4y
```

bug 7095: occasional failure of is.input

```
)clear all
```



```
--S 1 of 4
f n ==
  not empty?(u := Is(n, 2*m%)) => integer eval(m%, u)
  3 * n + 1
```

Type: Void

```
--R
```

```
--R
```

Type: Void

```
--E 1
```

```
)set stream showall on
```

```
--S 2 of 4
g(n:INT):STREAM(INT) == generate(f, n)
```

Function declaration g : Integer -> Stream Integer has been added to workspace.

Type: Void

```
--R
```

```
--R  Function declaration g : Integer -> Stream Integer has been added to
--R      workspace.
```

```
--R
```

Type: Void

```
--E 2
```

```
--S 3 of 4
s := g 27
```

There are 3 exposed and 0 unexposed library operations named generate having 2 argument(s) but none was determined to be applicable. Use HyperDoc Browse, or issue
)display op generate
 to learn more about the available operations. Perhaps package-calling the operation or using coercions on the arguments will allow you to apply the operation.
 Cannot find a definition or applicable library operation named generate with argument type(s)

Variable f
 Integer

Perhaps you should use "@" to indicate the required return type, or "\$" to specify which version of the function you need.
 AXIOM will attempt to step through and interpret the code.
 Compiling function g with type Integer -> Stream Integer
 There are 3 exposed and 0 unexposed library operations named generate having 2 argument(s) but none was determined to be applicable. Use HyperDoc Browse, or issue
)display op generate
 to learn more about the available operations. Perhaps

package-calling the operation or using coercions on the arguments will allow you to apply the operation.

Daly Bug

Cannot find a definition or applicable library operation named
generate with argument type(s)

Variable f
Integer

Perhaps you should use "@" to indicate the required return type,
or "\$" to specify which version of the function you need.

```
--R
--R   Compiling function f with type Integer -> Integer
--R   Compiling function g with type Integer -> Stream Integer
--R
--R   (3)  [27,82,41,124,62,31,94,47,142,71,...]
--R
--R                                                    Type: Stream Integer
--E 3
```

todo 300: pending/besselk.input.pamphlet

todo 300: pending/besselk.input.pamphlet

bug 7092: .axiom.input screws up builds

bug 7092: .axiom.input screws up builds

bug 7091: bootstrap bugs

```
Only in .: boo1.input
Only in .: boo2.input
Only in .: boo3.input
Only in .: boo4.input
Only in .: boo5.input
Only in .: boo6.input
Only in .: boo-cat.input
Only in .: boo-dom1.input
Only in .: boo-dom2.input
Only in .: boo-nilcat.spad
Only in .: boo-nildom.spad
Only in .: boo-pack1.input
```

```
Only in .: boo-pack2.input
Only in .: bootstrap.spad
Only in .: boot-sum.in
```

bug 7088/347: bug in map\$Set

```
-- bug in map$Set (issue 347)
A:Set Integer:=set [-2,-1,0]
B:Set Integer:=set [0,1,4]
C:=map(x +-> x^2,A)
test(C=B)
```

bug 7087/312: Bug in sqrt() for UnivariateTaylorSeries (issue 312)

```
-- Bug in sqrt() for UnivariateTaylorSeries (issue 312)
UTSx := UTS(FRAC INT,x,0)
monx2 := monomial(1,2)$UTSx
sqrt(monx2)
```

bug 7086/340: 'divisors\$IntegerNumberTheoryFunctions' incorrect

```
-- issue 340 'divisors$IntegerNumberTheoryFunctions' incorrect for units,
-- i.e., 1, -1
divisors 1
divisors (-1)
```

bug 7085/15: integration

```
integrate(1/(1+z^4), z=0..1)::Complex Float
```

bug 7084/293: integration

```
integrate (1/(1+x^4),x = %minusInfinity..%plusInfinity)
```

bug 7083/314: integration

```
integrate(2*atan(x)/(1+x^2),x)
```

bug 7081/184: argument

```
argument (-%i)
```

bug 7078/114: function fail

```
dom:=(INT->INT)
g(f:dom):dom== x+-(f(x))^2
p(x:INT):INT == x-1
q:= g p
```

bug 7077/156: maketaylor

```
)clear all
)set function compile off
u := operator 'u;
multidiff(expr, variable, count) == if count > 0 then D(multidiff(expr, variable, count-1), variable)
maketaylor(expr) == series( n +-> multidiff(expr, x, n), x=0)
maketaylor(u(x))
)set function compile on
```

bug 7076/176: D(factor)

```
p := -x*y^2+x*y+x^3-x^2
D(factor(p),x)
```

bug 7075/182: round

```
round(3.77623)
```

```
round(-3.77623)
```

bug 7074/186: compose

```
)clear all
E := Expression Integer
F := E -> E
compose(f:F, g:F):F == (e:E):E +-> f(g(e))
p(e:E):E == 2*e+1
q(e:E):E == 3*e+1
compose(p,q)
```

bug 7071/211: D(product)

```
)clear all
D(product((1-q^(n-i))/(1-q^(m-i)),i=0..m-1),q)
f:=operator 'f;
D(product(f(i,q),i=0..m),q)
```

bug 7070/278: simplifyLog

```
simplifyLog(2*sqrt(2)*log(2))
```

bug 7069/83: DMP

```
1::DMP([x],FRAC INT)::POLY FRAC INT
1::DMP([x],INT)::POLY INT
vv := 1::DMP([x],FRAC INT)
uu := vv ::POLY FRAC INT
1 * uu
```

bug 7068/371: trace fails

```
iprint("Release the hounds!")$IPRNTPK
for i in 1..10 repeat (iprint$IPRNTPK)(i::String)
```

```
-- tracing sanity check
)trace Integer )math
-1
)trace Integer )off
```

bug 7067/102: solve(sinh(z)=cosh(z),z)

```
solve(sinh(z)=cosh(z), z)
```

bug 7066/187: missing exports

```
(INT, INT)
Tuple Any

-- missing exports
MyFD := DirectProduct(3, PrimeField(100000000000000006579))
myv := [3, 2, 1]::MyFD
ii := lookup(myv)
index(ii)$MyFD
ii := lookup([1, 1, 1]::MyFD)
index(ii)$MyFD
ii := lookup([1, 1, 0]::MyFD)
index(ii)$MyFD
ii := lookup([1, 2, 3]::MyFD)
index(ii)$MyFD
hash([3, 2, 1]::MyFD)

MyFD:= Product(PrimeField(100000000000000006579), PrimeField(257))
myv := makeprod(1, 1)$MyFD
ii := lookup(myv)
index(ii)$MyFD
ii := lookup(makeprod(2, 1)$MyFD)
index(ii)$MyFD
ii := lookup(makeprod(1, 2)$MyFD)
index(ii)$MyFD
ii := lookup(makeprod(2, 128)$MyFD)
index(ii)$MyFD
hash(myv)
```

bug 7064/258: every?

```
R==>RectangularMatrix(2,4, PF 2)
every?(zero?, [lookup(index(i)$R)$R - i for i in 1..2^8])
```

bug 7063/198: zero?

```
zero?(complexIntegrate(%i/2*e^(-%i*x^2),x)-_
      (sqrt %pi * sqrt(%i * log e)*erf(x*sqrt(%i*log e)))/(4*log e))

zero?(complexIntegrate(-%i/2*e^(%i*x^2),x)-_
      (sqrt %pi * sqrt(-%i * log e)*erf(x*sqrt(-%i*log e)))/(4*log e))

zero?(complexIntegrate(log(x)/(x-1),x)+dilog x)

zero?(complexIntegrate(2*log(x)/(x-1),x)+2*dilog x)
```

bug 7062/408: abs

```
abs(Gamma(1/5::EXPR INT)::EXPR DFLOAT - 4.59084) < 0.00001
abs(digamma(1/5::EXPR INT)::EXPR DFLOAT+5.28904) < 0.000001
abs(besselJ(1,1::EXPR INT)::EXPR DFLOAT - 0.440051) < 0.000001
-- fails, because airyAi yields complex results even for real arguments
abs(airyAi(1/5::EXPR INT)::EXPR DFLOAT - 0.303703) < 0.00001
```

bug 7061: integrate sqrt

```
integrate(sqrt(z+sqrt(1+z))/(1+z)^2,z=0..%plusInfinity,"noPole")
```

$$(12) \quad \frac{-5 \operatorname{atan}(-) + 4}{4}$$

Type: Union(f1: OrderedCompletion Expression Integer,...)

(3) 0.1956236140 0839451649

Type: Expression Float

but MMA6/Maple11/Derive gives

$$1+5*\operatorname{atan}(1/2)/2 = 2.1591190$$

bug 7060: integration bug (different answer in fricas)

bug 7060: integration bug (different answer in fricas)

```
-> ex1:=integrate(sqrt(1+sqrt(1+z))/(1+z^2),z);
```

Type: Union(Expression Integer,...)

```
-> ex2:=eval(ex1,z=0);
```

Type: Expression Integer

```
-> numeric ex2
```

```
1.0243175471 308082709
```

Type: Float

fricas returns:

```
-2.5898455408 37987845
```

curiously

```
ex3:=limit(ex1,z=%plusInfinity)
```

```
(11) 0
```

Type: Union(OrderedCompletion Expression Integer,...)

but fricas doesn't return???

bug 7059: integration bug (different answer in fricas)

```
integrate(sqrt(1+sqrt(1+z))/(1+z^2),z=0..%plusInfinity,"noPole")::EXPR Float
```

gives the wrong answer:

(3) - 1.0243175471 308082709

Type: Expression Float

the correct answer:

(5*atan(3)-5*atan(1/3)+4)/4

$$(4) \frac{5\operatorname{atan}(3) - 5\operatorname{atan}\left(\frac{1}{3}\right) + 4}{4}$$

Type: Expression Integer

(5) -> numeric %

(5) 2.1591190225 020152905

Type: Float

(6) ->

bug 7058: calling AXIOMsys segfaults

bug 7058: calling AXIOMsys from anywhere on the path without giving full path name sometimes causes a segfault

which AXIOMsys

/research2/test/mnt/fedora5/bin/AXIOMsys

[root@localhost test]# AXIOMsys

AXIOM Computer Algebra System

Version: Axiom (Sept 2007)

Timestamp: Tuesday October 16, 2007 at 09:21:36

Issue)copyright to view copyright notices.

Issue)summary for a summary of useful system commands.

Issue)quit to leave AXIOM and return to shell.

Re-reading compress.daase Re-reading interp.daase

Re-reading operation.daase

Re-reading category.daase

Re-reading browse.daase

(1) ->)lisp (bye)

[root@localhost test]# echo \$AXIOM

/research2/test/mnt/fedora5

[root@localhost test]# AXIOMsys

```
Issue )copyright to view copyright notices.
Issue )summary for a summary of useful system commands.
Issue )quit to leave AXIOM and return to shell.
```

```
[root@localhost test]# strace AXIOMsys
execve("/research2/test/mnt/fedora5/bin/AXIOMsys", ["AXIOMsys"], [/ * 39 vars */]) = 0
brk(0) = 0x96a1000
mmap2(NULL, 4096, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0xb7f73000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
open("/etc/ld.so.cache", O_RDONLY) = 3
fstat64(3, {st_mode=S_IFREG|0644, st_size=67432, ...}) = 0
mmap2(NULL, 67432, PROT_READ, MAP_PRIVATE, 3, 0) = 0xb7f62000
close(3) = 0
open("/lib/libm.so.6", O_RDONLY) = 3
read(3, "\177ELF\1\1\1\0\0\0\0\0\0\0\3\0\3\0\1\0\0\0\Cv\000"... , 512) = 512
fstat64(3, {st_mode=S_IFREG|0755, st_size=199700, ...}) = 0
mmap2(0x761000, 147584, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x761000
mmap2(0x784000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x22) = 0x784000
close(3) = 0
open("/usr/lib/sse2/libgmp.so.3", O_RDONLY) = 3
read(3, "\177ELF\1\1\1\0\0\0\0\0\0\0\3\0\3\0\1\0\0\0\0fv\000"... , 512) = 512
fstat64(3, {st_mode=S_IFREG|0644, st_size=842637, ...}) = 0
mmap2(0x761000, 186848, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x110000
mmap2(0x13d000, 4096, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2c) = 0x13d000
close(3) = 0
open("/usr/lib/libreadline.so.5", O_RDONLY) = 3
read(3, "\177ELF\1\1\1\0\0\0\0\0\0\0\3\0\3\0\1\0\0\0\0D\000"... , 512) = 512
fstat64(3, {st_mode=S_IFREG|0755, st_size=197832, ...}) = 0
mmap2(0x7b9000, 199036, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7b9000
mmap2(0x7e5000, 16384, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2c) = 0x7e5000
mmap2(0x7e9000, 2428, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7e9000
close(3) = 0
```

```

open("/usr/lib/libncurses.so.5", O_RDONLY) = 3
read(3, "\177ELF\1\1\1\0\0\0\0\0\0\0\0\3\0\3\0\1\0\0\0 \265\35"... , 512) = 512
fstat64(3, {st_mode=S_IFREG|0755, st_size=304280, ...}) = 0
mmap2(0x71cd000, 296964, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x71cd000
mmap2(0x720d000, 32768, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x40) = 0x720d000
mmap2(0x7215000, 2052, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7215000
close(3) = 0
open("/lib/libc.so.6", O_RDONLY) = 3
read(3, "\177ELF\1\1\1\0\0\0\0\0\0\0\0\3\0\3\0\1\0\0\0\n\31d\000"... , 512) = 512
fstat64(3, {st_mode=S_IFREG|0755, st_size=1528292, ...}) = 0
mmap2(NULL, 4096, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0xb7f61000
mmap2(0x62c000, 1254780, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x62c000
mmap2(0x758000, 16384, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x12b) = 0x758000
mmap2(0x75c000, 9596, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x75c000
close(3) = 0
open("/lib/libdl.so.2", O_RDONLY) = 3
read(3, "\177ELF\1\1\1\0\0\0\0\0\0\0\0\3\0\3\0\1\0\0\0@214x\000"... , 512) = 512
fstat64(3, {st_mode=S_IFREG|0755, st_size=16352, ...}) = 0
mmap2(0x788000, 12412, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x788000
mmap2(0x78a000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1) = 0x78a000
close(3) = 0
mmap2(NULL, 4096, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0xb7f60000
set_thread_area({entry_number:-1 -> 6, base_addr:0xb7f606b0, limit:1048575, seg_32bit:1, contents:0, read_e
mprotect(0x78a000, 4096, PROT_READ) = 0
mprotect(0x758000, 12288, PROT_READ) = 0
mprotect(0x784000, 4096, PROT_READ) = 0
mprotect(0x628000, 4096, PROT_READ) = 0
munmap(0xb7f62000, 67432) = 0
personality(0xffffffff /* PER_??? */) = 0
personality(0x40000 /* PER_??? */) = 0
execve("AXIOMsys", ["AXIOMsys"], [/* 39 vars */]) = -1 ENOENT (No such file or directory)
getpid() = 25855
stat64("/proc/25855/exe", {st_mode=S_IFREG|0755, st_size=16188532, ...}) = 0
lstat64("/proc", {st_mode=S_IFDIR|0555, st_size=0, ...}) = 0
lstat64("/proc/25855", {st_mode=S_IFDIR|0555, st_size=0, ...}) = 0
lstat64("/proc/25855/exe", {st_mode=S_IFLNK|0777, st_size=0, ...}) = 0
readlink("/proc/25855/exe", "/research2/test/mnt/fedora5/bin/AXIOMsys", 4096) = 40
lstat64("/research2", {st_mode=S_IFDIR|0777, st_size=4096, ...}) = 0
lstat64("/research2/test", {st_mode=S_IFDIR|0755, st_size=4096, ...}) = 0
lstat64("/research2/test/mnt", {st_mode=S_IFDIR|0755, st_size=4096, ...}) = 0
lstat64("/research2/test/mnt/fedora5", {st_mode=S_IFDIR|0755, st_size=4096, ...}) = 0
lstat64("/research2/test/mnt/fedora5/bin", {st_mode=S_IFDIR|0755, st_size=4096, ...}) = 0
lstat64("/research2/test/mnt/fedora5/bin/AXIOMsys", {st_mode=S_IFREG|0755, st_size=16188532, ...}) = 0
rt_sigaction(SIGSEGV, {0x807ae70, [SEGV], SA_RESTART}, {SIG_DFL}, 8) = 0
getrlimit(RLIMIT_STACK, {rlim_cur=10240*1024, rlim_max=RLIM_INFINITY}) = 0
setrlimit(RLIMIT_STACK, {rlim_cur=8192*1024, rlim_max=RLIM_INFINITY}) = 0
getrlimit(RLIMIT_DATA, {rlim_cur=RLIM_INFINITY, rlim_max=RLIM_INFINITY}) = 0
mprotect(0x8403000, 10555392, PROT_READ|PROT_WRITE|PROT_EXEC) = 0
rt_sigaction(SIGSEGV, {0x807ae70, [SEGV], SA_RESTART}, {0x807ae70, [SEGV], SA_RESTART}, 8) = 0
getrlimit(RLIMIT_DATA, {rlim_cur=RLIM_INFINITY, rlim_max=RLIM_INFINITY}) = 0

```

```

brk(0)                                = 0x96a1000
brk(0xc765000)                        = 0xc765000
rt_sigaction(SIGFPE, {0x80b3740, [FPE], SA_RESTART|SA_SIGINFO}, NULL, 8) = 0
rt_sigaction(SIGPIPE, {0x80b37f0, [PIPE], SA_RESTART|SA_SIGINFO}, NULL, 8) = 0
rt_sigaction(SIGINT, {0x80b3410, [INT], SA_RESTART|SA_SIGINFO}, NULL, 8) = 0
rt_sigaction(SIGUSR1, {0x80b3410, [USR1], SA_RESTART|SA_SIGINFO}, NULL, 8) = 0
rt_sigaction(SIGIO, {0x80b3410, [IO], SA_RESTART|SA_SIGINFO}, NULL, 8) = 0
rt_sigaction(SIGALRM, {0x80b3410, [ALRM], SA_RESTART|SA_SIGINFO}, NULL, 8) = 0
ioctl(0, SNDCTL_TMR_TIMEBASE or TCGETS, {B38400 opost isig icanon -echo ...}) = 0
--- SIGSEGV (Segmentation fault) @ 0 (0) ---
rt_sigaction(SIGSEGV, {0x807ae70, [SEGV], SA_RESTART}, {0x807ae70, [SEGV], SA_RESTART}, 8) = 0
--- SIGSEGV (Segmentation fault) @ 0 (0) ---
+++ killed by SIGSEGV +++
Process 25855 detached

```

```

[root@localhost test]# echo $AXIOM
/research2/test/mnt/fedora5

```

```

[root@localhost test]# echo $PATH
/research2/test/mnt/fedora5/bin:/usr/kerberos/sbin:/usr/kerberos/bin:/usr/local/sbin:/usr/local/

```

```

[root@localhost test]# uname -a
Linux localhost.localdomain 2.6.18-1.2239.fc5 #1 Fri Nov 10 13:04:06 EST 2006 i686 i686 i386 GNU/

```

bug 7056:)browse has a bug

```

)browse
    Your argument list is not valid.

```

bug 7055: cd doesn't start from the current location

```

)cd pending

>> System error:
Can't change the current directory to "/usr/local/axiom/mnt/fedora5/NIL/".

```

bug 7053: close(viewport) dumps core

```

bug 7053: close(viewport) dumps core

```

bug 7051: is.input intermittent failures

```

)set message test on

)set message auto off

)clear all

--S 1 of 4
f n == ( not empty?(u := Is(n, 2*m%)) => integer eval(m%, u) ; 3 * n + 1 )

                                                    Type: Void
--R
--R                                                    Type: Void
--E 1

)set stream showall on

--S 2 of 4
g(n:INT):STREAM(INT) == generate(f, n)

Function declaration g : Integer -> Stream Integer has been added to
workspace.
                                                    Type: Void
--R
--R Function declaration g : Integer -> Stream Integer has been added to
--R workspace.
--R                                                    Type: Void
--E 2

--S 3 of 4
s := g 27

There are 3 exposed and 0 unexposed library operations named
generate having 2 argument(s) but none was determined to be
applicable. Use HyperDoc Browse, or issue
)display op generate
to learn more about the available operations. Perhaps
package-calling the operation or using coercions on the arguments
will allow you to apply the operation.
Cannot find a definition or applicable library operation named
generate with argument type(s)
Variable f
Integer

```

```

    Perhaps you should use "@" to indicate the required return type,
    or "$" to specify which version of the function you need.
  AXIOM will attempt to step through and interpret the code.
  Compiling function g with type Integer -> Stream Integer
  There are 3 exposed and 0 unexposed library operations named
    generate having 2 argument(s) but none was determined to be
    applicable. Use HyperDoc Browse, or issue
      )display op generate
    to learn more about the available operations. Perhaps
    package-calling the operation or using coercions on the arguments
    will allow you to apply the operation.

Daly Bug
  Cannot find a definition or applicable library operation named
    generate with argument type(s)
      Variable f
      Integer

    Perhaps you should use "@" to indicate the required return type,
    or "$" to specify which version of the function you need.
--R
--R   Compiling function f with type Integer -> Integer
--R   Compiling function g with type Integer -> Stream Integer
--R
--R   (3)  [27,82,41,124,62,31,94,47,142,71,...]
--R                                             Type: Stream Integer
--E 3

--S 4 of 4
extend(s, 150)

  There are 11 exposed and 0 unexposed library operations named extend
  having 2 argument(s) but none was determined to be applicable.
  Use HyperDoc Browse, or issue
    )display op extend
  to learn more about the available operations. Perhaps
  package-calling the operation or using coercions on the arguments
  will allow you to apply the operation.

Daly Bug
  Cannot find a definition or applicable library operation named
    extend with argument type(s)
      Variable s
      PositiveInteger

    Perhaps you should use "@" to indicate the required return type,
    or "$" to specify which version of the function you need.
--R
--R
--R   (4)

```

[illegible]

```

integrate(e^u/u,u=%minusInfinity..-1,"noPole")

(19) potentialPole
                                         Type: Union(pole: potentialPole,...)
(20) -> integrate(e^u/u,u=%minusInfinity..-1,"noPole")
      Loading /usr/local/axiom/mnt/fedora5/algebra/RDETR.o for package
      TranscendentalRischDE

(20) "failed"
                                         Type: Union(fail: failed,...)
(21) -> integrate(e^u/u,u)

(21) Ei(u log(e))
                                         Type: Union(Expression Integer,...)
(22) -> Ei(13)

```

(22) Ei(13)

Type: Expression Integer

(23) ->

bug 7048: cannot simplify expression using $e^{\hat{x} \log e}$

integrate(e^x, x)

(18)
$$\frac{x \log(e) \%e}{\log(e)}$$

Type: Union(Expression Integer,...)

bug 7047: Gamma returns a random garbage value

Gamma 1000/1000

(7) i.nfE+0

Type: DoubleFloat

bug 7046: radix has wrong ragits

```
;;; radix(5/24,39) ==> 0 . 8 4 34
;;;
;;;
;;; radix(5/24,40) ==> 0 . 8 D
;;;
;;;
;;; radix(5/24,43) ==> 0 . 8 41
;;;
;;; radix(35,36) ==> Z
;;;
;;; radix(36,37) ==> Error ... index out of range
;;;
;;; radix(10,16) ==> A
;;;
```


idea:)example command taken from regression test chunks

- 1) identify (using --S comment tags) the functions being demonstrated
- 2) move the regression chunks near the example function in the algebra
- 3) dynamically pull the chunks from the sources
- 4) remove the regression test comment prefix

bug 7044: why?

generic 16 making /research2/test/int/input/iprntpk.input from /research2/test/src/input/iprntpk.input.pamp

bug 7043: FAILED 1 of 139 stanzas file kamke3

```

MISMATCH
expected:"      (9216a  - 7168a )b - 2048a  + 2048a"
got:"      (18432a  - 57344a )b - 2048a  + 8192a"
MISMATCH
expected:"      8      6      9      7"
got:"      8      6      9      7"
MISMATCH
expected:"      (- 20736a  + 34048a  - 6400a )b + (4608a  - 22016a  + 9728a )b"
got:"      (- 41472a  + 272384a  - 204800a )b + (4608a  - 88064a  + 155648a )b"
MISMATCH
expected:"      8      6      4 3      9      7      5 2"
got:"      8      6      4 3      9      7      5 2"
MISMATCH
expected:"      (- 3072a  + 45312a  - 35840a  + 2048a )b"
got:"      (- 3072a  + 181248a  - 573440a  + 131072a )b"
MISMATCH
expected:"      9      7      5      3 4"
got:"      9      7      5      3 4"
MISMATCH
expected:"      (13824a  - 61824a  + 25984a  - 256a )b"
got:"      (27648a  - 494592a  + 831488a  - 32768a )b"
MISMATCH
expected:"      8      6      4      2 5"
got:"      8      6      4      2 5"
MISMATCH
expected:"      (512a  - 28416a  + 56736a  - 12800a )b"
got:"      (512a  - 113664a  + 907776a  - 819200a )b"
MISMATCH
expected:"      9      7      5      3 6"
got:"      9      7      5      3 6"

```

```

MISMATCH
expected:"      (- 2304a  + 34944a  - 35664a  + 4096a )b"
got:"      (- 4608a  + 279552a  - 1141248a  + 524288a )b"
MISMATCH
expected:"      8      6      4      2  7"
got:"      8      6      4      2  7"
MISMATCH
expected:"      (4608a  - 28224a  + 15168a  - 768a)b"
got:"      (18432a  - 451584a  + 970752a  - 196608a)b"
MISMATCH
expected:"      7      5      3      8"
got:"      7      5      3      8"
MISMATCH
expected:"      (- 5376a  + 15456a  - 4176a  + 64)b"
got:"      (- 43008a  + 494592a  - 534528a  + 32768)b"
MISMATCH
expected:"      6      4      2      9"
got:"      6      4      2      9"
MISMATCH
expected:"      (- 2016a  + 1368a  - 48)b  + (4032a  - 5712a  + 672a)b"
got:"      (- 64512a  + 175104a  - 24576)b  + (64512a  - 365568a  + 172032a)b"
MISMATCH
expected:"      4      2      11      5      3      10"
got:"      4      2      11      5      3      10"
MISMATCH
expected:"      - b  + 18a b  + (- 144a  + 12)b  + (672a  - 192a)b"
got:"      - 512b  + 4608a b  + (- 18432a  + 6144)b  + (43008a  - 49152a)b"
MISMATCH
expected:"      15      14      2      13      3      12"
got:"      15      14      2      13      3      12"
MISMATCH
expected:"      (82944a  - 50176a  - 10240a )b - 18432a  + 14336a  + 4096a"
got:"      (165888a  - 401408a  - 327680a )b - 18432a  + 57344a  + 65536a"
MISMATCH
expected:"      9      7      5      10      8      6"
got:"      9      7      5      10      8      6"
MISMATCH
expected:"      (24576a  - 196608a  + 82432a  + 9216a )b"
got:"      (24576a  - 786432a  + 1318912a  + 589824a )b"
MISMATCH
expected:"      10      8      6      4  2"
got:"      10      8      6      4  2"
MISMATCH
expected:"      (- 110592a  + 301056a  - 80640a  - 3584a )b"
got:"      (- 221184a  + 2408448a  - 2580480a  - 458752a )b"
MISMATCH
expected:"      9      7      5      3  3"
got:"      9      7      5      3  3"
MISMATCH
expected:"      (- 6144a  + 230400a  - 309888a  + 49664a  + 512a )b"

```

```

got:"          (- 6144a  + 921600a  - 4958208a  + 3178496a  + 131072a )b"
MISMATCH
expected:"          10          8          6          4          2  4"
got:"          10          8          6          4          2  4"
MISMATCH
expected:"          (27648a  - 290304a  + 215616a  - 18944a )b"
got:"          (55296a  - 2322432a  + 6899712a  - 2424832a )b"
MISMATCH
expected:"          9          7          5          3  5"
got:"          9          7          5          3  5"
MISMATCH
expected:"          (- 55296a  + 241920a  - 100032a  + 4096a )b"
got:"          (- 221184a  + 3870720a  - 6402048a  + 1048576a )b"
MISMATCH
expected:"          8          6          4          2  6"
got:"          8          6          4          2  6"
MISMATCH
expected:"          (64512a  - 137088a  + 29664a  - 384a)b"
got:"          (516096a  - 4386816a  + 3796992a  - 196608a)b"
MISMATCH
expected:"          7          5          3          7"
got:"          7          5          3          7"
MISMATCH
expected:"          (24192a  - 12960a  + 384a)b  + (- 48384a  + 52416a  - 5088a )b"
got:"          (- 774144a  + 3354624a  - 1302528a )b"
MISMATCH
expected:"          5          3          9          6          4          2  8"
got:"          6          4          2  8"
MISMATCH
expected:"          12a b  - 216a b  + (1728a  - 120a)b  + (- 8064a  + 1872a )b"
got:"          (- 516096a  + 479232a )b  + (774144a  - 1658880a  + 196608a)b"
MISMATCH
expected:"          13          2 12          3          11          4          2 10"
got:"          4          2 10          5          3          9"
MISMATCH
expected:"          + "
got:"          + "
MISMATCH
expected:"          c"
got:"          6144a b  - 55296a b  + (221184a  - 61440a)b"
MISMATCH
expected:"          2"
got:"          13          2 12          3          11"
MISMATCH
expected:"          *"
got:"          + "
MISMATCH
expected:"          - 14336a  + 2048a"
got:"          c"
MISMATCH

```

```

expected:"          7          5"
got:"          2"
MISMATCH
expected:"          +  "
got:"          *"
MISMATCH
expected:"          (221184a  - 215040a  + 35840a  - 3072a )b - 49152a  + 61440a"
got:"          245760a  - 229376a  + 131072a"
MISMATCH
expected:"          10          8          6          4          11          9"
got:"          9          7          5"
MISMATCH
expected:"          (24576a  - 479232a  + 336384a  - 37376a  + 1536a )b"
got:"          (442368a  - 1720320a  + 1146880a  - 393216a )b - 49152a"
MISMATCH
expected:"          11          9          7          5          3  2"
got:"          10          8          6          4          11"
MISMATCH
expected:"          (- 110592a  + 645120a  - 303360a  + 20224a  - 256a )b"
got:"          (24576a  - 1916928a  + 5382144a  - 2392064a  + 393216a )b"
MISMATCH
expected:"          10          8          6          4          2  3"
got:"          11          9          7          5          3  2"
MISMATCH
expected:"          (221184a  - 580608a  + 168960a  - 5632a )b"
got:"          (- 221184a  + 5160960a  - 9707520a  + 2588672a  - 131072a )b"
MISMATCH
expected:"          9          7          5          3  4"
got:"          10          8          6          4          2  3"
MISMATCH
expected:"          (- 258048a  + 354816a  - 57600a  + 640a )b"
got:"          (884736a  - 9289728a  + 10813440a  - 1441792a )b"
MISMATCH
expected:"          8          6          4          2  5"
got:"          9          7          5          3  4"
MISMATCH
expected:"          (193536a  - 145152a  + 11040a )b"
got:"          (- 2064384a  + 11354112a  - 7372800a  + 327680a )b"
MISMATCH
expected:"          7          5          3  6"
got:"          8          6          4          2  5"
MISMATCH
expected:"          (32256a  - 5760a )b  + (- 96768a  + 38016a  - 912a )b"
got:"          (3096576a  - 9289728a  + 2826240a )b"
MISMATCH
expected:"          5          3  8          6          4          2  7"
got:"          7          5          3  6"
MISMATCH
expected:"          - 48a b  + 864a b  + (- 6912a  + 384a )b"
got:"          (- 3096576a  + 4866048a  - 466944a )b"

```

```

MISMATCH
expected:"          2 11      3 10      4      2 9"
got:"          6      4      2 7"
MISMATCH
expected:" + "
got:" + "
MISMATCH
expected:" c"
got:" (2064384a - 1474560a )b"
MISMATCH
expected:" 3"
got:" 5      3 8"
MISMATCH
expected:" *"
got:" + "
MISMATCH
expected:" - 18432a + 2048a"
got:" - 24576a b + 221184a b + (- 884736a + 196608a )b"
MISMATCH
expected:"          8      6"
got:"          2 11      3 10      4      2 9"
MISMATCH
expected:" + "
got:" + "
MISMATCH
expected:" (147456a - 172032a + 46080a - 3072a )b - 32768a + 49152a"
got:" c"
MISMATCH
expected:"          11      9      7      5      12      10"
got:" 3"
MISMATCH
expected:" + "
got:" *"
MISMATCH
expected:" (- 294912a + 258048a - 46080a + 1536a )b"
got:" 196608a - 294912a + 131072a"
MISMATCH
expected:"          10      8      6      4 2"
got:"          10      8      6"
MISMATCH
expected:" (344064a - 215040a + 23040a - 256a )b"
got:" (294912a - 1376256a + 1474560a - 393216a )b - 32768a"
MISMATCH
expected:"          9      7      5      3 3"
got:"          11      9      7      5      12"
MISMATCH
expected:" (- 258048a + 107520a - 5760a )b"
got:" (- 1179648a + 4128768a - 2949120a + 393216a )b"
MISMATCH
expected:"          8      6      4 4"

```



```

expected:"      (10752a - 90240a + 34944a - 256a )b"
got:"      64a c - 48a b c + 12a b c - a b"
MISMATCH
expected:"      8      6      4      2 4"
got:"      11 3      10 2 2      9 4      8 6"
MISMATCH
expected:"      + "
got:"      /"
MISMATCH
expected:"      (- 43008a + 109440a - 21504a )b"
got:"      3072a - 4096a"
MISMATCH
expected:"      7      5      3 5"
got:"      8      6"
MISMATCH
expected:"      (- 2304a + 77952a - 87120a + 8448a )b"
got:"      (- 11520a + 107520a - 53248a )b + (- 24576a + 24576a )b"
MISMATCH
expected:"      8      6      4      2 6"
got:"      8      6      4 2      7      5"
MISMATCH
expected:"      (9216a - 83328a + 45600a - 1920a)b"
got:"      (92160a - 301056a + 49152a )b"
MISMATCH
expected:"      7      5      3      7"
got:"      7      5      3 3"
MISMATCH
expected:"      (- 16128a + 57120a - 15120a + 192)b"
got:"      (10752a - 360960a + 559104a - 16384a )b"
MISMATCH
expected:"      6      4      2      8"
got:"      8      6      4      2 4"
MISMATCH
expected:"      (- 10080a + 7224a - 240)b + (16128a - 25536a + 2880a)b"
got:"      (- 86016a + 875520a - 688128a )b"
MISMATCH
expected:"      4      2      10      5      3      9"
got:"      7      5      3 5"
MISMATCH
expected:"      - 9b + 144a b + (- 1008a + 84)b + (4032a - 1176a)b"
got:"      (- 2304a + 311808a - 1393920a + 540672a )b"
MISMATCH
expected:"      14      13      2      12      3      11"
got:"      8      6      4      2 6"
MISMATCH
expected:"      c"
got:"      (18432a - 666624a + 1459200a - 245760a)b"
MISMATCH
expected:"      *"
got:"      7      5      3      7"

```

```

MISMATCH
expected:"          (- 184320a  + 64512a  + 4096a )b + 46080a  - 21504a  - 2048a"
got:"      + "
MISMATCH
expected:"          8          6          4          9          7          5"
got:"      (- 64512a  + 913920a  - 967680a  + 49152)b"
MISMATCH
expected:"      + "
got:"          6          4          2          8"
MISMATCH
expected:"      (- 86016a  + 399360a  - 91392a  - 2560a )b"
got:"      + "
MISMATCH
expected:"          9          7          5          3  2"
got:"      (129024a  - 817152a  + 368640a)b"
MISMATCH
expected:"      + "
got:"          5          3          9"
MISMATCH
expected:"      (344064a  - 552960a  + 75264a  + 512a )b"
got:"      + "
MISMATCH
expected:"          8          6          4          2  3"
got:"      (129024a  - 150528a)b  + (- 161280a  + 462336a  - 61440)b"
MISMATCH
expected:"      + "
got:"          3          11          4          2          10"
MISMATCH
expected:"      (27648a  - 634368a  + 498240a  - 36864a )b"
got:"      + "
MISMATCH
expected:"          9          7          5          3  4"
got:"      - 2304b  + 18432a b  + (- 64512a  + 21504)b"
MISMATCH
expected:"      + "
got:"          14          13          2          12"
MISMATCH
expected:"      (- 110592a  + 698880a  - 289920a  + 9984a )b"
got:"      + "
MISMATCH
expected:"          8          6          4          2  5"
got:"      c"
MISMATCH
expected:"      + "
got:"      *"
MISMATCH
expected:"      (193536a  - 497280a  + 105120a  - 1152a)b"
got:"      (- 368640a  + 516096a  + 131072a )b + 46080a  - 86016a  - 32768a"
MISMATCH
expected:"          7          5          3          6"

```



```

got:"
MISMATCH
expected:"      8      6      4      9      7      5"
got:"      (- 193536a + 231168a - 21600a )b"
MISMATCH
expected:"      6      4      2 7"
got:"      9      7      5      3 2"
MISMATCH
expected:"      (- 48384a + 11424a )b + (120960a - 67872a + 1920a)b"
got:"      (688128a - 4423680a + 2408448a + 65536a )b"
MISMATCH
expected:"      4      2 9      5      3      8"
got:"      8      6      4      2 3"
MISMATCH
expected:"      108a b - 1728a b + (12096a - 840a)b"
got:"      (27648a - 2537472a + 7971840a - 2359296a )b"
MISMATCH
expected:"      12      2 11      3      10"
got:"      9      7      5      3 4"
MISMATCH
expected:"      + "
got:"      + "
MISMATCH
expected:"      c"
got:"      (- 221184a + 5591040a - 9277440a + 1277952a )b"
MISMATCH
expected:"      2"
got:"      8      6      4      2 5"
MISMATCH
expected:"      *"
got:"      + "
MISMATCH
expected:"      - 153600a + 21504a - 1024a"
got:"      (774144a - 7956480a + 6727680a - 294912a)b"
MISMATCH
expected:"      8      6      4"
got:"      7      5      3      6"
MISMATCH
expected:"      (- 688128a + 460800a - 43008a + 1024a )b + 172032a"
got:"      (- 1548288a + 7397376a - 2764800a )b"
MISMATCH
expected:"      9      7      5      3      10"
got:"      6      4      2 7"
MISMATCH
expected:"      (- 110592a + 1333248a - 610560a + 34560a - 256a )b"
got:"      (1935360a - 4343808a + 491520a)b"
MISMATCH
expected:"      10      8      6      4      2 2"
got:"      5      3      8"
MISMATCH

```

```

expected:"      (442368a - 1591296a + 453120a - 13056a )b"
got:"      (- 1548288a + 1462272a )b"
MISMATCH
expected:"      9      7      5      3 3"
got:"      4      2 9"
MISMATCH
expected:"      (- 774144a + 1236480a - 195840a + 1920a )b"
got:"      27648a b - 221184a b + (774144a - 215040a)b"
MISMATCH
expected:"      8      6      4      2 4"
got:"      12      2 11      3      10"
MISMATCH
expected:"      + "
got:"      + "
MISMATCH
expected:"      (774144a - 623616a + 46080a )b"
got:"      c"
MISMATCH
expected:"      7      5      3 5"
got:"      2"
MISMATCH
expected:"      + "
got:"      *"
MISMATCH
expected:"      (193536a - 34944a )b + (- 483840a + 196224a - 4560a )b"
got:"      - 614400a + 344064a - 65536a"
MISMATCH
expected:"      5      3 7      6      4      2 6"
got:"      8      6      4"
MISMATCH
expected:"      - 432a b + 6912a b + (- 48384a + 2688a )b"
got:"      (- 1376256a + 3686400a - 1376256a + 131072a )b + 172032a"
MISMATCH
expected:"      2 10      3 9      4      2 8"
got:"      9      7      5      3      10"
MISMATCH
expected:"      + "
got:"      + "
MISMATCH
expected:"      c"
got:"      (- 110592a + 5332992a - 9768960a + 2211840a - 65536a )b"
MISMATCH
expected:"      3"
got:"      10      8      6      4      2 2"
MISMATCH
expected:"      *"
got:"      + "
MISMATCH
expected:"      - 172032a + 46080a - 3072a"
got:"      (884736a - 12730368a + 14499840a - 1671168a )b"

```

MISMATCH
 expected:"
 9 7 5"
 got:"
 9 7 5 3 3"
 MISMATCH
 expected:"
 (- 589824a + 516096a - 92160a + 3072a)b + 147456a"
 got:"
 (- 3096576a + 19783680a - 12533760a + 491520a)b"
 MISMATCH
 expected:"
 10 8 6 4 11"
 got:"
 8 6 4 2 4"
 MISMATCH
 expected:"
 (1032192a - 645120a + 69120a - 768a)b"
 got:"
 (6193152a - 19955712a + 5898240a)b"
 MISMATCH
 expected:"
 9 7 5 3 2"
 got:"
 7 5 3 5"
 MISMATCH
 expected:"
 (- 1032192a + 430080a - 23040a)b"
 got:"
 (- 7741440a + 12558336a - 1167360a)b"
 MISMATCH
 expected:"
 8 6 4 3"
 got:"
 6 4 2 6"
 MISMATCH
 expected:"
 (- 258048a + 32256a)b + (645120a - 161280a + 2880a)b"
 got:"
 (6193152a - 4472832a)b"
 MISMATCH
 expected:"
 6 4 5 7 5 3 4"
 got:"
 5 3 7"
 MISMATCH
 expected:"
 576a b - 9216a b + (64512a - 2688a)b"
 got:"
 - 110592a b + 884736a b + (- 3096576a + 688128a)b"
 MISMATCH
 expected:"
 3 8 4 7 5 3 6"
 got:"
 2 10 3 9 4 2 8"
 MISMATCH
 expected:" + "
 got:" + "
 MISMATCH
 expected:" ?"
 got:" c"
 MISMATCH
 expected:" 2"
 got:" 3"
 MISMATCH
 expected:" *"
 got:" *"
 MISMATCH
 expected:"
 64a c - 48a b c + 12a b c - a b"
 got:"
 - 688128a + 737280a - 196608a"
 MISMATCH
 expected:"
 10 3 9 2 2 8 4 7 6"

```

got:"          9          7          5"
MISMATCH
expected:"      /"
got:"          + "
MISMATCH
expected:"      (- 40320a + 13440a )b + (11520a - 10752a )b + 5376a b - 1536a"
got:"          (- 1179648a + 4128768a - 2949120a + 393216a )b + 147456a"
MISMATCH
expected:"          6          4 3          7          5 2          6          7"
got:"          10          8          6          4          11"
MISMATCH
expected:"      + "
got:"          + "
MISMATCH
expected:"      (56448a - 74400a + 5376a )b + (- 16128a + 70080a - 10752a )b"
got:"          (4128768a - 10321920a + 4423680a - 196608a )b"
MISMATCH
expected:"          6          4          2 5          7          5          3 4"
got:"          9          7          5          3 2"
MISMATCH
expected:"      + "
got:"          + "
MISMATCH
expected:"      (4608a - 88704a + 49920a - 1536a)b"
got:"          (- 8257536a + 13762560a - 2949120a )b"
MISMATCH
expected:"          7          5          3          6"
got:"          8          6          4 3"
MISMATCH
expected:"      + "
got:"          + "
MISMATCH
expected:"      (- 16128a + 80640a - 20640a + 192)b"
got:"          (10321920a - 10321920a + 737280a )b"
MISMATCH
expected:"          6          4          2          7"
got:"          7          5          3 4"
MISMATCH
expected:"      + "
got:"          + "
MISMATCH
expected:"      (- 20160a + 15624a - 480)b + (24192a - 45360a + 4800a)b"
got:"          (- 8257536a + 4128768a )b"
MISMATCH
expected:"          4          2          9          5          3          8"
got:"          6          4 5"
MISMATCH
expected:"      + "
got:"          + "
MISMATCH

```

```

expected:"      - 36b  + 504a b  + (- 3024a  + 252)b  + (10080a  - 3024a)b"
got:"          147456a b  - 1179648a b  + (4128768a  - 688128a )b"
MISMATCH
expected:"          13      12      2      11      3      10"
got:"          3 8      4 7      5      3 6"
MISMATCH
expected:"      + "
got:"      + "
MISMATCH
expected:"          c"
got:"      ?"
MISMATCH
expected:"          *"
got:"      2"
MISMATCH
expected:"          - 46080a  + 10752a"
got:"      *"
MISMATCH
expected:"          8      6"
got:"      64a  c  - 48a b c  + 12a b c  - a b"
MISMATCH
expected:"      + "
got:"      10 3      9 2 2      8 4      7 6"
MISMATCH
expected:"          (129024a  - 318720a  + 32256a )b  + (161280a  - 26880a )b"
got:"      /"
MISMATCH
expected:"          8      6      4 2      7      5"
got:"      (11520a  - 43008a )b  + 10752a b  - 1536a"
MISMATCH
expected:"      + "
got:"          7      5 2      6      7"
MISMATCH
expected:"          (- 451584a  + 393600a  - 21504a )b"
got:"      + "
MISMATCH
expected:"          7      5      3 3"
got:"      (- 16128a  + 280320a  - 172032a )b  + (- 80640a  + 107520a )b"
MISMATCH
expected:"      + "
got:"          7      5      3 4      6      4 3"
MISMATCH
expected:"          (- 55296a  + 725760a  - 301440a  + 7680a )b"
got:"      + "
MISMATCH
expected:"          8      6      4      2 4"
got:"      (112896a  - 595200a  + 172032a )b"
MISMATCH
expected:"      + "
got:"          6      4      2 5"

```



```

got: "          c"
MISMATCH
expected: "      + "
got: "      *"
MISMATCH
expected: "          (221184a - 1548288a + 418560a - 9216a )b"
got: "          - 46080a + 43008a"
MISMATCH
expected: "          9          7          5          3 2"
got: "          8          6"
MISMATCH
expected: "          (- 774144a + 1612800a - 243840a + 1920a )b"
got: "          (129024a - 1274880a + 516096a )b + (322560a - 215040a )b"
MISMATCH
expected: "          8          6          4          2 3"
got: "          8          6          4 2          7          5"
MISMATCH
expected: "          (1161216a - 1048320a + 73920a )b"
got: "          (- 903168a + 3148800a - 688128a )b"
MISMATCH
expected: "          7          5          3 4"
got: "          7          5          3 3"
MISMATCH
expected: "          (483840a - 88704a )b + (- 967680a + 411264a - 9120a )b"
got: "          (- 55296a + 2903040a - 4823040a + 491520a )b"
MISMATCH
expected: "          5          3 6          6          4          2 5"
got: "          8          6          4          2 4"
MISMATCH
expected: "          - 1728a b + 24192a b + (- 145152a + 8064a )b"
got: "          (387072a - 5483520a + 4454400a - 147456a)b"
MISMATCH
expected: "          2 9          3 8          4          2 7"
got: "          7          5          3          5"
MISMATCH
expected: "      + "
got: "      + "
MISMATCH
expected: "          c"
got: "          (- 1161216a + 6451200a - 2273280a )b"
MISMATCH
expected: "          3"
got: "          6          4          2 6"
MISMATCH
expected: "      *"
got: "      + "
MISMATCH
expected: "          258048a - 46080a + 1536a"
got: "          (1935360a - 4644864a + 491520a)b"
MISMATCH

```

```

expected:"          8          6          4"
got:"          5          3          7"
MISMATCH
expected:"          (1032192a - 645120a + 69120a - 768a )b - 294912a"
got:"          (- 1935360a + 1870848a )b"
MISMATCH
expected:"          9          7          5          3          10"
got:"          4          2  8"
MISMATCH
expected:"          (- 1548288a + 645120a - 34560a )b"
got:"          55296a b - 387072a b + (1161216a - 322560a)b"
MISMATCH
expected:"          8          6          4  2"
got:"          11          2 10          3          9"
MISMATCH
expected:"          + "
got:"          + "
MISMATCH
expected:"          (- 645120a + 80640a )b + (1290240a - 322560a + 5760a )b"
got:"          c"
MISMATCH
expected:"          6          4  4          7          5          3  3"
got:"          2"
MISMATCH
expected:"          + "
got:"          *"
MISMATCH
expected:"          2304a b - 32256a b + (193536a - 8064a )b"
got:"          - 172032a"
MISMATCH
expected:"          3  7          4  6          5          3  5"
got:"          5"
MISMATCH
expected:"          + "
got:"          + "
MISMATCH
expected:"          ?"
got:"          (1806336a - 3072000a + 516096a )b - 258048a + 614400a"
MISMATCH
expected:"          3"
got:"          8          6          4          9          7"
MISMATCH
expected:"          *"
got:"          + "
MISMATCH
expected:"          64a c - 48a b c + 12a b c - a b"
got:"          (221184a - 6193152a + 6696960a - 589824a )b"
MISMATCH
expected:"          9  3          8  2  2          7  4          6  6"
got:"          9          7          5          3  2"

```



```

MISMATCH
expected:"      /"
got:"      + "
MISMATCH
expected:"      (17280a - 1536a )b + (- 5760a + 1408a )b - 768a b + 256a"
got:"      (- 1548288a + 12902400a - 7802880a + 245760a )b"
MISMATCH
expected:"      5      3 3      6      4 2      5      6"
got:"      8      6      4      2 3"
MISMATCH
expected:"      + "
got:"      + "
MISMATCH
expected:"      (- 40320a + 24000a - 384a)b + (13440a - 26400a + 1024a )b"
got:"      (4644864a - 16773120a + 4730880a )b"
MISMATCH
expected:"      5      3      5      6      4      2 4"
got:"      7      5      3 4"
MISMATCH
expected:"      + "
got:"      + "
MISMATCH
expected:"      (- 5376a + 53760a - 12960a + 64)b"
got:"      (- 7741440a + 13160448a - 1167360a )b"
MISMATCH
expected:"      6      4      2      6"
got:"      6      4      2 5"
MISMATCH
expected:"      + "
got:"      + "
MISMATCH
expected:"      (- 20160a + 17640a - 480)b + (16128a - 40320a + 3840a)b"
got:"      (7741440a - 5677056a )b"
MISMATCH
expected:"      4      2      8      5      3      7"
got:"      5      3 6"
MISMATCH
expected:"      + "
got:"      + "
MISMATCH
expected:"      - 84b + 1008a b + (- 5040a + 420)b + (13440a - 4200a)b"
got:"      - 221184a b + 1548288a b + (- 4644864a + 1032192a )b"
MISMATCH
expected:"      12      11      2      10      3      9"
got:"      2 9      3 8      4      2 7"
MISMATCH
expected:"      *"
got:"      3"
MISMATCH
expected:"      23040a - 1792a"

```

```

got:"          *"  
MISMATCH  
expected:"              7           5"  
got:"      1032192a   - 737280a   + 98304a"  
MISMATCH  
expected:"    + "  
got:"              8           6           4"  
MISMATCH  
expected:"      (- 107520a   + 124800a   - 3584a )b   + (- 69120a   + 3584a )b"  
got:"    + "  
MISMATCH  
expected:"              7           5           3 2           6           4"  
got:"      (2064384a   - 5160960a   + 2211840a   - 98304a )b   - 294912a"  
MISMATCH  
expected:"    + "  
got:"              9           7           5           3           10"  
MISMATCH  
expected:"      (322560a   - 134400a   + 1792a )b"  
got:"    + "  
MISMATCH  
expected:"              6           4           2 3"  
got:"      (- 6193152a   + 10321920a   - 2211840a )b"  
MISMATCH  
expected:"    + "  
got:"              8           6           4 2"  
MISMATCH  
expected:"      (64512a   - 443520a   + 83520a   - 384a)b"  
got:"    + "  
MISMATCH  
expected:"              7           5           3           4"  
got:"      (10321920a   - 10321920a   + 737280a )b"  
MISMATCH  
expected:"    + "  
got:"              7           5           3 3"  
MISMATCH  
expected:"      (- 193536a   + 349440a   - 27840a )b"  
got:"    + "  
MISMATCH  
expected:"              6           4           2 5"  
got:"      (- 10321920a   + 5160960a )b"  
MISMATCH  
expected:"    + "  
got:"              6           4 4"  
MISMATCH  
expected:"      (- 161280a   + 40320a )b   + (241920a   - 161280a   + 3840a)b"  
got:"    + "  
MISMATCH  
expected:"              4           2 7           5           3           6"  
got:"      294912a b   - 2064384a b   + (6193152a   - 1032192a )b"  
MISMATCH

```

```

expected:"          + "
got:"          3 7          4 6          5          3 5"
MISMATCH
expected:"          1008a b  - 12096a b  + (60480a  - 4200a)b"
got:"          + "
MISMATCH
expected:"          10          2 9          3          8"
got:"          ?"
MISMATCH
expected:"          + "
got:"          3"
MISMATCH
expected:"          c"
got:"          *"
MISMATCH
expected:"          2"
got:"          64a c  - 48a b c  + 12a b c  - a b"
MISMATCH
expected:"          *"
got:"          9 3          8 2 2          7 4          6 6"
MISMATCH
expected:"          (- 645120a  + 153600a  - 1792a )b + 215040a  - 76800a  + 1792a"
got:"          /"
MISMATCH
expected:"          7          5          3          8          6          4"
got:"          (- 5760a  + 5632a )b  - 1536a b + 256a"
MISMATCH
expected:"          + "
got:"          6          4 2          5          6"
MISMATCH
expected:"          (- 258048a  + 967680a  - 132480a  + 640a )b"
got:"          + "
MISMATCH
expected:"          8          6          4          2 2"
got:"          (13440a  - 105600a  + 16384a )b  + (34560a  - 12288a )b"
MISMATCH
expected:"          + "
got:"          6          4          2 4          5          3 3"
MISMATCH
expected:"          (774144a  - 860160a  + 55680a )b"
got:"          + "
MISMATCH
expected:"          7          5          3 3"
got:"          (- 80640a  + 192000a  - 12288a)b"
MISMATCH
expected:"          + "
got:"          5          3          5"
MISMATCH
expected:"          (645120a  - 120960a )b  + (- 967680a  + 443520a  - 9120a )b"
got:"          + "

```

MISMATCH
 expected: "
$$(- 5376a^5 + 215040a^3 - 207360a^5 + 4096)b^4 - 4032a^2b^6 + 48384ab^4 + (- 241920a^2 + 13440a^2)b^6$$
"
 got: "
$$(- 5376a^5 + 215040a^3 - 207360a^5 + 4096)b^4 - 4032a^2b^6 + 48384ab^4 + (- 241920a^2 + 13440a^2)b^6$$
"
 MISMATCH
 expected: "
$$(32256a^2 - 322560a^3 + 122880a^4)b^2$$
"
 got: "
$$(32256a^2 - 322560a^3 + 122880a^4)b^2$$
"
 MISMATCH
 expected: "
$$(107520a^3 - 134400a^3)b + (- 80640a^4 + 282240a^2 - 30720)b^8$$
"
 got: "
$$(107520a^3 - 134400a^3)b + (- 80640a^4 + 282240a^2 - 30720)b^8$$
"
 MISMATCH
 expected: "
$$23040a^5 - 256a^3$$
"
 got: "
$$23040a^5 - 256a^3$$
"
 MISMATCH
 expected: "
$$(- 5376b^5 + 32256a^3b + (- 80640a^4 + 26880)b^10$$
"
 got: "
$$(- 5376b^5 + 32256a^3b + (- 80640a^4 + 26880)b^10$$
"
 MISMATCH
 expected: "
$$(- 1032192a^8 + 430080a^6 - 23040a^4)b + 344064a^9 - 215040a^7$$
"
 got: "
$$(- 1032192a^8 + 430080a^6 - 23040a^4)b + 344064a^9 - 215040a^7$$
"
 MISMATCH
 expected: "
$$(- 860160a^6 + 107520a^4)b + (1290240a^7 - 322560a^5 + 5760a^3)b^2$$
"
 got: "
$$(- 860160a^6 + 107520a^4)b + (1290240a^7 - 322560a^5 + 5760a^3)b^2$$
"
 MISMATCH
 expected: "
$$5376a^6b - 64512a^4b + (322560a^7 - 13440a^5)b^2$$
"
 got: "
$$5376a^6b - 64512a^4b + (322560a^7 - 13440a^5)b^2$$
"

```

got:"          (- 107520a + 499200a - 57344a )b + (- 138240a + 28672a )b"
MISMATCH
expected:"          3 6          4 5          5          3 4"
got:"          7          5          3 2          6          4"
MISMATCH
expected:" + "
got:"          + "
MISMATCH
expected:" ?"
got:"          (645120a - 1075200a + 57344a )b"
MISMATCH
expected:" 4"
got:"          6          4          2 3"
MISMATCH
expected:" *"
got:"          + "
MISMATCH
expected:" 16a c - 8a b c + a b"
got:"          (64512a - 1774080a + 1336320a - 24576a)b"
MISMATCH
expected:" 7 2          6 2          5 4"
got:"          7          5          3          4"
MISMATCH
expected:" /"
got:"          + "
MISMATCH
expected:" (- 16800a + 3600a )b + (6720a - 4800a )b + 3600a b - 1440a"
got:"          (- 387072a + 2795520a - 890880a )b"
MISMATCH
expected:"          4          2 3          5          3 2          4          5"
got:"          6          4          2 5"
MISMATCH
expected:" + "
got:"          + "
MISMATCH
expected:" (10080a - 10920a + 240)b + (- 4032a + 18480a - 1440a)b"
got:"          (967680a - 2580480a + 245760a)b"
MISMATCH
expected:"          4          2          5          5          3          4"
got:"          5          3          6"
MISMATCH
expected:" + "
got:"          + "
MISMATCH
expected:" 126b - 1260a b + (5040a - 420)b + (- 10080a + 3360a)b"
got:"          (- 1290240a + 1290240a )b"
MISMATCH
expected:"          9          8          2          7          3          6"
got:"          4          2 7"
MISMATCH

```

```

expected:"      + "
got:"          + "
MISMATCH
expected:"      c"
got:"          64512a b   - 387072a b   + (967680a   - 268800a)b"
MISMATCH
expected:"      *"
got:"          10          2 9          3          8"
MISMATCH
expected:"          4800a"
got:"      + "
MISMATCH
expected:"          4"
got:"      c"
MISMATCH
expected:"      + "
got:"          2"
MISMATCH
expected:"          (32256a   - 80640a   + 4320a )b   + (67200a   - 7200a )b - 26880a"
got:"      *"
MISMATCH
expected:"          6          4          2 2          5          3          6"
got:"          28672a"
MISMATCH
expected:"      + "
got:"          4"
MISMATCH
expected:"          (80640a   - 18480a )b   + (- 80640a   + 53760a   - 960a)b"
got:"      + "
MISMATCH
expected:"          4          2 4          5          3          3"
got:"          (- 1290240a   + 1228800a   - 57344a )b + 215040a   - 307200a"
MISMATCH
expected:"      + "
got:"          7          5          3          8          6"
MISMATCH
expected:"          - 1008a b   + 10080a b   + (- 40320a   + 2520a)b"
got:"      + "
MISMATCH
expected:"          7          2 6          3          5"
got:"          (- 258048a   + 3870720a   - 2119680a   + 40960a )b"
MISMATCH
expected:"      + "
got:"          8          6          4          2 2"
MISMATCH
expected:"      c"
got:"      + "
MISMATCH
expected:"      2"
got:"          (1548288a   - 6881280a   + 1781760a )b"

```

MISMATCH
 expected:" *" 7 5 3 3"
 got:"
 MISMATCH
 expected:" - 64512a + 26880a - 1440a"
 got:" + "
 MISMATCH
 expected:" 7 5 3"
 got:" (- 3870720a + 7096320a - 583680a)b"
 MISMATCH
 expected:" + "
 got:" 6 4 2 4"
 MISMATCH
 expected:" (- 161280a + 20160a)b + (161280a - 40320a + 720a)b"
 got:" + "
 MISMATCH
 expected:" 5 3 2 6 4 2"
 got:" (5160960a - 3870720a)b"
 MISMATCH
 expected:" + "
 got:" 5 3 5"
 MISMATCH
 expected:" 2016a b - 20160a b + (80640a - 3360a)b"
 got:" + "
 MISMATCH
 expected:" 2 5 3 4 4 2 3"
 got:" - 258048a b + 1548288a b + (- 3870720a + 860160a)b"
 MISMATCH
 expected:" + "
 got:" 2 8 3 7 4 2 6"
 MISMATCH
 expected:" ?"
 got:" + "
 MISMATCH
 expected:" 5"
 got:" c"
 MISMATCH
 expected:" *" 3"
 got:"
 MISMATCH
 expected:" 16a c - 8a b c + a b"
 got:" *"
 MISMATCH
 expected:" 6 2 5 2 4 4"
 got:" 368640a - 16384a"
 MISMATCH
 expected:" /"
 got:" 5 3"
 MISMATCH
 expected:" - 288a b + 144a"

```

got: "          + "
MISMATCH
expected: "          3          4"
got: "          (- 2064384a + 3440640a - 737280a )b + 344064a - 860160a"
MISMATCH
expected: "          + "
got: "          8          6          4          9          7"
MISMATCH
expected: "          (2016a - 3528a + 48)b + (4032a - 192a)b + (- 2016a + 336a )b"
got: "          + "
MISMATCH
expected: "          4          2          4          3          3          4          2 2"
got: "          (5160960a - 5160960a + 368640a )b"
MISMATCH
expected: "          + "
got: "          7          5          3 2"
MISMATCH
expected: "          126b - 1008a b + (3024a - 252)b + (- 4032a + 1512a)b"
got: "          + "
MISMATCH
expected: "          8          7          2          6          3          5"
got: "          (- 6881280a + 3440640a )b"
MISMATCH
expected: "          + "
got: "          6          4 3"
MISMATCH
expected: "          c"
got: "          + "
MISMATCH
expected: "          *"
got: "          344064a b - 2064384a b + (5160960a - 860160a )b"
MISMATCH
expected: "          (- 16128a + 480a )b + 8064a - 480a"
got: "          3 6          4 5          5          3 4"
MISMATCH
expected: "          4          2          5          3"
got: "          + "
MISMATCH
expected: "          + "
got: "          ?"
MISMATCH
expected: "          (32256a - 8064a )b + (- 16128a + 16128a - 192a)b"
got: "          4"
MISMATCH
expected: "          4          2 3          5          3          2"
got: "          *"
MISMATCH
expected: "          + "
got: "          16a c - 8a b c + a b"
MISMATCH

```



```

expected:"          - 1008a b + 8064a b + (- 24192a + 1512a)b"
got:"          7 2      6 2      5 4"
MISMATCH
expected:"                                6          2 5          3          4"
got:"      /"
MISMATCH
expected:"      + "
got:"      (- 33600a + 28800a )b + (6720a - 19200a )b + 7200a b - 1440a"
MISMATCH
expected:"      c"
got:"          4          2 3          5          3 2          4          5"
MISMATCH
expected:"      2"
got:"      + "
MISMATCH
expected:"      *"
got:"      (20160a - 87360a + 7680)b + (- 4032a + 73920a - 23040a)b"
MISMATCH
expected:"      (- 64512a + 8064a )b + 32256a - 8064a + 144a"
got:"          4          2          5          5          3          4"
MISMATCH
expected:"          5          3          6          4          2"
got:"      + "
MISMATCH
expected:"      + "
got:"      4032b - 20160a b + (40320a - 13440)b + (- 40320a + 53760a)b"
MISMATCH
expected:"      2016a b - 16128a b + (48384a - 2016a )b"
got:"          9          8          2          7          3          6"
MISMATCH
expected:"          2 4          3 3          4          2 2"
got:"      + "
MISMATCH
expected:"      + "
got:"      c"
MISMATCH
expected:"      ?"
got:"      *"
MISMATCH
expected:"      6"
got:"      - 26880a + 19200a"
MISMATCH
expected:"      *"
got:"          6          4"
MISMATCH
expected:"      4a c - a b"
got:"      + "
MISMATCH
expected:"      4          3 2"
got:"      (32256a - 322560a + 69120a )b + (134400a - 57600a )b"

```

```

MISMATCH
expected:"      /"
got:"          6          4          2 2          5          3"
MISMATCH
expected:"      504a b  + (- 1008a  + 84)b  + (672a  - 336a)b  + 504a b - 336a"
got:"      + "
MISMATCH
expected:"          4          2          3          3          2          2          3"
got:"      (322560a  - 295680a )b  + (- 161280a  + 430080a  - 30720a)b"
MISMATCH
expected:"      + "
got:"          4          2 4          5          3          3"
MISMATCH
expected:"      (336a b  - 2016a b  + (4032a  - 168a)b - 2688a  + 336a )c - 84b"
got:"      + "
MISMATCH
expected:"          3          2 2          3          4          2          5"
got:"      - 32256a b  + 161280a b  + (- 322560a  + 80640a)b"
MISMATCH
expected:"      + "
got:"          7          2 6          3          5"
MISMATCH
expected:"      ?"
got:"      + "
MISMATCH
expected:"      7"
got:"      c"
MISMATCH
expected:"      *"
got:"          2"
MISMATCH
expected:"      4a c - a b"
got:"      *"
MISMATCH
expected:"          3          2 2"
got:"      - 64512a  + 107520a  - 23040a"
MISMATCH
expected:"      /"
got:"          7          5          3"
MISMATCH
expected:"      (- 144a  + 12)b  - 24a b + 24a"
got:"      + "
MISMATCH
expected:"          2          2          2"
got:"      (- 645120a  + 322560a )b  + (322560a  - 322560a  + 23040a )b"
MISMATCH
expected:"      + "
got:"          5          3 2          6          4          2"
MISMATCH
expected:"      (144a b  - 576a b + 576a  - 24a)c - 36b  + 144a b"

```

```

got:"          + "
MISMATCH
expected:"          2      2      3      4      3"
got:"          64512a b  - 322560a b  + (645120a  - 107520a )b"
MISMATCH
expected:" + "
got:"          2 5      3 4      4      2 3"
MISMATCH
expected:"          a"
got:" + "
MISMATCH
expected:" ? + ----- ?"
got:" ?"
MISMATCH
expected:" 9 9b - 18a 8"
got:" 5"
MISMATCH
expected:" WARNING (genufact): No known algorithm to factor"
got:" *"
MISMATCH
expected:" , trying square-free."
got:" 16a c  - 8a b c + a b"
MISMATCH
expected:" 64a  c  - 48a  b c  + 12a  b c - a b"
got:" 6 2      5 2      4 4"
MISMATCH
expected:" 12 3      11 2 2      10 4      9 6"
got:" /"
MISMATCH
expected:" /"
got:" (- 2016a  + 1344a )b  - 576a b + 144a"
MISMATCH
expected:" - 64b"
got:" 4      2 2      3      4"
MISMATCH
expected:" 9"
got:" + "
MISMATCH
expected:" + "
got:" (2016a  - 14112a  + 768)b  + (8064a  - 1536a)b"
MISMATCH
expected:" (- 12a b  + 120a b  - 384a b  + 384a b )c + b  - 12b  + 48b"
got:" 4      2      4      3      3"
MISMATCH
expected:" 13      11      9      7      15      13      11"
got:" + "
MISMATCH
expected:" + "
got:" 2016b  - 8064a b  + (12096a  - 4032)b  + (- 8064a  + 12096a)b"
MISMATCH

```

```

expected:"      (48a b  - 384a b  + 912a b  - 640a b  + 256a b )c"
got:"          8      7      2      6      3      5"
MISMATCH
expected:"      2 11      2 9      2 7      2 5      2 3 2"
got:"      + "
MISMATCH
expected:"      + "
got:"      c"
MISMATCH
expected:"      (- 64a b  + 384a b  - 576a b  + 256a b )c"
got:"      *"
MISMATCH
expected:"      3 9      3 7      3 5      3 3 3"
got:"      (- 32256a  + 3840a )b + 8064a  - 1920a"
MISMATCH
expected:"      + "
got:"      4      2      5      3"
MISMATCH
expected:"      ?"
got:"      + "
MISMATCH
expected:"      *"
got:"      (64512a  - 64512a )b  + (- 16128a  + 64512a  - 3072a)b"
MISMATCH
expected:"      64a  c  - 48a  b c  + 12a  b c  - a b"
got:"          4      2 3      5      3      2"
MISMATCH
expected:"      11 3      10 2 2      9 4      8 6"
got:"      + "
MISMATCH
expected:"      /"
got:"      - 16128a b  + 64512a b  + (- 96768a  + 24192a)b"
MISMATCH
expected:"      - 240b  + 192b"
got:"          6      2 5      3      4"
MISMATCH
expected:"      10      8"
got:"      + "
MISMATCH
expected:"      + "
got:"      c"
MISMATCH
expected:"      (108a b  - 840a b  + 1920a b  - 1152a b )c - 9b  + 84b"
got:"      2"
MISMATCH
expected:"      12      10      8      6      14      12"
got:"      *"
MISMATCH
expected:"      + "
got:"      (- 129024a  + 64512a )b + 32256a  - 32256a  + 2304a"

```

```

MISMATCH
expected:"          (- 432a b  + 2688a b  - 4560a b  + 1920a b  - 256a b )c"
got:"              5          3          6          4          2"
MISMATCH
expected:"          2 10          2 8          2 6          2 4          2 2 2"
got:"          + "
MISMATCH
expected:"          + "
got:"          32256a b  - 129024a b  + (193536a  - 32256a )b"
MISMATCH
expected:"          (576a b  - 2688a b  + 2880a b  - 768a b )c"
got:"              2 4          3 3          4          2 2"
MISMATCH
expected:"          3 8          3 6          3 4          3 2 3"
got:"          + "
MISMATCH
expected:"          + "
got:"          ?"
MISMATCH
expected:"          ?"
got:"          6"
MISMATCH
expected:"          2"
got:"          *"
MISMATCH
expected:"          *"
got:"          4a c - a b"
MISMATCH
expected:"          64a  c  - 48a b c  + 12a b c - a b"
got:"          4      3 2"
MISMATCH
expected:"          10 3          9 2 2          8 4          7 6"
got:"          /"
MISMATCH
expected:"          /"
got:"          1008a b - 336a"
MISMATCH
expected:"          480b  - 192b"
got:"          2          3"
MISMATCH
expected:"          9          7"
got:"          + "
MISMATCH
expected:"          + "
got:"          - 672b  + 2016a b  + (- 2016a  + 672)b  + (672a  - 1344a)b"
MISMATCH
expected:"          (- 432a b  + 2520a b  - 3840a b  + 1152a b )c + 36b  - 252b"
got:"              5          4          2          3          3          2"
MISMATCH
expected:"          11          9          7          5          13          11"

```

```

got: "      + "
MISMATCH
expected: "      + "
got: "      (2688a b - 8064a b + (8064a - 1344a)b - 2688a + 1344a )c"
MISMATCH
expected: "      (1728a b - 8064a b + 9120a b - 1920a b )c"
got: "      3      2 2      3      4      2"
MISMATCH
expected: "      2 9      2 7      2 5      2 3 2"
got: "      + "
MISMATCH
expected: "      + "
got: "      ?"
MISMATCH
expected: "      (- 2304a b + 8064a b - 5760a b + 768a b)c"
got: "      7"
MISMATCH
expected: "      3 7      3 5      3 3      3 3"
got: "      *"
MISMATCH
expected: "      + "
got: "      4a c - a b"
MISMATCH
expected: "      ?"
got: "      3      2 2"
MISMATCH
expected: "      3"
got: "      /"
MISMATCH
expected: "      *"
got: "      (- 144a + 48)b - 48a b + 24a"
MISMATCH
expected: "      64a c - 48a b c + 12a b c - a b"
got: "      2      2      2"
MISMATCH
expected: "      9 3      8 2 2      7 4      6 6"
got: "      + "
MISMATCH
expected: "      /"
got: "      (576a b - 1152a b + 576a - 96a)c - 144b + 288a b"
MISMATCH
expected: "      - 480b + 64b"
got: "      2      2      3      4      3"
MISMATCH
expected: "      8      6"
got: "      + "
MISMATCH
expected: "      + "
got: "      a"
MISMATCH

```

```

expected:"          (1008a b  - 4200a b  + 3840a b  - 384a b )c - 84b  + 420b"
got:"          ?  + ----- ?"
MISMATCH
expected:"          10          8          6          4          12          10"
got:"          9  18b - 18a  8"
MISMATCH
expected:"          + "
got:"          WARNING (genufact): No known algorithm to factor"
MISMATCH
expected:"          (- 4032a b  + 13440a b  - 9120a b  + 640a b )c"
got:"          , trying square-free."
MISMATCH
expected:"          2 8          2 6          2 4          2 2  2"
got:"          64a  c  - 48a  b c  + 12a  b c  - a b"
MISMATCH
expected:"          + "
got:"          12 3          11 2 2          10 4          9 6"
MISMATCH
expected:"          (5376a b  - 13440a b  + 5760a b  - 256a )c"
got:"          /"
MISMATCH
expected:"          3 6          3 4          3 2          3 3"
got:"          - 64b"
MISMATCH
expected:"          + "
got:"          9"
MISMATCH
expected:"          ?"
got:"          + "
MISMATCH
expected:"          4"
got:"          (- 12a b  + 120a b  - 384a b  + 384a b )c + b  - 12b  + 48b"
MISMATCH
expected:"          *"
got:"          13          11          9          7          15          13          11"
MISMATCH
expected:"          16a c  - 8a b c + a b"
got:"          + "
MISMATCH
expected:"          7 2          6 2          5 4"
got:"          (48a b  - 384a b  + 912a b  - 640a b  + 256a b )c"
MISMATCH
expected:"          /"
got:"          2 11          2 9          2 7          2 5          2 3  2"
MISMATCH
expected:"          (1008a b  - 2520a b  + 960a b )c - 126b  + 420b  - 240b"
got:"          + "
MISMATCH
expected:"          7          5          3          9          7          5"
got:"          (- 64a b  + 384a b  - 576a b  + 256a b )c"

```

```

MISMATCH
expected:"          + "
got:"          3 9          3 7          3 5          3 3 3"
MISMATCH
expected:"          (- 2016a b + 3360a b - 720a b)c"
got:"          + "
MISMATCH
expected:"          2 5          2 3          2 2"
got:"          ?"
MISMATCH
expected:"          + "
got:"          *"
MISMATCH
expected:"          ?"
got:"          64a c - 48a b c + 12a b c - a b"
MISMATCH
expected:"          5"
got:"          11 3          10 2 2          9 4          8 6"
MISMATCH
expected:"          *"
got:"          /"
MISMATCH
expected:"          16a c - 8a b c + a b"
got:"          - 240b + 192b"
MISMATCH
expected:"          6 2          5 2          4 4"
got:"          10          8"
MISMATCH
expected:"          /"
got:"          + "
MISMATCH
expected:"          (- 1008a b + 1512a b - 192a b )c + 126b - 252b + 48b"
got:"          (108a b - 840a b + 1920a b - 1152a b )c - 9b + 84b"
MISMATCH
expected:"          6          4          2          8          6          4"
got:"          12          10          8          6          14          12"
MISMATCH
expected:"          (2016a b - 2016a b + 144a )c"
got:"          (- 432a b + 2688a b - 4560a b + 1920a b - 256a b )c"
MISMATCH
expected:"          2 4          2 2          2 2"
got:"          2 10          2 8          2 6          2 4          2 2 2"
MISMATCH
expected:"          + "
got:"          + "
MISMATCH
expected:"          4a c - a b"
got:"          (576a b - 2688a b + 2880a b - 768a b )c"
MISMATCH
expected:"          4          3 2"

```



```

got:"          3 8          3 6          3 4          3 2 3"
MISMATCH
expected:"          ----- ?"
got:"      + "
MISMATCH
expected:"      (- 336a b + 168a b)c + 84b - 84b 6"
got:"      ?"
MISMATCH
expected:"          3          5          3"
got:"      2"
MISMATCH
expected:"      + "
got:"      *"
MISMATCH
expected:"          4a c - a b"
got:"      64a c - 48a b c + 12a b c - a b"
MISMATCH
expected:"          a          3          2 2"
got:"      10 3          9 2 2          8 4          7 6"
MISMATCH
expected:"      ? - -- ? + ----- ?"
got:"      /"
MISMATCH
expected:"      9 9b 8 (144a b - 24a)c - 36b + 12b 7"
got:"      480b - 192b"
MISMATCH
expected:"          2          4          2"
got:"      9          7"
MISMATCH
expected:"      WARNING (genufact): No known algorithm to factor"
got:"      + "
MISMATCH
expected:"          4a c - a b          4a c - a b"
got:"      (- 432a b + 2520a b - 3840a b + 1152a b )c + 36b - 252b"
MISMATCH
expected:"          3          2 2          5          4 2"
got:"          11          9          7          5          13          11"
MISMATCH
expected:"      ? + ----- ? + -----, trying square-free."
got:"      + "
MISMATCH
expected:"      4 - 4a c + 2b - 4a b + 4a 2 - b + 4a b - 4a"
got:"      (1728a b - 8064a b + 9120a b - 1920a b )c"
MISMATCH
expected:"          2          2          2          2"
got:"          2 9          2 7          2 5          2 3 2"
MISMATCH
expected:"      WARNING (genufact): No known algorithm to factor"
got:"      + "
MISMATCH

```

```

expected:"          4a c - a b          4a c - a b"
got:"          (- 2304a b + 8064a b - 5760a b + 768a b)c"
MISMATCH
expected:"          3      2 2      5      4 2"
got:"          3 7      3 5      3 3      3 3"
MISMATCH
expected:"      ? + ----- ? - -----, trying square-free."
got:"      + "
MISMATCH
expected:"      4      - 4a c + 2b      2      b"
got:"      ?"
MISMATCH
expected:"          2      2"
got:"      3"
MISMATCH
expected:"      WARNING (genufact): No known algorithm to factor"
got:"      *"
MISMATCH
expected:" "
got:"          64a c - 48a b c + 12a b c - a b"
FAILED kamke3 85 of 139

```

bug 7040: warning: missing sentinel in function call

```

edible.c: In function main:
edible.c:226: warning: missing sentinel in function call

```

bug 7039: suggest explicit braces to avoid ambiguous

```

/research2/test/int/hyper/spadint.c: In function print_source_to_string1:
/research2/test/int/hyper/spadint.c:990: warning: suggest explicit braces to avoid ambiguous els

```

bug 7038: suggest explicit braces to avoid ambiguous

```

/research2/test/int/hyper/spadint.c: In function print_to_string1:
/research2/test/int/hyper/spadint.c:503: warning: suggest explicit braces to avoid ambiguous els
/research2/test/int/hyper/spadint.c:511: warning: suggest explicit braces to avoid ambiguous els

```

bug 7037: passing arg 2 makes pointer from integer

```
/research2/test/int/hyper/parse-types.c: In function parse_condnode:
/research2/test/int/hyper/parse-types.c:159: warning: passing argument 2 of httperror makes pointer from integer
```

bug 7036: dereferencing type-punned pointer will break rules

```
/research2/test/int/hyper/event.c: In function handle_event:
/research2/test/int/hyper/event.c:244: warning: dereferencing type-punned pointer will break strict-aliasing
```

bug 7035: defined but not used

```
/research2/test/int/sman/sman.c:752: warning: clean_up_terminal defined but not used
```

bug 7034: session.c defined but not used

```
/research2/test/int/sman/session.c:58: warning: pr defined but not used
```

section

```
bug 7033:
/research2/test/obj/fedora5/graph/gdraws/gfun.o: In function 'PSInit':gfun.c:(.text+0x237f): warning: the use of uninitialized values
/research2/test/obj/fedora5/graph/gdraws/gfun.o: In function 'PSGlobalInit':gfun.c:(.text+0xd21): warning: the use of uninitialized values
```

section

```
bug 7032:
/research2/test/src/graph/view3d/static.h:42: warning: labels defined but not used
```

section

bug 7031:

```
/research2/test/int/graph/view3d/surface3d.c: In function drawRenderedPolygon:  
/research2/test/int/graph/view3d/surface3d.c:392: warning: col_rgb.r may be used uninitialized i  
/research2/test/int/graph/view3d/surface3d.c:392: warning: col_rgb.g may be used uninitialized i  
/research2/test/int/graph/view3d/surface3d.c:392: warning: col_rgb.b may be used uninitialized i
```

section

bug 7030:

```
/research2/test/int/graph/view3d/stuff3d.c: In function norm_dist:  
/research2/test/int/graph/view3d/stuff3d.c:126: warning: pert.z is used uninitialized in this fu
```

section

bug 7029:

```
/research2/test/int/graph/view3d/process3d.c: In function processEvents:  
/research2/test/int/graph/view3d/process3d.c:855: warning: linearMouseXY.x may be used uninitial  
/research2/test/int/graph/view3d/process3d.c:855: warning: linearMouseXY.y may be used uninitial  
/research2/test/int/graph/view3d/process3d.c:855: warning: mouseXY.x may be used uninitialized i  
/research2/test/int/graph/view3d/process3d.c:855: warning: mouseXY.y may be used uninitialized i
```

section

bug 7028:

```
/research2/test/int/graph/view3d/control3d.c: In function getControlXY:  
/research2/test/int/graph/view3d/control3d.c:798: warning: cXY.putX may be used uninitialized in  
/research2/test/int/graph/view3d/control3d.c:798: warning: cXY.putY may be used uninitialized in
```

section

bug 7027:

```
/research2/test/obj/fedora5/graph/gdraws/gfun.o: In function 'PSInit':gfun.c:(.text+0x237f): warning: the use of `tmpnam' is dangerous, better use `mkstemp'
/research2/test/obj/fedora5/graph/gdraws/gfun.o: In function 'PSGlobalInit':gfun.c:(.text+0xd21): warning: the use of `tmpnam' is dangerous, better use `mkstemp'
```

section

bug 7026:

```
/research2/test/int/graph/view2d/process2d.c: In function processEvents:
/research2/test/int/graph/view2d/process2d.c:491: warning: mouseXY.x may be used uninitialized in this function
/research2/test/int/graph/view2d/process2d.c:491: warning: mouseXY.y may be used uninitialized in this function
```

section

bug 7025:

```
/research2/test/int/graph/view2d/control2d.c: In function getControlXY:
/research2/test/int/graph/view2d/control2d.c:367: warning: cXY.putX may be used uninitialized in this function
/research2/test/int/graph/view2d/control2d.c:367: warning: cXY.putY may be used uninitialized in this function
```

section

bug 7024: edible.c:226: warning: missing sentinel in function call

section

```
bug 7022: NewDistributedMultivariatePolynomial does not exist
Hyperdoc bug Topics>Polynomials>Specific Types>Polynomial
likely this is XDistributedPolynomial... change all refs in help
```

section

bug 7021: vertical bar is not implemented properly.

```
m1:=matrix [[1,2,1],[2,1,-2],[1,-2,4]]
eigenvalues(m1)
should be
```

$$[5, \lambda^2 \mid \lambda^2 - 5]$$

but instead prints as:

$$[5, \lambda^2 \mid \lambda^2 - 5]$$

section

bug 7018: mathml does not render "failed" properly

```
limit(x*log(x),x=0)
[leftHandLimit=~failed~,rightHandLimit=0]
```

section

bug 7017: time screws the regression tests

```
add )set mes time off
to all regression tests
```

section

bug 7013: DirectProduct harms matrix multiply by constant

(9) -> k

$$(9) \begin{pmatrix} +2 & -1 & 0 \\ -1 & 2 & -1 \\ +0 & -1 & 2 \end{pmatrix}$$

Type: SquareMatrix(3,Fraction Integer)

(10) -> k*4

(10) [4,0,4]

Type: DirectProduct(3,Fraction Integer)

section

bug 7012 (SF/1795259): Error printing lists
fixed in OpenAxiom by changing:

```
aggcat.spad
array1.spad
list.spad
tree.spad
```

)abbrev domain F00 foo

```
foo(): with
  makeFoo: () -> %
  coerce: % -> OutputForm
  == add
  makeFoo() == "a foo" pretend %

  coerce x == (x pretend String)::OutputForm
```

)co foo.spad

Type: List foo

(7) -> makeFoo()

(7) "a foo"

Type: foo

(8) -> [makeFoo()]

LISP output:
(a foo)

Type: List foo

The reason is that the List functor is defined
to have a coercion to OutputForm only if its argument
is of type SetCategory. This is clearly wrong. There is
no reason to associate prettyprinting with SetCategory
membership.

changing the signature line to read:
foo(): SetCategory with
causes

Type: List foo

(10) -> [makeFoo()]

```
(10) ["a foo"]
```

Type: List foo

section

```
bug7009 (SF/1790563): fatal error (fixed in openaxiom rev 135)
```

```
(1) -> Insn == Enumeration(New, Move, Line, Close)
```

Type: Void

```
(2) -> Insn
```

```
Compiling body of rule Insn to compute value of type Domain
```

```
>> System error:
```

```
Caught fatal error [memory may be damaged]
```

```
in i-output.boot function outputDomainConstructor change
```

```
if VECTORP form then form := devaluate form
```

```
if VECTORP CAR form then form := devaluate form
```

section

```
bug7008: never completes
```

```
y:=operator 'y
```

```
ode360 := D(y(x),x)*cos(a*y(x))-b*(1-c*cos(a*y(x)))*_
          sqrt(cos(a*y(x))**2-(1-c*cos(a*y(x))))
```

```
solve(ode360,y,x)
```

```
IDEA: write a new axiom output for
```

```
X+Y+Z
```

```
where X=
```

```
where Y=
```

```
where Z=
```

```
and it linebreaks properly
```

section

```
bug7007: never completes
```



```

y:=operator 'y
ode184 := (a*x**2+b*x+c)**2*(D(y(x),x)+y(x)**2) + A
ode184a:=solve(ode184,y,x)

```

section

```

bug7006: never completes
y:=operator 'y
ode182 := x*(x**3-1)*D(y(x),x) - 2*x*y(x)**2 + y(x) + x**2
ode182a:=solve(ode182,y,x)

```

section

```

bug7005: infinite loop trying to factor
y:=operator 'y
ode162 := (x-a)*(x-b)*D(y(x),x) + y(x)**2 + k*(y(x)+x-a)*(y(x)+x-b)
ode162a:=solve(ode162,y,x)

```

```

WARNING (genufact): No known algorithm to factor
      3      2      2      2
      ? - 3? + (- k + 3)? + k - 1, trying square-free.

```

section

```

bug7004: plugging the particular solution into the ODE using eval fails
         to take the derivative. And eval will not let you use y'(x)
y:=operator 'y
ode2 := D(y(x),x) + a*y(x) - c*exp(b*x)
ode2a:=solve(ode2,y,x)
mm:=ode2a.particular
eval(ode2,y(x)=mm)           <= should be zero
D(mm,x)-c*exp(b*x)+a*mm

```

section

bug7003:

```
ode73 := D(y(x),x) - ((a3*x**3+a2*x**2+a1*x+a0)/_
                      (a3*y(x)**3+a2*y(x)**2+a1*y(x)+a0))**(2/3)
```

$$(4) \quad y'(x) - \frac{a_3 x^3 + a_2 x^2 + a_1 x + a_0}{\sqrt[3]{a_3 y(x)^3 + a_2 y(x)^2 + a_1 y(x) + a_0}}$$

Type: Expression Integer

```
solve(ode73,y,x)
```

```
>> Error detected within library code:
Table construction failed in MLIFT
```

section

bug7002: infinite loop

```
ode59 := D(y(x),x) - a*sqrt(y(x)**2+1) - b
solve(ode59,y,x)
```

section

bug7001:

```
B0 n == matrix [[(if i=j+1 and odd? j then -1 else _
                  if i=j-1 and odd? i then 1 else 0) _
                  for j in 1..n] for i in 1..n]
```

```
PfChar(lambda, A) ==
```

```
  n := nrows A
  (n = 2) => lambda^2 + A.(1,2)
  M := subMatrix(A, 3, n, 3, n)
  r := subMatrix(A, 1, 1, 3, n)
  s := subMatrix(A, 3, n, 2, 2)
```

```

p := PfChar(lambda, M)
d := degree(p, lambda)

B := B0(n-2)
C := r*B
g := [(C*s).(1,1), A.(1,2), 1]
if d >= 4 then
  B := M*B
  for i in 4..d by 2 repeat
    C := C*B
    g := cons((C*s).(1,1), g)
  g := reverse! g

res := 0
for i in 0..d by 2 for j in 2..d+2 repeat
  c := coefficient(p, lambda, i)
  for e in first(g, j) for k in 2..-d by -2 repeat
    res := res + c * e * lambda^(k+i)

res

pfaffian A == eval(PfChar(1, A), 1=0)

makepfaff(seq,n) ==
  m:= matrix [[(if i<j then (seq.(j-i)) _
                else if i>j then -(seq.(i-j))
                else 0) for j in 1..n] for i in 1..n]

seq:=[2 for i in 1..]
mp:=[makepfaff(seq,k) for k in 2..10 by 2]
[pfaffian mp.i for i in 1..5]

```

section

question: can we represent portions of n-dimensional space in provisos?

section

```

bug7000:
m0:=[[0,0],[0,0]]
m1:Matrix(Integer):=[[0,3],[-3,0]]

```

```

m2:Matrix(Integer):=[[0,5],[-5,0]]
m3:Matrix(Integer):=[[0,7],[-7,0]]
mn:Matrix(Integer):=[[m1,m0,m0],[m0,m2,m0],[m0,m0,m3]]

```

section

```

bug102:
solve(sinh(z)=cosh(z),z)

>> Error detected within library code:
No identity element for reduce of empty list using operation
append

```

section

```

bug104:
solve(cos(z)/sin(z)=0,z)

(2) []
Type: List Equation Expression Integer
should be %pi/2

(3) -> solve(cot(z)=0,z)

(3) [z=  $\frac{\%pi}{2}$ ]
Type: List Equation Expression Integer

```

section

```

why do these differ?
diff -Naur ./viewdef.spad.pamphlet /research/may07/build-improvements/src/algebra/viewdef.spad.pam
diff -Naur ./view3d.spad.pamphlet /research/may07/build-improvements/src/algebra/view3d.spad.pam
diff -Naur ./view2d.spad.pamphlet /research/may07/build-improvements/src/algebra/view2d.spad.pam

```

section

```

--- src/algebra/intrf.spad.pamphlet 2007-04-27 21:29:46.000000000 -0400
+++ /research/may07/build-improvements/src/algebra/intrf.spad.pamphlet 2007-04-27 21:34:07.000000000 -0400
@@ -478,7 +478,7 @@
                                else notelm := notelm + monomial(missing, n)
                                p := reductum p
                                zero? notelm => [answr, coef0]
-                                [answr, notelm + monomial(coef0, 0)]
+                                [answr, notelm]

-- f is either 0 or of the form p(t)/(1 + t**2)**n
-- returns either

```

section

```

--- src/algebra/intfact.spad.pamphlet 2007-04-27 21:29:46.000000000 -0400
+++ /research/may07/build-improvements/src/algebra/intfact.spad.pamphlet 2007-04-27 21:34:07.000000000 -0400
@@ -401,51 +401,27 @@
    r:I := 1
    q:I := 1
    G:I := 1
-    l:I
-    k:I
    until G > 1 repeat
        x := y
-        ys := y
        for i in 1..convert(r)@Integer repeat
            y := (y*y+5::I) rem n
            q := (q*abs(x-y)) rem n
-        k := 0::I
-        G := gcd(q,n)
+        k:I := 0
        until (k>=r) or (G>1) repeat
            ys := y
            for i in 1..convert(min(m,r-k))@Integer repeat
                y := (y*y+5::I) rem n
                q := (q*abs(x-y)) rem n
+                q := q*abs(x-y) rem n
            G := gcd(q,n)
            k := k+m
-        k := k + r
        r := 2*r

```

```

        if G=n then
-       l := k - m
-       G := 1::I
        until G>1 repeat
            ys := (ys*ys+5::I) rem n
            G := gcd(abs(x-ys),n)
-       l := l + 1
-       if G = n then
-       y := x0
-       x := x0
-       for i in 1..convert(l)@Integer repeat
-       y := (y*y+5::I) rem n
-       G := gcd(abs(x-y), n)
-       until G>1 repeat
-       y := (y*y+5::I) rem n
-       x := (x*x+5::I) rem n
-       G := gcd(abs(x-y), n)
        G=n => "failed"
        G

-   PollardSmallFactor20(n:I):Union(I,"failed") ==
-   for i in 1..20 repeat
-       r := PollardSmallFactor n
-       r case I => return r
-   r
-
    BasicSieve(r, lim) ==
        l:List(I) :=
            [1::I,2::I,2::I,4::I,2::I,4::I,2::I,4::I,6::I,2::I,6::I]
@@ -494,7 +470,7 @@
        (y:=perfectSqrt (x**2-n)) case I =>
            insert_!(x+y,a,c)
            insert_!(x-y,a,c)
-       (d := PollardSmallFactor20 n) case I =>
+       (d := PollardSmallFactor n) case I =>
            for m in 0.. while zero?(n rem d) repeat n := n quo d
            insert_!(d, a, m * c)
            if n > 1 then insert_!(n, a, c)

```

section

```

--- src/algebra/gaussian.spad.pamphlet 2007-04-27 21:29:46.000000000 -0400
+++ /research/may07/build-improvements/src/algebra/gaussian.spad.pamphlet 2007-04-27 21:34:07.00
@@ -367,18 +367,10 @@
        argument x == atan2loc(imag x, real x)

```

```

else
  if R has RadicalCategory then
  -   argument x ==
  -     n1 := sqrt(norm(x))
  -     x1 := real(x) + n1
  -     (2::R)*atan(imag(x) * recip(x1)::R)
  -
  -   else
  -     -- Emulate sqrt using exp and log
  -     argument x ==
  -       n1 := exp(half*log(norm(x)))
  -       x1 := real(x) + n1
  -       (2::R)*atan(imag(x) * recip(x1)::R)
+   -- Not ordered so dictate two quadrants
+   argument x ==
+     zero? real x => pi()$R * half
+     atan(imag(x) * recip(real x)::R)

pi() == pi()$R :: %

```

section

```

--- src/algebra/free.spad.pamphlet 2007-04-27 21:29:46.000000000 -0400
+++ /research/may07/build-improvements/src/algebra/free.spad.pamphlet 2007-04-27 21:34:07.000000000 -0400
@@ -531,21 +531,21 @@
  inmax l ==
    mx := first l
    for t in rest l repeat
  -   if mx.gen < t.gen then mx := t
+   if t.gen > mx.gen then mx := t
  mx

  -   -- lexicographic order
  a < b ==
    zero? a =>
      zero? b => false
  -   0 < (inmax terms b).exp
+   (inmax terms b).exp > 0
  ta := inmax terms a
  zero? b => ta.exp < 0
+   ta := inmax terms a
  tb := inmax terms b
  -   ta.gen < tb.gen => 0 < tb.exp
  -   tb.gen < ta.gen => ta.exp < 0
+   ta.gen < tb.gen => true

```

```

+      ta.gen > tb.gen => false
+      ta.exp < tb.exp => true
-      tb.exp < ta.exp => false
+      ta.exp > tb.exp => false
      lc := ta.exp * ta.gen
      (a - lc) < (b - lc)

```

section

```

--- src/algebra/files.spad.pamphlet 2007-04-27 21:29:46.000000000 -0400
+++ /research/may07/build-improvements/src/algebra/files.spad.pamphlet 2007-04-27 21:34:07.000000000
@@ -389,8 +389,10 @@
      defstream(fn: Name, mode: IOMode): FileState ==
        mode = "input" =>
+      not readable? fn => error ["File is not readable", fn]
        RDEFINSTREAM(fn::String)$Lisp
        mode = "output" =>
+      not writable? fn => error ["File is not writable", fn]
        RDEFOUTSTREAM(fn::String)$Lisp
        error ["IO mode must be input or output", mode]

@@ -407,7 +409,9 @@
      mode = "either" =>
        exists? fname =>
          open(fname, "input")
-      reopen_!(open(fname, "output"), "input")
+      writable? fname =>
+      reopen_!(open(fname, "output"), "input")
+      error "File does not exist and cannot be created"
      [fname, defstream(fname, mode), mode]
      reopen_!(f, mode) ==
        close_! f

@@ -502,10 +506,6 @@
      ++ \spad{lib.k := v} saves the value \spad{v} in the library
      ++ \spad{lib}. It can later be extracted using the key \spad{k}.

-      close_!: % -> %
-      ++ close!(f) returns the library f closed to input and output.
-
-
      == KeyedAccessFile(Any) add
      Rep := KeyedAccessFile(Any)
      library f == open f

```


section

```

--- src/algebra/efstruc.spad.pamphlet 2007-04-27 21:29:46.000000000 -0400
+++ /research/may07/build-improvements/src/algebra/efstruc.spad.pamphlet 2007-04-27 21:34:07.000000000 -0400
@@ -2,7 +2,7 @@
\usepackage{axiom}
\begin{document}
\title{\$SPAD/src/algebra efstruc.spad}
-\author{Manuel Bronstein, Waldek Hebisch}
+\author{Manuel Bronstein}
\maketitle
\begin{abstract}
\end{abstract}
@@ -101,9 +101,9 @@
<<package EFSTRUC ElementaryFunctionStructurePackage>>=
)abbrev package EFSTRUC ElementaryFunctionStructurePackage
++ Risch structure theorem
-++ Author: Manuel Bronstein, Waldek Hebisch
+++ Author: Manuel Bronstein
++ Date Created: 1987
-++ Date Last Updated: 9 October 2006
+++ Date Last Updated: 16 August 1995
++ Description:
++ ElementaryFunctionStructurePackage provides functions to test the
++ algebraic independence of various elementary functions, using the
@@ -346,102 +346,9 @@
return false
true

```

section

- 1) the patch correct problem in goodCoef, there are other bugs in integrator, I will work on them separately.
- 2) AFAICS goodCoef is an optimization, it should be safe to return "failed" always (but since we take different path in the other part we hit different bugs). I tried to preserve optimization done by goodCoef whenever it is safe.
- 3) I would like to use higher level constructs, but I kept hitting into problems with SPAD compiler, so finally I used a rather low level way
- 4) without the patch the AXIOM recurses infinitely handling the few integrals below (first I previously gave in the bug report). With the patch the first two are correctly handled, the other two quickly hit into unimplemented part of Risch algorithm:

section

```

integrate(
  simplify(
    D((log((x)+1)+a)
      / (2*(((x*(exp(exp(x)/2)))-(a*(x+exp(-x/2))))*(2/x))-3)),
    x),
  x)

```

section

```

integrate(
  simplify(
    D((((
      (sqrt((sqrt((x)/(x)))/(a)))
      / (sqrt(x)))
      * (((x)-(1))*(1))-(1)))
      * (a))
      * ((log((x)
      * ((a)/(log(sqrt((x)/((x)*(x)*(x)))))))
      + ((x)/(sqrt(2)))*(exp(exp(x))))
      + ((x)-(exp(2))))
      + ((1)*(2)), x)),
    x)

```

This fails with ‘Function not supported by Risch d.e.’

section

```

integrate(
  simplify(
    D((((((a)/(sqrt((sqrt(x))-(x))))+(x))*(x))+(1))
      * (((((1)-(log(x)))+(1))*(exp(exp(2))))*(2))
      * (((1)*((log((a)*(1)))*(a)))
      * (exp(((exp(x))*(x)
      - (exp(x))))))
      * (a), x)),
    x)

```

This fails with ‘intef: failed1 - cannot handle that integrand’

section

```

integrate(
  simplify(
    D((((((x)+(log(x)))*((x)/(x))/(sqrt(x))))
      / ((a)*(x)))
      - (sqrt(sqrt(((exp(x))/((x)-(sqrt(x))))-(log(x))))))
      - (x))
    - (((exp((sqrt(log(1)))*(x)*((x)/(log(x))))))
      / (exp((1)+(2)-((x)-(x))))))
      * (((x)/(1))-(log(x)))*(1))+(((2)+(a))/(exp(sqrt(x))))))
    / (2), x)),
  x)

```

If k1 is part of k2 we should not express k1 in terms of k2
 (otherwise we would get infinite recursion).
 Below we impose a stronger condition: we require
 height(k1) to be maximal

```

<<package EFSTRUC ElementaryFunctionStructurePackage>>=
goodCoef(v, l, s) ==
  h:NonNegativeInteger := 0
  j:Integer := 0
  ll : List K := [];
  for k in l repeat
    if (is?(k, "log"::SY) or is?(k, "exp"::SY)
      or is?(k, "tan"::SY) or is?(k, "atan"::SY)) then
      ll := [k, :ll]
      h := h + 1
  not (h = (maxIndex(v) - minIndex(v) + 1)) => "failed"
  h := 0
  ll := reverse(ll)
  for i in minIndex v .. maxIndex v for k in ll repeat
    h1 := height(k)
    if (h1 > h) then
      j := i
      h := h1
  for i in minIndex v .. maxIndex v for k in ll repeat
    is?(k, s) and (i >= j) and
  for i in minIndex v .. maxIndex v for k in l repeat
    is?(k, s) and
      ((r:=recip(qelt(v,i))) case Q) and
      (retractIfCan(r::Q)@Union(Z, "failed") case Z)
      and gdCoef?(qelt(v, i), v) => return([i, k])

```

section

```

--- src/algebra/draw.spad.pamphlet 2007-04-27 21:29:46.000000000 -0400
+++ /research/may07/build-improvements/src/algebra/draw.spad.pamphlet 2007-04-27 21:34:07.000000
@@ -318,10 +318,9 @@
    myTrap1: (SF-> SF, SF) -> SF
    myTrap1(ff:SF-> SF, f:SF):SF ==
      s := trapNumericErrors(ff(f))$Lisp :: Union(SF, "failed")
-      s case "failed" => 0
+      s case "failed" => _$NaNvalue$Lisp
      r:=s::SF
-      r >max()$SF => max()$SF
-      r < min()$SF => min()$SF
+      r >max()$SF or r < min()$SF => _$NaNvalue$Lisp
      r

    makePt2: (SF,SF) -> Point SF
@@ -494,10 +493,9 @@
    myTrap2: ((SF, SF) -> SF, SF, SF) -> SF
    myTrap2(ff:(SF, SF) -> SF, u:SF, v:SF):SF ==
      s := trapNumericErrors(ff(u, v))$Lisp :: Union(SF, "failed")
-      s case "failed" => 0
+      s case "failed" => _$NaNvalue$Lisp
      r:SF := s::SF
-      r >max()$SF => max()$SF
-      r < min()$SF => min()$SF
+      r >max()$SF or r < min()$SF => _$NaNvalue$Lisp
      r

    recolor(ptFunc,colFunc) ==

```

section

```

--- src/algebra/plot3d.spad.pamphlet 2007-04-27 21:29:46.000000000 -0400
+++ /research/may07/build-improvements/src/algebra/plot3d.spad.pamphlet 2007-04-27 21:34:07.0000
@@ -192,11 +192,11 @@
    select(l,f,g) ==
      m := f first l
-      -- if (EQL(m, _$NaNvalue$Lisp)$Lisp) then m := 0
+      if (EQL(m, _$NaNvalue$Lisp)$Lisp) then m := 0
-      if (EQL(m, _$NaNvalue$Lisp)$Lisp) then m := 0
-      for p in rest l repeat m := g(m,fp)
      for p in rest l repeat

```

```

        fp : F := f p
-      -- if (EQL(fp, _$NaNvalue$Lisp)$Lisp) then fp := 0
+      if (EQL(fp, _$NaNvalue$Lisp)$Lisp) then fp := 0
        m := g(m,fp)
    m

@@ -447,7 +447,7 @@
    myTrap(ff:F-> F, f:F):F ==
        s := trapNumericErrors(ff(f))$Lisp :: Union(F, "failed")
        if (s) case "failed" then
-          r:F := 0
+          r:F := _$NaNvalue$Lisp
        else
            r:F := s
    r

```

section

```

--- src/algebra/numtheor.spad.pamphlet 2007-04-27 21:29:46.000000000 -0400
+++ /research/may07/build-improvements/src/algebra/numtheor.spad.pamphlet 2007-04-27 21:34:07.000000000 -0400
@@ -432,50 +432,18 @@
        for entry in factors factor n repeat
            r := ((entry.factor - 1) / $RN entry.factor) * r
        numer(n * r)
-@

-<<package INTHEORY IntegerNumberTheoryFunctions>>=
    divisors n ==
-    oldList : List Integer := [1]
+    oldList : List Integer := concat(1, nil())
    for f in factors factor n repeat
-    newList : List Integer := oldList
-    for k in 1..f.exponent repeat
+    newList : List Integer := nil()
+    for k in 0..f.exponent repeat
        pow := f.factor ** k
        for m in oldList repeat
-            newList := concat(pow * m, newList)
+            newList := concat(pow * m, newList)
        oldList := newList
+    sort(#1 < #2, newList)

-    sort(#1 < #2, oldList)
-@
-
-[[divisors]] generates the list of positive divisors of an integer. We proceed

```

-in an inductive manner: `[[oldList]]` contains the list of all possible divisors
 -containing only the first few factors of `[[n]]`. We then prepend all possible
 -divisors containing the new factor `[[f]]` by multiplying the `[[oldList]]` with
 -all possible positive powers of `[[f]]`.

```
-
-Up to [[patch--50]] the last line read
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
\subsection{section}
```

```
-    sort(#1 < #2, newList)
-
```

-which fails when `[[n]]` equals 1 or -1 , since in this case `[[factors factor n]]` returns the empty list, and thus the loop in which `[[newlist]]` is initialised is never entered. (issue 340) -
 -Furthermore, we took the opportunity to make the function slightly more -efficient. Up to `[[patch-50]]`, we had

section

```
...
    newList : List Integer := []
    for k in 0..f.exponent repeat
    ...
```

Thus, the `[[oldList]]` was copied element by element every time a new factor was added. Since we `\emph{prepend}` the new elements of `[[newList]]`, the two list can safely share memory.

```
<<package INTHEORY IntegerNumberTheoryFunctions>>=
    numberOfDivisors n ==
        n = 0 => 0
        */[1+entry.exponent for entry in factors factor n]
```

section

```
--- src/algebra/newdata.spad.pamphlet 2007-04-27 21:29:46.000000000 -0400
+++ /research/may07/build-improvements/src/algebra/newdata.spad.pamphlet 2007-04-27 21:34:07.000
@@ -10,21 +10,6 @@
\tableofcontents
\eject
\subsection{package IPRNTPK InternalPrintPackage}
```

-Putting (or omitting) the final call to {\bf FORCE-OUTPUT} in {\bf iprint}
 -is controversial:

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
\subsection{section}
\begin{verbatim}
```

Waldek: iprint is used to print (regular) triangular sets.

Unconditional flush defeats buffering optimizations, so IMHO
 it should be done only for some streams (for example streams
 connected to terminals).

Flushing output should be done when the output is complete,
 while iprint clearly is used to compose bigger units from small parts.

Tim: unless the princ contains a newline there is no guarantee that
 the output will appear. if the output is intended to be a prompt,
 for instance, which does not contain a newline then the flush
 is needed to force the output to appear.

Deleting the call to flush breaks the existing semantics of the package.

```
<<package IPRNTPK InternalPrintPackage>>=
)abbrev package IPRNTPK InternalPrintPackage
++ Author: Themos Tsikas
@@ -49,7 +34,7 @@
  Implementation == add
    iprint(s:String) ==
      PRINC(coerce(s)@Symbol)$Lisp
-      FORCE_-OUTPUT$Lisp
+      FLUSH()$Lisp
```

section

```
--- src/algebra/manip.spad.pamphlet 2007-04-27 21:29:46.000000000 -0400
+++ /research/may07/build-improvements/src/algebra/manip.spad.pamphlet 2007-04-27 21:34:07.000000000 -0400
@@ -623,13 +623,10 @@
  -- like to combine it with a log term.
  terms :List F := [simplifyLog(term) for term in termList::List(F)]
  exprs :List F := []
-
- nterms :List F := []
-
- for term in terms repeat
-   if retractIfCan(term)@Union(FPR,"failed") case FPR then
-     exprs := cons(term, exprs)
-   else
-     nterms := cons(term, nterms)
-
- terms := nterms
+
+ for i in 1..#terms repeat
+   if retractIfCan(terms.i)@Union(FPR,"failed") case FPR then
```

```

+         exprs := cons(terms.i,exprs)
+         terms := delete!(terms,i)
      if not empty? exprs then
        foundLog := false
        i : NonNegativeInteger := 0

```

section

```

--- src/algebra/plot.spad.pamphlet 2007-04-27 21:29:46.000000000 -0400
+++ /research/may07/build-improvements/src/algebra/plot.spad.pamphlet 2007-04-27 21:34:07.000000
@@ -477,10 +477,9 @@
      myTrap: (F-> F, F) -> F
      myTrap(ff:F-> F, f:F):F ==
        s := trapNumericErrors(ff(f))$Lisp :: Union(F, "failed")
-      s case "failed" => 0
+      s case "failed" => _$NaNvalue$Lisp
      r:F:=s:F
-      r > max()$F => max()$F
-      r < min()$F => min()$F
+      r > max()$F or r < min()$F => _$NaNvalue$Lisp
      r

      plot(f:F -> F,xRange:R) ==

```

section

```

Files src/algebra/sttaylor.spad.pamphlet and /research/may07/build-improvements/src/algebra/stta
--- src/algebra/sttaylor.spad.pamphlet 2007-04-27 21:29:46.000000000 -0400
+++ /research/may07/build-improvements/src/algebra/sttaylor.spad.pamphlet 2007-04-27 21:34:07.00
@@ -434,8 +434,6 @@
      (ord := (order exquo denom(rn))) case "failed" =>
        error "***: rational power does not exist"
      co := frst x
-      if ord > 0 and rn < 0 then
-        error "***: negative power does not exist"
      (invCo := recip co) case "failed" =>
        error "*** rational power of coefficient undefined"
      -- This error message is misleading, isn't it? see sups.spad/cRationalPower
@@ -450,7 +448,6 @@

      RATPOWERS => co**rn * YS(powerrn(rn,(invCo :: A) * x,#1))
      error "*** rational power of coefficient undefined"

```



```

-      monom(1,(ord :: I) * numer(rn)) * power

      if A has Field then
        mapdiv(x,y) == delay

\subsection{domain DFLOAT DoubleFloat}
Greg Vanuxem has added some functionality to allow the user to modify
the printed format of floating point numbers. The format of the numbers
@@ -422,7 +846,7 @@
      ++ (that is, \spad{ |(r-f)/f| < b*(-n)}).

== add
-   format: String := "~A"
+   format: String := "~G"
      MER ==> Record(MANTISSA:Integer,EXPONENT:Integer)

      manexp: % -> MER
@@ -481,17 +905,16 @@
      base() = 2 => precision()
      base() = 16 => 4*precision()
      wholePart(precision()*log2(base()))::PositiveInteger
-   max()      == MOST_-POSITIVE_-DOUBLE_-FLOAT$Lisp
-   min()      == MOST_-NEGATIVE_-DOUBLE_-FLOAT$Lisp
+   max()      == MOST_-POSITIVE_-LONG_-FLOAT$Lisp
+   min()      == MOST_-NEGATIVE_-LONG_-FLOAT$Lisp
      order(a) == precision() + exponent a - 1
-   0          == FLOAT(0$Lisp,MOST_-POSITIVE_-DOUBLE_-FLOAT$Lisp)$Lisp
-   1          == FLOAT(1$Lisp,MOST_-POSITIVE_-DOUBLE_-FLOAT$Lisp)$Lisp
+   0          == FLOAT(0$Lisp,MOST_-POSITIVE_-LONG_-FLOAT$Lisp)$Lisp
+   1          == FLOAT(1$Lisp,MOST_-POSITIVE_-LONG_-FLOAT$Lisp)$Lisp
      -- rational approximation to e accurate to 23 digits
-   exp1()     == FLOAT(534625820200,MOST_-POSITIVE_-DOUBLE_-FLOAT$Lisp)$Lisp / FLOAT(196677847971,MOST_-POSITIVE_-DOUBLE_-FLOAT$Lisp)$Lisp
+   exp1()     == FLOAT(534625820200,MOST_-POSITIVE_-LONG_-FLOAT$Lisp)$Lisp / FLOAT(196677847971,MOST_-POSITIVE_-LONG_-FLOAT$Lisp)$Lisp
      pi()      == PI$Lisp
      coerce(x:%):OutputForm ==
-   x >= 0 => message(FORMAT(NIL$Lisp,format,x)$Lisp pretend String)
-   - (message(FORMAT(NIL$Lisp,format,-x)$Lisp pretend String))
+   outputForm(FORMAT(NIL$Lisp,format,x)$Lisp pretend DoubleFloat)
      convert(x:%):InputForm == convert(x pretend DoubleFloat)$InputForm
      x < y      == (x<y)$Lisp
      - x        == (-x)$Lisp
@@ -507,7 +930,7 @@
      log10 x    == checkComplex log(x)$Lisp
      x:% ** i:Integer == EXPT(x,i)$Lisp
      x:% ** y:%   == checkComplex EXPT(x,y)$Lisp
-   coerce(i:Integer):% == FLOAT(i,MOST_-POSITIVE_-DOUBLE_-FLOAT$Lisp)$Lisp
+   coerce(i:Integer):% == FLOAT(i,MOST_-POSITIVE_-LONG_-FLOAT$Lisp)$Lisp
      exp x      == EXP(x)$Lisp
      log x      == checkComplex LN(x)$Lisp
      log2 x     == checkComplex LOG2(x)$Lisp

```

```

@@ -650,7 +1073,286 @@
    x ** (n::% / d::%)

```

section

```

--- src/algebra/sets.spad.pamphlet 2007-04-27 21:29:46.000000000 -0400
+++ /research/may07/build-improvements/src/algebra/sets.spad.pamphlet 2007-04-27 21:34:07.000000
@@ -22,14 +22,14 @@
++ Keywords:
++ References:
++ Description:
-++ A set over a domain S models the usual mathematical notion of a finite set
-++ of elements from S.
+++ A set over a domain D models the usual mathematical notion of a finite set
+++ of elements from D.
++ Sets are unordered collections of distinct elements
++ (that is, order and duplication does not matter).
++ The notation \spad{set [a,b,c]} can be used to create
++ a set and the usual operations such as union and intersection are available
++ to form new sets.
-++ If S has OrderdSet, \Language{} maintains the entries in
+++ In our implementation, \Language{} maintains the entries in
++ sorted order. Specifically, the parts function returns the entries
++ as a list in ascending order and
++ the extract operation returns the maximum entry.
@@ -59,6 +59,10 @@

    map(f, s) == map_!(f,copy s)

+   map_!(f,s) ==
+       map_!(f,s)$Rep
+       removeDuplicates_! s
+
    reduce(f, s) == reduce(f, s)$Rep

    reduce(f, s, x) == reduce(f, s, x)$Rep
@@ -75,11 +79,6 @@
    max s == inspect s
    min s == (empty? s => error "Empty set"; s(minIndex s))

-   map_!(f,s) ==
-       map_!(f,s)$Rep
-       sort_!(s)$Rep
-       removeDuplicates_! s
-
    construct l ==

```

```

        zero?(n := #l) => empty()
        a := new(n, first l)
@@ -188,10 +187,6 @@
        k := inc k
        s

```

```

-      map_!(f,s) ==
-      map_!(f,s)$Rep
-      removeDuplicates_! s
-

```

section

```

--- src/algebra/polycat.spad.pamphlet 2007-04-27 21:29:46.000000000 -0400
+++ /research/may07/build-improvements/src/algebra/polycat.spad.pamphlet 2007-04-27 21:34:07.000000000 -0400
@@ -162,7 +162,7 @@
        x/r == map(#1/r,x)
        if R has IntegralDomain then
            x exquo r ==
-            zero? r => error "Division by 0"
+            -- probably not a very good definition in most special cases
            zero? x => 0
            ans:% :=0
            t:=leadingCoefficient x exquo r
@@ -568,10 +568,8 @@
        unit(s := squareFree p) * */[f.factor for f in factors s]
        content(p,v) == content univariate(p,v)
        primitivePart p ==
-        zero? p => p
-        unitNormal((p exquo content p) ::%).canonical
        primitivePart(p,v) ==
-        zero? p => p
-        unitNormal((p exquo content(p,v)) ::%).canonical
        if R has OrderedSet then
            p:% < q:% ==
@@ -608,7 +606,2117 @@
            p)$PolynomialCategoryLifting(E,VarSet,R,%,InputForm)

```

section

```

--- src/algebra/combfunc.spad.pamphlet 2007-04-27 21:29:46.000000000 -0400

```

```

+++ /research/may07/build-improvements/src/algebra/combfunc.spad.pamphlet 2007-04-27 21:34:07.00
@@ -1,5 +1,5 @@
\documentclass{article}
-\usepackage{axiom, amsmath, amsfonst}
+\usepackage{axiom}
\begin{document}
\title{\$SPAD/src/algebra combfunc.spad}
\author{Manuel Bronstein, Martin Rubey}
@@ -88,53 +88,6 @@
    binomial : (F, F) -> F
    ++ binomial(n, r) returns the number of subsets of r objects
    ++ taken among n objects, i.e. n!/(r! * (n-r)!);
-@
-
-We currently simplify binomial coefficients only for non-negative integral
-second argument, using the formula
-$$$ \binom{n}{k} = \frac{1}{k!} \prod_{i=0}^{k-1} (n-i), $$$
-except if the second argument is symbolic: in this case  $[\text{binomial}(n,n)]$  is
-simplified to one.
-
-Note that there are at least two different ways to define binomial coefficients
-for negative integral second argument. One way, particular suitable for
-combinatorics, is to set the binomial coefficient equal to zero for negative
-second argument. This is, partially, also the approach taken in
- $[\text{combinat.spad}]$ , where we find
-
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
\subsection{section}
\begin{verbatim}

    binomial(n, m) ==
      n < 0 or m < 0 or m > n => 0
      m = 0 => 1

Of course, here  $[[n]]$  and  $[[m]]$  are integers. This definition agrees with the
recurrence

$$$ \binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}. $$$

Alternatively, one can use the formula
$$$ \binom{n}{k} = \frac{\Gamma(n+1)}{\Gamma(k+1)\Gamma(n-k+1)}, $$$
and leave the case where  $k \in \mathbb{Z}$ ,  $n \in \mathbb{Z}$  and  $k \leq n < 0$ 
undefined, since the limit does not exist in this case:

Since we then have that  $n-k+1 \geq 1$ ,  $\Gamma(n-k+1)$  is finite. So it is
sufficient to consider  $\frac{\Gamma(n+1)}{\Gamma(k+1)}$ . On the one hand, we
have
$$$ \lim_{n \rightarrow \infty} \lim_{k \rightarrow \infty} \frac{\Gamma(n+1)}{\Gamma(k+1)} = 0, $$$
since for any non-integral  $n_0$ ,  $\Gamma(n_0+1)$  is finite. On the other

```

hand,

$$\lim_{k \rightarrow \infty} \frac{\Gamma(n_0+1)}{\Gamma(k_0+1)}$$
does not exist, since for non-integral k_0 , $\Gamma(k_0+1)$ is finite while $\Gamma(n_0+1)$ is unbounded.

However, since for $k \in \mathbb{Z}$, $n \in \mathbb{Z}$ and $0 < k < n$ both definitions agree, one could also combine them. This is what, for example, Mathematica does. It seems that MuPAD sets $\text{binomial}(n, n) = 1$ for all arguments $[n]$, and returns $\text{binomial}(-2, n)$ unevaluated. Provisos may help here.

```
<<package COMBF CombinatorialFunction>>=
  permutation: (F, F) -> F
    ++ permutation(n, r) returns the number of permutations of
    ++ n objects taken r at a time, i.e. n!/(n-r)!;
@@ -569,33 +522,17 @@

  if R has RetractableTo(Z) and F has Algebra(Fraction(Z)) then
    iibinom l ==
-      (s:=retractIfCan(second l)@Union(R,"failed")) case R and
-      (t:=retractIfCan(s)@Union(Z,"failed")) case Z and t>0 =>
-        ans:=1::F
-        for i in 0..t-1 repeat
-          ans:=ans*(first l - i::R::F)
-        (1/factorial t) * ans
-      (s:=retractIfCan(first l-second l)@Union(R,"failed")) case R and
-      (t:=retractIfCan(s)@Union(Z,"failed")) case Z and t>0 =>
-        ans:=1::F
-        for i in 1..t repeat
-          ans:=ans*(second l+i::R::F)
-        (1/factorial t) * ans
+      (t:=retractIfCan(s)@Union(Z,"failed")) case Z and s>0=>
+        ans:=1::F
+        for i in 1..t repeat
+          ans:=ans*(second l+i::R::F)
+        (1/factorial t) * ans
+      (r1 := retractIfCan(first l)@Union(R,"failed")) case "failed" or
+      (r2 := retractIfCan(second l)@Union(R,"failed")) case "failed"
+        => iibinom l
    binomial(r1::R, r2::R)::F

-@
-
-[[iibinom]] checks those cases in which the binomial coefficient may be
-evaluated explicitly. Note that up to [[patch--51]], the case where the second
-argument is a positive integer was not checked.(Issue~\#336) Currently, the
-naive iterative algorithm is used to calculate the coefficient, there is room
-for improvement here.
-
-<<package COMBF CombinatorialFunction>>=
```

```

-
      else
        iibinom l ==
          (r1 := retractIfCan(first l)@Union(R,"failed")) case "failed" or

```

section

```

maybe this fixes bug 100?
--- ./defintrf.spad.pamphlet 2007-04-27 21:29:46.000000000 -0400
+++ /research/may07/build-improvements/src/algebra/defintrf.spad.pamphlet 2007-04-27 21:34:07.00
@@ -182,10 +182,7 @@
      ["min"/[t.left for t in l], i.halfinf.endpoint]
      l := [u::REC for t in l | (u := refine(p, t, bounds)) case REC]
      incl? => l
-      -- greg, FIXME dirty workaround
-      -- select_!(keeprec?(i.halfinf.endpoint, #1), l)
-      ep := i.halfinf.endpoint
-      select_!(keeprec?(ep, #1), l)
+      select_!(keeprec?(i.halfinf.endpoint, #1), l)
+      error "findRealZero: should not happen"

      checkBudan(p, a, b, incl?) ==

```

section

Find who fixes bug 100 in wh-sandbox

section

Remove any bootstrap code that is not actually needed

section

fix src/algebra/exprode.spad.pamphlet

```
diffRhsK(k, g) ==
  h := univariate(g, k)
  (degree(number h) <= 1) and ground? denom h =>
    zero? coefficient(number h, 1) =>
      error "EXPRODE:diffRhsK:Division by zero"
    - coefficient(number h, 0) / coefficient(number h, 1)
  error "Improper differential equation"
```

section

rewrite trig functions using a simplest notion supplied by user "fructer"

section

fix file write bug regression result 0 of 0

section

change)help to

- (0) handle abbrev and synonym
- (1) show all topics
- (2) use min prefix
- (3) be case sensitive Eg library vs Library
- (4))help name)function to show the docs on a function
- (5))view file should fire up a dvi reader on the appropriate dvi file
- (6) create help files on the fly from axiom commands so help expands as the user works
- (8))drag should auto-create help for newly drag-and-dropped files
- (9))document should update help
- (10))weave should update help
- (11))lib should update help
- (12))co should update help

section

make sure that `savesystem` works

section

develop a "where" syntax, eg. `a_b` where `a=2` and `b=3`

section

get configure from `oaxiom`, run on mac and windows

section

```
document ThreeDimensionalViewport, e.g. to generate pixmaps:
port:=draw(5*besselJ(0,sqrt(x^2+y^2)),x=-20..20,y=-20..20)
drawStyle(port,"shade")
outlineRender(port,"on")
write(port,"temp",["pixmap"])
write(port,"temp1",["postscript"])
```

section

document how to document algebra with input files

section

apply fracas patch 32

section

(\end occurred inside a group at level 1)

```
### simple group (level 1) entered at line 17255 ({)
### bottom level
(see the transcript file for additional information)
Output written on book.dvi (1134 pages, 2650492 bytes).
```

section

```
)spool intlf.output
this file should be in REGRESS, not OUTS
```

section

```
special note:
  compiling exported coerce : String -> $
    FNAME;coerce;$4 is replaced by PARSE-NAMESTRING
```

section

```
remove member function from boot:
  Files /research/may07/wh-sandbox/src/boot/typrops.boot.pamphlet and silver/src/boot/typrops.boot.pamphlet c
```

section

package handling:

Files /research/may07/wh-sandbox/src/boot/ptyout.boot.pamphlet and silver/src/boot/ptyout.boot.

Files /research/may07/wh-sandbox/src/boot/typars.boot.pamphlet and silver/src/boot/typars.boot.

section

no idea:

Files /research/may07/wh-sandbox/src/boot/tytree1.boot.pamphlet and silver/src/boot/tytree1.bo

Files /research/may07/wh-sandbox/src/boot/btscan2.boot.pamphlet and silver/src/boot/btscan2.bo

Files /research/may07/wh-sandbox/src/boot/btpile2.boot.pamphlet and silver/src/boot/btpile2.bo

section

call graph documentation

Files /research/may07/wh-sandbox/src/boot/btinc12.boot.pamphlet and silver/src/boot/btinc12.bo

section

fix htsearch

Files /research/may07/wh-sandbox/src/hyper/pages/man0.ht and silver/src/hyper/pages/man0.ht dif

section

no idea

Files /research/may07/wh-sandbox/src/hyper/pages/MSET.ht and silver/src/hyper/pages/MSET.ht dif

section

dead file?

Only in silver/src/interp: nag-e02b.boot.pamphlet

Only in silver/src/interp: postprop.lisp.pamphlet

section

missing \$ERASE function?

section

call to (OBEY "ERASE ... is nonsense

section

document PLEQN

section

gclweb latex rewrite
needs chunk environment
needs standalone pass

section

find/fix factor bug
factor 3474749660383 = 1303 * 2666730361 = 1303 * 16927 * 157543

section

```
rewrite graphics
  into single file
  into lisp
```

section

```
rewrite hyperdoc
  into single file
  into lisp
```

section

```
split out X11 functionality
```

section

```
clean up latex files
```

section

```
check libs
  src/algebra/Makefile libcheck
```

section

```
interp.exposed seems to have disappeared.
src/interp/exposed.lsp is involved.
)lisp (libcheck "absolute path to algebra")
the interp.exposed file is missing from int/algebra
```

also fix bookvol4 documentation for this

section

```
gclweb weave
\spadcommand{a+b=c} -> (spadcommand ("a+b=c"))
```

section

run ./configure everywhere, make new Makefile stanzas

section

radix(36,37) fails

section

Jon Claerbout (Stanford University) as quoted in
 Buckheit, Jonathan B., Donoho, David L. "WaveLab and Reproducible Research"
<http://www.stat.stanford.edu/~donoho/Reports/1995/wavelab.pdf>
 "An article about computational science in a scientific publication is not the
 scholarship itself, it is merely advertising of the scholarship. The actual
 scholarship is the complete software development environment and the complete
 set of instructions which generated the figures."

section

```
initializing nrlib RPOLCAT- for RecursivePolynomialCategory&
Warning: RPOLCAT-;exactQuo has a duplicate definition in this file
```

Warning: RPOLCAT-;ZToR has a duplicate definition in this file
 Warning: RPOLCAT-;PZToPR has a duplicate definition in this file

section

Compiling POLYCAT-.lsp.
 Warning: PSETCAT-;exactQuo has a duplicate definition in this file

section

add Stephen's patch:
 width-fix.patch

section

Merge Martin's Algebra:
 find Martins algebra examples on the wiki

SUPEXPR FAMR2 NEWTON UPFS GOPT GUESSF1 UTSSOL FFFG UFPS1 GOPT0 EXPRSOL
 FFFGF RECOP GUESS GUESSINT GUESSP GUESSF

fffg ssolve recop mantepse

arxiv.org/MartinAlgebra.pdf

section

asq and databases:
 modify asq to add -csvdump switch which outputs the databases as
 comma separated values

section

in src/interp/bootlex.lisp the `|$PrettyPrint|` variable seems to allow defuns to be prettyprinted.

```
(defun print-defun (name body)
  (let* ((sp (assoc 'vmlisp::compiler-output-stream vmlisp::optionlist))
        (st (if sp (cdr sp) *standard-output*)))
    (if (and (is-console st) (symbolp name) (fboundp name)
            (not (compiled-function-p (symbol-function name))))
        (compile name))
    (when (or |$PrettyPrint| (not (is-console st)))
      (print-full body st) (force-output st))))
```

section

```
unused functions in metalex?
-; (trace skip-blanks)
-; (trace get-special-token)
-; (trace token-lookahead-type)
-; (trace make-adjustable-string)
-; (trace print-package)
-; (trace get-number-token)
-
-(trace next-META-line)
-(trace kill-comments)
-(trace kill-trailing-blanks)
-(trace get-META-token)
-(trace get-identifier-token)
-(trace get-string-token)
-(trace get-bstring-token)
-(trace make-defun)
-(trace print-fluids)
-(trace set-prefix)
-(trace print-rule)
-(trace meta-meta-error-handler)
```

section

4 arguments instead of 3 in build algebra

section

```
duplicate definition in file
PSETCAT-;exactQuo
REGSET;decompose
PFO;cmult
RPOLCAT-;exactQuo
RPOLCAT-;ZToR
RPOLCAT-;PZToPR
```

section

```
unify
make tangle accept both syntaxes
```

section

```
sbcl
port
```

section

```
STRING
.. String of cat
(|StringCategory|)    has no  hash : % -> Integer
    finalizing nrlib STRING
    Processing String for Browser database:
--->/research2/test0819/mnt/fedora5/../../src/algebra/STRING.spad-->String(): Missing Description
```


section

```

Compiling npextras.lisp.
; (DEFUN COMPILER::CMPNOTE ...) is being compiled.
;; Warning: The variable X is not used.
; (IN-PACKAGE 'BOOTTRAN ...) is being compiled.
;; Warning: The package operation (IN-PACKAGE 'BOOTTRAN :USE
                                '("LISP" "SYSTEM")) was in a bad place.

```

section

```

Compiling /research2/test0819/obj/fedora5/interp/parsing.lisp.
; (DEFUN ESCAPE-KEYWORDS ...) is being compiled.
;; The variable KEYWORDS is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN /EMBED-Q ...) is being compiled.
;; The variable /EMBEDNAMES is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN IOCLEAR ...) is being compiled.
;; Warning: The variable IN is not used.
;; Warning: The variable OUT is not used.

```

section

```

Compiling /research2/test0819/obj/fedora5/interp/clam.lisp.
; (DEFUN |consForHashLookup| ...) is being compiled.
;; The variable |$hashNode| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |mkCircularCountAlist| ...) is being compiled.
;; The variable |$reportFavoritesIfNumber| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |remHashEntriesWith0Count,fn| ...) is being compiled.
;; The variable |$hashTable| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |remHashEntriesWith0Count| ...) is being compiled.
;; The variable |remHashEntriesWith0Count,fn| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |initCache| ...) is being compiled.
;; The variable |$failed| is undefined.
;; The compiler will assume this variable is a global.

```

```

; (DEFUN |addToSlam| ...) is being compiled.
;; The variable |$mutableDomain| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |recordInstantiation1| ...) is being compiled.
;; The variable |$insideCoerceInteractive| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$instantCoerceCount| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$insideCanCoerceFrom| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$instantCanCoerceCount| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$insideEvalMmCondIfTrue| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$instantMmCondCount| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |listTruncate| ...) is being compiled.
;; The variable |$op| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/obj/fedora5/interp/slam.lsp.
; (DEFUN |reportFunctionCompilation| ...) is being compiled.
;; The variable |$compiledOpNameList| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$minivectorNames| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$compilingInputFile| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$minivectorCode| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$minivector| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$compileRecurrence| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |mkCircularAlist| ...) is being compiled.
;; The variable |$failed| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |compileRecurrenceRelation| ...) is being compiled.
;; The variable |$TriangleVariableList| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable |junk| is not used.
; (DEFUN |clearAllSlams,fn| ...) is being compiled.
;; The variable |$functorDependencyAlist| is undefined.
;; The compiler will assume this variable is a global.

```

```
; (DEFMACRO |clearSlam| ...) is being compiled.
;; Warning: The variable #:G2668 is not used.
```

section

```
Compiling /research2/test0819/obj/fedora5/interp/g-boot.lsp.
; (DEFUN |bootAbsorbSEQsAndPROGNs,flatten| ...) is being compiled.
;; The variable |$labelsForGO| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |defLetForm| ...) is being compiled.
;; The variable $LET is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |defLET1| ...) is being compiled.
;; The variable |$letGenVarCounter| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |defLET2| ...) is being compiled.
;; The variable |$inDefIS| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |defIS1| ...) is being compiled.
;; The variable |$inDefLET| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$isGenVarCounter| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
Compiling /research2/test0819/obj/fedora5/interp/c-util.lsp.
; (DEFUN |continue| ...) is being compiled.
;; The variable |$x| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$m| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$f| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |level| ...) is being compiled.
;; The variable |$level| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |displayComp| ...) is being compiled.
;; The variable |$bright| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$dim| is undefined.
;; The compiler will assume this variable is a global.
```

```

;; The variable |$op| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$s| is undefined.
;; The compiler will assume this variable is a global.
;; The variable $X is undefined.
;; The compiler will assume this variable is a global.
;; The variable $M is undefined.
;; The compiler will assume this variable is a global.
;; The variable $F is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |intersectionContour,modeCompare| ...) is being compiled.
;; The variable |$var| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |isAlmostSimple,setAssignment| ...) is being compiled.
;; The variable |$assignmentList| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |stackSemanticError| ...) is being compiled.
;; The variable |$initCapsuleErrorCount| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |stackMessage| ...) is being compiled.
;; The variable |$compErrorMessageStack| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |?modemaps| ...) is being compiled.
;; The variable |$CapsuleModemapFrame| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |traverse,traverseInner| ...) is being compiled.
;; The variable |$seen| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$notseen| is undefined.
;; The compiler will assume this variable is a global.
;; The variable SET-PA-SPEC is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/obj/fedora5/interp/g-util.lsp.
; (DEFUN |reshape| ...) is being compiled.
;; Warning: The variable |b| is not used.
; (DEFUN |update| ...) is being compiled.
;; The variable /VERSION is undefined.
;; The compiler will assume this variable is a global.
;; The variable /WSNAME is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |spadThrow| ...) is being compiled.
;; The variable |$interpOnly| is undefined.
;; The compiler will assume this variable is a global.

```

```
;; The variable |$mapName| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |semchkPropList| ...) is being compiled.
;; Warning: The variable |val| is not used.
; (DEFUN |leftTrim| ...) is being compiled.
;; The variable |$blank| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
Compiling /research2/test0819/int/interp/vmlisp.lisp.
; (DEFUN UNEMBED ...) is being compiled.
;; Warning: The variable #:G163615 is not used.
;; Warning: The variable #:G163896 is not used.
; (IN-PACKAGE 'BOOT) is being compiled.
;; Warning: The package operation (IN-PACKAGE 'BOOT) was in a bad place.
```

section

```
Compiling /research2/test0819/int/interp/buildom.clisp.
; (DEFUN |MappingEqual| ...) is being compiled.
;; Warning: The variable |dom| is not used.
; (DEFUN |MappingPrint| ...) is being compiled.
;; Warning: The variable |dom| is not used.
; (DEFUN |coerceMap2E| ...) is being compiled.
;; The variable |$testingSystem| is undefined.
;; The compiler will assume this variable is a global.
; (DEFMACRO |Enumeration| ...) is being compiled.
;; Warning: The variable #:G154733 is not used.
; (DEFUN |EnumEqual| ...) is being compiled.
;; Warning: The variable |dom| is not used.
; (DEFMACRO |RecordCategory| ...) is being compiled.
;; Warning: The variable #:G154763 is not used.
; (DEFMACRO |EnumerationCategory| ...) is being compiled.
;; Warning: The variable #:G154770 is not used.
; (DEFMACRO |UnionCategory| ...) is being compiled.
;; Warning: The variable #:G154777 is not used.
; (DEFUN |mkMappingFunList| ...) is being compiled.
;; Warning: The variable |mapForm| is not used.
```

section

```

Compiling /research2/test0819/int/interp/cattable.clisp.
; (DEFUN |showCategoryTable| ...) is being compiled.
;; The variable *HASCATEGORY-HASH* is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |displayCategoryTable| ...) is being compiled.
;; The variable |$ct| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |genCategoryTable| ...) is being compiled.
;; The variable *ANCESTORS-HASH* is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$nonLisplibDomains| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$noCategoryDomains| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |simpHasPred,simp| ...) is being compiled.
;; The variable |$hasArgs| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |simpHasAttribute| ...) is being compiled.
;; The variable |$domain| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$predvec| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |moreGeneralCategoryPredicate| ...) is being compiled.
;; Warning: The variable |id| is not used.
; (DEFUN |catPairUnion| ...) is being compiled.
;; Warning: The variable |op| is not used.
;; Warning: The variable |cat| is not used.
; (DEFUN |categoryParts,exportsOf| ...) is being compiled.
;; The variable |$constlist| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |categoryParts,build| ...) is being compiled.
;; The variable |$oplist| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$attrlist| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |categoryParts| ...) is being compiled.
;; The variable |$TriangleVariableList| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |compressSexpr| ...) is being compiled.
;; The variable |$found| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |updateCategoryTable| ...) is being compiled.
;; The variable |$newcompMode| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$NRTflag| is undefined.
;; The compiler will assume this variable is a global.

```

```

; (DEFUN |updateCategoryTableForDomain| ...) is being compiled.
;; The variable |$doNotCompressHashTableIfTrue| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |clearCategoryTable1| ...) is being compiled.
;; The variable |$cname| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable |val| is not used.

```

section

```

Compiling /research2/test0819/int/interp/clammed.clisp.
; (DEFUN |canCoerceFrom| ...) is being compiled.
;; The variable |canCoerceFrom;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |canCoerce| ...) is being compiled.
;; The variable |canCoerce;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |coerceConvertMmSelection| ...) is being compiled.
;; The variable |coerceConvertMmSelection;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |hasFileProperty| ...) is being compiled.
;; The variable |hasFileProperty;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |isValidType| ...) is being compiled.
;; The variable |isValidType;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |isValidType;| ...) is being compiled.
;; The variable |$QuotientField| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |selectMms1| ...) is being compiled.
;; The variable |selectMms1;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |resolveTT| ...) is being compiled.
;; The variable |resolveTT;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |isLegitimateMode| ...) is being compiled.
;; The variable |isLegitimateMode;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |underDomainOf| ...) is being compiled.
;; The variable |underDomainOf;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```
Compiling /research2/test0819/int/interp/compress.clisp.
; (DEFUN |minimalise,HashCheck| ...) is being compiled.
;; The variable |$hash| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
Compiling /research2/test0819/int/interp/format.clisp.
; (DEFUN |formatOperation| ...) is being compiled.
;; The variable |Undef| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$commentedOps| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |constructorName| ...) is being compiled.
;; The variable |$abbreviateTypes| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |formWrapId| ...) is being compiled.
;; The variable |$formatSigAsTeX| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |formArguments2String,fn| ...) is being compiled.
;; The variable |$OutputForm| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |formDecl2String| ...) is being compiled.
;; The variable |$permitWhere| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |formJoin1| ...) is being compiled.
;; The variable |$abbreviateJoin| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$declVar| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |tuple2String| ...) is being compiled.
;; Warning: The variable |x| is not used.
```

section

```
Compiling /research2/test0819/int/interp/g-boot.clisp.
; (DEFUN |bootAbsorbSEQsAndPROGNs,flatten| ...) is being compiled.
;; The variable |$labelsForGO| is undefined.
```



```
;; The compiler will assume this variable is a global.
; (DEFUN |defLetForm| ...) is being compiled.
;; The variable $LET is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |defLET1| ...) is being compiled.
;; The variable |$letGenVarCounter| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |defLET2| ...) is being compiled.
;; The variable |$inDefIS| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |defIS1| ...) is being compiled.
;; The variable |$inDefLET| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$isGenVarCounter| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
Compiling /research2/test0819/int/interp/g-cndata.clisp.
; (DEFUN |mkLowerCaseConTable| ...) is being compiled.
;; The variable |$lowerCaseConTb| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |attribute?| ...) is being compiled.
;; The variable *ATTRIBUTES* is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |installConstructor| ...) is being compiled.
;; Warning: The variable |type| is not used.
; (DEFUN |constructorAbbreviationErrorCheck| ...) is being compiled.
;; Warning: The variable |errmsg| is not used.
; (DEFUN |abbreviationError| ...) is being compiled.
;; Warning: The variable |abb| is not used.
; (DEFUN |condAbbrev| ...) is being compiled.
;; Warning: The variable |arg| is not used.
```

section

```
Compiling /research2/test0819/int/interp/g-error.clisp.
; (DEFUN |argumentDataError| ...) is being compiled.
;; The variable |$AlgebraError| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |errorSupervisor1| ...) is being compiled.
;; The variable |$SystemError| is undefined.
```

```

;; The compiler will assume this variable is a global.
;; The variable |$UserError| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |returnToTopLevel| ...) is being compiled.
;; The variable CHR is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |returnToReader| ...) is being compiled.
;; The variable |$ReadingFile| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |sayErrorly1| ...) is being compiled.
;; The variable |$testingSystem| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$testingErrorPrefix| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |throwMessage| ...) is being compiled.
;; The variable |$mapName| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$printMsgsToFile| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/g-timer.clisp.
; (DEFUN |peekTimedName| ...) is being compiled.
;; The variable |$timedNameStack| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |makeLongTimeString| ...) is being compiled.
;; The variable |$printTimeIfTrue| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |makeLongSpaceString| ...) is being compiled.
;; The variable |$printStorageIfTrue| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |computeElapsedTime| ...) is being compiled.
;; The variable |$oldElapsedGCTime| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$oldElapsedTime| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |computeElapsedSpace| ...) is being compiled.
;; The variable |$oldElapsedSpace| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |timedOptimization| ...) is being compiled.
;; The variable |$reportOptimization| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/g-util.clisp.
; (DEFUN |reshape| ...) is being compiled.
;; Warning: The variable |b| is not used.
; (DEFUN |update| ...) is being compiled.
;; The variable /VERSION is undefined.
;; The compiler will assume this variable is a global.
;; The variable /WSNAME is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |spadThrow| ...) is being compiled.
;; The variable |$interpOnly| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$mapName| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |semchkPropList| ...) is being compiled.
;; Warning: The variable |val| is not used.
; (DEFUN |leftTrim| ...) is being compiled.
;; The variable |$blank| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |isDomain| ...) is being compiled.
;; The variable |$domainTypeTokens| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/i-analy.clisp.
; (DEFUN |getBasicMode| ...) is being compiled.
;; The variable |$useIntegerSubdomain| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |getBasicMode0| ...) is being compiled.
;; The variable |$DoubleFloat| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |getMinimalVariableTower| ...) is being compiled.
;; The variable |$univariateDomains| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$multivariateDomains| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |polyVarList| ...) is being compiled.
;; The variable |$QuotientField| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |pushDownTargetInfo| ...) is being compiled.
;; The variable |$OutputForm| is undefined.
;; The compiler will assume this variable is a global.

```

```

;; The variable |$Any| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$immediateDataSymbol| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |bottomUp| ...) is being compiled.
;; The variable |$localVars| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |bottomUpDefaultCompile| ...) is being compiled.
;; Warning: The variable |isSub| is not used.
; (DEFUN |interpRewriteRule| ...) is being compiled.
;; Warning: The variable |expr| is not used.
; (DEFUN |bottomUpForm| ...) is being compiled.
;; The variable |$inRetract| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |bottomUpFormTuple| ...) is being compiled.
;; Warning: The variable |t| is not used.
;; Warning: The variable |argModeSetList| is not used.
; (DEFUN |printableArgModeSetList| ...) is being compiled.
;; The variable |$origArgModeSetList| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |bottomUpForm0| ...) is being compiled.
;; The variable |$HTCompanionWindowID| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/i-code.clisp.
; (DEFUN |intCodeGenCOERCE| ...) is being compiled.
;; The variable |$mapName| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$OutputForm| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |wrapMapBodyWithCatch| ...) is being compiled.
;; The variable |$mapThrowCount| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/i-coerce.clisp.
; (DEFUN |retract1| ...) is being compiled.
;; The variable |$SingleInteger| is undefined.
;; The compiler will assume this variable is a global.

```

```

; (DEFUN |retract2Specialization| ...) is being compiled.
;; The variable |$Any| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$QuotientField| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |coerceRetract| ...) is being compiled.
;; The variable |$OutputForm| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$coerceFailure| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |canCoerceTopMatching| ...) is being compiled.
;; The variable |$univariateDomains| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$multivariateDomains| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |canCoerceLocal| ...) is being compiled.
;; The variable |$CoerceTable| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |canCoerceCommute| ...) is being compiled.
;; The variable |$CommuteTable| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |canConvertByFunction| ...) is being compiled.
;; The variable |$useConvertForCoercions| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |coerceInteractive| ...) is being compiled.
;; The variable |$mapName| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |coerceInt1| ...) is being compiled.
;; The variable |$AnonymousFunction| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |getSubDomainPredicate| ...) is being compiled.
;; The variable |$superHash| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/i-coerfn.clisp.
; (DEFUN |Expr2Complex| ...) is being compiled.
;; The variable |$DoubleFloat| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN I2EI ...) is being compiled.
;; Warning: The variable |source| is not used.
;; Warning: The variable |target| is not used.
; (DEFUN I2OI ...) is being compiled.
;; Warning: The variable |source| is not used.
;; Warning: The variable |target| is not used.

```

```

; (DEFUN I2PI ...) is being compiled.
;; Warning: The variable |source| is not used.
;; Warning: The variable |target| is not used.
; (DEFUN I2NNI ...) is being compiled.
;; Warning: The variable |source| is not used.
;; Warning: The variable |target| is not used.
; (DEFUN OV2SE ...) is being compiled.
;; Warning: The variable |target| is not used.
; (DEFUN |OV2Sy| ...) is being compiled.
;; Warning: The variable |target| is not used.
; (DEFUN |Rn2F| ...) is being compiled.
;; Warning: The variable |source| is not used.
;; Warning: The variable |target| is not used.
; (DEFUN |Qf2domain| ...) is being compiled.
;; The variable |$QuotientField| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |Sy2OV| ...) is being compiled.
;; Warning: The variable |source| is not used.
; (DEFUN |Sy2Var| ...) is being compiled.
;; Warning: The variable |source| is not used.
; (DEFUN |commuteComplex| ...) is being compiled.
;; Warning: The variable |source| is not used.
; (DEFUN |commuteQuaternion| ...) is being compiled.
;; Warning: The variable |source| is not used.
; (DEFUN |commuteFraction| ...) is being compiled.
;; Warning: The variable |source| is not used.
; (DEFUN |commuteSparseUnivariatePolynomial| ...) is being compiled.
;; Warning: The variable |source| is not used.

```

section

```

Compiling /research2/test0819/int/interp/i-eval.clisp.
; (DEFUN |evaluateType1| ...) is being compiled.
;; Warning: The variable |x| is not used.
;; Warning: The variable |m| is not used.
; (DEFUN |throwEvalTypeMsg| ...) is being compiled.
;; The variable |$noEvalTypeMsg| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |evalForm| ...) is being compiled.
;; The variable |$localVars| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$NRTmonitorIfTrue| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |getMappingArgValue| ...) is being compiled.
;; Warning: The variable |t| is not used.
; (DEFUN |getArgValueComp| ...) is being compiled.

```

```
;; The variable |$mapName| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |failCheck| ...) is being compiled.
;; The variable |$coerceFailure| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
Compiling /research2/test0819/int/interp/i-funsel.clisp.
; (DEFUN |selectMms| ...) is being compiled.
;; The variable |$AnonymousFunction| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |defaultTarget| ...) is being compiled.
;; The variable |$DoubleFloat| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$Any| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$QuotientField| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |defaultTargetFE| ...) is being compiled.
;; The variable |$FunctionalExpression| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |getLocalMms| ...) is being compiled.
;; The variable |$useIntegerSubdomain| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$Coerce| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |mmCost0| ...) is being compiled.
;; Warning: The variable |name| is not used.
; (DEFUN |getFunctionFromDomain| ...) is being compiled.
;; The variable |$nonLisplibDomains| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |allOrMatchingMms| ...) is being compiled.
;; Warning: The variable |tar| is not used.
; (DEFUN |findFunctionInDomain1| ...) is being compiled.
;; The variable |$SubDom| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |matchMmSig| ...) is being compiled.
;; The variable |$RTC| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |selectMmsGen,exact?| ...) is being compiled.
;; Warning: The variable |tar| is not used.
; (DEFUN |selectMmsGen,matchMms| ...) is being compiled.
;; The variable |$Subst| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |matchTypes| ...) is being compiled.
```

```

;; The variable |$SymbolType| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable |t1| is not used.
; (DEFUN |evalMmFreeFunction| ...) is being compiled.
;; Warning: The variable |op| is not used.
;; Warning: The variable |mmC| is not used.
; (DEFUN |evalMmCat| ...) is being compiled.
;; Warning: The variable |sig| is not used.
; (DEFUN |evalMmCat1| ...) is being compiled.
;; The variable |$hope| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |hasCateSpecialNew| ...) is being compiled.
;; The variable |$ComplexInteger| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |hasCaty| ...) is being compiled.
;; The variable |$domPvar| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/i-intern.clisp.
; (DEFUN |mkAtreeWithSrcPos| ...) is being compiled.
;; The variable |$useParserSrcPos| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |transferSrcPosInfo| ...) is being compiled.
;; The variable |$transferParserSrcPos| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |mkAtree1| ...) is being compiled.
;; The variable |$immediateDataSymbol| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |transferPropsToNode,transfer| ...) is being compiled.
;; The variable |$localVars| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |atree2Tree1| ...) is being compiled.
;; The variable |$OutputForm| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$mapName| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |get1| ...) is being compiled.
;; The variable |$CapsuleModemapFrame| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |get2| ...) is being compiled.
;; Warning: The variable |e| is not used.
; (DEFUN |objEnv| ...) is being compiled.
;; Warning: The variable |obj| is not used.

```


section

```

Compiling /research2/test0819/int/interp/i-map.clisp.
; (DEFUN |makeInternalMapName| ...) is being compiled.
;; The variable |$interpreterFrameRing| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$specialMapNameSuffix| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |addDefMap| ...) is being compiled.
;; The variable |$mapName| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |addMap| ...) is being compiled.
;; Warning: The variable |x| is not used.
; (DEFUN |sayDroppingFunctions| ...) is being compiled.
;; The variable |$displayDroppedMap| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |mkFormalArg| ...) is being compiled.
;; The variable |$s1| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |clearDependencies| ...) is being compiled.
;; Warning: The variable |clearLocalModemapsIfTrue| is not used.
; (DEFUN |displayRule| ...) is being compiled.
;; Warning: The variable |op| is not used.
; (DEFUN |outputFormat| ...) is being compiled.
;; The variable |$OutputForm| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |analyzeMap| ...) is being compiled.
;; The variable |$analyzingMapList| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |analyzeMap0| ...) is being compiled.
;; The variable |$mapList| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |compFailure| ...) is being compiled.
;; The variable |$useCoerceOrCroak| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$reportInterpOnly| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |interpMap| ...) is being compiled.
;; The variable |$timedNameStack| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |analyzeDeclaredMap| ...) is being compiled.
;; The variable |$mapTarget| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |genMapCode| ...) is being compiled.
;; The variable |$whereCacheList| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$localVars| is undefined.
;; The compiler will assume this variable is a global.

```

```

; (DEFUN |compileCoerceMap| ...) is being compiled.
;; The variable |$minivectorNames| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$compilingInputFile| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$minivectorCode| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$minivector| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |analyzeNonRecursiveMap| ...) is being compiled.
;; The variable |$mapThrowCount| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |mkLocalVar| ...) is being compiled.
;; Warning: The variable |op| is not used.
; (DEFUN |isFreeVar| ...) is being compiled.
;; The variable |$freeVars| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |mkFreeVar| ...) is being compiled.
;; Warning: The variable |op| is not used.

```

section

```

Compiling /research2/test0819/int/interp/i-output.clisp.
; (DEFUN |specialChar| ...) is being compiled.
;; The variable |$specialCharacterAlist| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$specialCharacters| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |appChar| ...) is being compiled.
;; The variable |$highlightDelta| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |outputTran| ...) is being compiled.
;; The variable |$DoubleFloat| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$fractionDisplayType| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |outputMapTran| ...) is being compiled.
;; The variable |$op| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |mkSuperSub| ...) is being compiled.
;; The variable |$linearFormatScripts| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |appSum| ...) is being compiled.
;; Warning: The variable |arg| is not used.
; (DEFUN |isRationalNumber| ...) is being compiled.
;; Warning: The variable |x| is not used.

```

```

; (DEFUN |widthSC| ...) is being compiled.
;; Warning: The variable |u| is not used.
; (DEFUN |maprin| ...) is being compiled.
;; The variable |$demoFlag| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |maprinChk| ...) is being compiled.
;; The variable |$MatrixList| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$collectOutput| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |PushMatrix| ...) is being compiled.
;; The variable |$MatrixCount| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |overbarApp| ...) is being compiled.
;; The variable UNDERBAR is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |concatTrouble,fixUp| ...) is being compiled.
;; The variable |$addBlankIfTrue| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |formulaFormat| ...) is being compiled.
;; The variable |$OutputForm| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |texFormat| ...) is being compiled.
;; The variable |$texOutputStream| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |mathmlFormat| ...) is being compiled.
;; The variable |$mathmlOutputStream| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |output| ...) is being compiled.
;; The variable |$mathmlFormat| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |outputNumber| ...) is being compiled.
;; The variable |$outputLines| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |charyTop| ...) is being compiled.
;; The variable |$testOutputLineFlag| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$testOutputLineList| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |charySemiColon| ...) is being compiled.
;; Warning: The variable |v| is not used.
; (DEFUN |charyMinus| ...) is being compiled.
;; Warning: The variable |u| is not used.
; (DEFUN |charyBinary| ...) is being compiled.
;; Warning: The variable |u| is not used.
; (DEFUN |charyEquatnum| ...) is being compiled.
;; Warning: The variable |v| is not used.
; (DEFUN |eq0| ...) is being compiled.
;; Warning: The variable |u| is not used.
; (DEFUN |nothingWidth| ...) is being compiled.

```

```

;; Warning: The variable |x| is not used.
; (DEFUN |nothingSuper| ...) is being compiled.
;; Warning: The variable |x| is not used.
; (DEFUN |nothingSub| ...) is being compiled.
;; Warning: The variable |x| is not used.
; (DEFUN |nothingApp| ...) is being compiled.
;; Warning: The variable |u| is not used.
;; Warning: The variable |x| is not used.
;; Warning: The variable |y| is not used.
; (DEFUN |remWidth| ...) is being compiled.
;; The variable |remWidth| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |transcomparg| ...) is being compiled.
;; The variable STANDARGLIST is undefined.
;; The compiler will assume this variable is a global.
;; The variable FRLIS* is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |maPrin| ...) is being compiled.
;; The variable |$runTestFlag| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$mkTestFlag| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$mkTestOutputStack| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/i-resolv.clisp.
; (DEFUN |resolveTypeListAny| ...) is being compiled.
;; The variable |$Any| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |acceptableTypesToResolve1| ...) is being compiled.
;; The variable |$DoubleFloat| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$QuotientField| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |resolveTTRed1| ...) is being compiled.
;; The variable |$Res| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |getConditionalCategoryOfType1| ...) is being compiled.
;; The variable |$TriangleVariableList| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |resolveTM1| ...) is being compiled.
;; The variable |$Subst| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$Coerce| is undefined.

```

```

;; The compiler will assume this variable is a global.
; (DEFUN |resolveTMTaggedUnion| ...) is being compiled.
;; Warning: The variable |t| is not used.
; (DEFUN |resolveTMSpecial| ...) is being compiled.
;; The variable |$AnonymousFunction| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |resolveTMEq2| ...) is being compiled.
;; Warning: The variable |cm| is not used.
; (DEFUN |resolveTMRed| ...) is being compiled.
;; The variable |$ResMode| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |constructM| ...) is being compiled.
;; The variable |$FunctionalExpression| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/i-spec1.clisp.
; (DEFUN |evalUntargetedADEF| ...) is being compiled.
;; The variable |$AnonymousFunction| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable |types| is not used.
; (DEFUN |evalTargetedADEF| ...) is being compiled.
;; The variable |$anonymousMapCounter| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |mkInterpTargetedADEF| ...) is being compiled.
;; The variable |$mapName| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |compileADEFBody| ...) is being compiled.
;; The variable |$compiledOpNameList| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$minivectorNames| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$compilingInputFile| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$minivectorCode| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$minivector| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$freeVariables| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$boundVariables| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |upAlgExtension| ...) is being compiled.
;; The variable |$printTypeIfTrue| is undefined.
;; The compiler will assume this variable is a global.

```

```

; (DEFUN |upTARGET| ...) is being compiled.
;; The variable |$localVars| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |upCOLLECT| ...) is being compiled.
;; The variable |$compilingLoop| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |upCOLLECT1| ...) is being compiled.
;; The variable |$interpOnly| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |falseFun| ...) is being compiled.
;; Warning: The variable |x| is not used.
; (DEFUN |interpIter| ...) is being compiled.
;; The variable |$indexVars| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$indexTypes| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |interpCOLLECTbodyIter| ...) is being compiled.
;; The variable |$collectTypeList| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |collectOneStream| ...) is being compiled.
;; Warning: The variable |t| is not used.
; (DEFUN |collectSeveralStreams| ...) is being compiled.
;; Warning: The variable |t| is not used.
; (DEFUN |mkAndApplyZippedPredicates| ...) is being compiled.
;; The variable |$indexList| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |mkIterZippedFun| ...) is being compiled.
;; The variable |$index| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable |indexList| is not used.
; (DEFUN |replaceSymbols| ...) is being compiled.
;; The variable |$declaredMode| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |upNullList| ...) is being compiled.
;; Warning: The variable |l| is not used.
; (DEFUN |replaceSharps| ...) is being compiled.
;; The variable |$TriangleVariableList| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |containsPolynomial| ...) is being compiled.
;; The variable |$univariateDomains| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$multivariateDomains| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/i-spec2.clisp.
; (DEFUN |upDollar| ...) is being compiled.
;; The variable |$localVars| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$immediateDataSymbol| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$OutputForm| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |uperror| ...) is being compiled.
;; The variable |$mapName| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |upfreeWithType| ...) is being compiled.
;; Warning: The variable |type| is not used.
; (DEFUN |uplocalWithType| ...) is being compiled.
;; Warning: The variable |type| is not used.
; (DEFUN |compileIF| ...) is being compiled.
;; The variable |$declaredMode| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable |cond| is not used.
; (DEFUN |evalIF| ...) is being compiled.
;; The variable |$lastLineInSEQ| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |evalis| ...) is being compiled.
;; The variable |$opIsIs| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |evalIsPredicate| ...) is being compiled.
;; Warning: The variable |mode| is not used.
; (DEFUN |isPatMatch| ...) is being compiled.
;; The variable |$subs| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |upiterate| ...) is being compiled.
;; The variable |$repeatBodyLabel| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$iterateCount| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |upbreak| ...) is being compiled.
;; The variable |$repeatLabel| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$breakCount| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |getInterpMacroNames| ...) is being compiled.
;; The variable |$InterpreterMacroAlist| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |isInterpMacro| ...) is being compiled.
;; The variable |$specialOps| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |upREPEAT| ...) is being compiled.
;; The variable |$compilingLoop| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |upREPEAT1| ...) is being compiled.

```

```

;; The variable |$interpOnly| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |upreturn| ...) is being compiled.
;; The variable |$mapTarget| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$mapThrowCount| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |upNullTuple| ...) is being compiled.
;; Warning: The variable |l| is not used.

```

section

```

Compiling /research2/test0819/int/interp/i-util.clisp.
; (DEFUN MKPROMPT ...) is being compiled.
;; The variable |$interpreterFrameName| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |Zeros| ...) is being compiled.
;; The variable |$ZeroVecCache| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |variableNumber| ...) is being compiled.
;; The variable |$variableNumberAlist| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |newType?| ...) is being compiled.
;; Warning: The variable |t| is not used.
; (DEFUN |Undef| ...) is being compiled.
;; The variable |Undef| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |addModemap| ...) is being compiled.
;; The variable |$CapsuleModemapFrame| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |orderUnionEntries| ...) is being compiled.
;; The variable |$newCompilerUnionFlag| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/lisplib.clisp.
; (DEFUN |hasFilePropertyNoCache| ...) is being compiled.
;; Warning: The variable |abbrev| is not used.
; (DEFUN |loadLibNoUpdate| ...) is being compiled.
;; Warning: The variable |libName| is not used.
; (DEFUN |autoLoad| ...) is being compiled.

```



```

;; Warning: The variable |abb| is not used.
; (DEFUN |compileConstructorLib| ...) is being compiled.
;; The variable /FN is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |compConLib1| ...) is being compiled.
;; The variable /EDITFILE is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$listingDirectory| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |compDefineLisplib| ...) is being compiled.
;; The variable |$compileDocumentation| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$libraryDirectory| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$newConlist| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |initializeLisplib| ...) is being compiled.
;; The variable ERRORS is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$lisplibAbbreviation| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$lisplibAncestors| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$lisplibOpAlist| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$lisplibSuperDomain| is undefined.
;; The compiler will assume this variable is a global.
;; The variable /MAJOR-VERSION is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |finalizeLisplib| ...) is being compiled.
;; The variable |$lisplibCategory| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$lisplibAttributes| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$lisplibPredicates| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$lisplibParents| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$lisplibSlot1| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$profileCompiler| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |transformOperationAlist| ...) is being compiled.
;; The variable |$functionLocations| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |getSlotFromDomain| ...) is being compiled.
;; The variable |Undef| is undefined.
;; The compiler will assume this variable is a global.

```

section

```
Compiling /research2/test0819/int/interp/match.clisp.
; (DEFUN |patternCheck,subWild| ...) is being compiled.
;; The variable |$oldWild| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$wildCard| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
Compiling /research2/test0819/int/interp/msg.clisp.
; (DEFUN |putDatabaseStuff| ...) is being compiled.
;; The variable |aL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |posPointers| ...) is being compiled.
;; The variable |getMsgFTTag| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |setMsgCatlessAttr| ...) is being compiled.
;; The variable |catless| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
Compiling /research2/test0819/int/interp/msgdb.clisp.
; (DEFUN |substituteSegmentedMsg| ...) is being compiled.
;; The variable |$texFormatting| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |noBlankBeforeP| ...) is being compiled.
;; The variable |$msgdbNoBlanksBeforeGroup| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$msgdbListPrims| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |noBlankAfterP| ...) is being compiled.
;; The variable |$msgdbNoBlanksAfterGroup| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |throwPatternMsg| ...) is being compiled.
;; The variable |$testingSystem| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$testingErrorPrefix| is undefined.
```

```

;; The compiler will assume this variable is a global.
; (DEFUN |sayKeyedMsgLocal| ...) is being compiled.
;; The variable |$displayMsgNumber| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$printMsgsToFile| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |returnStLFromKey| ...) is being compiled.
;; The variable |$msgDatabaseName| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |keyedMsgCompFailure| ...) is being compiled.
;; The variable |$useCoerceOrCroak| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$Coerce| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$reportInterpOnly| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |throwKeyedMsgCannotCoerceWithValue| ...) is being compiled.
;; The variable |$OutputForm| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |spadStartUpMsgs| ...) is being compiled.
;; The variable *BUILD-VERSION* is undefined.
;; The compiler will assume this variable is a global.
;; The variable *YEARWEEK* is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$opSysName| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$msgAlist| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |brightPrint0| ...) is being compiled.
;; The variable $MARG is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |escapeSpecialChars| ...) is being compiled.
;; The variable |$htCharAlist| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$htSpecialChars| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/newfort.clisp.
; (DEFUN |newFortranTempVar| ...) is being compiled.
;; The variable |$exp2FortTempVarIndex| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$defaultFortranType| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |exp2FortOptimize| ...) is being compiled.

```

```

;; The variable |$fortranOptimizationLevel| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |beenHere| ...) is being compiled.
;; The variable |$fortCsList| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |exp2FortOptimizeCS1,pushCsStacks| ...) is being compiled.
;; The variable |$fortCsExprStack| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$fortCsFuncStack| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |exp2FortOptimizeCS1,popCsStacks| ...) is being compiled.
;; Warning: The variable |x| is not used.
; (DEFUN |exp2FortOptimizeCS1| ...) is being compiled.
;; The variable |$fortCsHash| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |exp2FortOptimizeArray| ...) is being compiled.
;; The variable |$fortName| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$exprStack| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |fortran2Lines1| ...) is being compiled.
;; The variable |$fortIndent| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$fortLength| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |fortError1| ...) is being compiled.
;; The variable |$fortError| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |exp2FortSpecial| ...) is being compiled.
;; The variable |$fortranArrayStartingIndex| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |fortranifyIntrinsicFunctionName| ...) is being compiled.
;; The variable |$useIntrinsicFunctions| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$intrinsic| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$fortranPrecision| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |indentFortLevel| ...) is being compiled.
;; The variable |$maximumFortranExpressionLength| is undefined.
;; The compiler will assume this variable is a global.
; (DEFMACRO |changeExprLength| ...) is being compiled.
;; Warning: The variable #:G155164 is not used.
; (DEFMACRO |nameLen| ...) is being compiled.
;; Warning: The variable #:G155334 is not used.
; (DEFUN |currentSP| ...) is being compiled.
;; The variable |$currentSubprogram| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |fortPre1| ...) is being compiled.
;; The variable |$fortInts2Floats| is undefined.

```

```
;; The compiler will assume this variable is a global.
; (DEFUN |segment| ...) is being compiled.
;; The variable |$fortranSegment| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
Compiling /research2/test0819/int/interp/nrunfast.clisp.
; (DEFUN |initNewWorld| ...) is being compiled.
;; The variable |$NRTflag| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$NRTvec| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$NRTmakeCompactDirect| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$NRTquick| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$NRTmakeShortDirect| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$newWorld| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$monitorNewWorld| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$NRTmonitorIfTrue| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$doNotCompressHashTableIfTrue| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |replaceGoGetSlot| ...) is being compiled.
;; The variable |$returnNowhereFromGoGet| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |newLookupInTable| ...) is being compiled.
;; The variable |$lookupDefaults| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |opIsHasCat| ...) is being compiled.
;; The variable |$hasCatOpHash| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |hashNewLookupInTable| ...) is being compiled.
;; The variable |$hashOp1| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$hashOp0| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |hashNewLookupInCategories| ...) is being compiled.
;; The variable |$Slot1DataBase| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |getNewDefaultPackage| ...) is being compiled.
;; The variable |packageVec| is undefined.
```

```

;; The compiler will assume this variable is a global.
;; The variable |catVec| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable |dom| is not used.
;; Warning: The variable |dollar| is not used.
; (DEFUN |lazyMatchArg2| ...) is being compiled.
;; The variable |$isDefaultingPackage| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |newExpandGoGetTypeSlot| ...) is being compiled.
;; Warning: The variable |dollar| is not used.
; (DEFUN |sigDomainVal| ...) is being compiled.
;; Warning: The variable |dollar| is not used.

```

section

```

Compiling /research2/test0819/int/interp/nrungo.clisp.
; (DEFUN |basicLookup| ...) is being compiled.
;; The variable |$hashOp0| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$hashOp1| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$hashOpApply| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$hashOpSet| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$hashSeg| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |basicLookupCheckDefaults| ...) is being compiled.
;; The variable |$lookupDefaults| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable |domain| is not used.
; (DEFUN |lookupInTable| ...) is being compiled.
;; The variable |$predVector| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |lookupInCategories| ...) is being compiled.
;; The variable |$Slot1DataBase| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NRTcompileEvalForm| ...) is being compiled.
;; The variable |$insideCompileBodyIfTrue| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NRTgetMinivectorIndex| ...) is being compiled.
;; The variable |$minivector| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$compilingInputFile| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$minivectorCode| is undefined.

```

```
;; The compiler will assume this variable is a global.
; (DEFUN |mkDiffAssoc| ...) is being compiled.
;; The variable |$TriangleVariableList| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
Compiling /research2/test0819/int/interp/nrunopt.clisp.
; (DEFUN |getInfovecCode| ...) is being compiled.
;; The variable |$template| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$NRTslot1Info| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$lookupFunction| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |makeDomainTemplate| ...) is being compiled.
;; The variable |$byteVec| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |makeGoGetSlot| ...) is being compiled.
;; The variable |$byteAddress| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |makeCompactDirect| ...) is being compiled.
;; The variable |$NRTslot1PredicateList| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |makeCompactDirect1,fn| ...) is being compiled.
;; The variable |$isOpPackageName| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |makeCompactDirect1| ...) is being compiled.
;; The variable |$byteVecAcc| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |orderBySubsumption| ...) is being compiled.
;; The variable |$op| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NRTgenInitialAttributeAlist| ...) is being compiled.
;; The variable |$lisplibAttributes| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$pairlis| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NRTgenFinalAttributeAlist| ...) is being compiled.
;; The variable |$NRTattributeAlist| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |makePredicateBitVector| ...) is being compiled.
;; The variable |$insideCategoryPackageIfTrue| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$categoryPredicateList| is undefined.
;; The compiler will assume this variable is a global.
```

```

;; The variable |$predGensymAlist| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$lisplibPredicates| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |augmentPredCode| ...) is being compiled.
;; The variable $ is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NRTmakeCategoryAlist| ...) is being compiled.
;; The variable |$uncondAlist| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$condAlist| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NRTcatCompare| ...) is being compiled.
;; The variable |$levelAlist| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |depthAssoc| ...) is being compiled.
;; The variable |$depthAssocCache| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |getCodeVector| ...) is being compiled.
;; The variable |$infovec| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |dcOpTable| ...) is being compiled.
;; Warning: The variable |i| is not used.
; (DEFUN |dcOpPrint| ...) is being compiled.
;; The variable |$predvec| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |extendsCategory| ...) is being compiled.
;; The variable |$why| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |templateVal| ...) is being compiled.
;; Warning: The variable |domform| is not used.

```

section

```

Compiling /research2/test0819/int/interp/record.clisp.
; (DEFUN |inputFile2RecordFile| ...) is being compiled.
;; The variable |$testStream| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |recordAndPrintTest| ...) is being compiled.
;; The variable |$mkTestOutputStack| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$mkTestInputStack| is undefined.
;; The compiler will assume this variable is a global.

```


section

```

Compiling /research2/test0819/int/interp/rulesets.clisp.
; (DEFUN |createResolveTTRules| ...) is being compiled.
;; The variable |$mpolyTTRules| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$Res| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$nameList| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$abList| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$generalTTRules| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |createResolveTMRules| ...) is being compiled.
;; The variable |$mpolyTMRules| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$ResMode| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$generalTMRules| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |createTypeEquivRules| ...) is being compiled.
;; The variable |$TypeEQ| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$TypeEqui| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |initializeRuleSets| ...) is being compiled.
;; The variable |$newResolveAbbreviations| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$ruleSetsInitialized| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/server.clisp.
; (DEFUN |serverReadLine| ...) is being compiled.
;; The variable |$EndServerSession| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$NeedToSignalSessionManager| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$SessionManager| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$EndOfOutput| is undefined.
;; The compiler will assume this variable is a global.

```

```

;; The variable |$CallInterp| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$CreateFrame| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$frameAlist| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$frameNumber| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$currentFrameNum| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$CreateFrameAnswer| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$SwitchFrames| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$EndSession| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$LispCommand| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$sockBufferLength| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$MenuServer| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$QuietSpadCommand| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$SpadCommand| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$NonSmanSession| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$KillLispSystem| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |parseAndInterpret| ...) is being compiled.
;; The variable |$useNewParser| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |serverLoop| ...) is being compiled.
;; The variable |$Prompt| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/sfsfun.clisp.
; (DEFUN |logH| ...) is being compiled.
;; Warning: The variable |z| is not used.
; (DEFUN |BesselIBackRecur| ...) is being compiled.
;; Warning: The variable |largev| is not used.

```

section

```

Compiling /research2/test0819/int/interp/slam.clisp.
; (DEFUN |reportFunctionCompilation| ...) is being compiled.
;; The variable |$compiledOpNameList| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$minivectorNames| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$compilingInputFile| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$minivectorCode| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$minivector| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$compileRecurrence| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |mkCircularAlist| ...) is being compiled.
;; The variable |$failed| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |compileRecurrenceRelation| ...) is being compiled.
;; The variable |$TriangleVariableList| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable |junk| is not used.
; (DEFUN |clearAllSlams,fn| ...) is being compiled.
;; The variable |$functorDependencyAlist| is undefined.
;; The compiler will assume this variable is a global.
; (DEFMACRO |clearSlam| ...) is being compiled.
;; Warning: The variable #:G154929 is not used.

```

section

```

Compiling /research2/test0819/int/interp/template.clisp.
; (DEFUN |putPredHash| ...) is being compiled.
;; The variable |$predHash| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$predVectorFrontier| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$predVector| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |mkSigPredVectors| ...) is being compiled.
;; The variable |$consDB| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NRTdescendCodeTran| ...) is being compiled.
;; The variable |$template| is undefined.

```

```

;; The compiler will assume this variable is a global.
; (DEFUN |measure| ...) is being compiled.
;; The variable |SparseUnivariatePolynomial| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |SparseUnivariatePolynomial;opDirect| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |SparseUnivariatePolynomial;template| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |measureCommon,fn| ...) is being compiled.
;; The variable HASH is undefined.
;; The compiler will assume this variable is a global.
;; The variable TABLE is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$table| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/fortcall.clisp.
; (DEFUN |makeFort1| ...) is being compiled.
;; The variable |$fortranDirectory| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |makeFortranFun| ...) is being compiled.
;; The variable |$fortranLibraries| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |writeCFile| ...) is being compiled.
;; The variable |$addUnderscoreToFortranNames| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |makeAspGenerators1| ...) is being compiled.
;; Warning: The variable |type| is not used.
; (DEFUN |makeCompilation| ...) is being compiled.
;; The variable |$fortranCompilerName| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |spadify| ...) is being compiled.
;; The variable RESULTS is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |prepareResults,defaultValue| ...) is being compiled.
;; The variable |shortZero| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |longZero| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |generateDataName| ...) is being compiled.
;; The variable |$fortranTmpDir| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |nagCall| ...) is being compiled.
;; The variable |$nagMessages| is undefined.

```

```
;; The compiler will assume this variable is a global.
;; The variable |$nagHost| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$fortPersistence| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
Compiling /research2/test0819/int/interp/util.lisp.
; (DEFUN MAKE-DIRECTORY ...) is being compiled.
;; The variable $SPADROOT is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN INTERP-MAKE-DIRECTORY ...) is being compiled.
;; The variable $CURRENT-DIRECTORY is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN TRANSLATE ...) is being compiled.
;; Warning: The variable FN is not used.
; (DEFUN MAKE-DEPSYS ...) is being compiled.
;; Warning: The variable LSP is not used.
;; Warning: The variable SRC is not used.
;; Warning: The variable INT is not used.
;; Warning: The variable MNT is not used.
; (IN-PACKAGE "BOOTTRAN") is being compiled.
;; Warning: The package operation (IN-PACKAGE "BOOTTRAN") was in a bad place.
; (DEFUN BOOTTOCL ...) is being compiled.
;; Warning: The variable FN is not used.
; (IN-PACKAGE "BOOT") is being compiled.
;; Warning: The package operation (IN-PACKAGE "BOOT") was in a bad place.
; (DEFUN BUILD-INTERPSYS ...) is being compiled.
;; The variable COMP-FUNCTIONS is undefined.
;; The compiler will assume this variable is a global.
;; The variable PARSE-FUNCTIONS is undefined.
;; The compiler will assume this variable is a global.
;; The variable BROWSE-FUNCTIONS is undefined.
;; The compiler will assume this variable is a global.
;; The variable TRANSLATE-FUNCTIONS is undefined.
;; The compiler will assume this variable is a global.
;; The variable NAGBR-FUNCTIONS is undefined.
;; The compiler will assume this variable is a global.
;; The variable ASAUTO-FUNCTIONS is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable LSP is not used.
;; Warning: The variable SRC is not used.
;; Warning: The variable INT is not used.
;; Warning: The variable OBJ is not used.
;; Warning: The variable MNT is not used.
```

```

;; Warning: The variable SYS is not used.
; (IN-PACKAGE "COMPILER") is being compiled.
;; Warning: The package operation (IN-PACKAGE "COMPILER") was in a bad place.
; (IN-PACKAGE "BOOT") is being compiled.
;; Warning: The package operation (IN-PACKAGE "BOOT") was in a bad place.
; (DEFUN MAKELIB ...) is being compiled.
;; Warning: The variable NOOPTIMIZE is not used.
; (DEFUN LIBCHECK ...) is being compiled.
;; The variable ABBREVS is undefined.
;; The compiler will assume this variable is a global.
;; The variable CONSTRUCTORS is undefined.
;; The compiler will assume this variable is a global.
;; The variable SRCABBREVS is undefined.
;; The compiler will assume this variable is a global.
;; The variable SRCCONSTRUCTORS is undefined.
;; The compiler will assume this variable is a global.
;; The variable SPADS is undefined.
;; The compiler will assume this variable is a global.
;; The variable SHORT is undefined.
;; The compiler will assume this variable is a global.
;; The variable LONG is undefined.
;; The compiler will assume this variable is a global.
;; The variable POINT is undefined.
;; The compiler will assume this variable is a global.
;; The variable MARK is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable END is not used.
;; Warning: The variable START is not used.
;; Warning: The variable IN is not used.
;; Warning: The variable INTERP is not used.
;; Warning: The variable END is not used.
;; Warning: The variable START is not used.
;; Warning: The variable IN is not used.
;; Warning: The variable INTERP is not used.

```

section

```

Compiling /research2/test0819/int/interp/c-util.clisp.
; (DEFUN |continue| ...) is being compiled.
;; The variable |$x| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$m| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$f| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |level| ...) is being compiled.

```

```

;; The variable |$level| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |displayComp| ...) is being compiled.
;; The variable |$bright| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$dim| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$op| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$s| is undefined.
;; The compiler will assume this variable is a global.
;; The variable $X is undefined.
;; The compiler will assume this variable is a global.
;; The variable $M is undefined.
;; The compiler will assume this variable is a global.
;; The variable $F is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |intersectionContour,modeCompare| ...) is being compiled.
;; The variable |$var| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |isAlmostSimple,setAssignment| ...) is being compiled.
;; The variable |$assignmentList| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |stackSemanticError| ...) is being compiled.
;; The variable |$initCapsuleErrorCount| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |stackMessage| ...) is being compiled.
;; The variable |$compErrorMessageStack| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |?modemaps| ...) is being compiled.
;; The variable |$CapsuleModemapFrame| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |traverse,traverseInner| ...) is being compiled.
;; The variable |$seen| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$notseen| is undefined.
;; The compiler will assume this variable is a global.
;; The variable SET-PA-SPEC is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/profile.clisp.
; (DEFUN |profileWrite| ...) is being compiled.
;; The variable |$profileAlist| is undefined.
;; The compiler will assume this variable is a global.

```

```
; (DEFUN |profileTran| ...) is being compiled.
;; The variable |$profileHash| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |profileRecord| ...) is being compiled.
;; The variable |$op| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$signatureOfForm| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
Compiling /research2/test0819/int/interp/category.clisp.
; (DEFUN |SigListUnion| ...) is being compiled.
;; The variable |$NewCatVec| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable |e| is not used.
; (DEFUN |SourceLevelSubset| ...) is being compiled.
;; The variable |$noSubsumption| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$noSubsets| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
Compiling /research2/test0819/int/interp/funcator.clisp.
; (DEFUN |DomainPrint1| ...) is being compiled.
;; The variable |$Sublis| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$WhereList| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |DPname| ...) is being compiled.
;; The variable |$WhereCounter| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NewbFVectorCopy| ...) is being compiled.
;; The variable |Undef| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |setVector12| ...) is being compiled.
;; The variable |$extraParms| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$domainShell| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |setVector4| ...) is being compiled.
```



```

;; The variable |$HackSlot4| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$getDomainCode| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |setVector4Onecat,Supplementaries| ...) is being compiled.
;; The variable |$supplementaries| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |setVector4part3| ...) is being compiled.
;; The variable |$epilogue| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |mkVectorWithDeferral| ...) is being compiled.
;; The variable |$ConstantAssignments| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |DescendCodeAdd1| ...) is being compiled.
;; The variable |$addFormLhs| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |DescendCode| ...) is being compiled.
;; The variable |$packagesUsed| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$locals| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |ProcessCond| ...) is being compiled.
;; The variable |$pairlis| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$NRtslot1PredicateList| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable |viewassoc| is not used.
; (DEFUN |SetFunctionSlots| ...) is being compiled.
;; The variable |$catvecList| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$MissingFunctionInfo| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |LookUpSigSlots| ...) is being compiled.
;; The variable |$insideCategoryPackageIfTrue| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |CheckVector| ...) is being compiled.
;; The variable |$catNames| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$CheckVectorList| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |InvestigateConditions,pessimise| ...) is being compiled.
;; The variable |$Conditions| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |InvestigateConditions| ...) is being compiled.
;; Warning: The variable |u| is not used.
; (DEFUN |resolvePatternVars| ...) is being compiled.
;; The variable |$TriangleVariableList| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/info.clisp.
; (DEFUN |addInfo| ...) is being compiled.
;; The variable |$Information| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |actOnInfo| ...) is being compiled.
;; The variable |$functorLocalParameters| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/iterator.clisp.
; (DEFUN |getIdentity| ...) is being compiled.
;; Warning: The variable |e| is not used.
; (DEFUN |compIterator| ...) is being compiled.
;; The variable |$until| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/nruncomp.clisp.
; (DEFUN |NRTaddDeltaCode| ...) is being compiled.
;; The variable |$catvecList| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$NRTbase| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$NRTdeltaList| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$NRTdeltaListComp| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$template| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$NRTaddForm| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$addForm| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |optDeltaEntry| ...) is being compiled.
;; The variable |$killOptimizeIfTrue| is undefined.
;; The compiler will assume this variable is a global.

```

```

;; The variable |$optimizableConstructorNames| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |genDeltaEntry| ...) is being compiled.
;; The variable |$profileCompiler| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$functorLocalParameters| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$NRTdeltaLength| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NRTassocIndex| ...) is being compiled.
;; The variable |$found| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NRTgetLocalIndex1| ...) is being compiled.
;; Warning: The variable |killBindingIfTrue| is not used.
; (DEFUN |NRTgetAddForm| ...) is being compiled.
;; The variable |$Slot1DataBase| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NRTassignCapsuleFunctionSlot| ...) is being compiled.
;; The variable |$insideCategoryPackageIfTrue| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NRTisExported?| ...) is being compiled.
;; The variable |$domainShell| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |consDomainName| ...) is being compiled.
;; The variable |$devaluateList| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |buildFunctor| ...) is being compiled.
;; The variable |$NRTvec| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$uncondAlist| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$condAlist| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$NRTslot1PredicateList| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$setelt| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$checkVectorList| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$NRTdomainFormList| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$getDomainCode| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NRTvectorCopy| ...) is being compiled.
;; Warning: The variable |cacheName| is not used.
;; Warning: The variable |domName| is not used.
; (DEFUN |NRTsetVector4| ...) is being compiled.
;; The variable |$lisplibCategoriesExtended| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NRTsetVector4Part2| ...) is being compiled.

```

```

;; The variable |$pairlis| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NRTsetVector4a| ...) is being compiled.
;; The variable |$uncondList| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$condList| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NRTmakeSlot1| ...) is being compiled.
;; The variable |$NRTmakeCompactDirect| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable |domainShell| is not used.
; (DEFUN |NRTmakeSlot1Info| ...) is being compiled.
;; The variable |$lisplibOpAlist| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$NRTderivedTargetIfTrue| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NRToptimizeHas| ...) is being compiled.
;; The variable |$hasCategoryAlist| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NRTaddToSlam| ...) is being compiled.
;; The variable |$mutableDomain| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |changeDirectoryInSlot1,fn| ...) is being compiled.
;; The variable |$lastPred| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$newEnv| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |genSlotSig| ...) is being compiled.
;; Warning: The variable |pred| is not used.
; (DEFUN |NRTputInHead| ...) is being compiled.
;; The variable |$elt| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/htcheck.clisp.
; (DEFUN |buildHtMacroTable| ...) is being compiled.
;; The variable |$htMacroTable| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$primitiveHtCommands| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/ht-util.clisp.
; (DEFUN |httpDestroyPage| ...) is being compiled.
;; The variable |$activePageList| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |iht| ...) is being compiled.
;; The variable |$newPage| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$htLineList| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |bcHt| ...) is being compiled.
;; The variable |$curPage| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |doDoitButton| ...) is being compiled.
;; Warning: The variable |htPage| is not used.
; (DEFUN |typeCheckInputAreas| ...) is being compiled.
;; The variable |$bcParseOnly| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |parseAndEval1| ...) is being compiled.
;; The variable |$useNewParser| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |htEscapeString| ...) is being compiled.
;; The variable |$funnyQuote| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$funnyBacks| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/br-con.clisp.
; (DEFUN |conPageFastPath| ...) is being compiled.
;; The variable |$lowerCaseConTb| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |kxPage| ...) is being compiled.
;; Warning: The variable |htPage| is not used.
; (DEFUN |kiPage| ...) is being compiled.
;; The variable |$conformsAreDomains| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable |junk| is not used.
; (DEFUN |kePage| ...) is being compiled.
;; Warning: The variable |junk| is not used.
; (DEFUN |ksPage| ...) is being compiled.
;; Warning: The variable |junk| is not used.

```

```

; (DEFUN |dbSearchOrder| ...) is being compiled.
;; The variable |$predvec| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |kcPage| ...) is being compiled.
;; The variable |$defaultPackageNamesHT| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable |junk| is not used.
; (DEFUN |kcpPage| ...) is being compiled.
;; Warning: The variable |junk| is not used.
; (DEFUN |kcaPage| ...) is being compiled.
;; Warning: The variable |junk| is not used.
; (DEFUN |kcdPage| ...) is being compiled.
;; Warning: The variable |junk| is not used.
; (DEFUN |kcdoPage| ...) is being compiled.
;; Warning: The variable |junk| is not used.
; (DEFUN |kcaPage1| ...) is being compiled.
;; Warning: The variable |article| is not used.
; (DEFUN |kccPage| ...) is being compiled.
;; Warning: The variable |junk| is not used.
; (DEFUN |kcdePage| ...) is being compiled.
;; Warning: The variable |junk| is not used.
; (DEFUN |kcuPage| ...) is being compiled.
;; Warning: The variable |junk| is not used.
; (DEFUN |kcnPage| ...) is being compiled.
;; The variable |conname| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable |junk| is not used.
; (DEFUN |conOpPage| ...) is being compiled.
;; Warning: The variable |conform| is not used.
; (DEFUN |conOpPage1| ...) is being compiled.
;; The variable |$Primitives| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |dbConstructorDoc,hn| ...) is being compiled.
;; The variable |$sig| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$args| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |dbConstructorDoc,gn| ...) is being compiled.
;; The variable |$op| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |dbDocTable| ...) is being compiled.
;; The variable |$docTableHash| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |dbAddDocTable| ...) is being compiled.
;; The variable |$docTable| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |dbGetDocTable,hn| ...) is being compiled.
;; The variable |$which| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$conform| is undefined.

```

```

;; The compiler will assume this variable is a global.
; (DEFUN |kTestPred| ...) is being compiled.
;; The variable |$domain| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |dbAddChainDomain| ...) is being compiled.
;; The variable |$infovec| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |dbPresentCons| ...) is being compiled.
;; The variable |$includeUnexposed?| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$exposedOnlyIfTrue| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |dbConsExposureMessage| ...) is being compiled.
;; The variable |$atLeastOneUnexposed| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |dbShowConsDoc1| ...) is being compiled.
;; The variable |$TriangleVariableList| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |dbSelectCon| ...) is being compiled.
;; Warning: The variable |which| is not used.

```

section

```

Compiling /research2/test0819/int/interp/topics.clisp.
; (DEFUN |mkTopicHashTable| ...) is being compiled.
;; The variable |$defaultsHash| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$topicsDefaults| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$conTopicHash| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$topicHash| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$groupAssoc| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$topicIndex| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |skipBlanks| ...) is being compiled.
;; The variable |$charBlank| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |topicCode| ...) is being compiled.
;; The variable |$topicSynonyms| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |td| ...) is being compiled.
;; The variable |$topicClasses| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/interop.clisp.
; (DEFUN |DNameToSExpr1| ...) is being compiled.
;; The variable |DNameStringID| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |DNameToSExpr| ...) is being compiled.
;; The variable |DNameOtherID| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |SExprToDName| ...) is being compiled.
;; The variable |DNameApplyID| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |DNameTupleID| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |runOldAxiomFunctor| ...) is being compiled.
;; The variable |$oldAxiomPreCategoryDispatch| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |makeLazyOldAxiomDispatchDomain| ...) is being compiled.
;; The variable |$attributeDispatch| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$lazyOldAxiomDomainDispatch| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |makeOldAxiomDispatchDomain| ...) is being compiled.
;; The variable |$oldAxiomDomainDispatch| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |lazyOldAxiomDomainLookupExport| ...) is being compiled.
;; Warning: The variable |env| is not used.
; (DEFUN |lazyOldAxiomDomainHashCode| ...) is being compiled.
;; Warning: The variable |env| is not used.
; (DEFUN |lazyOldAxiomDomainDevaluate| ...) is being compiled.
;; Warning: The variable |env| is not used.
; (DEFUN |lazyOldAxiomAddChild| ...) is being compiled.
;; Warning: The variable |kid| is not used.
;; Warning: The variable |env| is not used.
; (DEFUN |oldAxiomPreCategoryBuild| ...) is being compiled.
;; The variable |$oldAxiomCategoryDispatch| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable |env| is not used.
; (DEFUN |oldAxiomPreCategoryHashCode| ...) is being compiled.
;; Warning: The variable |env| is not used.
; (DEFUN |oldAxiomCategoryDefaultPackage| ...) is being compiled.
;; Warning: The variable |dom| is not used.
; (DEFUN |oldAxiomPreCategoryDevaluate| ...) is being compiled.
;; The variable T$ is undefined.
;; The compiler will assume this variable is a global.

```



```

;; Warning: The variable |env| is not used.
; (DEFUN |oldAxiomCategoryDevaluate| ...) is being compiled.
;; Warning: The variable |env| is not used.
; (DEFUN |oldAxiomCategoryLookupExport| ...) is being compiled.
;; Warning: The variable |box| is not used.
;; Warning: The variable |env| is not used.
; (DEFUN |oldAxiomCategoryParentCount| ...) is being compiled.
;; Warning: The variable |env| is not used.
; (DEFUN |oldAxiomCategoryNthParent| ...) is being compiled.
;; Warning: The variable |env| is not used.
; (DEFUN |oldAxiomCategoryHashCode| ...) is being compiled.
;; Warning: The variable |env| is not used.
; (DEFUN |attributeDevaluate| ...) is being compiled.
;; Warning: The variable |env| is not used.
; (DEFUN |attributeLookupExport| ...) is being compiled.
;; Warning: The variable |box| is not used.
;; Warning: The variable |env| is not used.
; (DEFUN |attributeHashCode| ...) is being compiled.
;; Warning: The variable |env| is not used.
; (DEFUN |attributeCategoryBuild| ...) is being compiled.
;; Warning: The variable |dom| is not used.
;; Warning: The variable |env| is not used.
; (DEFUN |attributeCategoryParentCount| ...) is being compiled.
;; Warning: The variable |attrObj| is not used.
;; Warning: The variable |env| is not used.
; (DEFUN |attributeNthParent| ...) is being compiled.
;; Warning: The variable |attrObj| is not used.
;; Warning: The variable |env| is not used.
; (DEFUN |oldAxiomDomainLookupExport| ...) is being compiled.
;; The variable |$hashOp1| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$hashOp0| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$hashOpApply| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$hashOpSet| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$hashSeg| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable |env| is not used.
; (DEFUN |oldAxiomDomainHashCode| ...) is being compiled.
;; Warning: The variable |env| is not used.
; (DEFUN |oldAxiomDomainHasCategory| ...) is being compiled.
;; Warning: The variable |env| is not used.
; (DEFUN |oldAxiomDomainDevaluate| ...) is being compiled.
;; Warning: The variable |env| is not used.
; (DEFUN |oldAxiomAddChild| ...) is being compiled.
;; Warning: The variable |child| is not used.
;; Warning: The variable |env| is not used.
; (DEFUN |coerceConvertMmSelection| ...) is being compiled.

```

```
;; The variable |coerceConvertMmSelection;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
Compiling /research2/test0819/int/interp/patches.lisp.
; (DEFUN CATCHALL ...) is being compiled.
;; Warning: The variable B is not used.
; (DEFUN CLEAR-HIGHLIGHT ...) is being compiled.
;; The variable |$specialCharacters| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN RESET-HIGHLIGHT ...) is being compiled.
;; The variable |$saveHighlight| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$saveSpecialchars| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |cd| ...) is being compiled.
;; The variable $CURRENT-DIRECTORY is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN TOPLEVEL ...) is being compiled.
;; Warning: The variable FOO is not used.
; (DEFUN /RF ...) is being compiled.
;; Warning: The variable FOO is not used.
; (DEFUN /RQ ...) is being compiled.
;; Warning: The variable FOO is not used.
; (DEFUN |/RQ,LIB| ...) is being compiled.
;; Warning: The variable FOO is not used.
; (DEFUN /RF-1 ...) is being compiled.
;; The variable /EDITFILE is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$useNewParser| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN /EF ...) is being compiled.
;; Warning: The variable FOO is not used.
; (DEFUN AKCL-VERSION ...) is being compiled.
;; The variable SYSTEM:*AKCL-VERSION* is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN SHAREDITEMS ...) is being compiled.
;; Warning: The variable X is not used.
; (DEFUN WHOCALLED ...) is being compiled.
;; Warning: The variable N is not used.
; (SHADOW 'MAP) is being compiled.
;; Warning: The package operation (SHADOW 'MAP) was in a bad place.
; (DEFUN |spadtestValueHook| ...) is being compiled.
;; Warning: The variable VAL is not used.
;; Warning: The variable TYPE is not used.
```

```

; (DEFUN |testError| ...) is being compiled.
;; Warning: The variable ERROTYPE is not used.
;; Warning: The variable ERROVALUE is not used.
; (DEFUN |bootFind| ...) is being compiled.
;; Warning: The variable WORD is not used.
; (DEFUN |fetchKeyedMsg| ...) is being compiled.
;; The variable |$defaultMsgDatabaseName| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |setViewportProcess| ...) is being compiled.
;; The variable |$ViewportProcessToWatch| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |waitForViewport| ...) is being compiled.
;; The variable |$MenuServer| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN PRINT-XDR-STREAM ...) is being compiled.
;; Warning: The variable Z is not used.
; (DEFUN |xdrOpen| ...) is being compiled.
;; Warning: The variable DIR is not used.
; (DEFUN |clearParserMacro| ...) is being compiled.
;; The variable |$pfMacros| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN /VERSIONCHECK ...) is being compiled.
;; The variable /MAJOR-VERSION is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/hashcode.clisp.
; (DEFUN |hashType| ...) is being compiled.
;; The variable |$VoidHash| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$DomainsWithoutLisplibs| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |hashString| ...) is being compiled.
;; The variable |$hashModulus| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling /research2/test0819/int/interp/nag-c02.clisp.
; (DEFUN |c02affSolve| ...) is being compiled.

```

```
;; The variable |$bcParseOnly| is undefined.  
;; The compiler will assume this variable is a global.
```

section

```
Compiling /research2/test0819/int/interp/nag-c05.clisp.  
; (DEFUN |c05nbfSolve| ...) is being compiled.  
;; The variable |$bcParseOnly| is undefined.  
;; The compiler will assume this variable is a global.
```

section

```
Compiling /research2/test0819/int/interp/nag-c06.clisp.  
; (DEFUN |c06eafSolve| ...) is being compiled.  
;; The variable |$bcParseOnly| is undefined.  
;; The compiler will assume this variable is a global.
```

section

```
Compiling /research2/test0819/int/interp/nag-d01.clisp.  
; (DEFUN |d01ajfGen| ...) is being compiled.  
;; The variable |$bcParseOnly| is undefined.  
;; The compiler will assume this variable is a global.
```

section

```
Compiling /research2/test0819/int/interp/nag-d02.clisp.  
; (DEFUN |d02bbfSolve| ...) is being compiled.  
;; The variable |$bcParseOnly| is undefined.  
;; The compiler will assume this variable is a global.
```

section

```
Compiling /research2/test0819/int/interp/nag-d03.clisp.  
; (DEFUN |d03edfSolve| ...) is being compiled.  
;; The variable |$bcParseOnly| is undefined.  
;; The compiler will assume this variable is a global.
```

section

```
Compiling /research2/test0819/int/interp/nag-e01.clisp.  
; (DEFUN |e01bafSolve| ...) is being compiled.  
;; The variable |$bcParseOnly| is undefined.  
;; The compiler will assume this variable is a global.
```

section

```
Compiling /research2/test0819/int/interp/nag-e02.clisp.  
; (DEFUN |e02adfSolve| ...) is being compiled.  
;; The variable |$bcParseOnly| is undefined.  
;; The compiler will assume this variable is a global.
```

section

```
Compiling /research2/test0819/int/interp/nag-e04.clisp.  
; (DEFUN |e04dgrSolve| ...) is being compiled.  
;; The variable |$bcParseOnly| is undefined.  
;; The compiler will assume this variable is a global.
```

section

```
Compiling /research2/test0819/int/interp/nag-f01.clisp.  
; (DEFUN |f01brfSolve| ...) is being compiled.
```

```

;; The variable |$bcParseOnly| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |f01qcfDefaultSolve| ...) is being compiled.
;; Warning: The variable |lda| is not used.
; (DEFUN |f01qdfDefaultSolve| ...) is being compiled.
;; Warning: The variable |lda| is not used.
;; Warning: The variable |ldb| is not used.
; (DEFUN |f01qefDefaultSolve| ...) is being compiled.
;; Warning: The variable |lda| is not used.
; (DEFUN |f01rdfDefaultSolve| ...) is being compiled.
;; Warning: The variable |lda| is not used.
;; Warning: The variable |ldb| is not used.
; (DEFUN |f01refDefaultSolve| ...) is being compiled.
;; Warning: The variable |lda| is not used.

```

section

```

Compiling /research2/test0819/int/interp/nag-f02.clisp.
; (DEFUN |f02aafSolve| ...) is being compiled.
;; The variable |$bcParseOnly| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |f02aafDefaultSolve| ...) is being compiled.
;; Warning: The variable |ia| is not used.
; (DEFUN |f02abfDefaultSolve| ...) is being compiled.
;; Warning: The variable |ia| is not used.
;; Warning: The variable |iv| is not used.
; (DEFUN |f02adfDefaultSolve| ...) is being compiled.
;; Warning: The variable |ia| is not used.
;; Warning: The variable |ib| is not used.
; (DEFUN |f02aefDefaultSolve| ...) is being compiled.
;; Warning: The variable |ia| is not used.
;; Warning: The variable |ib| is not used.
;; Warning: The variable |iv| is not used.
; (DEFUN |f02affDefaultSolve| ...) is being compiled.
;; Warning: The variable |ia| is not used.
; (DEFUN |f02agfDefaultSolve| ...) is being compiled.
;; Warning: The variable |ia| is not used.
;; Warning: The variable |ivr| is not used.
;; Warning: The variable |ivi| is not used.
; (DEFUN |f02ajfDefaultSolve| ...) is being compiled.
;; Warning: The variable |iar| is not used.
;; Warning: The variable |iai| is not used.
; (DEFUN |f02akfDefaultSolve| ...) is being compiled.
;; Warning: The variable |iar| is not used.
;; Warning: The variable |iai| is not used.
;; Warning: The variable |ivr| is not used.

```

```
;; Warning: The variable |livi| is not used.
; (DEFUN |f02awfDefaultSolve| ...) is being compiled.
;; Warning: The variable |liar| is not used.
;; Warning: The variable |liai| is not used.
; (DEFUN |f02axfDefaultSolve| ...) is being compiled.
;; Warning: The variable |liar| is not used.
;; Warning: The variable |liai| is not used.
;; Warning: The variable |livr| is not used.
;; Warning: The variable |livi| is not used.
; (DEFUN |f02bbfDefaultSolve| ...) is being compiled.
;; Warning: The variable |lia| is not used.
;; Warning: The variable |liv| is not used.
; (DEFUN |f02bjfDefaultSolve| ...) is being compiled.
;; Warning: The variable |lia| is not used.
;; Warning: The variable |lib| is not used.
;; Warning: The variable |liv| is not used.
```

section

```
Compiling /research2/test0819/int/interp/nag-f04.clisp.
; (DEFUN |f04adfSolve| ...) is being compiled.
;; The variable |$bcParseOnly| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |f04atfDefaultSolve| ...) is being compiled.
;; Warning: The variable |iaa| is not used.
; (DEFUN |f04jgfDefaultSolve| ...) is being compiled.
;; Warning: The variable |nra| is not used.
;; Warning: The variable |lwork| is not used.
```

section

```
Compiling /research2/test0819/int/interp/nag-f07.clisp.
; (DEFUN |f07adfSolve| ...) is being compiled.
;; The variable |$bcParseOnly| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |f07adfDefaultSolve| ...) is being compiled.
;; Warning: The variable |lda| is not used.
```

section

```
Compiling /research2/test0819/int/interp/nag-s.clisp.  
; (DEFUN |s17dcfGen| ...) is being compiled.  
;; The variable |$bcParseOnly| is undefined.  
;; The compiler will assume this variable is a global.
```

section

```
Compiling ABELMON.lsp.  
; (DEFUN |AbelianMonoid| ...) is being compiled.  
;; The variable |AbelianMonoid;AL| is undefined.  
;; The compiler will assume this variable is a global.  
;; Warning: The variable #:G82597 is not used.
```

section

```
Compiling ABELSG.lsp.  
; (DEFUN |AbelianSemiGroup| ...) is being compiled.  
;; The variable |AbelianSemiGroup;AL| is undefined.  
;; The compiler will assume this variable is a global.  
;; Warning: The variable #:G82568 is not used.
```

section

```
Compiling ALAGG.lsp.  
; (DEFUN |AssociationListAggregate| ...) is being compiled.  
;; The variable |AssociationListAggregate;AL| is undefined.  
;; The compiler will assume this variable is a global.  
; (DEFUN |AssociationListAggregate;| ...) is being compiled.  
;; The variable |AssociationListAggregate;CAT| is undefined.  
;; The compiler will assume this variable is a global.
```


section

```

Compiling BOOLEAN.lsp.
; (DEFUN |BOOLEAN;test;2$;1| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |BOOLEAN;nt| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |BOOLEAN;true;$;3| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |BOOLEAN>false;$;4| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |BOOLEAN;not;2$;5| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |BOOLEAN;^;2$;6| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |BOOLEAN;~;2$;7| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |BOOLEAN;and;3$;8| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |BOOLEAN;/\;3$;9| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |BOOLEAN;or;3$;10| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |BOOLEAN;\;/;3$;11| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |BOOLEAN;=;3$;15| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |BOOLEAN;implies;3$;16| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |BOOLEAN;<;3$;17| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |BOOLEAN;size;Nni;18| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |BOOLEAN;lookup;$Pi;20| ...) is being compiled.
;; Warning: The variable $ is not used.

```

section

```

Compiling CABMON.lsp.
; (DEFUN |CancellationAbelianMonoid| ...) is being compiled.
;; The variable |CancellationAbelianMonoid;AL| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable #:G82646 is not used.

```

section

```

Compiling CHAR.lsp.
; (DEFUN |CHAR;=;2$B;1| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |CHAR;<;2$B;2| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |CHAR;size;Nni;3| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |CHAR;ord;$I;7| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |CHAR;space;$;9| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |CHAR;quote;$;10| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |CHAR;escape;$;11| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |CHAR;latex;$S;19| ...) is being compiled.
;; Warning: The variable $ is not used.

```

section

```

Compiling CLAGG.lsp.
; (DEFUN |Collection| ...) is being compiled.
;; The variable |Collection;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |Collection;| ...) is being compiled.
;; The variable |Collection;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling COMRING.lsp.
; (DEFUN |CommutativeRing| ...) is being compiled.
;; The variable |CommutativeRing;AL| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable #:G82892 is not used.

```

section

```

Compiling DFLOAT.lsp.
; (DEFUN |DFLOAT;checkComplex| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;base;Pi;6| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;precision;Pi;9| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;max;$;11| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;min;$;12| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;Zero;$;14| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;One;$;15| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;exp1;$;16| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;pi;$;17| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;<;2$B;20| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;-;2$;21| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;+;3$;22| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;-;3$;23| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;*;3$;24| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;*;I2$;25| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;max;3$;26| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;min;3$;27| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;=;2$B;28| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;/;$I$;29| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;**;$I$;32| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;coerce;I$;34| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;exp;2$;35| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;sin;2$;38| ...) is being compiled.

```

```

;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;cos;2$;39| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;tan;2$;40| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;cot;2$;41| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;sec;2$;42| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;csc;2$;43| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;atan;2$;46| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;acot;2$;48| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;sinh;2$;50| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;cosh;2$;51| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;tanh;2$;52| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;csch;2$;53| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;coth;2$;54| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;sech;2$;55| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;asinh;2$;56| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;acsch;2$;59| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;/;3$;62| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;negative?;$B;63| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;zero?;$B;64| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;hash;$I;65| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;recip;$U;66| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;differentiate;2$;67| ...) is being compiled.
;; Warning: The variable |x| is not used.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;wholePart;$I;70| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;float;2IPi$;71| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;convert;2$;72| ...) is being compiled.
;; Warning: The variable $ is not used.

```

```
; (DEFUN |DFLOAT;atan;3$;75| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;retract;$I;78| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;retractIfCan;$U;79| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |DFLOAT;abs;2$;81| ...) is being compiled.
;; Warning: The variable $ is not used.
```

section

```
Compiling DIFRING.lsp.
; (DEFUN |DifferentialRing| ...) is being compiled.
;; The variable |DifferentialRing;AL| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable #:G84565 is not used.
```

section

```
Compiling DIVRING.lsp.
; (DEFUN |DivisionRing| ...) is being compiled.
;; The variable |DivisionRing;AL| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable #:G84035 is not used.
```

section

```
Compiling ENTIRER.lsp.
; (DEFUN |EntireRing| ...) is being compiled.
;; The variable |EntireRing;AL| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable #:G82841 is not used.
```

section

```
Compiling ES.lsp.  
; (DEFUN |ExpressionSpace| ...) is being compiled.  
;; The variable |ExpressionSpace;AL| is undefined.  
;; The compiler will assume this variable is a global.  
;; Warning: The variable #:G82344 is not used.
```

section

```
Compiling EUCDOM.lsp.  
; (DEFUN |EuclideanDomain| ...) is being compiled.  
;; The variable |EuclideanDomain;AL| is undefined.  
;; The compiler will assume this variable is a global.  
;; Warning: The variable #:G83585 is not used.
```

section

```
Compiling FFIELDC.lsp.  
; (DEFUN |FiniteFieldCategory| ...) is being compiled.  
;; The variable |FiniteFieldCategory;AL| is undefined.  
;; The compiler will assume this variable is a global.  
;; Warning: The variable #:G83129 is not used.
```

section

```
Compiling FFIELDC-.lsp.  
; (DEFUN |FFIELDC-;differentiate;2S;1| ...) is being compiled.  
;; Warning: The variable |x| is not used.
```

section

```
Compiling FPS.lsp.  
; (DEFUN |FloatingPointSystem| ...) is being compiled.  
;; The variable |FloatingPointSystem;AL| is undefined.
```

```
;; The compiler will assume this variable is a global.
;; Warning: The variable #:G105645 is not used.
```

section

```
Compiling GCDDOM.lsp.
; (DEFUN |GcdDomain| ...) is being compiled.
;; The variable |GcdDomain;AL| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable #:G83171 is not used.
```

section

```
Compiling HOAGG.lsp.
; (DEFUN |HomogeneousAggregate| ...) is being compiled.
;; The variable |HomogeneousAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |HomogeneousAggregate;| ...) is being compiled.
;; The variable |HomogeneousAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
Compiling ILLIST.lsp.
; (DEFUN |ILLIST;#;$Nni;1| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |ILLIST;concat;S2$;2| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |ILLIST;eq?;2$B;3| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |ILLIST;first;$S;4| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |ILLIST;elt;$firstS;5| ...) is being compiled.
;; Warning: The variable G101995 is not used.
;; Warning: The variable $ is not used.
; (DEFUN |ILLIST;empty;$;6| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |ILLIST;empty?;$B;7| ...) is being compiled.
```

```

;; Warning: The variable $ is not used.
; (DEFUN |ILIST;rest;2$;8| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |ILIST;elt;$rest$;9| ...) is being compiled.
;; Warning: The variable G102000 is not used.
;; Warning: The variable $ is not used.
; (DEFUN |ILIST;setelt;$first2S;11| ...) is being compiled.
;; Warning: The variable G102005 is not used.
; (DEFUN |ILIST;setelt;$rest2$;13| ...) is being compiled.
;; Warning: The variable G102010 is not used.
; (DEFUN |ILIST;construct;L$;14| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |ILIST;parts;$L;15| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |ILIST;reverse!;2$;16| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |ILIST;reverse;2$;17| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |ILIST;minIndex;$I;18| ...) is being compiled.
;; Warning: The variable |x| is not used.
; (DEFUN |ILIST;rest;$Nni$;19| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |ILIST;merge!;M3$;28| ...) is being compiled.
;; Warning: The variable $ is not used.

```

section

```

Compiling INS.lsp.
; (DEFUN |IntegerNumberSystem| ...) is being compiled.
;; The variable |IntegerNumberSystem;AL| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable #:G1068 is not used.

```

section

```

Compiling INS-.lsp.
; (DEFUN |INS-;characteristic;Nni;1| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INS-;differentiate;2S;2| ...) is being compiled.
;; Warning: The variable |x| is not used.
; (DEFUN |INS-;copy;2S;5| ...) is being compiled.
;; Warning: The variable $ is not used.

```



```
; (DEFUN |INS-;rational?;SB;8| ...) is being compiled.
;; Warning: The variable |x| is not used.
;; Warning: The variable $ is not used.
```

section

Compiling INT.lsp.

```
; (DEFUN |INT;zero?;$B;6| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;Zero;$;7| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;One;$;8| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;base;$;9| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;copy;2$;10| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;inc;2$;11| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;dec;2$;12| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;hash;2$;13| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;negative?;$B;14| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;coerce;2$;16| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;convert;2$;17| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;length;2$;18| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;addmod;4$;19| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;submod;4$;20| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;mulmod;4$;21| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;convert;$Df;23| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;convert;$S;25| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;latex;$S;26| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;positiveRemainder;3$;27| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;reducedSystem;2M;28| ...) is being compiled.
```

```

;; Warning: The variable $ is not used.
; (DEFUN |INT;reducedSystem;MVR;29| ...) is being compiled.
;; Warning: The variable |v| is not used.
;; Warning: The variable $ is not used.
; (DEFUN |INT;abs;2$;30| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;random;$;31| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;random;2$;32| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;=;2$B;33| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;<;2$B;34| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;-;2$;35| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;+;3$;36| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;-;3$;37| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;*;3$;38| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;*;3$;39| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;**;$Nni$;40| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;odd?;$B;41| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;max;3$;42| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;min;3$;43| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;divide;2$R;44| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;quo;3$;45| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;rem;3$;46| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;shift;3$;47| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;exquo;2$U;48| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;recip;$U;49| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;gcd;3$;50| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;unitNormal;$R;51| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |INT;unitCanonical;2$;52| ...) is being compiled.
;; Warning: The variable $ is not used.

```

section

```

Compiling INTDOM.lsp.
; (DEFUN |IntegralDomain| ...) is being compiled.
;; The variable |IntegralDomain;AL| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable #:G83060 is not used.

```

section

```

Compiling ISTRING.lsp.
; (DEFUN |ISTRING;new;NniC$;1| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |ISTRING;empty;$;2| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |ISTRING;empty?;$B;3| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |ISTRING;#;$Nni;4| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |ISTRING;=;2$B;5| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |ISTRING;<;2$B;6| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |ISTRING;concat;3$;7| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |ISTRING;copy;2$;8| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |ISTRING;minIndex;$I;11| ...) is being compiled.
;; Warning: The variable |s| is not used.
; (DEFUN |ISTRING;latex;$S;14| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |ISTRING;match;2$CNni;33| ...) is being compiled.
;; Warning: The variable $ is not used.

```

section

```

Compiling LIST.lsp.

```

```
; (DEFUN |LIST:nil;$;1| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |LIST:null;$B;2| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |LIST:cons;S2$;3| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |LIST:append;3$;4| ...) is being compiled.
;; Warning: The variable $ is not used.
```

section

```
Compiling LNAGG.lsp.
; (DEFUN |LinearAggregate| ...) is being compiled.
;; The variable |LinearAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |LinearAggregate;| ...) is being compiled.
;; The variable |LinearAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
Compiling LSAGG.lsp.
; (DEFUN |ListAggregate| ...) is being compiled.
;; The variable |ListAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |ListAggregate;| ...) is being compiled.
;; The variable |ListAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
Compiling MONOID.lsp.
; (DEFUN |Monoid| ...) is being compiled.
;; The variable |Monoid;AL| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable #:G82434 is not used.
```

section

```

Compiling MTSCAT.lsp.
; (DEFUN |MultivariateTaylorSeriesCategory| ...) is being compiled.
;; The variable |MultivariateTaylorSeriesCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |MultivariateTaylorSeriesCategory;| ...) is being compiled.
;; The variable |MultivariateTaylorSeriesCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling NNI.lsp.
; (DEFUN |NNI;sup;3$;1| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |NNI;shift;$I$;2| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |NNI;subtractIfCan;2$U;3| ...) is being compiled.
;; Warning: The variable $ is not used.

```

section

```

Compiling OINTDOM.lsp.
; (DEFUN |OrderedIntegralDomain| ...) is being compiled.
;; The variable |OrderedIntegralDomain;AL| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable #:G84531 is not used.

```

section

```

Compiling ORDRING.lsp.
; (DEFUN |OrderedRing| ...) is being compiled.
;; The variable |OrderedRing;AL| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable #:G84457 is not used.

```

section

```

Compiling OUTFORM.lsp.
; (DEFUN |OUTFORM;print;$V;1| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |OUTFORM;=;2$B;4| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |OUTFORM;coerce;2$;6| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |OUTFORM;outputForm;I$;7| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |OUTFORM;outputForm;$S;8| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |OUTFORM;outputForm;Df$;9| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |OUTFORM;sform| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |OUTFORM;eform| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |OUTFORM;iform| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |OUTFORM;width;$I;14| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |OUTFORM;height;$I;15| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |OUTFORM;subHeight;$I;16| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |OUTFORM;superHeight;$I;17| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |OUTFORM;height;I;18| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |OUTFORM;width;I;19| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |OUTFORM;infix?;$B;72| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |OUTFORM;elt;$L$;73| ...) is being compiled.
;; Warning: The variable $ is not used.

```

section

```

Compiling PRIMARR.lsp.
; (DEFUN |PRIMARR;#;$Nni;1| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |PRIMARR;minIndex;$I;2| ...) is being compiled.

```

```
;; Warning: The variable |x| is not used.
;; Warning: The variable $ is not used.
; (DEFUN |PRIMARR;empty;$;3| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |PRIMARR;qelt;$IS;5| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |PRIMARR;elt;$IS;6| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |PRIMARR;qsetelt!;$I2S;7| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |PRIMARR;setelt;$I2S;8| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |PRIMARR;fill!;$S$;9| ...) is being compiled.
;; Warning: The variable $ is not used.
```

section

```
Compiling POLYCAT.lsp.
; (DEFUN |PolynomialCategory| ...) is being compiled.
;; The variable |PolynomialCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |PolynomialCategory;| ...) is being compiled.
;; The variable |PolynomialCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
Compiling PSETCAT.lsp.
; (DEFUN |PolynomialSetCategory| ...) is being compiled.
;; The variable |PolynomialSetCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |PolynomialSetCategory;| ...) is being compiled.
;; The variable |PolynomialSetCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
Compiling PSETCAT-.lsp.
```

End of Pass 1.

Warning: PSETCAT-;exactQuo has a duplicate definition in this file

section

Compiling QFCAT.lsp.

```
; (DEFUN |QuotientFieldCategory| ...) is being compiled.
;; The variable |QuotientFieldCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |QuotientFieldCategory;| ...) is being compiled.
;; The variable |QuotientFieldCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

Compiling RCAGG.lsp.

```
; (DEFUN |RecursiveAggregate| ...) is being compiled.
;; The variable |RecursiveAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |RecursiveAggregate;| ...) is being compiled.
;; The variable |RecursiveAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

Compiling RCAGG-.lsp.

```
; (DEFUN |RCAGG-;elt;AvalueS;1| ...) is being compiled.
;; Warning: The variable G84515 is not used.
; (DEFUN |RCAGG-;setelt;Avalue2S;2| ...) is being compiled.
;; Warning: The variable G84517 is not used.
```

section

Compiling REF.lsp.


```
; (DEFUN |REF;=;2$B;1| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |REF;ref;$S;2| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |REF;elt;$S;3| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |REF;setelt;$2S;4| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |REF;deref;$S;5| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |REF;setref;$2S;6| ...) is being compiled.
;; Warning: The variable $ is not used.
```

section

```
Compiling RING.lsp.
; (DEFUN |Ring| ...) is being compiled.
;; The variable |Ring;AL| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable #:G82789 is not used.
```

section

```
Compiling RNG.lsp.
; (DEFUN |Rng| ...) is being compiled.
;; The variable |Rng;AL| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable #:G82722 is not used.
```

section

```
Compiling RNS.lsp.
; (DEFUN |RealNumberSystem| ...) is being compiled.
;; The variable |RealNumberSystem;AL| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable #:G105478 is not used.
```

section

```

Compiling RNS-.lsp.
; (DEFUN |RNS-;characteristic;Nni;1| ...) is being compiled.
;; Warning: The variable $ is not used.

```

section

```

Compiling SETAGG.lsp.
; (DEFUN |SetAggregate| ...) is being compiled.
;; The variable |SetAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |SetAggregate;| ...) is being compiled.
;; The variable |SetAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling SETCAT.lsp.
; (DEFUN |SetCategory| ...) is being compiled.
;; The variable |SetCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable #:G82359 is not used.

```

section

```

Compiling SETCAT-.lsp.
; (DEFUN |SETCAT-;hash;SSi;1| ...) is being compiled.
;; Warning: The variable |s| is not used.
;; Warning: The variable $ is not used.
; (DEFUN |SETCAT-;latex;SS;2| ...) is being compiled.
;; Warning: The variable |s| is not used.
;; Warning: The variable $ is not used.

```

section

```

Compiling SINT.lsp.
; (DEFUN |SINT;reducedSystem;MM;6| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;convert;$I;8| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;Zero;$;10| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;One;$;11| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;base;$;12| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;max;$;13| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;min;$;14| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;=;2$B;15| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;~;2$;16| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;not;2$;17| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;/\;3$;18| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;\;/;3$;19| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;Not;2$;20| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;And;3$;21| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;Or;3$;22| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;xor;3$;23| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;<;2$B;24| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;inc;2$;25| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;dec;2$;26| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;-;2$;27| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;+;3$;28| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;-;3$;29| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;*;3$;30| ...) is being compiled.

```

```

;; Warning: The variable $ is not used.
; (DEFUN |SINT;quo;3$;32| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;rem;3$;33| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;divide;2$R;34| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;gcd;3$;35| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;abs;2$;36| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;odd?;$B;37| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;zero?;$B;38| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;max;3$;39| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;min;3$;40| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;hash;2$;41| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;length;2$;42| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;shift;3$;43| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;mulmod;4$;44| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;addmod;4$;45| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;submod;4$;46| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;negative?;$B;47| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;reducedSystem;MVR;48| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;positiveRemainder;3$;49| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;coerce;I$;50| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;random;2$;52| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SINT;unitNormal;$R;53| ...) is being compiled.
;; Warning: The variable $ is not used.

```

section

```

Compiling STAGG-.lsp.
; (DEFUN |STAGG-;c2| ...) is being compiled.
;; Warning: The variable |r| is not used.

```

section

```

Compiling SYMBOL.lsp.
; (DEFUN |SYMBOL;convert;2$;7| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SYMBOL;coerce;$;8| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SYMBOL;=;2$B;9| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SYMBOL;<;2$B;10| ...) is being compiled.
;; Warning: The variable $ is not used.
; (DEFUN |SYMBOL;scripted?;$B;30| ...) is being compiled.
;; Warning: The variable $ is not used.

```

section

```

Compiling TSETCAT.lsp.
; (DEFUN |TriangularSetCategory| ...) is being compiled.
;; The variable |TriangularSetCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |TriangularSetCategory;| ...) is being compiled.
;; The variable |TriangularSetCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling UFD.lsp.
; (DEFUN |UniqueFactorizationDomain| ...) is being compiled.
;; The variable |UniqueFactorizationDomain;AL| is undefined.
;; The compiler will assume this variable is a global.
;; Warning: The variable #:G83334 is not used.

```

section

```

Compiling ULSCAT.lsp.
; (DEFUN |UnivariateLaurentSeriesCategory| ...) is being compiled.
;; The variable |UnivariateLaurentSeriesCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |UnivariateLaurentSeriesCategory;| ...) is being compiled.
;; The variable |UnivariateLaurentSeriesCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling UPOLYC.lsp.
; (DEFUN |UnivariatePolynomialCategory| ...) is being compiled.
;; The variable |UnivariatePolynomialCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |UnivariatePolynomialCategory;| ...) is being compiled.
;; The variable |UnivariatePolynomialCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling UPOLYC-.lsp.
; (DEFUN |UPOLYC-;degree;SSaosNni;2| ...) is being compiled.
;; Warning: The variable |v| is not used.
; (DEFUN |UPOLYC-;eval;SSaos2S;6| ...) is being compiled.
;; Warning: The variable |v| is not used.
; (DEFUN |UPOLYC-;eval;SSaosRS;8| ...) is being compiled.
;; Warning: The variable |v| is not used.
; (DEFUN |UPOLYC-;minimumDegree;SSaosNni;11| ...) is being compiled.
;; Warning: The variable |v| is not used.
; (DEFUN |UPOLYC-;monomial;SSaosNniS;13| ...) is being compiled.
;; Warning: The variable |v| is not used.
; (DEFUN |UPOLYC-;coerce;SaosS;14| ...) is being compiled.
;; Warning: The variable |v| is not used.
; (DEFUN |UPOLYC-;content;SSaosS;30| ...) is being compiled.
;; Warning: The variable |v| is not used.
; (DEFUN |UPOLYC-;differentiate;SSaosS;38| ...) is being compiled.
;; Warning: The variable |v| is not used.

```

section

```

Compiling URAGG.lsp.
; (DEFUN |UnaryRecursiveAggregate| ...) is being compiled.
;; The variable |UnaryRecursiveAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |UnaryRecursiveAggregate;| ...) is being compiled.
;; The variable |UnaryRecursiveAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

Compiling URAGG-.lsp.
; (DEFUN |URAGG-;elt;AfirstS;1| ...) is being compiled.
;; Warning: The variable G84610 is not used.
; (DEFUN |URAGG-;elt;AlastS;2| ...) is being compiled.
;; Warning: The variable G84612 is not used.
; (DEFUN |URAGG-;elt;ArestA;3| ...) is being compiled.
;; Warning: The variable G84614 is not used.
; (DEFUN |URAGG-;setelt;Afirst2S;25| ...) is being compiled.
;; Warning: The variable G84713 is not used.
; (DEFUN |URAGG-;setelt;Alast2S;26| ...) is being compiled.
;; Warning: The variable G84715 is not used.
; (DEFUN |URAGG-;setelt;Arest2A;27| ...) is being compiled.
;; Warning: The variable G84717 is not used.

```

section

```

      finalizing nrlib AHYP
; (DEFUN |ArcHyperbolicFunctionCategory| ...) is being compiled.
;; The variable |ArcHyperbolicFunctionCategory;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

      finalizing nrlib ATTREG

```

```
; (DEFUN |AttributeRegistry| ...) is being compiled.
;; The variable |AttributeRegistry;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib CFCAT
; (DEFUN |CombinatorialFunctionCategory| ...) is being compiled.
;; The variable |CombinatorialFunctionCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib ELTAB
; (DEFUN |Eltable| ...) is being compiled.
;; The variable |Eltable;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |Eltable;| ...) is being compiled.
;; The variable |Eltable;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib KOERCE
; (DEFUN |CoercibleTo| ...) is being compiled.
;; The variable |CoercibleTo;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |CoercibleTo;| ...) is being compiled.
;; The variable |CoercibleTo;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib KONVERT
```



```
; (DEFUN |ConvertibleTo| ...) is being compiled.
;; The variable |ConvertibleTo;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |ConvertibleTo;| ...) is being compiled.
;; The variable |ConvertibleTo;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib OM
; (DEFUN |OpenMath| ...) is being compiled.
;; The variable |OpenMath;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib PTRANFN
; (DEFUN |PartialTranscendentalFunctions| ...) is being compiled.
;; The variable |PartialTranscendentalFunctions;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |PartialTranscendentalFunctions;| ...) is being compiled.
;; The variable |PartialTranscendentalFunctions;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib SPFCAT
; (DEFUN |SpecialFunctionCategory| ...) is being compiled.
;; The variable |SpecialFunctionCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib TYPE
```

```
; (DEFUN |Type| ...) is being compiled.
;; The variable |Type;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib COMBOPC
; (DEFUN |CombinatorialOpsCategory| ...) is being compiled.
;; The variable |CombinatorialOpsCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib FORTCAT
; (DEFUN |FortranProgramCategory| ...) is being compiled.
;; The variable |FortranProgramCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
```

```
-----non extending category-----
.. InfiniteTuple #1 of cat
(|Join| (|CoercibleTo| (|OutputForm|)) (CATEGORY |domain| (SIGNATURE |map| ($ (|Mapping| |#1| |#
```

section

```
(|LazyStreamAggregate| |#1|) finalizing nrlib ITUPLE
```

section

```
finalizing nrlib PATAB
; (DEFUN |Patternable| ...) is being compiled.
;; The variable |Patternable;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |Patternable;| ...) is being compiled.
;; The variable |Patternable;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```

    finalizing nrlib PPCURVE
; (DEFUN |PlottablePlaneCurveCategory| ...) is being compiled.
;; The variable |PlottablePlaneCurveCategory;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib PSCURVE
; (DEFUN |PlottableSpaceCurveCategory| ...) is being compiled.
;; The variable |PlottableSpaceCurveCategory;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib REAL
; (DEFUN |RealConstant| ...) is being compiled.
;; The variable |RealConstant;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib RETRACT
; (DEFUN |RetractableTo| ...) is being compiled.
;; The variable |RetractableTo;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |RetractableTo;| ...) is being compiled.
;; The variable |RetractableTo;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```
      finalizing nrlib SEGCAT
; (DEFUN |SegmentCategory| ...) is being compiled.
;; The variable |SegmentCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |SegmentCategory;| ...) is being compiled.
;; The variable |SegmentCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
      finalizing nrlib FMC
; (DEFUN |FortranMatrixCategory| ...) is being compiled.
;; The variable |FortranMatrixCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
      finalizing nrlib FMFUN
; (DEFUN |FortranMatrixFunctionCategory| ...) is being compiled.
;; The variable |FortranMatrixFunctionCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
      finalizing nrlib FORTFN
; (DEFUN |FortranFunctionCategory| ...) is being compiled.
;; The variable |FortranFunctionCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```

      finalizing nrlib FVC
; (DEFUN |FortranVectorCategory| ...) is being compiled.
;; The variable |FortranVectorCategory;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

      finalizing nrlib FVFUN
; (DEFUN |FortranVectorFunctionCategory| ...) is being compiled.
;; The variable |FortranVectorFunctionCategory;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

      finalizing nrlib SEGXCAT
; (DEFUN |SegmentExpansionCategory| ...) is being compiled.
;; The variable |SegmentExpansionCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |SegmentExpansionCategory;| ...) is being compiled.
;; The variable |SegmentExpansionCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

      finalizing nrlib AGG
; (DEFUN |Aggregate| ...) is being compiled.
;; The variable |Aggregate;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

      finalizing nrlib BASTYPE
; (DEFUN |BasicType| ...) is being compiled.

```

```
;; The variable |BasicType;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib ELTAGG
; (DEFUN |EltableAggregate| ...) is being compiled.
;; The variable |EltableAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |EltableAggregate;| ...) is being compiled.
;; The variable |EltableAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib FAMONC
; (DEFUN |FreeAbelianMonoidCategory| ...) is being compiled.
;; The variable |FreeAbelianMonoidCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |FreeAbelianMonoidCategory;| ...) is being compiled.
;; The variable |FreeAbelianMonoidCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib FILECAT
; (DEFUN |FileCategory| ...) is being compiled.
;; The variable |FileCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |FileCategory;| ...) is being compiled.
;; The variable |FileCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```

    finalizing nrlib FINITE
; (DEFUN |Finite| ...) is being compiled.
;; The variable |Finite;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib FNCAT
; (DEFUN |FileNameCategory| ...) is being compiled.
;; The variable |FileNameCategory;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib IDPC
; (DEFUN |IndexedDirectProductCategory| ...) is being compiled.
;; The variable |IndexedDirectProductCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |IndexedDirectProductCategory;| ...) is being compiled.
;; The variable |IndexedDirectProductCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib IEVALAB
; (DEFUN |InnerEvalable| ...) is being compiled.
;; The variable |InnerEvalable;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |InnerEvalable;| ...) is being compiled.
;; The variable |InnerEvalable;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib LMODULE
; (DEFUN |LeftModule| ...) is being compiled.
;; The variable |LeftModule;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |LeftModule;| ...) is being compiled.
;; The variable |LeftModule;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib LOGIC
; (DEFUN |Logic| ...) is being compiled.
;; The variable |Logic;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib MONAD
; (DEFUN |Monad| ...) is being compiled.
;; The variable |Monad;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib NIPROB
Warnings:
    [1] coerce:  nia has no value
    [2] coerce:  mdnia has no value
    [3] retract:  nia has no value
    [4] retract:  mdnia has no value

```

section


```

    finalizing nrlib NUMINT
; (DEFUN |NumericalIntegrationCategory| ...) is being compiled.
;; The variable |NumericalIntegrationCategory;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib ODECAT
; (DEFUN |OrdinaryDifferentialEquationsSolverCategory| ...) is being compiled.
;; The variable |OrdinaryDifferentialEquationsSolverCategory;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib OPTCAT
; (DEFUN |NumericalOptimizationCategory| ...) is being compiled.
;; The variable |NumericalOptimizationCategory;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib OPTPROB
Warnings:
  [1] coerce:  noa has no value
  [2] coerce:  lsa has no value
  [3] retract: noa has no value
  [4] retract: lsa has no value

```

section

```

    finalizing nrlib ORDSET
; (DEFUN |OrderedSet| ...) is being compiled.
;; The variable |OrderedSet;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

finalizing nrlib PATMAB
; (DEFUN |PatternMatchable| ...) is being compiled.
;; The variable |PatternMatchable;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |PatternMatchable;| ...) is being compiled.
;; The variable |PatternMatchable;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

finalizing nrlib PDECAT
; (DEFUN |PartialDifferentialEquationsSolverCategory| ...) is being compiled.
;; The variable |PartialDifferentialEquationsSolverCategory;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

finalizing nrlib PDEPROB
--->/research2/test0819/mnt/fedora5/../../src/algebra/PDEPROB.spad-->NumericalPDEProblem(constru
--->/research2/test0819/mnt/fedora5/../../src/algebra/PDEPROB.spad-->NumericalPDEProblem(): Miss

REPSQ abbreviates package RepeatedSquaring
(PUSH (QUOTE |RepeatedSquaring|) |$mutableDomains|)
Value = (|RepeatedSquaring|)

REPDB abbreviates package RepeatedDoubling
(PUSH (QUOTE |RepeatedDoubling|) |$mutableDomains|)
Value = (|RepeatedDoubling|)

```

section

```

    finalizing nrlib RMODULE
; (DEFUN |RightModule| ...) is being compiled.
;; The variable |RightModule;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |RightModule;| ...) is being compiled.
;; The variable |RightModule;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib SEXCAT
; (DEFUN |SExpressionCategory| ...) is being compiled.
;; The variable |SExpressionCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |SExpressionCategory;| ...) is being compiled.
;; The variable |SExpressionCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib SPACEC
; (DEFUN |ThreeSpaceCategory| ...) is being compiled.
;; The variable |ThreeSpaceCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |ThreeSpaceCategory;| ...) is being compiled.
;; The variable |ThreeSpaceCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib STEP
; (DEFUN |StepThrough| ...) is being compiled.
;; The variable |StepThrough;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```
finalizing nrlib ATRIG
; (DEFUN |ArcTrigonometricFunctionCategory| ...) is being compiled.
;; The variable |ArcTrigonometricFunctionCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib BMODULE
; (DEFUN |BiModule| ...) is being compiled.
;; The variable |BiModule;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |BiModule;| ...) is being compiled.
;; The variable |BiModule;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib CACHSET
; (DEFUN |CachableSet| ...) is being compiled.
;; The variable |CachableSet;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib CHARNZ
; (DEFUN |CharacteristicNonZero| ...) is being compiled.
;; The variable |CharacteristicNonZero;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```

    finalizing nrlib CHARZ
; (DEFUN |CharacteristicZero| ...) is being compiled.
;; The variable |CharacteristicZero;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib DVARCAT
; (DEFUN |DifferentialVariableCategory| ...) is being compiled.
;; The variable |DifferentialVariableCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |DifferentialVariableCategory;| ...) is being compiled.
;; The variable |DifferentialVariableCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib ELEMFUN
; (DEFUN |ElementaryFunctionCategory| ...) is being compiled.
;; The variable |ElementaryFunctionCategory;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib EVALAB
; (DEFUN |Evalable| ...) is being compiled.
;; The variable |Evalable;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |Evalable;| ...) is being compiled.
;; The variable |Evalable;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib FEVALAB
; (DEFUN |FullyEvalableOver| ...) is being compiled.
;; The variable |FullyEvalableOver;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |FullyEvalableOver;| ...) is being compiled.
;; The variable |FullyEvalableOver;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib FPATMAB
; (DEFUN |FullyPatternMatchable| ...) is being compiled.
;; The variable |FullyPatternMatchable;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |FullyPatternMatchable;| ...) is being compiled.
;; The variable |FullyPatternMatchable;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib GROUP
; (DEFUN |Group| ...) is being compiled.
;; The variable |Group;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib IXAGG
; (DEFUN |IndexedAggregate| ...) is being compiled.
;; The variable |IndexedAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |IndexedAggregate;| ...) is being compiled.
;; The variable |IndexedAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

section

```
finalizing nrlib MONADWU
; (DEFUN |MonadWithUnit| ...) is being compiled.
;; The variable |MonadWithUnit;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib NARNG
; (DEFUN |NonAssociativeRng| ...) is being compiled.
;; The variable |NonAssociativeRng;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib OASGP
; (DEFUN |OrderedAbelianSemiGroup| ...) is being compiled.
;; The variable |OrderedAbelianSemiGroup;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib ORDFIN
; (DEFUN |OrderedFinite| ...) is being compiled.
;; The variable |OrderedFinite;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib ORDMON
; (DEFUN |OrderedMonoid| ...) is being compiled.
```



```
;; The variable |OrderedMonoid;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib PATMATCH
```

```
Warnings:
```

```
[1] ist: not known that (SetCategory) is of mode (CATEGORY package (SIGNATURE is? ((Boolean) Subject I
```

section

```
finalizing nrlib PERMCAT
```

```
; (DEFUN |PermutationCategory| ...) is being compiled.
;; The variable |PermutationCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |PermutationCategory;| ...) is being compiled.
;; The variable |PermutationCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib PDRING
```

```
; (DEFUN |PartialDifferentialRing| ...) is being compiled.
;; The variable |PartialDifferentialRing;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |PartialDifferentialRing;| ...) is being compiled.
;; The variable |PartialDifferentialRing;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib TRIGCAT
```

```
; (DEFUN |TrigonometricFunctionCategory| ...) is being compiled.
;; The variable |TrigonometricFunctionCategory;AL| is undefined.
```

```
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib BGAGG
; (DEFUN |BagAggregate| ...) is being compiled.
;; The variable |BagAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |BagAggregate;| ...) is being compiled.
;; The variable |BagAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib BRAGG
; (DEFUN |BinaryRecursiveAggregate| ...) is being compiled.
;; The variable |BinaryRecursiveAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |BinaryRecursiveAggregate;| ...) is being compiled.
;; The variable |BinaryRecursiveAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib DIFEXT
; (DEFUN |DifferentialExtension| ...) is being compiled.
;; The variable |DifferentialExtension;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |DifferentialExtension;| ...) is being compiled.
;; The variable |DifferentialExtension;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```

    finalizing nrlib DLAGG
; (DEFUN |DoublyLinkedAggregate| ...) is being compiled.
;; The variable |DoublyLinkedAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |DoublyLinkedAggregate;| ...) is being compiled.
;; The variable |DoublyLinkedAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib ELAGG
; (DEFUN |ExtensibleLinearAggregate| ...) is being compiled.
;; The variable |ExtensibleLinearAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |ExtensibleLinearAggregate;| ...) is being compiled.
;; The variable |ExtensibleLinearAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib ES2
Warnings:
    [1] map:  IN has no value
    [2] map:  x has no value

```

section

```

    finalizing nrlib GRMOD
; (DEFUN |GradedModule| ...) is being compiled.
;; The variable |GradedModule;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |GradedModule;| ...) is being compiled.
;; The variable |GradedModule;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib HYPCAT
; (DEFUN |HyperbolicFunctionCategory| ...) is being compiled.
;; The variable |HyperbolicFunctionCategory;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib MODRING
Warnings:
    [1] coerce: signature of lhs not unique: R$ chosen

```

section

```

    finalizing nrlib MODULE
; (DEFUN |Module| ...) is being compiled.
;; The variable |Module;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |Module;| ...) is being compiled.
;; The variable |Module;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib NASRING
; (DEFUN |NonAssociativeRing| ...) is being compiled.
;; The variable |NonAssociativeRing;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib ZMOD
; (DEFUN |IntegerMod;| ...) is being compiled.
;; The variable IDENTITY is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib ALGEBRA
; (DEFUN |Algebra| ...) is being compiled.
;; The variable |Algebra;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |Algebra;| ...) is being compiled.
;; The variable |Algebra;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib BTCAT
; (DEFUN |BinaryTreeCategory| ...) is being compiled.
;; The variable |BinaryTreeCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |BinaryTreeCategory;| ...) is being compiled.
;; The variable |BinaryTreeCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib FMCAT
; (DEFUN |FreeModuleCat| ...) is being compiled.
;; The variable |FreeModuleCat;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |FreeModuleCat;| ...) is being compiled.
;; The variable |FreeModuleCat;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

-----non extending category-----

```

.. InnerFreeAbelianMonoid(#1,#2,#3) of cat
(|FreeAbelianMonoidCategory| |#1| |#2|)  has no  outputForm : (%,(OutputForm,OutputForm) -> OutputForm),

```

section

```
finalizing nrlib IFAMON
```

section

```
finalizing nrlib GRALG
; (DEFUN |GradedAlgebra| ...) is being compiled.
;; The variable |GradedAlgebra;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |GradedAlgebra;| ...) is being compiled.
;; The variable |GradedAlgebra;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib OCAMON
; (DEFUN |OrderedCancellationAbelianMonoid| ...) is being compiled.
;; The variable |OrderedCancellationAbelianMonoid;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib PRQAGG
; (DEFUN |PriorityQueueAggregate| ...) is being compiled.
;; The variable |PriorityQueueAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |PriorityQueueAggregate;| ...) is being compiled.
;; The variable |PriorityQueueAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```

    finalizing nrlib QUAGG
; (DEFUN |QueueAggregate| ...) is being compiled.
;; The variable |QueueAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |QueueAggregate;| ...) is being compiled.
;; The variable |QueueAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib SKAGG
; (DEFUN |StackAggregate| ...) is being compiled.
;; The variable |StackAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |StackAggregate;| ...) is being compiled.
;; The variable |StackAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

-----non extending category-----

```

.. BinarySearchTree #1 of cat
(|Join| (|BinaryTreeCategory| |#1|) (CATEGORY |domain| (ATTRIBUTE |shallowlyMutable|) (ATTRIBUTE |finiteAgg|

```

-----non extending category-----

```

.. BinaryTournament #1 of cat
(|Join| (|BinaryTreeCategory| |#1|) (CATEGORY |domain| (ATTRIBUTE |shallowlyMutable|) (SIGNATURE |binaryTou

```

section

```

    finalizing nrlib CARD
Warnings:
  [1] **: signature of lhs not unique: $$$ chosen
  [2] **: :(NonNegativeInteger) -- should replace by pretend

```

section

```

    finalizing nrlib DQAGG
; (DEFUN |DequeueAggregate| ...) is being compiled.

```

```
;; The variable |DequeueAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |DequeueAggregate;| ...) is being compiled.
;; The variable |DequeueAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib FACTFUNC
Warnings:
[1] nthRoot: radi has no value
```

section

```
finalizing nrlib FMTC
; (DEFUN |FortranMachineTypeCategory| ...) is being compiled.
;; The variable |FortranMachineTypeCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib MLO
; (DEFUN |MonogenicLinearOperator| ...) is being compiled.
;; The variable |MonogenicLinearOperator;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |MonogenicLinearOperator;| ...) is being compiled.
;; The variable |MonogenicLinearOperator;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib NAALG
; (DEFUN |NonAssociativeAlgebra| ...) is being compiled.
;; The variable |NonAssociativeAlgebra;AL| is undefined.
```



```
;; The compiler will assume this variable is a global.
; (DEFUN |NonAssociativeAlgebra;| ...) is being compiled.
;; The variable |NonAssociativeAlgebra;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib NEWTON
---->NewtonInterpolation((newton ((SparseUnivariatePolynomial F) (List F)))): Not documented!!!!
```

section

```
finalizing nrlib OAGROUP
; (DEFUN |OrderedAbelianGroup| ...) is being compiled.
;; The variable |OrderedAbelianGroup;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib OAMONS
; (DEFUN |OrderedAbelianMonoidSup| ...) is being compiled.
;; The variable |OrderedAbelianMonoidSup;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib PID
; (DEFUN |PrincipalIdealDomain| ...) is being compiled.
;; The variable |PrincipalIdealDomain;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```

    finalizing nrlib XALG
; (DEFUN |XAlgebra| ...) is being compiled.
;; The variable |XAlgebra;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |XAlgebra;| ...) is being compiled.
;; The variable |XAlgebra;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib AMR
; (DEFUN |AbelianMonoidRing| ...) is being compiled.
;; The variable |AbelianMonoidRing;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |AbelianMonoidRing;| ...) is being compiled.
;; The variable |AbelianMonoidRing;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib FAGROUP
Warnings:
    [1] <:  exp has no value
    [2] <:  gen has no value

```

section

```

    finalizing nrlib FIELD-
Warnings:
    [1] squareFree: not known that (IntegralDomain) is of mode (CATEGORY domain (SIGNATURE / (

```

section

```

    finalizing nrlib FIELD
; (DEFUN |Field| ...) is being compiled.
;; The variable |Field;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib FLAGG
; (DEFUN |FiniteLinearAggregate| ...) is being compiled.
;; The variable |FiniteLinearAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |FiniteLinearAggregate;| ...) is being compiled.
;; The variable |FiniteLinearAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib FLINEXP-
Warnings:
    [1] reducedSystem: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE reducedSystem ((Matrix

```

section

```

    finalizing nrlib FLINEXP
; (DEFUN |FullyLinearlyExplicitRingOver| ...) is being compiled.
;; The variable |FullyLinearlyExplicitRingOver;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |FullyLinearlyExplicitRingOver;| ...) is being compiled.
;; The variable |FullyLinearlyExplicitRingOver;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

      finalizing nrlib FRETRCT
; (DEFUN |FullyRetractableTo| ...) is being compiled.
;; The variable |FullyRetractableTo;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |FullyRetractableTo;| ...) is being compiled.
;; The variable |FullyRetractableTo;CAT| is undefined.
;; The compiler will assume this variable is a global.

-----non extending category-----
.. FourierSeries(#1,#2) of cat
(|Join| (|Algebra| |#1|) (CATEGORY |domain| (IF (|has| |#2| (ATTRIBUTE |canonical|)) (IF (|has|

```

section

```

(|IndexedDirectProductCategory| |#1| (|FourierComponent| |#2|))      finalizing nrlib FSERIES

```

section

```

      finalizing nrlib FT
Semantic Errors:
  [1] void is not a known type

```

section

```

      finalizing nrlib IDPAG
Warnings:
  [1] -: res has no value
  [2] -: endcell has no value

-----non extending category-----
.. LocalAlgebra(#1,#2,#3) of cat
(|Join| (|Algebra| |#2|) (CATEGORY |domain| (IF (|has| |#1| (|OrderedRing|)) (ATTRIBUTE (|Ordere

```

section

```
(IF (|has| |#1| (|OrderedAbelianGroup|)) (ATTRIBUTE (|OrderedAbelianGroup|)) |noBranch|) finalizing nrlib
```

section

```
finalizing nrlib OML0
Warnings:
  [1] op: :$ -- should replace by @
  [2] po: :P -- should replace by pretend
  [3] *: :P -- should replace by pretend
  [4] coerce: :P -- should replace by pretend
```

section

```
finalizing nrlib PADICCT
; (DEFUN |PAdicIntegerCategory| ...) is being compiled.
;; The variable |PAdicIntegerCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |PAdicIntegerCategory;| ...) is being compiled.
;; The variable |PAdicIntegerCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib RADCAT
; (DEFUN |RadicalCategory| ...) is being compiled.
;; The variable |RadicalCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```

      finalizing nrlib XFALG
; (DEFUN |XFreeAlgebra| ...) is being compiled.
;; The variable |XFreeAlgebra;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |XFreeAlgebra;| ...) is being compiled.
;; The variable |XFreeAlgebra;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

      finalizing nrlib A1AGG
; (DEFUN |OneDimensionalArrayAggregate| ...) is being compiled.
;; The variable |OneDimensionalArrayAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |OneDimensionalArrayAggregate;| ...) is being compiled.
;; The variable |OneDimensionalArrayAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

      finalizing nrlib ARR2CAT
; (DEFUN |TwoDimensionalArrayCategory| ...) is being compiled.
;; The variable |TwoDimensionalArrayCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |TwoDimensionalArrayCategory;| ...) is being compiled.
;; The variable |TwoDimensionalArrayCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

      finalizing nrlib ASP34
Semantic Errors:
      [1] coerce: code is BOTH a variable and a literal

Warnings:
      [1] IFLAG has no value
      [2] N has no value
      [3] X has no value

```

```

[4] Y has no value
[5] LRWORK has no value
[6] LIWORK has no value
[7] RWORK has no value
[8] IWORK has no value
[9] coerce: I has no value
[10] coerce: J has no value
[11] coerce: W1 has no value
[12] coerce: W2 has no value
[13] coerce: MS has no value

```

-----non extending category-----

.. BalancedBinaryTree #1 of cat

(|Join| (|BinaryTreeCategory| |#1|) (CATEGORY |domain| (ATTRIBUTE |finiteAggregate|) (ATTRIBUTE |shallowlyM

section

finalizing nrlib BFUNCT

Processing BasicFunctions for Browser database:

--->/research2/test0819/mnt/fedora5/../../src/algebra/BFUNCT.spad-->BasicFunctions(constructor): Not document

--->/research2/test0819/mnt/fedora5/../../src/algebra/BFUNCT.spad-->BasicFunctions(): Missing Description

section

finalizing nrlib BTREE

Warnings:

```

[1] empty: pretend$ -- should replace by @
[2] node: :Rep -- should replace by pretend
[3] setvalue!: :Rep -- should replace by pretend
[4] setleft!: :Rep -- should replace by pretend
[5] setright!: :(List (Tree S)) -- should replace by pretend

```

-----non extending category-----

.. Dequeue #1 of cat

(|Join| (|DequeueAggregate| |#1|) (CATEGORY |domain| (SIGNATURE |dequeue| (\$ (|List| |#1|)))) has no q

-----non extending category-----

.. DataList #1 of cat

(|Join| (|ListAggregate| |#1|) (CATEGORY |domain| (SIGNATURE |coerce| (\$ (|List| |#1|))) (SIGNATURE |coerce

section

```

      finalizing nrlib DRAWCX
--->/research2/test0819/mnt/fedora5/../../src/algebra/DRAWCX.spad-->DrawComplex(constructor): No
--->/research2/test0819/mnt/fedora5/../../src/algebra/DRAWCX.spad-->DrawComplex(): Missing Descr

-----non extending category-----
.. d01gbfAnnaType of cat
(|NumericalIntegrationCategory|)  has no

```

section

```

(|TableAggregate| (|Symbol|) (|Any|))  finalizing nrlib D01GBFA

-----non extending category-----
.. d02ejfAnnaType of cat
(|OrdinaryDifferentialEquationsSolverCategory|)  has no

```

section

```

(|TableAggregate| (|Symbol|) (|Any|))  finalizing nrlib D02EJFA

-----non extending category-----
.. d03fafAnnaType of cat
(|PartialDifferentialEquationsSolverCategory|)  has no

```

section

```

(|TableAggregate| (|Symbol|) (|Any|))  finalizing nrlib D03FAFA

```

section

```

      finalizing nrlib FAMR
; (DEFUN |FiniteAbelianMonoidRing| ...) is being compiled.
;; The variable |FiniteAbelianMonoidRing;AL| is undefined.

```



```
;; The compiler will assume this variable is a global.
; (DEFUN |FiniteAbelianMonoidRing;| ...) is being compiled.
;; The variable |FiniteAbelianMonoidRing;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

```
FLASORT abbreviates package FiniteLinearAggregateSort
(PUSH (QUOTE |FiniteLinearAggregateSort|) |$mutableDomains|)
Value = (|FiniteLinearAggregateSort|)
```

```
-----non extending category-----
```

```
.. FreeGroup #1 of cat
```

```
(|Join| (|Group|) (|RetractableTo| |#1|) (CATEGORY |domain| (SIGNATURE * ($ |#1| $)) (SIGNATURE * ($ $ |#1|
```

section

```
finalizing nrlib FM1
```

```
Warnings:
```

```
[1] ListOfTerms: :(List (Record (: k S) (: c R))) -- should replace by pretend
```

```
-----non extending category-----
```

```
.. FreeModule1(#1,#2) of cat
```

```
(|Join| (|FreeModuleCat| |#1| |#2|) (CATEGORY |domain| (SIGNATURE * ($ |#2| |#1|)))) has no
```

section

```
(|IndexedDirectProductCategory| |#1| |#2|) finalizing nrlib FM1
```

section

```
finalizing nrlib FPC
```

```
; (DEFUN |FieldOfPrimeCharacteristic| ...) is being compiled.
```

```
;; The variable |FieldOfPrimeCharacteristic;AL| is undefined.
```

```
;; The compiler will assume this variable is a global.
```

section

```

finalizing nrlib FMONOID
Warnings:
[1] overlap:  l has no value
[2] divide:   l has no value
[3] hclf:    h has no value
[4] lquo:    gen has no value
[5] lquo:    exp has no value

```

section

```

finalizing nrlib FMONOID
-----non extending category-----
.. FreeMonoid #1 of cat
(|Join| (|Monoid|) (|RetractableTo| |#1|) (CATEGORY |domain| (SIGNATURE * ($ |#1| $)) (SIGNATURE

```

section

```

finalizing nrlib IPADIC
Warnings:
[1] padic: pretend$ -- should replace by @
[2] =: st has no value
[3] intToPAdic: digit has no value
[4] intToPAdic: carry has no value
[5] intPlusPAdic: digit has no value
[6] intPlusPAdic: carry has no value
[7] intMinusPAdic: digit has no value
[8] intMinusPAdic: carry has no value
[9] plusAux: digit has no value
[10] plusAux: carry has no value
[11] minusAux: digit has no value
[12] minusAux: carry has no value
[13] intMult: digit has no value
[14] intMult: carry has no value
[15] timesAux: digit has no value
[16] timesAux: carry has no value
[17] coerce: pretend(Integer) -- should replace by @
[18] coerce: 1 has no value

-----non extending category-----
.. LieExponentials(#1,#2,#3) of cat
(|Join| (|Group|) (CATEGORY |domain| (SIGNATURE |exp| ($ (|LiePolynomial| |#1| |#2|)))) (SIGNATUR

```

section

```
(|XPolynomialsCat| |#1| |#2|)    finalizing nrlib LEXP
```

section

```
    finalizing nrlib LIECAT
; (DEFUN |LieAlgebra| ...) is being compiled.
;; The variable |LieAlgebra;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |LieAlgebra;| ...) is being compiled.
;; The variable |LieAlgebra;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
    finalizing nrlib LMOPS
Warnings:
  [1] mapExpon:  ans has no value
  [2] outputForm: The conditional modes (List (OutputForm)) and $ conflict
```

section

```
    finalizing nrlib LZSTAGG
; (DEFUN |LazyStreamAggregate| ...) is being compiled.
;; The variable |LazyStreamAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |LazyStreamAggregate;| ...) is being compiled.
;; The variable |LazyStreamAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```

finalizing nrlib MOEBIUS
Semantic Errors:
[1] a is BOTH a variable and a literal
[2] b is BOTH a variable and a literal
[3] c is BOTH a variable and a literal
[4] d is BOTH a variable and a literal
[5] eval: a is BOTH a variable and a literal
[6] eval: b is BOTH a variable and a literal
[7] eval: c is BOTH a variable and a literal
[8] eval: d is BOTH a variable and a literal
[9] proportional?: a is BOTH a variable and a literal
[10] proportional?: b is BOTH a variable and a literal
[11] proportional?: c is BOTH a variable and a literal
[12] proportional?: d is BOTH a variable and a literal

Warnings:
[1] recip: signature of lhs not unique: $$ chosen

-----non extending category-----
.. ModuleOperator(#1,#2) of cat
(|Join| (|Ring|) (|RetractableTo| |#1|) (|RetractableTo| (|BasicOperator|)) (|Eltable| |#2| |#2|

```

section

```
(|Module| (|Integer|))      finalizing nrlib MODOP
```

section

```

finalizing nrlib MRING
Warnings:
[1] index: ans has no value
[2] retractIfCan: coef has no value
[3] retractIfCan: monom has no value
[4] recip: monom has no value
[5] recip: coef has no value
[6] coerce: coef has no value
[7] coerce: monom has no value
[8] =: coef has no value
[9] =: monom has no value
[10] sortAndAdd: m has no value
[11] sortAndAdd: res has no value

```

-----non extending category-----

.. OrderedFreeMonoid #1 of cat

(|Join| (|OrderedMonoid|) (|RetractableTo| |#1|) (CATEGORY |domain| (SIGNATURE * (\$ |#1| \$)) (SIGNATURE * (

section

finalizing nrlib ONECOMP

; (DEFUN |OnePointCompletion;| ...) is being compiled.

;; The variable IDENTITY is undefined.

;; The compiler will assume this variable is a global.

section

finalizing nrlib ORDCOMP

Warnings:

- [1] retract: fin has no value
- [2] retractIfCan: fin has no value
- [3] coerce: fin has no value
- [4] coerce: inf has no value
- [5] whatInfinity: inf has no value
- [6] =: inf has no value
- [7] =: fin has no value
- [8] *: inf has no value
- [9] *: fin has no value
- [10] -: inf has no value
- [11] -: fin has no value
- [12] +: inf has no value
- [13] +: fin has no value
- [14] recip: fin has no value
- [15] <: inf has no value
- [16] <: fin has no value

section

finalizing nrlib ORDCOMP

; (DEFUN |OrderedCompletion;| ...) is being compiled.

;; The variable IDENTITY is undefined.

;; The compiler will assume this variable is a global.

section

```

finalizing nrlib OREPCAT
; (DEFUN |UnivariateSkewPolynomialCategory| ...) is being compiled.
;; The variable |UnivariateSkewPolynomialCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |UnivariateSkewPolynomialCategory;| ...) is being compiled.
;; The variable |UnivariateSkewPolynomialCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

finalizing nrlib PENDTREE
Warnings:
[1] ptree: pretend$ -- should replace by @
[2] ptree: :Rep -- should replace by pretend
[3] ptree: :$ -- should replace by @
[4] =: :Rep -- should replace by pretend

```

section

```

finalizing nrlib PFR
Warnings:
[1] LessThan: :(Boolean) -- should replace by pretend
[2] multiplyFracTerms: :Rep -- should replace by pretend
[3] multiplyFracTerms: c has no value
[4] compactFraction: :(NonNegativeInteger) -- should replace by pretend
[5] compactFraction: s has no value
[6] compactFraction: bf has no value

```

section

```

finalizing nrlib PMDOWN

```

Warnings:

```
[1] patternMatch: not known that (SetCategory) is of mode (CATEGORY package (SIGNATURE fixPredicate (
```

section

finalizing nrlib PRTITION

Warnings:

```
[1] coerce: signature of lhs not unique: (List (Integer))$ chosen
[2] conjugate: pretendRep -- should replace by @
[3] coerce: pretendRep -- should replace by @
[4] pdct: pretendRep -- should replace by @
```

section

finalizing nrlib PMLSAGG

Warnings:

```
[1] match: not known that (SetCategory) is of mode (CATEGORY package (SIGNATURE patternMatch ((Pattern
```

section

finalizing nrlib PSCAT

```
; (DEFUN |PowerSeriesCategory| ...) is being compiled.
;; The variable |PowerSeriesCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |PowerSeriesCategory;| ...) is being compiled.
;; The variable |PowerSeriesCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

-----non extending category-----

.. QuadraticForm(#1,#2) of cat

```
(|Join| (|AbelianGroup|) (CATEGORY |domain| (SIGNATURE |quadraticForm| ($ (|SquareMatrix| |#1| |#2|))) (SIG
```

section

```
(|SquareMatrixCategory| |#1| |#2| (|DirectProduct| |#1| |#2|) (|DirectProduct| |#1| |#2|)) finalizing nr
```

-----non extending category-----

.. Queue #1 of cat

(|Join| (|QueueAggregate| |#1|) (CATEGORY |domain| (SIGNATURE |queue| (\$ (|List| |#1|)))))) has

section

(|StackAggregate| |#1|) finalizing nrlib QUEUE

section

finalizing nrlib SEXOF

Warnings:

[1] convert: pretend\$ -- should replace by @

section

finalizing nrlib STTAYLOR

Warnings:

[1] powern: order has no value

section

finalizing nrlib TRANFUN

; (DEFUN |TranscendentalFunctionCategory| ...) is being compiled.

;; The variable |TranscendentalFunctionCategory;AL| is undefined.

;; The compiler will assume this variable is a global.

section

finalizing nrlib VSPACE


```
; (DEFUN |VectorSpace| ...) is being compiled.
;; The variable |VectorSpace;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |VectorSpace;| ...) is being compiled.
;; The variable |VectorSpace;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib XPOLYC
; (DEFUN |XPolynomialsCat| ...) is being compiled.
;; The variable |XPolynomialsCat;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |XPolynomialsCat;| ...) is being compiled.
;; The variable |XPolynomialsCat;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib XPR
Warnings:
  [1] quasiRegular?: k has no value
  [2] *: k has no value
  [3] *: c has no value

-----non extending category-----
.. XPolynomialRing(#1,#2) of cat
(|Join| (|Ring|) (|XAlgebra| |#1|) (|FreeModuleCat| |#1| |#2|) (CATEGORY |domain| (SIGNATURE * ($ $ |#1|)))
```

section

```
finalizing nrlib COMBINAT
Warnings:
  [1] binomial: s has no value
  [2] binomial: b has no value

-----non extending category-----
.. d01fcfAnnaType of cat
```

```
(|NumericalIntegrationCategory|)    has no
```

section

```
(|TableAggregate| (|Symbol|) (|Any|))    finalizing nrlib D01FCFA
```

```
-----non extending category-----
```

```
.. e04mbfAnnaType of cat
```

```
(|NumericalOptimizationCategory|)    has no
```

section

```
(|TableAggregate| (|Symbol|) (|Any|))    finalizing nrlib E04MBFA
```

section

```
    finalizing nrlib FLALG
; (DEFUN |FreeLieAlgebra| ...) is being compiled.
;; The variable |FreeLieAlgebra;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |FreeLieAlgebra;| ...) is being compiled.
;; The variable |FreeLieAlgebra;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

```
-----non extending category-----
```

```
.. Heap #1 of cat
```

```
(|Join| (|PriorityQueueAggregate| |#1|) (CATEGORY |domain| (SIGNATURE |heap| ($ (|List| |#1|))))
```

section

```
(|OneDimensionalArrayAggregate| |#1|)    finalizing nrlib HEAP
```

section

```
finalizing nrlib IFARRAY
Warnings:
  [1] remove!: k has no value
  [2] select!: k has no value
```

section

```
finalizing nrlib INTCAT
; (DEFUN |IntervalCategory| ...) is being compiled.
;; The variable |IntervalCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |IntervalCategory;| ...) is being compiled.
;; The variable |IntervalCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib INTHEORY
Warnings:
  [1] fibonacci: f2 has no value
  [2] jacobi: j has no value
  [3] eulerPhi: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE bernoulli ((Fra
```

section

```
finalizing nrlib LFCAT
; (DEFUN |LiouvillianFunctionCategory| ...) is being compiled.
;; The variable |LiouvillianFunctionCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib LODOCAT
```

```
; (DEFUN |LinearOrdinaryDifferentialOperatorCategory| ...) is being compiled.
;; The variable |LinearOrdinaryDifferentialOperatorCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |LinearOrdinaryDifferentialOperatorCategory;| ...) is being compiled.
;; The variable |LinearOrdinaryDifferentialOperatorCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib LWORD
Warnings:
[1] factor1: d has no value
[2] LyndonWordsList1: :(Integer) -- should replace by pretend
[3] LyndonWordsList1: lbase1 has no value

-----non extending category-----
.. LyndonWord #1 of cat
(|Join| (|OrderedSet|) (|RetractableTo| |#1|) (CATEGORY |domain| (SIGNATURE |retractable?| ((|Bo
```

section

```
finalizing nrlib LWORD
-->/research2/test0819/mnt/fedora5/../../src/algebra/LWORD.spad-->LyndonWord(): Spurious commen
```

section

```
finalizing nrlib MATCAT
; (DEFUN |MatrixCategory| ...) is being compiled.
;; The variable |MatrixCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |MatrixCategory;| ...) is being compiled.
;; The variable |MatrixCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

-----non extending category-----
.. SparseUnivariateSkewPolynomial(#1,#2,#3) of cat
(|Join| (|UnivariateSkewPolynomialCategory| |#1|) (CATEGORY |domain| (SIGNATURE |outputForm| ((|
```

section

```
(|UnivariatePolynomialCategory| |#1|)      finalizing nrlib ORESUP
      Processing SparseUnivariateSkewPolynomial for Browser database:
```

```
-----non extending category-----
```

```
.. UnivariateSkewPolynomial(#1,#2,#3,#4) of cat
```

```
(|Join| (|UnivariateSkewPolynomialCategory| |#2|) (CATEGORY |domain| (SIGNATURE |coerce| ($ (|Variable| |#1|
```

section

```
finalizing nrlib PLOT3D
```

```
Warnings:
```

- [1] select: fp has no value
- [2] rangeRefine: c has no value
- [3] rangeRefine: q has no value
- [4] rangeRefine: NUMFUNEVALS has no value
- [5] refine: curves has no value
- [6] plot: curves has no value

section

```
finalizing nrlib PLOT3D
```

```
; (DEFUN |PLOT3D;select| ...) is being compiled.
```

```
;; The variable |$NaNvalue| is undefined.
```

```
;; The compiler will assume this variable is a global.
```

```
; (DEFUN |PLOT3D;myTrap| ...) is being compiled.
```

```
;; The variable |$numericFailure| is undefined.
```

```
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib PR
```

```
Warnings:
```

- [1] times!: endcell has no value
- [2] times!: res has no value

```

[3] addm!: newcell has no value
[4] addm!: res has no value
[5] addm!: endcell has no value
[6] **: pretend(NonNegativeInteger) -- should replace by @
[7] ^: pretend(NonNegativeInteger) -- should replace by @
[8] unitNormal: lcf has no value
[9] unitCanonical: lcf has no value
[10] fmecg: rout has no value

```

-----non extending category-----

```

.. PolynomialRing(#1,#2) of cat
(|Join| (|FiniteAbelianMonoidRing| |#1| |#2|) (CATEGORY |domain| (IF (|has| |#1| (|IntegralDomain|

```

section

```

(|IndexedDirectProductCategory| |#1| |#2|)    finalizing nrlib PR

```

section

```

    finalizing nrlib SRAGG
; (DEFUN |StringAggregate| ...) is being compiled.
;; The variable |StringAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

finalizing nrlib STREAM
Warnings:
[1] pretendS -- should replace by @
[2] showAllElements: not known that (LazyStreamAggregate S) is of mode (CATEGORY domain (S
[3] remove: signature of lhs not unique: $(Mapping (Boolean) S)$ chosen
[4] lazyEval: :(Mapping $) -- should replace by pretend
[5] expand!: d has no value

```

section

```

    finalizing nrlib STREAM
; (DEFUN |STREAM;showAll?;B;12| ...) is being compiled.
;; The variable |$streamsShowAll| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |Stream;| ...) is being compiled.
;; The variable |$NullStream| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$NonNullStream| is undefined.
;; The compiler will assume this variable is a global.

-----non extending category-----
.. TaylorSeries #1 of cat
(|Join| (|MultivariateTaylorSeriesCategory| |#1| (|Symbol|)) (CATEGORY |domain| (SIGNATURE |coefficient| ((

```

section

```

    finalizing nrlib UPSCAT
; (DEFUN |UnivariatePowerSeriesCategory| ...) is being compiled.
;; The variable |UnivariatePowerSeriesCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |UnivariatePowerSeriesCategory;| ...) is being compiled.
;; The variable |UnivariatePowerSeriesCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib VECTCAT
; (DEFUN |VectorCategory| ...) is being compiled.
;; The variable |VectorCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |VectorCategory;| ...) is being compiled.
;; The variable |VectorCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

-----non extending category-----
.. XDistributedPolynomial(#1,#2) of cat
(|Join| (|FreeModuleCat| |#2| (|OrderedFreeMonoid| |#1|)) (|XPolynomialsCat| |#1| |#2|)) has no #? : % .

```

section

```

    finalizing nrlib XF
; (DEFUN |ExtensionField| ...) is being compiled.
;; The variable |ExtensionField;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |ExtensionField;| ...) is being compiled.
;; The variable |ExtensionField;CAT| is undefined.
;; The compiler will assume this variable is a global.

-----non extending category-----
.. XPBWPolynomial(#1,#2) of cat
(|Join| (|XPolynomialsCat| |#1| |#2|) (|FreeModuleCat| |#2| (|PoincareBirkhoffWittLyndonBasis| |#1|)))
--->/research2/test0819/mnt/fedora5/../../src/algebra/XRPOLY.spad-->XRecursivePolynomial(): Spu

-----non extending category-----
.. Bits of cat
(|Join| (|BitAggregate|) (CATEGORY |domain| (SIGNATURE |bits| ($ (|NonNegativeInteger|) (|Boolean|))))))

-----non extending category-----
.. LiePolynomial(#1,#2) of cat
(|Join| (|FreeLieAlgebra| |#1| |#2|) (|FreeModuleCat| |#2| (|LyndonWord| |#1|))) (CATEGORY |domain|)

```

section

```

    finalizing nrlib PTCAT
; (DEFUN |PointCategory| ...) is being compiled.
;; The variable |PointCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |PointCategory;| ...) is being compiled.
;; The variable |PointCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib STRICAT
; (DEFUN |StringCategory| ...) is being compiled.
;; The variable |StringCategory;AL| is undefined.
;; The compiler will assume this variable is a global.

```


section

```
finalizing nrlib CARTEN
Warnings:
  [1] index2int:  n has no value
  [2] permsign!:  nTrans has no value
  [3] elt:  n has no value
```

section

```
finalizing nrlib CLIP
Warnings:
  [1] discardAndSplit:  ans has no value
  [2] clip:  yMax has no value
  [3] clip:  yMin has no value
```

-----non extending category-----

.. Database #1 of cat

```
(|Join| (|SetCategory|) (CATEGORY |domain| (SIGNATURE |elt| ($ $ (|QueryEquation|))) (SIGNATURE |elt| ((|Data
```

section

```
(|ListAggregate| |#1|)    finalizing nrlib DBASE
```

-----non extending category-----

.. DenavitHartenbergMatrix #1 of cat

```
(|Join| (|MatrixCategory| |#1| (|Vector| |#1|) (|Vector| |#1|)) (CATEGORY |domain| (SIGNATURE * ((|Point| |
```

section

```
finalizing nrlib DIOSP
Warnings:
  [1] dioSolve:  c has no value
```

section

```
finalizing nrlib DIRPCAT-
Warnings:
  [1] reducedSystem: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE coerce (S
```

section

```
finalizing nrlib DIRPCAT
; (DEFUN |DirectProductCategory| ...) is being compiled.
;; The variable |DirectProductCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |DirectProductCategory;| ...) is being compiled.
;; The variable |DirectProductCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

-----non extending category-----
.. d02bbfAnnaType of cat
(|OrdinaryDifferentialEquationsSolverCategory|) has no
```

section

```
(|TableAggregate| (|Symbol|) (|Any|)) finalizing nrlib D02BBFA
Processing d02bbfAnnaType for Browser database:

-----non extending category-----
.. d02bhfAnnaType of cat
(|OrdinaryDifferentialEquationsSolverCategory|) has no
```

section

```
(|TableAggregate| (|Symbol|) (|Any|)) finalizing nrlib D02BHFA
```

section

```
finalizing nrlib D02CJFA
Warnings:
  [1] ODEsolve: i has no value
```

```
-----non extending category-----
.. d02cjfAnnaType of cat
(|OrdinaryDifferentialEquationsSolverCategory|) has no
```

section

```
(|TableAggregate| (|Symbol|) (|Any|)) finalizing nrlib D02CJFA
```

section

```
finalizing nrlib FAXF-
Warnings:
  [1] extensionDegree: signature of lhs not unique: (PositiveInteger) chosen
  [2] minimalPolynomial: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE charthRoot ((Union
  [3] degree: signature of lhs not unique: (PositiveInteger)S chosen
```

section

```
finalizing nrlib FAXF
; (DEFUN |FiniteAlgebraicExtensionField| ...) is being compiled.
;; The variable |FiniteAlgebraicExtensionField;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |FiniteAlgebraicExtensionField;| ...) is being compiled.
;; The variable |FiniteAlgebraicExtensionField;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib FFPOLY2
Warnings:
```

```
[1] rootOfIrreduciblePoly: basispointer has no value
[2] rootOfIrreduciblePoly: beta has no value
```

section

```
finalizing nrlib FNLA
Warnings:
  [1] have: hi has no value

-----non extending category-----
.. FreeNilpotentLie(#1,#2,#3) of cat
(|Join| (|NonAssociativeAlgebra| |#3|) (CATEGORY |domain| (SIGNATURE |dimension| ((|NonNegativeI
```

section

```
(|IndexedDirectProductCategory| |#3| (|OrdSetInts|))      finalizing nrlib FNLA
```

section

```
finalizing nrlib IRSN
Warnings:
  [1] aIdInverse: :(NonNegativeInteger) -- should replace by @
  [2] signum: s has no value
  [3] signum: z has no value
  [4] testPermutation: n has no value
  [5] testPermutation: ok has no value
```

section

```
finalizing nrlib MHROWRED
Warnings:
  [1] non0: allZero has no value
  [2] non0: ans has no value
  [3] mkMat: STEP has no value
  [4] mkMat: i has no value
```

```

[5] diagSubMatrix:  IN has no value
[6] diagSubMatrix:  r has no value
[7] diagSubMatrix:  STEP has no value
[8] diagSubMatrix:  z has no value
[9] rowEchelonLocal: pivord has no value

```

section

```

finalizing nrlib NUMQUAD
Warnings:
  [1] romberg:  change has no value
  [2] simpson:  change has no value
  [3] trapezoidal:  change has no value
  [4] rombergo:  change has no value
  [5] simpsono:  change has no value
  [6] trapezoidal:  change has no value

```

section

```

finalizing nrlib ODESYS
Warnings:
  [1] backsolve:  part has no value

```

section

```

finalizing nrlib PERMAN
Warnings:
  [1] permanent3:  a has no value
  [2] permanent:  :(Integer) -- should replace by pretend
  [3] permanent:  :(PositiveInteger) -- should replace by pretend
  [4] permanent:  a has no value
  [5] permanent2:  :(Integer) -- should replace by pretend
  [6] permanent2:  :(PositiveInteger) -- should replace by pretend
  [7] permanent2:  :(NonNegativeInteger) -- should replace by pretend
  [8] permanent2:  a has no value

```

section

```

finalizing nrlib PERMAN
Processing Permanent for Browser database:
--->/research2/test0819/mnt/fedora5/../../src/algebra/PERMAN.spad-->Permanent((commutative (attr
-----permanent (R SM))-----
--->/research2/test0819/mnt/fedora5/../../src/algebra/PERMAN.spad-->Permanent((permanent (R SM))
--->/research2/test0819/mnt/fedora5/../../src/algebra/PERMAN.spad-->Permanent((permanent (R SM))
"\\spad{permanent(x)} computes the permanent of a square matrix \\spad{x}. The {\em permanent}

```

section

```

finalizing nrlib PFECAT-
Warnings:
[1] gcdPolynomial: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE charthRoot
[2] charthRoot: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE charthRoot ((
[3] solveLinearPolynomialEquation: not known that (Ring) is of mode (CATEGORY domain (SIGN

```

section

```

finalizing nrlib PFECAT
; (DEFUN |PolynomialFactorizationExplicit| ...) is being compiled.
;; The variable |PolynomialFactorizationExplicit;AL| is undefined.
;; The compiler will assume this variable is a global.

-----non extending category-----
.. Point #1 of cat
(|PointCategory| |#1|) has no vector : List #1 -> %

```

section

```

finalizing nrlib POINT

```

section

```
finalizing nrlib PSEUDLIN
Warnings:
  [1] normalForm0: B has no value
  [2] normalForm0: Binv has no value
```

section

```
finalizing nrlib REP2
Warnings:
  [1] cyclicSubmodule: basis has no value
  [2] standardBasisOfCyclicSubmodule: standardBasis has no value
  [3] splitInternal: submoduleRepresentation has no value
  [4] splitInternal: factormoduleRepresentation has no value
  [5] areEquivalent?: x0 has no value
  [6] areEquivalent?: x1 has no value
  [7] areEquivalent?: foundResult has no value
  [8] areEquivalent?: result has no value
  [9] areEquivalent?: transitionM has no value
  [10] isAbsolutelyIrreducible?: result has no value
  [11] meatAxe: :(PositiveInteger) -- should replace by pretend
  [12] meatAxe: x has no value
  [13] meatAxe: foundResult has no value
  [14] meatAxe: result has no value
  [15] meatAxe: signature of lhs not unique: (List (List (Matrix R)))(List (Matrix R))(Boolean) chosen
```

section

```
finalizing nrlib SETMN
Warnings:
  [1] elements: l has no value
  [2] replaceKthElement: found has no value
  [3] replaceKthElement: i has no value
```

section

```

finalizing nrlib STRING
Warnings:
  [1] OMwrite: pretend(String) -- should replace by @

-----non extending category-----
.. String of cat
(|StringCategory|)    has no  hash : % -> Integer

```

section

```

finalizing nrlib STRING
--->/research2/test0819/mnt/fedora5/../../src/algebra/STRING.spad-->String(): Missing Description

```

section

```

finalizing nrlib ASP1
Warnings:
  [1] X has no value

```

section

```

finalizing nrlib ASP10
Warnings:
  [1] P has no value
  [2] Q has no value
  [3] DQDL has no value
  [4] X has no value
  [5] ELAM has no value
  [6] JINT has no value

```

section

```

finalizing nrlib ASP24

```


Warnings:

- [1] N has no value
- [2] XC has no value
- [3] FC has no value
- [4] coerce: FC has no value

section

finalizing nrlib ASP4

Warnings:

- [1] NDIM has no value
- [2] X has no value

section

finalizing nrlib ASP50

Warnings:

- [1] M has no value
- [2] N has no value
- [3] XC has no value
- [4] FVECC has no value
- [5] I has no value
- [6] coerce: FVECC has no value

section

finalizing nrlib ASP6

Warnings:

- [1] N has no value
- [2] X has no value
- [3] FVEC has no value
- [4] IFLAG has no value
- [5] coerce: FVEC has no value

section

```
finalizing nrlib ASP73
Warnings:
  [1] X has no value
  [2] Y has no value
  [3] ALPHA has no value
  [4] BETA has no value
  [5] GAMMA has no value
  [6] DELTA has no value
  [7] EPSOLN has no value
  [8] PHI has no value
  [9] PSI has no value
```

section

```
finalizing nrlib BALFACT
Warnings:
  [1] balancedFactorisation: not known that (IntegralDomain) is of mode (CATEGORY package (S
-----non extending category-----
.. BinaryExpansion of cat
(|Join| (|QuotientFieldCategory| (|Integer|)) (CATEGORY |domain| (SIGNATURE |coerce| ((|Fraction
```

section

```
finalizing nrlib BOUNDZRO
Warnings:
  [1] bringDown: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE integer
  [2] qbound: bound has no value
```

section

```
finalizing nrlib CHVAR
Warnings:
  [1] eval: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE mkIntegr
```

```
[2] rootPoly: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE mkIntegral ((Re
```

section

```
finalizing nrlib CONTFRAC
Warnings:
[1] =: whole has no value
[2] =: den has no value
[3] partialQuotients: whole has no value
[4] extend: fract has no value
[5] complete: fract has no value
[6] iGenApproximants: num has no value
[7] iGenApproximants: den has no value
[8] iGenConvergents: num has no value
[9] iGenConvergents: den has no value
[10] coerce: 1 has no value
```

section

```
finalizing nrlib CONTFRAC
; (DEFUN |CONTFRAC;showAll?| ...) is being compiled.
;; The variable |$streamsShowAll| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib CYCLOTOM
Warnings:
[1] cyclotomic: 1 has no value
```

section

```
finalizing nrlib DDFACT
Warnings:
```

```

[1] notSqFr: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE facto
[2] exptMod: :FP -- should replace by pretend
[3] ddfact1: ddfact has no value
[4] separateFactors: n1 has no value
[5] separateFactors: ris has no value
[6] separateFactors: newaux has no value
[7] distdfact: factlist has no value

-----non extending category-----
.. DecimalExpansion of cat
(|Join| (|QuotientFieldCategory| (|Integer|)) (CATEGORY |domain| (SIGNATURE |coerce| ((|Fraction

```

section

```

finalizing nrlib DIOPS
; (DEFUN |DictionaryOperations| ...) is being compiled.
;; The variable |DictionaryOperations;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |DictionaryOperations;| ...) is being compiled.
;; The variable |DictionaryOperations;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

finalizing nrlib DIRPROD
Warnings:
[1] Zero: pretend$ -- should replace by @
[2] One: pretend$ -- should replace by @
[3] subtractIfCan: pretend$ -- should replace by @
[4] recip: pretend$ -- should replace by @
[5] unitVector: pretend$ -- should replace by @

-----non extending category-----
.. DirectProduct(#1,#2) of cat
(|DirectProductCategory| |#1| |#2|) has no

```

section

```

(|VectorCategory| |#2|) finalizing nrlib DIRPROD

```

section

```

    finalizing nrlib DIRPROD
; (DEFUN |DirectProduct;| ...) is being compiled.
;; The variable IDENTITY is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib DISPLAY
Warnings:
  [1] sayLength:  sum has no value

```

section

```

    finalizing nrlib DPOLCAT-
Warnings:
  [1] degree:  d has no value
  [2] weights:  ws has no value
  [3] initial: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE coerce (A A)) (SIGNATURE coerce (A A)))
  [4] eval:  IN has no value
  [5] eval:  s has no value
  [6] eval:  e has no value

```

section

```

    finalizing nrlib DPOLCAT
; (DEFUN |DifferentialPolynomialCategory| ...) is being compiled.
;; The variable |DifferentialPolynomialCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |DifferentialPolynomialCategory;| ...) is being compiled.
;; The variable |DifferentialPolynomialCategory;CAT| is undefined.

```

```
;; The compiler will assume this variable is a global.
```

```
-----non extending category-----
.. d01ajfAnnaType of cat
(|NumericalIntegrationCategory|)  has no
```

section

```
(|TableAggregate| (|Symbol|) (|Any|))  finalizing nrlib D01AJFA
```

```
-----non extending category-----
.. d01akfAnnaType of cat
(|NumericalIntegrationCategory|)  has no
```

section

```
(|TableAggregate| (|Symbol|) (|Any|))  finalizing nrlib D01AKFA
```

```
-----non extending category-----
.. d01alfAnnaType of cat
(|NumericalIntegrationCategory|)  has no
```

section

```
(|TableAggregate| (|Symbol|) (|Any|))  finalizing nrlib D01ALFA
```

```
-----non extending category-----
.. d01amfAnnaType of cat
(|NumericalIntegrationCategory|)  has no
```

section

```
(|TableAggregate| (|Symbol|) (|Any|))  finalizing nrlib D01AMFA
```

```
-----non extending category-----
.. d01apfAnnaType of cat
(|NumericalIntegrationCategory|)  has no
```

section

```
(|TableAggregate| (|Symbol|) (|Any|))    finalizing nrlib D01APFA

-----non extending category-----
.. d01aqfAnnaType of cat
(|NumericalIntegrationCategory|)    has no
```

section

```
(|TableAggregate| (|Symbol|) (|Any|))    finalizing nrlib D01AQFA
```

section

```
finalizing nrlib EQ
extension of ##1 to (Polynomial (Integer)) ignored
```

section

```
finalizing nrlib EQ
Semantic Errors:
  [1] factorAndSplit: rcf has two modes:

Warnings:
  [1] factorAndSplit: not known that (IntegralDomain) is of mode (CATEGORY domain (SIGNATURE factorAndSp
  [2] factorAndSplit: not known that (IntegralDomain) is of mode (CATEGORY S (SIGNATURE factor ((Factor
```

section

```
finalizing nrlib EQ
; (DEFUN |Equation;| ...) is being compiled.
;; The variable IDENTITY is undefined.
;; The compiler will assume this variable is a global.
```

```
-----non extending category-----
.. e04dgfAnnaType of cat
(|NumericalOptimizationCategory|)  has no
```

section

```
(|TableAggregate| (|Symbol|) (|Any|))  finalizing nrlib E04DGFA
```

```
-----non extending category-----
.. e04fdfAnnaType of cat
(|NumericalOptimizationCategory|)  has no
```

section

```
(|TableAggregate| (|Symbol|) (|Any|))  finalizing nrlib E04FDFA
```

```
-----non extending category-----
.. e04gcfAnnaType of cat
(|NumericalOptimizationCategory|)  has no
```

section

```
(|TableAggregate| (|Symbol|) (|Any|))  finalizing nrlib E04GCFA
```

```
-----non extending category-----
.. e04jafAnnaType of cat
(|NumericalOptimizationCategory|)  has no
```

section

```
(|TableAggregate| (|Symbol|) (|Any|))  finalizing nrlib E04JAFA
```

section

finalizing nrlib FFHOM

Warnings:

- [1] compare: equal has no value
- [2] initialize: mat has no value
- [3] unknown Functor code (error FFHOM: one extension degree must divide the other one)

section

finalizing nrlib FFPOLY

Warnings:

- [1] leastAffineMultiple: coeffVector has no value
- [2] primitive?: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE primitive? ((
- [3] nextSubset: restOfs has no value
- [4] nextSubset: noGap has no value
- [5] nextSubset: i has no value
- [6] nextIrreduciblePoly: fcopy has no value
- [7] nextIrreduciblePoly: s has no value
- [8] nextPrimitivePoly: fcopy has no value
- [9] nextPrimitivePoly: term has no value
- [10] nextPrimitivePoly: noGenerator has no value
- [11] nextPrimitivePoly: c has no value
- [12] nextPrimitivePoly: weight has no value
- [13] nextPrimitivePoly: s has no value
- [14] nextPrimitivePoly: l has no value
- [15] nextNormalPoly: fcopy has no value
- [16] nextNormalPoly: l has no value
- [17] nextNormalPoly: a has no value
- [18] nextNormalPoly: s has no value
- [19] nextNormalPrimitivePoly: fcopy has no value
- [20] nextNormalPrimitivePoly: term has no value
- [21] nextNormalPrimitivePoly: noGenerator has no value
- [22] nextNormalPrimitivePoly: c has no value
- [23] nextNormalPrimitivePoly: la has no value
- [24] nextNormalPrimitivePoly: a has no value
- [25] nextNormalPrimitivePoly: middlelookuplist has no value
- [26] nextNormalPrimitivePoly: middlepol has no value
- [27] nextNormalPrimitivePoly: weight has no value
- [28] nextNormalPrimitivePoly: s has no value
- [29] nextNormalPrimitivePoly: lc has no value
- [30] random: polRepr has no value

section

```

finalizing nrlib FGLMICPK
Warnings:
  [1] zeroDim?: not known that (Ring) is of mode (CATEGORY package (SIGNATURE zeroDimensiona
  [2] zeroDim?: lv has no value

```

section

```

finalizing nrlib FINAALG
; (DEFUN |FiniteRankNonAssociativeAlgebra| ...) is being compiled.
;; The variable |FiniteRankNonAssociativeAlgebra;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |FiniteRankNonAssociativeAlgebra;| ...) is being compiled.
;; The variable |FiniteRankNonAssociativeAlgebra;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

finalizing nrlib FINRALG
; (DEFUN |FiniteRankAlgebra| ...) is being compiled.
;; The variable |FiniteRankAlgebra;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |FiniteRankAlgebra;| ...) is being compiled.
;; The variable |FiniteRankAlgebra;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

finalizing nrlib FFF
Warnings:
  [1] createLowComplexityTable: k has no value
  [2] createLowComplexityTable: t1 has no value
  [3] createLowComplexityTable: a has no value
  [4] createLowComplexityTable: pretend(NonNegativeInteger) -- should replace by @
  [5] createLowComplexityTable: l has no value

```

```
[6] createMultiplicationTable: 1 has no value
```

```
FOP abbreviates package FortranOutputStackPackage
(SPADLET |$noSubsumption| NIL)
```

section

```
finalizing nrlib FORMULA
```

```
Warnings:
```

- [1] display: pretend(Integer) -- should replace by @
- [2] coerce: pretend(Integer) -- should replace by @
- [3] stringify: pretend(String) -- should replace by @
- [4] splitLong1: 1 has no value
- [5] splitLong1: 1s has no value
- [6] splitLong1: s has no value
- [7] formatSpecial: form has no value
- [8] formatPlex: s has no value
- [9] formatFormula: pretend(Boolean) -- should replace by @
- [10] formatFormula: :(Integer) -- should replace by pretend

```
FORT abbreviates package FortranPackage
(SPADLET |$noSubsumption| (QUOTE T))
```

section

```
finalizing nrlib FORT
```

```
Warnings:
```

- [1] linkToFortran: fst has no value

section

```
finalizing nrlib FRAC
```

```
Warnings:
```

- [1] OMwrite: pretend(String) -- should replace by @
- [2] factorPolynomial: den1 has no value
- [3] factorSquareFreePolynomial: den1 has no value

```
-----non extending category-----
```

```
.. Fraction #1 of cat
(|Join| (|QuotientFieldCategory| |#1|) (CATEGORY |package| (IF (|has| |#1| (|IntegerNumberSystem|
```

section

```
(IF (|has| |#1| (|OrderedRing|)) (ATTRIBUTE (|OrderedRing|)) |noBranch|)      finalizing nrlib FRA
```

section

```
finalizing nrlib FTEM
Warnings:
  [1] processTemplate: active has no value

-----non extending category-----
.. FortranTemplate of cat
(|Join| (|FileCategory| (|FileName|) (|String|)) (CATEGORY |package| (SIGNATURE |processTemplate
```

section

```
finalizing nrlib FTEM
; (DEFUN |FTEM;processTemplate;2Fn;7| ...) is being compiled.
;; The variable |$fortranOutputFile| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib GENMFACT
Warnings:
  [1] factor: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE factor
```

section

finalizing nrlib GENPGCD

Warnings:

- [1] gcdPolynomial: canonical has no value
- [2] gcdSameVariables: not known that (Ring) is of mode (CATEGORY package (SIGNATURE gcdPolynomial ((S
- [3] gcdSameVariables: up1 has no value
- [4] gcdSameVariables: up2 has no value
- [5] lift: not known that (Ring) is of mode (CATEGORY package (SIGNATURE gcdPolynomial ((SparseUnivari

section

finalizing nrlib GALFACTU

Warnings:

- [1] singleFactorBound: not known that (Ring) is of mode (CATEGORY F (SIGNATURE Gamma (F F)))

section

finalizing nrlib GBF

Warnings:

- [1] createGroebnerBases: nP has no value
- [2] createGroebnerBases: allReducedFactors has no value
- [3] factorGroebnerBasis: fctr has no value
- [4] groebnerFactorize: signature of lhs not unique: (List (List Dpol))(List Dpol)(List Dpol) chosen

section

finalizing nrlib GBINTERN

Warnings:

- [1] redPo: m has no value

section

finalizing nrlib GHENSEL

Warnings:

```

[1] genFact:  auxfl has no value
[2] HenselLift:  constp has no value
[3] HenselLift:  fln has no value
[4] completeHensel:  factlist has no value
[5] completeHensel:  finallist has no value
[6] completeHensel:  aux has no value
[7] completeHensel:  auxfl has no value

(|Module| |#6|)    extends
(|BiModule| |#6| |#6|)    but not
(|BiModule| |#2| |#2|) -----non extending category-----
.. GeneralModulePolynomial(#1,#2,#3,#4,#5,#6) of cat
(|Join| (|Module| |#6|) (|Module| |#2|) (CATEGORY |domain| (SIGNATURE |leadingCoefficient| (|#2|

```

section

```

(|IndexedDirectProductCategory| |#2| (|ModuleMonomial| |#3| |#4| |#5|))    finalizing nrllib GMOD

```

section

```

finalizing nrllib GRIMAGE
Warnings:
[1] makeGraphImage: signature of lhs not unique: $(List (List (Point (DoubleFloat)))) chos

```

section

```

finalizing nrllib GRIMAGE
--->/research2/test0819/mnt/fedora5/../../src/algebra/GRIMAGE.spad-->GraphImage((figureUnits (UN

```

section

```

finalizing nrllib GROESOL
Warnings:
[1] testPower: not known that (Ring) is of mode (CATEGORY package (SIGNATURE groebSolve ((
[2] testGenPos:  newlpol has no value

```

```
[3] groebSolve:  result has no value
```

```
-----non extending category-----
```

```
.. HexadecimalExpansion of cat
```

```
(|Join| (|QuotientFieldCategory| (|Integer|)) (CATEGORY |domain| (SIGNATURE |coerce| ((|Fraction| (|Integer|
```

section

```
finalizing nrlib HEUGCD
```

```
Warnings:
```

- [1] localgcd: flag has no value
- [2] localgcd: result has no value
- [3] internal: Cgcd has no value
- [4] internal: contgcd has no value
- [5] internal: ans has no value

section

```
finalizing nrlib ICARD
```

```
Warnings:
```

- [1] elt: pretend(String) -- should replace by @

section

```
finalizing nrlib IDECOMP
```

```
Warnings:
```

- [1] internalForm: not known that (PolynomialCategory (Fraction (Integer)) (DirectProduct nv (NonNegat
- [2] internalForm: not known that (PolynomialCategory (Fraction (Polynomial (Integer))) (DirectProduct
- [3] externalForm: not known that (PolynomialCategory (Fraction (Polynomial (Integer))) (DirectProduct
- [4] externalForm: not known that (PolynomialCategory (Fraction (Integer)) (DirectProduct nv (NonNegat
- [5] zeroRadComp: not known that (PolynomialCategory (Fraction (Polynomial (Integer))) (DirectProduct n
- [6] zeroRadComp: not known that (Ring) is of mode (CATEGORY package (SIGNATURE zeroDimPrime? ((Boolea
- [7] findvar: lmonicvar has no value
- [8] zeroPrimDecomp: not known that (PolynomialCategory (Fraction (Polynomial (Integer))) (DirectProdu
- [9] testPower: :(Integer) -- should replace by pretend
- [10] primaryDecomp: not known that (PolynomialCategory (Fraction (Integer)) (DirectProduct nv (NonNeg
- [11] contract: IN has no value
- [12] contract: vv has no value

```
[13] contract: not known that (PolynomialCategory (Fraction (Integer)) (DirectProduct nv (
```

section

```
finalizing nrlib IIARRAY2
Warnings:
[1] latex: s has no value
```

section

```
finalizing nrlib IMATLIN
Warnings:
[1] nullSpace: basis has no value
[2] determinant: ans has no value
[3] generalizedInverse: not known that (MatrixCategory FSUP VFSUP VFSUP) is of mode (CATEG
```

section

```
finalizing nrlib IMATQF
Warnings:
[1] nullSpace: not known that (FiniteLinearAggregate QF) is of mode (CATEGORY Col2 (ATTRIB
```

section

```
finalizing nrlib INMODGCD
--->/research2/test0819/mnt/fedora5/../../src/algebra/INMODGCD.spad-->InnerModularGcd(constructo
--->/research2/test0819/mnt/fedora5/../../src/algebra/INMODGCD.spad-->InnerModularGcd(): Missing
```

section

finalizing nrlib INNMFAC

Warnings:

- [1] supFactor: factorlist has no value
- [2] supFactor: irr has no value
- [3] supFactor: pow has no value
- [4] varChoose: not known that (Ring) is of mode (CATEGORY package (SIGNATURE factor ((Factored P) P (I
- [5] intChoose: unifact has no value
- [6] intChoose: int has no value
- [7] simplify: pol1 has no value
- [8] simplify: factorlist has no value
- [9] intfact: unifact has no value
- [10] intfact: lpol has no value
- [11] intfact: factfin has no value
- [12] mFactor: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE factor ((Factor
- [13] mFactor: flead has no value
- [14] mFactor: irr has no value
- [15] mFactor: pow has no value
- [16] mFactor: x has no value
- [17] mFactor: ffactor has no value
- [18] mFactor: lvar has no value
- [19] mFactor: factorlist has no value

section

finalizing nrlib INTSLPE

Warnings:

- [1] solveLinearPolynomialEquation: slpePrime has no value
- [2] solveLinearPolynomialEquation: oldtable has no value

section

finalizing nrlib INTTR

Warnings:

- [1] expintegratepoly: coef0 has no value
- [2] explimintfrac: logand has no value
- [3] explimintfrac: coeff has no value

section

```
finalizing nrlib ISUMP
```

```
Warnings:
```

```
[1] sum: not known that (Ring) is of mode (CATEGORY package (SIGNATURE sum ((Record (: num
```

section

```
finalizing nrlib LAUPOL
```

```
Warnings:
```

```
[1] not known that (Ring) is of mode (CATEGORY domain (SIGNATURE separate ((Record (: poly
```

section

```
finalizing nrlib LEADCDET
```

```
Warnings:
```

```
[1] distFact: c has no value
```

section

```
finalizing nrlib LGROBP
```

```
Warnings:
```

```
[1] totalex: result has no value
```

```
[2] minPol: :(NonNegativeInteger) -- should replace by pretend
```

```
[3] intcompBasis: part has no value
```

```
[4] linGenPos: :(PositiveInteger) -- should replace by pretend
```

```
[5] linGenPos: result has no value
```

```
[6] groebgen: :(NonNegativeInteger) -- should replace by pretend
```

```
[7] groebgen: not known that (Ring) is of mode (CATEGORY package (SIGNATURE linGenPos ((Re
```

section

```
finalizing nrlib LINDEP
```

```
Warnings:
```

```
[1] linearlyDependent?: not known that (Ring) is of mode (CATEGORY package (SIGNATURE line
```

section

```
finalizing nrlib LPEFRAC
```

```
Warnings:
```

```
[1] solveLinearPolynomialEquationByFractions: not known that (Ring) is of mode (CATEGORY package (SIGNATURE linSolve ((Record (: part
```

section

```
finalizing nrlib LSPP
```

```
Warnings:
```

```
[1] poly2vect: not known that (Ring) is of mode (CATEGORY package (SIGNATURE linSolve ((Record (: part
```

section

```
finalizing nrlib MATLIN
```

```
Warnings:
```

```
[1] minorDet: ans has no value
```

```
[2] rowEchelon: xnj has no value
```

section

```
finalizing nrlib MDDFACT
```

```
Semantic Errors:
```

```
[1] ddfactor: degree is BOTH a variable and a literal
```

```
[2] ddfact: degree is BOTH a variable and a literal
```

```
[3] ddFact: degree is BOTH a variable and a literal
```

```
[4] sepfact: degree is BOTH a variable and a literal
```

```
[5] separateFactors: degree is BOTH a variable and a literal
```

```
[6] sepFact1: degree is BOTH a variable and a literal
```

```
Warnings:
```

```
[1] ddfact: ans has no value
```

```
[2] ddfact: :(Integer) -- should replace by @
```

```

[3] sepFact1:  ans has no value
[4] sepFact1:  stack has no value
[5] probSplit:  :(NonNegativeInteger) -- should replace by pretend

```

section

```
finalizing nrlib MFINFACT
```

```
Warnings:
```

```

[1] pretendOV -- should replace by @
[2] supFactor: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE fac
[3] mFactor:  irr has no value
[4] mFactor:  pow has no value
[5] mFactor: not known that (Ring) is of mode (CATEGORY package (SIGNATURE factor ((Factor
[6] mFactor:  ffactor has no value
[7] mFactor:  factorlist has no value
[8] intfact:  unifact has no value
[9] intfact:  lpol has no value
[10] intfact:  factfin has no value
[11] intChoose: newm has no value
[12] intChoose: unifact has no value
[13] intChoose:  int has no value
[14] simplify: pol1 has no value
[15] simplify: factorlist has no value

```

section

```
finalizing nrlib MLIFT
```

```
Warnings:
```

```

[1] Use: import (GenExEuclid R (SparseUnivariatePolynomial R))
[2] Use: import (NPCoef (SparseUnivariatePolynomial R) E OV R P)
[3] Use: import (IntegerCombinatoricFunctions (Integer))
[4] lifting:  nplist has no value
[5] normalDerivM: not known that (Ring) is of mode (CATEGORY package (SIGNATURE corrPoly (

```

```

MODMON abbreviates domain ModMonic
(PUSH (QUOTE |ModMonic|) |$mutableDomains|)

```

section

```
finalizing nrlib MODMON
```

```
Warnings:
```

- [1] :(NonNegativeInteger) -- should replace by pretend
- [2] lift: pretendRep -- should replace by @
- [3] not known that (SetCategory) is of mode (CATEGORY domain (SIGNATURE frobenius (\$ \$)))
- [4] not known that (Ring) is of mode (CATEGORY domain (SIGNATURE frobenius (\$ \$)))
- [5] not known that (IntegralDomain) is of mode (CATEGORY domain (SIGNATURE frobenius (\$ \$)))

section

```
finalizing nrlib MONOTOOL
```

```
Warnings:
```

- [1] splitSquarefree: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE split ((
- [2] splitSquarefree: s has no value

section

```
finalizing nrlib MPCPF
```

```
Warnings:
```

- [1] factor: not known that (SIGNATURE variable ((Union \$ failed) (Symbol))) is of mode (CATEGORY domain

section

```
finalizing nrlib MPC3
```

```
Warnings:
```

- [1] map: not known that (Ring) is of mode (CATEGORY package (SIGNATURE map (PR2 (Mapping Vars2 Vars1)

section

```
finalizing nrlib MPRFF
```

```
Warnings:
```

- [1] factor: not known that (PolynomialCategory (Fraction (Integer)) (IndexedExponents (Symbol)) (Symbol
- [2] factor: pretend(Factored (Polynomial R)) -- should replace by @

```
[3] factor:  flist has no value
[4] pushdown: not known that (Ring) is of mode (CATEGORY package (SIGNATURE totalfract ((R
```

section

```
finalizing nrlib MULTSQFR
Semantic Errors:
  [1] squareFree:  ff has two modes:

Warnings:
  [1] nsqfree:  lcf has no value
  [2] nsqfree:  leadpol has no value
  [3] nsqfree:  sqlc has no value
  [4] nsqfree:  :(NonNegativeInteger) -- should replace by pretend
  [5] nsqfree:  sqlead has no value
  [6] nsqfree:  unitsq has no value
  [7] nsqfree:  sqdec has no value
  [8] squareFree: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE sq
  [9] squareFree:  fctr has no value
  [10] squareFree:  xpnt has no value
  [11] squareFree:  p has no value
  [12] squareFree:  result1 has no value
  [13] squareFree:  lvar has no value
  [14] squareFree: not known that (Ring) is of mode (CATEGORY package (SIGNATURE squareFree
  [15] intChoose: :R -- should replace by @
  [16] intChoose:  d1 has no value
  [17] coefChoose:  lcoef has no value
  [18] lift:  leadlist has no value
  [19] lift:  p0 has no value
  [20] lift:  p1 has no value
```

section

```
finalizing nrlib NPCOEF
Warnings:
  [1] npcoef:  :(NonNegativeInteger) -- should replace by pretend
  [2] npcoef:  ndet has no value
  [3] npcoef:  detufact has no value
  [4] npcoef:  ltodel has no value
  [5] npcoef:  detcoef has no value
  [6] check:  :(NonNegativeInteger) -- should replace by pretend
  [7] check:  doit has no value
```

```

[8] check:  cfu has no value
[9] check:  poselt has no value
[10] check:  pp has no value
[11] buildtable:  table has no value
[12] modify:  :(NonNegativeInteger) -- should replace by pretend
[13] modify:  lterase has no value

```

section

```

finalizing nrlib NSUP
Warnings:
  [1] subResultantGcd: not known that (UnivariatePolynomialCategory R) is of mode (CATEGORY domain (SIG
-----non extending category-----
.. NewSparseUnivariatePolynomial #1 of cat
(|Join| (|UnivariatePolynomialCategory| |#1|) (|CoercibleTo| (|SparseUnivariatePolynomial| |#1|)) (|Retract

```

section

```

finalizing nrlib ODEPRIM
Warnings:
  [1] indicialEquations:  eq has no value
  [2] NPMulambda:  lamb has no value
  [3] NPMulambda:  lf has no value

```

section

```

finalizing nrlib ODEPRRIC
Warnings:
  [1] leadingDenomRicDE:  ind has no value
  [2] constantCoefficientOperator:  ans has no value
  [3] innermax:  ans has no value
  [4] leadingCoefficientRicDE:  ind has no value
  [5] innerlb:  lb has no value

```

section

```
finalizing nrlib OMPKG
Warnings:
[1] OMreadStr: pretend(String) -- should replace by @
[2] OMlistCDs: pretend(List (String)) -- should replace by @
[3] OMlistSymbols: pretend(List (String)) -- should replace by @
```

section

```
finalizing nrlib OMSERVER
--->/research2/test0819/mnt/fedora5/../../src/algebra/OMSERVER.spad-->OpenMathServerPackage((OMs
"\spad{OMsend(c,{u})} attempts to output \\axiom{\\spad{u}} on \\axiom{\\spad{c}} in OpenMath."
```

section

```
finalizing nrlib PADICRC
Warnings:
[1] coerce: 1 has no value
```

section

```
finalizing nrlib PF
Warnings:
[1] unknown Functor code (error Argument to prime field must be a prime)
```

section

```
finalizing nrlib PFBR
Warnings:
[1] hensel: not known that (Ring) is of mode (CATEGORY package (SIGNATURE solveLinearPolyn
[2] hensel: foundFactors has no value
```



```

[3] factorSFBRLcUnitInner: not known that (Ring) is of mode (CATEGORY package (SIGNATURE solveLinearP
[4] factorSFBRLcUnitInner: fctr has no value
[5] chooseFSQViableSubstitutions: ppR has no value
[6] chooseSLPEViableSubstitutions: lpolysR has no value
[7] raise: :R -- should replace by pretend
[8] raise: :R -- should replace by @
[9] factorByRecursion: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE solveL
[10] factorByRecursion: :S -- should replace by @
[11] factorSquareFreeByRecursion: lcppPow has no value

```

section

finalizing nrlib PFBRU

Warnings:

```

[1] solveLinearPolynomialEquationByRecursion: not known that (Ring) is of mode (CATEGORY package (SIG
[2] solveLinearPolynomialEquationByRecursion: answer has no value
[3] hensel: foundFactors has no value
[4] chooseFSQViableSubstitutions: ppR has no value
[5] raise: :R -- should replace by pretend
[6] raise: :R -- should replace by @
[7] factorSFBRLcUnitInner: not known that (Ring) is of mode (CATEGORY package (SIGNATURE solveLinearP
[8] factorSFBRLcUnitInner: fctr has no value
[9] factorByRecursion: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE solveL
[10] factorByRecursion: :S -- should replace by pretend
[11] factorByRecursion: :S -- should replace by @
[12] factorSquareFreeByRecursion: lcppPow has no value

```

section

finalizing nrlib PGCD

Warnings:

```

[1] localgcd: s has no value
[2] gcdTermList: not known that (Ring) is of mode (CATEGORY package (SIGNATURE gcd (P P P)) (SIGNATUR
[3] lift: p0 has no value

```

section

finalizing nrlib PLEQN

Warnings:

- [1] regime: wcd has no value
- [2] regime: yzero has no value
- [3] bsolve: rksoln has no value
- [4] bsolve: lrec3 has no value
- [5] ParCondList: covered has no value
- [6] ParCondList: zro has no value
- [7] ParCondList: npc has no value
- [8] ParCondList: done has no value
- [9] ParCondList: rcl has no value
- [10] pr2dmp: pretendGR -- should replace by @
- [11] sqfree: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE psolv
- [12] ParCond: found has no value
- [13] ParCond: DetEqn has no value

section

finalizing nrlib PNTHEORY

Warnings:

- [1] cyclotomic: s has no value
- [2] cyclotomic: c has no value

section

finalizing nrlib POLUTIL

Warnings:

- [1] sylvesterSequence: res has no value
- [2] sturmVariationsOf: ll has no value

section

finalizing nrlib POLYCATQ

Warnings:

- [1] isPower: var has no value
- [2] P2UP: not known that (Ring) is of mode (CATEGORY package (SIGNATURE variables ((List V

section

```

finalizing nrlib POLYCATQ
Processing PolynomialCategoryQuotientFunctions for Browser database:
--->/research2/test0819/mnt/fedora5/../../src/algebra/POLYCATQ.spad-->PolynomialCategoryQuotientFunctions((
--->/research2/test0819/mnt/fedora5/../../src/algebra/POLYCATQ.spad-->PolynomialCategoryQuotientFunctions((
--->/research2/test0819/mnt/fedora5/../../src/algebra/POLYCATQ.spad-->PolynomialCategoryQuotientFunctions((

```

section

```

finalizing nrlib POLYLIFT
Warnings:
[1] map: not known that (Ring) is of mode (CATEGORY package (SIGNATURE map (S (Mapping S Vars) (Mapping

```

section

```

finalizing nrlib POLYLIFT
--->/research2/test0819/mnt/fedora5/../../src/algebra/POLYLIFT.spad-->PolynomialCategoryLifting((+ (% % %))
--->/research2/test0819/mnt/fedora5/../../src/algebra/POLYLIFT.spad-->PolynomialCategoryLifting((* (% % %))
--->/research2/test0819/mnt/fedora5/../../src/algebra/POLYLIFT.spad-->PolynomialCategoryLifting(** (% % (N

```

section

```

finalizing nrlib POLYROOT
Warnings:
[1] zroot: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE rroot ((Record (:

```

section

```

finalizing nrlib POLY2UP
Warnings:
[1] univariate: not known that (Ring) is of mode (CATEGORY package (SIGNATURE univariate ((Univariate)

```

section

```
finalizing nrlib PRS
Warnings:
[1] resultantnaif: a has no value
[2] resultantEuclideanif: a has no value
[3] semiResultantEuclideanif: a has no value
[4] chainSubResultants: L has no value
[5] schema: L has no value
```

section

```
finalizing nrlib PSQFR
Warnings:
[1] pPolRoot: not known that (Ring) is of mode (CATEGORY package (SIGNATURE squareFree ((F
[2] pthPower: isSq has no value
[3] pthPower: fctr has no value
[4] pthPower: xpnt has no value
[5] finSqFr: :(NonNegativeInteger) -- should replace by pretend
[6] finSqFr: cont1 has no value
```

section

```
finalizing nrlib PUSHVAR
Warnings:
[1] map: not known that (Ring) is of mode (CATEGORY package (SIGNATURE pushdown (PPR PPR O
```

section

```
finalizing nrlib QALGSET
Warnings:
[1] npoly: not known that (OrderedAbelianMonoid) is of mode (CATEGORY domain (IF (has (Non
[2] idealSimplify: not known that (OrderedAbelianMonoid) is of mode (CATEGORY domain (IF (
```

```
[3] idealSimplify: not known that (OrderedAbelianMonoidSup) is of mode (CATEGORY domain (IF (has (Non
[4] idealSimplify: not known that (PolynomialCategory R (Product (NonNegativeInteger) Expon) Var) is o
[5] idealSimplify: gb has no value
```

section

```
finalizing nrlib RADIX
```

```
Warnings:
```

```
[1] fractionPart: signature of lhs not unique: (Fraction (Integer))$ chosen
[2] coerce: le has no value
[3] radixFrac: c has no value
[4] unknown Functor code (error Radix base must be at least 2)
```

section

```
finalizing nrlib RATFACT
```

```
--->/research2/test0819/mnt/fedora5/../../src/algebra/RATFACT.spad-->RationalFactorize(constructor): Not do
--->/research2/test0819/mnt/fedora5/../../src/algebra/RATFACT.spad-->RationalFactorize(): Missing Descripti
```

section

```
finalizing nrlib RCFIELD-
```

```
Warnings:
```

```
[1] rootOf: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE sqrt (S (Integer))) (SIGNATUR
[2] allRootsOf: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE sqrt (S (Integer))) (SIG
```

section

```
finalizing nrlib RCFIELD
```

```
; (DEFUN |RealClosedField| ...) is being compiled.
;; The variable |RealClosedField;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib RDETR
Warnings:
[1] polyRDE: eq has no value
[2] SPDEncancel1: q has no value
[3] SPDEncancel2: q has no value
```

section

```
finalizing nrlib REALO
Warnings:
[1] makeSqfr: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE real
[2] Zero1: G has no value
[3] rootBound: i has no value
[4] transMultInv: :(NonNegativeInteger) -- should replace by pretend
[5] var: i has no value
```

section

```
finalizing nrlib REALSOLV
--->/research2/test0819/mnt/fedora5/../../src/algebra/REALSOLV.spad-->RealSolvePackage(construct
--->/research2/test0819/mnt/fedora5/../../src/algebra/REALSOLV.spad-->RealSolvePackage(): Missin
```

section

```
finalizing nrlib RESRING
Warnings:
[1] unknown Functor code (error the residue ring is the zero ring)
```

section

```
finalizing nrlib RETSOL
```

```
Warnings:
```

```
[1] PQIfCan: not known that (Ring) is of mode (CATEGORY package (SIGNATURE solveRetract ((List (List
```

section

```
finalizing nrlib RFFACTOR
```

```
Warnings:
```

```
[1] factorFraction: not known that (PolynomialCategory (Fraction (Integer)) (IndexedExponents (Symbol
```

section

```
finalizing nrlib RMACAT
```

```
; (DEFUN |RectangularMatrixCategory| ...) is being compiled.
;; The variable |RectangularMatrixCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |RectangularMatrixCategory;| ...) is being compiled.
;; The variable |RectangularMatrixCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib RRCC
```

```
; (DEFUN |RealRootCharacterizationCategory| ...) is being compiled.
;; The variable |RealRootCharacterizationCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |RealRootCharacterizationCategory;| ...) is being compiled.
;; The variable |RealRootCharacterizationCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib SHP
```

```
Warnings:
```

```

[1] subresultantSequenceNext: :(List (UnivariatePolynomial x R)) -- should replace by @
[2] subresultantSequenceInner: :(UnivariatePolynomial x R) -- should replace by @
[3] subresultantSequenceInner: :(List (UnivariatePolynomial x R)) -- should replace by @
[4] subresultantSequence: :(List (UnivariatePolynomial x R)) -- should replace by @
[5] polsth1: :(List (UnivariatePolynomial x R)) -- should replace by @
[6] polsth1: Listf has no value
[7] polsth2: :(List (UnivariatePolynomial x R)) -- should replace by @
[8] polsth3: :(List (UnivariatePolynomial x R)) -- should replace by @
[9] wfunctaux: :(List R) -- should replace by @
[10] wfunctaux: :(NonNegativeInteger) -- should replace by @
[11] wfunct: :(List R) -- should replace by @
[12] SturmHabicht: :(UnivariatePolynomial x R) -- should replace by @
[13] SturmHabichtMultiple: :(UnivariatePolynomial x R) -- should replace by @
[14] SturmHabichtMultiple: not known that (IntegralDomain) is of mode (CATEGORY domain (SI

```

section

finalizing nrlib SIGNRF

Warnings:

```

[1] psign: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE sign ((

```

section

finalizing nrlib SMP

Warnings:

```

[1] ^: pretend(NonNegativeInteger) -- should replace by @
[2] **: pretend(NonNegativeInteger) -- should replace by @
[3] exquo: The conditional modes (SparseUnivariatePolynomial $) and D conflict
[4] eval: IN has no value
[5] eval: val has no value

```

section

finalizing nrlib SOLVEFOR

Warnings:

```

[1] intsolve: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE solv

```


section

```
finalizing nrlib SOLVEFOR
--->/research2/test0819/mnt/fedora5/../../src/algebra/SOLVEFOR.spad-->PolynomialSolveByFormulas((** (% % (F
```

section

```
finalizing nrlib SPLTREE
compiling exported eq? : ($,$) -> Boolean
  SPLTREE;eq?;2$B;20 is replaced by errorin eq? from SPLTREE : la vache qui rit est-elle folle?
```

section

```
finalizing nrlib SPLTREE
Warnings:
  [1] splitNodeOf!: la has no value
```

section

```
finalizing nrlib STTFNC
compiling exported acos : Stream Coef -> Stream Coef
  STTFNC;acos;2S;12 is replaced by erroracos: acos undefined on this coefficient domain
compiling exported acot : Stream Coef -> Stream Coef
  STTFNC;acot;2S;13 is replaced by erroracot: acot undefined on this coefficient domain
compiling exported asec : Stream Coef -> Stream Coef
  STTFNC;asec;2S;14 is replaced by errorasec: asec undefined on this coefficient domain
compiling exported acsc : Stream Coef -> Stream Coef
  STTFNC;acsc;2S;15 is replaced by erroracsc: acsc undefined on this coefficient domain
compiling exported acosh : Stream Coef -> Stream Coef
  STTFNC;acosh;2S;24 is replaced by erroracosh: acosh undefined on this coefficient domain
compiling exported acoth : Stream Coef -> Stream Coef
  STTFNC;acoth;2S;25 is replaced by erroracoth: acoth undefined on this coefficient domain
compiling exported asech : Stream Coef -> Stream Coef
  STTFNC;asech;2S;26 is replaced by errorasech: asech undefined on this coefficient domain
compiling exported acsch : Stream Coef -> Stream Coef
  STTFNC;acsch;2S;27 is replaced by erroracsch: acsch undefined on this coefficient domain
```

section

```

finalizing nrlib SUP
Warnings:
  [1] **: pretend(NonNegativeInteger) -- should replace by @
  [2] ^: pretend(NonNegativeInteger) -- should replace by @
  [3] **: y has no value
  [4] pomopo!: rout has no value
  [5] exquo: signature of lhs not unique: (Union $ failed)$$ chosen
  [6] fmecg: rout has no value
  [7] pseudoRemainder: :(Integer) -- should replace by pretend
  [8] pseudoRemainder: :(NonNegativeInteger) -- should replace by pretend
  [9] elt: :(NonNegativeInteger) -- should replace by pretend

-----non extending category-----
.. SparseUnivariatePolynomial #1 of cat
(|Join| (|UnivariatePolynomialCategory| |#1|) (CATEGORY |domain| (SIGNATURE |outputForm| ((|Outp

```

section

```

(IF (|has| |#1| (|IntegralDomain|)) (IF (|has| (|NonNegativeInteger|) (|CancellationAbelianMonoi

```

section

```

finalizing nrlib SUPFRACF
Warnings:
  [1] squareFree: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE fa

```

section

```

finalizing nrlib TEX
Warnings:
  [1] display: pretend(Integer) -- should replace by @
  [2] coerce: pretend(Integer) -- should replace by @

```

```

[3] stringify: pretend(String) -- should replace by @
[4] splitLong1:  l has no value
[5] splitLong1:  ls has no value
[6] splitLong1:  s has no value
[7] formatSpecial:  form has no value
[8] formatSpecial:  prescript has no value
[9] formatPlex:  s has no value
[10] formatTex: pretend(Boolean) -- should replace by @

```

section

finalizing nrlib TEXTFILE

Warnings:

```
[1] endOfFile?: pretend(Boolean) -- should replace by @
```

section

finalizing nrlib TREE

Warnings:

```

[1] children:  node has no value
[2] setchildren!:  node has no value
[3] setchildren!: pretend$ -- should replace by @
[4] setvalue!:  node has no value
[5] count: signature of lhs not unique: (NonNegativeInteger)$ chosen
[6] value:  node has no value
[7] node?:  t has no value
[8] multipleOverbar: The conditional modes (String) and S conflict

```

section

finalizing nrlib TWOFACT

Warnings:

```

[1] generalSqFr: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE generalTwoFa
[2] generalSqFr:  flg has no value
[3] generalSqFr:  xpnt has no value
[4] generalSqFr: :(NonNegativeInteger) -- should replace by pretend
[5] generalSqFr:  fctr has no value
[6] generalTwoFactor:  flg has no value

```

```

[7] generalTwoFactor:  fctr has no value
[8] generalTwoFactor:  xpnt has no value
[9] generalTwoFactor:  ll has no value
[10] generalTwoFactor:  unitPart has no value
[11] twoFactor:  n has no value
[12] twoFactor:  look has no value
[13] twoFactor:  umv has no value
[14] twoFactor:  umex has no value

```

section

```

finalizing nrlib UNIFACT
Warnings:
[1] eisenstein: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE fa
[2] bound: :(NonNegativeInteger) -- should replace by pretend
[3] bound:  cbound has no value
[4] numFactors: pretend(Integer) -- should replace by @
[5] numFactors:  ans has no value
[6] choose:  qSave has no value
[7] choose:  ddSave has no value
[8] quadratic:  beta has no value
[9] quadratic:  alpha has no value
[10] subMinusX: :ZP -- should replace by pretend
[11] henselFact:  c has no value
[12] henselFact:  factorlist has no value
[13] henselFact: :(List ZP) -- should replace by pretend

-----non extending category-----
.. UnivariatePolynomial(#1,#2) of cat
(|Join| (|UnivariatePolynomialCategory| |#2|) (CATEGORY |domain| (SIGNATURE |coerce| ($ (|Variab

```

section

```

finalizing nrlib UPDECOMP
Warnings:
[1] leftFactorIfCan:  g has no value

```

section

```

    finalizing nrlib UPXSCAT
; (DEFUN |UnivariatePuisseuxSeriesCategory| ...) is being compiled.
;; The variable |UnivariatePuisseuxSeriesCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |UnivariatePuisseuxSeriesCategory;| ...) is being compiled.
;; The variable |UnivariatePuisseuxSeriesCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib UPSQFREE
Warnings:
  [1] BumInSepFFE: :(NonNegativeInteger) -- should replace by @
  [2] squareFree:  lffe has no value
  [3] squareFree:  dunit has no value

```

section

```

    finalizing nrlib VIEWDEF
Warnings:
  [1] viewWriteDefault:  thingsToWrite has no value

```

section

```

    finalizing nrlib WEIER
Warnings:
  [1] transback: :(Stream (Stream (Polynomial R))) -- should replace by pretend
  [2] transback: :(TaylorSeries R) -- should replace by pretend
  [3] transback: :(Stream (TaylorSeries R)) -- should replace by pretend
  [4] clikeUniv: not known that (Ring) is of mode (CATEGORY package (SIGNATURE crest ((Mapping (Stream
  [5] streamlikeUniv: :(NonNegativeInteger) -- should replace by pretend
  [6] sts2stst: :(Stream (NonNegativeInteger)) -- should replace by pretend
  [7] weier: :(TaylorSeries R) -- should replace by pretend
  [8] wei: :(Stream (Polynomial R)) -- should replace by pretend

```

section

```
finalizing nrlib WP
Warnings:
  [1] innercoerce:  ans has no value
  [2] unknown Functor code (error incompatible length lists in WeightedPolynomial)
```

section

```
finalizing nrlib DIAGG
; (DEFUN |Dictionary| ...) is being compiled.
;; The variable |Dictionary;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |Dictionary;| ...) is being compiled.
;; The variable |Dictionary;CAT| is undefined.
;; The compiler will assume this variable is a global.

-----non extending category-----
.. ExponentialOfUnivariatePuisseuxSeries(#1,#2,#3) of cat
(|Join| (|UnivariatePuisseuxSeriesCategory| |#1|) (|OrderedAbelianMonoid|) (CATEGORY |domain| (SI
```

section

```
(|UnivariatePuisseuxSeriesConstructorCategory| |#1| (|UnivariateLaurentSeries| |#1| |#2| |#3|))
```

section

```
finalizing nrlib FRAMALG-
Warnings:
  [1] coordinates: signature of lhs not unique: (Matrix R)(Vector S) chosen
```

section

```

    finalizing nrlib FRAMALG
; (DEFUN |FramedAlgebra| ...) is being compiled.
;; The variable |FramedAlgebra;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |FramedAlgebra;| ...) is being compiled.
;; The variable |FramedAlgebra;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib MDAGG
; (DEFUN |MultiDictionary| ...) is being compiled.
;; The variable |MultiDictionary;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |MultiDictionary;| ...) is being compiled.
;; The variable |MultiDictionary;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib PLOT
Warnings:
  [1] listBranches: newl has no value
  [2] listBranches: outList has no value
  [3] rangeRefine: c has no value
  [4] rangeRefine: q has no value
  [5] rangeRefine: NUMFUNVALS has no value
  [6] refine: curves has no value
  [7] plot: curves has no value
  [8] plot: t has no value
  [9] coerce: l has no value

```

section

```

    finalizing nrlib PLOT
; (DEFUN |PLOT;myTrap| ...) is being compiled.
;; The variable |$numericFailure| is undefined.
;; The compiler will assume this variable is a global.

```

```
;; The variable |$NaNvalue| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib SMATCAT-
```

```
Warnings:
```

```
[1] equation2R: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE ** (S S (Inte
```

section

```
finalizing nrlib SMATCAT
```

```
; (DEFUN |SquareMatrixCategory| ...) is being compiled.
;; The variable |SquareMatrixCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |SquareMatrixCategory;| ...) is being compiled.
;; The variable |SquareMatrixCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib UPXSCCA
```

```
; (DEFUN |UnivariatePuisseuxSeriesConstructorCategory| ...) is being compiled.
;; The variable |UnivariatePuisseuxSeriesConstructorCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |UnivariatePuisseuxSeriesConstructorCategory;| ...) is being compiled.
;; The variable |UnivariatePuisseuxSeriesConstructorCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib FFINTBAS
```

```
Warnings:
```

```
[1] squaredFactors: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATUR
```


section

```

    finalizing nrlib FSAGG
; (DEFUN |FiniteSetAggregate| ...) is being compiled.
;; The variable |FiniteSetAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |FiniteSetAggregate;| ...) is being compiled.
;; The variable |FiniteSetAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib KDAGG
; (DEFUN |KeyedDictionary| ...) is being compiled.
;; The variable |KeyedDictionary;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |KeyedDictionary;| ...) is being compiled.
;; The variable |KeyedDictionary;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib MSETAGG
; (DEFUN |MultisetAggregate| ...) is being compiled.
;; The variable |MultisetAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |MultisetAggregate;| ...) is being compiled.
;; The variable |MultisetAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib MONOGEN

```

```
; (DEFUN |MonogenicAlgebra| ...) is being compiled.
;; The variable |MonogenicAlgebra;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |MonogenicAlgebra;| ...) is being compiled.
;; The variable |MonogenicAlgebra;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib NFINTBAS
Warnings:
[1] wildPrimes:  ans has no value
[2] tameProduct:  ans has no value
[3] integralBasis:  runningRbden has no value
[4] iWildLocalIntegralBasis:  rbden has no value
```

section

```
finalizing nrlib SPACE3
Warnings:
[1] coerce:  llliPt has no value
```

section

```
finalizing nrlib CCLASS
Warnings:
[1] remove!:  signature of lhs not unique: $(Character)$ chosen
```

section

```
finalizing nrlib GALFACT
Warnings:
[1] eisensteinIrreducible?:  not known that (IntegralDomain) is of mode (CATEGORY package (
[2] modularFactor:  s has no value
```

```

[3] completeFactor:  g0 has no value
[4] completeFactor:  degg has no value
[5] completeFactor:   g has no value
[6] completeFactor:  ltrue has no value
[7] completeFactor:  level has no value
[8] divideSet:    l has no value
[9] btwFactor:   reverse? has no value
[10] btwFactor:  negativelc? has no value
[11] btwFactor:   lf has no value
[12] henselFact:  c has no value
[13] henselFact:  factorlist has no value
[14] btwFact:    c has no value
[15] btwFact:    factorlist has no value

```

section

```
finalizing nrlib IALGFACT
```

```
Warnings:
```

```
[1] factor: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE factor ((Factored
```

section

```
finalizing nrlib MMLFORM
```

```
Warnings:
```

```

[1] exprex:  s has no value
[2] formatSub:  j has no value
[3] formatSub:  s has no value
[4] formatSub1: s has no value
[5] formatPlex: s has no value

```

section

```
finalizing nrlib OMSAGG
```

```

; (DEFUN |OrderedMultisetAggregate| ...) is being compiled.
;; The variable |OrderedMultisetAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |OrderedMultisetAggregate;| ...) is being compiled.

```

```
;; The variable |OrderedMultisetAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib PERM
Warnings:
[1] rotateCycle: minpos has no value
[2] coerceToCycle: nextCycle has no value
[3] coercePreimagesImages: preImage has no value
[4] coercePreimagesImages: image has no value
```

section

```
finalizing nrlib PERMGRP
Warnings:
[1] shortenWord: newlw has no value
[2] orbitInternal: pos has no value
[3] orbitInternal: orbitList has no value
[4] bsgs1: j has no value
[5] bsgs1: out has no value
[6] bsgs1: outword has no value
[7] bsgs1: baseOfGroup has no value
[8] reduceGenerators: wordlist has no value
[9] bsgs: gpbases has no value
[10] bsgs: baseOfGroup has no value
[11] bsgs: wordProblem has no value
[12] bsgs: wordlist has no value
[13] bsgs: gporb has no value
[14] bsgs: noresult has no value
[15] bsgs: newBasePoint has no value
[16] bsgs: basePoint has no value
[17] memberInternal: supp has no value
[18] memberInternal: sgs has no value
[19] memberInternal: baseOfGroup has no value
[20] memberInternal: gporb has no value
[21] coerce: :(Symbol) -- should replace by pretend
```

section

```
finalizing nrlib PWFFINTB
Warnings:
  [1] listSquaredFactors:  ans has no value
```

section

```
finalizing nrlib SAE
Warnings:
  [1] lift: pretendRep -- should replace by @
  [2] index:  ans has no value
  [3] unknown Functor code (error Modulus cannot be made monic)
```

section

```
finalizing nrlib SGCF
Warnings:
  [1] unrankImproperPartitions0:  l has no value
  [2] unrankImproperPartitions1:  nonZeros has no value
  [3] subSet:  l has no value
  [4] nextLatticePermutation:  ready has no value
  [5] listYoungTableaus:  lattice has no value
```

section

```
finalizing nrlib TBAGG-
Warnings:
  [1] map!: signature of lhs not unique: S(Mapping Entry Entry)S chosen
```

section

```

    finalizing nrlib TBAGG
; (DEFUN |TableAggregate| ...) is being compiled.
;; The variable |TableAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |TableAggregate;| ...) is being compiled.
;; The variable |TableAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib VIEW3D
Warnings:
  [1] viewpoint:  Theta has no value

```

section

```

    finalizing nrlib VIEW3D
--->/research2/test0819/mnt/fedora5/../../src/algebra/VIEW3D.spad-->ThreeDimensionalViewport(con
--->/research2/test0819/mnt/fedora5/../../src/algebra/VIEW3D.spad-->ThreeDimensionalViewport():

```

section

```

    finalizing nrlib ALIST
Warnings:
  [1] latex:  s has no value
  [2] remove!:  key has no value

```

section

```

    finalizing nrlib HASHTBL
Warnings:
  [1] remove!:  pretendEntry -- should replace by @
  [2] search:  pretendEntry -- should replace by @

```

section

```
finalizing nrlib INTPACK
Warnings:
  [1] preAnalysis:  nia has no value
  [2] measureSpecific:  nia has no value
  [3] measureSpecific:  mdnia has no value
  [4] integrateSpecific:  nia has no value
  [5] integrateSpecific:  mdnia has no value
  [6] recoverAfterFail:  nia has no value
  [7] integrateArgs:  mdnia has no value
  [8] integrateArgs:  nia has no value
```

section

```
finalizing nrlib INTPACK
--->/research2/test0819/mnt/fedora5/../../src/algebra/INTPACK.spad-->AnnaNumericalIntegrationPackage(constr
--->/research2/test0819/mnt/fedora5/../../src/algebra/INTPACK.spad-->AnnaNumericalIntegrationPackage(): Mis
```

section

```
finalizing nrlib IPF
Warnings:
  [1] tableForDiscreteLogarithm: pretend(Table (PositiveInteger) (NonNegativeInteger)) -- should replace
  [2] initializeLog:  n has no value
  [3] coordinates: signature of lhs not unique: (Vector $)$ chosen
  [4] charthRoot: signature of lhs not unique: $$ chosen
```

section

```
finalizing nrlib ACF-
Warnings:
  [1] zeroOf: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE zerosOf ((List S) (SparseUni
```

section

```

finalizing nrlib ACF
; (DEFUN |AlgebraicallyClosedField| ...) is being compiled.
;; The variable |AlgebraicallyClosedField;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

finalizing nrlib ACPLLOT
Warnings:
[1] listPtsOnHorizBdry: pointList has no value
[2] listPtsOnVertBdry: pointList has no value
[3] listPtsInRect: pointList has no value
[4] makeOneVarSketch: sketchRoots has no value
[5] makeOneVarSketch: lf has no value
[6] makeOneVarSketch: rt has no value
[7] makeOneVarSketch: bt has no value
[8] makeOneVarSketch: tp has no value
[9] makeLineSketch: branch has no value
[10] makeLineSketch: lf has no value
[11] makeLineSketch: rt has no value
[12] makeLineSketch: bt has no value
[13] makeLineSketch: tp has no value
[14] makeRatFcnSketch: outList has no value
[15] makeRatFcnSketch: lf has no value
[16] makeRatFcnSketch: rt has no value
[17] makeRatFcnSketch: bt has no value
[18] makeRatFcnSketch: tp has no value
[19] makeRatFcnSketch: topList has no value
[20] makeRatFcnSketch: botList has no value
[21] listPtsOnLoop: pointList has no value
[22] computeNextPt: xPointList has no value
[23] computeNextPt: yPointList has no value

-----non extending category-----
.. AntiSymm(#1,#2) of cat
(|Join| (|LeftAlgebra| |#1|) (|RetractableTo| |#1|) (CATEGORY |domain| (SIGNATURE |leadingCoeffi

```

section


```
(|BiModule| |#1| |#1|)    finalizing nrlib ANTISYM
```

section

```
finalizing nrlib ANTISYM
Warnings:
  [1] retractable?: k has no value
  [2] retractIfCan: k has no value
  [3] *: k has no value
  [4] *: c has no value
  [5] *: z has no value
  [6] coerce: c has no value
  [7] coerce: k has no value
```

section

```
finalizing nrlib ASP12
Warnings:
  [1] MAXIT has no value
  [2] IFLAG has no value
  [3] ELAM has no value
  [4] FINFO has no value
```

section

```
finalizing nrlib ASP27
Warnings:
  [1] IFLAG has no value
  [2] N has no value
  [3] LRWORK has no value
  [4] LIWORK has no value
  [5] Z has no value
  [6] W has no value
  [7] RWORK has no value
  [8] IWORK has no value
```

section

```
finalizing nrlib ASP28
Warnings:
[1]  IFLAG has no value
[2]  N has no value
[3]  LRWORK has no value
[4]  LIWORK has no value
[5]  Z has no value
[6]  W has no value
[7]  RWORK has no value
[8]  IWORK has no value
```

section

```
finalizing nrlib ASP33
Warnings:
[1]  JINT has no value
[2]  X has no value
[3]  V has no value
```

section

```
finalizing nrlib ASP49
Warnings:
[1]  MODE has no value
[2]  N has no value
[3]  X has no value
[4]  OBJF has no value
[5]  OBJGRD has no value
[6]  NSTATE has no value
[7]  IUSER has no value
[8]  USER has no value
```

section

finalizing nrlib ASP55

Warnings:

- [1] MODE has no value
- [2] NCNLN has no value
- [3] N has no value
- [4] NROWJ has no value
- [5] NEEDC has no value
- [6] X has no value
- [7] C has no value
- [8] CJAC has no value
- [9] NSTATE has no value
- [10] IUSER has no value
- [11] USER has no value
- [12] makeCond: NEEDC has no value

section

finalizing nrlib ASP7

Warnings:

- [1] X has no value
- [2] Y has no value
- [3] F has no value
- [4] coerce: F has no value

section

finalizing nrlib ASP78

Warnings:

- [1] X has no value
- [2] G has no value
- [3] coerce: G has no value

section

finalizing nrlib ASP8

Semantic Errors:

- [1] coerce: code is BOTH a variable and a literal

Warnings:

- [1] COUNT has no value
- [2] XSOL has no value
- [3] N has no value
- [4] Y has no value
- [5] FORWRD has no value
- [6] RESULT has no value
- [7] M has no value
- [8] coerce: POINTS has no value
- [9] coerce: X02ALF has no value
- [10] coerce: RESULT has no value
- [11] coerce: COUNT has no value
- [12] coerce: XSOL has no value

section

finalizing nrlib ASP9

Semantic Errors:

- [1] coerce: code is BOTH a variable and a literal

Warnings:

- [1] X has no value
- [2] Y has no value
- [3] coerce: CHDUM1 has no value
- [4] coerce: CD02EJ has no value
- [5] coerce: AD02CJ has no value
- [6] coerce: GOPT1 has no value
- [7] coerce: GOPT2 has no value

section

finalizing nrlib COMPCAT-

Warnings:

- [1] solveLinearPolynomialEquation: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE))
- [2] factorPolynomial: not known that (IntegralDomain) is of mode (CATEGORY domain (SIGNATURE))
- [3] factorPolynomial: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE))
- [4] reducedSystem: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE))
- [5] patternMatch: not known that (SetCategory) is of mode (CATEGORY domain (SIGNATURE))

section

```

    finalizing nrlib COMPCAT
; (DEFUN |ComplexCategory| ...) is being compiled.
;; The variable |ComplexCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |ComplexCategory;| ...) is being compiled.
;; The variable |ComplexCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

finalizing nrlib DRAWCFUN
Warnings:
[1] drawPlot:  brans has no value
[2] drawPlot:  xValues has no value
[3] drawPlot:  yValues has no value
[4] draw:      fcn has no value
[5] makeObject: fcn has no value
[6] makeObject: pointsColored? has no value
[7] makeObject: col2 has no value
[8] makeObject: col3 has no value

```

section

```

    finalizing nrlib DRAWCFUN
; (DEFUN |DRAWCFUN;myTrap1| ...) is being compiled.
;; The variable |$numericFailure| is undefined.
;; The compiler will assume this variable is a global.

-----non extending category-----
.. d01anfAnnaType of cat
(|NumericalIntegrationCategory|)  has no

```

section

```

(|TableAggregate| (|Symbol|) (|Any|))    finalizing nrlib D01ANFA

```

```
-----non extending category-----
.. d01asfAnnaType of cat
(|NumericalIntegrationCategory|)  has no
```

section

```
(|TableAggregate| (|Symbol|) (|Any|))  finalizing nrlib D01ASFA
```

section

```
finalizing nrlib D03AGNT
--->/research2/test0819/mnt/fedora5/../../src/algebra/D03AGNT.spad-->d03AgentsPackage(constructo
--->/research2/test0819/mnt/fedora5/../../src/algebra/D03AGNT.spad-->d03AgentsPackage(): Missing
```

section

```
finalizing nrlib EP
Warnings:
[1] not known that (SIGNATURE variable ((Union $ failed) (Symbol))) is of mode (CATEGORY d
[2] charpol: :(PositiveInteger) -- should replace by pretend
[3] eigenvalues:  lrat has no value
[4] eigenvalues:  lsym has no value
[5] intAlgEig: not known that (Ring) is of mode (CATEGORY package (SIGNATURE characteristi
[6] eigenvectors:  ratSol has no value
[7] eigenvectors:  algSol has no value
```

section

```
finalizing nrlib E04AGNT
Warnings:
[1] optAttributes:  noa has no value
[2] optAttributes:  lsa has no value
[3] optAttributes:  s has no value
```

section

```

    finalizing nrlib E04AGNT
--->/research2/test0819/mnt/fedora5/../../src/algebra/E04AGNT.spad-->e04AgentsPackage(constructor): Not doc
--->/research2/test0819/mnt/fedora5/../../src/algebra/E04AGNT.spad-->e04AgentsPackage(): Missing Description

-----non extending category-----
.. FortranExpression(#1,#2,#3) of cat
(|Join| (|ExpressionSpace|) (|Algebra| |#3|) (|RetractableTo| |#3|) (|PartialDifferentialRing| (|Symbol|))

```

section

```

(|FunctionSpace| |#3|)    finalizing nrlib FEXPR

```

section

```

finalizing nrlib FEXPR
Warnings:
[1] checkSymbols:  IN has no value
[2] checkForNagOperators:  X01AAF has no value
[3] pi:  X01AAF has no value

```

section

```

finalizing nrlib FFCAT-
Warnings:
[1] rationalPoint?: not known that (IntegralDomain) is of mode (CATEGORY domain (SIGNATURE differentia
[2] primitivePart:  STEP has no value
[3] primitivePart:  i has no value
[4] repOrder:  nostart has no value
[5] kmin:  nostart has no value
[6] kmin:  k has no value
[7] integral?:  den has no value

```

section

```

finalizing nrlib FFCAT
; (DEFUN |FunctionFieldCategory| ...) is being compiled.
;; The variable |FunctionFieldCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |FunctionFieldCategory;| ...) is being compiled.
;; The variable |FunctionFieldCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

finalizing nrlib FFCGP
Warnings:
[1] pretend(NonNegativeInteger) -- should replace by @
[2] +: pretend(SingleInteger) -- should replace by @
[3] +: pretendRep -- should replace by @
[4] retractIfCan: pretendRep -- should replace by @
[5] inGroundField?: pretendRep -- should replace by @
[6] -: pretendRep -- should replace by @
[7] extensionDegree: signature of lhs not unique: (PositiveInteger) chosen
[8] unknown Functor code (error field too large for this representation)

```

section

```

finalizing nrlib FFNB
Warnings:
[1] degree: signature of lhs not unique: (PositiveInteger)$ chosen
[2] coerce: 1 has no value
[3] initializeLog: n has no value
[4] extensionDegree: signature of lhs not unique: (PositiveInteger) chosen

```

section

```

finalizing nrlib FFP
Warnings:

```



```

[1] degree: signature of lhs not unique: (PositiveInteger)$ chosen
[2] tableForDiscreteLogarithm: pretend(Table (PositiveInteger) (NonNegativeInteger)) -- should replace
[3] initializeLog: n has no value
[4] extensionDegree: signature of lhs not unique: (PositiveInteger) chosen

```

section

```
finalizing nrlib FLOAT
```

```
Warnings:
```

```

[1] OMwrite: pretend(String) -- should replace by @
[2] cos: s has no value
[3] exp: e1 has no value
[4] exp1: E has no value
[5] normalize: e has no value
[6] power: y has no value
[7] power10: y has no value

```

section

```
finalizing nrlib FPARFRAC
```

```
Warnings:
```

```

[1] fullPartialFraction: not known that (IntegralDomain) is of mode (CATEGORY domain (SIGNATURE + ($ U
[2] fullParFrac: ans has no value

```

section

```
finalizing nrlib FPARFRAC
```

```

--->/research2/test0819/mnt/fedora5/../../src/algebra/FPARFRAC.spad-->FullPartialFractionExpansion((fullPar
--->/research2/test0819/mnt/fedora5/../../src/algebra/FPARFRAC.spad-->FullPartialFractionExpansion((fullPar
"\spad{fullPartialFraction(f)} returns \spad{[p,{ [[j,{ Dj,{ Hj]...}]]} such that \spad{f = p(x) + \spad{

```

section

```
finalizing nrlib FR
```

```

Warnings:
[1] zero?:  fctr has no value
[2] eval:   IN has no value
[3] eval:   e has no value
[4] convert: fctr has no value
[5] convert: xpnt has no value
[6] exquo:  associate has no value
[7] exquo:  goodQuotient has no value

```

section

```

finalizing nrlib FRNAALG-
Warnings:
[1] leftRankPolynomial:  xx has no value
[2] rightRankPolynomial: xx has no value
[3] coordinates: signature of lhs not unique: (Matrix R)(Vector S) chosen

```

section

```

finalizing nrlib FRNAALG-
; (DEFUN |FRNAALG-;leftRankPolynomial;Sup;1| ...) is being compiled.
;; The variable |xx| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

finalizing nrlib FRNAALG
; (DEFUN |FramedNonAssociativeAlgebra| ...) is being compiled.
;; The variable |FramedNonAssociativeAlgebra;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |FramedNonAssociativeAlgebra;| ...) is being compiled.
;; The variable |FramedNonAssociativeAlgebra;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

finalizing nrlib FS-

Warnings:

- [1] coerce: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE * (S (PositiveInteger)
- [2] subs: IN has no value
- [3] subs: x has no value
- [4] variables: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE * (S (PositiveInteger)
- [5] eval: IN has no value
- [6] eval: f has no value
- [7] smprep: IN has no value
- [8] smprep: a has no value
- [9] smprep: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE * (S (PositiveInteger) S)) (S
- [10] diffdiff0: ans has no value
- [11] diffEval: k has no value
- [12] univariate: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE * (S (PositiveInteger) S
- [13] convert: not known that (IntegralDomain) is of mode (CATEGORY domain (SIGNATURE * (S (PositiveInteger)

section

finalizing nrlib FS

```
; (DEFUN |FunctionSpace| ...) is being compiled.
;; The variable |FunctionSpace;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |FunctionSpace;| ...) is being compiled.
;; The variable |FunctionSpace;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

finalizing nrlib FST

Warnings:

- [1] coerce: REAL has no value
- [2] coerce: INTEGER has no value
- [3] coerce: COMPLEX has no value
- [4] coerce: CHARACTER has no value
- [5] coerce: LOGICAL has no value
- [6] coerce: real has no value
- [7] coerce: integer has no value
- [8] coerce: complex has no value

```
[9] coerce: character has no value
[10] coerce: logical has no value
[11] coerce: upperDoubleComplexSymbol has no value
```

section

```
finalizing nrlib GDMP
Warnings:
[1] zero?: :Rep -- should replace by pretend
[2] listCoef: :Rep -- should replace by pretend
[3] numberOfMonomials: :Rep -- should replace by pretend
[4] monomial?: :Rep -- should replace by pretend
[5] maxNorm: m has no value
[6] variables: :(PositiveInteger) -- should replace by pretend
[7] univariate: deg has no value
[8] coerce: 1 has no value

-----non extending category-----
.. GeneralDistributedMultivariatePolynomial(#1,#2,#3) of cat
(|Join| (|PolynomialCategory| |#2| |#3| (|OrderedVariableList| |#1|)) (CATEGORY |domain| (SIGNATURE
```

section

```
(IF (|has| |#2| (|IntegralDomain|)) (IF (|has| |#3| (|CancellationAbelianMonoid|)) (SIGNATURE |f

-----non extending category-----
.. Pi of cat
(|Join| (|Field|) (|CharacteristicZero|) (|RetractableTo| (|Integer|)) (|RetractableTo| (|Fracti
```

section

```
(|QuotientFieldCategory| (|SparseUnivariatePolynomial| (|Integer|))) finalizing nrlib HACKPI
```

section

```
finalizing nrlib HACKPI
Warnings:
  [1] pi: pretend$ -- should replace by @
```

section

```
finalizing nrlib IDEAL
Warnings:
  [1] not known that (OrderedAbelianMonoidSup) is of mode (CATEGORY domain (IF (has (NonNegativeInteger)
  [2] leadterm: not known that (Ring) is of mode (CATEGORY package (SIGNATURE * ($ $ $)) (SIGNATURE **
  [3] contractGrob: not known that (PolynomialCategory F newExpon VarSet) is of mode (CATEGORY domain (
  [4] monomDim: monvar has no value
  [5] **: :(NonNegativeInteger) -- should replace by pretend
  [6] relationsIdeal: not known that (OrderedAbelianMonoid) is of mode (CATEGORY domain (IF (has Expon
  [7] relationsIdeal: not known that (OrderedAbelianMonoidSup) is of mode (CATEGORY domain (IF (has Exp
  [8] relationsIdeal: not known that (PolynomialCategory F nExponent VarSet1) is of mode (CATEGORY doma
  [9] relationsIdeal: :(Vector (NonNegativeInteger)) -- should replace by pretend
  [10] relationsIdeal: The conditional modes (Polynomial F) and nPoly conflict
```

section

```
finalizing nrlib INFORM
Warnings:
  [1] conv: pretend$ -- should replace by @
```

section

```
finalizing nrlib IR
Warnings:
  [1] nesimp: IN has no value
  [2] nesimp: ne has no value
```

section

```

finalizing nrlib ISUPS
Warnings:
  [1] cSin:  %sin has no value
  [2] cCos:  %cos has no value
  [3] sinhcos: %sin has no value
  [4] sinhcos: %cos has no value
  [5] cSinh:  %sinh has no value
  [6] cCosh:  %cosh has no value

-----non extending category-----
.. Library of cat
(|Join| (|TableAggregate| (|String|) (|Any|)) (CATEGORY |domain| (SIGNATURE |library| ($ (|FileM

```

section

```

(|FileCategory| (|FileName|) (|Record| (|:| |key| (|String|)) (|:| |entry| (|Any|))))    finaliz

```

section

```

finalizing nrlib LMDICT
--->/research2/test0819/mnt/fedora5/../../src/algebra/LMDICT.spad-->ListMultiDictionary(construc
--->/research2/test0819/mnt/fedora5/../../src/algebra/LMDICT.spad-->ListMultiDictionary(): Missi

```

section

```

finalizing nrlib LODOOPS
Warnings:
  [1] killer: not known that (Ring) is of mode (CATEGORY package (SIGNATURE symmetricProduct

```

section

```

finalizing nrlib MATRIX
Warnings:
  [1] positivePower: pretend(Matrix R) -- should replace by @
  [2] determinant: not known that (CommutativeRing) is of mode (CATEGORY R (ATTRIBUTE (commu

```

section

```
finalizing nrlib MKFLCFN
Warnings:
  [1] mkLisp: pretend(InputForm) -- should replace by @
```

section

```
finalizing nrlib MKFLCFN
; (DEFUN |MKFLCFN;mkDefun| ...) is being compiled.
;; The variable |$compileDontDefineFunctions| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib MSET
Warnings:
  [1] intersect:  n has no value
  [2] difference:  n has no value
```

section

```
finalizing nrlib M3D
Warnings:
  [1] matrixConcat3D:  retVal has no value

-----non extending category-----
.. ThreeDimensionalMatrix #1 of cat
(|Join| (|HomogeneousAggregate| |#1|) (CATEGORY |domain| (IF (|has| |#1| (|Ring|)) (PROGN (SIGNATURE |zeroM
```

section

```
(|OneDimensionalArrayAggregate| (|PrimitiveArray| (|PrimitiveArray| |#1|)))    finalizing nrlib
```

section

```
finalizing nrlib NAGC02
```

```
Warnings:
```

- [1] c02aff: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [2] c02agf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @

section

```
finalizing nrlib NAGC05
```

```
Warnings:
```

- [1] c05adf: fn has no value
- [2] c05adf: fp has no value
- [3] c05adf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [4] c05nbf: fn has no value
- [5] c05nbf: fp has no value
- [6] c05nbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [7] c05pbf: fn has no value
- [8] c05pbf: fp has no value
- [9] c05pbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @

section

```
finalizing nrlib NAGC06
```

```
Warnings:
```

- [1] c06eaf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [2] c06ebf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [3] c06ecf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [4] c06ekf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [5] c06fpf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [6] c06fqf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [7] c06frf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @


```
[8] c06fuf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[9] c06gbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[10] c06gcf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[11] c06gqf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[12] c06gsf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
```

section

finalizing nrlib NAGD03

Warnings:

```
[1] d03edf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[2] d03eef: fn has no value
[3] d03eef: fp has no value
[4] d03eef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[5] d03faf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
```

section

finalizing nrlib NAGE01

Warnings:

```
[1] e01baf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[2] e01bef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[3] e01bff: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[4] e01bgf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[5] e01bhf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[6] e01daf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[7] e01saf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[8] e01sbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[9] e01sef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[10] e01sff: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
```

section

finalizing nrlib NAGE02

Warnings:

```
[1] e02adf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[2] e02aef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[3] e02agf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
```

```

[4] e02ahf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[5] e02ajf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[6] e02akf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[7] e02baf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[8] e02bbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[9] e02bcf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[10] e02bdf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[11] e02bef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[12] e02daf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[13] e02dcf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[14] e02ddf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[15] e02def: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[16] e02dff: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[17] e02gaf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[18] e02zaf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by

```

section

finalizing nrlib NAGE04

Warnings:

```

[1] e04dgm: fn has no value
[2] e04dgm: fp has no value
[3] e04dgm: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[4] e04fdf: fn has no value
[5] e04fdf: fp has no value
[6] e04fdf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[7] e04gcm: fn has no value
[8] e04gcm: fp has no value
[9] e04gcm: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[10] e04jaf: fn has no value
[11] e04jaf: fp has no value
[12] e04jaf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[13] e04mbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[14] e04naf: fn has no value
[15] e04naf: fp has no value
[16] e04naf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[17] e04ucf: fn has no value
[18] e04ucf: fp has no value
[19] e04ucf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[20] e04ycf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by

```

section

finalizing nrlib NAGF07

Warnings:

- [1] f07adf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [2] f07aef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [3] f07fdf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [4] f07fef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @

section

finalizing nrlib NAGS

Warnings:

- [1] s01eaf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [2] s13aaf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [3] s13acf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [4] s13adf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [5] s14aaf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [6] s14abf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [7] s14baf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [8] s15adf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [9] s15aef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [10] s17acf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [11] s17adf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [12] s17aef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [13] s17aff: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [14] s17agf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [15] s17ahf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [16] s17ajf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [17] s17akf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [18] s17dcf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [19] s17def: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [20] s17dgf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [21] s17dhf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [22] s17dlf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [23] s18acf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [24] s18adf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [25] s18aef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [26] s18aff: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [27] s18dcf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [28] s18def: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [29] s19aaf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [30] s19abf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [31] s19acf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [32] s19adf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [33] s20acf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
- [34] s20adf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @

```

[35] s21baf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[36] s21bbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[37] s21bcf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[38] s21bdf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by

```

section

```

finalizing nrlib NAGSP
Warnings:
[1] getUniqueId:  uniqueId has no value

```

section

```

finalizing nrlib NAGSP
; (DEFUN |NAGSP;checkPrecision;B;1| ...) is being compiled.
;; The variable |$fortranPrecision| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$nagEnforceDouble| is undefined.
;; The compiler will assume this variable is a global.
;; The variable |$nagMessages| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NAGSP;fortranCompilerName;S;4| ...) is being compiled.
;; The variable |$fortranCompilerName| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NAGSP;fortranLinkerArgs;S;5| ...) is being compiled.
;; The variable |$fortranLibraries| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

finalizing nrlib NUMFMT
Warnings:
[1] ScanRoman:  nprens has no value

```

section

```

    finalizing nrlib OC
; (DEFUN |OctonionCategory| ...) is being compiled.
;; The variable |OctonionCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |OctonionCategory;| ...) is being compiled.
;; The variable |OctonionCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib ODEPACK
--->/research2/test0819/mnt/fedora5/../../src/algebra/ODEPACK.spad-->AnnaOrdinaryDifferentialEquationPackage
--->/research2/test0819/mnt/fedora5/../../src/algebra/ODEPACK.spad-->AnnaOrdinaryDifferentialEquationPackage

```

section

```

    finalizing nrlib ODERAT
Warnings:
[1] ratDsolve1: not known that (Ring) is of mode (CATEGORY package (SIGNATURE ratDsolve ((Record (: p
[2] ratDsolve0:  b has no value
[3] infMuLambda:  lamb has no value
[4] infMuLambda:  lf has no value

```

section

```

    finalizing nrlib OPTPACK
Warnings:
[1] constant:  noa has no value
[2] measureSpecific:  noa has no value
[3] measureSpecific:  lsa has no value
[4] optimizeSpecific:  noa has no value
[5] optimizeSpecific:  lsa has no value
[6] goodnessOfFit:  lsa has no value
[7] goodnessOfFit:  method has no value

```

```

[8] goodnessOfFit:  nameOfRoutine has no value
[9] goodnessOfFit:  w has no value
[10] goodnessOfFit:  objf has no value

```

section

```

finalizing nrlib OPTPACK
--->/research2/test0819/mnt/fedora5/../../src/algebra/OPTPACK.spad-->AnnaNumericalOptimizationPa
--->/research2/test0819/mnt/fedora5/../../src/algebra/OPTPACK.spad-->AnnaNumericalOptimizationPa

```

section

```

finalizing nrlib PATTERN
Warnings:
[1] quoted?:  sym has no value
[2] generic?:  sym has no value
[3] multiple?:  sym has no value
[4] optional?:  sym has no value
[5] =: pretend(Boolean) -- should replace by @
[6] isOp:  ker has no value
[7] variables:  exp has no value
[8] variables:  qot has no value
[9] variables:  ker has no value
[10] PAT20:  ret has no value
[11] PAT20:  sym has no value
[12] PAT20:  exp has no value
[13] PAT20:  qot has no value
[14] PAT20:  ker has no value
[15] patcopy:  ret has no value
[16] patcopy:  sym has no value
[17] patcopy:  ker has no value
[18] patcopy:  qot has no value
[19] patcopy:  exp has no value
[20] pateq?:  ret has no value
[21] pateq?:  qot has no value
[22] pateq?:  sym has no value
[23] pateq?:  ker has no value
[24] pateq?:  exp has no value

```

section

```

    finalizing nrlib PATTERN
; (DEFUN |Pattern;| ...) is being compiled.
;; The variable IDENTITY is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib PMKERNEL
Warnings:
  [1] patternMatchArg: not known that (SetCategory) is of mode (CATEGORY package (SIGNATURE patternMatchArg))
  [2] patternMatchInner: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE patternMatchInner))

```

section

```

    finalizing nrlib PRIMELT
Warnings:
  [1] triangularLinearIfCan: not known that (Ring) is of mode (CATEGORY package (SIGNATURE primitiveElement))

```

section

```

    finalizing nrlib QALGSET2
Warnings:
  [1] npoly: not known that (PolynomialCategory (Fraction (Integer)) (DirectProduct nv (NonNegativeInteger)))
  [2] radicalSimplify: not known that (PolynomialCategory (Fraction (Integer)) (DirectProduct nv (NonNegativeInteger)))
  [3] radicalSimplify: gb has no value

```

section

```

    finalizing nrlib REP1
Warnings:

```

```

[1] antisymmetricTensors: not known that (Ring) is of mode (CATEGORY R (ATTRIBUTE (commuta

-----non extending category-----
.. Result of cat
(|Join| (|TableAggregate| (|Symbol|) (|Any|)) (CATEGORY |package| (SIGNATURE |showScalarValues|

```

section

```

(ATTRIBUTE |finiteAggregate|)      finalizing nrlib RESULT

```

section

```

      finalizing nrlib QUATCAT
; (DEFUN |QuaternionCategory| ...) is being compiled.
;; The variable |QuaternionCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |QuaternionCategory;| ...) is being compiled.
;; The variable |QuaternionCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

      finalizing nrlib RFFACT
Warnings:
[1] likuniv: not known that (Ring) is of mode (CATEGORY package (SIGNATURE factor ((Factor

-----non extending category-----
.. RectangularMatrix(#1,#2,#3) of cat
(|Join| (|RectangularMatrixCategory| |#1| |#2| |#3| (|DirectProduct| |#2| |#3|) (|DirectProduct|

```

section

```

(|MatrixCategory| |#3| (|Vector| |#3|) (|Vector| |#3|))      finalizing nrlib RMATRIX

(|RealConstant|)      extends
(|ConvertibleTo| (|DoubleFloat|))      but not

```



```
(|ConvertibleTo| (|String|)) -----non extending category-----
```

```
.. RomanNumeral of cat
```

```
(|Join| (|IntegerNumberSystem|) (CATEGORY |domain| (ATTRIBUTE |canonical|) (ATTRIBUTE |canonicalsClosed|) (
```

section

```
(|ConvertibleTo| (|String|))      finalizing nrlib ROMAN
```

```
-----non extending category-----
```

```
.. RoutinesTable of cat
```

```
(|Join| (|TableAggregate| (|Symbol|) (|Any|)) (CATEGORY |domain| (SIGNATURE |concat| ($ $ $)) (SIGNATURE |r
```

section

```
finalizing nrlib RPOLCAT-
```

```
Warnings:
```

```
[1] mainCoefficients: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE coerce (S S)) (SIG
```

```
[2] convert: stpol has no value
```

section

```
finalizing nrlib RPOLCAT-
```

```
Warning: RPOLCAT-;exactQuo has a duplicate definition in this file
```

```
Warning: RPOLCAT-;ZToR has a duplicate definition in this file
```

```
Warning: RPOLCAT-;PZToPR has a duplicate definition in this file
```

section

```
finalizing nrlib RPOLCAT
```

```
; (DEFUN |RecursivePolynomialCategory| ...) is being compiled.
```

```
;; The variable |RecursivePolynomialCategory;AL| is undefined.
```

```
;; The compiler will assume this variable is a global.
```

```
; (DEFUN |RecursivePolynomialCategory;| ...) is being compiled.
```

```
;; The variable |RecursivePolynomialCategory;CAT| is undefined.
```

```
;; The compiler will assume this variable is a global.
```

section

```

finalizing nrlib SAOS
-->/research2/test0819/mnt/fedora5/../../src/algebra/SAOS.spad-->SingletonAsOrderedSet((create
-->/research2/test0819/mnt/fedora5/../../src/algebra/SAOS.spad-->SingletonAsOrderedSet((convert

-----non extending category-----
.. SquareMatrix(#1,#2) of cat
(|Join| (|SquareMatrixCategory| |#1| |#2| (|DirectProduct| |#1| |#2|) (|DirectProduct| |#1| |#2|)

```

section

```

(|MatrixCategory| |#2| (|Vector| |#2|) (|Vector| |#2|))    finalizing nrlib SQMATRIX

```

section

```

finalizing nrlib SWITCH
Warnings:
[1]  NULL has no value

```

section

```

finalizing nrlib SYMS
Semantic Errors:
[1]  Domain is not a known type
[2]  void is not a known type

```

section

```

finalizing nrlib SYMTAB

```

Warnings:

- [1] typeList: fst has no value
- [2] typeList2: fst has no value
- [3] newTypeLists: fst has no value
- [4] oForm2: S has no value
- [5] oForm2: P has no value
- [6] oForm: bounds has no value

section

finalizing nrlib SYSSOLP

Warnings:

- [1] makeEq: not known that (Ring) is of mode (CATEGORY package (SIGNATURE solve ((List (List (Equation
- [2] triangularSystems: not known that (SIGNATURE variable ((Union \$ failed) (Symbol))) is of mode (CA
- [3] triangularSystems: not known that (PolynomialCategory (Polynomial R) DP OV) is of mode (CATEGORY o
- [4] triangularSystems: not known that (PolynomialCategory (Polynomial R) (DirectProduct (# lv) (NonNeg
- [5] triangularSystems: not known that (PolynomialCategory (Polynomial R) (DirectProduct (call LENGTH

section

finalizing nrlib SYSSOLP

```
--->/research2/test0819/mnt/fedora5/../../src/algebra/SYSSOLP.spad-->SystemSolvePackage(constructor): Not d
--->/research2/test0819/mnt/fedora5/../../src/algebra/SYSSOLP.spad-->SystemSolvePackage(): Missing Descript
```

section

finalizing nrlib UTSCAT-

Warnings:

- [1] coerce: 1 has no value

section

finalizing nrlib UTSCAT

```
; (DEFUN |UnivariateTaylorSeriesCategory| ...) is being compiled.
```

```
;; The variable |UnivariateTaylorSeriesCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |UnivariateTaylorSeriesCategory;| ...) is being compiled.
;; The variable |UnivariateTaylorSeriesCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib WFFINTBS
Warnings:
[1] listSquaredFactors:  ans has no value
[2] iLocalIntegralBasis:  rbdn has no value
[3] integralBasis:  runningRbdn has no value
```

section

```
finalizing nrlib ACFS-
Warnings:
[1] zeroOf: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE zerosOf ((List S) S) S) S)
[2] zeroOf: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE zerosOf ((List S) S) S) S)
[3] not known that (Ring) is of mode (CATEGORY domain (SIGNATURE zerosOf ((List S) S) S) S)
```

section

```
finalizing nrlib ACFS
; (DEFUN |AlgebraicallyClosedFunctionSpace| ...) is being compiled.
;; The variable |AlgebraicallyClosedFunctionSpace;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |AlgebraicallyClosedFunctionSpace;| ...) is being compiled.
;; The variable |AlgebraicallyClosedFunctionSpace;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
[1] rootOf: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE rootOf (F (SparseUnivariates (Ring)) is of mode (CATEGORY package (SIGNATURE rootOf (F (SparseUnivariate)
[3] inrootof: not known that (Ring) is of mode (CATEGORY package (SIGNATURE rootOf (F (SparseUnivariate)
[4] hackroot: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE iroot (F R (Integer)).
[5] inroot0: rn has no value
[6] inroot0: coef has no value
[7] inroot0: radicand has no value
[8] inroot0: exponent has no value
[9] inroot0: rd has no value
```

```
finalizing nrlib ALGFACT
```

```
[1] allk: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE factor ((Factored UP) UP (L
[2] extend: not known that (Ring) is of mode (CATEGORY package (SIGNATURE factor ((Factored UP) UP (L
[3] fact: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE factor ((Factored UP) UP
[4] fact: not known that (SIGNATURE coerce ($ (SparseMultivariatePolynomial (Integer) (Kernel (Algebraic
[5] fact: not known that (SIGNATURE number ((SparseMultivariatePolynomial (Integer) (Kernel (Algebraic
[6] fact: not known that (SIGNATURE denom ((SparseMultivariatePolynomial (Integer) (Kernel (Algebraic
```

finalizing nrlib ALGMANIP

```
[1] ratDenom: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE rootSplit (F F)) (SIGNATURE
[2] ratPoly: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE rootSplit (F F)) (SIGNATURE
[3] ratPoly: not known that (SIGNATURE coerce ($ (SparseMultivariatePolynomial R (Kernel F)))) is of mode
[4] ratPoly: not known that (SIGNATURE number ((SparseMultivariatePolynomial R (Kernel F)) $)) is of mode
[5] ratPoly: not known that (SIGNATURE denom ((SparseMultivariatePolynomial R (Kernel F)) $)) is of mode
[6] innerRF: not known that (Ring) is of mode (CATEGORY package (SIGNATURE rootSplit (F F)) (SIGNATURE
```

```
finalizing nrlib ALGSC
```

Warnings:

```
[1] elt: :Rep -- should replace by pretend
[2] coerce: le has no value
```

```
-----non extending category-----
```

```
.. AlgebraGivenByStructuralConstants(#1,#2,#3,#4) of cat
(|Join| (|FramedNonAssociativeAlgebra| |#1|) (|LeftModule| (|SquareMatrix| |#2| |#1|)) (CATEGORY
```

section

```
(|DirectProductCategory| |#2| |#1|) finalizing nrlib ALGSC
```

```
(|RealConstant|) extends
(|ConvertibleTo| (|DoubleFloat|)) but not
(|ConvertibleTo| (|Complex| (|Float|))) -----non extending category-----
.. AlgebraicNumber of cat
(|Join| (|ExpressionSpace|) (|AlgebraicallyClosedField|) (|RetractableTo| (|Integer|)) (|Retract
```

section

```
finalizing nrlib AN
```

section

```
finalizing nrlib APPRULE
```

```
Warnings:
```

```
[1] splitRules: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE applyR
[2] rewrite: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE applyRule
[3] isitwithpred: not known that (SetCategory) is of mode (CATEGORY package (SIGNATURE app
```

section

```
finalizing nrlib ASP19
```

```
Warnings:
```

```
[1] M has no value
[2] N has no value
[3] LJC has no value
```

```

[4]  XC has no value
[5]  FVECC has no value
[6]  FJACC has no value
[7]  localAssign1: not known that (FiniteLinearAggregate (FortranExpression (construct) (construct (QU
[8]  coerce:  FJACC has no value

```

section

```

finalizing nrlib ASP20
Warnings:
  [1]  N has no value
  [2]  NROWH has no value
  [3]  NCOLH has no value
  [4]  JTHCOL has no value
  [5]  HESS has no value
  [6]  X has no value
  [7]  HX has no value

```

section

```

finalizing nrlib ASP30
Semantic Errors:
  [1]  coerce:  code is BOTH a variable and a literal

Warnings:
  [1]  MODE has no value
  [2]  M has no value
  [3]  N has no value
  [4]  LRWORK has no value
  [5]  LIWORK has no value
  [6]  X has no value
  [7]  Y has no value
  [8]  RWORK has no value
  [9]  IWORK has no value
  [10] IFAIL has no value
  [11] coerce:  A has no value

```

section

finalizing nrlib ASP31

Semantic Errors:

[1] coerce: code is BOTH a variable and a literal

Warnings:

[1] X has no value

[2] Y has no value

[3] localAssign: not known that (FiniteLinearAggregate (FortranExpression (construct (QUOT

[4] coerce: PW has no value

section

finalizing nrlib ASP35

Warnings:

[1] N has no value

[2] X has no value

[3] FVEC has no value

[4] LDFJAC has no value

[5] FJAC has no value

[6] IFLAG has no value

[7] localAssign1: not known that (FiniteLinearAggregate (FortranExpression (construct) (co

section

finalizing nrlib ASP41

Warnings:

[1] N has no value

[2] X has no value

[3] EPS has no value

[4] Y has no value

[5] F has no value

[6] localAssign1: not known that (FiniteLinearAggregate (FortranExpression (construct (QUO

[7] makeCodeThree: EPS has no value

section

finalizing nrlib ASP42

Warnings:

[1] EPS has no value

[2] N has no value

[3] YA has no value

[4] YB has no value

[5] BC has no value

[6] AJ has no value

[7] BJ has no value

[8] BCEP has no value

[9] localAssign1: not known that (FiniteLinearAggregate (FortranExpression (construct (QUOTE EPS)) (c

[10] makeCodeThree: EPS has no value

section

finalizing nrlib ASP74

Warnings:

[1] X has no value

[2] Y has no value

[3] A has no value

[4] B has no value

[5] C has no value

[6] IBND has no value

section

finalizing nrlib ASP77

Warnings:

[1] X has no value

[2] localAssign: not known that (FiniteLinearAggregate (FortranExpression (construct (QUOTE X)) (const

[3] coerce: F has no value

section

```
finalizing nrlib ASP80
Warnings:
[1] XL has no value
[2] XR has no value
[3] ELAM has no value
[4] YL has no value
[5] YR has no value
```

section

```
finalizing nrlib CINTSLPE
Warnings:
[1] solveLinearPolynomialEquation: slpePrime has no value
[2] solveLinearPolynomialEquation: oldtable has no value
```

section

```
finalizing nrlib COMPFACT
Warnings:
[1] conv: pris has no value
[2] conv: dris has no value
[3] backConv: fctr has no value
[4] backConv: xpnt has no value
[5] backConv: flg has no value
```

section

```
finalizing nrlib COMPLEX
Warnings:
[1] OMwrite: pretend(String) -- should replace by @
```

section

```
finalizing nrlib CMPLXRT
```

```
--->/research2/test0819/mnt/fedora5/../../src/algebra/CMPLXRT.spad-->ComplexRootPackage(constructor): Not d
--->/research2/test0819/mnt/fedora5/../../src/algebra/CMPLXRT.spad-->ComplexRootPackage(): Missing Descript.
```

section

```
finalizing nrlib CPMATCH
```

```
Warnings:
```

```
[1] patternMatch: not known that (SetCategory) is of mode (CATEGORY $ (SIGNATURE patternMatch ((Patter
```

section

```
finalizing nrlib CRFP
```

```
Warnings:
```

```
[1] pleskenSplit: split has no value
[2] startPolynomial: centerIsRoot has no value
[3] startPolynomial: fp has no value
[4] factor: result has no value
```

section

```
finalizing nrlib CTRIGMNP
```

```
Warnings:
```

```
[1] complexForm: not known that (RadicalCategory) is of mode (CATEGORY domain (IF (has R (IntegralDom
[2] complexForm: not known that (TranscendentalFunctionCategory) is of mode (CATEGORY domain (IF (has
[3] real?: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE complexNormalize (F F)
[4] complexElementary: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE complexNor
```

section

```
finalizing nrlib D01WGTS
```

```
Warnings:
```

```
[1] exprIsLogarithmicWeight: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE expr
[2] functionIsQuotient: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE exprHasWe
```

section

```

finalizing nrlib D02AGNT
Warnings:
  [1] stiffnessAndStabilityFactor: b has no value
  [2] stiffnessAndStabilityFactor: e has no value
  [3] stiffnessAndStabilityOfODE: Y has no value

-----non extending category-----
.. d03eefAnnaType of cat
(|PartialDifferentialEquationsSolverCategory|) has no

```

section

```

(|TableAggregate| (|Symbol|) (|Any|)) finalizing nrlib D03EEFA

-----non extending category-----
.. DeRhamComplex(#1,#2) of cat
(|Join| (|LeftAlgebra| (|Expression| |#1|)) (|RetractableTo| (|Expression| |#1|)) (CATEGORY |dom

```

section

```

finalizing nrlib DERHAM

```

section

```

finalizing nrlib DRAWCURV
Warnings:
  [1] draw: xRangeFloat has no value
  [2] draw: yRangeFloat has no value
  [3] draw: floatRanges has no value

-----non extending category-----
.. e04nafAnnaType of cat
(|NumericalOptimizationCategory|) has no

```

section

```
(|TableAggregate| (|Symbol|) (|Any|))    finalizing nrlib E04NAFA
```

```
-----non extending category-----
```

```
.. e04ucfAnnaType of cat
```

```
(|NumericalOptimizationCategory|)    has no
```

section

```
(|TableAggregate| (|Symbol|) (|Any|))    finalizing nrlib E04UCFA
```

section

```
finalizing nrlib EF
```

```
Warnings:
```

```
[1] pi: Pie has no value
```

```
[2] iisqrt1: isqrt1 has no value
```

```
[3] iisqrt2: isqrt2 has no value
```

```
[4] iisqrt3: isqrt3 has no value
```

```
[5] kernel: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE exp (F F)) (SIGNATURE
```

```
[6] ilog: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE exp (F F)) (SIGNATURE 1
```

section

```
finalizing nrlib EFSTRUC
```

```
Warnings:
```

```
[1] realElem: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE normalize (F F)) (S
```

```
[2] rtNormalize: func has no value
```

```
[3] tanQ: not known that (Ring) is of mode (CATEGORY package (SIGNATURE normalize (F F)) (SIGNATURE n
```

```
[4] rootNormalize0: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE normalize (F F)) (SIGNATURE
```

```
[5] ktoY: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE normalize (F F)) (SIGNATURE
```

```
[6] k2Elem: ez has no value
```

```
[7] k2Elem: iez has no value
```

```
[8] k2Elem: tz2 has no value
```

```
[9] tannosimp: den has no value
```

```
[10] tannosimp: num has no value
```

```

[11] expnosimp: den has no value
[12] expnosimp: num has no value
[13] rischNormalize: vec has no value
[14] rootKernelNormalize: vec has no value
[15] validExponential: IN has no value
[16] validExponential: g has no value

```

section

```

finalizing nrlib ESTOOLS
Warnings:
[1] isQuotient: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE f2df (
[2] numberOfOperations1: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATU

-----non extending category-----
.. ExponentialExpansion(#1,#2,#3,#4) of cat
(|Join| (|QuotientFieldCategory| (|UnivariatePuisseuxSeriesWithExponentialSingularity| |#1| |#2|

```

section

```

(IF (|has| (|UnivariatePuisseuxSeriesWithExponentialSingularity| |#1| |#2| |#3| |#4|) (|IntegerNu

```

section

```

finalizing nrlib EXPRODE
Warnings:
[1] diffRhs: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE seriesSol
[2] k2exquo: IN has no value
[3] k2exquo: f has no value
[4] div2exquo: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE seriesS
[5] diffRhsK: not known that (Ring) is of mode (CATEGORY package (SIGNATURE seriesSolve ((
[6] findEq: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE seriesSolv
[7] seriesSolve: IN has no value
[8] seriesSolve: deq has no value
[9] seriesSolve: y has no value

```

section

finalizing nrlib FC

Semantic Errors:

- [1] common: name is BOTH a variable and a literal
- [2] common: contents is BOTH a variable and a literal
- [3] forLoop: body is BOTH a variable and a literal

Warnings:

- [1] getCode: conditionalBranch has no value
- [2] getCode: assignmentBranch has no value
- [3] getCode: arrayAssignmentBranch has no value
- [4] getCode: conditionalBranch has no value
- [5] getCode: returnBranch has no value
- [6] getCode: blockBranch has no value
- [7] getCode: commentBranch has no value
- [8] getCode: callBranch has no value
- [9] getCode: forBranch has no value
- [10] getCode: labelBranch has no value
- [11] getCode: loopBranch has no value
- [12] getCode: printBranch has no value
- [13] getCode: commonBranch has no value

section

finalizing nrlib FC

```
; (DEFUN |FC;getContinue| ...) is being compiled.
;; The variable |$fortIndent| is undefined.
;; The compiler will assume this variable is a global.
```

section

finalizing nrlib FDIVCAT

```
; (DEFUN |FiniteDivisorCategory| ...) is being compiled.
;; The variable |FiniteDivisorCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |FiniteDivisorCategory;| ...) is being compiled.
;; The variable |FiniteDivisorCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib FSRED
```

```
Warnings:
```

- [1] bringDown: not known that (Ring) is of mode (CATEGORY package (SIGNATURE bringDown ((F
- [2] bringDown: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE bringDo

section

```
finalizing nrlib FSUPFACT
```

```
Warnings:
```

- [1] anfactor: overq has no value
- [2] anfactor: overan has no value
- [3] ffactor0: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE ffactor
- [4] ffactor0: not known that (Ring) is of mode (CATEGORY package (SIGNATURE ffactor ((Fact
- [5] qfactor: overq has no value

section

```
finalizing nrlib FSPECF
```

```
Warnings:
```

- [1] iiabs: not known that (Ring) is of mode (CATEGORY R (SIGNATURE abs (R R)))

section

```
finalizing nrlib FS2
```

```
Warnings:
```

- [1] map: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE map (B (Mappi

section

```
finalizing nrlib FS2UPS
```


Warnings:

- [1] exprToUPS: not known that (Ring) is of mode (CATEGORY package (SIGNATURE exprToUPS ((Union (: %ser
- [2] listToUPS: %series has no value
- [3] powerToUPS: %series has no value
- [4] kernelToUPS: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE exprToUPS ((Union (
- [5] nthRootToUPS: %series has no value
- [6] logToUPS: %series has no value
- [7] atancotToUPS: %series has no value
- [8] applyIfCan: %series has no value
- [9] powToUPS: %series has no value
- [10] newElem: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE exprToUPS ((Union (
- [11] opsInvolvingX: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE exprToUPS ((U
- [12] powerToGenUPS: %series has no value
- [13] nthRootToGenUPS: %series has no value
- [14] logToGenUPS: %series has no value
- [15] expToGenUPS: %series has no value
- [16] expGenUPS: %series has no value
- [17] atancotToGenUPS: %problem has no value
- [18] atancotToGenUPS: %series has no value
- [19] genUPSApplyIfCan: %series has no value
- [20] applyBddIfCan: %problem has no value
- [21] applyBddIfCan: %series has no value
- [22] powToGenUPS: %series has no value

section

finalizing nrlib GAUSSFAC

Warnings:

- [1] intfactor: unity has no value
- [2] factor: :(NonNegativeInteger) -- should replace by pretend
- [3] factor: result has no value
- [4] factor: unity has no value

section

finalizing nrlib GCNAALG

Warnings:

- [1] leftRankPolynomial: signature of lhs not unique: (SparseUnivariatePolynomial (Fraction (Polynomial
- [2] rightRankPolynomial: signature of lhs not unique: (SparseUnivariatePolynomial (Fraction (Polynomial
- [3] genericLeftNorm: rf has no value
- [4] genericRightNorm: rf has no value

section

```

finalizing nrlib GCNAALG
(|Module| (|Fraction| (|Polynomial| |#1|))) extends
(|LeftModule| (|Fraction| (|Polynomial| |#1|))) but not
(|LeftModule| (|SquareMatrix| |#2| (|Fraction| (|Polynomial| |#1|))))

```

section

```

finalizing nrlib GENUFACT
Warnings:
  [1] factor: not known that (UnivariatePolynomialCategory (Integer)) is of mode (CATEGORY d
  [2] factor: not known that (UnivariatePolynomialCategory (Fraction (Integer))) is of mode
  [3] factor: not known that (UnivariatePolynomialCategory (Complex (Integer))) is of mode (
  [4] factor: not known that (UnivariatePolynomialCategory (Complex (Fraction (Integer)))) i
  [5] factor: not known that (UnivariatePolynomialCategory (AlgebraicNumber)) is of mode (CA
  [6] factor: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE factor
-----non extending category-----
.. InnerAlgebraicNumber of cat
(|Join| (|ExpressionSpace|) (|AlgebraicallyClosedField|) (|RetractableTo| (|Integer|)) (|Retract

```

section

```

(|FunctionSpace| (|Integer|)) finalizing nrlib IAN

```

section

```

finalizing nrlib INEP
Warnings:
  [1] charpol: :(PositiveInteger) -- should replace by pretend
  [2] unknown Functor code (error unsupported matrix type)

```

section

```
finalizing nrlib INFSP
```

```
Warnings:
```

- [1] evaluate: not known that (Ring) is of mode (CATEGORY package (SIGNATURE innerSolve1 ((List F) (Sp
- [2] innerSolve1: not known that (UnivariatePolynomialCategory (Complex (Integer))) is of mode (CATEGOR
- [3] innerSolve1: pretend(List F) -- should replace by @
- [4] innerSolve: pretenddmp -- should replace by @
- [5] innerSolve: lq has no value
- [6] innerSolve: not known that (PolynomialCategory K (DirectProduct (# lv) (NonNegativeInteger)) OV) :
- [7] innerSolve: not known that (PolynomialCategory K (DirectProduct (call LENGTH lv) (NonNegativeInte
- [8] innerSolve: pretend(List dmp) -- should replace by @
- [9] innerSolve: listGen has no value
- [10] innerSolve: result has no value

section

```
finalizing nrlib INTAF
```

```
Warnings:
```

- [1] rootintegrate: not known that (Ring) is of mode (CATEGORY package (SIGNATURE algint ((Integrati
- [2] rootintegrate: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE algint ((Integ
- [3] UP2UPUP: not known that (Ring) is of mode (CATEGORY package (SIGNATURE algint ((IntegrationResult

section

```
finalizing nrlib INTALG
```

```
Warnings:
```

- [1] R2UP: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE algintegrate ((Integrat
- [2] R2UP: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE algintegrate ((Integ
- [3] R2UP: not known that (Ring) is of mode (CATEGORY package (SIGNATURE algintegrate ((IntegrationRes
- [4] univ: not known that (Ring) is of mode (CATEGORY package (SIGNATURE algintegrate ((IntegrationRes

section

```
finalizing nrlib INTEF
```

```
Warnings:
```

```

[1] tanint: not known that (Ring) is of mode (CATEGORY package (SIGNATURE lfextendedint ((
[2] tanint: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE lfextended
[3] unknownint: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE lfexte
[4] droponex: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE lfextend
[5] alglfint: not known that (Ring) is of mode (CATEGORY package (SIGNATURE lfextendedint

```

section

finalizing nrlib INTGO

Warnings:

```

[1] kerdiff: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE palgint0
[2] univ: not known that (Ring) is of mode (CATEGORY package (SIGNATURE palgint0 ((Integra
[3] univ: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE palgint0 ((I
[4] palgint0: not known that (Ring) is of mode (CATEGORY package (SIGNATURE palgint0 ((Int
[5] palgint0: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE palg
[6] palgrDE0: var has no value
[7] palgrDE0: coeff has no value

```

section

finalizing nrlib INTHERAL

Warnings:

```

[1] HermiteIntegrate: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNAT
[2] localsolve: coeff has no value

```

section

finalizing nrlib INTPAF

Warnings:

```

[1] UPUP2F0: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE palgint
[2] chv: not known that (Ring) is of mode (CATEGORY package (SIGNATURE palgint ((Integrati
[3] linearInXIfCan: not known that (Ring) is of mode (CATEGORY package (SIGNATURE palgint
[4] linearInXIfCan: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATUR
[5] prootintegrate1: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE p
[6] candidates: 1 has no value

```

section

finalizing nrlib INTPM

Warnings:

- [1] splitConstant: more than 1 modemap for: (One) with dc=F ==>(((F F) ((has R (SemiGroup)) (CONST F
- [2] matcherfei: not known that (SetCategory) is of mode (CATEGORY domain (SIGNATURE pmlntegrate ((Union
- [3] matchdilog: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE pmlntegrate ((Union
- [4] matchdilog0: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE pmlntegrate ((Union (Re
- [5] matchdilog0: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE pmlntegrate ((Union
- [6] matchli0: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE pmlntegrate ((Union (Record
- [7] matchsici: ci0? has no value
- [8] pmlntegrate: const has no value
- [9] pmlntegrate: nconst has no value
- [10] pmComplexintegrate: const has no value
- [11] pmComplexintegrate: nconst has no value
- [12] formula1: not known that (OrderedSet) is of mode (CATEGORY \$ (SIGNATURE pmlntegrate ((Union F fa

section

finalizing nrlib INTTOOLS

Warnings:

- [1] varselect: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE varselect ((List (I
- [2] vark: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE varselect ((List (Kerne
- [3] removeConstantTerm: not known that (OrderedSet) is of mode (CATEGORY \$ (SIGNATURE removeConstantT
- [4] removeConstantTerm: ans has no value
- [5] cont: not known that (Ring) is of mode (CATEGORY \$ (SIGNATURE mkPrim (F F (Symbol))))
- [6] cont: unit has no value
- [7] linearLog?: not known that (Ring) is of mode (CATEGORY \$ (SIGNATURE mkPrim (F F (Symbol))))
- [8] intPatternMatch: nl has no value

section

finalizing nrlib ITRIGMNP

Warnings:

- [1] FG2F: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE GF2FG (FG (Complex F)))
- [2] ker2explogs: IN has no value
- [3] ker2explogs: v has no value
- [4] ker2explogs: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE GF2FG (FG (Comp1
- [5] ker2explogs: a has no value
- [6] ker2trigs: IN has no value

```
[7] explogs2trigs: not known that (Ring) is of mode (CATEGORY package (SIGNATURE GF2FG (FG
```

section

```
finalizing nrlib KOVACIC
Warnings:
  [1] kovacic: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE kovac
```

section

```
finalizing nrlib LF
Warnings:
  [1] integrand: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE belong?
  [2] eqint: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE belong? ((B
```

section

```
finalizing nrlib LODOF
Warnings:
  [1] factor: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE factor
  [2] zro1: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE factor1 ((List (Lin
```

section

```
finalizing nrlib LSQM
Warnings:
  [1] not known that (NonAssociativeAlgebra R) is of mode (CATEGORY domain (SIGNATURE transp

-----non extending category-----
.. LieSquareMatrix(#1,#2) of cat
(|Join| (|SquareMatrixCategory| |#1| |#2| (|DirectProduct| |#1| |#2|) (|DirectProduct| |#1| |#2|
```

section

```
(|CoercibleTo| (|SquareMatrix| |#1| |#2|))    finalizing nrlib LSQM
```

section

```
finalizing nrlib OMEXPR
```

```
Warnings:
```

```
[1] outputOMFunction: %defint has no value
```

```
[2] outputOMFunction: %defsum has no value
```

```
[3] outputOMFunction: %defprod has no value
```

```
[4] outputOMFunction: %power has no value
```

```
[5] outputOMExpr: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE OMwrite ((String
```

```
[6] OMwrite: pretend(String) -- should replace by @
```

section

```
finalizing nrlib OMEXPR
```

```
--->/research2/test0819/mnt/fedora5/../../src/algebra/OMEXPR.spad-->ExpressionToOpenMath((OMwrite ((String)
```

```
--->/research2/test0819/mnt/fedora5/../../src/algebra/OMEXPR.spad-->ExpressionToOpenMath((OMwrite ((String)
```

```
--->/research2/test0819/mnt/fedora5/../../src/algebra/OMEXPR.spad-->ExpressionToOpenMath((OMwrite ((Void) (
```

```
--->/research2/test0819/mnt/fedora5/../../src/algebra/OMEXPR.spad-->ExpressionToOpenMath((OMwrite ((Void) (
```

```
-----non extending category-----
```

```
.. MachineComplex of cat
```

```
(|Join| (|FortranMachineTypeCategory|) (|ComplexCategory| (|MachineFloat|)) (CATEGORY |domain| (SIGNATURE |
```

section

```
(IF (|has| (|MachineFloat|) (|OpenMath|)) (ATTRIBUTE (|OpenMath|)) |noBranch|)    finalizing nrlib MCMPLX
```

section

```
finalizing nrlib MULTFACT
```

```
Warnings:
```

```
[1] factor: not known that (OrderedSet) is of mode (CATEGORY OV (SIGNATURE convert ((Symbol
```

section

```
finalizing nrlib NAGD01
```

```
Warnings:
```

```
[1] d01ajf: fn has no value
[2] d01ajf: fp has no value
[3] d01ajf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[4] d01akf: fn has no value
[5] d01akf: fp has no value
[6] d01akf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[7] d01alf: fn has no value
[8] d01alf: fp has no value
[9] d01alf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[10] d01amf: fn has no value
[11] d01amf: fp has no value
[12] d01amf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[13] d01anf: fn has no value
[14] d01anf: fp has no value
[15] d01anf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[16] d01apf: fn has no value
[17] d01apf: fp has no value
[18] d01apf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[19] d01aqf: fn has no value
[20] d01aqf: fp has no value
[21] d01aqf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[22] d01asf: fn has no value
[23] d01asf: fp has no value
[24] d01asf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[25] d01bbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[26] d01fcf: fn has no value
[27] d01fcf: fp has no value
[28] d01fcf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[29] d01gaf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
[30] d01gbf: fn has no value
[31] d01gbf: fp has no value
[32] d01gbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by
```

section


```

finalizing nrlib NAGD01
--->/research2/test0819/mnt/fedora5/../../src/algebra/NAGD01.spad-->NagIntegrationPackage((d01asf ((Result)
"\spad{d01asf(a,{omega},{key},{epsabs},{limlst},{lw},{liw},{ifail},{g})} calculates an approximation to the

```

section

```

finalizing nrlib NAGD02
Warnings:
[1] d02bbf: fn has no value
[2] d02bbf: fp has no value
[3] d02bbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[4] d02bhf: fn has no value
[5] d02bhf: fp has no value
[6] d02bhf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[7] d02cjf: fn has no value
[8] d02cjf: fp has no value
[9] d02cjf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[10] d02ejf: fn has no value
[11] d02ejf: fp has no value
[12] d02ejf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[13] d02gaf: fn has no value
[14] d02gaf: fp has no value
[15] d02gaf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[16] d02gbf: fn has no value
[17] d02gbf: fp has no value
[18] d02gbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[19] d02kef: fn has no value
[20] d02kef: fp has no value
[21] d02kef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[22] d02raf: fn has no value
[23] d02raf: fp has no value
[24] d02raf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @

```

section

```

finalizing nrlib NAGF01
Warnings:
[1] f01brf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[2] f01bsf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[3] f01maf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[4] f01mcf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @

```

```

[5] f01qcf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[6] f01qdf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[7] f01qef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[8] f01rcf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[9] f01rdf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[10] f01ref: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @

```

section

finalizing nrlib NAGF02

Warnings:

```

[1] f02aaf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[2] f02abf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[3] f02adf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[4] f02aef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[5] f02aff: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[6] f02agf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[7] f02ajf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[8] f02akf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[9] f02awf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[10] f02axf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[11] f02bbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[12] f02bjf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[13] f02fjf: fn has no value
[14] f02fjf: fp has no value
[15] f02fjf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[16] f02wef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[17] f02xef: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @

```

section

finalizing nrlib NAGF04

Warnings:

```

[1] f04adf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[2] f04arf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[3] f04asf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[4] f04atf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[5] f04axf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[6] f04faf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[7] f04jgf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[8] f04maf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[9] f04mbf: fn has no value

```

```

[10] f04mbf: fp has no value
[11] f04mbf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[12] f04mcf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @
[13] f04qaf: fn has no value
[14] f04qaf: fp has no value
[15] f04qaf: pretend(List (Record (: key (Symbol)) (: entry (Any)))) -- should replace by @

```

section

```
finalizing nrlib NLINSOL
```

Warnings:

```
[1] allRoots: not known that (Ring) is of mode (CATEGORY package (SIGNATURE solveInField ((List (List
```

section

```
finalizing nrlib NSMP
```

Warnings:

```
[1] monicModulo: mM has no value
```

section

```
finalizing nrlib ODERTRIC
```

Semantic Errors:

```
[1] RatODETools is not a known type
```

Warnings:

```

[1] ricDsolve: not known that (IntegralDomain) is of mode (CATEGORY package (SIGNATURE ricDsolve ((List (List
[2] ratsln: ls has no value
[3] ratsln: lv has no value
[4] ratsol: ans has no value
[5] polyRicDE: deg has no value
[6] polyRicDE: ans has no value
[7] newtonSolution: m has no value
[8] zro1: not known that (Ring) is of mode (CATEGORY domain (SIGNATURE ricDsolve ((List (Fraction UP))

```

section

```
finalizing nrlib PADE
Warnings:
  [1] padeInner:  plist has no value
  [2] padeInner:  alist has no value
```

section

```
finalizing nrlib PDEPACK
--->/research2/test0819/mnt/fedora5/../../src/algebra/PDEPACK.spad-->AnnaPartialDifferentialEqua
--->/research2/test0819/mnt/fedora5/../../src/algebra/PDEPACK.spad-->AnnaPartialDifferentialEqua
```

section

```
finalizing nrlib PFO
Warnings:
  [1] klist: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE order ((Uni
  [2] UPQ2F: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE order ((Uni
  [3] commonDen: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE order (
  [4] selectIntegers: not known that (Ring) is of mode (CATEGORY package (SIGNATURE order ((

Warning: PFO;cmult has a duplicate definition in this file
```

section

```
finalizing nrlib PMASSFS
Warnings:
  [1] ass: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE assert (F F (
```

section

finalizing nrlib PMFS

Warnings:

- [1] patternMatch: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE patternMatch ((
- [2] patternMatch: var has no value

section

finalizing nrlib PMPREDFS

Warnings:

- [1] suchThat: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE suchThat (F F (Mapp

section

finalizing nrlib PSETPK

Warnings:

- [1] selectPolynomials: gps has no value
- [2] selectPolynomials: bps has no value
- [3] selectOrPolynomials: gps has no value
- [4] selectOrPolynomials: bps has no value
- [5] selectAndPolynomials: gps has no value
- [6] selectAndPolynomials: bps has no value
- [7] certainlySubVariety?: polnum has no value
- [8] autoRemainder: newlp has no value
- [9] rewriteSetByReducingWithParticularGenerators: rs has no value
- [10] squareFreeFactors: not known that (IntegralDomain) is of mode (CATEGORY domain (SIGNATURE remove
- [11] univariatePolynomialsGcds: p has no value
- [12] univariatePolynomialsGcds: pInV has no value
- [13] univariatePolynomialsGcds: stop has no value
- [14] univariatePolynomialsGcds: lg has no value
- [15] removeRoughlyRedundantFactorsInContents: newcp has no value
- [16] removeRedundantFactorsInContents: newcp has no value
- [17] removeRedundantFactorsInPols: newcp has no value
- [18] removeRedundantFactorsInPols: newp has no value
- [19] unprotectedRemoveRedundantFactors: d has no value
- [20] removeRedundantFactors: toSee has no value
- [21] removeRedundantFactors: toSave has no value

section

```
finalizing nrlib RDEEF
```

```
Warnings:
```

- [1] rischDEalg: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE rischDE (Re
- [2] rischDEalg: not known that (Ring) is of mode (CATEGORY package (SIGNATURE rischDE (Re
- [3] normalise0: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE rischDE (Re
- [4] normalise0: not known that (Ring) is of mode (CATEGORY package (SIGNATURE rischDE (Re
- [5] polyDElog: limitedlogs has no value
- [6] polyDElog: more than 1 modemap for: (Zero) with dc=F ==>((F F) ((has R (AbelianSemiG
- [7] polyDElog: eq has no value
- [8] polyDEexp: eq has no value

section

```
finalizing nrlib RDEEFS
```

```
Warnings:
```

- [1] basecase: not known that (Ring) is of mode (CATEGORY package (SIGNATURE rischDEsys ((U
- [2] basecase: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE rischDEs
- [3] rischDEsys: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE rischD

section

```
finalizing nrlib RSETCAT-
```

```
Warnings:
```

- [1] intersect: toSave has no value

section

```
finalizing nrlib RSETCAT
```

```
; (DEFUN |RegularTriangularSetCategory| ...) is being compiled.
;; The variable |RegularTriangularSetCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |RegularTriangularSetCategory;| ...) is being compiled.
;; The variable |RegularTriangularSetCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib RULE
```

Warnings:

```
[1] F2Symbol: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE rule ($ F F)) (SIGNATURE rule ($ F F)))
```

```
[2] retractIfCan: not known that (OrderedSet) is of mode (CATEGORY domain (SIGNATURE rule ($ F F)) (S
```

section

finalizing nrlib SIMPAN

Warnings:

```
[1] simplify: not known that (TranscendentalFunctionCategory) is of mode (CATEGORY domain (IF (has (In
```

SFORT abbreviates domain SimpleFortranProgram
(SPADLET |\$noSubsumption| (QUOTE T))

section

```
finalizing nrlib SOLVESER
```

Warnings:

```
[1] unvectorise: not known that (OrderedSet) is of mode (CATEGORY package (SIGNATURE decomposeFunc ((I
```

```
[2] unvectorise: not known that (Ring) is of mode (CATEGORY package (SIGNATURE decomposeFunc ((Fracti
```

section

```
finalizing nrlib SUMFS
```

Warnings:

```
[1] sum: not known that (OrderedSet) is of mode (CATEGORY package
```

```
[2] notRF?: not known that (OrderedSet) is of mode (CATEGORY package
```

```
(|UnivariateTaylorSeriesCategory| |#1|) extends
(|UnivariatePowerSeriesCategory| |#1| (|NonNegativeInteger|)) but not
(|UnivariatePowerSeriesCategory| |#1| (|Integer|)) -----non
extending category-----
.. SparseUnivariateTaylorSeries(#1,#2,#3) of cat
(|Join| (|UnivariateTaylorSeriesCategory| |#1|) (CATEGORY |domain|
```

section

```
(|UnivariatePowerSeriesCategory| |#1| (|Integer|))    finalizing nrlib SUTS
```

section

```
finalizing nrlib TOOLSIGN
Warnings:
[1] nonQsign: pretend(AlgebraicNumber) -- should replace by @
[2] nonQsign: not known that (AlgebraicallyClosedField) is of mode
[3] nonQsign: not known that (TranscendentalFunctionCategory) is of
```

section

```
finalizing nrlib TRIGMNIP
Warnings:
[1] K2KG: not known that (RadicalCategory) is of mode (CATEGORY domain
[2] K2KG: not known that (TranscendentalFunctionCategory) is of mode
[3] K2KG: not known that (OrderedSet) is of mode (CATEGORY package
[4] real?: ker has no value
[5] complexKernels: not known that (OrderedSet) is of mode (CATEGORY
[6] complexKernels: lk has no value
[7] complexKernels: lv has no value
[8] complexNormalize: not known that (AlgebraicallyClosedField) is of
[9] complexNormalize: not known that (TranscendentalFunctionCategory)
```

section

```
finalizing nrlib TRMANIP
Warnings:
[1] logArgs: not known that (OrderedSet) is of mode (CATEGORY package
[2] logArgs: sum has no value
[3] logArgs: arg has no value
[4] simplifyLog1: exprs has no value
[5] simplifyLog1: terms has no value
[6] simplifyLog1: not known that (OrderedSet) is of mode (CATEGORY
```



```

[7] expandpow: not known that (OrderedSet) is of mode (CATEGORY
[8] termexp: exponent has no value
[9] htrigs: not known that (Ring) is of mode (CATEGORY package
[10] exlog: not known that (IntegralDomain) is of mode
[11] logexpand: IN has no value
[12] logexpand: x has no value
[13] kerexpand: IN has no value
[14] kerexpand: x has no value

```

section

```

finalizing nrlib ULSCCAT
; (DEFUN |UnivariateLaurentSeriesConstructorCategory| ...) is being compiled.
;; The variable |UnivariateLaurentSeriesConstructorCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |UnivariateLaurentSeriesConstructorCategory;| ...) is being compiled.
;; The variable |UnivariateLaurentSeriesConstructorCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

finalizing nrlib UPXSSING
Warnings:
[1] retractIfCan: signature of lhs not unique: (Union
(UnivariatePuisseuxSeries FE var cen) failed)$ chosen
[2] sortAndDiscardTerms: zeroTerms has no value
[3] sortAndDiscardTerms: infiniteTerms has no value
[4] sortAndDiscardTerms: failedTerms has no value
[5] dominantTermOnList: %list has no value
[6] dominantTermOnList: %order has no value

-----non extending category-----
.. UnivariatePuisseuxSeriesWithExponentialSingularity(#1,#2,#3,#4) of cat
(|Join| (|FiniteAbelianMonoidRing| (|UnivariatePuisseuxSeries| |#2| |#3| |#4|)

```

section

```

(IF (|has| (|UnivariatePuisseuxSeries| |#2| |#3| |#4|) (|IntegralDomain|))

```

section

```
finalizing nrlib WUTSET
Warnings:
[1] medialSetWithTrace: rs has no value
[2] medialSetWithTrace: contradiction has no value
[3] medialSetWithTrace: bs has no value
[4] characteristicSetUsingTrace: contradiction has no value
[5] characteristicSetUsingTrace: rs has no value
[6] characteristicSetUsingTrace: ms has no value
[7] zeroSetSplit: newlts has no value
```

section

```
finalizing nrlib DEFINTEF
Warnings:
[1] checkForPole: not known that (OrderedSet) is of mode (CATEGORY
[2] polyIfCan: not known that (Ring) is of mode (CATEGORY package
[3] polyIfCan: not known that (OrderedSet) is of mode (CATEGORY package
```

section

```
finalizing nrlib DFINTTLS
Warnings:
[1] findLimit: not known that (OrderedSet) is of mode (CATEGORY package
[2] mkLogPos: not known that (OrderedSet) is of mode (CATEGORY package
[3] checkForZero: not known that (Ring) is of mode (CATEGORY package
[4] findRealZero: fin has no value
[5] findRealZero: halfinf has no value
[6] var: i has no value
```

section

```
finalizing nrlib DEFINTRF
Warnings:
[1] nopole: not known that (TranscendentalFunctionCategory) is of mode
```

[2] nopole: not known that (AlgebraicallyClosedFunctionSpace R) is of

section

```
finalizing nrlib D01TRNS
Warnings:
  [1] transformFunction: not known that (TranscendentalFunctionCategory)

-----non extending category-----
.. d01TransformFunctionType of cat
(|NumericalIntegrationCategory|)  has no
```

section

```
(|TableAggregate| (|Symbol|) (|Any|))    finalizing nrlib D01TRNS
```

section

```
finalizing nrlib D01TRNS
--->/research2/test0819/mnt/fedora5/../../src/algebra/D01TRNS.spad
-->d01TransformFunctionType(): Missing Description
```

section

```
finalizing nrlib EFULS
Warnings:
  [1] tanIfCan: not known that (Algebra (Fraction (Integer))) is
```

section

```
finalizing nrlib ESCONT
```

Warnings:

```
[1] zerosOf: not known that (OrderedSet) is of mode (CATEGORY
```

section

```
finalizing nrlib ESCONT
-----non extending category-----
.. ExpertSystemContinuityPackage of cat
(CATEGORY |package| (SIGNATURE |getlo| ((|DoubleFloat|) (|Segment|
```

section

finalizing nrlib EXPR

Warnings:

```
[1] not known that (OrderedSet) is of mode (CATEGORY domain
[2] simplifyPower: not known that (OrderedSet) is of mode
[3] **: pretend(Integer) -- should replace by @
[4] **: not known that (OrderedSet) is of mode (CATEGORY domain
[5] <: not known that (OrderedSet) is of mode (CATEGORY domain
[6] numer: not known that (OrderedSet) is of mode (CATEGORY domain
[7] toprat: not known that (Field) is of mode (CATEGORY domain
[8] toprat: not known that (ExpressionSpace) is of mode
[9] toprat: not known that (SIGNATURE numer
[10] toprat: not known that (SIGNATURE denom
[11] toprat: not known that (SIGNATURE coerce ($)
[12] reducedSystem: not known that (Ring) is of mode (CATEGORY domain
[13] commonk0: not known that (OrderedSet) is of mode (CATEGORY domain
[14] rootOf: not known that (FunctionSpace R) is of mode (CATEGORY
[15] rootOf: not known that (Ring) is of mode (CATEGORY domain
[16] pi: not known that (FunctionSpace R) is of mode (CATEGORY domain
[17] pi: not known that (RadicalCategory) is of mode (CATEGORY domain
[18] abs: not known that (FunctionSpace R) is of mode (CATEGORY domain
[19] **: not known that (FunctionSpace R) is of mode (CATEGORY domain
[20] erf: not known that (FunctionSpace R) is of mode (CATEGORY domain
[21] erf: not known that (RadicalCategory) is of mode (CATEGORY domain
[22] erf: not known that (TranscendentalFunctionCategory) is of mode
[23] operator: not known that (FunctionSpace R) is of mode (CATEGORY
[24] operator: not known that (ExpressionSpace) is of mode (CATEGORY
[25] evl0: not known that (OrderedSet) is of mode (CATEGORY domain
[26] gcdPolynomial: not known that (GcdDomain) is of mode (CATEGORY
[27] factorPolynomial: not known that (OrderedSet) is of mode
[28] factorPolynomial: not known that (Ring) is of mode (CATEGORY
```

```

[29] coerce: not known that (OrderedSet) is of mode (CATEGORY domain
[30] retract: not known that (OrderedSet) is of mode (CATEGORY domain
[31] retractIfCan: not known that (OrderedSet) is of mode (CATEGORY
[32] k2expr: not known that (ExpressionSpace) is of mode (CATEGORY
[33] smp2expr: not known that (SetCategory) is of mode (CATEGORY
[34] smp2expr: not known that (SIGNATURE + ($ $ $)) is of mode
[35] smp2expr: not known that (SIGNATURE * ($ $ $)) is of mode
[36] smp2expr: not known that (SIGNATURE ** ($ $ (NonNegativeInteger)))
[37] smp2an: not known that (OrderedSet) is of mode (CATEGORY domain
[38] convert: not known that (OrderedSet) is of mode (CATEGORY domain
[39] eval: not known that (ConvertibleTo (InputForm)) is of mode
[40] patternMatch: not known that (FunctionSpace R) is of mode
[41] patternMatch: not known that (ConvertibleTo (Pattern (Integer)))
[42] patternMatch: not known that (PatternMatchable (Integer)) is of
[43] patternMatch: not known that (RetractableTo (Kernel $)) is of
[44] patternMatch: not known that (SetCategory) is of mode (CATEGORY
[45] patternMatch: not known that (ConvertibleTo (Pattern (Float)))
[46] patternMatch: not known that (PatternMatchable (Float)) is of
[47] isPlus: gen has no value
[48] not known that (Ring) is of mode (CATEGORY domain (SIGNATURE
[49] not known that (IntegralDomain) is of mode (CATEGORY domain

```

section

```

finalizing nrlib EXPR
; (DEFUN |Expression;| ...) is being compiled.
;; The variable IDENTITY is undefined.
;; The compiler will assume this variable is a global.

```

section

```

finalizing nrlib EXPR2UPS
Warnings:
[1] iTaylor: %problem has no value
[2] iTaylor: %series has no value
[3] taylor: more than 1 modemap for: (Zero) with dc=FE ==>
(((FE FE) ((has R (AbelianSemiGroup)) (CONST FE ($)))) ((FE FE)
(T (CONST FE ($))))))
[4] iLaurent: %problem has no value
[5] iLaurent: %series has no value
[6] laurent: more than 1 modemap for: (Zero) with dc=FE ==>
(((FE FE) ((has R (AbelianSemiGroup)) (CONST FE ($)))) ((FE FE)

```

```

(T (CONST FE ($))))
  [7] iPuisseux: %problem has no value
  [8] iPuisseux: %series has no value
  [9] puisseux: more than 1 modemap for: (Zero) with dc=FE ==>
(((FE FE) ((has R (AbelianSemiGroup)) (CONST FE ($)))) ((FE FE)
(T (CONST FE ($))))
  [10] iSeries: %problem has no value
  [11] iSeries: %series has no value
  [12] series: more than 1 modemap for: (Zero) with dc=FE ==>
(((FE FE) ((has R (AbelianSemiGroup)) (CONST FE ($)))) ((FE FE)
(T (CONST FE ($))))

```

section

```

finalizing nrlib FDIV
Warnings:
  [1] *: signature of lhs not unique: $(Integer)$ chosen

```

section

```

finalizing nrlib FDIV
; (DEFUN |FiniteDivisor;| ...) is being compiled.
;; The variable IDENTITY is undefined.
;; The compiler will assume this variable is a global.

```

section

```

finalizing nrlib FSCINT
Warnings:
  [1] K2KG: not known that (RadicalCategory) is of mode (CATEGORY domain
  [2] K2KG: not known that (TranscendentalFunctionCategory) is of mode
  [3] K2KG: not known that (OrderedSet) is of mode (CATEGORY package
  [4] internalIntegrate: not known that (OrderedSet) is of mode (CATEGORY
  [5] internalIntegrate: not known that (AlgebraicallyClosedField) is of
  [6] internalIntegrate: not known that (TranscendentalFunctionCategory)

```

section

finalizing nrlib FSINT

Warnings:

- [1] K2KG: not known that (RadicalCategory) is of mode (CATEGORY domain
- [2] K2KG: not known that (TranscendentalFunctionCategory) is of mode
- [3] K2KG: not known that (OrderedSet) is of mode (CATEGORY package
- [4] postSubst: not known that (Ring) is of mode (CATEGORY package
- [5] postSubst: not known that (OrderedSet) is of mode (CATEGORY
- [6] integrate: not known that (AlgebraicallyClosedField) is of mode
- [7] integrate: not known that (TranscendentalFunctionCategory) is of

section

finalizing nrlib FS2EXXP

Warnings:

- [1] newElem: not known that (OrderedSet) is of mode
- [2] k2Elem: not known that (OrderedSet) is of mode
- [3] iExprToXXP: %series has no value
- [4] listToXXP: %expansion has no value
- [5] powerToXXP: %expansion has no value
- [6] nthRootXXPIfCan: %problem has no value
- [7] nthRootXXPIfCan: %series has no value
- [8] nthRootToXXP: %problem has no value
- [9] nthRootToXXP: %expansion has no value
- [10] genPowerToXXP: %expansion has no value
- [11] genExp: %series has no value
- [12] exponential: %problem has no value
- [13] exponential: %series has no value
- [14] expToXXP: %expansion has no value
- [15] logToXXP: %expansion has no value
- [16] logToXXP: %problem has no value
- [17] logToXXP: %series has no value
- [18] applyIfCan: %expansion has no value
- [19] applyBddIfCan: %problem has no value
- [20] applyBddIfCan: %expansion has no value
- [21] opsInvolvingX: not known that (OrderedSet) is of mode
- [22] atancotToXXP: %problem has no value
- [23] atancotToXXP: %series has no value

-----non extending category-----

.. GeneralUnivariatePowerSeries(#1,#2,#3) of cat

(|Join| (|UnivariatePuisseuxSeriesCategory| |#1|) (CATEGORY |domain|

section

```
(|UnivariatePuisseuxSeriesConstructorCategory| |#1|
```

section

```
finalizing nrlib HELFDIV
Warnings:
[1] unknown Functor code (error HyperellipticFiniteDivisor:
curve must be hyperelliptic)
```

section

```
finalizing nrlib INVLAPLA
Warnings:
[1] ilt: not known that (Ring) is of mode (CATEGORY package
[2] ilt: not known that (OrderedSet) is of mode (CATEGORY package
[3] iltsqfr: not known that (IntegralDomain) is of mode (CATEGORY
[4] iltirred: not known that (Ring) is of mode (CATEGORY package
```

section

```
finalizing nrlib IR2F
Warnings:
[1] evenRoots: not known that (OrderedSet) is of mode
[2] ilog: not known that (OrderedSet) is of mode (CATEGORY package
[3] ilog: not known that (Ring) is of mode (CATEGORY package
[4] lg2func: not known that (Ring) is of mode (CATEGORY package
```

section

```
finalizing nrlib IRRF2F
```


Warnings:

- [1] expand: not known that (AlgebraicallyClosedFunctionSpace R)
- [2] expand: not known that (TranscendentalFunctionCategory) is of
- [3] integrate: not known that (AlgebraicallyClosedField) is of
- [4] integrate: not known that (TranscendentalFunctionCategory)

section

finalizing nrlib LAPLACE

Warnings:

- [1] algebraic?: not known that (OrderedSet) is of mode
- [2] isLinear: not known that (Ring) is of mode
- [3] isLinear: not known that (OrderedSet) is of mode
- [4] atn: d has no value
- [5] mkPlus: not known that (OrderedSet) is of mode
- [6] locallaplace: const has no value
- [7] locallaplace: nconst has no value

section

finalizing nrlib LIMITPS

Warnings:

- [1] firstNonLogPtr: not known that (OrderedSet) is of mode
- [2] complLimit: %series has no value
- [3] reallimit: %problem has no value
- [4] reallimit: %series has no value
- [5] reallimit: func has no value
- [6] reallimit: prob has no value
- [7] xxpLimit: %expansion has no value
- [8] limitPlus: %problem has no value
- [9] limitPlus: %series has no value
- [10] limitPlus: func has no value
- [11] limitPlus: prob has no value

section

finalizing nrlib LODEEF

Warnings:

```

[1] algSolve: not known that (OrderedSet) is of mode (CATEGORY
[2] algSolve: not known that (Ring) is of mode (CATEGORY package
[3] xpart: not known that (OrderedSet) is of mode (CATEGORY
[4] ulodo: not known that (Ring) is of mode (CATEGORY package

```

section

```

finalizing nrlib NODE1
Warnings:
[1] solve: not known that (OrderedSet) is of mode (CATEGORY package
[2] checkBernoulli: not known that (Ring) is of mode (CATEGORY package

```

section

```

finalizing nrlib ODECONST
Warnings:
[1] basisSqfr: not known that (Ring) is of mode

```

section

```

finalizing nrlib ODEINT
Warnings:
[1] expint: not known that (OrderedSet) is of mode
[2] expint: lrec has no value
[3] expint: exponent has no value
[4] isQlog: not known that (OrderedSet) is of mode

```

section

```

finalizing nrlib REP
Warnings:
[1] evalvect: not known that (OrderedSet) is of mode (CATEGORY
[2] gramschmidt: :(PositiveInteger) -- should replace by pretend
[3] gramschmidt: :RMR -- should replace by pretend

```

```
[4] gramschmidt: :(Matrix (Expression (Integer))) --
should replace by pretend
```

section

```
finalizing nrlib ULSCONS
Warnings:
[1] termsToOutputForm: 1 has no value
```

section

```
finalizing nrlib UPXS
Warnings:
[1] termsToOutputForm: 1 has no value

-----non extending category-----
.. UnivariateTaylorSeries(#1,#2,#3) of cat
(|Join| (|UnivariateTaylorSeriesCategory| |#1|) (CATEGORY |domain|
```

section

```
finalizing nrlib UTS
```

section

```
finalizing nrlib ASP29
Warnings:
[1] ISTATE has no value
[2] NEXTIT has no value
[3] NEVALS has no value
[4] NVECS has no value
[5] K has no value
[6] F has no value
[7] D has no value
```

section

```
finalizing nrlib COMBF
Warnings:
[1] **: not known that (OrderedSet) is of mode (CATEGORY package
[2] facts: not known that (OrderedSet) is of mode (CATEGORY package
[3] summand: not known that (OrderedSet) is of mode (CATEGORY package
[4] ipow: n has no value
```

section

```
finalizing nrlib D01AGNT
Warnings:
[1] continuousAtPoint?: not known that (AlgebraicallyClosedField)
[2] continuousAtPoint?: not known that (TranscendentalFunctionCategory)
[3] functionIsOscillatory: not known that (OrderedSet) is of mode
[4] singularitiesOf: str has no value
```

```
-----non extending category-----
.. d01AgentsPackage of cat
(CATEGORY |package| (SIGNATURE |rangeIsFinite|
```

section

```
finalizing nrlib D01AGNT
```

section

```
finalizing nrlib FSPRMELT
Warnings:
[1] F2P: not known that (OrderedSet) is of mode
[2] K2P: not known that (OrderedSet) is of mode
[3] primitiveElement: not known that (Ring) is of mode
[4] primitiveElement: not known that (OrderedSet) is of mode
[5] F2UP: not known that (Ring) is of mode (CATEGORY $
```

section

```

    finalizing nrlib NTSCAT
; (DEFUN |NormalizedTriangularSetCategory| ...) is being compiled.
;; The variable |NormalizedTriangularSetCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |NormalizedTriangularSetCategory;| ...) is being compiled.
;; The variable |NormalizedTriangularSetCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib REGSET
Warnings:
  [1] construct:  ts has no value
  [2] preprocess: lp1 has no value
  [3] preprocess: lp2 has no value
  [4] preprocess: lts has no value

```

section

```

    finalizing nrlib REGSET
Warning: REGSET;decompose has a duplicate definition in this file

-----non extending category-----
.. RegularChain(#1,#2) of cat
(|Join| (|RegularTriangularSetCategory| |#1| (|IndexedExponents|

```

section

```

    finalizing nrlib RGCHAIN

```

section

```

finalizing nrlib RSETGCD
Warnings:
[1] toseInvertibleSet:  toSave has no value
[2] prepareSubResAlgo:  toSave has no value
[3] internalLastSubResultant:  toReturn has no value

```

section

```

finalizing nrlib RSDCMPK
Warnings:
[1] algebraicDecompose:  llpwt has no value
[2] internalDecompose:  llpwt has no value
[3] internalDecompose:  lts has no value
[4] decompose:  toSave has no value
[5] upDateBranches:  branches1 has no value
[6] upDateBranches:  branches2 has no value

```

section

```

finalizing nrlib RSDCMPK

--->/research2/test0819/mnt/fedora5/../../src/algebra/RSDCMPK.spad
-->RegularSetDecompositionPackage((KrullNumber (N LP Split))):
Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/RSDCMPK.spad
-->RegularSetDecompositionPackage((numberOfVariables (N LP Split))):
Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/RSDCMPK.spad
-->RegularSetDecompositionPackage((algebraicDecompose ((Record
(: done Split) (: todo (List LpWT))) P TS B))): Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/RSDCMPK.spad
-->RegularSetDecompositionPackage((transcendentalDecompose ((Record
(: done Split) (: todo (List LpWT))) P TS N))): Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/RSDCMPK.spad
-->RegularSetDecompositionPackage((transcendentalDecompose ((Record
(: done Split) (: todo (List LpWT))) P TS))): Not documented!!!!

```

```

--->/research2/test0819/mnt/fedora5/../../src/algebra/RSDCMPK.spad
-->RegularSetDecompositionPackage((internalDecompose ((Record
(: done Split) (: todo (List LpWT))) P TS N B))): Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/RSDCMPK.spad
-->RegularSetDecompositionPackage((internalDecompose ((Record
(: done Split) (: todo (List LpWT))) P TS N))): Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/RSDCMPK.spad
-->RegularSetDecompositionPackage((internalDecompose ((Record
(: done Split) (: todo (List LpWT))) P TS))): Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/RSDCMPK.spad
-->RegularSetDecompositionPackage((decompose (Split LP Split B B))):
Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/RSDCMPK.spad
-->RegularSetDecompositionPackage((decompose (Split LP Split B B B B))):
Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/RSDCMPK.spad
-->RegularSetDecompositionPackage((updateBranches ((List LpWT) LP Split
(List LpWT) Wip N))): Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/RSDCMPK.spad
-->RegularSetDecompositionPackage((convert ((String) (Record (: val
(List P)) (: tower TS))))): Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/RSDCMPK.spad
-->RegularSetDecompositionPackage((printInfo ((Void) (List (Record
(: val (List P)) (: tower TS))) N))): Not documented!!!!

```

section

```

    finalizing nrlib SFRTCAT
; (DEFUN |SquareFreeRegularTriangularSetCategory| ...) is being compiled.
;; The variable |SquareFreeRegularTriangularSetCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |SquareFreeRegularTriangularSetCategory;| ...) is being compiled.
;; The variable |SquareFreeRegularTriangularSetCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```
finalizing nrlib SIGNEF
Warnings:
  [1] sign: not known that (OrderedSet) is of mode
  [2] smpsign: not known that (IntegralDomain) is of mode
```

section

```
finalizing nrlib SNTSCAT
; (DEFUN |SquareFreeNormalizedTriangularSetCategory| ...) is being compiled.
;; The variable |SquareFreeNormalizedTriangularSetCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |SquareFreeNormalizedTriangularSetCategory;| ...) is being compiled.
;; The variable |SquareFreeNormalizedTriangularSetCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib SOLVETRA
Warnings:
  [1] solveInner: not known that (OrderedSet) is of mode
  [2] solveInner: not known that (Ring) is of mode
  [3] tryToTrans: not known that (TranscendentalFunctionCategory) is of
  [4] tryToTrans: not known that (AlgebraicallyClosedField) is of mode
  [5] subsTan: not known that (IntegralDomain) is of mode
  [6] buildnexpr: anscoeff has no value
  [7] buildnexpr: ansmant has no value
  [8] combineLog: ans has no value
  [9] funcinv: not known that (OrderedSet) is of mode
```

section

```
finalizing nrlib SRDCMPK
Warnings:
  [1] algebraicDecompose: lts has no value
```



```

[2] internalDecompose:  llpwt has no value
[3] internalDecompose:  lts has no value
[4] decompose:  toSave has no value
[5] upDateBranches:  branches1 has no value
[6] upDateBranches:  branches2 has no value

```

section

```

finalizing nrlib SRDCMPK
Processing SquareFreeRegularSetDecompositionPackage for Browser database:

-->/research2/test0819/mnt/fedora5/../../src/algebra/SRDCMPK.spad
-->SquareFreeRegularSetDecompositionPackage((KrullNumber (N LP
-->Split))) : Not documented!!!!

-->/research2/test0819/mnt/fedora5/../../src/algebra/SRDCMPK.spad
-->SquareFreeRegularSetDecompositionPackage((numberOfVariables (N LP
-->Split))) : Not documented!!!!

-->/research2/test0819/mnt/fedora5/../../src/algebra/SRDCMPK.spad
-->SquareFreeRegularSetDecompositionPackage((algebraicDecompose
-->((Record (: done Split) (: todo (List LpWT))) P TS))) : Not
-->documented!!!!

-->/research2/test0819/mnt/fedora5/../../src/algebra/SRDCMPK.spad
-->SquareFreeRegularSetDecompositionPackage((transcendentalDecompose
-->((Record (: done Split) (: todo (List LpWT))) P TS N))) : Not
-->documented!!!!

-->/research2/test0819/mnt/fedora5/../../src/algebra/SRDCMPK.spad
-->SquareFreeRegularSetDecompositionPackage((transcendentalDecompose
-->((Record (: done Split) (: todo (List LpWT))) P TS))) : Not
-->documented!!!!

-->/research2/test0819/mnt/fedora5/../../src/algebra/SRDCMPK.spad
-->SquareFreeRegularSetDecompositionPackage((internalDecompose
-->((Record (: done Split) (: todo (List LpWT))) P TS N B))) : Not
-->documented!!!!

-->/research2/test0819/mnt/fedora5/../../src/algebra/SRDCMPK.spad
-->SquareFreeRegularSetDecompositionPackage((internalDecompose
-->((Record (: done Split) (: todo (List LpWT))) P TS N))) : Not
-->documented!!!!

-->/research2/test0819/mnt/fedora5/../../src/algebra/SRDCMPK.spad
-->SquareFreeRegularSetDecompositionPackage((internalDecompose

```

```

-->((Record (: done Split) (: todo (List LpWT))) P TS)): Not
-->documented!!!!

-->/research2/test0819/mnt/fedora5/../../src/algebra/SRDCMPK.spad
-->SquareFreeRegularSetDecompositionPackage((decompose (Split LP Split
-->B B))): Not documented!!!!

-->/research2/test0819/mnt/fedora5/../../src/algebra/SRDCMPK.spad
-->SquareFreeRegularSetDecompositionPackage((decompose (Split LP Split
-->B B B B B))): Not documented!!!!

-->/research2/test0819/mnt/fedora5/../../src/algebra/SRDCMPK.spad
-->SquareFreeRegularSetDecompositionPackage((updateBranches ((List
-->LpWT) LP Split (List LpWT) Wip N))): Not documented!!!!

-->/research2/test0819/mnt/fedora5/../../src/algebra/SRDCMPK.spad
-->SquareFreeRegularSetDecompositionPackage((convert ((String) (Record
-->(: val (List P)) (: tower TS))))): Not documented!!!!

-->/research2/test0819/mnt/fedora5/../../src/algebra/SRDCMPK.spad
-->SquareFreeRegularSetDecompositionPackage((printInfo ((Void) (List
-->(Record (: val (List P)) (: tower TS))) N))): Not documented!!!!

```

section

```

finalizing nrlib SREGSET
Warnings:
[1] construct:  ts has no value
[2] internalAugment:  lts has no value
[3] preprocess:  lp1 has no value
[4] preprocess:  lp2 has no value
[5] preprocess:  lts has no value

```

section

```

finalizing nrlib ZDSOLVE
Warnings:
[1] squareFree:  toSave has no value
[2] realSolve:  not known that (Ring) is of mode
[3] realSolve:  toSave has no value
[4] positiveSolve:  toSave has no value
[5] univariateSolve:  lq2 has no value

```

section

```
finalizing nrlib IRURPK
Warnings:
  [1] makeLinearAndMonic:  toSave has no value
  [2] rur:  toSave has no value
```

section

```
finalizing nrlib LEXTRIPK
Warnings:
  [1] trueVariables:  trueIs has no value
  [2] lexTriangular:  polnum has no value
  [3] lexTriangular:  toSave has no value
  [4] squareFreeLexTriangular:  polnum has no value
  [5] squareFreeLexTriangular:  toSave has no value
```

section

```
finalizing nrlib NORMPK
Semantic Errors:
  [1] recip:  hesrg has two modes:

Warnings:
  [1] normalizedAssociate:  mp has no value
  [2] normalizedAssociate:  tp has no value
```

section

```
finalizing nrlib QCMPACK
Warnings:
  [1] removeSuperfluousCases:  maxcases has no value
  [2] removeSuperfluousCases:  lpwt1 has no value
```

```

[3] removeSuperfluousCases: headmaxcases has no value
[4] removeSuperfluousCases: toSave has no value
[5] removeSuperfluousQuasiComponents: maxlts has no value
[6] removeSuperfluousQuasiComponents: headmaxlts has no value
[7] removeSuperfluousQuasiComponents: toSave has no value
[8] branchIfCan: polnum has no value
[9] prepareDecompose: branches has no value

```

section

```

finalizing nrlib RURPK
Warnings:
[1] rur: lq has no value
[2] rur: q has no value
[3] rur: toSave has no value
[4] rur: toReturn has no value

```

section

```

finalizing nrlib SFRGCD
Warnings:
[1] stosePrepareSubResAlgo: toSave has no value
[2] stoseInternalLastSubResultant: toReturn has no value
[3] stoseInvertibleSetsqfreq: toSave has no value
[4] stoseInvertibleSetreg: toSave has no value

```

section

```

finalizing nrlib SFRGCD
Processing SquareFreeRegularTriangularSetGcdPackage for Browser database:

--->/research2/test0819/mnt/fedora5/../../src/algebra/SFRGCD.spad-->
SquareFreeRegularTriangularSetGcdPackage((startTableGcd! ((Void) S S
S))) : Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/SFRGCD.spad
-->SquareFreeRegularTriangularSetGcdPackage((stopTableGcd!

```

```

((Void))) : Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/SFRGCD.spad
-->SquareFreeRegularTriangularSetGcdPackage((startTableInvSet!
  ((Void) S S S))) : Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/SFRGCD.spad
-->SquareFreeRegularTriangularSetGcdPackage((stopTableInvSet!
  ((Void))) : Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/SFRGCD.spad
-->SquareFreeRegularTriangularSetGcdPackage((stosePrepareSubResAlgo
  ((List LpWT) P P TS))) : Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/SFRGCD.spad
-->SquareFreeRegularTriangularSetGcdPackage((stoseInternalLastSubResultant
  ((List PWT) P P TS B B))) : Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/SFRGCD.spad
-->SquareFreeRegularTriangularSetGcdPackage((stoseInternalLastSubResultant
  ((List PWT) (List LpWT) V B))) : Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/SFRGCD.spad
-->SquareFreeRegularTriangularSetGcdPackage((stoseIntegralLastSubResultant
  ((List PWT) P P TS))) : Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/SFRGCD.spad
-->SquareFreeRegularTriangularSetGcdPackage((stoseLastSubResultant
  ((List PWT) P P TS))) : Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/SFRGCD.spad
-->SquareFreeRegularTriangularSetGcdPackage((stoseInvertible?
  (B P TS))) : Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/SFRGCD.spad
-->SquareFreeRegularTriangularSetGcdPackage((stoseInvertible?sqfreg
  ((List BWT) P TS))) : Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/SFRGCD.spad
-->SquareFreeRegularTriangularSetGcdPackage((stoseInvertibleSetsqfreg
  (Split P TS))) : Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/SFRGCD.spad
-->SquareFreeRegularTriangularSetGcdPackage((stoseInvertible?reg
  ((List BWT) P TS))) : Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/SFRGCD.spad
-->SquareFreeRegularTriangularSetGcdPackage((stoseInvertibleSetreg
  (Split P TS))) : Not documented!!!!

```

```

--->/research2/test0819/mnt/fedora5/../../src/algebra/SFRGCD.spad
-->SquareFreeRegularTriangularSetGcdPackage((stoseInvertible?
      ((List BWT) P TS))): Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/SFRGCD.spad
-->SquareFreeRegularTriangularSetGcdPackage((stoseInvertibleSet
      (Split P TS))): Not documented!!!!

--->/research2/test0819/mnt/fedora5/../../src/algebra/SFRGCD.spad
-->SquareFreeRegularTriangularSetGcdPackage((stoseSquareFreePart
      ((List PWT) P TS))): Not documented!!!!

```

section

```

finalizing nrlib SFQCPK
Warnings:
[1] removeSuperfluousCases: maxcases has no value
[2] removeSuperfluousCases: lpwt1 has no value
[3] removeSuperfluousCases: headmaxcases has no value
[4] removeSuperfluousCases: toSave has no value
[5] removeSuperfluousQuasiComponents: maxlts has no value
[6] removeSuperfluousQuasiComponents: headmaxlts has no value
[7] removeSuperfluousQuasiComponents: toSave has no value
[8] branchIfCan: polnum has no value
[9] prepareDecompose: branches has no value

```

section

```

finalizing nrlib ODEEF
Warnings:
[1] solve: not known that (OrderedSet) is of mode
[2] parseODE: n has no value
[3] parseODE: c has no value
[4] parseODE: k has no value
[5] getcoeff: not known that (OrderedSet) is of mode
[6] getcoeff: not known that (Ring) is of mode

```

section

```

finalizing nrlib RINTERP
Processing RationalInterpolation for Browser database:
--->-->RationalInterpolation((interpolate ((Fraction (Polynomial F))
(List F) (List F) (NonNegativeInteger) (NonNegativeInteger)))):
Not documented!!!!

```

section

```

finalizing nrlib ABELGRP
; (DEFUN |AbelianGroup| ...) is being compiled.
;; The variable |AbelianGroup;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

finalizing nrlib ABELMON
; (DEFUN |AbelianMonoid| ...) is being compiled.
;; The variable |AbelianMonoid;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

finalizing nrlib ABELSG
; (DEFUN |AbelianSemiGroup| ...) is being compiled.
;; The variable |AbelianSemiGroup;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

finalizing nrlib ALAGG
; (DEFUN |AssociationListAggregate| ...) is being compiled.
;; The variable |AssociationListAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.

```

```
; (DEFUN |AssociationListAggregate;| ...) is being compiled.  
;; The variable |AssociationListAggregate;CAT| is undefined.  
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib CABMON  
; (DEFUN |CancellationAbelianMonoid| ...) is being compiled.  
;; The variable |CancellationAbelianMonoid;AL| is undefined.  
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib COMRING  
; (DEFUN |CommutativeRing| ...) is being compiled.  
;; The variable |CommutativeRing;AL| is undefined.  
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib DIFRING  
; (DEFUN |DifferentialRing| ...) is being compiled.  
;; The variable |DifferentialRing;AL| is undefined.  
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib DIVRING  
; (DEFUN |DivisionRing| ...) is being compiled.  
;; The variable |DivisionRing;AL| is undefined.  
;; The compiler will assume this variable is a global.
```


section

```

    finalizing nrlib ENTIRER
; (DEFUN |EntireRing| ...) is being compiled.
;; The variable |EntireRing;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib ES-
Warnings:
  [1] tower: not known that (OrderedSet) is of mode
  [2] freeOf?: not known that (OrderedSet) is of mode
  [3] eval:  IN has no value
  [4] eval:  f has no value
  [5] eval:  s has no value
  [6] map:   IN has no value
  [7] map:   x has no value

```

section

```

    finalizing nrlib ES
; (DEFUN |ExpressionSpace| ...) is being compiled.
;; The variable |ExpressionSpace;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib EUCDOM-
Warnings:
  [1] principalIdeal:  coef1 has no value
  [2] principalIdeal:  coef2 has no value

```

section

```

    finalizing nrlib EUCDOM
; (DEFUN |EuclideanDomain| ...) is being compiled.
;; The variable |EuclideanDomain;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib FFIELDC-
Warnings:
  [1] conditionP: not known that (Ring) is of mode
  [2] order: signature of lhs not unique: (PositiveInteger)S chosen
  [3] order:  ord has no value
  [4] discreteLog:  disc1 has no value
  [5] discreteLog:  disclog has no value
  [6] discreteLog: not known that (IntegralDomain) is of mode
  [7] gcdPolynomial: not known that (Ring) is of mode

```

section

```

    finalizing nrlib FFIELDC
; (DEFUN |FiniteFieldCategory| ...) is being compiled.
;; The variable |FiniteFieldCategory;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib FPS
; (DEFUN |FloatingPointSystem| ...) is being compiled.
;; The variable |FloatingPointSystem;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```
finalizing nrlib GCDDOM-
Warnings:
  [1] gcdPolynomial: not known that (Ring) is of mode
```

section

```
finalizing nrlib GCDDOM
; (DEFUN |GcdDomain| ...) is being compiled.
;; The variable |GcdDomain;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib HOAGG
; (DEFUN |HomogeneousAggregate| ...) is being compiled.
;; The variable |HomogeneousAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |HomogeneousAggregate;| ...) is being compiled.
;; The variable |HomogeneousAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib ILIST
Warnings:
  [1] latex: s has no value
```

section

```
finalizing nrlib INS-
```

Warnings:

- [1] factor: not known that (IntegralDomain) is of mode
- [2] patternMatch: not known that (SetCategory) is of mode
- [3] powmod: y has no value

section

```
finalizing nrlib INS
; (DEFUN |IntegerNumberSystem| ...) is being compiled.
;; The variable |IntegerNumberSystem;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib INT
Warnings:
[1] OMwrite: pretend(String) -- should replace by @
[2] hash: signature of lhs not unique: $$ chosen
[3] factorPolynomial: not known that (UnivariatePolynomialCategory
[4] gcdPolynomial: not known that (UnivariatePolynomialCategory
```

section

```
finalizing nrlib INTDOM
; (DEFUN |IntegralDomain| ...) is being compiled.
;; The variable |IntegralDomain;AL| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib ISTRING
Warnings:
[1] split: j has no value
```

section

```

    finalizing nrlib LNAGG
; (DEFUN |LinearAggregate| ...) is being compiled.
;; The variable |LinearAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |LinearAggregate;| ...) is being compiled.
;; The variable |LinearAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib LSAGG
; (DEFUN |ListAggregate| ...) is being compiled.
;; The variable |ListAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |ListAggregate;| ...) is being compiled.
;; The variable |ListAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib MONOID
; (DEFUN |Monoid| ...) is being compiled.
;; The variable |Monoid;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib MTSCAT
; (DEFUN |MultivariateTaylorSeriesCategory| ...) is being compiled.
;; The variable |MultivariateTaylorSeriesCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |MultivariateTaylorSeriesCategory;| ...) is being compiled.
;; The variable |MultivariateTaylorSeriesCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

```
-----non extending category-----
.. NonNegativeInteger of cat
(|Join| (|OrderedAbelianMonoidSup|) (|Monoid|) (CATEGORY |domain| '

```

section

```
(|IntegerNumberSystem|)      finalizing nrlib NNI

```

section

```
      finalizing nrlib OINTDOM
; (DEFUN |OrderedIntegralDomain| ...) is being compiled.
;; The variable |OrderedIntegralDomain;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```
      finalizing nrlib ORDRING
; (DEFUN |OrderedRing| ...) is being compiled.
;; The variable |OrderedRing;AL| is undefined.
;; The compiler will assume this variable is a global.

```

```
-----non extending category-----
.. PositiveInteger of cat
(|Join| (|AbelianSemiGroup|) (|OrderedSet|) (|Monoid|) (CATEGORY |domain| (SIGNATURE |gcd| ($ $

```

section

```
(|OrderedAbelianMonoidSup|)    finalizing nrlib PI

```

section

finalizing nrlib POLYCAT-

Warnings:

- [1] eval: IN has no value
- [2] coefficient: not known that (Ring) is of mode
- [3] totalDegree: w has no value
- [4] reducedSystem: IN has no value
- [5] reducedSystem: r has no value
- [6] reducedSystem: not known that (Ring) is of mode
- [7] solveLinearPolynomialEquation: not known that (Ring) is of mode
- [8] factorPolynomial: not known that (Ring) is of mode
- [9] factor: not known that (IntegralDomain) is of mode
- [10] conditionP: :(Integer) --should replace by pretend
- [11] patternMatch: not known that (SetCategory) is of mode

section

finalizing nrlib POLYCAT

```
; (DEFUN |PolynomialCategory| ...) is being compiled.
;; The variable |PolynomialCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |PolynomialCategory;| ...) is being compiled.
;; The variable |PolynomialCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

finalizing nrlib PSETCAT-

Warnings:

- [1] collectUnder: lq has no value
- [2] collectUpper: lq has no value
- [3] collect: lq has no value
- [4] sort: us has no value
- [5] sort: vs has no value
- [6] sort: ws has no value
- [7] localTriangular?: q has no value
- [8] headRemainder: r has no value
- [9] rewriteIdealWithHeadRemainder: rs has no value
- [10] rewriteIdealWithRemainder: rs has no value

section

```
finalizing nrlib PSETCAT-
Warning: PSETCAT-;exactQuo has a duplicate definition in this file
```

section

```
finalizing nrlib PSETCAT
; (DEFUN |PolynomialSetCategory| ...) is being compiled.
;; The variable |PolynomialSetCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |PolynomialSetCategory;| ...) is being compiled.
;; The variable |PolynomialSetCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

```
finalizing nrlib QFCAT-
Warnings:
[1] reducedSystem: not known that (Ring) is of mode
[2] patternMatch: not known that (SetCategory) is of mode
```

section

```
finalizing nrlib QFCAT
; (DEFUN |QuotientFieldCategory| ...) is being compiled.
;; The variable |QuotientFieldCategory;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |QuotientFieldCategory;| ...) is being compiled.
;; The variable |QuotientFieldCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```


section

```

      finalizing nrlib RCAGG
; (DEFUN |RecursiveAggregate| ...) is being compiled.
;; The variable |RecursiveAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |RecursiveAggregate;| ...) is being compiled.
;; The variable |RecursiveAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

      finalizing nrlib RING
; (DEFUN |Ring| ...) is being compiled.
;; The variable |Ring;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

      finalizing nrlib RNG
; (DEFUN |Rng| ...) is being compiled.
;; The variable |Rng;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

      finalizing nrlib RNS-
Warnings:

```

```

      [1] patternMatch: not known that (SetCategory) is of mode (CATEGORY domain (SIGNATURE round (S S)) (S

```

section

```

      finalizing nrlib RNS
; (DEFUN |RealNumberSystem| ...) is being compiled.
;; The variable |RealNumberSystem;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

      finalizing nrlib SETAGG
; (DEFUN |SetAggregate| ...) is being compiled.
;; The variable |SetAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |SetAggregate;| ...) is being compiled.
;; The variable |SetAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

      finalizing nrlib SETCAT
; (DEFUN |SetCategory| ...) is being compiled.
;; The variable |SetCategory;AL| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

      finalizing nrlib SINT
Warnings:
      [1] hash: signature of lhs not unique: $$ chosen

```

section

```

      finalizing nrlib SINT
      (SPADLET |$noSubsets| NIL)
      Value = NIL

```

section

```

    finalizing nrlib STAGG
; (DEFUN |StreamAggregate| ...) is being compiled.
;; The variable |StreamAggregate;AL| is undefined.
;; The compiler will assume this variable is a global.
; (DEFUN |StreamAggregate;| ...) is being compiled.
;; The variable |StreamAggregate;CAT| is undefined.
;; The compiler will assume this variable is a global.

```

section

```

    finalizing nrlib SYMBOL
Warnings:
  [1] latex:  s has no value
  [2] latex:  sc has no value

```

section

```

    finalizing nrlib TSETCAT-
Warnings:
  [1] basicSet:  p has no value
  [2] basicSet:  gps has no value
  [3] basicSet:  bps has no value
  [4] initials:  lip has no value
  [5] initiallyReduced?:  red has no value
  [6] reduce: signature of lhs not unique: PPS(Mapping P P P)(Mapping (Boolean) P P) chosen
  [7] rewriteSetWithReduction:  rs has no value
  [8] select: signature of lhs not unique: (Union P failed)SV chosen
  [9] collectQuasiMonic:  newlp has no value

```

section

```

    finalizing nrlib TSETCAT
; (DEFUN |TriangularSetCategory| ...) is being compiled.
;; The variable |TriangularSetCategory;AL| is undefined.

```

```
;; The compiler will assume this variable is a global.
; (DEFUN |TriangularSetCategory;| ...) is being compiled.
;; The variable |TriangularSetCategory;CAT| is undefined.
;; The compiler will assume this variable is a global.
```

section

finalizing nrlib UFD-

Warnings:

```
[1] squareFreePart: not known that (IntegralDomain) is of mode (CATEGORY domain (SIGNATURE
```

section

```
finalizing nrlib UFD
; (DEFUN |UniqueFactorizationDomain| ...) is being compiled.
;; The variable |UniqueFactorizationDomain;AL| is undefined.
;; The compiler will assume this variable is a global.
```